Infinite dessin orders, projective resolutions and spectral determinants

Extended proofs, Schur functors and Auslander–Reiten calculations

Research working draft

November 6, 2026

Abstract

We work throughout with infinite-dimensional locally gentle algebras and completed surface orders arising from explicit cellular clean dessins on compact Riemann surfaces. We construct their projective resolutions, Cartan determinants and dual dessins; develop determinant semi-invariants, Schur modules and cyclic-quiver character kernels; and calculate actual lattice almost-split sequences and an affine-type diamond quotient. A constrained-path Fock algebra retains the gentle relations and has a compact extension with incidence boundary map. Its degree spectral triple and an inserted arithmetic Fredholm family are explicit. Jacobi sums provide theta-dessin cellular Frobenius determinants with a positive local metric; inert primes obstruct the same real two-dimensional carrier. Recent spectral approaches are compared with their actual hypotheses. Selected power-series, module and matrix statements are verified in Lean4. A derived branch functor repairs projective determinant multiplicity; tame descent recovers a hereditary order but forces multiplicities for full-deck coefficient kernels. We also give a requested-error adaptive theta/readout construction for three fixed completed Dirichlet functions, with executed rational training and explicit sampling budgets, and derive a cutoff-uniform original-family type-I remainder envelope, retaining its unresolved exponent loss. A classical quadratic pivot refines the cube-factor sector, and a joint Gauss identity preserves conductor phases. An ordinary all-character conductor comparison improves the unsigned raw envelope to $H+DH^{1/6}+(DH)^{2/3}$, while an executed finite arithmetic Fourier readout certifies correlation and signed-observable errors without asserting a varying-family signed saving. A moving common-profile construction aligns the actual coefficients on nonprincipal sixth-power rows, and an exact joint example rules out variance-only centering; the fixed-profile signed target remains open. An exact conductor-12 character twist and gcd–Möbius inversion give a separate bound for the signed coprime slice, with the former 7/6 comparison retained as history. A signed common-divisor zeta expansion and a complete-column exact laboratory retain every raw mask; weighted divisor certificates do not close the required asymptotic saving. An exact rough-prime support identity gives an explicit target-budget mask correction; the remaining coherent signed main term is still unestimated. A pinned fixed-order preprint theorem, applied to the physical sixth-power-free radicands, improves the global unsigned critical power to 9/8 and the $A_6 = 1$ signed slice to 13/12. Mean-retaining coherent reductions and conditional scalar-input diagnostics keep the fixed-profile signed target open. The arithmetic positive determinant comparison remains open; no proof of GRH is claimed. Primitive root normalization, all-component finite dual prefixes, actual-cutoff Schur transfer and a full equal-norm localization retain the coherent unequal-norm target as an open problem. An actual Gauss-weighted short-divisor tail admits a physical cubic-matrix estimate. A ramified theta construction now covers the actual fixed finite-character family, and a whole-factor partition bounds the original pure finite dual prefix with normalized power 101/144, retaining every ray, raw zero and finite cutoff. The infinite-frequency complement and full signed assembly remain open; the whole signed and unsigned exponents are unchanged. The paired cutoff analysis gives target-size estimates on stated pure and mixed selections of the signed equality–complement cross, with an exact ordered divisor identity.

Contents

1 Cellular dessins and their infinite orders . . . 8
2 Projective resolutions, dual dessins and determinants . . . 9
2.1 Auxiliary staircase resolutions over the order . . . 10
2.2 A monotone quantum comparison with exact finite ranks . . . 11
2.3 A correct DG model of the infinite node . . . 12
3 Determinant semi-invariants and Schur constructions . . . 13
3.1 Young tableaux and a cyclic representation stratum . . . 14
4 Lattice almost-split sequences and genuine diamond quotients . . . 15
4.1 Polynomial strings, completed lattices and the Fukaya category . . . 16
5 Leavitt relations, the constrained Fock algebra and spectrum . . . 17
5.1 A graph presentation and an actual pullback . . . 18
5.2 A separate diagonal-fiber C* completion retaining the pairing . . . 19
6 Arithmetic attachment, descent and the remaining comparison . . . 20
6.1 A constructive arithmetic benchmark on the theta dessin . . . 21
6.2 A genuine cellular carrier, and what still fails for the order . . . 21
6.3 Arbitrary projective actions: an obstruction and a genuine determinant repair . . . 22
6.4 Tame descent: a surviving order and a sharper obstruction . . . 24
6.5 Whole components, derived endpoints and determinant relations . . . 26
7 A genuine theta covering and its categorical limits . . . 26
8 Verification and research frontier . . . 28
9 Why an absolute coprime pair-square estimate cannot close the short rows . . . 29
10 A signed Möbius decomposition and its termwise obstruction . . . 30
11 A full-mask Hecke expansion for the type-I piece . . . 31
12 Actual renormalization retains the effective action . . . 33
12.1 Genuine dessin incidence models . . . 33
12.2 Quantum thermal effective Hamiltonians . . . 34
13 Linear relations and functorial branch filtrations . . . 34
14 A kernel-checked extension over the infinite dessin algebra . . . 35
15 Learned fields, width limits and quantum geometry . . . 36
16 Complementary physical datasets and checked dynamics . . . 38
S1 Expanded dessin, resolution and derived calculations . . . 39
S1.1 The cycle order in coordinates . . . 39
S1.2 The two-arm complex and its calculations . . . 39
S2 Semi-invariants, tableaux and explicit intertwiners . . . 40
S2.1 Derived information retained by a presentation . . . 40
S2.2 One Young operator and one LR decomposition in coordinates . . . 41
S2.3 Two precise functorial limitations . . . 42
S3 Full lattice and diamond diagrams . . . 42
S3.1 The nodal exact diagram . . . 42
S3.2 Chebyshev geometry and the exact quotient . . . 43
S3.3 Nine actual strings and all maps . . . 44
S3.4 The pure-injective quotient and what its dimension means . . . 45
S4 The operator extension in detail . . . 45
S4.1 A corrected cycle lift and the integer boundary map . . . 45
S4.2 Spectral domain and arithmetic trace . . . 46
S5 Abel determinant estimates and the arithmetic program . . . 46
S5.1 Precisely how classical and recent sources enter . . . 48
S5.2 A concrete sequence of remaining constructions . . . 48
S6 Graph pullbacks and completions with specified fibers . . . 49
S6.1 The complete Fock graph presentation . . . 49
S6.2 A genuine application of the supplied pullback theorem . . . 50
S6.3 Diagonal residue obstruction and the corrected pullback . . . 50

S6.4 Generation, faithfulness and the exact failure of extension . . . 51
S6.5 Invariant basis number and the Leavitt comparison . . . 52
S7 String finiteness and the smooth quotient surface . . . 52
S8 Arithmetic character sums and the cellular determinant . . . 53
S8.1 The uniform calculation and its exceptional cases . . . 53
S8.2 Deck descent and the actual cellular basis . . . 54
S8.3 An explicit chain lift and determinant cancellation . . . 55
S8.4 A decisive restriction at inert primes . . . 55
S8.5 The remaining order and global spectral obligations . . . 56
S9 The theta order interface beyond tensor insertion . . . 56
S9.1 Normal forms and the special fiber . . . 57
S9.2 All projectives and all linear endomorphisms . . . 57
S9.3 An actual integral determinant functor of multiplicity one . . . 58
S9.4 The exact rotation obstruction and its allowed escape . . . 59
S10 Explicit tame descent and the full-deck coefficient test . . . 59
S10.1 The action on genuine formal series . . . 60
S10.2 A projective symmetry survives on the endomorphism order . . . 60
S10.3 Determinants, Schur weights and the boundary correction . . . 61
S10.4 Why the local Jacobi plane does not descend by this repair . . . 62
S10.5 The full orbit is the normalization, and arbitrary kernels still fail . . . 62
S11 Literal formal and computational coverage . . . 63
S12 Explicit covering calculations and derived transport . . . 65
S13 Weighted profiles, exact arithmetic examples and the surviving task . . . 66
S14 Exact factorization, retained zeros and the failed Cauchy route . . . 67
S14.1 Proof of the regrouping with original coefficients . . . 67
S14.2 The two conductor scales are unchanged . . . 67
S14.3 Why a first generic Cauchy step does not supply the saving . . . 68
S14.4 Exact finite checks and formal boundaries . . . 68
S15 Primitive conductors, deleted factors and the scalar contour . . . 69
S15.1 Constructing the column character without losing its zeros . . . 69
S15.2 Why the half-plane above one half is available . . . 69
S15.3 Mellin inversion, residues and profile uniformity . . . 70
S15.4 Verification boundaries . . . 71
S16 Controlled errors and a trained full-action RG approximation . . . 71
S16.1 Executed finite calculations . . . 72
S16.2 A quantum-feature model trained on actual RG . . . 72
S16.3 Quantum TokenGT and the geometric error-correction boundary . . . 73
S16.4 An actual finite gentle cut for the categorical interface . . . 73
S17 Relation calculations, splitting functors and determinant limits . . . 74
S17.1 Correct graphs and stable spaces . . . 74
S17.2 Proof of the dessin-category equivalence . . . 74
S17.3 Functoriality and the actual filtration . . . 75
S17.4 Strings, bands, Hom and one non-rigid string . . . 75
S17.5 A determinant comparison with its exact limits . . . 76
S18 Normal forms and the full module action . . . 76
S18.1 Why the polynomial matrix algebra is the path algebra . . . 76
S18.2 Coefficient selectors without a truncation . . . 77
S18.3 Constructing every normal image from actual arrows . . . 78
S18.4 Spanning, multiplication and the inverse coordinate map . . . 79
S18.5 Exactness for arbitrary algebra elements . . . 79
S18.6 Formal and computational coverage . . . 80
S19 Physical generative models and their verification . . . 80
S19.1 From a dessin to an actual quantum RG target . . . 80
S19.2 Diffusion and exact versus approximate reverse controls . . . 81

S19.3 Flow matching with a measurable target velocity . . . 82
S19.4 Architecture, matched controls and actual GPU results . . . 82
S19.5 A coherent learned purification flow . . . 83
S19.6 Width, fields, RG and arithmetic obligations . . . 84
S20 Covers, paths, pants and quantum games . . . 85
S20.1 Which free group and which universal cover? . . . 85
S20.2 Actual infinite-algebra path modules and unbounded resolutions . . . 86
S20.3 Pants tensors, physical dilation and sewing boundaries . . . 87
S20.4 Quantum games and the Connes-embedding distinction . . . 88
S20.5 Tropical accelerator reference designs and reproducibility . . . 88
S21 Construction, curation and interpretation . . . 89
S21.1 A record is an evidence-bearing physical task . . . 89
S21.2 Effective material families and gap guards . . . 90
S21.3 Molecular geometry and forces . . . 90
S21.4 What the split certificate proves . . . 90
S21.5 Motion, charge modulation and exchange . . . 91
S21.6 Source-backed peptides and genuine boundaries . . . 92
S21.7 Circuits and geometric visualization . . . 92
S22 Molecular structure, complex transport and learned chemistry . . . 93
S23 Geometry, orbitals and generative time . . . 93
S23.1 Chemical connectivity and source secondary structure . . . 93
S23.2 What the complex transport animation computes . . . 93
S23.3 An analytic orbital benchmark with actual nodes . . . 94
S23.4 A useful periodic chemical target . . . 94
S23.5 Score diffusion on the torus . . . 95
S23.6 Wrapped-geodesic flow matching . . . 95
S23.7 Physical state preparation, RG and phase-conditioned design . . . 96
S23.8 What the quantum hybrid actually contributes . . . 96
S24 Learning objectives with explicit analytic guarantees . . . 97
S25 A certifiable analytic learning lane . . . 97
S25.1 Targets, fields and arithmetic scope . . . 97
S25.2 A finite holomorphic readout theorem . . . 97
S25.3 The infinite-width limit and its actual rate . . . 98
S25.4 An unconditional primitive-Dirichlet size oracle . . . 99
S25.5 Additional objectives and what they can certify . . . 100
S25.5.1 Contour-aware approximation and active learning. . . . 100
S25.5.2 Arithmetic and functional-equation residuals. . . . 100
S25.5.3 Operator and determinant refinement. . . . 100
S25.5.4 Certificate-guided moment and algebra learning. . . . 100
S25.5.5 Quantum kernels, width and proof search. . . . 100
S25.6 A decisive obstruction to a false universal guarantee . . . 101
S25.7 Exact rational constants without transcendental rounding . . . 101
S26 Learning nonlocal strategies and checking their certificates . . . 101
S27 Exact game calculations and the embedding boundary . . . 102
S27.1 CHSH bounds and a constructive tilted strategy . . . 102
S27.2 A full magic-square strategy . . . 102
S27.3 Which learning objectives carry proofs? . . . 103
S28 Hubs, spokes and the topology retained by a dessin . . . 104
S29 Coherent electron-density clouds in an explicit spatial basis . . . 105
S30 A finite, validated approximation result . . . 106
S31 What this does and does not certify . . . 107
S32 Validated primitive operations . . . 107
S33 Recomputing the compact growth constants . . . 108
S34 The completed-function integral and its tails . . . 109

S35 High-order integration with a certified remainder . . . 109
S36 Reproduction and remaining interfaces . . . 110
S37 Requested-error readouts from preserved certified labels . . . 110
S38 Objective semantics, precision floors and replay . . . 111
S39 Requested-error theta samples for three fixed conductors . . . 112
S40 Adaptive sampling parameters and precision termination . . . 114
S41 An end-to-end adaptive analytic readout . . . 116
S42 Termination, compression and the boundary of the guarantee . . . 117
S43 An arithmetic-family moment with every original row retained . . . 118
S43.1 The exact masked remainder and the remaining loss . . . 119
S44 Family identification, the approximate functional equation and masks . . . 120
S44.1 A finite comparison with the literal Kummer rows . . . 121
S44.2 Fixed twists, exact tame conductors and the squarefree sieve . . . 121
S44.3 Removing the row-dependent functional-equation weights . . . 122
S44.4 All-ideal summation and principal rows . . . 123
S44.5 Poisson transfer with an explicit constant . . . 124
S44.6 The separately scoped unitary infinity type . . . 124
S44.7 Keeping the full type-I mask during transfer . . . 125
S44.8 What remains open and what has been verified . . . 126
S45 Recombining the actual weighted cutoff . . . 127
S46 Euler packaging, coprimality and the analytic boundary . . . 128
S47 A cutoff-uniform envelope for the original masked remainder . . . 129
S48 All-ideal reciprocity and the surviving block obstruction . . . 130
S48.1 Exact recombination retains the primitive corrections . . . 131
S48.2 The powered squarefree input and all-ideal pivot . . . 131
S48.3 Literal rows, reciprocal orientation and principal count . . . 132
S48.4 What the envelope still fails to prove . . . 133
S49 High cutoffs, actual bilinear factors and a signed covariance test . . . 133
S50 Proof details for the high-window and two-cutoff tests . . . 135
S50.1 The coefficient lower bound and its precise limitation . . . 135
S50.2 What legal factor Cauchy actually proves . . . 135
S50.3 A smooth profile and the exact covariance computation . . . 136
S50.4 Double Perron without a divergent critical-strip series . . . 136
S51 A quantitative covariance budget without a new exponent . . . 137
S52 Proof and arithmetic boundary of the covariance budget . . . 138
S52.1 The positive-contraction calculation . . . 138
S52.2 The balanced-divisor second moment . . . 139
S52.3 What the commutator actually controls . . . 139
S52.4 The complete near-saturating face . . . 140
S53 The fourth-moment conductor and its full raw zero set . . . 140
S54 Scope of the raw fourth-moment reduction . . . 141
S55 A quadratic pivot on the cube factor . . . 143
S56 Proof and scope of the quadratic pivot . . . 144
S57 Joint completion before conductor absolute values . . . 145
S58 The surviving joint identity and its missing estimate . . . 145
S59 A radical-conductor refinement of the raw envelope . . . 146
S60 Proof of the ordinary-conductor refinement . . . 147
S60.1 Primitive Gauss sums at every conductor . . . 147
S60.2 The classical ordinary sieve in these lattice conventions . . . 148
S60.3 Raw zeros, tuple multiplicity and block transfer . . . 149
S60.4 Uniform coefficient and principal-residue assembly . . . 150
S61 Learning the actual signed arithmetic correlations . . . 150
S62 Finite phase features, gradients and exact verification . . . 151
S63 Why the common profile must stay fixed . . . 153

S64 Proof and joint-mean boundary of the moving-profile construction . . . 154
S64.1 Norm groups and primitive conductors . . . 154
S64.2 Exact alignment, with arbitrary overlapping sixth factors . . . 155
S64.3 The coprime-triple density and the selected $h = 1$ block . . . 156
S64.4 Projection preserves the coupled problem, not a saving . . . 157
S64.5 A nonzero orientation mean on actual nonprincipal rows . . . 157
S65 A bound for the signed coprime slice and an exact character twist . . . 158
S66 Proof of the signed-slice budget and its surviving analytic burden . . . 159
S66.1 Gcd inversion before an absolute bound . . . 159
S66.2 The correctly oriented Jacobi generator . . . 160
S66.3 The actual two-factor completion, not a new row kernel . . . 161
S66.4 A central reciprocal remains after the gauge . . . 162
S66.5 The fixed-profile remaining-sector diagnostic . . . 163
S67 The signed divisor mean and a complete-column laboratory . . . 164
S68 Proofs and limits of the signed-divisor development . . . 166
S68.1 Actual divisor incidence and a finite error certificate . . . 166
S68.2 A signed common-divisor Euler product . . . 167
S68.3 Finite Perron heights and the exact eight-state numerator . . . 167
S68.4 A uniform common-profile Mellin embedding of the actual mean . . . 169
S68.5 A nonzero short-state correction and the unproved step . . . 169
S68.6 Exact arithmetic and source-bound computational scope . . . 170
S69 A target-sized budget for rough-prime masks . . . 171
S70 Proof of the rough-prime mask budget . . . 173
S70.1 The common ideal divisor and the exact rough part . . . 173
S70.2 An exact disjoint error identity . . . 174
S70.3 Elementary ideal and coefficient estimates . . . 175
S70.4 The normalized row-mask budget . . . 176
S70.5 The residual intersection budget . . . 176
S70.6 The exact unit-residual correction and the clean main . . . 176
S70.7 Finite control scope . . . 177
S70.8 A coherent prime-substitution budget, and its remaining cost . . . 177
S71 The coherent signed main and its remaining inputs . . . 179
S72 Proofs for the coherent main reductions . . . 181
S72.1 Removing only the auxiliary smooth restriction . . . 181
S72.2 An exact two-jump masked Mertens functional . . . 182
S72.3 Deleted-prime transfer and its fixed-field subpower input . . . 183
S72.4 The joint matrix, raw zeros and its compression . . . 185
S72.5 Exact Stieltjes boundary and squarefunction criterion . . . 186
S72.6 Mellin Parseval retains equal-norm coherence . . . 186
S72.7 Proof of the inherited smooth-divisor localization . . . 187
S72.8 Proof of the conditional zero-free input diagnostic . . . 188
S73 A sieve comparison indexed by the physical radicand . . . 189
S74 Proof of the physical-row sieve application . . . 191
S74.1 The precise literature input and the primary generator . . . 191
S74.2 The complete raw row family and the sixth factor . . . 192
S74.3 The raw global sum and the principal assembly . . . 192
S74.4 The signed $A_6 = 1$ common-divisor form . . . 193
S74.5 The updated short certificate and its unresolved input . . . 194
S75 Coherent dual bands, actual cutoffs and arithmetic localization . . . 194
S75.1 The original form and the scope of the dual representation . . . 195
S75.2 Primitive normalization and an exact finite-cutoff kernel . . . 195
S75.3 What finite prefixes and actual divisor bands now satisfy . . . 197
S75.4 Coherent means that can be localized, and those that remain . . . 198
S76 Proofs for the coherent-band comparisons . . . 199

S76.1 Finite Fourier transforms and the primitive Gaussian . . . 199
S76.2 Transport to every original class and ray component . . . 201
S76.3 The sharp capacity, square-divisor signs and exact Poisson . . . 201
S76.4 The joint bilinear estimate behind finite prefixes . . . 202
S76.5 Finite-height Schur transfer, including the operator error . . . 204
S76.6 The cubic transposition in an identity divisor class . . . 205
S76.7 Classification and proof of the full equal-norm sector . . . 207
S76.8 A common literal modulus and the surviving ray mean . . . 208
S77 An arithmetic Gauss tail with the original finite cutoff . . . 210
S77.1 Original objects and a fixed common profile . . . 210
S77.2 The actual retained mean and its finite leading model . . . 211
S77.3 The root is an incomplete cubic Gauss weight . . . 212
S77.4 A signed short-divisor tail from a physical cubic matrix . . . 212
S77.5 What has been reduced, and what remains . . . 213
S77.6 The exact discrete top prefix, with every ray retained . . . 213
S78 Proofs for the finite-cutoff Gauss tail . . . 215
S78.1 The coherent main retains all sixth-part and ray data . . . 215
S78.2 Exact good-modulus Gauss factors and separate primitive S values . . . 216
S78.3 Matched finite endpoint models and a real-axis remainder . . . 217
S78.4 All source and ray phases give bounded multiplicative columns . . . 218
S78.5 Two exact cutoff boxes expose a physical cubic matrix . . . 219
S78.6 Cubic Cauchy before the distinguished-divisor triangle . . . 219
S78.7 Full pair support and the original finite k reassembly . . . 220
S78.8 The critical reduction is not a bound for its residual mean . . . 221
S78.9 Direct discrete frequencies require no principal-density approximation . . . 221
S79 Ramified theta transforms and the original pure dual prefix . . . 223
S79.1 The normalized object and its original support . . . 223
S79.2 Actual finite characters and a sharp ramified row . . . 224
S79.3 The full original coefficient and its finite prefix . . . 225
S80 Proofs of the ramified transform and full-coefficient prefix bound . . . 226
S80.1 Fixed residue data, angular order and raw evaluations . . . 227
S80.2 Known integral cusps and an explicit absolute coefficient norm . . . 228
S80.3 Exact rational reduction and the actual Kubota multiplier . . . 229
S80.4 Radial cusp constants, angular derivatives and the Mellin Jacobian . . . 230
S80.5 The integral-index primary projector and raw cube completion . . . 231
S80.6 The coarse dual bound, safe convexity and literal Perron endpoints . . . 232
S80.7 The actual annular coefficient and exact eleven-factor expansion . . . 233
S80.8 Whole-factor cubic matrices and the large-prefix part . . . 234
S80.9 The full original opposite-column mask in the small-prefix row . . . 234
S80.10 The exact discrete prefix and the final normalization . . . 235
S80.11 Verification scope and the remaining arithmetic comparison . . . 236
S81 Paired cutoffs, exact divisor signs and proved regimes . . . 236
S81.1 The selected pairing . . . 237
S81.2 Target bounds on the selected cross . . . 238
S81.3 Ordered assignment signs . . . 238
S82 Sharp insertion and ordered cutoff estimates . . . 239
S82.1 The middle subtraction and scalar comparisons . . . 239
S82.2 The sharp moment and the grouped sparse tail . . . 241
S82.3 Coupling the dual to the shorter ordered factor . . . 243
S82.4 Pure central and comparable regions . . . 244
S82.5 Pure long insertion for every erasure length . . . 244
S82.6 Pure long erasure and the intermediate gap . . . 245
S82.7 Pure short insertion over large erasures . . . 246
S82.8 Mixed ratios and the critical-capacity strip . . . 246

S83 Wide cutoff corrections and full-frequency compensation 247
S84 Derivations for the wide cutoff correction 248
S84.1 The strict band and its prime terms 248
S84.2 The fixed-intersection scalar estimate 249
S84.3 The complete partner sum and the equality prefix 250

1 Cellular dessins and their infinite orders

We start with the surface-order constructions of Schreiber [1, 2], retaining the original sources unchanged. The generic-resolution source [3] is background, not a substitute for exactness of a particular complex. All primary algebra examples below come from explicit clean Belyi maps. Finite-dimensional objects occur as modules or specified quotient strata of those infinite algebras, not as replacements for them. The initial arithmetic target is a primitive nonprincipal Dirichlet L-function. Finite-order Hecke characters, Dedekind zeta functions and general Artin representations require separate comparison theorems; Artin holomorphy is not assumed.

A clean dessin on a compact oriented surface is specified by a nonempty finite dart set H, a fixed-point-free involution α, and a permutation σ such that $\langle \alpha, \sigma \rangle$ is transitive. Contract the white vertices of degree two. The resulting cellular graph has vertices the σ-orbits, edges the α-orbits, and faces the ϕ-orbits, where

$$\tau = \sigma^{-1}, \qquad \phi = \alpha\sigma^{-1}, \qquad 2 - 2g = \#\,\mathrm{cyc}(\sigma) - |H|/2 + \#\,\mathrm{cyc}(\phi).$$

Write $[h] = \{h, \alpha h\}$. Our quiver has vertices these edges and arrows $a_h : [h] \to [\tau h]$. Paths are composed in travel order: ab traverses a first. Set

$$A = kQ/(a_h a_{\phi h} : h \in H), \quad J = A_{>0}, \quad \Lambda = \varprojlim A/J^m.$$

Each vertex has two incoming and two outgoing arrows. Each arrow has precisely one permitted successor $a_{\tau h}$ and one forbidden successor $a_{\phi h}$; these are distinct because α has no fixed points. The same holds for predecessors. This is the *locally gentle* condition. Every permitted cycle repeats indefinitely, so A is infinite dimensional. We reserve "gentle" without qualification for the convention additionally imposing finite dimension.

Proposition 1.1 (The actual split order)**.** *For a permitted cycle c of length ℓ, let B_c be the hereditary cycle order in $M_\ell(k[[u_c]])$: diagonal and forward matrix entries are in $k[[u_c]]$, and entries crossing the chosen cycle cut are in $u_c k[[u_c]]$. Then Λ is the suborder of $B = \prod_c B_c$ whose two diagonal constant terms labelled $h, \alpha h$ agree. With $R = k[[u]]$ acting by $u \mapsto (u_c I)_c$, it is finite free of rank $\sum_c \ell(c)^2$; the conductor contains uB.*

Proof. In a cycle order a path of length less than ℓ is the corresponding matrix unit, multiplied by u_c exactly when it crosses the cut; another full turn multiplies it by u_c. Thus completed paths in each permitted branch give all entries of B_c. An arrow belongs to only that branch, whereas $e_{[h]}$ has diagonal entries at both occurrences $h, \alpha h$. Hence the only condition on branchwise formal coefficients is the displayed equality of diagonal constants. This proves both injectivity and surjectivity of the path-normal-form identification. Multiplication respects it: switching branches across a paired edge is precisely a forbidden pair. The quotient B/Λ is killed by u, so the submodule has the same generic rank as B. It is torsion free and finite over the DVR R, hence free of that rank. The path and u-adic topologies agree: $uA \subset J$, and if L is the longest permitted cycle, any path of length at least $L + 1$ is u times a shorter nonempty path. Thus $J^{L+1} \subset uA$. This also justifies the completion used in the identification. □

The path-degree variable q below is not u: one full turn has path degree $\ell(c)$ but u-degree one. Formal, Laurent and operator-norm completions will likewise remain distinct.

Here are five explicit global inputs. Permutations list darts starting at zero; $E = |H|/2$.

Surface and Belyi map β	σ	α	$(g, \operatorname{rank}_R \Lambda)$
$\mathbb{P}^1$, $4z(1-z)$	$(0)(1)$	(01)	$(0, 2)$
$\mathbb{P}^1$, $(z+1)^2/(4z)$	(01)	(01)	$(0, 4)$
$y^2 = x^3 - 1$, x^3	$(012)(345)$	$(03)(14)(25)$	$(1, 18)$
$\mathbb{P}^1$, $4z^3/(1+z^3)^2$	$(012)(345)$	$(03)(15)(24)$	$(0, 18)$
$y^2 = x^3 - x$, x^2	(0123)	$(02)(13)$	$(1, 16)$

The first is the nodal interval: $A = k[x, y]/(xy)$ and $\Lambda = k[[x, y]]/(xy)$. The second has $A = k\langle x, y\rangle/(x^2, y^2)$, still infinite dimensional. The two theta dessins have the same two permitted three-cycles but different face cycles, respectively (051324) and $(04)(13)(25)$. The elliptic square has one permitted and one forbidden four-cycle. For example x has pole order two on the cubic $y^2 = x^3 - 1$; x^3 has ramification $(3, 3)$ above zero, $(2, 2, 2)$ above one since $x^3 - 1 = y^2$, and (6) above infinity. The rational theta map has passport $(3, 3), (2, 2, 2), (2, 2, 2)$. These ramification computations, and the analogous direct zeros/poles in the other rows, verify cellular Belyi inputs rather than merely abstract quivers.

2 Projective resolutions, dual dessins and determinants

Use right modules $P_i = e_i A$, $S_i = P_i/P_i J$. Internal shift $M\langle n\rangle$ places its old generator in degree n. Let $\mathcal{R}_n(i)$ be the *indexing words* of length n from i with every adjacent pair forbidden, including e_i for $n = 0$. Such a word is usually zero in A; it still labels a summand.

Theorem 2.1 (Exact resolutions)**.** *There is a minimal graded projective resolution, with $P_0(i) = e_i A$ and for $n \geq 1$,*

$$P_n(i) = \bigoplus_{p = a_1 \cdots a_n \in \mathcal{R}_n(i)} e_{t(a_n)} A\langle n\rangle, \qquad d_n(x)_p = a_n x \text{ in the summand } a_1 \cdots a_{n-1}.$$

The augmentation $P_0(i) \to S_i$ is the quotient. For the completed order the identical finite-rank projective terms and formal-series maps resolve the residue simple. Ext and Tor against residue simples count forbidden words with the prescribed endpoints. In the graded case Ext has matching internal shift n.

Proof. The maps are right linear and degree zero. Adjacent differentials multiply by a forbidden pair. The augmentation kernel is the direct sum of aA over initial arrows. For any arrow a, multiplication $e_{t(a)}A \to e_{s(a)}A$ kills exactly the permitted paths beginning with its forbidden successor $f(a)$. All other basis paths survive to distinct basis paths. Thus its kernel is $f(a)A$. For a fixed prefix, different last arrows have disjoint surviving supports; there is no cancellation between summands. This proves exactness inductively. For formal series the same assertion holds coefficient by coefficient and identifies the entire closed kernel, not just its dense polynomial subspace. Terms are idempotent projectives and all entries are arrows. Consequently applying Hom or tensor with a residue simple kills every differential, proving minimality and the Ext/Tor count. In A minimality is graded; it does not require that J be its ordinary Jacobson radical. □

The two arms never terminate in this closed class. Their labelled patterns follow ϕ-cycles. An ungraded positive syzygy tail can have a shorter period by permutation of summands: the nodal

interval has labelled period two but unordered tail period one; the elliptic theta has labelled period six and unordered tail period three. A simple need not be isomorphic to a positive syzygy; for the node it is R-torsion, whereas the positive syzygies are lattices.

Let $C_{ij}(q) = \sum_{d\geq 0} \dim_k(e_i A_d e_j) q^d$ and $E_{ij}(q) = \sum_{n\geq 0}(-1)^n \dim_k \operatorname{Ext}^n_A(S_i, S_j) q^n$. Degreewise Euler characteristic of the resolution gives $EC = I$: every coefficient is finite, because a summand in homological degree n starts in internal degree n. Constant terms are I, so the inverse is two-sided. Already signed $E(q)$ must not be evaluated at $-q$ again.

Proposition 2.2 (Dual dessin and cycle determinant)**.** *The complementary-relation algebra is another closed dessin algebra, with $\alpha^{\#} = \alpha$, $\sigma^{\#} = \phi^{-1}$, permitted successor ϕ, and forbidden successor τ. It exchanges graph vertices and faces and preserves genus. Its Cartan matrix satisfies $E(q) = C_{A^{\#}}(-q)$, with the opposite convention if required for Yoneda composition. For independent commuting arrow weights w_a,*

$$\det C_A(\mathbf{w}) = \frac{\prod_{f\in \mathrm{cyc}(\phi)}(1-(-1)^{|f|}w(f))}{\prod_{c\in \mathrm{cyc}(\sigma)}(1-w(c))}.$$

Proof. $\alpha(\sigma^{\#})^{-1} = \alpha\phi = \tau$, so the claimed relations and face data follow directly; the connectedness group is unchanged. To prove the determinant, put $S_{v,a} = w_a$ when $s(a) = v$, and $T_{a,v} = 1$ when $t(a) = v$, zero otherwise. On arrow states let $U_{a,b} = w_b$ for permitted adjacency and $V_{a,b} = w_b$ for forbidden adjacency. Then $TS = U + V$ and the path sum is $C = I + S(I-U)^{-1}T$. Sylvester's identity gives $\det C = \det(I+V)/\det(I-U)$. Each transition is a weighted permutation, whose cycle determinant is respectively $1-(-1)^{\ell}w(c)$ and $1-w(c)$. Everything holds in formal power series, hence as rational functions. This supplies a direct proof of the Bessenrodt–Holm formula in the closed dessin class [4]. □

The five univariate determinants are, in the order above,

$$\frac{1+q}{1-q}, \quad \frac{1+q}{1-q}, \quad \frac{1-q^6}{(1-q^3)^2}, \quad \frac{(1-q^2)^3}{(1-q^3)^2}, \quad 1.$$

The last cancellation does not imply finite global dimension: its forbidden resolution is infinite. These rational functions have cycle poles/zeros, not arithmetic prime Euler factors.

For clarity, a complex is a sequence with $d^2 = 0$, its homotopy category identifies maps differing by $dh + hd$, and $D(A)$ inverts maps inducing cohomology isomorphisms. A perfect complex is represented by a bounded complex of finite projectives. Our infinite simple resolutions compute derived Hom and tensor but are generally not perfect. The two-term presentation $[P_1 \to P_0]$ is perfect even if its kernel is nonzero. Quadratic linearity is a Koszul statement; it is not by itself an equivalence between all completed derived categories. Likewise a finite-dimensional gentle graded-curve theorem [6] cannot simply be transferred to these closed infinite orders. Here the surface walks explicitly compute the resolution endpoints: τ turns round a graph vertex and ϕ follows a face. For longer string objects all maps can be computed from the occurrence matrices; a universal geodesic-intersection classification for this completed category remains a separate theorem.

2.1 Auxiliary staircase resolutions over the order

The two filtration parameters below are not dessin arrows. Let $R = k[x, y]$ for a field k, and let $I = (g_0, \ldots, g_{m-1})$, $m \geq 2$, where $g_i = x^{a_i} y^{b_i}$,

$$a_0 > \cdots > a_{m-1} = 0, \qquad 0 = b_0 < \cdots < b_{m-1}.$$

The positive outer exponents make R/I finite length. Write $R(-a,-b)$ for a free generator born in bidegree (a,b).

Proposition 2.3 (Adjacent syzygies)**.** *A minimal bigraded free resolution of $M = R/I$ is*

$$0 \longrightarrow \bigoplus_{i=0}^{m-2} R(-a_i,-b_{i+1}) \xrightarrow{d_2} \bigoplus_{i=0}^{m-1} R(-a_i,-b_i) \xrightarrow{d_1} R \longrightarrow M \longrightarrow 0,$$

where $d_1(e_i) = g_i$ and $d_2(f_i) = -y^{b_{i+1}-b_i}e_i + x^{a_i-a_{i+1}}e_{i+1}$. Deleting row i from d_2 gives determinant $(-1)^i g_i$. The Hilbert series numerator is $1 - \sum_i s^{a_i}t^{b_i} + \sum_{i=0}^{m-2} s^{a_i}t^{b_{i+1}}$.

Proof. Each column maps to zero. In a relation $\sum h_i g_i = 0$, all terms except the last contain $x^{a_{m-2}}$. Coprimality with $y^{b_{m-1}}$ implies $h_{m-1} = x^{a_{m-2}}q$. Subtract $qd_2(f_{m-2})$ from the coefficient vector; its last component vanishes. Factor $x^{a_{m-2}}$ from the remaining generators. They form a smaller staircase with the same strict inequalities, ending in a pure y-power. Induction expresses the remaining relation in the preceding adjacent columns. The base of one generator has no relations, since R is a domain. Hence $\ker d_1 = \operatorname{im} d_2$. If $d_2(q) = 0$, its first row gives $-y^{b_1}q_0 = 0$; successive rows then force all $q_i = 0$. This proves injectivity. Every differential entry has positive degree, so reduction modulo (x,y) kills the differentials and the resolution is minimal. In the row-deleted matrix the unique nonzero permutation term uses the negative y entries before row i and the positive x entries after it. Their exponents telescope to b_i, a_i, respectively, and the sign is $(-1)^i$. Finally take the degreewise Euler characteristic; the free summands have Hilbert denominator $(1-s)(1-t)$. □

For $I = (x^2, xy, y^2)$ the underlying sequence is also kernel checked in Lean 4 over the actual ring MvPolynomial(Fin 2, k). The verification proves the kernel/image equalities, the stated ideal image, surjective quotient augmentation, projective free terms, and vanishing of both induced Hom differentials to the constant-coefficient residue field. It uses regular multiplication on the free modules, distinct from the residue action on k. General staircases, graded minimality and the surface-order coefficient cone remain separately scoped prose arguments.

At grade $\nu \in \mathbb{N}^2$, M_ν has dimension one exactly when no generator degree is componentwise at most ν. Its comparable-grade map has rank one exactly when both endpoint grades satisfy this condition. A CW realization takes one loop and attaches a disk once along it at each generator degree. The active indices form an interval: the x inequality selects a final interval, the y inequality an initial interval. With c active disks, the cellular map $k^c \to k$ has every entry one. Thus $\dim H_1 = \mathbf{1}_{c=0}$ and $\dim H_2 = \max(c-1, 0)$, the latter counting active adjacent syzygies. This auxiliary CW complex is not the dessin surface.

For the actual nodal order $\Lambda = k[[\xi,\eta]]/(\xi\eta)$ put $u = \xi + \eta$ and $E = \Lambda/u^2\Lambda$. The basis $(1, \xi, u, u\xi)$ of E satisfies $\xi^2 = u\xi$, $u^2 = 0$, $\eta = u - \xi$. Tensoring the resolution above with $0 \to \Lambda \xrightarrow{u^2} \Lambda \to E \to 0$ gives, over $\mathcal{B} = \Lambda[x,y]$, free ranks $(1, m+1, 2m-1, m-1)$. Indeed the scalar resolution stays exact after tensoring over the field; its degree-zero module is free over Λ on the standard monomials. Multiplication by u^2 is injective there and on all free terms, so the long exact homology sequence of its mapping cone gives positive-degree homology zero and degree-zero module $E \otimes_k M$. In contrast, as an R-module this object has ranks $4(1, m, m-1)$ in its minimal resolution. Neither finite-length E nor these parameter rings replace the infinite parent dessin algebra or its lattice category.

2.2 A monotone quantum comparison with exact finite ranks

Proposition 2.4 (One common channel)**.** *Let T be a positive trace-preserving linear map between finite complex matrix algebras. For density matrices ρ, σ, their trace distance $d(\rho,\sigma) = \frac{1}{2}\|\rho - \sigma\|_1$*

satisfies $d(T\rho, T\sigma) \le d(\rho, \sigma)$. For finitely many labelled states, keep fixed vertex birth grades e_i. At (r, e) include vertices with $e_i \le e$ and edges with distance at most r, and take the flag complex. Then T gives an identity-on-labels inclusion from the input to the output bifiltration, natural in (r, e).

Proof. Write the Hermitian difference as $A = A_+ - A_-$ with $A_\pm \ge 0$. Positivity, the trace norm triangle inequality and trace preservation give $\|TA\|_1 \le \|TA_+\|_1 + \|TA_-\|_1 = \operatorname{tr} A_+ + \operatorname{tr} A_- = \|A\|_1$. Thus every input edge is an output edge; the fixed births preserve vertices, and all cliques are preserved. Increasing either parameter preserves the simplices, and every map is the same label inclusion, so the squares commute. Complete positivity is sufficient but is not needed for this norm argument. □

For a finite inclusion $K \subset L$, work over a field and embed a basis Z_K of source one-cycles into the target edge space. If B_L is the matrix of target two-boundaries, the induced H_1 rank is

$$\rho_1(K, L) = \operatorname{rank}[B_L \; Z_K] - \operatorname{rank} B_L.$$

Its image is $(Z_K + \operatorname{im} B_L)/\operatorname{im} B_L$; source boundaries already map into target boundaries. This proves the formula, including empty complexes. Computing a flag two-skeleton suffices for H_0, H_1, but does not compute the full flag complex's H_2. Integer ceilings of scaled exact distances preserve the inclusion. For numerical distances, frozen integer labels define an exact combinatorial diagram only after their inequality is checked; this is not an interval certificate of the underlying quantum distances. None of these finite ranks identifies a completed arithmetic determinant or proves GRH.

2.3 A correct DG model of the infinite node

Use cohomological degrees and an uncurved differential of degree $+1$. An odd element ε with $d\varepsilon = 1$ contracts the underlying k-complex of every unital left DG module: $h(m) = \varepsilon m$ satisfies $dh + hd = 1$ by the graded Leibniz rule. Thus this example has zero cohomology, contrary to the original ground-field-resolution claim. An isomorphism on H^0 alone is also insufficient for a DG quasi-isomorphism; all degrees must be checked.

Proposition 2.5 (Nodal Koszul model)**.** *Let k be a field, $R = k[\xi, \eta]$ and $f = \xi\eta$ and $K = R \oplus R\varepsilon$, with $|\varepsilon| = -1$, $\varepsilon^2 = 0$, $dR = 0$ and $d\varepsilon = f$. The augmentation $K \to A = R/(\xi\eta)$, $a + b\varepsilon \mapsto [a]$, is a DG-algebra quasi-isomorphism onto the actual infinite nodal dessin algebra. Branch rescaling $\xi \mapsto q\xi$, $\eta \mapsto q^{-1}\eta$, $\varepsilon \mapsto \varepsilon$ for $q \in k^\times$ lifts to a DG automorphism.*

Proof. Multiplication is $(a + b\varepsilon)(c + e\varepsilon) = ac + (ae + bc)\varepsilon$. The differential kills degree zero and sends $b\varepsilon$ to fb. Its square is zero. For two odd inputs the Leibniz right side is $fbc\varepsilon - bfc\varepsilon = 0$; the other homogeneous cases follow by R-linearity. Multiplication by f is injective in the polynomial domain. Hence $H^{-1}(K) = 0$, $H^0(K) = R/(f)$, and all other groups vanish. The augmentation preserves products and kills boundaries. Rescaling fixes f, so its coefficientwise extension commutes with d and has inverse obtained by replacing q with q^{-1}. This is a one-relation Koszul construction [78, 79]. □

Replacing R by $k[[\xi, \eta]]$ gives the completed order: setting each variable separately to zero maps onto the branch pullback, and the kernel consists precisely of mixed monomials, hence equals $(\xi\eta)$. Rescaling is generally semilinear over the coefficient ring, rather than linear over the fixed diagonal DVR $k[[u]]$, $u = \xi + \eta$. It preserves that DVR as a subring exactly when $q^2 = 1$, and fixes it pointwise exactly when $q = 1$. No Hilbert norm or arithmetic spectrum follows.

The corresponding Lean file checks the actual exterior-line algebra, differential, sign-Leibniz, homotopy and polynomial ambient-resolution identities. Full graded/category/completion objects

remain separately scoped. The two theta dessins likewise require endpoint-sensitive Ext: self Ext^1 is zero in both; for the elliptic theta self Ext^3 has dimension two, while for the spherical theta positive even self Ext has dimension two. Outdegree alone does not count self extensions. Minimal A_∞ models use additional graded data [77, §3.3]; m_3 is not automatically a physical curvature.

3 Determinant semi-invariants and Schur constructions

Assume k algebraically closed of characteristic zero for invariant-ring statements. A finite-dimensional representation of the *infinite* A has spaces N_i, arrow matrices $N(a)$ satisfying the monomial relations, and base-change group $G = \prod_i \mathrm{GL}(N_i)$. With column vectors, $N(a) \mapsto g_{t(a)} N(a) g_{s(a)}^{-1}$. Let $P_0 = \bigoplus P_i^{a_i}$, $P_1 = \bigoplus P_i^{b_i}$, and $f : P_1 \to P_0$ be a finite path-matrix map. When $\sum a_i \dim N_i = \sum b_i \dim N_i$, define

$$c_f(N) = \det D_f(N), \qquad D_f(N) = \mathrm{Hom}_A(f, N).$$

It is intrinsically a determinant-line section; chosen bases give this scalar.

Proposition 3.1 (Weights and exact homological scope)**.** *$c_f(gN) = \prod_i \det(g_i)^{b_i - a_i} c_f(N)$. Restricted determinants of this form span all semi-invariants on a closed G-stable relation or nilpotence locus. Set $M = \operatorname{coker} f$. On the square locus,*

$$c_f(N) \neq 0 \quad \Longleftrightarrow \quad \mathrm{Hom}(M, N) = 0 \quad \Longleftrightarrow \quad \mathrm{Hom}(M, N) = \mathrm{Ext}^1(M, N) = 0.$$

If f is injective, $\operatorname{coker} D_f = \mathrm{Ext}^1(M, N)$ *and higher Ext vanish. Without injectivity neither that cokernel equality nor higher vanishing follows.*

Proof. Naturality gives $D_f(gN) = G_1 D_f(N) G_0^{-1}$, where G_j repeats the relevant g_i. Take determinants. The contragredient coordinate-ring action has the inverse character. Schofield–van den Bergh's spanning theorem allows oriented cycles [36, Theorem2.3]. Exactness of invariants for reductive G in characteristic zero lifts a fixed-weight function from any closed stable locus to the ambient representation space; restrict the spanning determinants. Injectivity of the path map need not survive that restriction. Finally $\ker D_f = \mathrm{Hom}(M, N)$. Writing $K = \operatorname{im} f$, the exact sequence with projective P_0 expresses $\mathrm{Ext}^1(M, N)$ as $\mathrm{Hom}(K, N)/\operatorname{im} \mathrm{Hom}(P_0, N)$; the epimorphism $P_1 \to K$ embeds this quotient in $\operatorname{coker} D_f$. A square finite matrix with zero kernel has zero cokernel. If f is injective it is the actual length-one resolution, giving the stronger assertion. □

For the node, $f = \times x$, $M = \Lambda/(x)$, $N = \Lambda/(x, y^2)$ give $\dim \operatorname{coker} D_f = 2$ but $\dim \mathrm{Ext}^1(M, N) = 1$. For the elliptic square name arrows so that $a, b : 0 \to 1$, $c, d : 1 \to 0$, permitted cycle *acbd* and forbidden cycle *adbc*. With $f = a$ and $N(a) = 1$, $N(b) = N(c) = N(d) = 0$, its Hom matrix is [1], yet the subsequent forbidden resolution gives $\mathrm{Ext}^{2,3}(M, N) = k$. Thus nonvanishing is not all-degree derived orthogonality.

Proposition 3.2 (What completion does to invariants)**.** *A continuous d-dimensional representation of Λ kills J^d. On the reduced nilpotent representation variety of fixed dimension vector every invariant regular function is constant. Nonzero-weight semi-invariants need not be constant.*

Proof. Continuity with discrete finite-dimensional target kills some J^m; its Loewy filtration has at most d nonzero strict steps. Choose vertexwise complements to this filtration. An arrow sends a layer p into layers $q \geq p+1$. Scaling layer p by z^p conjugates each component by z^{q-p}. At $z = 0$ the representation tends to the zero-arrow representation. Every invariant therefore has the same value at all points; reducedness and the Nullstellensatz give the ring assertion. In the square example the

scalar $N(a)$ varies on a nilpotent locus and has weight g_1/g_0, so semi-invariants survive. Formal entries in a projective map are evaluated on this locus by truncating modulo J^d, a legitimate finite polynomial operation. □

On the node, $u = x+y$ is regular and A is free of rank two over $k[u]$. The polynomial presentation $\times(u-\lambda)$ for $\lambda \neq 0$ is injective and its determinant on N is $\det(u_N - \lambda I)$, the characteristic polynomial of u_N up to the sign $(-1)^d$. But $u-\lambda$ is a unit in Λ, with its geometric-series inverse, so those modules do not extend continuously to the completion. An R-free lattice and its u-nilpotent quotients are different categories. On a lattice a separately specified endomorphism F has a meaningful relative characteristic polynomial over R; the scalar base action uI alone does not create Frobenius.

3.1 Young tableaux and a cyclic representation stratum

For a nonempty partition $\lambda \vdash m$, choose a numbering T of its boxes. On $V^{\otimes m}$ let $Y_T = R_T C_T$, the unnormalized row sum followed by the signed column sum. The classical identity $Y_T^2 = h_\lambda Y_T$, where h_λ is the hook product, makes Y_T/h_λ an idempotent in characteristic zero. Its image is $S_\lambda(V)$, one Schur copy, rather than the whole symmetric-group isotypic component [31, 32]. Its character is s_λ, computed by semistandard tableaux (weak rows, strict columns), and

$$\dim S_\lambda(k^d) = \prod_{(i,j)\in\lambda} \frac{d+j-i}{h_{ij}}.$$

Since $f^{\otimes m}$ commutes with the place permutations, it induces $S_\lambda(f)$ and respects composition and zero maps. Applying this at every vertex and arrow therefore gives a genuine representation of the monomial dessin algebra: $S_\lambda(N(b))S_\lambda(N(a)) = S_\lambda(0) = 0$ for a forbidden pair. The empty partition does not have this property: it sends zero arrow maps to the identity of k.

The Littlewood–Richardson rule gives natural intertwiners

$$S_\lambda(V) \otimes S_\mu(V) \cong \bigoplus_\nu S_\nu(V)^{\oplus c^\nu_{\lambda\mu}}.$$

Here $c^\nu_{\lambda\mu}$ counts semistandard tableaux of skew shape ν/λ and content μ whose row-reading word is Yamanouchi. Naturality makes the same decompositions intertwine all the arrows. The tensor operation is vertexwise with synchronous arrow maps, not $\otimes_A$ and not a Hopf-algebra tensor structure. There is no automatic tensor unit preserving these zero relations. These functors are not additive or exact; they cannot simply be applied degreewise to a projective resolution. They also need not preserve the fixed R-action on lattices.

In the elliptic theta set one of the two permitted three-cycle branches to zero. The other has arbitrary maps $X_1 : V_0 \to V_2$, $X_2 : V_2 \to V_1$, $X_3 : V_1 \to V_0$, with all spaces of dimension d. This is a closed representation stratum of our infinite dessin algebra. Put $M = X_3X_2X_1$.

Theorem 3.3 (Cyclic invariants and a local positive kernel)**.** *The invariant ring on this stratum is $k[c_1(M), \dots, c_d(M)]$, where c_j are characteristic coefficients. Its total-arrow-degree Hilbert series is $\prod_{j=1}^d (1-t^{3j})^{-1}$. Common multidegree (m,m,m) has basis $\chi_\lambda(M) = \operatorname{tr} S_\lambda(M)$, $\lambda \vdash m$, $\ell(\lambda) \le d$; other multidegrees have no invariants. Products obey the Littlewood–Richardson rule. Over $\mathbb{C}$, if $\|M_i\| \le R_0$ and $0 \le r$ and $rR_0^2 < 1$, the kernel*

$$K_r(M_i, M_j) = \det(I - rM_i \otimes \overline{M_j})^{-1} = \sum_\lambda r^{|\lambda|} \chi_\lambda(M_i)\overline{\chi_\lambda(M_j)}$$

is absolutely convergent and positive semidefinite.

Proof. Cycle-trace generation for quivers, including relation quotients, is the Le Bruyn–Procesi theorem [29, 30]. Every cycle here is a power of the full three-cycle, up to cyclic rotation. In characteristic zero Newton identities give the generators c_j. They are algebraically independent: take $X_1 = X_2 = I$ and arbitrary X_3. Scalar changes at the three vertices force equal arrow multidegrees. Restriction to that same section identifies degree (m, m, m) with degree-m symmetric polynomials in eigenvalues, whose Schur basis and product rule give the assertion. For the kernel use the ordinary Cauchy identity, first as a formal identity in eigenvalues. On Schur modules the operator norm of $S_\lambda(M)$ is at most $R_0^{|\lambda|}$, so its trace is bounded by $\dim S_\lambda(\mathbb{C}^d) R_0^{|\lambda|}$. The dimension Cauchy identity bounds the absolute series by $(1 - rR_0^2)^{-d^2}$. Each summand is a rank-one Gram kernel, proving positivity by absolute summation. □

The term for the empty partition is a constant scalar feature, not an allowed module-valued tensor unit. In dimension two,

$$(\operatorname{tr} M)^2 = \operatorname{tr} \operatorname{Sym}^2 M + \det M, \qquad \operatorname{tr} \operatorname{Sym}^2 M = \tfrac{1}{2}((\operatorname{tr} M)^2 + \operatorname{tr} M^2).$$

More generally $\det(I - zM) = \sum_j (-z)^j \operatorname{tr} \bigwedge^j M$ and its reciprocal is $\sum_m z^m \operatorname{tr} \operatorname{Sym}^m M$. These are actual determinant/Schur connections, not identifications with an arithmetic L-function. On continuous finite-length Λ-modules the monodromy is nilpotent and all positive-degree characters vanish. The displayed Cauchy positivity is local representation-theoretic positivity, not the global Weil form.

4 Lattice almost-split sequences and genuine diamond quotients

For the interval dessin let $X = (x)$, $Y = (y)$ in $\Lambda = k[[x, y]]/(xy)$, $R = k[[u]]$, $u = x + y$. A lattice is a finitely generated Λ-module free over R.

Theorem 4.1 (The nodal lattice calculation)**.** *Every lattice is $\Lambda^r \oplus X^a \oplus Y^b$. The almost-split sequences in this lattice category are*

$$0 \to Y \hookrightarrow \Lambda \xrightarrow{\times x} X \to 0, \qquad 0 \to X \hookrightarrow \Lambda \xrightarrow{\times y} Y \to 0.$$

In particular $\tau X = Y$, $\tau Y = X$. Multiplication by u between two copies of either row, followed by reduction modulo u, gives an actual nonzero exact three-by-three diagram.

Proof. Normalize M *inside its total quotient module*: $\overline{M} = BM$, $B = k[[x]] \times k[[y]]$. The conductor gives $u\overline{M} \subset M \subset \overline{M}$. Both branch projections of $W = M/u\overline{M} \subset k^p \oplus k^q$ are onto. Intersect W with the two factors; its remaining quotient identifies their two residual quotients. Separate bases for these pure factors and the common quotient put W into pure first lines, pure second lines and diagonal pairs. Lift branch bases over the DVRs. These are respectively the summands X, Y, Λ. Their endomorphism rings are local.

Exactness of the first sequence is $\operatorname{ann} x = Y$. It does not split: a map $X \to \Lambda$ lands in $\operatorname{ann} y = X$, and its composite with $\times x$ is a nonunit in $\operatorname{End} X = k[[x]]$. Every map $\Lambda \to X$ factors through this epimorphism, $\operatorname{Hom}(Y, X) = 0$, and every nonunit endomorphism of X is divisible by x and factors through $X \hookrightarrow \Lambda \to X$. Classification handles all finite sums; a unit X component would be a retraction. This is right almost splitness. The dual argument for nonsections out of Y proves left almost splitness. Endomorphisms fixing either structure map have constant term one, hence are units, proving minimality. Swap x, y for the other sequence. Finally all three terms are R-free. Multiplication by u is injective and reduction preserves exactness, proving the diagram assertion. □

The lattice AR quiver has arrows $Y \to \Lambda \to X$ and $X \to \Lambda \to Y$; the mesh composites are zero. Its projective vertex Λ is not the endpoint of another nonsplit AR sequence. AR meshes do not mean that all rows and columns of any drawn grid are short exact.

For a second, explicit family fix $n \geq 3$ and the rational Belyi map

$$\beta_n(z) = \frac{1}{1 - T_n(z)^2} = \frac{1}{(1 - z^2)U_{n-1}(z)^2},$$

where T_n, U_{n-1} are Chebyshev polynomials. It has passport $(2n), (2^n), (2^{n-1}, 1, 1)$. Its contracted dual graph is a path, identifying its genus-zero dessin. The associated infinite algebra has vertices $1, \ldots, n$, endpoint loops l, r, opposite arrows $a_i : i \to i + 1$, $b_i : i + 1 \to i$, and precisely the relations $l^2, r^2, a_i b_i, b_i a_i$. Its single permitted cycle is $l a_1 \cdots a_{n-1} r b_{n-1} \cdots b_1$; the order rank is $(2n)^2$.

Retain one arrow at each edge in an orientation Ω, and kill its opposite. The quotient is exactly the string algebra $H_n(\Omega)$ of Huang–Lin–Su [46]; the killed-arrow ideal contains J^{n+2}, so it is also an actual quotient of Λ_n. Inflation identifies its module category with the full abelian quotient-annihilator subcategory $\mathcal{C}_\Omega$ of Λ_n-modules. This is generally not extension closed in the whole order category. The imported AR property is intrinsic to $\mathcal{C}_\Omega$, not automatically an ambient lattice or finite-length AR property. The extended edition gives nine explicit string modules, all twelve maps and the four exact meshes from the supplied diamond resource, now situated in this actual dessin quotient.

The other supplied resources also have a precise home. For the interval order the subcategory annihilated by J^2 is $\operatorname{Mod} B_0$, $B_0 = \Lambda/J^2$. This is a domestic string quotient, not itself a gentle algebra. Inflation is full and preserves purity, and $\operatorname{Zg} B_0$ is a closed subspace of $\operatorname{Zg} \Lambda$. Ringel's infinite-word/pure-injective program and its proof by Puninski–Prest apply there [43, 44]; the isolated band xy^{-1} gives Krull–Gabriel and relative Cantor–Bendixson dimension two by [45, Theorem10.19]. This is not an ambient rank calculation for all Λ-modules. The parameter in $x = J_2, y = \lambda J_2$ survives even though all cycle traces vanish; the pp condition $\exists v \ (z = xv \ \wedge \ (y - \lambda x)v = 0)$ detects it. These modules are u-torsion and are not nonzero order lattices. Infinite-string and locally finite AR theorems must retain their distinct finiteness hypotheses; a surviving permitted cycle has infinitely many paths at a vertex.

4.1 Polynomial strings, completed lattices and the Fukaya category

Crawley–Boevey's theorem [38, Theorems1.1–1.4] applies directly to the infinite polynomial dessin algebra: its finitely controlled modules decompose into strings and finite-dimensional bands; its pointwise artinian modules additionally allow primitive injective bands. For our finite quiver, these conditions mean finitely generated and artinian respectively. The words may be infinite, but their modules have algebraic direct-sum bases. This is not a classification of every module or of arbitrary formal completions.

In fact no nonzero completed-order lattice, restricted to A, meets either finiteness condition. The polynomial A is finite over $k[u]$: remove full turns from sufficiently long paths as in Proposition 1.1. If an R-free lattice were finitely generated over A, it would be finite torsion free, hence free, over the PID $k[u]$. Multiplication by $1 - u$ would then fail to be onto, whereas it is a unit over $R = k[[u]]$. Also $M \supsetneq uM \supsetneq u^2M \supsetneq \cdots$ excludes artinianity. For a concrete indecomposable outside the ordinary string/band list take $k[[x]]$ with $y = 0$: its endomorphisms are multiplication by formal series, so its endomorphism ring is local. Its x is injective and nonsurjective, while $1 - x$ is invertible; these properties exclude every pure-x ordinary string or band. The extended edition gives the full word argument.

The Fukaya connection requires equally precise categories. Lekili–Polishchuk explicitly allow infinite-dimensional locally gentle algebras, but their equivalence is for *perfect dg modules* of a homologically smooth algebra, with specified grading, line field and stops [66, Definition3.1.1, Lemma3.1.3, Theorem3.2.2]. Every nonempty closed dessin here has both permitted and forbidden cycles, so its polynomial algebra is neither proper nor smooth by their criterion. This does not exclude appropriate singular or completed variants; it prevents a direct use of that particular theorem. Infinite word, graded curve and geodesic are not interchangeable without a surface realization and boundary/grading data.

There is a genuine positive example inside our setting: the polynomial interval quotient $A/(y) = k[x]$ is still infinite dimensional, is smooth, and has no forbidden cycle. Its perfect category has the partially wrapped annulus model with one stop total. In degree zero $\mathrm{Perf}(k[x]) = D^b(\mathrm{mod}_{\mathrm{fg}}\, k[x])$. Inflation of this quotient is fully faithful on ordinary modules, but not on derived categories: $\mathrm{Ext}^2_A(k[x], k[x]) = k$, whereas $\mathrm{Ext}^2_{k[x]}(k[x], k[x]) = 0$. The former follows from the alternating y, x resolution. Thus this quotient bridge does not identify the whole closed dessin's perfect category with the annulus category.

Finally, Leavitt path algebras *can* fail invariant basis number, but need not. The permitted-state Laurent algebras here have finite-dimensional matrix quotients, and therefore invariant basis number. The polynomial and completed dessin algebras have their vertex-simple quotients, giving the same conclusion. The complex Leavitt algebra of G_α below is a dense *-subalgebra of its graph C*-algebra, on that specified graph; it is not the original gentle algebra with ordinary CK1 imposed.

5 Leavitt relations, the constrained Fock algebra and spectrum

Work now over $\mathbb{C}$. Ordinary Leavitt and graph C* constructions [47, 48] cannot simply impose our original forbidden products: if $ab = 0$ and CK1 says $a^*a = p_{t(a)}$, then $0 = a^*ab = b$. Every arrow is a forbidden successor, hence every arrow and then every vertex is zero. CK2 is not needed. The ordinary graph-Toeplitz CK1 setting has the same collapse.

The valid permitted-arrow-state graph instead consists of the disjoint directed σ-cycles. Its Leavitt algebra is $\bigoplus_c M_{|c|}(k[z, z^{-1}])$, and its graph C* algebra is $\bigoplus_c M_{|c|}(C(\mathbb{T}))$. These forget the α-gluing; both theta dessins give the same answer. They are not the completed original order. Nor is the ordinary Toeplitz algebra of a cycle of length $\ell > 1$ equal to $M_\ell(\mathcal{T})$: their K_0 ranks are ℓ and one respectively [50].

Let $\mathcal{H} = \ell^2(\mathcal{P})$ for all permitted finite paths, including vacua e_v. Define $L_a\delta_p = \delta_{ap}$ when the prefix is permitted and composable, and zero otherwise. Let P_v project onto paths starting at v, $Q_a = L_aL_a^*$ and v_v onto the vacuum. Then

$$L_aL_{\phi(a)} = 0, \quad L_a^*L_a = P_{t(a)} - Q_{\phi(a)} = v_{t(a)} + Q_{\tau(a)}, \quad P_v - \sum_{s(a)=v} Q_a = v_v.$$

These follow on basis vectors. The different initial projection avoids CK1 collapse. Write $\mathcal{T}_A = C^*(P_v, L_a)$, a constrained-path Fock C* algebra in the monomial/subproduct setting [51, 52]. The polynomial A acts faithfully: applying a finite path sum to the sum of vacua recovers its coefficients. This representation need not extend to all formal Λ series: $\sum_{m\geq 0} c^m$ along a cycle would send its vacuum to a nonsquare-summable sum.

Theorem 5.1 (Compact extension and incidence index)**.** *Let $\mathcal{H}_v$ be the terminal-path sector. There is an exact sequence*

$$0 \longrightarrow \bigoplus_{v\in Q_0} \mathcal{K}(\mathcal{H}_v) \longrightarrow \mathcal{T}_A \longrightarrow \bigoplus_{c\in \mathrm{cyc}(\sigma)} M_{|c|}(C(\mathbb{T})) \longrightarrow 0.$$

With $\delta[u] = [1 - W^*W] - [1 - WW^*]$, its boundary map is $\delta(c) = -\sum_{h\in c} e_{[h]}$, minus the unsigned incidence map B_Γ of the cellular graph. Thus, for its E edges,

	$K_0(\mathcal{T}_A)$	$K_1(\mathcal{T}_A)$
Γ *bipartite*	$\mathbb{Z}^{E+1}$	$\mathbb{Z}$
Γ *nonbipartite*	$\mathbb{Z}^E \oplus \mathbb{Z}/2$	0.

Proof. All prefixes preserve the terminal sectors. For paths p, q ending at v, $L_p v_v L_q^* = |\delta_p\rangle\langle\delta_q|$. Therefore the vacuum ideal is exactly the displayed sum of compacts: it contains these matrix units and that sum is itself an ideal. In the quotient, L_a has initial projection $Q_{\tau(a)}$ and range Q_a, a graph CK family for the permitted-state cycles. The Q_a have infinite rank and nonzero quotient images; the length gauge action descends. Gauge-invariant uniqueness [48, Theorem2.1] identifies the quotient as claimed.

For a cycle c based at arrow a, lift its corner unitary by $W = L_c Q_a + (1 - Q_a)$. Then $W^*W = 1$ and $1 - WW^* = Q_a - L_c Q_a L_c^*$ projects onto the first $|c|$ paths starting with a. Their terminal vertices count exactly its incident darts, including twice for a loop. This proves the index formula. The six-term exact sequence gives $K_1 = \ker B_\Gamma$ and $0 \to \operatorname{coker} B_\Gamma \to K_0 \to \mathbb{Z}^{\#\,\mathrm{cyc}\,\sigma} \to 0$. The free last term splits abstractly. Spanning-tree integer elimination gives Smith entries 1 repeated $V - 1$ times and one zero in the bipartite case; an odd cycle instead adds the entry two. This proves the groups. □

The nodal interval has $K_0 = \mathbb{Z}^2$, $K_1 = \mathbb{Z}$; the single-loop sphere has $\mathbb{Z} \oplus \mathbb{Z}/2$, zero. Both theta dessins have $\mathbb{Z}^4$, $\mathbb{Z}$. Coarse K-theory does not recover genus or the arithmetic factors.

5.1 A graph presentation and an actual pullback

The Fock algebra itself has a graph presentation, on a different graph from the original gentle quiver. Let G_α have vertices q_h for $h \in H$ and sinks b_e for $e \in Q_0$, and edges

$$s_h : q_h \to q_{\tau h}, \qquad t_h : q_h \to b_{[\tau h]}.$$

Proposition 5.2. *$C^*(G_\alpha) \cong \mathcal{T}_A$, by $q_h \mapsto Q_{a_h}$, $b_e \mapsto v_e$, $s_h \mapsto L_{a_h} Q_{a_{\tau h}}$ and $t_h \mapsto L_{a_h} v_{[\tau h]}$. The integer graph relation at h is $q_h - q_{\tau h} - b_{[\tau h]}$. Eliminating these relations recovers $K_1 = \ker B_\Gamma$ and $K_0 \cong \operatorname{coker} B_\Gamma \oplus \mathbb{Z}^{\#\,\mathrm{cyc}\,\sigma}$.*

Proof. The two edges at q_h have orthogonal initial pieces and orthogonal ranges summing to Q_{a_h}, by the Fock initial-projection identity. All vertex projections are nonzero. The length gauge acts with degree one on both edge families, so gauge-invariant uniqueness applies. They generate since $L_{a_h} = s_h + t_h$ and $P_e = v_e + \sum_{[h]=e} Q_{a_h}$. In the integer matrix a kernel coefficient is constant around each permitted cycle; its sink equations are exactly $B_\Gamma c = 0$. In the cokernel, eliminate all but one state per cycle; its remaining relation is the sum of its incident sink generators. The remaining cycle generators are free, proving the stated groups. □

For the nodal interval, G_α has two loop vertices, each with an exit to the common vacuum sink. Chirvasitu–Hajac–Tobolski's actual pullback theorem [65, Theorem3.4] gives

$$\{T \in \mathcal{T}_A : \pi(T) \in \mathbb{C}^2\} \cong C^*(E_1), \qquad \mathbb{C}^2 \hookrightarrow C(\mathbb{T})^2 \text{ by constants},$$

where E_1 has the same three vertices and countably many arrows from each loop vertex to the sink, with no loops. The arrow numbered $n \geq 0$ maps to loop^n followed by exit. This algebra is AF, with $K_0 = \mathbb{Z}^3$, $K_1 = 0$. The extended edition verifies every hypothesis. The theorem does not turn

arbitrary cycle-vertex identifications into pullbacks: its first graph is acyclic and its path functor must cover all pointed paths. Its sink-extension Theorem4.3 has further injection and path-coverage requirements.

5.2 A separate diagonal-fiber C* completion retaining the pairing

The formal order's residue gluing cannot be copied by taking diagonals of arbitrary full matrices: $\operatorname{diag}(E_{12}E_{21}) \neq \operatorname{diag}(E_{12})\operatorname{diag}(E_{21})$. For $\ell > 1$ there is no nonzero *-homomorphism from $M_\ell(C(\mathbb{T}))$ to a commutative C*-algebra, as restriction to its constant matrix units shows. Restrict the distinguished fiber first. Define

$$D_\ell = \{f \in C(\mathbb{T}, M_\ell(\mathbb{C})) : f(1) \text{ is diagonal}\}, \qquad \mathcal{D}_H = \Big(\prod_c D_{|c|}\Big) \underset{\mathbb{C}^H}{\times} \mathbb{C}^{Q_0},$$

where the first map evaluates diagonal entries at one and the second repeats λ_e at the two darts paired by α.

Theorem 5.3 (A faithful polynomial pullback model)**.** *There is a split exact sequence*

$$0 \to \bigoplus_c C_0(\mathbb{T} \setminus \{1\}, M_{|c|}) \to \mathcal{D}_H \to \mathbb{C}^{Q_0} \to 0,$$

so $K_0(\mathcal{D}_H) = \mathbb{Z}^{|Q_0|}$ and $K_1(\mathcal{D}_H) = \mathbb{Z}^{\#\operatorname{cyc}\sigma}$. The polynomial dessin algebra embeds faithfully and generates this C-algebra with its adjoints. This embedding does not extend to a unital algebra map from all of Λ.*

Proof. The section consists of the constant paired diagonal projections. The kernel is the displayed vanishing fiber, a sum of matrix suspensions, giving the groups by split exactness and suspension. For an explicit embedding set $g(z) = |1 - z|^2/4$, choose one cut arrow per permitted cycle, and send a_h to $g(z)z^{\epsilon_h}E_{h,\tau h}$ in its branch, where $\epsilon_h = 1$ at the cut and zero otherwise. Send e_e to the constant paired diagonal projection. Every forbidden product is zero. A full cycle at h maps to $w_c(z)E_{hh}$, $w_c = g^{|c|}z$. This scalar function is injective on the circle: away from its unique zero at one its argument recovers z. Continuous functional calculus therefore supplies every $f(z)E_{hh}$ with $f(1) = 0$. Multiplying by path images and approximating division away from one supplies every vanishing off-diagonal entry. Together with the constant paired diagonals these generate $\mathcal{D}_H$. Path normal forms in a fixed entry are a nonzero prefactor times a polynomial in w_c; its infinite range proves independence. Evaluation at one first separates the zero-length coefficients. Finally for $z_0 \neq 1$, $1 - w_c(z_0)^{-1}c_h$ is a unit in the formal order, but its proposed image is singular at z_0. A unital map preserves units, proving nonextension. □

This is the C* algebra generated by a faithful polynomial representation and its adjoints, distinct from the completed order and from $\mathcal{T}_A$. The norm closure of the polynomial image alone need not contain those adjoints. Their K groups already differ. The pairing belongs to the labelled presentation; an unmarked isomorphism class need not recover the ribbon ordering. No arithmetic descent, Frobenius action or completed-L positive determinant is inferred from this additional valid gluing model.

Proposition 5.4 (The degree spectral triple)**.** *Define $N\delta_p = |p|\delta_p$ on $\{\xi : \sum_p |p|^2|\xi_p|^2 < \infty\}$. With the polynomial creation/annihilation *-algebra this is an ungraded spectral triple, p-summable exactly for $p > 1$. With N_+^{-s} zero on the kernel,*

$$\operatorname{Tr} e^{-tN} = E + \frac{2E}{e^t - 1}, \quad \operatorname{Tr} N_+^{-s} = 2E\zeta(s), \quad \operatorname{Tr}\big(\chi(N)N_+^{-s}\big) = 2EL(s,\chi)$$

for $t > 0$ and $\operatorname{Re} s > 1$ in the last two formulas, where χ is an externally chosen Dirichlet character.

Proof. There are E vacua and $2E$ paths of every positive length: each starting arrow has a unique continuation. The real diagonal operator is self-adjoint on the stated maximal domain; finite supports are a core and its resolvent is compact. Prefixes preserve the domain and $[N, L_a] = L_a$, $[N, L_a^*] = -L_a^*$, so polynomial commutators are bounded. The trace formulas and summability reduce to the corresponding scalar series. The character multiplier is bounded; membership in $\mathcal{T}_A$ is not asserted. □

Length parity anticommutes with the creators but commutes with N, so it is not an even-spectral-triple grading. The ordinary odd Fredholm class of this positive N is trivial: $N(1+N^2)^{-1/2} - I$ is compact. The spectrum depends only on E; zeros of the scalar zeta trace are not eigenvalues of N.

6 Arithmetic attachment, descent and the remaining comparison

The compact ideal gives a precise home for an arithmetic determinant, with no claim that the arithmetic is derived from the dessin. Fix one terminal sector. Its two paths of each length $n \geq 1$ give orthonormal vectors $\xi_n = (\delta_{p_{n,1}} + \delta_{p_{n,2}})/\sqrt{2}$ and an isometry $U : \ell^2(\mathbb{N}_{\geq 1}) \to \mathcal{H}_v$.

For a nonprincipal character χ put $A_n = \sum_{j \leq n} \chi(j)$, $B_0 = \sup |A_n| \leq q_\chi$, and, on $2^k \leq n < 2^{k+1}$, set

$$w_n = \sqrt{n}(k+1), \quad u_n = A_n/w_n, \quad v_n(s) = w_n(n^{-s} - (n+1)^{-s}), \quad \varphi_s(z) = \sum_n v_n(s) z_n - z_1.$$

Here $u_1 = 1$. For $\operatorname{Re} s > 0$, $u, v(s) \in \ell^2$ and the pairing is complex linear, with no conjugation of s. Define

$$\widetilde{K_s} = U(u \otimes \varphi_s)U^*, \qquad \det_F(I + \widetilde{K_s}) = L(s, \chi).$$

Indeed dyadic blocks give $\|u\|^2 \leq 2B_0^2$, and on a compact set $\operatorname{Re} s \geq \sigma_0 > 0$, $|s| \leq M$, the squared v tail is bounded by $M^2 \sum_{k \geq J} (k+1)^2 2^{-2\sigma_0 k}$. Thus the rank-one family is trace-norm holomorphic. Its determinant is $1 + \varphi_s(u)$, equal by Abel summation to L. Replacing v by $P_N v$ and adding $A_N(N+1)^{-s} e_1^*$ to the functional gives exactly every raw partial sum S_N, with compact-uniform trace-norm convergence; $\|UXU^*\|_1 = \|X\|_1$. For primitive χ, completion replaces the functional by $g(s) \sum v_n(s) z_n - z_1$, with the standard nonzero gamma factor g, rather than multiplying the whole old functional. The extended edition supplies the quantitative estimates.

This inserts a scalar encoding into one actual dessin compact ideal; it is not known to be A-linear, a derived chain map or Frobenius compatible. A common fixed positive metric cannot make this same fixed-range family symmetric for all critical heights: symmetry forces its functionals to be proportional, whereas the coefficient ratio at heights $0, t \neq 0$ is asymptotic to $(\sigma + it)n^{-it}/\sigma$, with two opposite subsequential limits. Different positive realizations remain possible.

Proposition 6.1 (Complex dessin data do not determine arithmetic descent)**.** *The rational Belyi pairs $E_d : y^2 = x^3 - d$, $\beta_d = x^3/d$, for $d = 1, 2$, are isomorphic over $\mathbb{C}$ but have distinct Frobenius polynomials at the good prime seven: $1 - 4T + 7T^2$ and $1 - T + 7T^2$.*

Proof. Choose $a^3 = d$, $b^2 = d$. The projective isomorphism $(x, y) \mapsto (ax, by)$ from E_1 to E_d preserves the Belyi map. The degree-six clean passports are the elliptic theta passport already computed. Directly counting y for $x = 0, \ldots, 6$ gives respectively $(0, 1, 1, 0, 1, 0, 0)$ and $(0, 0, 0, 2, 0, 2, 2)$. Including infinity yields four and seven points, hence $a_7 = 8 - \#E_d(\mathbb{F}_7) = 4, 1$. Both discriminants $-432d^2$ are nonzero modulo seven. □

Any invariant of the complex dessin or its undecorated order/Fock algebra alone is the same for these two inputs, so cannot recover both factors without descent data. This does not obstruct a theory including a specified rational model and compatible Galois action. Elliptic H^1 has weight one and unshifted center $s = 1$, unlike Dirichlet center $1/2$; the example is not a counterexample to GRH.

6.1 A constructive arithmetic benchmark on the theta dessin

The preceding twists can be distinguished constructively once their finite-field descent data are retained. This closes the elementary local-factor computation, not the global positive-determinant problem. Let $\kappa = \mathbb{F}_q$, $\operatorname{char}\kappa \neq 2, 3$ and $d \in \kappa^*$. Put $E_d : y^2 = x^3 - d$ and $a_q(d) = q + 1 - \#E_d(\kappa)$. For $q \equiv 1 \pmod 3$, fix a nontrivial cubic character ρ and the quadratic character η, both extended by zero at zero. Define

$$J = \sum_{u \in \kappa} \rho(u)\eta(1-u), \qquad j = -\eta(-1)J, \qquad \lambda_d = \eta(d)\rho(d), \quad \pi_d = \lambda_d j.$$

Theorem 6.2 (Uniform local factor)**.** *For every such prime power and nonzero d,*

$$a_q(d) = \pi_d + \overline{\pi_d}, \quad |\pi_d|^2 = q, \quad P_d(T) := 1 - a_q(d)T + qT^2 = (1 - \pi_d T)(1 - \overline{\pi_d} T).$$

If $q \equiv 2 \pmod 3$, instead $a_q(d) = 0$. The split-case values lie in $\mathbb{Z}[\omega]$, $\omega^2 + \omega + 1 = 0$.

Proof. Cube fibers have size $1 + \rho(t) + \overline{\rho(t)}$, including at zero; the first term is the literal constant one, not the zero-extended trivial character. Counting square fibers gives $a_q(d) = -\sum_x \eta(x^3 - d)$. Grouping by $t = x^3$ removes the constant term since $\sum_t \eta(t - d) = 0$. Substitution $t = du$ gives $\sum_t \rho(t)\eta(t - d) = \rho(d)\eta(-d)J$; the other term is its conjugate. This proves the signed trace formula.

For a nontrivial additive character ψ, write $G(\chi) = \sum_x \chi(x)\psi(x)$. Character orthogonality gives $|G(\chi)|^2 = q$ for nontrivial χ: substitute $x = uy$ in its squared norm; the inner sum over $y \neq 0$ is $q - 1$ at $u = 1$ and -1 otherwise. Grouping $G(\rho)G(\eta)$ by $x + y$ gives $JG(\rho\eta)$, the zero-sum fiber vanishing since $\rho\eta$ is nontrivial. The three Gauss norms yield $|J|^2 = q$. Neither purity nor RH was assumed. The remaining signs and character values have modulus one. For $q \equiv 2 \pmod 3$ cubing is bijective, so the quadratic-character sum is zero. Finally all split-case summands are signs times cube roots of unity. These are the classical Gauss–Jacobi methods [68, pp.499–502], with our normalization proved explicitly. □

At $q = 7$, choose $\rho(3) = \omega$. Then $j = 3 + 2\omega$, $\lambda_2 = \omega^2$, and $\pi_2 = -1 - 3\omega$, giving the two factors already counted. At $q = 13$, $\rho(2) = \omega$ gives $J = 1 + 4\omega$ but $j = -1 - 4\omega$; this detects the essential sign hidden by the first example.

6.2 A genuine cellular carrier, and what still fails for the order

Assume $q \equiv 1 \pmod 3$ in this cellular construction. The elliptic theta dessin has cellular chains $C_2 = \mathbb{Q}$, $C_1 = \mathbb{Q}^3$, $C_0 = \mathbb{Q}^2$. Orient the three edges from v_+ to v_-; then $\partial_1(e_i) = v_- - v_+$ and $\partial_2 = 0$. Thus H_1 is the sum-zero plane, with basis $e_0 - e_1, e_1 - e_2$. Rotation σ acts there by

$$C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \qquad G = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

G is the actual Euclidean edge Gram matrix in this basis. The deck involution α acts as $-I$. Consequently the six-element deck group $\langle \sigma, \alpha \rangle$ acts on this homology and permutes the original quiver and its entire projective resolution.

Write $\pi_d = A_d + B_d\omega$. Its multiplication action on this *actual cellular* plane is

$$M_d = A_d I + B_d C = \begin{pmatrix} A_d & -B_d \\ B_d & A_d - B_d \end{pmatrix}, \qquad \det(I - TM_d) = P_d(T), \qquad M_d^t G M_d = qG.$$

This constructs the local matrix from finite-field character sums and the dessin's deck action, rather than fitting its trace. Positivity follows from $2x^2 - 2xy + 2y^2 = (x-y)^2 + x^2 + y^2$. The normalized matrix is G-unitary, not asserted self-adjoint. The extended edition supplies a rational cellular chain lift, its exact determinant cancellation, and Kummer descent compatibility on homology. Its odd-over-even determinant ratio is the classical finite-field curve zeta function

$$Z(E_d/\kappa, T) = \frac{P_d(T)}{(1-T)(1-qT)}.$$

Identification with the geometric ℓ-adic Frobenius polynomial uses the standard elliptic trace and determinant theorem; a canonical integral comparison or an order-derived Frobenius functor has not been constructed here.

There is also a precise obstruction to extending this same real two-dimensional carrier to $q \equiv 2 \pmod 3$. Frobenius then conjugates the cubic deck action. The relation $MC = C^2 M$ forces $M = \left(\begin{smallmatrix} a & b \\ a+b & -a \end{smallmatrix}\right)$, whose determinant is $-a^2 - ab - b^2 \le 0$, contradicting the required positive determinant q. This excludes that real carrier with its fixed deck action, not complex or enlarged Hilbert–Pólya constructions. The extended edition gives a compatible complex local replacement.

For the following insertion test use the characteristic-zero theta order $\Lambda_{\mathbb{Q}}$ over $R_{\mathbb{Q}} = \mathbb{Q}[[u]]$, with residue simple $S_e = \mathbb{Q}$. The limitation is concrete. The deck action has finite order and cannot alone supply the norm-$\sqrt{q}$ Jacobi coefficient. Inserting M_d into $S_e \otimes \mathbb{Q}^2$ gives an actual residue-module endomorphism with determinant P_d, but that simple is not perfect over the order and choosing it breaks full deck symmetry. Inserting the same matrix $\Lambda_{\mathbb{Q}}$-linearly into the regular projective instead gives the $R_{\mathbb{Q}}$-determinant P_d^{18}, or P_d^6 on $e\Lambda_{\mathbb{Q}} \otimes \mathbb{Q}^2$. Taking a root is not an established determinant-line comparison; a deck-semilinear lift is a different map. These surviving multiplicity/equivariance obligations prevent promoting the local computation to the required arithmetic order functor.

For the global comparison fix a nonzero rational d and use its good reductions. The finite-field curve zeta has center $1/2$ after $T = q^{-s}$. The *global elliptic* L-function over $\mathbb{Q}$ has unshifted center one, whereas Dirichlet L-functions have center $1/2$. Local weight-one purity does not determine either global zero set. The elementary local bound proves absolute convergence of the elliptic Euler product on $\operatorname{Re} s > 3/2$; the new local identities give no determinant-preserving continuation with the desired global positive spectrum.

6.3 Arbitrary projective actions: an obstruction and a genuine determinant repair

The preceding tensor test has a stronger replacement. Keep the characteristic-zero completed elliptic theta order Λ over $R = \mathbb{Q}[[u]]$, and write $K = \mathbb{Q}((u))$. All determinants in the next theorem are *ordinary underlying-R determinants*; they are not reduced norms.

Theorem 6.3 (Projective determinant multiplicity)**.** *Let P be any finitely generated projective right Λ-module and $f \in \operatorname{End}_\Lambda(P)$. Then $P \simeq \bigoplus_{i=0}^{2} (e_i\Lambda)^{n_i}$. Let $A_i \in M_{n_i}(\mathbb{Q})$ be the induced maps on its three top multiplicity spaces. Then*

$$\det_R(1 - Tf)\Big|_{u=0} = \left(\prod_{i=0}^{2} \det(1 - TA_i)\right)^6.$$

For any bounded complex of these projectives and any Λ-linear chain endomorphism, its alternating determinant ratio specializes to a sixth power in $\mathbb{Q}(T)$. Generically, the ordinary ratio is a cube in $K(T)$.

Proof. The arrow ideal is the Jacobson radical: its elements admit geometric-series inverses after subtracting from one, and the quotient is $\mathbb{Q}^3$. A projective's top lifts to a map from the displayed sum; Nakayama gives surjectivity, projectivity splits it, and another application of Nakayama kills the kernel.

Put $B = \Lambda/u\Lambda$. Each e_iB has path layers of dimensions $1, 2, 2, 1$: the identity, two length-one paths, two length-two paths, and one length-three path. The two full cycles based at i add to ue_i, so become negatives; all length-four paths vanish. A positive-length entry of a projective homomorphism strictly raises path length. Consequently the associated graded map has six copies of $A = \operatorname{diag}(A_0, A_1, A_2)$, giving the formula by triangular block determinants. Base change of determinants justifies specialization. Multiplicativity along filtrations is ordinary linear algebra, consistent with [69, Tags0FJ9,0GSX]; no purity assumption enters.

Each term's determinant has constant term one in T, so every denominator has nonzero specialization. Taking the finite alternating product therefore preserves the sixth power, even after cancellation. Finally $\Lambda_K = M_3(K) \times M_3(K)$; a projective's branch modules are sums of three-dimensional simples. A commuting endomorphism acts on each multiplicity space, repeating every factor three times. This proves the generic assertion. □

Thus no such ordinary perfect-complex ratio can equal the constant-in-u quadratic $P_d(T)$, or the full local curve zeta $P_d(T)/((1-T)(1-qT))$. For P_d, the valuation at infinity is -2, not a multiple of six. For the zeta ratio, $P_d(1) = q + 1 - a_q > 0$ by the local bound $|a_q| \le 2\sqrt{q} < q + 1$, so its pole at $T = 1$ is simple. This also covers square prime powers where the numerator may have a double root. It is an obstruction for the specified ordinary determinant and linear category, not all spectral or determinant theories.

There is nevertheless a *genuine integral functor* removing this multiplicity. Choose a permitted branch and its cycle cut. Its projection $\Lambda \to B_+ \subset M_3(R)$ makes the column lattice $W_+ = R^3$ a left Λ-module. Define

$$\Phi_+ : \operatorname{Perf}(\Lambda) \longrightarrow \operatorname{Perf}(R), \qquad P^\bullet \longmapsto P^\bullet \otimes_\Lambda W_+.$$

Bounded projectives compute this derived tensor functor; flatness of W_+ on all modules is unnecessary. Since $e_iW_+ \simeq R$, its determinant on $e_i\Lambda \otimes_{\mathbb{Q}} V$ with external action $1 \otimes M_d$ is exactly P_d, whereas on the regular module it is P_d^3. This is a defined change of determinant functor, not an arbitrary polynomial root. It does not construct the arithmetic action independently: the already computed M_d is still the input.

The price of this repair is precise. Branch swap α exchanges W_+ and W_-; pairing them yields P_d^2 on that basic-projective input. The chosen branch determinant cannot intertwine α on all endomorphisms: the corner unit $e_i + c_i^+$ acting on $e_i\Lambda$, with c_i^+ a plus full cycle, acts on its determinant by $1 + u$, whereas its α-image acts by one. Also no rank-one-per-vertex column lattice supported on one branch admits an R-linear lift of the rotation σ fixing u. Its three nonzero arrow scalars have nonnegative u-valuations summing to one; an invertible cyclic lift would make all three equal. This contradiction concerns that lattice, not every determinant-level rotation comparison. Ramification $u = v^3$ *and a new lattice* with all three arrow scalars v admits such a rotation, changing the stipulated base/lattice data. A descent-compatible arithmetic determinant retaining the required symmetry and a global positive realization remains open.

6.4 Tame descent: a surviving order and a sharper obstruction

We now test the proposed ramified repair rather than assume that its rotation descends. Extend coefficients explicitly to $k = \mathbb{Q}(\omega)$, $\omega^2 + \omega + 1 = 0$. Put $R = k[[u]]$, $S = k[[v]]$, $u = v^3$, and let $g(v) = \omega v$, fixing k. On the balanced plus column $W = S^3$, let $C\epsilon_i = \epsilon_{i+1}$ and

$$\Theta = Dg, \qquad D = \operatorname{diag}(1, \omega^2, \omega).$$

The plus arrows act by $vE_{i,i-1}$, and minus arrows by zero. Direct calculation gives $\Theta^3 = 1$, compatibility of Θ with this Λ_R-action, and

$$\Theta C = \omega^2 C\Theta, \qquad L = W^\Theta = R\epsilon_0 \oplus Rv\epsilon_1 \oplus Rv^2\epsilon_2.$$

Thus invariants restore precisely the original cut lattice. The extension $R \to S$ is finite *free* of rank three and faithfully flat; the group quotient is not a torsor at $v = 0$. The comparison $S \otimes_R L \to W$ has matrix $\operatorname{diag}(1, v, v^2)$, determinant u and cokernel $S/(v) \oplus S/(v^2)$. Hence this semilinear group action alone is not effective descent of the column to R. This is the explicit local root-stack situation, consistent with the trivial-stabilizer criterion for vector-bundle descent [70, Theorem10.3]; no stack theorem is needed for the displayed calculation.

Two positive constructions survive. The determinant line $\bigwedge_S^3 W$ has trivial D and C actions and does descend, but the natural map

$$\det_R(L) \longrightarrow (\det_S W)^\Theta$$

has image uR inside R, not an isomorphism. Determinant and coarse invariants cannot be interchanged while dropping this nonunit comparison. Also the *carrier* endomorphism algebra $\operatorname{End}_S(W)^\Theta$, conjugated to the invariant basis, is exactly the hereditary order with entries R on and below the diagonal and uR above. Its rotation descends as $\operatorname{Ad}(A)$, where

$$A = \begin{pmatrix} 0 & 0 & u \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad A^3 = uI.$$

The algebra automorphism has order three although A is not an integral unit. This is not $\operatorname{End}_\Lambda(W)$ or a descended arithmetic cohomology group.

The natural arithmetic attempt fails: $aI + bC$ does not commute with Θ when $b \neq 0$, as in both prime-seven Jacobi examples. The original cellular sum-zero plane is not Θ-stable either. Replacing C by vC does give the descended operator A on L, but

$$\det(1 - T(aI + bA)) = (1 - aT)^3 - ub^3T^3,$$

whose special fiber is $(1 - aT)^3$, not the required local quadratic. These carrier maps are not automatically Λ-linear arithmetic endomorphisms.

There is a stronger category-level conclusion.

Theorem 6.4 (Full-deck coefficient kernels)**.** *Let W_0 be a finite R-free left Λ_R-module on which the central u acts by the given scalar. Suppose it has invertible R-linear intertwiners lifting the order automorphisms σ and α. Then, for some $m \geq 0$, every e_iW_0 has rank $6m$. For any finite projective P and Λ_R-linear $f : P \to P$, let A_i be its top blocks. The genuine coefficient functor $\Phi_{W_0} = - \otimes^{\mathbf{L}}_\Lambda W_0$ satisfies*

$$\det_R(1 - T\Phi_{W_0}(f))\Big|_{u=0} = \left(\prod_i \det_k(1 - TA_i)\right)^{6m}.$$

Bounded perfect ratios retain this exponent. Thus this whole class of full-deck, fixed-center finite free coefficient kernels cannot supply either the single local quadratic or the full local curve zeta.

Proof. In the generic fiber intersect W_0 with its two central branch summands to obtain finite free integral lattices $W_\pm$; an integral direct-sum decomposition is not assumed. Rotation preserves them and branch swap exchanges them. On a branch let the common vertex rank be r. The three arrow matrices have cyclic product uI_r. Their determinant valuations are nonnegative, sum to r, and rotation intertwiners make them equal. Therefore $3 \mid r$. Branch swap makes the two ranks equal, so the full vertex rank is $6m$.

For the determinant formula, filter W_0/uW_0 by the arrow radical, whose fourth power vanishes. A positive-length entry of f raises the filtration; a top block A_i repeats across its i-th graded pieces a total of $\dim_k e_i(W_0/uW_0) = 6m$ times. Triangular determinants give the assertion. Finite alternating products preserve it. Quadratic degree and the uncancelled simple pole of the full local zeta exclude the targets as before. □

This theorem is sharp: $W_0 = \Lambda_R$ has vertex rank six. It requires neither left projectivity nor left flatness of a general kernel. It excludes this coefficient category, not arbitrary semilinear correspondences, twisted determinant lines or infinite spectral constructions. The extended edition identifies the full character-orbit repair with the hereditary normalization and explains why Schur commutator cancellation is insufficient. All determinant multiplicities here are relative to the stated k-valued base; restricting a rational quadratic factor to the former $\mathbb{Q}$-base squares it. No arithmetic descent or global positive determinant has been recovered by this ramification.

The recent NCG literature supplies concrete comparison targets, with strong hypotheses. Connes–Consani construct finite abelian covers of the adelic Riemann sector with prime-orbit length $\log p$ and arithmetic Frobenius monodromy [53, Theorem3.12]. The prolate/local-factor and moment constructions [54, 55] give particular finite-prime analytic models, not a uniform positive comparison for all completed L-functions. The zeta-spectral-triple determinant theorem [56, Theorem5.10] assumes a specified restricted Weil form, simple even bottom eigenfunction and boundary normalization. Its infinite comparison is still an additional step. The distributional real-zero theorem [57, Theorem6.1] concerns a particular interval quadratic form, essential self-adjointness on its trigonometric core and an even simple ground state; a Fock vacuum alone verifies none of those arithmetic identifications. Deninger's periodic-packet construction and real-cohomology obstruction [60, 62] further specify, rather than remove, these issues. Full theorem-level reading coverage and exact versions are recorded alongside this draft.

A complete approach must construct arithmetic descent and prime actions on actual order/derived or analytic objects, prove determinant-line and local-factor compatibility, control regularization and conductor/gamma normalization, and establish a positive realization for that *same* function. Finite-field cohomological trace formulas and Deligne purity supply the classical model [20, 21]; alternating duality alone is not positivity or purity. In a number-field construction these are obligations, not imported conclusions.

For a sharply testable final analytic target, put $F_\chi(z) = \Lambda(1/2 + iz, \chi)$. Suppose entire functions $h_m(z)\det(zI - H_m)$, with H_m finite self-adjoint and h_m entire nowhere zero, converge locally uniformly on $\mathbb{C}$ to this nonzero completed function. Hurwitz on every disk disjoint from $\mathbb{R}$ then proves all its zeros real, hence the Dirichlet critical-line assertion. Establishing this convergence with independently constructed H_m is exactly the missing load-bearing claim; a rank-one encoding or a reformulation does not establish it. Hecke, Dedekind and Artin extensions must state their analytic scope anew.

6.5 Whole components, derived endpoints and determinant relations

The full node lattice category has precisely the three indecomposables X, Λ, Y and arrows $X \to \Lambda \to Y$, $Y \to \Lambda \to X$. Its stable category has only X, Y, no irreducible arrows, and suspension exchanging them; its AR triangles have zero middle term. For the polynomial node at a smooth support $x = \lambda + t$, $y = 0$, $\lambda \neq 0$, the modules $T_r = k[t]/t^r$ form an ambient homogeneous tube:

$$0 \longrightarrow T_r \xrightarrow{(\pi,t)} T_{r-1} \oplus T_{r+1} \xrightarrow{(t,-\pi)} T_r \longrightarrow 0, \quad T_0 = 0, \quad \tau T_r = T_r.$$

The shift changes the derived placement, not the quasi-length. These nonzero-central-support objects do not extend to continuous finite-length modules of the arrow-adic completion. At its singular point even the candidate pure-branch tube fails ambient almost-splitness: $T_2^y \to k$ cannot factor through $T_2^x \to k$, since every intervening map lands in the x-socle, killed by the last map.

The nine HLS strings above sit in an entire intrinsic quotient component $T_{p,q}$, $(p, q) \in \mathbb{Z}^2$, with arrows increasing one coordinate and $\tau T_{p,q} = T_{p-1,q-1}$ [46]. Its central object $E = M(ab)$ has infinite projective dimension: restriction to the first corner gives the dual-number simple with its infinite loop resolution, while restricted projectives are free or zero. Its module AR sequence therefore yields a distinguished derived triangle but not a derived AR triangle ending at E. Ringel's four-quarter wind-wheel tile [43], with infinite-word diagonals and mixed center, is a different category's quilt; it is not silently identified with that HLS component or the Belyi surface.

There is no general equation "AR mesh equals polynomial semi-invariant syzygy." The intrinsic $k(1 \to 2)$ quotient of the infinite Chebyshev algebra has AR row $0 \to S_2 \to E \to S_1 \to 0$ and a free presentation determinant z on dimension $(1, 1)$, so its semi-invariant ring $k[z]$ has no generator relation. A genuine framed elliptic-square stratum has $Z = [x_1, x_2, y_1, y_2] \in \mathrm{Mat}_{2\times 4}$ with six minors and relation

$$p_{12}p_{34} - p_{13}p_{24} + p_{14}p_{23} = 0.$$

These minors have the same determinant character for the target GL_2 with framed source; they are not all semi-invariants for the full unframed vertex group. In contrast, the nodal lattice AR row gives a genuine relative determinant identity over R: multiplication by $a + bx$ on the basis $(y, 1)$ has matrix $\left(\begin{smallmatrix} a & -b \\ 0 & a+ub \end{smallmatrix}\right)$ and characteristic factor $(1 - Ta)(1 - T(a + ub))$. The endpoints have infinite order-projective dimension but are R-perfect.

Finally, the three-vertex theta algebra has no unital surjection onto a six-vertex radical-square-zero circular algebra: its semisimple image has at most three primitive idempotent images. Duplicating face vertex spaces instead gives an exact faithful unfolding which is not full; it does not inherit quotient or AR-transfer claims. Zwara Hom-control means that the Hom-dimension defect is bilinear on K_0, not merely a covering direct-sum formula [72, 73]. The book supplies explicit fixed-dimension counterexamples, corrected circular-complex incidence resolutions, complete drawn component windows and all category hypotheses. These additions do not supply the arithmetic determinant comparison.

7 A genuine theta covering and its categorical limits

Use column matrices, hence covariant quiver representations (left modules); opposite categories translate the right-module convention in [74]. Let A be the infinite elliptic theta dessin algebra: vertices $i \in \mathbb{Z}/3$, arrows $a_i, b_i : i \to i + 1$, and all mixed length-two paths zero. For the arrow ideal J and $L \geq 2$ put $B_L = A/J^L$. This is a finite string quotient of our actual infinite gentle algebra; for $L > 2$ its length-L relations prevent treating it as quadratic gentle.

Proposition 7.1 (A locally bounded integer cover)**.** *The category $\widetilde{B}_L$ with vertices $r \in \mathbb{Z}$, two arrows $a_r, b_r : r \to r+1$, and the same mixed and length-L zero relations, has orbit B_L under $r \mapsto r+3$. For finite-support representations, push-down F is exact and*

$$\operatorname{Hom}_{B_L}(FM, FN) \simeq \bigoplus_{g \in \mathbb{Z}} \operatorname{Hom}_{\widetilde{B}_L}(M, T_g N), \qquad (T_g N)_r = N_{r-3g}.$$

Only finitely many summands are nonzero. In general F is neither full, dense, nor Hom-controlled in Zwara's sense.

Proof. A nonzero path is constant-branch with length less than L. Thus the incoming and outgoing total Hom dimensions at a cover vertex are $2L-1$, and $\dim B_L = 3(2L-1)$. Paths and relations lift uniquely; the deck action is free and has the stated orbit. Push-down sums spaces of equal residue. A morphism decomposes uniquely into blocks with target lift index minus source index equal to $-3g$. The arrow commutation equations preserve that difference, proving the formula; finite supports bound the possible differences. Direct sums of vector spaces preserve exact sequences. The following explicit witnesses prove the three failures. □

For $L = 5$, let P be the cover interval on $0, 1, 2, 3, 4$ with all its a-maps one and b-maps zero, and let S be the sum of the five simples at the same vertices. They have the same K_0 class, but

	$\dim \operatorname{End}_{\widetilde{B}_5}$	$\dim \operatorname{End}_{B_5}(F(-))$	defect
S	5	9	4
P	1	2	1

Indeed the pushed-down simple multiplicities are $(2, 2, 1)$, whereas $\operatorname{End}_{B_5}(FP) = kI \oplus kX^3$ for its single nilpotent five-step chain X. The extra endomorphism X^3 proves non-fullness; the unequal defects at the same class exclude a bilinear K_0 defect. To use Zwara's finite-algebra definition literally, restrict the source to the finite convex interval $[0, 4]$, then extend by zero before push-down. This exact functor has the same endomorphism spaces and K_0 comparison.

Proposition 7.2 (An indecomposable module outside this cover image)**.** *Over a characteristic-zero field, B_3 has a nine-dimensional indecomposable representation which is not the push-down of any finite-support $\widetilde{B}_3$ representation.*

Proof. Label its basis by $0, \dots, 8$ with vertex residues $(0, 1, 0, 1, 2, 1, 2, 0, 2)$, and use matrix units E_{ij} acting on columns. Set

$$X = E_{10} + E_{32} + E_{43} + E_{65} + E_{76} + E_{08},$$
$$Y = E_{12} + E_{45} + E_{78}.$$

Their restrictions between adjacent vertex spaces give the a_i, b_i maps. They satisfy $XY = YX = X^3 = Y^3 = 0$, hence every length-three path is zero. The endomorphism ring preserving vertex spaces is $kI \oplus \mathfrak{r}$, where

$$\mathfrak{r} = \langle E_{02} + E_{13}, E_{15}, E_{35} + E_{46}, E_{48}, E_{68} + E_{70}, E_{72} \rangle, \qquad \mathfrak{r}^3 = 0.$$

Solving the arrow commutation equations gives exactly these seven basis elements. The ring is local: an element $cI + n$ is invertible exactly when $c \neq 0$, by the finite geometric series. It has no nontrivial idempotents, so the representation is indecomposable.

A finite-support lift would define an Euler operator D acting by r on the summand at lift vertex r. Then $[D, X] = X$ and $[D, Y] = Y$. Every displayed nonzero arrow is the unique nonzero entry in

its row and column in its own matrix. Its commutator entry is therefore $D_{ii} - D_{jj}$. Add the six X equations and subtract the three Y equations. Each diagonal term cancels, leaving $0 = 6 - 3 = 3$, a contradiction. This argument allows arbitrary D, not just a diagonal matrix in the displayed basis. □

Corollary 7.3 (Higher-genus and derived non-density). For the clean dessin with rotations $\sigma = (0 \cdots m-1)(m \cdots 2m-1)$ and $\alpha = \prod_{i=0}^{m-1}(i\, i+m)$, odd $m \geq 3$, and genus $(m-1)/2$, the analogous A_m/J^3 has a $3m$-dimensional representation outside its integer-cover image in characteristic zero. Its bounded-derived push-down is not dense either.

Proof. Use basis $v_0, \ldots, v_{3m-1}$, assign vertices $i, i+1, i$ to $v_{3i}, v_{3i+1}, v_{3i+2}$ modulo m, and set

$$Xv_{3i} = v_{3i+1}, \quad Xv_{3i+2} = v_{3i+3}, \quad Yv_{3i+2} = v_{3i+1},$$

with indices modulo $3m$ and all other values zero. Again mixed products and cubes vanish. An Euler operator would yield $2m - m = m = 0$ by the same telescoping calculation, impossible in characteristic zero. If a bounded finite-support complex pushed to this module in degree zero, exactness would make its zeroth cohomology a module lift, a contradiction. □

The source's derived result is a *precovering* in the stated bounded categories [74, Theorem 4.28]; density remains an additional condition. Neither this result nor the GLS equivariant classification [75] identifies our completed order with a GLS algebra. Graded surface/Fukaya models [76] likewise require their own dg category, stops and grading. These categorical tools do not construct the independent positive arithmetic determinant.

8 Verification and research frontier

The two nodal/invariant Lean files construct the node as an actual subring of $k[[X]] \times k[[X]]$ with equal constants, prove its annihilators, regular common parameter, augmentation, alternating projective resolution and a bijective pair of power-series coordinates. A second file proves CK1 collapse under the stated permutation/coverage hypotheses, determinant base change, actual 2×2 symmetric-square functoriality and relation preservation, a Schur character identity and the fixed-parameter lattice failure. These thirty printed theorem types have only standard Lean foundations. They do not formalize general dessins, abstract Ext/AR/Schur–Weyl, C* exactness, Fredholm determinants, arithmetic descent or GRH. Earlier formal layers remain preserved with their literal scopes.

Exact-rational diagnostics independently check the displayed dessins, path identities, Schur and Young examples and the nine-object grid. These finite checks supplement the proofs, rather than prove infinite spectral or arithmetic assertions. The next arithmetic target is an exact descent/Frobenius-compatible construction on the labelled order/Fock data, with a determinant-preserving positive comparison. The local Schur kernel and the simple degree spectrum alone do not supply it. All source clones and earlier research are retained; the earlier general focused PDFs are archived at the preceding commit.

Ten further declarations in `DessinPullbackComponents.lean` check actual matrix-unit identities, nonmultiplicativity of raw diagonal extraction, impossibility of a unital two-state matrix residue into a nonzero commutative ring, multiplicativity on diagonal fibers, and genuine equal-image pullback subring/projection components. The analytic suite then contained187printed declarations at unchanged pins. The circle-function C* model, functional calculus, generation, graph uniqueness

and K-theory remain prose/external-theorem arguments, not hidden assumptions of these component checks.

Sixteen additional declarations in `ThetaJacobi.lean` check genuine finite-field MulChar Jacobi conjugation, norm and rescaling with explicit nontriviality hypotheses; cubic–quadratic noncancellation; the cyclotomic multiplication determinant and positive Gram identity; and actual theta cellular boundary, homology-inclusion, restriction and rational chain-map matrices. The full analytic suite now contains203declarations. The point-count proof, deck/Kummer comparison, cohomological trace interpretation and global arithmetic comparison are not thereby fully formalized.

The final three `ThetaJacobi.lean` declarations verify the actual conjugation matrix identities and the real two-dimensional inert-prime determinant obstruction. They do not assume any unknown zero location or exclude complex/enlarged spectral realizations.

`ThetaProjectiveMultiplicity.lean` adds five declarations. Arbitrary upper-block matrices with six identical diagonal blocks have the sixth-power determinant and characteristic-factor formulas. A literal power-series constant-coefficient map specializes matrix determinants, for each scalar parameter; the coefficientwise polynomial/order interface remains prose. A nonzero quadratic cannot be a rational-function sixth power, proved by polynomial degrees. Three genuine power series with product X cannot have the cyclically equivalent residue-vanishing conditions forced by invertible rank-one arrow intertwiners. The derivation of the path blocks from the full theta order, projective decomposition, derived branch functor and deck intertwiners is not fully formalized. The complete analytic suite now has208printed declarations; no hidden arithmetic positivity assumption occurs.

`ThetaTameDescent.lean` adds twelve printed component declarations (and one auxiliary polynomial lemma). They use genuine formal-power-series rescaling and its coefficient operator, proving the coefficient action, invariant-coefficient vanishing and the tame operator's cube. Actual three-state matrices satisfy the clock/rotation commutator, comparison determinant and closed-fiber equations, weighted-cycle cube and full characteristic polynomial, noncommutation of a nonconstant rotation polynomial, trivial whole-column determinant actions, and the same-operator comparison for vC. The classification of all fixed series by residues, full order action and hereditary invariant algebra, stack interpretation, valuation/rank theorem, general coefficient-functor filtration and GRH are not thereby fully formalized. No unknown positivity premise is assumed.

9 Why an absolute coprime pair-square estimate cannot close the short rows

Keep the original sextic family

$$M_u(D) = D^{-1/2} \sum_n \mu(n)\nu(n) W(Nn/D)\chi_n(u), \qquad \chi_n(u) = (u/n)_6,$$

with squarefree primary column ideals outside the fixed excluded set, their full annular coefficients (fix W supported in $[1,2]$), and all nonzero Eisenstein element rows $Nu \le H$. The cubic symbol is χ_n^2, not χ_n. The following classical frame inequality [80] gives a new obstruction to one proposed estimate in this arithmetic program.

Proposition 9.1 (Coprime pair-square obstruction)**.** *Let A be an R-by-N complex matrix with $|A_{ui}| = 1$, and put $K_{ij} = \sum_u A_{ui}\overline{A_{uj}}$. Then*

$$\sum_{i \ne j} |K_{ij}|^2 \ge NR(N-R).$$

For the original character kernels, let $B_1(H,D)$ be the sum of $|K_H(n,m)|^2$ over all distinct coprime squarefree column ideals in $[D,2D]$. At $H=D^{3/4}$,

$$B_1(H,D) \gg \frac{D^{11/4}}{(\log D)^2}.$$

In particular the proposed sufficient input $B_1(H,D) \ll H^2 D^{\varepsilon}$ fails for each fixed $\varepsilon < 5/4$. This does not refute the signed Möbius mean square.

Proof. Set $F_{uv} = \sum_i A_{ui}\overline{A_{vi}}$. Reordering four finite sums proves $\sum_{i,j} |K_{ij}|^2 = \sum_{u,v} |F_{uv}|^2$. The row diagonal contributes RN^2; the column diagonal contributes NR^2. Removing the latter and retaining the former gives the inequality.

Restrict the column pairs in the nonnegative sum B_1 to distinct prime ideals p, q of norm in $[D,2D]$, outside the same fixed set. If $H < D$, no such prime divides a nonzero row element: divisibility would imply $Nu \geq Np$. Thus all original zero masks are exactly one on this ball, $|\chi_p(u)| = 1$, and the residual primitive finite character conductor is pq, of norm between D^2 and $4D^2$. All these pairs have common divisor1. The lattice count gives $R = (2\pi/\sqrt{3})H + O(\sqrt{H}+1)$. The classical unconditional fixed-field prime-ideal theorem gives $N \sim D/\log D$; this is an external analytic input, not a GRH hypothesis. At the stated endpoint $R = o(N)$, and the bound is therefore $(1-o(1))N^2R \gg D^{11/4}/(\log D)^2$. □

To see the exact failed step, write $c_n = \mu(n)\nu(n)W(Nn/D)$ and $C_D = \sum_n |c_n|^2 = O(D)$. Cauchy over coprime pairs bounds the absolute signed contribution by $D^{-1}C_D B_1^{1/2}$. The hoped-for pair-square upper bound would have closed this particular route, but is false. A large absolute pair-square mass is not a lower bound on the quadratic form with the fixed c_n. Nor is the prime part a positive summand of the full Möbius energy: prime–composite cross terms can cancel. The original low-gcd estimate and the positive arithmetic determinant bridge remain open.

10 A signed Möbius decomposition and its termwise obstruction

Keep the original squarefree good-ideal family, its fixed ν, its full $W(Nn/D)$ and every zero-extended sextic symbol. In ideal Dirichlet convolution, write $\zeta_0(n) = 1$ for every good ideal and let δ be the convolution unit. For $1 \leq U < D$ put $a_U = \mu \mathbf{1}_{Nn \leq U}$ and $b_U = \mu - a_U$. The classical Vaughan algebra gives

$$\mu = 2a_U - a_U * a_U * \zeta_0 + b_U * b_U * \zeta_0. \tag{1}$$

Indeed $\mu * \zeta_0 = \delta$, and expansion of the last product proves the identity without an analytic assumption. This is a classical decomposition, not a new cancellation theorem; compare [87, Lemma4.1].

On squarefree n all factors in $n = dc = abc$ are pairwise coprime. Define the actual, nonarbitrary coefficient functions

$$\beta_U(d) = \sum_{\substack{ab=d \\ Na, Nb \leq U}} \mu(a)\mu(b), \qquad g_U(s) = \sum_{\substack{b \mid s \\ Nb > U}} \mu(b).$$

For squarefree d, $\beta_U(d)$ is $\mu(d)$ times the number of the displayed ordered factorizations. For

$Ns > U$, $g_U(s) = -\sum_{b|s,\, Nb\le U} \mu(b)$. The two regrouped pieces are

$$I_u = D^{-1/2} \sum_{\substack{d,c \text{ squarefree} \\ (d,c)=1}} \beta_U(d)\nu(dc)\chi_d(u)\chi_c(u)W(NdNc/D),$$

$$II_u = D^{-1/2} \sum_{\substack{a,s \text{ squarefree} \\ Na,Ns>U,\,(a,s)=1}} \mu(a)g_U(s)\nu(as)\chi_a(u)\chi_s(u)W(NaNs/D).$$

Then $M_u = -I_u + II_u$. All ideals here remain outside the same fixed excluded set. The short $2a_U$ term vanishes because $U < D$ and W is supported in $[1,2]$. No phase or coprimality mask was removed.

Proposition 10.1 (An all-cutoff separate-piece bound is false)**.** *Take the allowed principal ν, $U = 1$, and a fixed nonnegative smooth W supported in $[1,2]$, bounded below by a positive constant on a closed subinterval of $(1,2)$ of positive length. For each of the two pieces,*

$$\sum_{0<Nu\le H} |I_u|^2 \gg_W \frac{D}{\log^2 D}, \qquad \sum_{0<Nu\le H} |II_u|^2 \gg_W \frac{D}{\log^2 D} \quad (H \ge 1).$$

Consequently a proposed bound HD^ε for each piece at $H = D^{3/4}$ fails for $\varepsilon < 1/4$. In the exact original notation $H(HD)^\varepsilon$, it fails for $\varepsilon < 1/7$. This refutes an all-cutoff termwise strengthening, not the original signed mean square or a specified moving-cutoff argument.

Proof. At $U = 1$, $a_U = \delta$, so $a_U * a_U * \zeta_0 = \zeta_0$ and $b_U * b_U * \zeta_0 = \mu + \zeta_0 - 2\delta$. On the annulus the two coefficients are 1 and $1 + \mu(n)$, respectively. On squarefree ideals the latter is zero or two. At the actual unit row $u = 1$, every original character equals one. A fixed-field prime-ideal theorem gives $\gg D/\log D$ good prime columns in the subinterval, so $I_1 \gg_W \sqrt{D}/\log D$. Fix a good prime ideal q_0. Products $q_0 p$ with varying distinct good primes in the rescaled subinterval have $\mu(q_0 p) = 1$. Their nonnegative contributions give $II_1 \gg_W \sqrt{D}/\log D$. Keeping that one row in each nonnegative energy proves the result. The prime-ideal theorem is a classical unconditional external input, with the field and q_0 fixed. □

The factorization does not shorten the primitive row conductor: for squarefree coprime a, s it is as, not a or s separately. For two resulting columns n, n' with common factor h, the oscillatory conductor is nn'/h^2 and a zero-preserving period is nn'/h. The signed low-gcd estimate and the independent positive arithmetic determinant comparison therefore remain open.

11 A full-mask Hecke expansion for the type-I piece

We now continue the signed decomposition through its actual squarefree inner sum. Fix $K = \mathbb{Q}(\sqrt{-3})$, the finite-order Hecke character ν, and the same finite excluded prime set S, containing primes above 6 and the conductor of ν. For $u \in \mathcal{O}_K \setminus \{0\}$, let ψ_u be the primitive Hecke character induced by $\nu(n)(u/n)_6$ away from $S \cup \mathrm{supp}(u)$. It is defined before, and independently of, the type-I divisor d. The abelian Kummer extension $K(u^{1/6})/K$ constructs the sextic factor; multiplying by ν gives the stated character. This uses classical abelian reciprocity, not general Artin holomorphy. Write $\psi_{u,2}$ for the primitive character inducing its square and $\mathfrak{f}_u, \mathfrak{f}_{u,2}$ for the two primitive conductors. For a squarefree good ideal d put

$$T = T(u,d) = S \cup \mathrm{supp}(u) \cup \mathrm{supp}(d), \qquad R_T = \prod_{p\in T} Np, \quad Q_u = |D_K| N\mathfrak{f}_u.$$

All primes in T are deleted even if absent from either primitive conductor.

Theorem 11.1 (Restoration and a uniform scalar contour bound)**.** *For $\Re s > 1$ the original type-I inner Dirichlet series is*

$$F_{u,d}(s) = \sum_{\substack{c \text{ squarefree} \\ (c,T)=1}} \frac{\psi_u(c)}{(Nc)^s} = \frac{L(s,\psi_u)}{L(2s,\psi_{u,2})}\, C_T(s), \tag{2}$$

$$C_T(s) = \frac{\displaystyle\prod_{p\in T,\ p\nmid \mathfrak{f}_u} (1-\psi_u(p)(Np)^{-s})}{\displaystyle\prod_{p\in T,\ p\nmid \mathfrak{f}_{u,2}} (1-\psi_{u,2}(p)(Np)^{-2s})}. \tag{3}$$

It continues meromorphically to $\Re s > 1/2$, with only a possible simple pole at $s = 1$ when $\psi_u = 1$. In that case its residue is

$$\rho_T = \frac{\operatorname{res}_{s=1}\zeta_K(s)}{\zeta_K(2)} \prod_{p\in T} (1+(Np)^{-1})^{-1}.$$

Fix $0 < \delta < 1/2$, $\varepsilon > 0$ and $W \in C_c^\infty((0,\infty))$ supported in $[1,2]$. Set

$$\mathcal{M}_4(W) = \sum_{j=0}^{4} \sup_x |(x\partial_x)^j W(x)|.$$

Uniformly for $Y \geq 1$, u and d,

$$\sum_{\substack{c \text{ squarefree} \\ (c,T)=1}} \psi_u(c) W(Nc/Y) = \mathbf{1}_{\psi_u=1}\rho_T Y \widehat{W}(1) + O_{\nu,S,\delta,\varepsilon}\Big(\mathcal{M}_4(W) Y^{1/2+\delta} Q_u^{1/4-\delta/2+\varepsilon} R_T^{\varepsilon}\Big). \tag{4}$$

Here $\widehat{W}(s) = \int_0^\infty W(x)x^{s-1}\,dx$. The continuation and functional equation of degree-one Hecke functions are classical external inputs; the contour argument is given below.

At a good prime with $v_p(u) = 3$, ψ_u is ramified while its square is unramified; the second product in (3) must restore a factor invisible to squaring the zero-extended primitive value. At $v_p(u) = 6$ both characters can forget the prime, but the original row still vanishes on columns divisible by it. In particular, for principal ν, $u = \pi^3$ and $u = \pi^6$ at $N\pi = 7$, the deleted local factors are

	primitive ratio at $s = 1$	restoration	full masked factor
$v_p(u) = 3$	48/49	49/48	1
$v_p(u) = 6$	8/7	7/8	1

These are exact local calculations, not numerical evidence about zeros.

For $1 \leq U$, $U^2 \leq D$, insert $Y = D/Nd$ into (4) and retain the actual $\beta_U(d)$ and outer $\nu(d)\chi_d(u)$. Classical tame Kummer conductors give $Q_u \ll_\nu Nu$. Thus the normalized I_u equals its displayed principal-residue sum plus an error E_u with

$$|E_u| \ll_{\nu,S,\delta,\varepsilon} \mathcal{M}_4(W) D^\delta U^{1-2\delta+\varepsilon} (Nu)^{1/4-\delta/2+\varepsilon}. \tag{5}$$

The resulting elementary energy envelope is only

$$\sum_{0<Nu\leq H} |E_u|^2 \ll \mathcal{M}_4(W)^2 D^{2\delta} U^{2-4\delta+\varepsilon} H^{3/2-\delta+\varepsilon}. \tag{6}$$

It does not prove the short-row target of order H up to small losses. This is an insufficient upper bound, not a lower bound on the actual error or a counterexample to coefficient-sensitive averaging. The original signed mean square and positive determinant bridge stay open.

12 Actual renormalization retains the effective action

A compression score does not specify an RG transformation. Following the real-space formulation of Wilson and Kogut [81], we specify the eliminated variables, effective action, normalization and scale map. Our exact finite decimation is distinct from an approximate tensor-network truncation such as Levin–Nave TRG [82].

Definition 12.1 (Finite classical RG step)**.** Let the fine configuration be $(s,t) \in S \times E$, with finite nonempty sets S, E, and dimensionless log Boltzmann weight $V(s,t) = -\beta H(s,t)$. Define

$$(\mathcal{R}V)(s) = \log \sum_{t \in E} e^{V(s,t)}, \qquad Z(V) = \sum_{s,t} e^{V(s,t)}.$$

The full action includes its additive constant. A truncation or a learned approximation is a further operation with its own error, not this identity.

Proposition 12.2 (Partition, observables and composition)**.** *One has $Z(\mathcal{R}V) = Z(V)$, and the normalized expectation of every observable depending only on s agrees in the two descriptions. Eliminating E_1 and then E_2 equals eliminating $E_1 \times E_2$ at once, with the same tensor labels and inverse temperature.*

Proof. Exponentiating the definition gives $e^{\mathcal{R}V(s)} = \sum_t e^{V(s,t)}$. Sum over s, optionally multiplying by an observable $O(s)$, and divide both sides by the same strictly positive Z. Reordering finite sums proves the composition statement. No projection onto pair interactions is used. □

For an even homogeneous periodic Ising chain, write the transfer matrix as $T_{ab} = \exp(C + Kab + h(a+b)/2)$, $a, b \in \{-1, 1\}$. Its partition function is $\operatorname{tr} T^N$. Summing alternate spins replaces T by T^2 and N by $N/2$; the physical spacing doubles. Writing $P = T^2$, its positive entries determine

$$K' = \tfrac{1}{4} \log \frac{P_{++}P_{--}}{P_{+-}^2}, \quad h' = \tfrac{1}{2} \log \frac{P_{++}}{P_{--}}, \quad C' = \tfrac{1}{4} \log(P_{++}P_{--}P_{+-}^2).$$

These three parameters reproduce every entry of P, so no normalization is lost. At $h = 0$, $\tanh K' = \tanh^2 K$ and the eigenvalue-ratio correlation length of the homogeneous extension in blocked units halves. Its product with the spacing stays fixed. The fixed points in $t = \tanh K$ are $t = 0, 1$; the latter is zero temperature, not a positive-temperature transition. Odd dessin examples below use the full-action definition, not this even-chain shortcut.

12.1 Genuine dessin incidence models

For odd m, the clean dessin $y^2 = x^m - 1$, $\beta = x^m$ has two black vertices, m white vertices and one face; its genus is $(m-1)/2$. Its incidence graph is $K_{2,m}$ with $2m$ clean edges. Its infinite locally gentle order quiver instead has m vertices and completed DVR-rank $2m^2$. We place physical spins $B_+, B_-, s_0, \ldots, s_{m-1}$ on the incidence vertices; this is a specified finite spin model, not a faithful representation of that infinite algebra or order. The abstract incidence graph alone does not recover the ribbon embedding or genus.

With $\eta_j = (-1)^j$ and fixed nonuniform fields h_j, choose

$$V = J_+ B_+ \sum_j s_j + J_- B_- \sum_j \eta_j s_j + bB_+ - \tfrac{3}{5} bB_- + \sum_j h_j s_j.$$

Summing the two black spins gives the *entire* effective action

$$V'(s) = \sum_j h_j s_j + \log\left[2\cosh\left(b + J_+ \sum_j s_j\right)\right] + \log\left[2\cosh\left(-\tfrac{3}{5}b + J_- \sum_j \eta_j s_j\right)\right].$$

Every one of the $2m$ physical edges is present, although its couplings belong to a tied three-parameter family. The Walsh expansion $V'(s) = \sum_{A\subseteq\{0,\dots,m-1\}} c_A \prod_{j\in A} s_j$, with $c_A = 2^{-m}\sum_s V'(s)\prod_{j\in A} s_j$, is unique by Boolean orthogonality. It retains the constant $c_\varnothing$ and genuinely generated higher-body interactions. Our genus-two and genus-four examples use $m = 5, 9$ and continue by exact nested marginalization. Dropping all terms of degree greater than two is displayed as a measured approximation.

12.2 Quantum thermal effective Hamiltonians

For a finite Hermitian Hamiltonian H on nonzero finite-dimensional $\mathcal{H}_S \otimes \mathcal{H}_E$ and $\beta > 0$, define

$$W' = \operatorname{Tr}_E e^{-\beta H}, \qquad H_{\text{eff}}(\beta) = -\beta^{-1}\log W'.$$

The matrix W' is strictly positive: for $v \neq 0$, sum the strictly positive quadratic forms of $v \otimes e_j$ over an orthonormal basis of $\mathcal{H}_E$. Its Hermitian spectral logarithm is therefore defined. The identity $e^{-\beta H_{\text{eff}}} = W'$ proves partition and retained *equal-time* observable preservation by the partial-trace property. Repeated traces compose. The full Pauli expansion, including its identity coefficient, is retained. It is generally nonlocal and depends on β; this statement does not preserve all fine real-time dynamics or supply a local zero-temperature/thermodynamic RG fixed point.

`FiniteRG.lean` checks nine actual finite scalar marginal, exponential-action, partition, observable, composition, error-sandwich and transfer-trace statements. The quantum spectral-log and complete thermodynamic/field-theory interfaces are not formalized by these lemmas. These physics constructions give no arithmetic determinant comparison or new GRH conclusion.

13 Linear relations and functorial branch filtrations

The original linear-relations chapter mixes graphs, partial relations and stable kernels. We keep these different objects separate. For a field k, a linear relation is a subspace $C \subseteq V \oplus V$; its inverse swaps coordinates and $DC = \{(x,z) : \exists y, (x,y) \in C, (y,z) \in D\}$. The full graph $\Gamma_T = \{(x, Tx) : x \in V\}$ always has dimension $\dim V$, including pairs with zero output. In particular $\Gamma_T^n(0) = 0$, whereas $(\Gamma_T^{-1})^n(0) = \ker T^n$. Thus the source's $\kappa(\Gamma_T) = \bigcup_n \Gamma_T^n(0)$ is zero, not the ordinary stable kernel.

For odd $m \geq 3$, use the actual clean dessin of $y^2 = x^m - 1$, $\beta = x^m$, on the compact smooth curve, in characteristic zero. Its genus is $(m-1)/2$; its infinite locally gentle algebra A_m has vertices $i \in \mathbb{Z}/m$, arrows $a_i, b_i : i \to i-1$, and mixed consecutive branch products zero. Below paths act in their direction of travel on left representations over any coefficient field k. Let $\mathcal{E}_m$ be the full category of finite-dimensional representations with every $b_i = 0$ and a_i invertible for $i \neq 0$.

Theorem 13.1 (A concrete relation category on an infinite dessin algebra)**.** *The category $\mathcal{E}_m$ is equivalent to finite-dimensional pairs (V,T) with intertwining maps. A normal form $N_m(T)$ puts V at each vertex, $a_0 = T$, and all other $a_i = 1$. For $n = \dim V$,*

$$V = K(T) \oplus I(T), \qquad K(T) = \ker T^n, \quad I(T) = \operatorname{im} T^n.$$

These summands and their kernel/image filtrations are functorial. The first restriction is nilpotent and the second invertible. The central full-turn parameter u acts as T. A finite $N_m(T)$ extends

continuously to the u-adically completed order precisely when T is nilpotent. Such modules are torsion, not nonzero order lattices.

Proof. Transport all vertex spaces to vertex zero along the invertible arrows; the remaining map is the full-turn monodromy T. Arrow-compatible maps become exactly $hT = T'h$. Finite kernel/image chains stabilize by step n. If $v = T^n w$ and $T^n v = 0$, then $w \in \ker T^{2n} = \ker T^n$, hence $v = 0$; rank-nullity proves the direct sum. Stability gives the two restrictions, and $hT^r = T'^r h$ proves functoriality. Continuity on a finite-dimensional discrete space means some u^r kills it. Conversely nilpotence makes every formal series act by a finite sum. This proves the completion assertion. □

Over an algebraically closed k, the indecomposable branch pairs are $J_r(\lambda)$. The value $\lambda = 1$ is permitted for a polynomial band; it is not a forbidden deformation parameter. The nilpotent blocks are strings, but are not all rigid: $N_m(J_2(0))$ is a nonsplit self-extension of $N_m(J_1(0))$. The Hom dimension between two such Jordan pairs is $\min(r, s)$ for equal eigenvalues and zero otherwise. These assertions concern this specified full subcategory, not all infinite modules.

If L transports once along the a-arrows on $\bigoplus_i V$, then

$$\det(1 - zL) = \det(1 - z^m T). \tag{7}$$

Eliminating the identity arrow blocks leaves the final monodromy block; the eliminated identity blocks have determinant one. For $J_r(\lambda)$ the answer is $(1 - \lambda z^m)^r$. Nilpotent strings therefore contribute one, and arbitrary bands have no forced unit-circle zero location. This determinant is an actual cyclic-quiver invariant; it is not the missing arithmetic positive determinant comparison.

14 A kernel-checked extension over the infinite dessin algebra

The preceding branch example can be realized on every element of the infinite algebra, rather than verified only on sampled arrow matrices. Use the complex genus-two clean dessin $y^2 = x^5 - 1$, $\beta = x^5$, and any coefficient field k. Put $H \subseteq M_5(k[t])$ equal to the matrices whose reduction modulo t is upper triangular. Let

$$B = H \times_{k^5} H,$$

where each map to k^5 takes the constant diagonal. Equivalently, B is generated inside $M_5(k[t])^2$ by

$$e_i = (E_{ii}, E_{ii}), \quad a_i = (w_i E_{i-1,i}, 0), \quad b_i = (0, w_i E_{i-1,i}), \qquad w_0 = t, \quad w_i = 1 \ (i \neq 0).$$

Indices are modulo five. Matrix multiplication acts on left column modules. The path-normal-form proof below identifies B with A_5; this is a polynomial algebra, not an arrow-truncated finite quotient.

Theorem 14.1 (An ambient module extension)**.** *Let $D = k[\epsilon]/(\epsilon^2)$, $M_1 = k^5$ and $M_2 = D^5$. For $(C(t), D'(t)) \in B$, act on M_1 by $C(0)$ and on M_2 by $C(\epsilon)$. These are unital B-modules, and*

$$0 \longrightarrow M_1 \xrightarrow{\ i\ } M_2 \xrightarrow{\ p\ } M_1 \longrightarrow 0, \qquad i(x) = \epsilon x, \quad p(x + \epsilon y) = x,$$

is exact and nonsplit. Moreover B is infinite dimensional over k. The assertions hold in every characteristic, including two.

Proof. Evaluation of polynomials gives both actions. Multiplying $C(\epsilon) = C_0 + \epsilon C_1$ proves B-linearity of i, p; their kernel and image follow coordinatewise. If s were a B-linear section, the constant coefficient of $(se_0)_0$ would be one. But a_0 kills M_1, while $(a_0 s e_0)_4 = \epsilon (se_0)_0$ has nonzero ϵ coefficient, a contradiction. Finally the permitted turn $c = a_1 a_2 a_3 a_4 a_0$ has first component tE_{00}. The unital algebra map $k[t] \to B$, $f \mapsto f(c)$, is injective: its $(0,0)$ first-component entry is $f(t)$. Thus B cannot be finite dimensional. □

In the preceding notation these modules are $N_5(J_1(0))$ and $N_5(J_2(0))$. They are finite torsion modules over an infinite algebra; they are not nonzero lattices over its completed surface order. The nonsplit sequence defines a nonzero Yoneda extension class in the ambient module category. No assertion that it is almost split in that entire category is needed or inferred. In particular it does not supply an arithmetic spectral determinant or a zero-location theorem.

Theorem 14.2 (Unbounded normal-image faithfulness)**.** *In $M_5(k[t])^2$, index the stationary images by i and the positive images by (b,i,j,q), where $b \in \{a,b\}$, $0 \le i,j < 5$, $q \ge 0$ and $i < j$ or $q > 0$. The positive image is $t^q E_{ij}$ in factor b, with zero in the other factor. Its path length is $5q+j-i > 0$. All these images, including the shared stationary images, are linearly independent over every field. Every finite normal expansion has uniquely recoverable coefficients, without a bound on degree or path length.*

Proof. Read a stationary coefficient from the first factor's constant diagonal; read a positive coefficient from the indicated factor, entry and degree. The index restriction ensures that no positive image contributes to a constant diagonal. These linear selectors are biorthogonal to the images, so applying each selector to a finite expansion recovers its coefficient. □

This theorem is now kernel checked for actual polynomial matrix pairs. The membership and spanning steps are supplied by the next theorem. Comparison with a separately constructed path quotient remains a distinct algebraic formalization task.

Theorem 14.3 (Normal coordinates in the generated algebra)**.** *Let $A \subseteq M_5(k[t])^2$ be the algebra generated by the displayed vertices and arrows, and let $\mathcal{K}$ be the normal index set of Theorem 14.2. Every normal image belongs to A. The actual A-valued finite expansion*

$$E : k^{(\mathcal{K})} \longrightarrow A, \qquad c \longmapsto \sum_{x \in \mathcal{K}} c_x v_x$$

is a linear isomorphism, whose inverse is $(E^{-1}a)_x = \lambda_x(a)$. Thus the normal images form a basis of the infinite generated algebra, not merely an independent family in its ambient matrix space.

Proof. Write $P_b(i,j)$ for the shortest positive branch walk from j to i, using a full turn when $i = j$. Its image is $t^d E_{ij}$ in branch b, where $d = 0$ if $i < j$ and $d = 1$ otherwise. The actual arrow product $P_b(i,i)^n P_b(i,j)$ has image $t^{n+d}E_{ij}$ for every $n \ge 0$. Admissibility gives $q \ge d$, so $n = q - d$ proves membership.

The normal span contains the generators and $1 = \sum e_i$ and is multiplicatively closed: stationary factors select endpoints; positive factors either multiply to zero or to $t^{q+r}E_{il}$ in the same branch. When $q + r = 0$, the two strict upper inequalities imply $i < l$, so this output slot is still permitted. Adjoin induction gives spanning of A. The previous coefficient selectors prove uniqueness and the inverse formula. □

All membership, spanning and finite-expansion assertions in this theorem are now Lean checked over arbitrary fields. The formal linear equivalence is not asserted to be an algebra map from an independently defined path quotient, nor a statement about completed orders, derived categories or GRH.

15 Learned fields, width limits and quantum geometry

Learned functions enter this program as candidate generators, physical surrogates and control laws. A network may suggest an arithmetic metric, operator or gluing datum; a successful fit does not

establish its positivity, analytic domain or identification with a completed L-function. Those are independent mathematical obligations. This section gives explicit interfaces and executed examples rather than treating "quantum", "tropical" or "infinite width" as a substitute for that comparison.

Proposition 15.1 (A specified quantum-feature width limit)**.** *Let $x \in \mathbb{R}^d$, $w \sim N(0, I_d)$ and $b \sim \mathrm{Unif}[0, 2\pi]$ be independent, and set $\phi_{w,b}(x) = \sqrt{2}\langle Z\rangle_{R_Y(w\cdot x+b)|0\rangle} = \sqrt{2}\cos(w \cdot x + b)$. For independent features,*

$$K_m(x, y) = m^{-1} \sum_{r=1}^{m} \phi_r(x)\phi_r(y) \longrightarrow K(x, y) = e^{-\|x-y\|^2/2}$$

almost surely on every specified finite set of pairs. Every K_m is a positive semidefinite Gram kernel. For a readout $f_a(x) = m^{-1/2} \sum_r a_r\phi_r(x)$ with only a trained, its tangent kernel is exactly K_m. These statements do not assert the same limit for a jointly trained finite quantum transformer.

Proof. Uniform-phase integration gives $\mathbb{E}_b[2\cos(u + b)\cos(v + b)] = \cos(u - v)$. The Gaussian characteristic function then gives K. Products are bounded by two, so the strong law applies to each pair and hence any finite union. For any finite coefficient vector c, the Gram quadratic form is $m^{-1} \sum_r |\sum_i c_i\phi_r(x_i)|^2 \geq 0$. Differentiating with respect to a_r gives the final assertion. □

The frozen-kernel squared-loss flow has residual $r(t) = \exp(-K_m t)r(0)$ when the loss normalization is absorbed into time. Null directions do not decay. A complex feature space instead uses the Hermitian kernel $\mathbb{E}[\phi(x)\overline{\phi(y)}]$. A finite coefficient field has neither this Gaussian law nor an ordered positive Hilbert metric; changing algebraic coefficients is a separate operation.

Proposition 15.2 (Tropical limit and a cancellation boundary)**.** *For real $s_1, \ldots, s_n$ and $\tau > 0$,*

$$0 \leq \tau \log \sum_j e^{s_j/\tau} - \max_j s_j \leq \tau \log n.$$

An attention maximum therefore has a controlled log-sum-exp approximation. It does not by itself define a tropicalization of a variety or an HMS functor. In a valued coefficient field, valuation of a sum can increase by cancellation.

Proof. After subtracting $M = \max s_j$, the exponential sum lies between one and n. For the second statement, both $1+T$ and $-1+T$ have valuation zero in $\mathbb{Q}((T))$, whereas their sum $2T$ has valuation one. In characteristic two that sum is zero. Thus replacing every addition by its tropical minimum without tracking leading coefficients loses information. □

Kontsevich–Soibelman constructions require valued analytic and integral-affine gluing data; appropriate gentle/Fukaya comparisons require specified categories, gradings, stops and smoothness hypotheses [92, 91, 93]. Learned coefficients can propose such data, but the compatibility equations and arithmetic comparison still need proofs.

Executed physical examples. On the actual clean genus-two dessin, put seven quantum spins on its $K_{2,5}$ incidence graph and compute a normalized Gibbs state. Exact white-spin partial trace gives a two-spin quantum RG marginal. Twelve RTX4090 runs train four-qubit-feature diffusion and flow-matching models for its smoothed Z-measurement law: two objectives, entangled/product circuit controls, and three seeds. They have 5,810 nominal parameters and 2,000 steps each. A separate four-qubit variational purification flow learns the full known density interpolation, with

48 controls. Its minimum fidelity on a 33-point grid is 0.999903; this is not a uniform theorem or universal noise inverse. The matched controls show no consistent entanglement advantage.

Actual cuStateVec/cuTensorNet execution also covers 7-, 11- and 23-qubit finite circuits on genus-two, genus-four and genus-ten dessin incidence graphs. The 23-qubit case has 8,388,608 amplitudes and 447 gates. These larger circuits are untrained; the small cases have independent NumPy comparisons, while the largest has normalization and inverse-circuit checks. The exact environment and all failed/preliminary records are preserved.

Geometric and hardware interfaces. For the contracted m-edge dessin, the graph fundamental group is F_{m-1}, and its universal graph cover is the m-regular tree. The theta case $m = 3$ has a three-regular cover, distinct from the four-regular Cayley graph of F_2. Neither is the coefficient field $\mathbb{F}_2$ or the universal cover of the filled compact surface. Explicit path modules over the infinite genus-two algebra give Hom, Ext and Tor computations; Frobenius pants tensors give exact TQFT gluing, with a heralded physical implementation. CHSH gives a finite game benchmark, distinct from the unrestricted correlation separation in MIP*=RE [99]. Three executable tropical hardware references and a source-pinned chip-design survey complement the circuits. They are arithmetic/control designs, not synthesized chips, validated CFT/string dualities or a proof of GRH.

16 Complementary physical datasets and checked dynamics

The learned program now has two additional, separately constructed corpora: 16,384 effective-material response records and22,951 molecular conformer/force records in their default configurations. Time samples are not counted as independent systems. Source-linked and dynamical configurations are additional. The active HolonForge/Aster and Mora-3 repositories were read only; no data, vocabulary, checkpoint, source enrollment or training process was changed.

The material corpus covers SSH, Kitaev, QWZ, Rice–Mele, Harper and actual clean-dessin classical RG models. The molecular corpus includes designed amides and peptides, with explicit-H MMFF94 calculations, geometric persistence, and method-labelled conformations. Forty-eight existing Aster MP20 structures, 32 existing Mora PubChem identities and eight public PDB structures supply separately labelled anchors. Their new computed responses are not new experimental observations, DFT labels, binding affinities or CISS measurements.

All conformer, gauge, intervention and task descendants share a partition. Specified physical/chemical similarity edges are closed transitively before component-level stratification. Large components are kept intact even when 80/10/10 proportions or rare-class coverage become impossible. This gives a finite, threshold-relative separation statement; it cannot guarantee every kind of statistical independence or prevent overfitting. Source anchors are train only. Molecular anchors are also checked against every default heldout using the same chemical and applicable geometric thresholds; conflicting parent groups are excluded. Global exposure to the models' previous corpora is a separate evaluation gate.

For a number-conserving one-particle Hamiltonian H, write $P_{ij} = \langle c_j^\dagger c_i \rangle$ and $U(t) = e^{-itH}$. The data use $P(t) = U(t)P(0)U(t)^*$, occupations $n_i = P_{ii}$ and outgoing currents $J_{ij} = -2\,\mathrm{Im}(H_{ij}P_{ji})$. Then $\dot{n}_i = -\sum_j J_{ij}$, directly from $i\dot{P} = [H, P]$. Paired BdG evolution includes its independently computed condensate source; it is not falsely treated as electron-number-conserving normal transport. An exact interacting Hubbard dimer supplements Gaussian exchange correlations. Molecular nuclear dynamics uses MMFF94, with a separately labelled, uncalibrated effective π-orbital model for charge modulation.

Parquet configurations retain condition, target and evaluation roles, source versions/hashes,

units, failures and audit evidence. The local dataset explorer shows actual structures, currents, time evolution, graph wavelets and checked circuit compilations. These computations support candidate discovery and model validation; the independent arithmetic determinant/zero-location bridge remains open. No new whole-chapter Lean formalization is asserted.

S1 Expanded dessin, resolution and derived calculations

S1.1 The cycle order in coordinates

Number a permitted cycle in travel order $0, \dots, \ell-1$. Map an arrow $i \to i+1$ to $E_{i,i+1}$ for $i < \ell-1$, and the final arrow to $u_c E_{\ell-1,0}$. A path from i to j then gives $u_c^r E_{ij}$, where r counts cut crossings. These are linearly independent path normal forms, and every positive-length permitted path has exactly one such expression on exactly one branch. The full-cycle sum in a branch is $u_c I$. At a paired edge its two diagonal occurrences come from a single original idempotent; no positive-length coefficients are identified. This proves the order description including self-pairings within one cycle. Although the suborder is generically a product of full matrix algebras, it is not their unglued product at the special fiber.

For the interval this becomes the literal fiber product

$$\Lambda = \{(f,g) \in k[[X]]^2 : f(0) = g(0)\}, \qquad x = (X,0),\ y = (0,X),\ u = (X,X).$$

Every element has a unique expression $a(u) + b(u)x$: take $a = g$ and $b = (f-g)/X$. Thus it is R-free on $1, x$ with $x^2 = ux$ and $y = u - x$. The annihilator of x consists of pairs $(0,g)$ with $g(0) = 0$, exactly $y\Lambda$; the other branch is symmetric. Multiplication by u is injective, coefficientwise. These are actual formal power-series statements, not conclusions from finite truncations.

S1.2 The two-arm complex and its calculations

For an arrow a write $f(a)$ for its unique forbidden successor. The summand in $P_n(i)$ indexed by the initial arrow a has endpoint $t(f^{n-1}(a))$. Its map multiplies by that last arrow. Thus the resolution can be computed by iterating a finite permutation, while the projectives themselves remain infinite-dimensional. The Hom complex against a module N is the direct sum of the endpoint spaces with differentials given by the same arrow matrices. Tensor against a left module instead uses the same path entries acting on the left. No rank or exactness claim for an arbitrary N is inferred by counting endpoints alone.

For the nodal residue field the maps are particularly explicit:

$$\cdots \xrightarrow{\operatorname{diag}(x,y)} \Lambda^2 \xrightarrow{\operatorname{diag}(y,x)} \Lambda^2 \xrightarrow{(x\ \ y)} \Lambda \longrightarrow k \longrightarrow 0.$$

If $xz_1 + yz_2 = 0$, its first branch says the first branch of z_1 is zero, and its second branch says the second branch of z_2 is zero. Hence $z_1 = yw_1$, $z_2 = xw_2$. The same annihilator calculation proves the alternating kernels in all degrees. Projectivity is genuine: the terms are regular modules or their finite direct sums. It is not a finite free resolution, despite the period of its positive tail.

For the elliptic theta the quiver vertices are $v_0 = \{0,3\}$, $v_1 = \{1,4\}$, $v_2 = \{2,5\}$. The arrows a_0, a_3 go $v_0 \to v_2$, a_2, a_5 go $v_2 \to v_1$, and a_1, a_4 go $v_1 \to v_0$. The two forbidden arms from v_0 are

$$a_0, a_5, a_1, a_3, a_2, a_4, \dots, \qquad a_3, a_2, a_4, a_0, a_5, a_1, \dots.$$

Their top pairs are $(v_2, v_2), (v_1, v_1), (v_0, v_0)$, repeated. The three steps exchange the arms. Their distinct top pairs exclude shorter unordered positive periods. The spherical theta instead has

forbidden two-cycles and successive top pairs $(v_2, v_1), (v_0, v_0)$, giving period two. Thus even identical hereditary overorders need not give identical resolutions.

Quadratic Koszul duality here has a directly checkable local meaning: relations complement the two possible successors. The dual dessin replaces the graph rotation by ϕ^{-1}. In the Ext algebra against A/J, concatenating the forbidden-word basis gives the product when the join is forbidden; a permitted join gives zero. One can see this by lifting a length-r basis class to the resolution: discard the prescribed initial r letters and identify the remaining summand with its suffix resolution, taking zero on other prefixes. These prefix-discard comparison maps commute with the suffix differentials, with the standard shift signs when viewed as shifted complexes. The concatenated indexing word exists exactly when its join is forbidden; evaluation of the two basis classes on that word gives one. If the join is permitted there is no such indexing word and the product is zero. Composition therefore concatenates indexing words, with the chosen right-module/Yoneda opposite convention. This is a graded algebra statement; it neither makes the unbounded resolution perfect nor constructs an equivalence of all formal-completion derived categories.

For an arbitrary bounded projective complex $P^\bullet$, its determinant line over a commutative coefficient ring is $\bigotimes_n (\det P^n)^{(-1)^n}$, with determinant duals for negative exponents. A chain automorphism acts on this line; a quasi-isomorphism gives the corresponding determinant-functor isomorphism. For a general endomorphism F, the expression for $1 - zF$ uses the locus where it is invertible and gives a rational determinant-line action/trivialization, not an everywhere invertible line map at singular parameters. Additivity is compatible with exact triangles under the usual determinant-functor conventions [17, 18]. In our order context an R-linear determinant uses the finite free underlying R-modules; it must specify a separate endomorphism. Determinants of a differential presentation, alternating determinants of a chain action, regularized analytic determinants and the Cartan path matrix are four distinct constructions. Their equality requires a proof, not their shared name.

The geometric model literature [6, 8, 9] is useful for identifying which crossings and gradings would have to be represented by a functor. To apply it here one must specify the category of polynomial strings, completed lattices or finite-length quotients; construct its surface/curve objects; and prove that degree-r intersections compute the Hom group into shift r, with local systems and boundary exceptions included. We have supplied explicit walks and maps in the examples, not claimed this general completed-order theorem. On a finite word, a map can already be computed rigorously without such a theorem: assign a coefficient matrix between equal-vertex occurrences and solve $F_{t(a)} N(a) = N'(a) F_{s(a)}$ for every arrow. The diamond calculations below implement exactly this system.

S2 Semi-invariants, tableaux and explicit intertwiners

S2.1 Derived information retained by a presentation

Let $K = \operatorname{im} f$. The injection $\operatorname{Hom}(K, N) \to \operatorname{Hom}(P_1, N)$ descends to the Ext injection used in Theorem 3.1; its kernel modulo $\operatorname{im} \operatorname{Hom}(P_0, N)$ is zero because the map $P_1 \to K$ is onto. The missing part of the cokernel detects maps on P_1 not descending through K. This explains exactly why an injective resolution map is needed for its full Euler interpretation.

In the node example the resolution of $M = \Lambda/(x)$ has maps $x, y, x, y, \ldots$. On $N = k[y]/(y^2)$ the Hom cochain maps are $0, y, 0, y, \ldots$. Thus $\operatorname{Hom}(M, N) = N$ and $\operatorname{Ext}^1(M, N) = \ker y$ have dimensions two and one; the cokernel of the first zero map is two-dimensional. In the elliptic-square example the first four Hom maps are $1, 0, 0, 0$, periodically. Hence degrees zero and one vanish,

degrees two and three are k, and degree four vanishes. This also makes clear why a nonzero scalar semi-invariant does not give a derived orthogonality theorem.

With coordinate action $(gF)(N) = F(g^{-1}N)$ the Schofield weight is $a - b$, the negative of the point-evaluation weight. For a single arrow it is the same elementary distinction as the action on $\mathrm{Hom}(V, W)$ versus on its coordinate functions $V \otimes W^*$. The Cauchy decomposition for coordinate functions is

$$\mathrm{Sym}^m(V \otimes W^*) \cong \bigoplus_{\lambda \vdash m} S_\lambda(V) \otimes S_\lambda(W)^*.$$

A relation quotient can kill these components; its ambient multiplicities do not descend unchanged. Acyclic-quiver semi-invariant/LR results such as [35] retain their acyclicity assumptions. The explicit two-arrow slice below is acyclic, whereas the whole theta quiver is not.

S2.2 One Young operator and one LR decomposition in coordinates

For $\lambda = (2, 1)$ number the first row $1, 2$ and second row 3. Then

$$Y = (I + P_{12})(I - P_{13}), \quad Y^2 = 3Y, \quad \dim \mathrm{im}(Y \mid (k^2)^{\otimes 3}) = 2.$$

The hook product is three; its Specht dimension is $3!/3 = 2$, also two here for a different reason. The normalized projector extracts one $S_{(2,1)}$ copy. The trace $\mathrm{tr}((Y/3)M^{\otimes 3})$ is therefore the Schur character, not twice that character. This fixes the common row/column normalization ambiguity.

Write $t = \mathrm{tr}\, M$, $\Delta = \det M$ for a two-dimensional endomorphism. Its complete characters obey $h_0 = 1$, $h_1 = t$, $h_m = t h_{m-1} - \Delta h_{m-2}$, and

$$\chi_{(a,b)}(M) = \Delta^b h_{a-b}(M), \quad a \geq b \geq 0.$$

In particular $h_2 = t^2 - \Delta$, $h_3 = t^3 - 2t\Delta$, and $\chi_{(2,1)} = t\Delta$. Thus $t h_2 = h_3 + \chi_{(2,1)}$ is the LR identity $V \otimes \mathrm{Sym}^2 V \cong \mathrm{Sym}^3 V \oplus S_{(2,1)}V$, with $S_{(2,1)}V \cong \det V \otimes V$.

An explicit natural splitting avoids relying just on characters. In characteristic zero multiplication $m : V \otimes \mathrm{Sym}^2 V \to \mathrm{Sym}^3 V$ has natural section

$$s(v_1 v_2 v_3) = \tfrac{1}{3} \sum_{i=1}^{3} v_i \otimes \prod_{j \neq i} v_j.$$

The map $j : \bigwedge^2 V \otimes V \to V \otimes \mathrm{Sym}^2 V$ sends $(v \wedge w) \otimes z$ to $v \otimes wz - w \otimes vz$. In dimension two it is injective: for a basis e_1, e_2 its images at $z = e_1, e_2$ have distinct independent coordinates. They lie in $\ker m$, whose dimension is $6 - 4 = 2$. Hence (s, j) is the required isomorphism. Tensorial definitions make both maps commute with every linear endomorphism, including singular arrow maps. This is a genuine LR intertwiner on our dessin representation stratum.

For a concrete theta representation take

$$X_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad X_2 = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad X_3 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Then $M = X_3 X_2 X_1 = \left(\begin{smallmatrix} 2 & 2 \\ 2 & 3 \end{smallmatrix}\right)$, $t = 5$, $\Delta = 2$, $h_2 = 23$, $h_3 = 105$ and $\chi_{(2,1)} = 10$. The identity is $115 = 105 + 10$. Since the other permitted branch is zero, every forbidden product is zero; this is an actual representation of the infinite elliptic dessin algebra.

In the invertible-arrow open stratum one can gauge X_1, X_2 to the identity, leaving conjugation of M. The invariant-ring proof in the core works also at singular arrows by cycle-trace generation,

not by assuming that this gauge choice extends there. The degree-one monodromy coefficients have total arrow degree three; this is why the Hilbert-series denominator uses $3j$, not j.

For the Cauchy kernel choose the Schur subspace inside the tensor power with its inherited Hilbert norm. $M^{\otimes m}$ preserves it and has norm at most $\|M\|^m$. This proves the trace bound for nonnormal matrices as well as diagonal ones. Formal Cauchy identities extend by polynomial continuity in matrix entries; the stated absolute bound then justifies analytic summation. Gram positivity does not assert entrywise positivity or positivity of an unrelated arithmetic form. For three exact test matrices M, $\left(\begin{smallmatrix}1&1\\0&2\end{smallmatrix}\right)$, $\left(\begin{smallmatrix}0&1\\-1&1\end{smallmatrix}\right)$, the choice $r = 1/100$ satisfies $rR_0^2 \le 21/100 < 1$ using Frobenius norms, and all seven principal minors of the rational kernel matrix are nonnegative.

S2.3 Two precise functorial limitations

At any actual vertex simple S of A, the split exact sequence $0 \to S \to S \oplus S \to S \to 0$ uses first inclusion and second projection. Termwise Sym^2 has dimensions $1, 3, 1$ at that vertex; its middle kernel contains the mixed monomial as well as the first square, but its first image contains only the first square. Middle homology is S. Thus even split exactness is not preserved. Genuine derived Schur constructions instead use mixed tensor degrees, signed total differentials and suitable flat/projective replacements [32, 33]; their membership in a particular order category must still be checked.

The R-lattice $M = Re_x \oplus Re_y$ with $x = \mathrm{diag}(u, 0)$, $y = \mathrm{diag}(0, u)$ has $x + y = uI$. On $\mathrm{Sym}^2_R M$ the images are $\mathrm{diag}(u^2, 0, 0)$ and $\mathrm{diag}(0, 0, u^2)$. Their sum kills the mixed vector, hence equals neither uI nor $u^2 I$. This is a specific failure to retain the original central action, not a failure of the polynomial monomial representation functor. Even a common ramified base substitution does not repair this mixed vector.

Semi-invariants retain useful moduli on nilpotent modules. In the theta quiver keep only $a_0, a_3 : v_0 \to v_2$, put their scalar values $(x, y) = (1, \lambda)$ and all other arrows zero. The v_1 simple splits off. Every length-two path acts by zero, so these are continuous completed modules. The two scalars have the same weight g_{v_2}/g_{v_0}, and y/x is a rational invariant on $x \neq 0$, giving the projective pencil $[x : y]$. It is not a regular invariant on the whole affine locus, whose orbit closures meet zero.

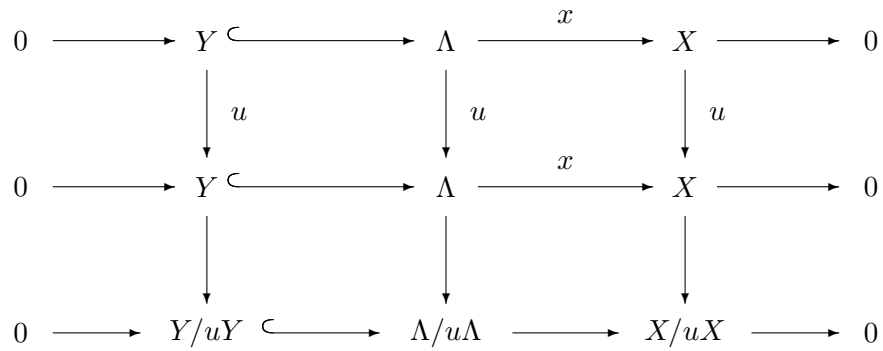
For the interval quotient take $x = J_2$, $y = \lambda J_2$. With its top and socle specified, the separated Kronecker coordinates are again $(1, \lambda)$; the relation to the same-weight ratio is exact. The pp formula in the core is stronger than merely noticing a ratio: on M_λ its solution set is $\mathrm{im}\, x$, whereas on M_μ, $\mu \neq \lambda$, the equation $(\mu - \lambda)xv = 0$ forces $z = xv = 0$. Different parameters are therefore distinguishable by actual positive-primitive conditions even though every cycle trace is zero.

S3 Full lattice and diamond diagrams

S3.1 The nodal exact diagram

The normalization proof of Theorem 4.1 uses BM inside the generic module. Replacing it without justification by $B \otimes_\Lambda M$ could introduce torsion and would not prove the same statement. Once the subdirect residue subspace is decomposed, the branch basis changes lift over complete DVRs; taking inverse images modulo the conductor gives the direct decomposition, not just a decomposition of residues.

Here is the promised diagram, with the terminal short exact column for every position:



Each vertical column is completed by zero above and below. In the bottom row identify $Y/uY = k$ via generator y, $X/uX = k$ via x, and $\Lambda/u\Lambda = k[\epsilon]/(\epsilon^2)$ with $x = \epsilon$, $y = -\epsilon$. Its first map is $1 \mapsto -\epsilon$, its second the augmentation. It is nonsplit as a dual-number sequence. This explicit order reduction, not an unrelated finite algebra, explains its appearance.

For a further Ext calculation, resolve $X \cong \Lambda/(y)$ with successive maps $y, x, \ldots$. On Y the first is multiplication by y and the second zero, so $\mathrm{Ext}^1(X, Y) = Y/yY = k$; on X the first is zero and the second injective, so $\mathrm{Ext}^1(X, X) = 0$. Interchanging branches gives the other two cases. The two nonsplit lattice rows realize the nonzero cross classes. Their Hom spaces are finite over R, generally infinite over k.

S3.2 Chebyshev geometry and the exact quotient

The identity $1 - T_n(z)^2 = (1 - z^2)U_{n-1}(z)^2$ follows from $z = \cos\theta$ and the sine/cosine multiple-angle formulas, hence as a polynomial identity. $T_n' = nU_{n-1}$. Therefore β_n has its only zero at infinity, of order $2n$; its n points over one are double since $\beta_n - 1 = T_n^2/(1 - T_n^2)$; and its poles are simple at ± 1 and double at the $n - 1$ interior roots of U_{n-1}. The derivative has no additional critical values. The total ramification is $(2n - 1) + n + (n - 1) = 4n - 2$, as required for a degree-$2n$ rational Belyi map. The dual graph has $n + 1$ vertices of degrees $1, 1, 2, \ldots, 2$, and n edges; it is a connected path. Its planar ribbon embedding is unique, so this identifies the particular nested-loop dessin, not only its passport.

In the quotient retaining orientation Ω, any surviving path traverses each chain edge at most once and either endpoint loop at most once. Its length is at most $n + 1$; thus the killed-arrow ideal contains J^{n+2}. The quotient of A_n and of its completion is the same $H_n(\Omega)$. For $n = 3$ with forward chain arrows, its path counts in lengths zero through four are $3, 4, 3, 2, 1$, total thirteen. These counts describe a quotient representation category of the infinite order, not a change of primary object.

Inflation is full because the two algebras have exactly the same actions on these annihilated objects. It preserves internal exact sequences. It does not preserve arbitrary ambient extension closure: the two adjacent simples are annihilated by the killed reverse arrow, but the two-dimensional extension having only that arrow nonzero is not. Likewise endpoint-local freeness over $k[l]/l^2$ or $k[r]/r^2$ in [46] is not R-freeness. To transfer an almost-split property to a larger category one must prove the requisite approximation/factorization statements; locally finite quiver criteria [40, 41, 42]

do not apply just because each vertex has two arrows. The permitted cycles violate interval/Hom finiteness.

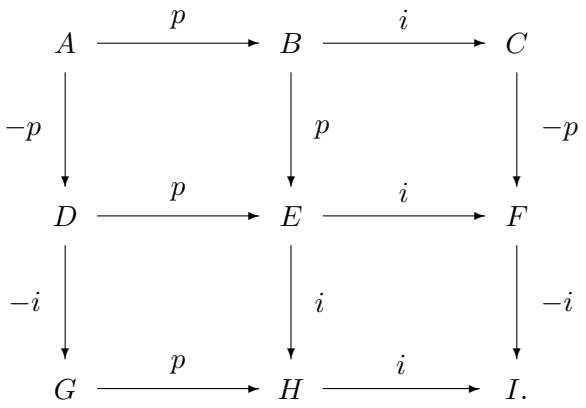
S3.3 Nine actual strings and all maps

Take $n = 3$ with $1 \xrightarrow{a} 2 \xrightarrow{b} 3$, loops l, r and $l^2 = r^2 = 0$ in the quotient. Here b denotes the retained forward a_2, not a reverse arrow of the original doubled chain. The following words are in travel order. A superscript $-$ means a formal inverse in a word, not an inverse algebra element.

Object	word	start	dimension vector
A	$l^-abrb^-a^-l^-$	1	$(4,2,2)$
B	l^-ab	1	$(2,1,1)$
C	l^-abr^-	1	$(2,1,2)$
D	$abrb^-a^-l^-$	1	$(3,2,2)$
E	ab	1	$(1,1,1)$
F	abr^-	1	$(1,1,2)$
G	$r^-b^-a^-labrb^-a^-l^-$	3	$(4,3,4)$
H	$r^-b^-a^-lab$	3	$(2,2,3)$
I	$r^-b^-a^-labr^-$	3	$(2,2,4)$

A basis vector is assigned to each visited position $0, \ldots, \ell(w)$ at its vertex. A direct letter sends its preceding vector to its following vector; an inverse letter sends the following vector to the preceding vector. Sum these elementary maps for each arrow and put all unmentioned entries zero. This defines every matrix, including repeated loops, and verifies the relations. Set the killed reverse arrows to zero to obtain actual Λ_3-modules.

Their signed grid is



Horizontal projections keep the initial word positions and kill the remaining positions. Horizontal inclusions append the final r^-, keeping old positions. Upper vertical projections delete the initial l^-: old position $j \geq 1$ maps to position $j - 1$, and old position zero maps to zero. Lower vertical inclusions prepend $r^-b^-a^-l$: position j maps to $j + 4$. Multiply these maps by the indicated signs. For every arrow these formulas commute with the string action, including the two positions adjacent to a cut or join; those are exactly where the loop inverse and chosen orientation matter.

Every small square has two length-two composites whose sum is zero. The exact sequences are

$$0 \to A \to B \oplus D \to E \to 0, \qquad 0 \to B \to C \oplus E \to F \to 0,$$
$$0 \to D \to E \oplus G \to H \to 0, \qquad 0 \to E \to F \oplus H \to I \to 0.$$

For example the maps out of A retain its initial positions $0, 1, 2, 3$ in B and delete its first position in D, so their joint kernel is zero. The maps to E are onto, and the dimension vectors give $\dim A + \dim E = \dim B + \dim D$ vertexwise. The zero composite plus this equality proves exactness. The other three use precisely the same keep/delete or append/prepend rules; their vertexwise dimension equalities and jointly injective/surjective maps prove exactness. The shipped exact checker constructs all matrices and verifies these assertions independently over $\mathbb{Q}$.

The stronger almost-split property is imported by identifying $E = w = ab$ as the minimal string of type $(2, 2)$ and these exact hook/cohook words with [46, Proposition3.12], using that paper's Propositions2.6/2.9 and Lemma3.10. Opposite-module conventions are reconciled by reversing travel/composition conventions; the closed doubled quiver is isomorphic to its opposite by exchanging the two arrows at each chain edge. Thus intrinsically $\tau E = A$, $\tau F = B$, $\tau H = D$, $\tau I = E$. The rows and columns of this grid are *not* generally short exact, unlike the nodal diagram. Two small cautions in that source are not needed here: a tube-rank label in Example3.9(3) is inconsistent with its listed mouth, and a general homogeneous-tube statement must not be read as a converse classifying every such object as a band. Neither is used to import these four meshes.

S3.4 The pure-injective quotient and what its dimension means

For $B_0 = \Lambda/J^2 = k\langle x, y\rangle/(x, y)^2$, words alternate direct and inverse letters because every direct/direct and inverse/inverse adjacent pair is forbidden. Up to rotation and inverse its only primitive band is xy^{-1}; the bridge quiver has two oriented band vertices with no connecting nontrivial bridge. The longest path has length zero. The domestic string results [44, 45] therefore give relative dimension $0 + 2 = 2$, not an automatic dimension three for every one-domestic algebra. Finite-dimensional points have relative CB rank zero; the appropriate Prüfer/adic and one-sided periodic string points have rank one; the unique generic point has rank two. These statements concern the quotient category with its inherited pp structure.

Indeed a pp formula is a finite system of linear equations with existentially quantified unknowns. On a module annihilated by J^2 every coefficient reduces to B_0, so purity and pure-injectivity are preserved in both directions under inflation. The condition $MJ^2 = 0$ is a definable closed condition, giving the closed Ziegler subspace. Its relative ranks need not be ranks in the entire Λ spectrum. This subcategory is not extension closed: $\Lambda/(x^3, y)$ gives an extension of objects killed by J^2 that is not itself killed by J^2. Since u^2 annihilates every object of the subcategory, it contains no nonzero R-free lattice. Ringel's representation-theoretic node construction is also not a Morita equivalence identifying the geometric nodal order with the Kronecker algebra. Infinite-dimensional polynomial string classifications [38] retain finitely-controlled or pointwise-artinian assumptions; an arbitrary complete lattice does not automatically satisfy them as an algebraic A-module.

S4 The operator extension in detail

S4.1 A corrected cycle lift and the integer boundary map

In the permitted-state graph each length-ℓ cycle has a full matrix-corner Laurent/circle coordinate. Its chosen K_1 generator is the full cycle at one state plus the complementary identity. In Fock space

$L_c^* L_c = Q_a + v_{s(a)}$. Thus the tempting lift $L_c + (1 - Q_a)$ has overlapping initial pieces and is not an isometry. The corrected lift in Theorem 5.1 first restricts to Q_a:

$$W = L_c Q_a + (1 - Q_a), \quad W^* W = I, \quad I - WW^* = Q_a - L_c Q_a L_c^*.$$

A path starting with a and length greater than ℓ has uniquely the full prefix c followed by another path starting with a. The missing range consists exactly of lengths $1, \dots, \ell$. Its terminal multiset is the original graph's incident-edge multiset at that cycle vertex. These finite defect vectors all lie in terminal compact ideals; their classes give the claimed unsigned incidence entries. The sign convention affects only the overall sign of the boundary map.

Here is the Smith calculation without spectral assumptions. Choose a spanning tree of the connected underlying graph and one root. By changing signs on one bipartition of the tree, its $E \times V$ incidence submatrix becomes the ordinary oriented tree incidence, with $V - 1$ unit pivots. After eliminating these pivots, a remaining column represents the common root coordinate, and each nontree edge evaluates on it as zero when its fundamental cycle is even, or ± 2 when odd. A loop is the latter case. If all are even the graph is bipartite and this column is zero; otherwise its entries have gcd two, giving the final pivot two. Remaining rows are free. Consequently the kernel/cokernel, then the six-term sequence, give the K groups stated in the core. No theorem reconstructing a cyclic embedding from those groups is asserted.

The compact ideal is block diagonal, not $\mathcal{K}(\mathcal{H})$: prefixes never move terminal vertices, so cross-terminal rank-one maps do not belong to the displayed ideal. This is why the arithmetic insertion must choose one terminal sector. A graph/Cuntz boundary quotient kills the vacua and hence this ideal, so that insertion need not survive passage to the quotient.

S4.2 Spectral domain and arithmetic trace

A vector belongs to the adjoint domain of the diagonal N precisely when its coefficientwise product by the real sequence $|p|$ is square summable; this proves self-adjointness on the given domain. The truncations by length prove finite supports are a core. Resolvent coefficients $(|p| - i)^{-1}$ tend to zero and each length space is finite dimensional, giving compactness. Prefix and deletion shift the length by at most one, so preserve the domain and prove the commutator formulas there. Leibniz then handles the dense polynomial *-algebra.

Every terminal sector has one vacuum and two paths at each length $n \geq 1$, even when some graph endpoints coincide: distinct formal paths remain distinct basis vectors. Thus restriction to a sector still has a simple vacuum, but it need not be a faithful representation of the whole concrete algebra. The trace of e^{-tN^2} is $E + 2E \sum_{n \geq 1} e^{-tn^2}$, distinct from the semigroup trace of N in the core. The trace identity with $\chi(N)$ is only on $\operatorname{Re} s > 1$, where the product is trace class. Analytic continuation of the scalar L-function does not make that same operator product trace class on $\operatorname{Re} s > 0$.

S5 Abel determinant estimates and the arithmetic program

For completeness the inserted family has a full elementary convergence proof. Since complete nonprincipal character periods sum to zero, $|A_n| \leq B_0 \leq q_\chi$. On $2^k \leq n < 2^{k+1}$,

$$\sum |u_n|^2 \leq \frac{B_0^2}{(k+1)^2}, \qquad \sum |v_n(s)|^2 \leq M^2 (k+1)^2 r^k, \quad r = 2^{-2\sigma_0} < 1.$$

The second follows from $n^{-s}-(n+1)^{-s}=s\int_n^{n+1}x^{-s-1}\,dx$. Define

$$Q_J(r)=r^J\left(\frac{(J+1)^2}{1-r}+\frac{2(J+1)r}{(1-r)^2}+\frac{r(1+r)}{(1-r)^3}\right).$$

Expanding $(J+j+1)^2$ in the three geometric moment sums proves $Q_J=\sum_{k\ge J}(k+1)^2r^k\to 0$. Locally uniform ℓ^2 convergence of the holomorphic finite-coordinate vectors proves Banach holomorphy. The linear functional $z\mapsto\sum v_nz_n$ has norm $\|v\|$; this is the complex-linear coefficient pairing, not a conjugate-linear dependence on s.

Write $\mathcal{I}(v)(z)=\sum_n v_nz_n$ for the coefficient functional, distinct from the arrow ideal J. For $N_J=2^J-1$, $J\ge 1$, let $K_N=u\otimes(\mathcal{I}(P_Nv)-e_1^*)$. Then

$$\|K-K_{N_J}\|_1\le\sqrt{2}B_0M\sqrt{Q_J(r)}.$$

Finite summation by parts gives

$$T_N=\det_F(I+K_N)=\sum_{n\le N}A_n(n^{-s}-(n+1)^{-s})=S_N-A_N(N+1)^{-s}.$$

Adding the last endpoint times e_1^* to the functional recovers S_N exactly because $u_1=1$. Its trace-norm error is at most

$$\sqrt{2}B_0(M\sqrt{Q_J(r)}+B_0(N_J+1)^{-\sigma_0}).$$

For arbitrary cutoffs use the last whole dyadic block for an enclosing tail. Galerkin compressions also converge: $\|u-P_{N_J}u\|\le B_0/\sqrt{J}$. With $C_K=1+M\sqrt{(1+r)/(1-r)^3}$,

$$\|K-P_{N_J}KP_{N_J}\|_1\le\frac{B_0C_K}{\sqrt{J}}+\sqrt{2}B_0M\sqrt{Q_J(r)}.$$

The raw endpoint adds the same endpoint bound as before. Isometric insertion by U preserves nonzero singular values, every norm estimate, and the Fredholm determinant. Everything is in one compact ideal of the actual infinite dessin Fock algebra.

For primitive parity a use $g(s)=(q_\chi/\pi)^{(s+a)/2}\Gamma((s+a)/2)$, holomorphic nonzero on $\operatorname{Re}s>0$. The completed functional is $g(s)\mathcal{I}(v)-e_1^*$, and the completed raw functional is $g(s)(\mathcal{I}(P_Nv)+A_N(N+1)^{-s}e_1^*)-e_1^*$. Their determinants are exactly gL and gS_N. Scaling the entire old perturbation would instead give $1+g(L-1)$, the wrong normalization. Trace-class background is standard [26], while these special rank-one identities follow directly from $\det_F(I+u\otimes\varphi)=1+\varphi(u)$.

The metric obstruction is likewise elementary. With inner product conjugate-linear in its first variable, symmetry of $T=u\otimes\varphi$ implies

$$\varphi(z)\langle u,u\rangle=\overline{\varphi(u)}\langle u,z\rangle.$$

A nonzero φ cannot have $\varphi(u)=0$; thus all symmetric maps with that range have proportional functionals. For $n\ge 2$ the coefficient ratios at $\sigma+it$ and σ satisfy

$$\frac{v_n(\sigma+it)}{v_n(\sigma)}=\frac{\sigma+it}{\sigma}\,n^{-it}(1+O_{\sigma,t}(1/n)).$$

Integers nearest $\exp(2\pi k/|t|)$ and $\exp((2\pi k+\pi)/|t|)$ give opposite nonzero subsequential limits. Hence no common positive metric symmetrizes this family for all critical heights. Restricting any proposed metric on the inserted family to $U\ell^2$ gives the same contradiction. An independently constructed positive comparison is still a possible different route.

S5.1 Precisely how classical and recent sources enter

Over a finite field the Grothendieck trace-formula architecture writes a zeta function as alternating determinants on finite-dimensional cohomology; Deligne's purity supplies the weights of eigenvalues [22, 20, 21]. If a weight-w Frobenius is $q^{w/2}$ times a positive-metric unitary, each eigenvalue has modulus $q^{w/2}$. The norm conclusion is immediate; constructing that form or proving purity is the deep step. Symplectic/alternating duality $F^t J F = q^w J$ alone only pairs reciprocal eigenvalues and does not give their individual moduli. An order resolution or Schur-positive character expansion alone cannot replace this step in number-field cohomology.

The recent primary comparisons used here are specific:

- [53, Theorem3.12, pp.15–17] gives finite abelian covers with local orbit length $\log p$ and genuine arithmetic monodromy. It suggests what a descent-equipped order functor should match; it does not identify our u or a face cycle with a prime without such a functor.

- [54, Propositions4.1–4.3, Theorem4.6] builds finite-prime local models with archimedean Fourier factors. Its Hilbertian isomorphisms are not an asserted uniformly controlled common isometry for our arithmetic approximants. [55, Proposition3.1] gives determinacy of a particular local moment problem; positive finite Hankel matrices are not the missing global Weil inequality.

- [56, Theorem3.6] gives discreteness/lower boundedness for the stated restricted Weil form. Theorem5.10, pp.23–24, additionally assumes a finite truncation, $\lambda > 1$, a simple even bottom eigenfunction ξ and $\delta_N \xi = 1$. Its quotient metric is positive and its regularized determinant is $-i\lambda^{-iz}\widehat{\xi}(z)$. Passing to the infinite arithmetic comparison, including the ground-state and convergence conditions in its §8, is not automatic.

- [57, Theorem6.1, p.13] assumes a specified real distribution on an interval, the corresponding trigonometric-core form, essential self-adjointness, and an isolated simple lowest eigenvalue with reflection-even eigenfunction. Its Fourier transform then has only real zeros. Our degree operator's vacuum and length parity are not those data.

- [60, Theorem6.1] gives prime-indexed periodic packets in its dynamical construction, not one proved orbit per prime with the full desired cohomological trace formula. [62] isolates an obstruction to a particular real-coefficient cohomology with full functorial Artin multiplicities; it is not a theorem excluding every Hilbert–Pólya operator. The current survey [58] distinguishes remaining steps. The recent Picard-monoid construction [59] is not already a polarized abelian variety with a number-field purity theorem. [63] uses RH as an input in the cited spectral identification, so cannot be used circularly as its proof.

These are targeted theorem/proof readings, not claims that every page of every cited work has been audited. The preserved review records specify actual page coverage and hashes, including inaccessible original PDFs where the modern primary proof was used instead.

S5.2 A concrete sequence of remaining constructions

First choose a number-field model of a Belyi pair, its finite covers and conductor/ramification data. Construct actions or exact correspondences on the labelled order/module or analytic categories that recover the local prime Frobenius polynomials. The elliptic twists show why omitting this datum loses information. Test at more than one arithmetic model of the same complex dessin, not just its topological passport.

Second specify a finite projective or otherwise determinant-admissible complex and its *independent* arithmetic endomorphism. Prove exact-sequence/determinant-line compatibility. Use semi-invariants to encode presentation degeneracies, and Schur/exterior characters to calculate local factors, retaining every central-action and nilpotence distinction above. General derived Schur replacements require their own construction; termwise Schur on the resolution fails the exhibited split sequence test.

Third prove the same local factors, gamma terms and conductor give a globally convergent or correctly regularized determinant. Trace-norm convergence must be established for the chosen analytic perturbations; strong convergence is insufficient in general. The inserted Abel family supplies this topology for an elementary scalar encoding, but has neither the independent arithmetic spectrum nor the positive metric needed for zero location.

Finally produce an independently proved positive form or self-adjoint determinant-preserving realization. One acceptance criterion is the Hurwitz target in the core; another is equality with the precise arithmetic quadratic form in the recent spectral theorems, with all their domain, ground-state, normalization and limit hypotheses verified. The local Cauchy kernel, K-group computation and graph spectral dimension alone meet none of these final comparison hypotheses. No remaining condition is silently asserted true here. Dirichlet, Hecke, Dedekind and Artin scopes must remain individually recorded.

S6 Graph pullbacks and completions with specified fibers

S6.1 The complete Fock graph presentation

For graph C* conventions the edge operator has initial projection at its range vertex and final projection under its source vertex. Put

$$S_h = L_{a_h} Q_{a_{\tau h}}, \qquad T_h = L_{a_h} v_{[\tau h]}.$$

The two right-hand projections are orthogonal subprojections of $L_{a_h}^* L_{a_h} = Q_{a_{\tau h}} + v_{[\tau h]}$. Thus $S_h^* S_h = Q_{a_{\tau h}}$, $T_h^* T_h = v_{[\tau h]}$, and $S_h^* T_h = 0$. For distinct h the prefix ranges are orthogonal, so every other cross-edge CK1 product is zero. Their final projections sum to Q_{a_h}. The vacua have no outgoing edges. The resulting graph has $|H| + |Q_0|$ vertices and $2|H|$ edges, with cycles only among its state vertices and exits to the paired sinks. Gauge covariance and nonzero vertices prove faithfulness as well as surjectivity. This explains precisely how a graph C* algebra is obtained without imposing the incompatible original-quiver CK1 relations.

The graph integer map has one column for each regular vertex h and no sink columns:

$$d : \mathbb{Z}^H \longrightarrow \mathbb{Z}^H \oplus \mathbb{Z}^{Q_0}, \qquad d(e_h) = e_h - e_{\tau h} - b_{[\tau h]}.$$

The state part of $d(x) = 0$ says $x_h = x_{\tau^{-1} h}$; the sink part then says the constant cycle coefficients are in $\ker B_\Gamma$. For the cokernel choose a base state on each cycle and successively use $q_h = q_{\tau h} + b_{[\tau h]}$ to eliminate all other states. Closing the cycle leaves $\sum_{h \in c} b_{[h]} = 0$, while its base state remains free. Hence

$$\operatorname{coker} d \cong \mathbb{Z}^{\#\,\mathrm{cyc}\,\sigma} \oplus \operatorname{coker} B_\Gamma.$$

This gives an independent integral check of the previous compact-extension computation. The full labelled graph, diagonal projections and edge families carry more information than those coarse groups.

S6.2 A genuine application of the supplied pullback theorem

For the interval dessin let $E_2 = G_\alpha$ have loop vertices q_x, q_y, common sink b, loops s_x, s_y and exits t_x, t_y. Remove b to obtain F_2, two disjoint loop graphs. Its quotient is exactly $C(\mathbb{T})^2$. Let E_1 have the same three vertices and edges $e_{x,n} : q_x \to b$, $e_{y,n} : q_y \to b$, $n \geq 0$, and no other edges. Remove b to obtain the edgeless F_1, whose algebra is $\mathbb{C}^2$.

The inclusions $F_i \subset E_i$ are admissible in the precise sense of [65, Definition3.1]: the removed sink set is hereditary/saturated; all edges ending at retained vertices are retained; no vertex emits infinitely many edges into the removed set and a positive finite number outside it. For E_1 the latter outside count is zero, which is allowed. Define a path functor identity on objects by

$$e_{x,n} \longmapsto s_x^n t_x, \qquad e_{y,n} \longmapsto s_y^n t_y.$$

Images of distinct positive edges are prefix incomparable: at the first different length one has an exit and the other a loop. Source vertices and identities cause no exception. The image consists of all positive pointed paths in E_2, whose last edge is not a short loop; vertices are included separately. The first graph is acyclic and the removed sink has no short loop in the second. All hypotheses of Theorem3.4 are now verified, giving

$$C^*(E_1) \cong \mathcal{T}_A \underset{C(\mathbb{T})^2}{\times} \mathbb{C}^2.$$

The quotient of $C^*(E_1)$ by its sink ideal is $\mathbb{C}^2$, and its sink ideal is compact operators on the paths ending at b. The graph is acyclic, hence its algebra is AF. There are no regular vertices in E_1: the two sources are infinite emitters and b is a sink. The graph K matrix has no columns and three rows, yielding $K_0 = \mathbb{Z}^3$ and $K_1 = 0$.

This is a positive instance of the new source inside our infinite dessin framework. It is a constants-symbol subalgebra of the Fock model, not the entire Fock algebra or the formal order. For a permitted cycle of length greater than one, its full closed path is itself pointed in this paper's convention, so an object-preserving acyclic path functor cannot cover all such paths. That simple obstruction prevents copying this example to every cycle by analogy. The sink-extension Theorem4.3 also requires a specified original pullback, injective gluing maps, annihilation of glued vertex projections and coverage of all paths ending at the gluing sinks. A two-to-one identification of sink occurrences within one branch does not satisfy these assumptions without another construction.

S6.3 Diagonal residue obstruction and the corrected pullback

Suppose one tries to glue full matrix-circle blocks by their diagonal values. In M_2, $E_{12}E_{21} = E_{11}$, although both factors have zero diagonal. Thus the proposed boundary map is not a homomorphism. For $\ell > 1$, more generally any *-homomorphism from the full matrix-circle algebra to a commutative C*-algebra has equal images of $E_{ii} = E_{ij}E_{ji}$ and $E_{jj} = E_{ji}E_{ij}$. These images are both orthogonal and equal, hence zero; summing the diagonal units kills the image of one and then the whole homomorphism. This works for nonunital *-maps as well: a nonzero image of one would give a nonzero unital corner map. The failure concerns full matrix-circle blocks, not the already-restricted fiber algebra D_ℓ.

Evaluation $D_\ell \to \mathbb{C}^\ell$ is a genuine surjective *-homomorphism since at one both factors are diagonal. Its kernel is $C_0(\mathbb{T} \setminus \{1\}, M_\ell)$. Pulling it back along the repeated-coordinate embedding $\Delta_\alpha : \mathbb{C}^{Q_0} \to \mathbb{C}^H$ gives exactly the fiber conditions of $\mathcal{D}_H$. The paired constant diagonal matrices give a unital *-section. Hence this is a split C*-extension, not a claim deduced from a merely

commutative diagram of nonmultiplicative maps. Since $\mathbb{T} \setminus \{1\} \cong \mathbb{R}$, each kernel block is a matrix suspension of $\mathbb{C}$, with $K_0 = 0$, $K_1 = \mathbb{Z}$. Split exactness now proves Theorem 5.3.

The pairing is retained in this specified presentation, not recovered from the unmarked isomorphism class. In particular arbitrary constant permutation matrices can relabel a branch's diagonal positions, so the spherical and elliptic theta examples have isomorphic unmarked $\mathcal{D}_H$ despite different ribbon orderings. Keep the labelled arrows, cycle cuts and distinguished fiber as additional data when using it as a dessin model. Also the central branch unitary zI_ℓ represents ℓ times the primitive corner generator in K_1, not a new independent generator per matrix entry.

S6.4 Generation, faithfulness and the exact failure of extension

Let ρ denote the polynomial representation in Theorem 5.3. Its vertex images are orthogonal projections summing to one. Every arrow image lies under the correct source and range vertex projections. A forbidden product vanishes because the intermediate state indices are τh and $\alpha\tau h$, which are distinct. Thus ρ is an actual algebra map, not a fitted characteristic polynomial.

For a full turn starting at h, $\rho(c_h) = w_c E_{hh}$ with $w_c(z) = g(z)^{|c|}z$. It is a normal element of $\mathcal{D}_H$. The function w_c vanishes exactly at one; away from that point $w_c/|w_c| = z$. Therefore it is injective on all of $\mathbb{T}$, and a homeomorphism to its image. Nonunital continuous functional calculus at $\rho(c_h)$, using functions zero at zero, gives every fE_{hh} with $f \in C(\mathbb{T})$, $f(1) = 0$. The zero condition makes the result vanish in all other matrix entries and branches.

To obtain an off-diagonal entry choose the unique shortest permitted path from state h to j within its cycle, of length $r > 0$ and cut-crossing count $\kappa \in \{0, 1\}$. Its image is $g^r z^\kappa E_{hj}$. Given $f(1) = 0$, multiply it on the left by the already available diagonal element

$$\frac{f(z)z^{-\kappa}}{g(z)^r + \varepsilon} E_{hh}, \qquad \varepsilon > 0.$$

The product is $fg^r/(g^r + \varepsilon)E_{hj}$ and converges uniformly to fE_{hj}. Indeed near one $|f|$ is small, and on the complement of that neighborhood g^r has a positive minimum. These two bounds prove uniform convergence without assuming division by a function vanishing at the fiber. Thus the generated algebra contains the whole kernel ideal. Its quotient contains the entire paired diagonal algebra, so it is $\mathcal{D}_H$.

For injectivity let a finite linear combination of path images vanish. At one, all positive paths vanish, forcing the coefficients of the original vertex idempotents to vanish. Fix a state-matrix entry in a branch. Its remaining paths have lengths $r + m\ell$ and images

$$g^r z^\kappa \sum_m a_m (g^\ell z)^m E_{hj},$$

with a possible initial zero coefficient if $r = 0$. On $\mathbb{T} \setminus \{1\}$ the prefactor is nonzero, and w_c has infinite range, so the polynomial has infinitely many distinct zeros. All coefficients are zero. Repeating by entries proves faithfulness.

No boundedness improvement repairs full formal extension. Pick any $z_0 \neq 1$, let $\lambda = w_c(z_0)^{-1}$, and consider $1 - \lambda c_h$. Its inverse $\sum_{m \geq 0} \lambda^m c_h^m$ exists in the arrow-adic order. The matrix image at z_0 has a zero eigenvalue on state h, so cannot be invertible. A unital algebra homomorphism would preserve the inverse identity, an immediate contradiction. This proves nonextension without any continuity assumption. It does not rely on the false claim that a norm greater than one alone forces a geometric-series denominator to be singular. Scaling g to have norm at most one still does not allow arbitrary formal series.

The two valid operator completions are structurally different. In the bipartite case $K_0(\mathcal{T}_A) = \mathbb{Z}^{E+1}$ differs from $K_0(\mathcal{D}_H) = \mathbb{Z}^E$; in the nonbipartite case the former has the extra $\mathbb{Z}/2$ torsion. Neither is automatically the norm completion of the arrow-adic topology. No invariant constructed solely from these complex objects distinguishes the rational twists already exhibited; arithmetic descent remains a separate input.

S6.5 Invariant basis number and the Leavitt comparison

If a unital k-algebra R_1 has a unital representation on a nonzero finite-dimensional vector space V, an isomorphism $R_1^m \cong R_1^n$ induces an isomorphism $V^m \cong V^n$ by tensoring. Comparing dimensions gives $m = n$. This proves invariant basis number without requiring the representation to be faithful. Every original dessin algebra and completed order has a vertex-simple representation, and each permitted-state Laurent algebra has evaluation onto a nonzero matrix block. The Leavitt algebra of G_α has the same finite-dimensional representation after killing the vacuum sinks and evaluating one boundary cycle. Thus all of these examples have invariant basis number. Leavitt's free-rank phenomena are important elsewhere but cannot be assumed in this class simply from the name LPA.

For the specified finite graph G_α, the usual complex Leavitt path algebra embeds in $C^*(G_\alpha)$ with norm-dense image, using its canonical generators and involution [67, Theorem7.3]. Those generators are the split S_h, T_h, not the unsplit original gentle arrows equipped with ordinary CK1. This is the exact algebraic/analytic twin relationship applicable here.

S7 String finiteness and the smooth quotient surface

For a finite quiver, finitely controlled implies finitely generated: choose one finitely generated submodule containing each vertex space and sum the finitely many choices. Conversely the module itself works. Stabilization at each of finitely many vertices likewise gives ordinary artinianity. Crawley–Boevey's polynomial theorem therefore applies directly to our finitely generated and artinian polynomial modules, with its primitive injective bands in the latter case. It does not assert a classification of every possible algebraic module.

The formal branch $M = k[[x]]$, $y = 0$, gives a precise counterexample to that stronger assertion. An A-endomorphism F preserves each $x^n M$. For the polynomial truncation f_n of f, one has $F(f) - f_n F(1) \in x^n M$. Separatedness and all n imply $F(f) = fF(1)$. Hence $\mathrm{End}_A(M) = k[[x]]$ is local, so M is indecomposable. If an ordinary word module has $y = 0$, its word cannot contain y or y^{-1}, since such a letter would create a nonzero basis map. A reduced word using only $x^{\pm 1}$ cannot change direction. Its nonperiodic choices are finite strings, the free ray $k[x]$, and the opposite locally nilpotent ray. The first and last fail injectivity of x, and the free ray fails invertibility of $1 - x$. A two-sided pure-x word is periodic, and in every associated band x acts invertibly. On M, x is injective but not onto. None of these possibilities equals M. This includes arbitrary nonzero indecomposable Laurent coefficient modules, not just finite bands.

For a general nonzero completed lattice, finite generation over A would imply finite generation over $k[u]$. Every nonzero polynomial in u is a power of u times a unit of $k[[u]]$, so it acts injectively. The lattice would be a nonzero finite free $k[u]$-module. Multiplication by $1 - u$ on such a free module has a nonzero cokernel, but the same map is invertible over $k[[u]]$. This proves the contradiction in the core. The strictly descending u-adic chain separately excludes artinianity. These are actual failed hypotheses, not merely unverified applicability.

Lekili–Polishchuk's precise convention takes $D(A)$ to mean perfect dg modules. Their locally gentle definition allows infinite dimension, and their smoothness criterion says no forbidden cycle; properness says no permitted cycle. Both fail for every nonempty closed dessin here. Their

Theorem3.2.2 therefore does not apply to the full polynomial dessin algebra, even though it applies to suitable infinite-dimensional quotients. The OPS finite-dimensional geometric-intersection theorem has its own grading, local-system, boundary and exceptional coincident-curve conditions. An unrestricted statement that every gentle/string derived object is an ungraded geodesic crossing is not justified by either source.

For the interval quotient $B = A/(y) = k[x]$, global dimension one follows from its PID structure: every submodule of a finite free module is free, giving finite projective resolutions of all finitely generated modules. Thus $\mathrm{Perf}(B) = D^b(\mathrm{mod}_{\mathrm{fg}}\, B)$ in grading zero. Its permitted loop gives one unmarked boundary component and its length-one forbidden thread x gives one stop on the other boundary, the annulus of the smooth partially wrapped model. This construction uses the infinite polynomial quotient, not the completed $k[[x]]$ category.

Finally the quotient does not give a derived fully faithful embedding by ordinary inflation. Resolve B over A by the alternating maps $y, x, y, x, \ldots$. Hom into B gives $0, x, 0, x, \ldots$, so $\mathrm{Ext}^{2r}_A(B, B) = B/xB = k$ for every $r \geq 1$, whereas all positive self-Ext over B vanish. Consequently its inflation is not derived fully faithful, and the perfect B-module B becomes an infinite-projective-dimension A-module. A surface functor transferring these categories must account for that difference explicitly.

S8 Arithmetic character sums and the cellular determinant

S8.1 The uniform calculation and its exceptional cases

In the finite-field calculation write $k = \mathbb{F}_q$; the final order-insertion paragraphs specify a separate characteristic-zero coefficient field. The curve is nonsingular because its discriminant is $-432d^2 \neq 0$, and has one point at infinity. Thus counting $1 + \eta(x^3 - d)$ in each affine fiber proves the trace formula in Theorem 6.2 directly. The multiplicative group of k is cyclic. When $3 \mid q - 1$, a nonzero cube has three roots and a noncube none, detected by $1 + \rho(t) + \bar{\rho}(t)$. At zero this expression is one, using the *constant* one. This convention differs from mathlib's zero-extended trivial multiplicative character, whose value there is zero. The proof and formal components never replace the constant by that character.

The substitution is valid including $u = 0, 1$ because the characters are zero extended:

$$\sum_t \rho(t)\eta(t - d) = \sum_u \rho(du)\eta(du - d) = \rho(d)\eta(-d) \sum_u \rho(u)\eta(1 - u).$$

Multiplying by the initial minus sign gives exactly $\pi_d + \bar{\pi}_d$. Changing ρ to ρ^{-1} conjugates both j and λ_d, so does not change the unordered roots or the integer trace. Characteristics two and three and $d = 0$ are excluded throughout. For a residue characteristic $p \equiv 2 \pmod 3$, odd extension degrees have trace zero by cube bijectivity, whereas even degrees lie in the split case; no claim of zero trace for all extensions of that characteristic is made.

For completeness, choose a nonzero $\mathbb{F}_p$-linear functional on k and compose it with $t \mapsto e^{2\pi i t/p}$ to obtain ψ. For a nontrivial multiplicative character χ,

$$\begin{aligned} |G(\chi)|^2 &= \sum_{u \neq 0} \chi(u) \sum_{y \neq 0} \psi((u - 1)y) \\ &= (q - 1) - \sum_{u \neq 0,1} \chi(u) = q. \end{aligned}$$

Both orthogonality identities follow by translating a finite character sum by an element on which its character is not one. The $z = 0$ terms in $G(\chi)G(\theta)$ sum to $\theta(-1) \sum_x (\chi\theta)(x)$, zero if the product

is nontrivial. For $z \neq 0$ set $x = zu$, $y = z(1-u)$ to obtain $J(\chi,\theta)G(\chi\theta)$. Coprime orders three and two ensure our product character is nontrivial. Taking norms proves $|J|^2 = q$, rather than presuming a desired positivity. This is the classical calculation reviewed directly in [68]; no claim of novelty is made.

At seven the five nonzero summands are $-\omega^2, -\omega, \omega, -\omega^2, 1$, so $J = 3 + 2\omega$. Multiplication by $\lambda_2 = \omega^2$ gives $-1 - 3\omega$. Their matrices are

$$\begin{pmatrix} 3 & -2 \\ 2 & 1 \end{pmatrix}, \qquad \begin{pmatrix} -1 & 3 \\ -3 & 2 \end{pmatrix}.$$

Both have determinant seven and the same positive Gram form; their traces are four and one. The arithmetic distinction between the rational models now comes from explicit finite-field characters, while their complex dessin is unchanged. Exact prime-field diagnostics test every nonzero twist for the listed nineteen primes; the theorem itself is uniform in prime powers and rests on the proof, not that finite list.

S8.2 Deck descent and the actual cellular basis

The CM cellular construction in this and the next subsection assumes $q \equiv 1 \pmod 3$. The cover $\beta = x^3/d$ has deck transformations $(x,y) \mapsto (\zeta_3 x, y)$ and $(x,y) \mapsto (x,-y)$, generating $C_3 \times C_2 = C_6$. Label their actions on darts by $\sigma = (012)(345)$ and $\alpha = (03)(14)(25)$; they commute. Their product is $(0\,4\,2\,3\,1\,5)$. They preserve the edge pairing and both successor permutations. Consequently they define algebra automorphisms, preserve J and its completion and fix the common u. The projective-resolution maps are compatible since $\theta_g(a_h x) = a_{gh}\theta_g(x)$. Twisting module actions gives an exact functor and the corresponding relabelled resolution, without an assumption of arithmetic Frobenius.

Over an algebraic closure choose $a^3 = d$, $b^2 = d$ and $\varphi(x,y) = (ax, by)$ from E_1 to E_d. For the q-power Frobenius morphisms,

$$\varphi^{-1} \circ F_d \circ \varphi = h_d \circ F_1, \qquad h_d(x,y) = (a^{q-1}x, b^{q-1}y).$$

When $q \equiv 1 \pmod 6$, the first coordinate is the cubic residue $d^{(q-1)/3}$ and the second $\eta(d)$. Choose the embedding of cube roots defining ρ consistently. On the deck eigenspace labelled by dx/y, the *deck* character is their ratio, $\rho(d)/\eta(d) = \lambda_d$. This labels the twist, not a nonzero Frobenius action on differentials in characteristic p: that pullback is zero. An inverse convention for arithmetic Galois Frobenius changes the action convention; our displayed formula uses the geometric curve morphism explicitly. The character-sum polynomial fixes its trace without an unstated convention.

Orient the three geometric edges as in the core. Each occurs twice with opposite orientation in the single face, so $\partial_2 = 0$; ∂_1 has columns $(-1,1)^t$. The homology basis $b_0 = e_0 - e_1$, $b_1 = e_1 - e_2$ has Gram matrix G inherited from the orthonormal cellular edges. Rotation sends $b_0 \mapsto b_1$, $b_1 \mapsto -b_0 - b_1$, exactly C. The involution swaps vertices and reverses each edge, hence acts as $-I$ on this plane. Thus multiplication by $A + B\omega$ is literally $AI + BC$, not merely an abstract matrix with the correct trace. Expanding its determinant gives the norm $A^2 - AB + B^2$, and multiplication of this norm proves $M^t G M = qG$. The displayed positive quadratic decomposition proves the metric independently of unknown zeros.

S8.3 An explicit chain lift and determinant cancellation

Let P be the three-cycle permutation on C_1. Write $\pi_d = A + B\omega$ and $\gamma = A + B$. Define

$$F_1 = AI + BP, \qquad F_0 = \frac{1}{2}\begin{pmatrix} 1+\gamma & 1-\gamma \\ 1-\gamma & 1+\gamma \end{pmatrix}, \qquad F_2 = q.$$

These are rational cellular maps. The sum-zero plane of F_1 has restriction M_d, while the edge-sum line has eigenvalue γ. On C_0 the vertex-sum line has eigenvalue one and the boundary line eigenvalue γ. It follows directly that $F_0\partial_1 = \partial_1 F_1$; the other chain condition is zero. The induced maps on H_0, H_1, H_2 have characteristic factors $1 - T$, $P_d(T)$, $1 - qT$. In particular

$$\det(I - TF_0) = (1 - T)(1 - \gamma T), \qquad \det(I - TF_1) = (1 - \gamma T)P_d(T).$$

The contractible boundary factor cancels, giving

$$\frac{\det(I - TF_1)}{\det(I - TF_0)\det(I - TF_2)} = \frac{P_d(T)}{(1 - T)(1 - qT)}.$$

This equality holds as a formal rational identity and by cancellation at apparent intermediate singularities. The alternating determinant with exponent $(-1)^i$ is its inverse; signs are not changed silently. The standard elliptic trace and determinant theorem identifies the right side with the finite-field curve zeta function [22, 20]. The elementary proof above supplies its actual trace and norm here; it does not independently reprove the general cohomological trace formalism.

This lift also has the required descent character on H_1. Write the base Jacobi value as $j = A_1 + B_1\omega$. Then h_d acts there by $\eta(d)C^r$ when $\rho(d) = \omega^r$, so twisting $A_1 I + B_1 C$ multiplies its eigenvalue j by λ_d, giving $M_d = A_d I + B_d C$. Reduction modulo $C^2 + C + I = 0$ gives the coefficients of π_d. If the corresponding full three-edge polynomials differ by $c(I + P + P^2)$, the coefficients are $c = 0, \eta(d)B_1, \eta(d)A_1$ for $r = 0, 1, 2$ respectively; they have the same homology action and differ only on the contractible boundary pair. An explicit homotopy sends $v_- - v_+$ to $c(e_0 + e_1 + e_2)$ and the vertex sum to zero. Its $\partial H + H\partial$ is the edge difference and the boundary scalar difference $3c$. Thus this normalization is compatible up to rational chain homotopy. It is not asserted to arise from a continuous Frobenius map of the complex Riemann surface, or to give a canonical integral Tate-lattice comparison.

S8.4 A decisive restriction at inert primes

For $q \equiv 2 \pmod 3$, Frobenius sends ζ_3 to ζ_3^{-1}. Thus its cohomological relation with the cubic deck action requires $MC = C^2 M$ (the pullback convention yields the equivalent reversed relation). On a real two-dimensional carrier with our fixed C, comparing entries forces

$$M = \begin{pmatrix} a & b \\ a+b & -a \end{pmatrix}, \qquad \det M = -a^2 - ab - b^2 = -\tfrac{1}{2}(a^2 + b^2 + (a+b)^2) \le 0.$$

The genuine elliptic Frobenius determinant is $q > 0$, so such a real carrier is impossible. This is an elementary instance of the real-coefficient cohomology issue discussed by Deninger [62]. It does not rule out a complex realization, a larger real space, a different coefficient theory or an infinite-dimensional height operator. In particular it is not a refutation of an unrestricted Hilbert–Pólya approach.

There is a local complex replacement, with its limitations explicit. The conjugation matrix

$$K = \begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix} \quad \text{satisfies} \quad K^2 = I,\ KC = C^2K,\ K^tGK = G.$$

Set $F = i\sqrt{q}\,K$ on the complexification. Then $\operatorname{tr} F = 0$, $\det F = q$, $F^2 = -qI$ and $F^*GF = qG$, giving the correct trace-zero local factor. This is a two-dimensional complex, hence four-dimensional real, local model. This is a one-q model; independently choosing $i\sqrt{q}K$ for each extension field does not give Frobenius-power compatibility. Such a family would need to start from one F_p and prove the comparison for its powers. The inserted $i\sqrt{q}$ is not a canonical rational or integral Tate comparison or a construction of a global positive determinant. The fixed real two-dimensional model has genuinely failed; its surviving split-prime construction and this complex alternative must not be conflated.

S8.5 The remaining order and global spectral obligations

The deck automorphism has finite order, so over characteristic zero its eigenvalues are roots of unity. Its permutation action alone cannot supply a weight-one norm $\sqrt{q} > 1$. The independently computed Jacobi coefficient is essential. The above construction closes the specified local cellular benchmark, not a reconstruction of arithmetic from bare complex dessin data.

For these order-insertion tests take $\Lambda = \Lambda_{\mathbb{Q}}$ over $R = \mathbb{Q}[[u]]$, with rational residue simples and $V_d = \mathbb{Q}^2$. This is distinct from reducing the matrix in characteristic p, where its determinant p becomes zero. There are two concrete failures of a naive order lift. First, a residue simple S_e is annihilated by all arrows; $S_e \otimes V_d$ with independent endomorphism $I \otimes M_d$ has the desired underlying finite-$\mathbb{Q}$ determinant. But its projective dimension is infinite by the nonterminating forbidden resolution, and σ permutes all three vertex labels, so choosing S_e does not preserve full deck symmetry. This is not a perfect-order determinant realization.

Second, $\Lambda \otimes V_d$ is free of rank two over Λ, but the underlying R-rank of Λ is eighteen. In an R-basis, the Λ-*linear* $I \otimes M_d$ is eighteen copies of M_d, hence its determinant is P_d^{18}. On $e\Lambda$ the rank is six: each of its two hereditary branches contributes a row of length three, with finite conductor gluing changing no rank. This gives P_d^6. Restoring all three residue vertices similarly gives P_d^3 for this untwisted tensor action. A genuine deck-semilinear map permutes factors as well and need not have these powers; it requires its own comparison. No justified determinant-line root or multiplicity-removing functor is supplied here.

Finally fix a nonzero rational d when forming its global Euler product from good reductions. Local purity cannot be multiplied into a global zero-location proof. Since each reciprocal root has modulus $\sqrt{p}$, the product of good elliptic factors is absolutely convergent on $\operatorname{Re} s > 3/2$. The local zeros at $\operatorname{Re} s = 1/2$ are outside that initial product argument. The finite-field curve zeta's center is $1/2$; the global elliptic L-function's unshifted center is one, or $1/2$ only after a half-unit shift. Neither is an identification with the primitive Dirichlet function targeted earlier. Ramified factors, archimedean normalization, adelic descent and the global determinant-preserving positive spectral limit remain separate obligations.

S9 The theta order interface beyond tensor insertion

We prove the multiplicity theorem and construct its actual functorial escape, retaining the completed infinite order as the primary algebra. The finite algebra used below is its specified special fiber, not a replacement primary example.

S9.1 Normal forms and the special fiber

Index the three quiver vertices by $i \in \mathbb{Z}/3$. Let $a_i^{\pm} : i \to i-1$ be the two permitted branches. Only paths of constant sign survive. Write $p_{i,r}^{\pm}$ for their length-r paths and $c_i^{\pm} = p_{i,3}^{\pm}$. The split order model gives

$$ue_i = c_i^+ + c_i^-, \qquad p_{i,r+3}^{\pm} = up_{i,r}^{\pm} \quad (r \geq 1).$$

The second identity follows because multiplying by the opposite full cycle switches branch and vanishes. An R-basis of $e_i\Lambda$ is

$$e_i,\ a_i^+, a_i^-,\ p_{i,2}^+, p_{i,2}^-,\ c_i^+.$$

These span by the two identities and completion. They are independent in the two hereditary branch orders: off-diagonal entries distinguish the four shorter paths; the remaining combination $xe_i + yc_i^+$ has paired diagonal entries $(x + uy, x)$, forcing $x = y = 0$. This is an actual power-series basis, not just a generic rank calculation.

In $B = \Lambda/u\Lambda$, the same eighteen basis vectors remain. A product of nonempty paths is zero if branches switch or the total length exceeds three; the length-three minus path is the negative of the plus path. A length-four path is u times a length-one path, hence zero. The radical $\mathfrak{r}$ is the positive-length ideal, with $\mathfrak{r}^4 = 0$ and $B/\mathfrak{r} = \mathbb{Q}^3$. Each e_iB has radical-layer dimensions $(1, 2, 2, 1)$ and has two composition factors at each vertex, so the Cartan matrix is

$$C_B = \begin{pmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}.$$

The Cartan matrix alone does not establish the endomorphism formula; the path filtration supplies that step.

S9.2 All projectives and all linear endomorphisms

Let J be the completed arrow ideal. Every $1 - x$, $x \in J$, has its convergent formal inverse; the same applies to $1 - yx$ for arbitrary $y \in \Lambda$. Thus J is contained in the Jacobson radical, and $\Lambda/J = \mathbb{Q}^3$ implies equality. If P is finitely generated projective, choose a basis of each multiplicity space in P/PJ and lift its vectors into Pe_i. This gives

$$Q = \bigoplus_i (e_i\Lambda)^{n_i} \longrightarrow P$$

inducing an isomorphism on tops. Nakayama makes this surjective. Its split kernel N is finitely generated and satisfies $N/NJ = 0$, hence vanishes by Nakayama. This proves the decomposition for every such projective without an unproved freeness assumption over Λ.

An entry $e_i\Lambda \to e_j\Lambda$ is left multiplication by an element of $e_j\Lambda e_i$. Its degree-zero part is possible only when $i = j$ and is a scalar multiple of e_i. For f on P, write its top blocks A_i and set $A = \operatorname{diag}_i A_i$. The remainder has positive path length. On P/uP it strictly raises the path filtration. At each of the two length-one and two length-two slots, and the length-zero and length-three slots, the degree-preserving map is exactly A. In an increasing-degree basis the matrix is lower block triangular. For the formal upper-block convention order the six slots instead as

$$3+,\ 2+,\ 2-,\ 1+,\ 1-,\ 0.$$

There are six diagonal copies of A, allowing arbitrary positive-length off-diagonal entries. Hence

$$\det_{\mathbb{Q}}(1 - T\overline{f}) = \det_{\mathbb{Q}}(1 - TA)^6.$$

Specializing the finite R-matrix determinant coefficientwise gives Theorem 6.3. No simultaneous diagonalizability, tensor form of f, arithmetic purity or deck equivariance is required.

For a bounded complex $P^\bullet$ with a linear chain map $f^\bullet$, fix either parity convention and write

$$D_R(P^\bullet, f^\bullet; T) = \prod_j \det_R(1 - Tf^j)^{(-1)^j}.$$

The inverse convention gives the reciprocal and the same obstruction. Each factor has constant coefficient one, so its reduction is nonzero. Specialization of this displayed rational expression is defined and equals

$$\left[\prod_{j,i} \det_{\mathbb{Q}}(1 - TA_i^j)^{(-1)^j}\right]^6.$$

Cancellations cannot change the divisibility of divisor multiplicities. In particular adding contractible projective pairs cannot repair it. Here a perfect endomorphism means a bounded-projective chain representative; no statement about an arbitrary infinite resolution or nonperfect lattice is inferred.

Generically put $N = \sum_i n_i$. As a right module over $M_3(K) \times M_3(K)$, P_K is the sum of N copies of each branch's three-dimensional row simple. Any commuting map has matrices $F_+, F_- \in M_N(K)$ on their multiplicity spaces, so

$$\det_K(1 - Tf_K) = \big(\det(1 - TF_+)\det(1 - TF_-)\big)^3.$$

For example, left multiplication by c_i^+ on $e_i\Lambda$ has ordinary characteristic determinant $(1 - uT)^3$, so a sixth power is forced only at the special fiber. This also shows why generic reduced determinants and ordinary determinants must not be conflated. We do not need an integral choice of either generic matrix to derive the cube obstruction.

For the local numerator, suppose a sixth power $A(T)^6/B(T)^6$ equals a nonzero quadratic $P(T)$ with $B \neq 0$. Polynomial degrees would give $6 \deg A = 2 + 6 \deg B$, impossible. This includes repeated roots. For the complete curve zeta, $q > 1$ and $|a_q| \leq 2\sqrt{q}$ imply $P(1) > 0$ and $P(1/q) = P(1)/q > 0$. Its two denominator roots are distinct uncancelled simple poles, again impossible for a sixth power. These arguments do not assume global RH or Artin holomorphy.

S9.3 An actual integral determinant functor of multiplicity one

Project to one hereditary cycle order $B_+ \subset M_3(R)$. Its standard column $W_+ = R^3$ is a left Λ-module with $e_iW_+ = R\epsilon_i$. More explicitly a_i^+ acts by $u^{w_i}E_{i,i-1}$, with $(w_0, w_1, w_2) = (1, 0, 0)$ for one cut, while every a_i^- acts by zero. These matrices respect travel-order multiplication as written: $E_{ij}E_{jk} = E_{ik}$. Each plus full cycle is uE_{ii}, so the common central u acts by uI. Nonempty paths tend to zero u-adically, giving the action of the *full completed order*, not just its polynomial subalgebra.

This (Λ, R)-bimodule defines $- \otimes^{\mathbf{L}}_{\Lambda} W_+$. On a bounded complex of finite projectives it is computed termwise, and its terms are finite free over R. An acyclic bounded projective complex is split contractible, and tensor preserves that contraction. Thus it descends to a genuine triangulated functor $\Phi_+ : \mathrm{Perf}(\Lambda) \to \mathrm{Perf}(R)$. No assertion of exactness on all un-derived modules or flatness of W_+ is needed. In fact W_+ has infinite projective dimension as a left Λ-module: Λ is noetherian, being finite over the central noetherian ring R, and W_+ is finitely generated. A finite projective resolution could thus be chosen with finite terms. Every left projective term has R-rank divisible by six by the same top argument, whereas W_+ has rank three, contradicting the alternating rank identity. This does not prevent tensoring bounded right-projective complexes.

The isomorphism $e_i\Lambda \otimes_\Lambda W_+ \to e_iW_+$ sends $e_i\lambda \otimes w \mapsto e_i\lambda w$. Its inverse is $e_iw \mapsto e_i \otimes e_iw$, proving rank one directly. Consequently $e_i\Lambda \otimes_{\mathbb{Q}} V$, with action $1 \otimes M$, maps to $R \otimes_{\mathbb{Q}} V$ with action M. Its characteristic determinant is $\det(1 - TM)$ *once*. On the regular input, the three vertex lines survive and the factor appears three times. A full symmetric vertex orbit therefore is not silently replaced by a chosen vertex.

Different cycle cuts give different integral column lattices. Their generic modules are isomorphic, and branch endomorphism characteristic determinants agree by generic conjugation; equality over K implies equality over R. Nevertheless an equivariant lattice is stronger data. Branch selection also changes the functor: for the unit $f = e_i + c_i^+$ on $e_i\Lambda$, the plus determinant is $1 + u$ and its characteristic factor is $1 - (1 + u)T$. Under α, c_i^+ becomes c_i^-, whose plus action is zero; the determinant becomes one and the characteristic factor $1 - T$. Since these scalar automorphisms differ, no natural isomorphism of determinant lines can make this chosen branch functor α-invariant for all endomorphisms. Pairing both branch functors makes this branch-swap data available but doubles the local factor on the basic-projective tensor input.

S9.4 The exact rotation obstruction and its allowed escape

Consider any left Λ-lattice supported on the plus branch, free of rank one at every e_i, and with central u acting as multiplication by u. Write its plus arrow scalars as $a_0, a_1, a_2 \in R$. Their product is u, so they are nonzero and their nonnegative valuations sum to one. A strict R-linear lift of σ that intertwines the order rotation permutes the three vertex lines by units. Its arrow-intertwining identities make the three arrow valuations equal. Thus $3m = 1$ for an integer $m \geq 0$, impossible. The argument already excludes an invertible intertwiner; imposing that its cube be one cannot help.

There is an elementary coefficient proof of the same obstruction. Intertwining by units makes vanishing of the three arrow constant coefficients equivalent. Their product has constant coefficient zero, so all three vanish. Factoring u from two of them would make $a_0a_1a_2$ divisible by u^2, contradicting that it equals u. This is the literal power-series component checked in Lean.

After the explicit ramified change $R' = \mathbb{Q}[[v]]$, $u = v^3$, *choose a new column lattice* with all three plus arrow maps $vE_{i,i-1}$. Cyclic permutation of the basis is an order-three intertwiner fixing v. The full-cycle relation is now $v^3 = u$ as required. Merely base-changing the old cut lattice gives arrow valuations $(3, 0, 0)$ and does not produce that linearization. This construction changes the base and lattice and does not establish an unramified descent comparison. It also still needs both branches to address α.

The rotation obstruction is about actual central-fixed rank-one branch lattices, not every possible determinant-level natural transformation. The branch characteristic determinants may be rotation-invariant through their generic reduced interpretation even without a lattice lift. Neither this theorem nor the ordinary-determinant obstruction excludes semilinear correspondences, other determinant categories or infinite analytic models.

The next arithmetic task is therefore narrower and more concrete: choose a branch functor, a ramified lattice or a different equivariant replacement only after proving the required descent comparison, and construct its prime action from arithmetic data rather than feed in the already known M_d. Bad primes, integral/Tate comparison, local-to-global conductor and gamma factors, and a positive spectral limit for the same completed function remain unresolved.

S10 Explicit tame descent and the full-deck coefficient test

The preceding ramified column is a precisely specified object of the original infinite theta-order framework. We compute its descent instead of treating a semilinear group action as automatically

effective integral descent. All constructions below are over the new *mathematical* coefficient field $k = \mathbb{Q}(\omega)$; they do not replace the original rational model or its source history.

S10.1 The action on genuine formal series

Write $R = k[[u]]$, $S = k[[v]]$ with $u = v^3$. Every $f(v) \in S$ has a unique expansion $f_0(u) + vf_1(u) + v^2 f_2(u)$. Hence S is finite free of rank three over R. Let $g(f)(v) = f(\omega v)$. On $W = S^3$ with basis $\epsilon_0, \epsilon_1, \epsilon_2$, put

$$\Theta(f_i\epsilon_i) = \omega^{-i} f_i(\omega v)\epsilon_i, \qquad C\epsilon_i = \epsilon_{i+1} \quad (i \bmod 3).$$

The plus arrow a_i^+ acts by $vE_{i,i-1}$, the minus arrow by zero. The cyclic product is uE_{ii} and arbitrarily long permitted paths tend to zero v-adically, so this is a module over the full completed order. For $i = 0$, the coefficient in the target under Θa_i^+ is ωv and in the source under $a_i^+\Theta$ it is $v\omega^{-2}$; these agree. The same calculation for $i = 1, 2$ proves compatibility with every arrow and therefore the completed algebra. Also $\Theta^3 = 1$.

For all coefficients, not just a truncated diagnostic,

$$[v^n](\Theta w)_i = \omega^{n-i}[v^n]w_i.$$

Thus invariance is equivalent to support in $n \equiv i \pmod 3$, and

$$L = W^\Theta = \bigoplus_{i=0}^{2} Rb_i, \qquad b_i = v^i\epsilon_i.$$

The arrows on this basis have scalar valuations $(1, 0, 0)$, since $a_0^+ b_2 = ub_0$, $a_1^+ b_0 = b_1$ and $a_2^+ b_1 = b_2$. Tame invariants return the old cut column rather than a new strict unramified rotation carrier.

The extension $R \subset S$ is faithfully flat. Its group quotient has a fixed point at $v = 0$, however, and is not a μ_3-torsor there: $S \otimes_R S$ is not the product of the three group graphs at that point. Semilinear μ_3 data are therefore not by themselves the full effective fpqc descent data for a vector bundle on $\operatorname{Spec} R$. The adjunction comparison is explicitly

$$S \otimes_R L \longrightarrow W, \qquad \operatorname{diag}(1, v, v^2),$$

with determinant $v^3 = u$, cokernel $S/(v) \oplus S/(v^2)$ and closed-fiber rank one. This proves its failure to be an isomorphism directly. The quotient stack $[\operatorname{Spec} S/\mu_3]$ retains the stabilizer at zero; its coarse space is $\operatorname{Spec} R$. This local root construction is described in [71]. Alper's vector-bundle criterion [70, Theorem10.3, pp.2386–2387] distinguishes precisely this nontrivial closed-fiber action from effective coarse descent. We use these sources for interpretation; all displayed module comparisons have already been calculated explicitly.

S10.2 A projective symmetry survives on the endomorphism order

Directly on the three basis vectors,

$$\Theta C = \omega^{-1} C\Theta = \omega^2 C\Theta.$$

Thus the rotation does not preserve L. The corrected vC does preserve it, since the factor ω from $g(v)$ cancels the commutator. In the invariant basis it has matrix A from the core, with $A^3 = uI$ and $\det A = u$. This is generically invertible and gives an order-three projective rotation after passing to the fraction field; it is not an invertible integral lattice intertwiner.

There is nevertheless a true algebra-level escape. Let $F = (f_{ij})$ be an arbitrary S-linear endomorphism of the *carrier* W, not assumed Λ-linear. Its tame conjugate has entries $\omega^{j-i} g(f_{ij})$. Invariance therefore means

$$f_{ij} \in v^{[i-j]_3} R,$$

where $[r]_3 \in \{0, 1, 2\}$ is the least residue. After replacing ϵ_i by $b_i = v^i \epsilon_i$, the entry becomes

$$v^{[i-j]_3+j-i} R = \begin{cases} R & i \geq j, \\ uR & i < j. \end{cases}$$

Consequently $\mathrm{End}_S(W)^\Theta$ is exactly the hereditary three-cycle order B_+ in the previously fixed cut convention. Multiplication is preserved by conjugation in the generic matrix algebra; no entrywise projection is being misused as a ring map.

The scalar commutator cancels under conjugation. Hence $\mathrm{Ad}(C)$ commutes with tame conjugation, descends to B_+, and is represented there by $\mathrm{Ad}(A)$ because v is central. Since $A^3 = uI$, this algebra automorphism has order three and preserves the hereditary order despite A^{-1} not lying in its unit group. This explains the surviving rotation of the branch order; it does not construct a rotated integral column or a Frobenius endomorphism. The other branch is identical and the paired-diagonal residue gluing still defines the original Λ_R inside $B_+ \times B_-$. Its label automorphisms remain the already constructed dessin symmetries.

S10.3 Determinants, Schur weights and the boundary correction

On $\det_S W = \bigwedge_S^3 W$, both $\det D = 1$ and $\det C = 1$. Thus this line has the scalar action g and genuinely descends to the trivial R-line with trivial rotation. This is a real positive determinant-level result. On the other hand,

$$b_0 \wedge b_1 \wedge b_2 = v^3(\epsilon_0 \wedge \epsilon_1 \wedge \epsilon_2),$$

so the natural comparison

$$\det_R(W^\Theta) \longrightarrow (\det_S W)^\Theta$$

is multiplication by u, with cokernel $R/(u)$. Both lines are abstractly free of rank one, but their given comparison is not an isomorphism. It records the divisor of the ramified lattice change and cannot be discarded when claiming a determinant-preserving integral comparison. For the *same* operator vC, the scalar on both determinant lines is u; naturality is respected. Comparing $\det C = 1$ with $\det A = u$ without the extra factor v would instead compare different operators.

For an induced nonzero homogeneous tensor/Schur carrier of degree d, the same functorial calculation gives commutator ω^{-d}. The two induced invertible operators commute exactly when $3 \mid d$. This is a group-action statement, not sufficient for coarse vector-bundle descent: $\mathrm{Sym}^3(S^3)$ has closed-fiber inertia multiplicities $(4, 3, 3)$ for the three characters, so its inertia is still nontrivial even though the commutator vanishes. A mixed tensor $W^{\otimes a} \otimes (W^*)^{\otimes b}$ has net exponent $a - b$; $\mathrm{End}_S(W)$ is the concrete net-zero escape computed above.

These carrier functors are not automatically functors on unital Λ-modules. Applying global $\bigwedge^3$ to the rank-one vertex idempotents kills all three, although their sum is the identity before applying this nonadditive functor. Vertexwise $\bigwedge^3$ of the rank-one column is zero. Vertexwise Sym^d for $d > 0$ does retain the monomial zero relations but changes the full-cycle scalar from u to u^d. It leaves the fixed central-action problem unless that change of base is included. None of the Schur cancellations supplies the missing two-dimensional arithmetic carrier.

S10.4 Why the local Jacobi plane does not descend by this repair

On constant vectors Θ acts by D. It sends the sum-zero cycle $\epsilon_0 - \epsilon_1$ to $\epsilon_0 - \omega^2\epsilon_1$, whose coordinate sum is nonzero. It also does not preserve the edge-sum line. Thus this order column is not already a tame descent of the earlier cellular chain complex or its H_1.

For the split local coefficient $\pi = a + b\omega$, the natural rotation polynomial on the balanced carrier is $F = aI + bC$. Since g fixes $a, b \in k$, its commutator is

$$\Theta F - F\Theta = b(\omega^2 - 1)C\Theta.$$

It is nonzero when $b \neq 0$. In particular both prime-seven twists $(a, b) = (3, 2), (-1, -3)$ fail this test. Prime powers with $b = 0$ are not excluded by this particular calculation; that exceptional commutation does not prove a global comparison.

Replacing C by vC gives an invariant carrier map, whose matrix on L is $aI + bA$. Direct determinant expansion gives

$$\det(1 - T(aI + bA)) = (1 - aT)^3 - ub^3T^3.$$

At $u = 0$ this is $(1 - aT)^3$, not $P_d(T)$, which has degree two. The weighted replacement therefore does not recover the established local factor. Neither C nor vC is generally Λ-linear: they intertwine its rotation. Their sums with the identity have not thereby become endomorphisms in the ordinary order-module category. They are legitimate carrier tests, with exactly the failures just exhibited.

S10.5 The full orbit is the normalization, and arbitrary kernels still fail

Twist the tame action by the three characters: $\Theta_r = \omega^r\Theta$, $r \in \mathbb{Z}/3$. Its fixed lattice is

$$L_r = \bigoplus_i Rv^{[i-r]_3}\epsilon_i,$$

with the cut at arrow r. Rotation gives an actual R-linear isomorphism $C : L_r \to L_{r+1}$, taking each displayed basis vector to the corresponding next one. Thus the direct sum of all three cuts has a strict rotation, at the cost of vertex rank three. Before invariants the orbit is S-rank nine and underlying R-rank twenty-seven; its invariant lattice is R-rank nine. The original orbit S-lattice still has nontrivial inertia at zero and need not be the base change of these invariants.

There is an explicit integral identification of the invariant orbit. In the standard hereditary order B_+, the j-th column has entries $u^{t_{ij}}R$ with $t_{ij} = 1$ for $i < j$ and zero otherwise. In its natural column basis the arrow coefficient has exponent

$$\delta_{i0} + t_{i-1,j} - t_{ij} = \delta_{ij}.$$

Its cut is therefore exactly j, identifying B_+e_j with L_j. Consequently $\bigoplus_r L_r \simeq B_+$ as left Λ_R-modules. Adding the exchanged minus-branch orbit gives the hereditary normalization $\widetilde{\Lambda} = B_+ \times B_-$, with full deck action and vertex rank six. On a basic projective with external arithmetic matrix this yields P_d^6.

In fact this normalization replacement restores the *ordinary determinant for every linear endomorphism*. For a finite projective P, the inclusion $P \to P \otimes_\Lambda \widetilde{\Lambda}$ becomes an isomorphism over $K = k((u))$, since $\widetilde{\Lambda}_K = \Lambda_K$. Its extended endomorphism and the original f therefore have the same generic characteristic determinant. Both polynomials have coefficients in R, which injects into K, so they agree already over R. The bounded-complex ratios also agree. This symmetry repair returns the previously refuted ordinary determinant, not merely its external tensor exponent.

For completeness we now prove Theorem 6.4 for arbitrary finite free kernels, without assuming they split integrally into branches or are projective over the order. Let

$$p_\pm = u^{-1} \sum_i c_i^\pm \in \Lambda_K.$$

These are central generic branch idempotents. Put $W_\pm = W_0 \cap p_\pm(W_0 \otimes_R K)$. They are finite free R-lattices spanning those generic summands, stable under Λ_R. Rotation fixes them, branch swap exchanges them. Their vertex ranks are constant on each branch: the arrow maps become invertible over K because a full cycle acts by u. Write these ranks r_+, r_-. No equality $W_0 = W_+ \oplus W_-$ over R has been assumed.

On the plus branch choose integral vertex bases. The three square arrow matrices X_0, X_1, X_2 have product uI_{r_+} around every cycle. Thus

$$\sum_i \mathrm{val}_u(\det X_i) = r_+.$$

The rotation lift is an invertible matrix between each pair of vertex lattices, hence its determinant is an R-unit. Its intertwining identities make all three arrow determinant valuations equal. Therefore $r_+ = 3m_+$; similarly $r_- = 3m_-$. The branch-swap lift gives $r_+ = r_-$, so generic rank additivity gives $\mathrm{rank}_R\, e_i W_0 = 6m$. Only invertible intertwiners were used; imposing the full deck group relations is stronger than necessary.

For the general determinant statement reduce modulo u and filter each vertex by $e_i J^a(W_0/uW_0)$. Since $J^4 \subset u\Lambda$, this is a finite filtration. In $\Phi_{W_0}(P) = \bigoplus_i (e_i W_0)^{n_i}$, positive-length entries of f raise the filtration, while the degree-zero matrix A_i repeats on its i-th graded pieces. Their total dimension is $6m$. Multiplying the graded characteristic determinants proves the asserted power for *every* f. Flatness of the kernel as a left Λ-module was not used; bounded right projectives compute the derived functor and give finite free R-outputs. Denominator constant terms one permit specialization of every bounded alternating ratio. The regular left kernel Λ_R attains $m = 1$, making the rank bound sharp.

The remaining arithmetic problem therefore requires a different hypothesis: a genuinely twisted determinant target, a specified non-linear/semilinear correspondence with its own comparison, or an infinite analytic realization. One cannot continue searching inside this closed finite-free, fixed-center, full-deck kernel class for a missing small estimate. The local purity and descent computations supply no global positive spectral limit. Moreover $k/\mathbb{Q}$ has degree two: ordinary restriction of scalars sends a rational local factor P_d to P_d^2. A relative k-determinant must not silently be identified with the former rational determinant. Finite-field curve zeta, global elliptic L-functions and Dirichlet GRH keep their distinct scopes and centers.

S11 Literal formal and computational coverage

`DessinNodeOrder.lean` defines the fiber-product subring of two genuine power-series rings. Its twenty printed declarations establish diagonal injectivity, nonzero branches, branch zero products, factorizations and annihilator identities, regular u, augmentation surjectivity and exactness, bijective power-series coordinates, square-zero alternating maps, three kernel-equals-range statements and projectivity of the actual regular/product terms. The coordinate statement is a bijection; the file does not yet package it as an R-linear equivalence or declare the general order freeness theorem.

`DessinInvariantComponents.lean` has ten printed declarations: three CK1-collapse components, determinant change of bases with an actual inverse, symmetric-square zero, identity,

composition and relation preservation, the degree-two Schur character identity, and the nodal central-parameter obstruction. Its symmetric-square matrix is

$$\mathrm{Sym}^2\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^2 & ab & b^2 \\ 2ac & ad+bc & 2bd \\ c^2 & cd & d^2 \end{pmatrix}.$$

This is the ordinary monomial basis convention; composition is proved universally over any commutative ring. The full project suite prints types and foundational dependencies; no theorem about general LR decompositions, abstract derived equivalences, C* K-theory or GRH is inferred from these component proofs.

The exact Python checks use fixed rational matrices and exhaustive finite sets, with no random seeds. They construct the displayed permutations and paths, evaluate rational Cartan identities, check formal Euler products through a stated degree, verify unrestricted-output Fock basis identities on bounded input sets, compute incidence Smith data, construct all nine string representations and signed maps, and test Young/LR/Cauchy examples and the seven-element-field counts. Those diagnostics are reproducible finite corroboration. The universal infinite claims rest on the proofs or explicitly cited theorems, and the arithmetic comparison remains open.

The ten additional `DessinPullbackComponents.lean` statements use literal two-state matrix units and a subring of a product of arbitrary rings, with equal images under two ring homomorphisms. The section hypothesis of its projection-surjectivity theorem is explicit. It does not construct a topological C*-pullback or assert that a commutative diagram is automatically a pullback. The no-residue theorem is universal algebra, with a nontrivial commutative target; it is not an assumed arithmetic positivity premise.

`ThetaJacobi.lean` adds sixteen declarations, with actual finite fields and MulChar rather than a hypothetical arithmetic kernel. Its norm theorem invokes mathlib's proved Jacobi inverse-product identity and derives characteristic inequality from finiteness; all three character nontrivialities are explicit. Other declarations verify the rescaling substitution, CM multiplication/polynomial/Gram positivity components, exact prime-seven matrix polynomials, and actual 2-by-3 boundary/3-by-2 cellular inclusion, homology restriction, three-edge determinant and rational vertex chain-map equations. It does not formalize elliptic point counting or an order-to-Frobenius functor.

The final three `ThetaJacobi.lean` declarations verify the actual conjugation matrix identities and the real two-dimensional inert-prime determinant obstruction. They do not assume any unknown zero location or exclude complex/enlarged spectral realizations.

`ThetaProjectiveMultiplicity.lean` adds five declarations. Arbitrary upper-block matrices with six identical diagonal blocks have the sixth-power determinant and characteristic-factor formulas. A literal power-series constant-coefficient map specializes matrix determinants, for each scalar parameter; the coefficientwise polynomial/order interface remains prose. A nonzero quadratic cannot be a rational-function sixth power, proved by polynomial degrees. Three genuine power series with product X cannot have the cyclically equivalent residue-vanishing conditions forced by invertible rank-one arrow intertwiners. The derivation of the path blocks from the full theta order, projective decomposition, derived branch functor and deck intertwiners is not fully formalized. The complete analytic suite now has208printed declarations; no hidden arithmetic positivity assumption occurs.

`ThetaTameDescent.lean` adds twelve printed component declarations (and one auxiliary polynomial lemma). They use genuine formal-power-series rescaling and its coefficient operator, proving the coefficient action, invariant-coefficient vanishing and the tame operator's cube. Actual three-state matrices satisfy the clock/rotation commutator, comparison determinant and closed-fiber equations,

weighted-cycle cube and full characteristic polynomial, noncommutation of a nonconstant rotation polynomial, trivial whole-column determinant actions, and the same-operator comparison for vC. The classification of all fixed series by residues, full order action and hereditary invariant algebra, stack interpretation, valuation/rank theorem, general coefficient-functor filtration and GRH are not thereby fully formalized. No unknown positivity premise is assumed.

S12 Explicit covering calculations and derived transport

Here are the linear details behind Proposition 7.2. Write $x_i = F_{ii}$ for diagonal entries of an endomorphism F preserving the three vertex spaces. Solving $FX = XF$, $FY = YF$ leaves a common diagonal c and six free entries. More explicitly its only off-diagonal entries are

$$F_{02} = F_{13} = t_0, \quad F_{15} = t_1, \quad F_{35} = F_{46} = t_2, \quad F_{48} = t_3, \quad F_{68} = F_{70} = t_4, \quad F_{72} = t_5.$$

All others vanish. This can be checked column by column using the displayed partial permutation matrices X, Y: their nonzero columns specify all constraints, and the zero columns force the remaining entries to zero. If $r_0, \dots, r_5$ are the six radical generators in the displayed order, the only nonzero products are

$$r_0 r_2 = r_1, \qquad r_2 r_4 = r_3, \qquad r_4 r_0 = r_5.$$

Every product of three radical generators vanishes. Thus the listed space is a local algebra, not merely a seven-dimensional vector-space count.

For clarity, the Euler equations used in the obstruction are

$$\begin{aligned} &d_1 - d_0 = 1, && d_3 - d_2 = 1, && d_4 - d_3 = 1,\\ &d_6 - d_5 = 1, && d_7 - d_6 = 1, && d_0 - d_8 = 1,\\ &d_1 - d_2 = 1, && d_4 - d_5 = 1, && d_7 - d_8 = 1, \end{aligned}$$

where $d_i = D_{ii}$. Subtract the last row from the first two rows after summing: the left side is zero and the right side is three. This is an identity in every characteristic-zero coefficient field. In characteristic three this particular obstruction disappears; no conclusion there is asserted. The closed walk

$$a_0 b_0^{-1} a_0\, a_1 b_1^{-1} a_1\, a_2 b_2^{-1} a_2$$

explains the computation: six forward steps and three backward steps have net lift displacement three. Arbitrary changes of basis cannot remove the Euler contradiction. This is an algebraic lift obstruction, not an assertion about the topological fundamental group of the compact torus.

The Hom formula also has a constructive perfect-complex version. Each cover representable is finite support and pushes to its corresponding vertex projective. For a bounded complex $P^\bullet$ of finite sums of such representables and a bounded finite-support complex $N^\bullet$, decompose the entire Hom complex by deck displacement. The differential preserves that displacement, and only finitely many displacements occur across the finitely many degrees. Taking cohomology gives

$$\operatorname{Hom}_{D(B_L)}(FP^\bullet, FN^\bullet[q]) \simeq \bigoplus_g \operatorname{Hom}_{D(\widetilde{B}_L)}(P^\bullet, T_g N^\bullet[q]).$$

Bounded projective complexes compute these derived Hom spaces, which justifies the equality rather than assuming every module perfect. This elementary proof does not assert an equivalence of entire bounded derived categories or cover objects of infinite projective dimension. The broader precovering statement has its exact hypotheses in [74, Theorem 4.28].

Finally the branch involution $\sigma(a_i) = b_i$, $\sigma(b_i) = a_i$ preserves B_L. In characteristic different from two one can study $B_L \# k[\mathbb{Z}/2]$ through equivariant representations. For the nine-dimensional module above, $\operatorname{rank} X = 6$ and $\operatorname{rank} Y = 3$, so $M \not\cong \sigma M$. The sum $M \oplus \sigma M$ has canonical swap equivariance. This is an explicit orbit construction; it neither chooses a nonexistent fixed-object linearization nor identifies this skew group algebra with the particular GLS algebras in [75].

`CoveringBand.lean` verifies the actual matrices' mixed products, cubic nilpotence and absence of an arbitrary Euler matrix. The general covering, module endomorphism classification, derived functors and the surface/Fukaya equivalence are not fully formalized by these three matrix statements. Exact rational code additionally checks thirty-six Hom pairs, the full radical multiplication and the two Hom-control defects.

S13 Weighted profiles, exact arithmetic examples and the surviving task

The finite matrix proof also retains an arbitrary row profile. Suppose $|A_{ui}|^2 = w_u$ for every column, set $S = \sum_u w_u$ and $Q = \sum_u w_u^2$. The row diagonals are $N w_u$ and column diagonals are S, so the same identity proves

$$\sum_{i \neq j} |K_{ij}|^2 \geq N^2 Q - N S^2.$$

This is a finite statement. Applying it to an infinite Schwartz tail would require its convergence and every newly encountered prime collision; these are not discarded. If a weighted pair-square attempt includes the annular W, one can instead choose a closed subannulus on which a nonzero $|W|$ is bounded below. Its prime count still has order $D/\log D$. The fixed ν has unit modulus outside its conductor. These facts do not replace the original coefficients in a signed estimate.

Here are exact sextic computations with all nonzero elements of the Eisenstein norm ball, and both prime ideals above each split rational prime in the column annulus. These kernel diagnostics use $\nu = 1$ and exclude only primes above2,3; they do not compute a signed Möbius energy. Values are integer norms in $\mathbb{Z}[\omega]$.

D	H	R	N	$\sum_{i \neq j} \lvert K_{ij} \rvert^2$	$NR(N-R)$
128	4	18	22	6840	1584
1024	9	36	138	651672	506736
4096	81	300	456	60080328	21340800
256	64	240	40	363672	−1920000

The final row has $H = D^{3/4}$ but $N < R$; its negative lower bound is correctly vacuous. The asymptotic theorem follows from the proof, not an extrapolation of these examples. For $p \equiv 1 \pmod 3$ and $w^2 + w + 1 = 0$ in $\mathbf{F}_p$, evaluate $a + b\omega$ as $a + bw$ and take its $(p-1)/6$ power, lifting the six roots via $\zeta_6 = 1 + \omega$. Every zero is tested before a unit value is assigned. For instance $1 + 3\omega$ has norm7 and vanishes at $\omega = 2$ modulo7, so the unimodularity hypothesis cannot be used beyond its stated domain.

`ShortRowPairBarrier.lean` verifies the finite frame identity, row-diagonal bound, weighted inequality and unimodular specialization, with actual complex sums and explicit hypotheses. It does not formalize the ideal-character comparison or prime-ideal theorem. The deterministic integer checker also verifies weighted profiles and256 scalar pair sums independently of its matrix computation. No numerical model or learned coefficient is used.

The next signed route must retain μ, ν, W before Cauchy. An exact starting identity in ideal Dirichlet convolution is

$$\mu = 2a - a * a * \mathbf{1} + b * b * \mathbf{1}, \qquad a = \mu\, \mathbf{1}_{Nn \le U}, \quad b = \mu - a.$$

Indeed $\mu * \mathbf{1} = \delta$ implies $(\mu - a) * (\mu - a) * \mathbf{1} = \mu - 2a + a * a * \mathbf{1}$. On squarefree n, each divisor factorization retains pairwise coprimality; the original row zeros still multiply along it. Estimating these type-I/type-II kernels with their true conductors is a new open arithmetic task. The identity alone supplies no cancellation or GRH proof.

S14 Exact factorization, retained zeros and the failed Cauchy route

S14.1 Proof of the regrouping with original coefficients

The monoid consists of nonzero integral ideals prime to the fixed excluded set S. Every divisor of a good ideal is good, so the identity $\mu * \zeta_0 = \delta$ holds on this same monoid. Expanding convolution in (1) gives finite divisor sums. Multiply the identity by $\mu(n)^2 \nu(n) \chi_n(u) W(Nn/D)$ and sum. The extra squarefree indicator does not change $\mu(n)$, since $\mu(n)^3 = \mu(n)$. On its support $n = abc$ forces each factor to be squarefree and any two to be coprime. Sextic multiplicativity, including zeros, gives $\chi_n(u) = \chi_a(u)\chi_b(u)\chi_c(u)$; a collision with any factor makes both sides zero. The fixed character also satisfies $\nu(n) = \nu(a)\nu(b)\nu(c)$ on these good ideals.

In the small-factor piece group $d = ab$. The remaining variable c is squarefree and coprime to d; $Nd \le U^2$ is necessary, but not sufficient for a nonzero $\beta_U(d)$: its specified truncated factorizations must still exist. For each such factorization on squarefree d, $\mu(a)\mu(b) = \mu(d)$, proving the coefficient formula. In the large-factor piece group $s = bc$. The condition $Nb > U$ implies $Ns > U$, and summing that divisor gives exactly $g_U(s)$. Since $Ns > U \ge 1$, s is not the unit ideal; its complete Möbius divisor sum vanishes, proving the alternative formula for g_U. This regrouping never replaces the two-variable weight by $W(Ns/D)$, never deletes $(a, s) = 1$, and never replaces the resulting coefficients by a favorable arbitrary family.

One can instead keep the unrestricted convolution identity on nonsquarefree columns, but then both individual pieces and their local conductor description change. For $U = 1$ and a good prime p, the column p^2 contributes one to each of $a_U * a_U * \zeta_0$ and $b_U * b_U * \zeta_0$, even though $\mu(p^2) = 0$; these cancel only after the pieces are recombined. Its squared sextic symbol has a cubic local component. Thus dropping total squarefreeness is not a harmless piecewise simplification, although the full unrestricted algebraic identity still holds.

S14.2 The two conductor scales are unchanged

For a squarefree $n = as$ with $(a, s) = 1$, every prime of n has one nontrivial sextic local component in the row variable. Hence its primitive finite conductor is the ideal as; $NaNs \asymp D$ on the column support. On multiplying a column by the conjugate of another, the local exponent is zero at common primes, 1 at a prime of n/h, and -1 at a prime of n'/h. The common prime still imposes its original zero on rows. Writing $n = hr$, $n' = ht$ therefore gives exactly

$$\chi_n(u)\overline{\chi_{n'}(u)} = \mathbf{1}_{(u,h)=1}\chi_r(u)\overline{\chi_t(u)}, \qquad q = rt, \quad \mathfrak{l} = hrt.$$

Here q is the primitive oscillatory conductor and $\mathfrak{l}$ a period of the full zero extension. These are ideals, not necessarily squarefree rational norms. In particular, conjugate primes over one split rational prime are distinct coprime ideal factors. Full defining moduli of ν and primary conventions remain fixed in S; this row-conductor calculation does not assert a new uniform reciprocity theorem in the column variable. In the coprime bulk $Nq \asymp D^2$ remains large.

S14.3 Why a first generic Cauchy step does not supply the saving

On one type-II dyad let $Na \asymp A$, $Ns \asymp B$, $AB \asymp D$. Keep the full profile and coprimality restrictions in

$$X_a(u) = \nu(a)\chi_a(u) \sum_{\substack{s \text{ squarefree} \\ Ns \asymp B,\,(a,s)=1}} g_U(s)\nu(s)\chi_s(u)W(NaNs/D).$$

The exact elementary Cauchy inequality is

$$\sum_{0<Nu\le H} \left| D^{-1/2} \sum_{Na\asymp A} \mu(a)X_a(u) \right|^2 \le \frac{\sum_{Na\asymp A}|\mu(a)|^2}{D} \sum_{Na\asymp A} \sum_{0<Nu\le H} |X_a(u)|^2.$$

The a-mask has been kept inside X_a. Even if one granted an optimistic uniform slice bound $(H+B)\sum_{Ns\asymp B}|g_U(s)|^2 D^\varepsilon$, the ordinary divisor estimate $|g_U(s)| \le \tau(s)$ and the fixed-field ideal count would only yield the envelope

$$A(H+B)D^\varepsilon \asymp (AH+D)D^\varepsilon.$$

That hypothetical slice estimate is not established here. This is an insufficiency of the generic upper-bound calculation, not a lower bound on the signed energy. At balanced $A, B \asymp D^{1/2}$ and $H = D^{3/4}$, the AH term loses $D^{1/2}$ relative to the target; the D term also exceeds H. A sharper use of the actual g_U, cross-factor constraints or covariance between the original pieces is required. This calculation does not refute all bilinear, moving-cutoff or dispersion methods.

The fixed-$U = 1$ lower bounds in Proposition 10.1 are different: they are genuine energy lower bounds for the separate pieces. If $W \ge w_0 > 0$ on $[c_1, c_2] \subset (1,2)$, count primes in $[c_1 D, c_2 D]$ for I_1 and in $[c_1 D/Nq_0, c_2 D/Nq_0]$ for II_1. Removing S and q_0 removes finitely many primes. The prime-ideal theorem gives positive constants times $D/\log D$ in both cases; the latter contribution has coefficient two. Because every coefficient of II_1 is nonnegative, keeping these products is valid. This is unlike isolating the prime part of the full signed Möbius energy, where uncontrolled cross terms remain. For the exact requested bound $H(HD)^\varepsilon$ at $H = D^{3/4}$, the competing power is $D^{3/4+7\varepsilon/4}$, giving the threshold $\varepsilon < 1/7$. Replacing $(HD)^\varepsilon$ by D^ε reparametrizes the loss and gives the equivalent $1/4$ formulation. The positivity argument is specific to $U = 1$. For a concrete control, the primary element $7 - 21\omega$ has norm $637 = 7 \cdot 7 \cdot 13$ and three distinct ideal prime factors. At $U = 7$ its type-I coefficient is -1, its type-II coefficient is -2, and its Möbius value is -1. Thus the separate pieces need not have nonnegative coefficients at another cutoff; no moving-cutoff obstruction follows from the preceding positivity proof.

S14.4 Exact finite checks and formal boundaries

The deterministic checker represents every good ideal of norm at most 1024 once by its primary generator, including inert prime ideals and squarefree products of three distinct prime ideals. It checks 1535 coefficient/regrouping cases at five cutoffs, retains 19680 encountered original zeros in actual sextic sums, and verifies 900 pair-conductor exponent ledgers. It evaluates principal ν and the fixed Kummer character $\nu(n) = \chi_n(2)$ on the same good ideals. The finite diagnostic weight is the rational profile $((Nn - D)(2D - Nn))^2/D^4$ on $[D, 2D]$, zero elsewhere; it is not claimed to be the smooth analytic weight of the theorem. Every identity is algebraic for arbitrary finite weights; these samples do not prove seminorm or scale-supremum estimates.

For example, at $D = 128$, $H = 16$, principal ν and $U = 1$, the unit-row normalized sums have the following exact common scaling:

$$\begin{array}{c|r} D^{9/2}M_1 & -122721831 \\ D^{9/2}I_1 & 314726661 \\ D^{9/2}II_1 & 192004830 \end{array}$$

Their exact difference is $II_1 - I_1 = M_1$. The lower-bound theorem uses the prime-ideal proof, not extrapolation from these small numbers. An independent no-production-import oracle also checks convolution, squarefree regrouping and weighted sextic sums.

`MobiusSplit.lean` proves eight actual statements: the inverse-ring identity, natural-number arithmetic truncation/tail, the Möbius Dirichlet identity, the unit cutoff, its exact tail convolution and coefficient, and nonnegativity. The natural-number Dirichlet ring is an actual arithmetic object; it is not the ideal convolution ring. Ideal arithmetic, sextic/Hecke character construction, analytic prime counting and the unproved signed mean square remain prose or named classical inputs. No new zero-free region, full GRH proof or positive determinant bridge has been obtained.

S15 Primitive conductors, deleted factors and the scalar contour

S15.1 Constructing the column character without losing its zeros

Let $\alpha^6 = u$. Since K contains μ_6, the extension $K(\alpha)/K$ is abelian, with character $g \mapsto g(\alpha)/\alpha$. At $p \nmid 6u$, arithmetic Frobenius gives the residue $u^{(Np-1)/6}$, exactly the source's sextic symbol. Abelian reciprocity makes it a finite-order Hecke character; multiply by ν and take its primitive inducing character. The classical reciprocity input is [89, Theorem 0.8 and Chapter I,§1]. The finite-order archimedean character is trivial, since $\mathbb{C}^\times$ is connected.

At $p \nmid 6$, write $u = \pi^e v$ with v a local unit. A sixth root of v lies in an unramified extension: its residue root is separable and lifts by Hensel's lemma. Inertia acts on a sixth root of π through μ_6, so on α the inertia character has order $6/\gcd(6, e)$. It is trivial exactly when $6 \mid e$; its square is trivial exactly when $3 \mid e$. A nontrivial tame one-dimensional character has conductor exponent one. Since ν is unramified outside S, it cannot cancel this inertia there. At primes above 6, the groups $K_p^\times/(K_p^\times)^6$ are finite; their finitely many characters have a uniform maximum conductor. Consequently

$$\mathfrak{f}_u \mid \mathfrak{c}_\nu \prod_{p \mid u,\ p \nmid 6} p, \qquad Q_u \ll_\nu Nu.$$

The fixed ideal $\mathfrak{c}_\nu$ includes the wild and fixed ν factors. No assertion identifies $\mathfrak{f}_u$ with the larger full zero modulus.

For $p \notin T$ put $z = \psi_u(p)(Np)^{-s}$. Its squarefree factor $1 + z = (1 - z^2)/(1 - z)$ proves the ratio using the *same* deleted set in numerator and denominator. Restoring the primitive Euler factors yields (3). At a prime where the primitive square becomes unramified its value cannot be obtained by squaring the sentinel zero $\psi_u(p) = 0$. For principal ν, $u = \pi^3$ has a quadratic Kummer character whose primitive square is globally principal. Its local value in that square is one, not zero. For $u = \pi^6$ both primitive characters are principal, although the full family still deletes p. These two cases prove the local table directly.

S15.2 Why the half-plane above one half is available

For a character η write $L^T(s, \eta)$ for its Euler product with all primes of T removed. For $\sigma > 1/2$ the product $1/L^T(2s, \psi_{u,2})$ converges absolutely and is bounded by $\zeta_K(2\sigma)$, uniformly in the character,

T and $t = \Im s$. Every deleted finite factor is nonzero there. Therefore

$$F_{u,d}(s) = L(s,\psi_u) \prod_{p\in T, p\nmid \mathfrak{f}_u} (1-\psi_u(p)(Np)^{-s})/L^T(2s,\psi_{u,2})$$

has just the stated possible principal pole. This uses absolute convergence at $2\sigma > 1$, not a conjectural zero-free assertion on the critical line. For any fixed $\sigma > 0$ and $\eta > 0$,

$$\prod_{p\in T}(1+(Np)^{-\sigma}) \ll_{\sigma,\eta} R_T^{\eta}.$$

Indeed $1 + q^{-\sigma} \le q^{\eta}$ for all sufficiently large q; the remaining finitely many prime ideals give a uniform fixed constant. This controls every moving deleted prime; the denominator has already been bounded by the full convergent zeta product.

We use the classical completed Hecke functional equation in the form

$$\Lambda(s,\psi_u) = Q_u^{s/2}(2\pi)^{-s}\Gamma(s)L(s,\psi_u), \qquad \Lambda(s,\psi_u) = w_u\Lambda(1-s,\overline{\psi_u}), \quad |w_u| = 1.$$

This is a named external theorem, not a newly formalized analytic result; compare [88, Theorem 4.3, Corollary 4.4]. The cited notes sketch some of Tate's calculations; their incomplete displayed epsilon formulas are not used as independently checked constants here. At $\Re s = 1+\eta$, absolute convergence bounds L by $\zeta_K(1+\eta)$. The functional equation and Stirling's formula give on $\Re s = -\eta$ the bound $Q_u^{1/2+\eta}(1+|t|)^{1+2\eta}$ up to a fixed constant. Phragmén–Lindelöf on this strip therefore gives, for fixed $0 < \sigma < 1$,

$$|L(\sigma+it,\psi_u)| \ll_{\sigma,\eta} Q_u^{(1-\sigma)/2+\eta}(1+|t|)^{1-\sigma+2\eta}.$$

For the principal character apply the strip argument to $(s-1)L(s,1)/(s+1)$, then divide back on the fixed line $\sigma < 1$. The constants need not be uniform as $\sigma \to 1$. Polynomial vertical strip growth and finite order, as supplied by classical Hecke theory, justify this application. No Lindelöf or subconvexity bound is inserted.

S15.3 Mellin inversion, residues and profile uniformity

Initially, absolute convergence and compact smooth support give

$$S_{u,d}(Y) = \frac{1}{2\pi i}\int_{(1+\eta)} \widehat{W}(s) Y^s F_{u,d}(s)\,ds.$$

Four integrations by parts using $x\partial_x$ give $|\widehat{W}(\sigma+it)| \ll_{\sigma} \mathcal{M}_4(W)(1+|t|)^{-4}$. Choose $0 < \eta < \min(\varepsilon/8, 1/8)$, decreasing it further when distributing the small losses, and shift to $\sigma = 1/2+\delta$. The convergent reciprocal Euler product, the finite factor bound and convexity above control the complete integrand. Their polynomial t growth is integrable against this Mellin decay and makes the horizontal edges tend to zero. The only crossed pole is $s = 1$ when $\psi_u = 1$; evaluating its finite factors gives exactly ρ_T. Taking absolute values on the new vertical line proves (4), after renaming the small losses. This also proves uniformity over any family of profiles with bounded $\mathcal{M}_4$; it does not interchange a profile supremum with an arithmetic mean square.

In the original I_u, the inner profile is exactly $W(Nc/Y)$ with $Y = D/Nd$; no d-dependent derivative enlargement occurs. On the support of β_U, $Nd \le U^2 \le D$, hence $Y \ge 1$. Also $R_T \le C_S Nu\,Nd$. The principal term is explicitly

$$P_u = \mathbf{1}_{\psi_u=1}\sqrt{D}\,\widehat{W}(1) \sum_{\substack{d \text{ squarefree}\\(d,S)=1}} \frac{\beta_U(d)\nu(d)\chi_d(u)}{Nd}\rho_{T(u,d)}.$$

This sum retains all original phases and zeros; no positivity of it is assumed. For the error use the actual truncated factorizations, not a free coefficient bound:

$$\sum_{d \text{ squarefree}} \frac{|\beta_U(d)|}{(Nd)^{1/2+\delta-\eta}} \le \left(\sum_{Na \le U} (Na)^{-1/2-\delta+\eta} \right)^2 \ll_{\delta,\eta} U^{1-2\delta+2\eta}.$$

The fixed-field ideal count is $O_K(X)$; dyadic summation proves the last inequality for sufficiently small $\eta < \delta + 1/2$. Inserting this and the conductor/radical bounds proves (5). The elementary lattice count $\#\{u : 0 < Nu \le H\} = O_K(H)$ then gives (6). Rows are not divided by units.

At $H = D^{3/4}$ and $U = D^a$ with $0 \le a \le 1/2$, the power of the resulting error envelope *relative to* H is

$$\frac{3}{8} + \frac{5}{4}\delta + (2 - 4\delta)a$$

up to the arbitrarily small losses. It remains positive for every $0 < \delta < 1/2$. Thus this scalar convexity argument alone supplies no short-row saving. This is a limitation of the estimate just derived, not an obstruction to averaging the numerators over Kummer rows, retaining signed coefficients, or exploiting covariance with type II.

The seminorm uniformity is not a proof of the original scale supremum. For each independently chosen scale D' the displayed outer argument requires $U^2 \le D'$. If a fixed cutoff is retained while D' shrinks, then eventually even $D' \le U$: the short $2a_U$ term in the Vaughan identity can meet the support and must be restored. Neither that term nor the covariance with type II may be omitted in a supremum or reassembly.

S15.4 Verification boundaries

`HeckeTypeI.lean` checks seven actual complex local/finite-product statements, including independent primitive coefficient arrays and their full deleted-factor restoration. It does not identify an ideal Hecke series with an abstract finite product or formalize a Mellin contour. The exact checker includes split and inert prime ideals, actual sextic phases, conductor drops, sixth-power rows and original mask zeros. The analytic theorem above uses classical abelian reciprocity, Hecke continuation/functional equation and standard complex-analytic tools as external inputs. Full ideal-character/formal analytic objects, the original signed low-gcd mean square and every GRH completion obligation remain open.

S16 Controlled errors and a trained full-action RG approximation

Proposition S16.1 (Classical action-error propagation)**.** *If $|\widehat{V}(s,t) - V(s,t)| \le \varepsilon$ at every finite configuration, then $|\mathcal{R}\widehat{V}(s) - \mathcal{R}V(s)| \le \varepsilon$. The partition ratio lies in $[e^{-\varepsilon}, e^{\varepsilon}]$. Every further* exact *marginalization preserves the same action bound.*

Proof. The pointwise inequalities $e^{-\varepsilon}e^{V} \le e^{\widehat{V}} \le e^{\varepsilon}e^{V}$ survive finite summation. Taking logarithms proves the first assertion; summing also over the retained variables proves the second. Iterate the same argument. For normalized measures the likelihood ratio lies between $e^{-2\varepsilon}$ and $e^{2\varepsilon}$, so a bounded observable error is at most $\|O\|_\infty \min(2, e^{2\varepsilon} - 1)$. Per-step learned errors would add; a single initial error does not grow under subsequent exact elimination. This scalar proof is not an assertion that the matrix exponential is operator-monotone. □

S16.1 Executed finite calculations

The implementation checks full enumeration against the closed black-spin formula, Walsh reconstruction, raw partition equality, retained-spin magnetizations and two-stage versus direct elimination. At the quantum $\Gamma = 0$, $\beta = 1$ limit, every Pauli coefficient agrees with the classical effective action, with the required minus sign between energy and logweight. At nonzero transverse field the model is

$$H = -J_+ Z_+ \sum_j Z_j - J_- Z_- \sum_j \eta_j Z_j - bZ_+ + \tfrac{3}{5} bZ_- - \sum_j h_j Z_j - \Gamma \sum_i X_i.$$

At general β, the classical zero-transverse-field limit uses βV, including the fixed white fields, rather than scaling only the three varied controls. For $m = 5$, its full dimension is 128 and the first retained dimension is 32. Every one of the 1024 five-spin Pauli coefficients is computed; later retained dimensions are 8,4,2. Spectral-log/exponential and full Pauli reconstruction residuals, exact-trace composition and retained-observable comparisons are recorded separately. Floating-point residuals are numerical checks of the proved finite identities, not interval certificates.

The local laboratory exposes every scale, coupling/field controls for the classical model, selected generated interaction supports on actual dessin radial lifts, the even-chain K, h, C trajectory and physical correlation length, and three fixed quantum matrix calculations. Live quantum parameter sliders are not offered for uncomputed Hamiltonians. Pair truncation carries the identity term and reports both partition and probability/density errors. A late-scale floating-point loss of the small eigenvalue is avoided by transporting the exact log-eigenratio coordinate $\log r \mapsto 2 \log r$; this is explicitly the transfer RG identity, not a newly diagonalized value.

S16.2 A quantum-feature model trained on actual RG

The learner targets all 32 unnormalized logweights of the $7 \to 5$ spin $m = 5$ decimation. Its three inputs are (J_+, J_-, b), with ranges $[0.15, 1.1]^2 \times [-0.6, 0.6]$. Fixed fields are $h_j = 0.045 \cos(1.7j) + 0.02j/5$. Seed 20261103 fixes 160 input samples, a 120/40 train/test split, and 40 OOD samples with $J_\pm \in [1.15, 1.5]$, $b \in [-0.8, 0.8]$. The three feature qubits encode the couplings by $R_y(1.5J_+) \otimes R_y(1.5J_-) \otimes R_y(2b)|000\rangle$. Two local R_y, R_z layers, with a CZ triangle after each, contain 12 nominal optimizer angles. The final three R_z phases commute with the measured Z products and are structurally inactive for this loss; they still change the displayed state phases. We count them as nominal parameters, not additional effective expressive directions. Seven nonempty Pauli-Z products feed an affine 8×32 readout. There are 268 parameters. Feature registers are not the seven physical spin sites or five quiver vertices.

The training objective is full-action MSE, including the constant; KL alone would ignore that constant. A training-only ridge fit initializes the head, then 120 Adam updates optimize both circuit and head. Actual PennyLane analytic expectation calls implement the feature circuit. Parameter-shift derivatives of each feature are combined with the classical head gradient; full-loss finite differences agree to 6.46×10^{-11}. All data, parameters, intermediate checkpoints and complete loss curves are preserved.

The held-out full-action MSE is approximately 0.00985, mean $D_{\mathrm{KL}}(p_{\mathrm{exact}} \| p_{\mathrm{learned}}) = 0.00300$, mean TV 0.0290, and mean absolute log-partition error 0.0610. A classical quadratic-feature baseline with the same 256 readout parameters obtains 0.00613,0.00240,0.0233 and 0.0138, respectively. OOD action MSE increases to 0.267 for the quantum hybrid, versus 0.0555 for the baseline. The quantum-feature constant MAE is 0.0444 and higher-body coefficient RMSE is 0.00900, versus 0.0124 and 0.0130 for the classical baseline; different metrics favor different aspects. The aggregate results show no quantum advantage. They expose genuine full-action approximation errors rather than substituting state-compression success for RG.

Each saved example computes its maximum error over all 32 coarse states; the proposition then applies to its subsequent exact marginalizations. These per-example finite bounds are not uniform over the input parameter box. The synchronized architecture inspector shows literal gate outputs, Pauli features, head, logweights, generated interactions and partition value, linked to the exact genus-two dessin/order context. No full order representation, continuum universality, material-specific accuracy or arithmetic zero-location bridge is inferred.

S16.3 Quantum TokenGT and the geometric error-correction boundary

A further executed node/edge-token hybrid uses the same exact RG teacher, with frozen three-qubit features, trained Q/K/V and a full-action head. Softmax, componentwise max-plus and hybrid variants are compared with a matched classical feature model; the latter has lower held-out action error. An explicit intervention-code/change variational MI objective learns concept directions; a 61-candidate latent forest and a held-out analogy transfer are scored on exact physical observables. Rips H_1 is actually reduced over $\mathbf{F}_2$, and a continuous Gaussian GFlowNet trains a trajectory-balance objective with specified root/Lebesgue measures. Empirical residuals do not prove exact terminal sampling, semantic reasoning or categorical equivalence.

The physical bit-flip repetition channel is separately executed and has $p_L = 3p^2 - 2p^3$ under ideal recovery. A genuine cellular CSS calculation on the genus-two one-face dessin gives $[[10, 4, 1]]$: vertex-incidence rank 6, zero face boundary, X-distance 1 and Z-distance 4. Thus this cellulation provides no useful protection merely from its genus. These finite layers, full formulas, proofs, parameters and scope are developed in the research book and interactive laboratories; the physical models are not faithful representations of the infinite completed order and supply no GRH bridge.

S16.4 An actual finite gentle cut for the categorical interface

On the genuine $m = 5$ dessin quiver write $a_i, b_i : i \to i - 1$ with indices modulo 5; cross-branch consecutive pairs are forbidden. Remove both arrows with source 0. The surviving algebra B has vertices 0 through 4, arrows a_i, b_i for $i = 1, \dots, 4$, and relations $a_i b_{i-1} = b_i a_{i-1} = 0$ for $i = 2, \dots, 4$. It is an explicit finite gentle quotient, not the completed order itself.

Every surviving positive path stays on one branch and has length at most 4. For length $\ell = 1, \dots, 4$ there are $2(5 - \ell)$ paths, while length 0 contributes five idempotents. Therefore $\dim_k B = 5 + 2(4 + 3 + 2 + 1) = 25$. The forbidden-chain resolution already proved for monomial gentle algebras has two alternating chains of length n from vertex i to $i - n$ when $1 \le n \le i$, and none for $n > i$. Applying Hom to simples kills its arrow differentials. Hence $\mathrm{pd}_B S_i = i$ and $\mathrm{Ext}^4_B(S_4, S_0) \cong k^2$. The split finite-dimensional simple-module criterion gives $\mathrm{gldim}\, B = 4$.

Every full branch cycle crosses a removed arrow, so the common completed order parameter u maps to zero. Thus B is a finite u-torsion quotient, not a DVR-free surface order. The difference is visible in derived Hom: S_0 is projective over B, whereas in the original completed order the two forbidden arms of length 5 return to vertex 0 and give $\mathrm{Ext}^5_\Lambda(S_0, S_0) \cong k^2$. The exact restriction along $\Lambda \twoheadrightarrow B$ is consequently not derived fully faithful. This supplies a concrete cut on which to investigate graded gentle/Fukaya models, while proving why its equivalence cannot simply be transported to the original completed order. Boundary/stops/line-field data and an actual module-valued neural decoder would still have to be supplied. Exact path and endpoint enumeration accompanies the prose proof; this cut/category comparison is not claimed wholly Lean formalized.

S17 Relation calculations, splitting functors and determinant limits

S17.1 Correct graphs and stable spaces

In the original basis let $fe_1 = e_2$, $fe_2 = e_3$, $fe_3 = 0$ and $ge_1 = 0$, $ge_2 = e_1$, $ge_3 = e_2$. Then

$$\Gamma_f = \langle (e_1, e_2), (e_2, e_3), (e_3, 0) \rangle,$$
$$\Gamma_{f^2} = \langle (e_1, e_3), (e_2, 0), (e_3, 0) \rangle,$$
$$\Gamma_{gf^2} = \langle (e_1, e_2), (e_2, 0), (e_3, 0) \rangle.$$

Each graph has dimension three. Omitting zero-output generators changes the relation into a partial graph and changes its powers and domain. The following are subspaces of V, not lists of pairs in $V \oplus V$:

T	$\bigcup_r \ker T^r$	$\bigcap_r \operatorname{im} T^r$
f, g	V	0
$gf = \operatorname{diag}(1,1,0)$	ke_3	$ke_1 \oplus ke_2$
$fg = \operatorname{diag}(0,1,1)$	ke_1	$ke_2 \oplus ke_3$

For all four full graphs $\bigcup_r \Gamma_T^r(0) = 0$. This corrects both the source's graph convention and its interchanged gf, fg kernel entries. For an arbitrary multivalued relation there need not even be a linear map on its domain; its regular-part theorem requires the appropriate relation subquotients and finiteness conditions. We do not replace those hypotheses by the graph calculation. For comparison, in [38, Section 4, Lemmas 4.2,4.5–4.6] write $C' = \bigcup_n C^n(0)$ and let C'' consist of vectors with a compatible infinite backward chain. In finite dimension $C'' = \bigcap_n C^n V$; this equality is not automatic for arbitrary infinite-dimensional spaces. The regular quotient is $C^\sharp / C^\flat$, where

$$C^\sharp = C'' \cap (C^{-1})'', \qquad C^\flat = (C'' \cap (C^{-1})') + (C' \cap (C^{-1})'').$$

The automorphism statement uses this quotient for an arbitrary relation. The splitting statement additionally requires its finite-dimensionality, rather than the source's unqualified ι/κ expression.

S17.2 Proof of the dessin-category equivalence

Write $a_i : V_i \to V_{i-1}$ for $i = 1, \dots, m-1$ and $p_i = a_1 \cdots a_i : V_i \to V_0$, with $p_0 = 1$; the rightmost map acts first. Every p_i is an isomorphism. Conjugating the arrow maps by these identifications makes $a_i = 1$ for $i \neq 0$, and makes $T = p_{m-1} a_0$ at the remaining arrow. All b-relations vanish. An intertwiner is transported to maps h_i with $h_i = h_{i-1}$ for $i \neq 0$ and $h_{m-1} T = T' h_0$. Hence every $h_i = h$ and $hT = T'h$. This proves full faithfulness and essential surjectivity, with inverse given by $(V, T) \mapsto N_m(T)$. It is also an exact equivalence: kernels and cokernels of intertwiners are the usual invariant kernels/cokernels; the identity arrow maps remain identities. Finite dimension is a condition on modules, while A_m itself is infinite dimensional through its arbitrarily long permitted branch paths.

The sum of the full permitted turns on both branches is central in this closed dessin algebra. On the chosen modules the b-turns vanish and each a-turn equals T, so u acts as T at every vertex. If $T^n = 0$, the action factors through $A_m/(u^n)$ and extends to the completion. If T has a nonzero invertible Fitting summand, no power of u kills that summand, precluding continuity. For $T = J_r(\lambda)$ with $\lambda \neq 0$, the obstruction is stronger: $u - \lambda$ is a unit of $k[[u]]$ but its rth power annihilates the proposed module, impossible for a nonzero unital module. This rules out even an unqualified algebraic extension of that band, not just a chosen topology.

S17.3 Functoriality and the actual filtration

Kernel dimensions are nondecreasing and at most n. Once two successive kernels agree, applying T^{-1} shows that all later kernels agree. Thus they stabilize by n; rank-nullity gives image stabilization as well. On $K(T)$, $T^n = 0$. On $I(T)$, $T(I(T)) = I(T)$; finite dimension makes this a bijection. If $hT = T'h$, induction gives $hT^r = T'^r h$, hence

$$h(\ker T^r) \subseteq \ker T'^r, \qquad h(\operatorname{im} T^r) \subseteq \operatorname{im} T'^r.$$

This proves functoriality even when $\dim V \neq \dim V'$: choose a common stabilization exponent for the two finite objects. The decomposition projection onto $K(T)$ along $I(T)$ is canonical and commutes with all intertwiners; the existence of a Jordan basis is not needed. Finite dimension is essential. On $V = k[t]$, multiplication by t has stable kernel zero and stable image $\bigcap_r t^r k[t] = 0$, although $V \neq 0$. The same construction $N_m(V, t)$ is an actual infinite-dimensional module of A_m. Thus the finite Fitting split cannot be extended to every module of our infinite algebra merely by repeating its notation.

Let $k_r = \dim \ker T^r$, with $k_0 = 0$, and form $G_r = \ker T^r / \ker T^{r-1}$. For a nilpotent Jordan block of size s, $k_r = \min(r, s)$, so $\dim G_r$ counts the blocks of size at least r, and $2k_r - k_{r-1} - k_{r+1}$ counts blocks of size exactly r. For $J_3(0) \oplus J_1(0) \oplus J_2(1)$ the stable dimensions are 4 and 2; the kernel dimensions begin $0, 2, 3, 4, 4$. The G_r are functorial subquotients. Their dimensions do not specify a natural choice of basis or split each filtration step.

Proposition S17.1 (What the source's splitting-functor assumptions imply)**.** *Let $\mathcal{A}, \mathcal{B}_i$ be additive categories with finite biproducts. Suppose additive functors $S_i : \mathcal{B}_i \to \mathcal{A}$ and $F_i : \mathcal{A} \to \mathcal{B}_i$ satisfy $F_i S_i \cong 1$, $F_i S_j \cong 0$ for $i \neq j$; only finitely many $F_i M$ are nonzero for each M; the F_i jointly reflect isomorphisms; and maps $\gamma_{i,M} : S_i F_i M \to M$ have invertible F_i-images. Then $M \cong \bigoplus_i S_i F_i M$. Every nonzero indecomposable is $S_i B$ for a unique index and an indecomposable B up to isomorphism.*

Proof. Sum the finitely many nonzero $\gamma_{i,M}$ to a map γ_M. After applying F_j, all off-diagonal summands vanish, leaving the assumed isomorphism $F_j \gamma_{j,M}$, or $0 \to 0$. Joint reflection therefore makes γ_M invertible. For an indecomposable M, exactly one of the nonzero summands can occur. Conversely, if $S_i B = X \oplus Y$ and B is indecomposable, applying F_i makes one summand vanish, say $F_i X = 0$; every other $F_j X$ also vanishes. Joint reflection applied to $0 \to X$ gives $X = 0$ up to isomorphism. Finally apply F_i to an isomorphism $S_i B \cong S_j B'$ to recover both index and object. □

The direction of γ matters: it is into M, contrary to one source diagram. Local finiteness is objectwise in i, not a finiteness assertion over all objects of a category. Abelian categories require kernels, cokernels and the usual exactness axioms; addition is on Hom groups, not on an alleged group of quiver vertices. The abstract assumptions above do not assert that the family F_i is faithful on arbitrary morphisms. Our Fitting example is concrete: $\mathcal{B}_0$ consists of nilpotent pairs, $\mathcal{B}_\times$ of invertible pairs, S are inclusions and $F_0, F_\times$ the two stable summands.

S17.4 Strings, bands, Hom and one non-rigid string

In this branch category, Jordan normal form classifies indecomposables over an algebraically closed field. Over a general field replace Jordan eigenvalues by powers of monic irreducible polynomials. A Jordan block $J_r(0)$ is one directed string with mr basis visits and $mr - 1$ nonzero successor arrows; $J_r(\lambda)$ for $\lambda \neq 0$ is a permitted cycle band. Formal inverse letters are reverse traversals, not inverse elements of A_m. Strings and their reverses describe the same string module under the usual isomorphism convention. Arbitrary infinite modules are not all classified by this finite calculation; the actual finiteness hypotheses in [38] remain essential.

Identify a Jordan pair with $k[t]/(t-\lambda)^r$. An intertwiner from size r to size s is determined by the image of 1, annihilated by $(t-\lambda)^r$. For equal eigenvalues this subspace has dimension $\min(r,s)$. For different eigenvalues, Bezout's identity makes $t-\lambda$ invertible on the target, so only zero remains. Full faithfulness above transfers this Hom computation to A_m. The sequence $0 \to J_1(0) \to J_2(0) \to J_1(0) \to 0$ is not split: with $J_2(0)e_1 = e_2$, a lift of the quotient generator has nonzero first coordinate and is not killed by $J_2(0)$. Its image under N_m is a nonsplit ambient-A_m self-extension. Thus even the string $N_m(J_1(0))$ is not rigid. No assertion that this exact full subcategory is derived fully faithful is needed for the nonsplitting argument.

For this particular closed A_m, every oriented or inverse letter has exactly two admissible successors: continue on its branch in the same orientation, or switch branch and reverse orientation. Cancellation and forbidden pairs, including inverse forbidden pairs, remove the other choices. A fixed ordering of these choices identifies each first-letter cylinder with $\{0,1\}^{\mathbb{N}}$. The whole infinite-word space is a finite disjoint union of Cantor spaces, hence itself Cantor. There are no terminating words inside that space to remove. Arbitrary gentle quivers can have isolated or eventually forced paths, so this conclusion is not universal. Algebraic covers retain the lifted zero-relations and deck equivariance data; arbitrary periodic cover representations are not automatically bands or a unitary time evolution.

S17.5 A determinant comparison with its exact limits

On $\bigoplus_i V_i$, $L^m = \operatorname{diag}(T,\dots,T)$. For a direct determinant proof of (7), successively eliminate the blocks 1 at vertices $1,\dots,m-1$ in $1-zL$. Block row additions preserve determinant and leave $1-z^mT$ at vertex zero. This works over $k[z]$ without inverting z or T; the zero-size case gives one on both sides. The Fitting splitting separates a nilpotent factor 1 and the determinant on the invertible summand. For instance $m=5$, $T=J_2(2)$ gives $(1-2z^5)^2$, whose complex roots have modulus $2^{-1/5}$, despite being a genuine genus-two dessin module.

Under conjugation of T this is an invariant polynomial. With independent source/target bases a presentation determinant is instead a semi-invariant with its determinant-character weight; those actions must not be conflated. The polynomial equality does not construct a positive Hermitian metric, an arithmetic Euler factor, a regularized determinant or a zero-preserving map to completed L-functions. Functorial kernel filtrations similarly do not construct an equivalence of derived/Fukaya categories. A fixed graded surface category and a determinant-compatible arithmetic realization remain separate obligations.

The exact rational checker evaluates actual genus-two/four quivers, Hom equations, full-turn matrices, both filtration directions and the nonsplit extension. `lean-analytic/LinearRelations.lean` proves full graph identities, semiconjugacy and filtration preservation, actual finite-space Fitting splitting/stabilization and unit-power annihilation. It does not package the entire dessin category, Jordan classification, general splitting-functor proposition or determinant proof in Lean.

S18 Normal forms and the full module action

S18.1 Why the polynomial matrix algebra is the path algebra

With our convention a path from j to i acts as E_{ij}; the rightmost arrow acts first. Every nonzero positive path uses only the a-branch or only the b-branch, because mixed consecutive arrows are the defining quadratic zero relations. Along either branch the path of length ℓ starting at j is unique. Write

$$\ell = j - i + 5q, \qquad 0 \le i, j < 5.$$

Its image is $t^q E_{ij}$ in the chosen factor and zero in the other. For $i < j$, $q \geq 0$; for $i > j$, $q \geq 1$; for $i = j$ and positive length, $q \geq 1$. The stationary path is the shared (E_{ii}, E_{ii}).

These images are linearly independent: off-diagonal matrix units, positive diagonal monomials, branch factors and degrees are independent; the shared constant diagonal units are independent of all positive paths. They span $H \times_{k^5} H$, since upper entries permit all polynomials, lower entries are divisible by t, and only the constant diagonals of the two factors must agree. Products of paths agree with matrix products. Thus the arrow assignment induces a bijective algebra map from $kQ/(a_i b_{i+1}, b_i a_{i+1})$ to B. This argument includes arbitrarily long paths; it imposes no nilpotence relation on a full turn.

The central parameter u, the sum of all full turns on both branches, has image (tI_5, tI_5). Completing in this parameter is a separate operation. The polynomial algebra and its completion are not identified by the normal-form calculation. The finite modules below are annihilated by u^2, so their continuous completed actions require only a finite Taylor polynomial.

S18.2 Coefficient selectors without a truncation

Here is an explicit proof of Theorem 14.2 and the coordinate convention used by the implementation. Let

$$\mathcal{S} = \{(i, j, q) : 0 \leq i, j < 5,\ q \in \mathbb{N},\ i < j \text{ or } q > 0\}, \qquad \mathcal{K} = \{0, \dots, 4\} \sqcup (\{a, b\} \times \mathcal{S}).$$

For a stationary index set $v_i = (E_{ii}, E_{ii})$; for a positive index set $v_{a,i,j,q} = (t^q E_{ij}, 0)$ and $v_{b,i,j,q} = (0, t^q E_{ij})$. These are elements of the ambient polynomial matrix-pair space, with no degree cutoff. For $M = (M_a, M_b)$ define

$$\lambda_i(M) = [t^0](M_a)_{ii}, \qquad \lambda_{b,i,j,q}(M) = [t^q](M_b)_{ij}.$$

Here the subscript b in the second formula denotes either branch. Every λ_x is k-linear, since extraction of a fixed matrix entry and polynomial coefficient is linear.

We verify $\lambda_x(v_y) = \delta_{xy}$, including the cases at degree zero. For two stationary indices this is the ordinary diagonal matrix-unit identity. A stationary selector vanishes on a positive image in the other factor. In its own factor it could see a constant diagonal only if $i = j$ and $q = 0$, which the definition of $\mathcal{S}$ excludes. Conversely, a positive selector applied to a stationary image could be nonzero only at precisely the same forbidden slot. For two positive indices, different branches have disjoint supports. In one branch the matrix entry forces both endpoints to agree and the monomial coefficient forces the degrees to agree. These conditions give equality of the full indices. The witness of the condition defining $\mathcal{S}$ carries no extra algebraic data. This proves biorthogonality in every characteristic, using no division.

For any finitely supported $c : \mathcal{K} \to k$, linearity now gives

$$\lambda_x \left(\sum_{y \in \mathcal{K}} c_y v_y \right) = c_x.$$

If the sum vanishes, every c_x vanishes; if two sums agree, their coefficients agree. This proves linear independence and injectivity of normal expansion on the actual k-module of finitely supported functions. It does not replace finite support by a convergent or formal infinite sum.

The length formula also contains useful information beyond independence. If $i < j$ then $5q + j > i$ even at $q = 0$; otherwise $q > 0$ and $i \leq 4$ give the same strict inequality. Hence $\ell = 5q + j - i$ is a positive integer, and $\ell + i = 5q + j$. For fixed starting vertex j and length ℓ, two such equations

imply $i - i' = 5(q - q')$. Since $i, i' \in \{0, \dots, 4\}$, this forces $i = i'$ and then $q = q'$. In Lean the subtraction is natural subtraction; positivity is proved before using the rearranged equation.

For example the a-path of length 12 from vertex 4 ends at vertex 2 and has image $(t^2 E_{24}, 0)$. The length-4 a-path from 3 to 4 has image $(tE_{43}, 0)$. Their product, with the rightmost path first, is $(t^3 E_{23}, 0)$, of length 16. Replacing the second path by its b-branch version makes the product zero. Degree-zero upper entries such as $(E_{24}, 0)$ are permitted, but $(E_{22}, 0)$ is not a stationary image: the stationary element is the shared pair (E_{22}, E_{22}). This last distinction prevents a spurious extra idempotent in the proposed basis.

The selector theorem alone gives the faithfulness part of the intended quotient comparison. The next construction proves membership and spanning inside the generated algebra without assuming that comparison. A separately defined path algebra, mixed-zero quotient and its universal algebra map still require formal packaging. Completion, module-category and almost-split universal properties require their own subsequent constructions. None of these finite-support identities supplies an arithmetic self-adjoint determinant or controls zeta zeros.

S18.3 Constructing every normal image from actual arrows

We now prove Theorem 14.3. This closes the membership and spanning gap left by ambient coefficient recovery. Let $p(j) = j - 1$ modulo five and, inside the actual generated algebra, define

$$W_b(0, j) = e_j, \qquad W_b(n + 1, j) = W_b(n, p(j))\, b_j.$$

In this formula b_j denotes the arrow in the chosen branch; it may be an a-arrow. The rightmost arrow acts first. This recursive definition is an element of A for every n, because it uses only its generators and multiplication. Set

$$d_{ij} = \begin{cases} 0 & i < j, \\ 1 & i \geq j, \end{cases} \quad L_{ij} = j - i + 5d_{ij}, \quad P_b(i, j) = W_b(L_{ij}, j), \quad C_b(i) = P_b(i, i).$$

The integer L_{ij} lies between one and five. In particular $P_b(i, i)$ is a positive full turn, whereas $W_b(0, i)$ is the shared stationary idempotent. These two elements are not interchangeable.

Directly following the arrows, the path $P_b(i, j)$ crosses the weighted arrow b_0 exactly d_{ij} times. Its image is therefore $t^{d_{ij}} E_{ij}$ in factor b, with zero in the other factor. The Lean proof checks all five-by-five endpoint pairs for each of the two branches; this is a finite verification of a finite combinatorial case split, not a bound on permitted path lengths. For all $n \in \mathbb{N}$, induction gives

$$\mathrm{view}\big(C_b(i)^n P_b(i, j)\big) = t^{n + d_{ij}} E_{ij} \quad \text{in factor } b.$$

For $n = 0$ the power is the unit of the whole two-factor algebra, so multiplication by $P_b(i, j)$ still selects the correct branch. The induction step uses $C_b(i)$ with image tE_{ii} and $E_{ii} E_{ij} = E_{ij}$. No assertion that a branch-only degree-zero diagonal belongs to the algebra is needed. Since a permitted slot satisfies $q \geq d_{ij}$, the choice $n = q - d_{ij}$ constructs its preimage in A. This proves membership at every polynomial degree.

For clarity, if $i = 2$, $j = 4$, $q = 2$, then $P_a(2, 4) = a_3 a_4$ and $C_a(2)^2 P_a(2, 4)$ realizes $(t^2 E_{24}, 0)$ by a length-12 arrow word. For $i = 4$, $j = 0$, $q = 1$, we have $P_a(4, 0) = a_0$ and the exponent $q - d_{40}$ is zero. The corresponding exponent must never be replaced by a negative power or by an inverse arrow. The construction works over $\mathbb{F}_2$ as well as characteristic zero.

S18.4 Spanning, multiplication and the inverse coordinate map

Let S be the k-linear span of all normal images in the ambient space. Membership just proved gives $S \subseteq A$. We prove the reverse inclusion by showing that S is closed under the operations used to generate A. The product rules are

$$e_i e_j = \delta_{ij} e_i, \qquad e_r v_{b,i,j,q} = \delta_{ri} v_{b,i,j,q}, \qquad v_{b,i,j,q} e_r = \delta_{jr} v_{b,i,j,q},$$

and for two positive images

$$v_{b,i,j,q} v_{c,r,l,s} = \begin{cases} v_{b,i,l,q+s} & b = c \text{ and } j = r, \\ 0 & \text{otherwise.} \end{cases}$$

The displayed nonzero output is always an admissible positive image. If $q + s > 0$ this follows from its degree. If $q + s = 0$, nonnegativity gives $q = s = 0$, so the input conditions imply $i < j = r < l$. Thus $i < l$ and the output is admissible even at degree zero. This explains why closure cannot be justified merely by ignoring the lower-triangular constraints.

By bilinearity the product of two finite linear combinations lies in S. Every vertex is one of its normal images, and each arrow is a permitted image with degree zero except the wrap arrow, which has degree one. The sum of the shared vertices is the unit. Scalar multiples of this unit belong to S. Induction on the algebra generated by the arrows and vertices therefore gives $A \subseteq S$. In Lean this is actual `Algebra.adjoin_induction`, not a predicate that assumes the desired normal form for every element. Consequently $A = S$ as submodules of the polynomial matrix-pair space.

Now lift each v_x to an element $\widetilde{v}_x \in A$ using its proved membership. Their images under the injective inclusion are the already independent ambient images, so the lifts are independent. The equality $A = S$ gives surjectivity of finite expansion; independence gives injectivity. This constructs the linear equivalence E on the genuine generated algebra. For any $a \in A$, apply the ambient coefficient recovery identity to $E^{-1}a$ to obtain

$$(E^{-1}a)_x = \lambda_x(\mathrm{view}(a)).$$

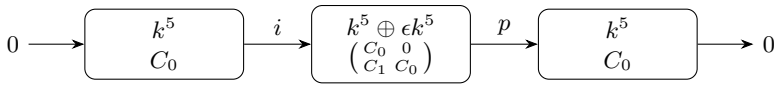
The formula holds for every element of A, not only on a sampled list of paths or generators. Its finite support follows from the equivalence with finitely supported functions. No convergence of an infinite expansion is asserted. Although the product table has been used to prove span closure, we have not yet supplied the formal universal map from a separately constructed bound-quiver path quotient. The distinction between a linear coordinate equivalence and that algebraic quotient interface is retained.

S18.5 Exactness for arbitrary algebra elements

In coordinates $M_2 = k^5 \oplus \epsilon k^5$ let $C(t) = \sum_{r \geq 0} C_r t^r$ be the first component of an arbitrary B-element. Its action is

$$(x, y) \longmapsto (C_0 x, C_1 x + C_0 y), \qquad \rho_2(C) = \begin{pmatrix} C_0 & 0 \\ C_1 & C_0 \end{pmatrix}.$$

Evaluation is multiplicative: the constant coefficient of a product is $C_0 C_0'$ and the linear coefficient is $C_0 C_1' + C_1 C_0'$. Consequently the formula is a representation of the entire algebra. The second branch acts by zero but has not been removed from the algebra. Both maps in the sequence commute with this general block action, not merely with a finite test list of generators.



Every vector in $\ker p$ is $(0, y) = i(y)$, i is injective, and p is surjective by $x = p(x, 0)$. The section obstruction uses only a_0: it is zero on M_1, while on M_2 it sends the coordinate-zero constant coefficient to the coordinate-four infinitesimal part. A section commuting with this arrow would force $1 = 0$. No algebraic closure, eigenvalue classification, division by two or positivity premise occurs.

S18.6 Formal and computational coverage

The file `lean-analytic/BranchExtension.lean` defines B as the actual generated subalgebra of polynomial matrix pairs, proves its infinite dimensionality through injective polynomial evaluation, and constructs i, p as B-linear maps. The original 21 printed theorem types and axiom reports cover the generators, infinite embedding, actions, exactness and absence of a module section. Eight additional reports in `NormalPath` cover slot extensionality, positive length, the length equation, recovery from start and length, biorthogonality, coefficient recovery, linear independence and injective normal expansion. Ten further printed reports now establish actual generated-algebra membership, equality with the normal span, lifted independence, bijective finite expansion and its explicit inverse coefficients. The normal-form isomorphism with the separately constructed path quotient above is proved in prose; it is not packaged as a Lean categorical equivalence. Nor is a Yoneda-Ext object or the universal factorization property of an almost-split sequence formalized by these maps alone.

The exact checker `examples/relations/check_extension.py` independently uses unbounded normal paths and polynomial matrix pairs. It checks 405 distinct paths through length 40, 7,225 products, 30 distinct loop powers and 80 whole-polynomial module actions. The full generator system for a hypothetical section has coefficient rank 49 and augmented rank 50 over $\mathbb{Q}, \mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5$. These finite diagnostics complement the symbolic proof; their finite cutoffs do not establish infinite dimensionality by themselves.

The separate exact checker `examples/relations/check_normal_paths.py` checks 475 normal images at degrees up to 1,000,003 (length 5,000,019), all 225,625 selector pairs, 1,030 direct arrow steps, 15,625 composable-or-zero product comparisons and 200 finite expansions over $\mathbb{Q}, \mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5$. Invalid constant slots are rejected. These are finite diagnostics of the code and conventions; the unbounded assertion is established by the Lean theorem, not by the size of the tests.

A further deterministic checker, `examples/relations/check_normal_algebra.py`, constructs all 50 shortest arrow words and 375 actual preimages, including degree 1,000,003, using sparse binary powers of the full turns. It checks 5,625 product round trips and 200 exact expansions over the four stated fields. Three ambient matrices with forbidden constant shapes fail reconstruction, as required. These diagnostics test the implementation independently of finite degree bounds; the general membership/span/equivalence result is the Lean theorem, not an extrapolation from these samples.

S19 Physical generative models and their verification

S19.1 From a dessin to an actual quantum RG target

Take $y^2 = x^5 - 1$, $\beta = x^5$. Its clean incidence graph has two black vertices and five white vertices, hence seven physical spin sites and ten incidence edges. These differ from the five quiver vertices

and the infinite-dimensional polynomial algebra. The physical Hilbert space below is finite; no faithful action of that whole algebra on it is assumed.

Let X_q, Z_q be Pauli matrices on physical site q, with the black sites numbered zero and one and the white sites $2 + j$. Our executed Hamiltonian is

$$\begin{aligned} H = & -.50 \sum_{j=0}^{4} Z_0 Z_{2+j} - .35 \sum_{j=0}^{4} (-1)^j Z_1 Z_{2+j} \\ & - .55 \sum_{q=0}^{6} X_q - .12 Z_0 + .072 Z_1 - \sum_{j=0}^{4} h_j Z_{2+j}. \end{aligned}$$

Here $h_j = .045 \cos(1.7j) + .02j/5$. All coefficients are dimensionless and inverse temperature is one. Diagonalizing the actual 128×128 Hermitian matrix gives $\rho = e^{-H}/\operatorname{tr}(e^{-H})$. We retain the two black spins:

$$\rho_R = \operatorname{Tr}_{\text{white}} \rho, \qquad (\rho_R)_{ab} = \sum_{j=0}^{31} \rho_{aj,bj}.$$

This is an actual coarse quantum state, with trace one and positive spectrum. An independent Pauli–Kronecker NumPy reconstruction agrees with the saved matrix to $2.04 \cdot 10^{-10}$. Its minimum eigenvalue is approximately 0.0806414, and its off-diagonal Frobenius norm is 0.300551. There is therefore quantum coherence that a probability histogram alone cannot describe. No chemical or experimental material dataset is involved: this is an exactly specified finite quantum-matter benchmark.

Let $p_s = (\rho_R)_{ss}$ for $s \in \{\pm 1\}^2$. The hybrid samplers learn

$$p_\sigma(x) = \sum_s p_s \mathcal{N}(x; s, \sigma^2 I_2), \qquad \sigma = .2.$$

This is a continuous dequantization of the measured spin law. A sign measurement of these samples differs from p by at most $2\Phi(-1/\sigma)$ in total variation, by a coordinatewise sign-flip coupling and the union bound. This is below $6 \cdot 10^{-7}$ here. It is small compared with finite-sampling error, but it is not zero. The samplers do not recover phases or the entire ρ_R. For example, ρ_R and $(Z \otimes I)\rho_R(Z \otimes I)$ have the same diagonal but different density matrices.

S19.2 Diffusion and exact versus approximate reverse controls

For $T = 64$ use the cosine cumulative noise schedule, with endpoint offset .008, clipped step variances between .0001 and .999. Write $\alpha_k = 1 - \beta_k$ and $\bar{\alpha}_k = \prod_{j \le k} \alpha_j$. The forward noising law is

$$x_k = \sqrt{\bar{\alpha}_k}\, x_0 + \sqrt{1 - \bar{\alpha}_k}\, \epsilon, \qquad \epsilon \sim N(0, I_2).$$

A network $\epsilon_\theta(x_k, k/T)$ minimizes the ordinary mean squared noise-prediction error. It is a classical stochastic sampler containing quantum-circuit features, not an implemented coherent quantum channel. The executed reverse update uses

$$\mu_\theta = \frac{x_k - \beta_k \epsilon_\theta / \sqrt{1 - \bar{\alpha}_k}}{\sqrt{\alpha_k}}, \qquad \widetilde{\beta}_k = \beta_k \frac{1 - \bar{\alpha}_{k-1}}{1 - \bar{\alpha}_k}, \qquad x_{k-1} = \mu_\theta + \sqrt{\widetilde{\beta}_k}\, z.$$

The last step is deterministic. Even an exact conditional noise predictor does not make this Gaussian reverse kernel the exact reverse mixture at finite T: posterior uncertainty of the mixture component

remains. The oracle-noise control therefore retains a reverse-kernel approximation, finite schedule and Monte Carlo error, not just a numerical integration error.

There is also an exact, known-teacher reverse-mixture control. Conditional on center s, put $v_k = \bar{\alpha}_k\sigma^2 + 1 - \bar{\alpha}_k$ and $m_{k,s} = \sqrt{\bar{\alpha}_k}s$. At a supplied x_k, draw the center with posterior weights proportional to $p_s \exp(-\|x_k - m_{k,s}\|^2/(2v_k))$. Then draw

$$x_{k-1} \mid x_k, s \sim N\left(m_{k-1,s} + \frac{\sqrt{\alpha_k}v_{k-1}}{v_k}(x_k - \sqrt{\alpha_k}m_{k-1,s}), \frac{v_{k-1}\beta_k}{v_k}I_2\right).$$

These formulas follow by conditioning a jointly Gaussian pair with covariance $\sqrt{\alpha_k}v_{k-1}I_2$. Starting from the exact forward terminal mixture and composing these reverse conditionals recovers p_σ exactly in distribution, up to numerical and sampling error. It uses the full teacher, so it is a diagnostic rather than a learned solution.

S19.3 Flow matching with a measurable target velocity

Independently sample $z \sim N(0, I_2)$ and $x_1 \sim p_\sigma$, set $x_t = (1-t)z + tx_1$, and train $v_\theta(x_t, t)$ toward $x_1 - z$. This is the conditional flow-matching regression objective. Conditional on center s, its time-t mean and variance are

$$ts, \qquad v_t = (1-t)^2 + t^2\sigma^2.$$

Let $\pi_s(x,t)$ be the corresponding mixture posterior. The exact conditional velocity is

$$v_*(x,t) = \sum_s \pi_s(x,t)\left[s + \frac{t\sigma^2 - (1-t)}{v_t}(x - ts)\right].$$

Indeed $\mathrm{Cov}(x_1 - z, x_t \mid s) = (t\sigma^2 - (1-t))I_2$; Gaussian conditioning gives the displayed formula. The positive variance v_t includes both endpoints. The code verifies this formula and the diffusion-score formula by independent finite differences of a separately written mixture log density. Learned sampling uses 128 Heun steps. Oracle-velocity sampling isolates finite integration and sampling error for this known smooth density; it is not an upstream QFM reproduction [94].

S19.4 Architecture, matched controls and actual GPU results

The neural input is $[x_0, x_1, t, \sin 2\pi t, \cos 2\pi t, \sin 4\pi t, \cos 4\pi t]$. An affine encoder maps its seven channels to four angles through $\pi\tanh$. A two-layer four-qubit circuit uses angle-reuploading R_Y, R_Z gates and, for the entangled architecture, a CNOT ring. Fifteen commuting Z-string expectations are concatenated with the seven input channels. A $22 \to 64 \to 64 \to 2$ SiLU head predicts noise or velocity. The product control uses the same nominal parameters and removes the entanglers. Final diagonal phase rotations are invisible to the all-Z readout; equality of nominal parameter counts does not establish equality of effective Fisher capacity.

Every run has 5,810 nominal parameters, batch size 512 and 2,000 Adam steps. There are 16,384 training draws and 4,096 disjoint held-out draws, with seeds 20261104, 20261105 and 20261106. The four architectures/objectives were rerun at every seed. The following figures are mean and sample standard deviation across those three runs, not confidence intervals or asymptotic theorems.

Objective	circuit	sign-law TV	sliced W_1
Diffusion	product	$.0380 \pm .0113$	$.0768 \pm .0306$
Diffusion	entangled	$.0451 \pm .0133$	$.0939 \pm .0183$
Flow	product	$.0217 \pm .0079$	$.0460 \pm .0077$
Flow	entangled	$.0293 \pm .0130$	$.0522 \pm .0159$

The second metric uses sixteen fixed projection directions and an independent teacher draw. The experiment demonstrates learning but no consistent quantum or entanglement advantage; the product control has lower mean error on both reported metrics. No physical quantum processor, shot noise, industrial chemical prediction, OOD material transfer or complexity speedup was tested. All checkpoints, seeds, histories, runtimes and teacher controls are retained.

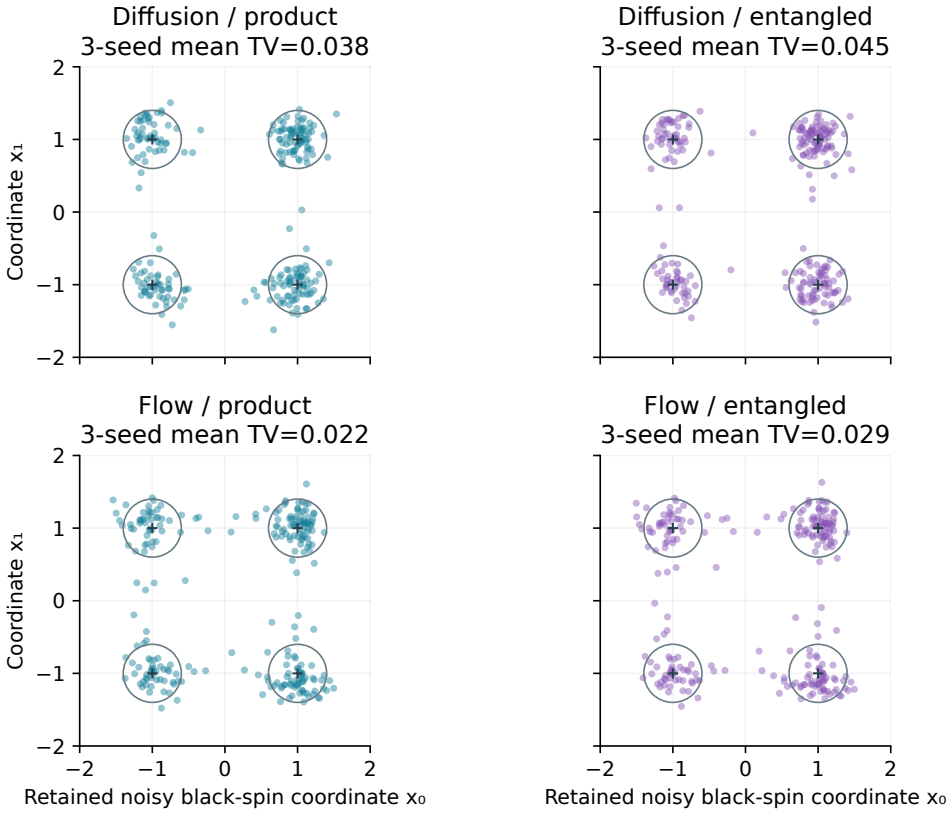


Figure 1: Actual final samples from seed20261104, with exact mixture centers and two-standard-deviation circles. Titles report mean sign-law TV over three seeds. The product control is competitive; no entanglement advantage is inferred.

S19.5 A coherent learned purification flow

We separately implement quantum evolution, without identifying it with the classical samplers. Exact depolarizing diffusion is

$$D_t(\rho_R) = e^{-t}\rho_R + (1 - e^{-t})I_4/4.$$

It preserves trace and positivity and satisfies $D_s D_t = D_{s+t}$ by direct substitution. Its linear inverse on all matrices is not positive: $D_t^{-1}(P) = e^t P - (e^t - 1)I_4/4$ has a negative eigenvalue on the orthogonal complement of any rank-one projection P when $t > 0$. Thus a learned inverse cannot be a universal deterministic physical channel.

For a constructive state-preparation benchmark, take the family $R_t = (1 - t)I_4/4 + t\rho_R$, $0 \le t \le 1$, and its canonical purification $|\Psi_t\rangle = \mathrm{vec}\sqrt{R_t}$ in $\mathbb{C}^4 \otimes \mathbb{C}^4$. It is normalized because $\|\sqrt{R_t}\|_F^2 = \mathrm{tr}\, R_t = 1$. The initial purification is two Bell pairs. We train a six-block four-qubit

circuit: each block conjugates local R_Y rotations by a fixed CNOT network and its inverse; angles are $t\theta + t^2\omega$. All blocks are identity at the control time zero. The 48 learned controls are trained against full known purification states, using the fidelity loss. This ideal supervision is stronger than access to measured samples and is accounted for explicitly.

An RTX4090 run of 2,000 steps, batch 64, gives minimum fidelity 0.999903 on a 33-point grid and maximum reduced-density Frobenius error 0.009642. The reduced state is obtained by an actual ancilla partial trace and passes trace/positivity checks. These are grid diagnostics, not a uniform or out-of-family guarantee. The construction is a learned coherent control path for known targets, not a universally learned inverse to depolarization. Its full 76-gate replay and intermediate complex states are exported to the same numerical architecture inspector as the other models.

S19.6 Width, fields, RG and arithmetic obligations

Proposition 15.1 fixes which width is being increased: the number of independent one-qubit features, with only their readout trained. At initialization let $a_r \sim N(0,1)$ be independent standard centered readout coefficients, independent of the frozen features. Conditional finite-dimensional outputs are then exactly centered Gaussian with covariance K_m. The almost-sure kernel convergence implies convergence of these finite-dimensional characteristic functions to the Gaussian law with covariance K. This proves the specified finite-dimensional GP limit. It does not prove tightness in a function-space norm, parameter-feature learning, a many-qubit limit or a continuum QFT. For frozen K_m, squared-loss gradient flow gives $\dot{r} = -K_m r$, whose eigencomponents decay at their kernel eigenvalues. A zero eigenvalue leaves a nonzero residual unchanged.

The executable width experiment fits the existing exact 32-configuration *classical* black-spin-decimated action on genus two, including its constant. This is a second true RG teacher, distinct from the quantum white-spin partial trace above. With 96 parameter triples for training, 32 for testing, eight feature seeds and ridge 10^{-5}, widths 16 and 4,096 give mean relative kernel errors 0.2990 and 0.01726 and mean full-action test MSE 0.02005 and 0.000743. The limiting RBF readout is also evaluated. Finite numerical convergence supports the implementation; the proposition supplies the limit proof. Quantum NTK literature requires its own parameterization, ensemble and lazy/large-width hypotheses [91, 90].

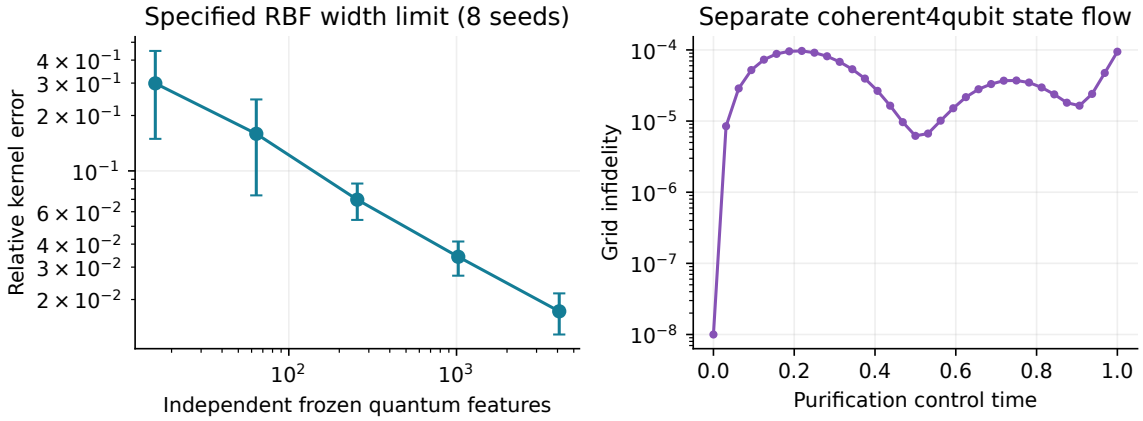


Figure 2: Left: eight-seed frozen-feature kernel errors, with sample standard deviations. Right: the separate learned coherent purification flow, evaluated on33time points. These are distinct width and quantum-control experiments.

For complex embeddings the Hermitian Gram form, realification $A + iB \mapsto \left(\begin{smallmatrix} A & -B \\ B & A \end{smallmatrix}\right)$, and conjugation must be respected. An arbitrary real network on twice as many channels need not be complex linear. In characteristic two, $1+1 = 0$; an embedding into real vectors cannot be a unital field homomorphism preserving that addition. Finite-field path algebras and cellular/CSS computations are still meaningful, but Gaussian integration, ordered Gibbs weights and real backpropagation do not transfer by renaming the field. An explicit discrete learner over $\mathbb{F}_p$, $p = 2, 3, 5$, uses features $1, x, y, xy$ and coefficients searched in the finite field to learn $f(x, y) = xy$. All p^2 inputs are verified; an affine-only learner has nonzero minimum error. The trained map compiles to the permutation unitary $U_f|x, y, z\rangle = |x, y, z + f(x, y) \bmod p\rangle$ on a complex Hilbert space of dimension p^3. Retaining x, y makes its inverse subtraction explicit. This is reversible encoding of a field-valued polynomial learner, not a field embedding into complex amplitudes or backpropagation in $\mathbb{F}_p$. Its training is exact finite coefficient search. A tropical semiring is not a field; coefficient cancellation, valuation signs and the underlying valued algebra must be specified.

For GRH, each architecture offers an experimentally testable role: quantum features can parameterize candidate metrics or probe observables; graph tokens can encode actual local incidence, masks and monodromy; tropical attention can search extremal valuations or active constraints; flow/diffusion can propose continuous operator/control parameters; RG can organize a genuine hierarchy of coarse actions. None of these roles proves that its Gram matrix is the arithmetic positive metric. One must still construct the arithmetic object with correct conductor, gamma factors, zero divisor and domain, prove compact-uniform control and zero location, and keep Dirichlet, Hecke, Dedekind and Artin hypotheses separate. A learned candidate is accepted only after its exact defining equations and analytic estimates are supplied.

S20 Covers, paths, pants and quantum games

S20.1 Which free group and which universal cover?

Write $F_2 = \langle x, y\rangle$, $\mathbb{F}_2$ and $\widehat{F}_2$ for the free group, coefficient field and profinite completion, respectively. For a theta graph with vertices A, B and edges a, b, c, an explicit cover has vertices $(g, A), (g, B)$, $g \in F_2$, and edges

$$a : (g, A) - (g, B), \qquad b : (g, A) - (gx, B), \qquad c : (g, A) - (gy, B).$$

Left multiplication is the deck action. Contracting all a-edges produces the usual four-regular Cayley tree of F_2; before contraction the graph is three-regular. Equivalently, vertices are reduced nonbacktracking edge paths from A, with an immediately reversed edge cancelled. This gives a tree and unique path lifting in every length. Its radius-r patch has $1 + 3\sum_{k=0}^{r-1} 2^k$ vertices, hence 766 at radius eight. With a planar ribbon structure the abstract graph is a spine of a topological pair of pants. The actual $m = 3$ dessin instead has the ribbon structure of a once-punctured torus. Forgetting the cyclic orders identifies the abstract graphs, not these thickenings. Graph distance is not automatically hyperbolic geodesic length.

For the contracted clean m-edge dessin there are two vertices and m parallel edges, so π_1 is F_{m-1} and the analogous universal tree is m-regular. The new examples include $m = 5, 9$, of genus two and four. Reinserting clean white bivalent vertices subdivides its edges. Attaching the face to recover the compact surface introduces a relation; the closed genus-two surface has its surface group and a disk universal cover, not one of these graph trees. Separately, $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ has fundamental group F_2. The unbranched cover induced by a clean dessin removes all preimages of those three values. For our degree-$2m$ map there are $m + 3$ punctures; its Euler characteristic is $-2m$, not that

of the dessin's one-skeleton. Thus the occurrences of F_2 in monodromy, graph covers and arithmetic profinite theory must not be silently identified.

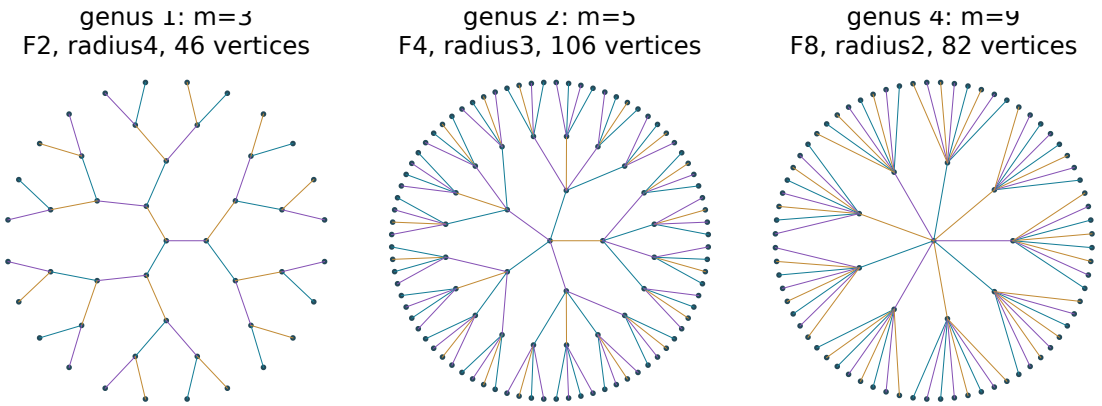


Figure 3: Finite windows of universal covers of contracted actual clean dessins. These are schematic graph trees, not uniformizing embeddings of the compact surfaces. Clean white bivalent vertices are suppressed.

For finite-supported representations of a suitably graded bound quiver, pushdown maps split into deck-homogeneous blocks. The arrow equations preserve the degree, and monomial zero relations are homogeneous. The resulting Hom direct sum controls all deck pieces, not just degree one. For an explicit graph-only diagnostic two lifted edge intervals have upstairs endomorphism dimension two, but downstairs endomorphism dimension four: the deck pieces have dimensions $2+1+1$. This graph-quiver calculation is *not* substituted for a dessin-algebra module example. The actual infinite polynomial module examples below supply that part. A Hom-controlled functor needs its specified exceptional/source/target subcategory and control statement; exactness, a deck-sum formula or equal dimensions alone is not an equivalence or an AR universality theorem. A derived comparison must control all $\mathrm{Hom}(X, Y[n])$ and composition, together with an actual derived/enhanced functor.

S20.2 Actual infinite-algebra path modules and unbounded resolutions

To fix conventions, let A_{col} be the earlier rightmost-arrow-first polynomial matrix algebra acting on left columns. Here we use travel-order path multiplication and write $A_{\mathrm{trav}} = A_{\mathrm{col}}^{\mathrm{op}}$ for the prose-identified bound-quiver algebra. Right A_{trav} representations therefore encode the earlier covariant left-column representations. There are vertices $i \in \mathbb{Z}/5$ and arrows $a_i, b_i : i \to i-1$, with

$$a_i b_{i-1} = b_i a_{i-1} = 0.$$

Both pure branch paths survive at every length. No arrow-truncated finite algebra is being used. We work with ordinary polynomial modules, not with all completed lattices or all infinite strings.

Put $P_i = e_i A_{\mathrm{trav}}$ and let S_i be its residue simple. The augmentation kernel is $a_i A \oplus b_i A$. Multiplication by an arrow kills exactly the opposite-branch positive paths at its target, while each surviving path has a distinct normal image. Its kernel is therefore the right ideal generated by its unique forbidden successor, without cancellation. Iteration yields a projective resolution with

$$P_0 = P_i, \qquad P_n = P_{i-n} \oplus P_{i-n} \quad (n \geq 1).$$

The two arms are indexed by alternating-branch relation words. The *differential entry is the last arrow*, not the product of the zero relation word. This kernel description proves exactness in every

degree. Applying Hom to a simple kills every arrow entry, hence

$$\dim \mathrm{Hom}(S_i, S_j) = \mathbf{1}_{i=j}, \qquad \dim \mathrm{Ext}^n(S_i, S_j) = 2\mathbf{1}_{j=i-n} \quad (n \geq 1).$$

Tensoring with the left residue simple gives the same Tor dimensions. A graded version needs its internal degree shifts. In particular the residue simples have infinite projective dimension in this ambient algebra.

Let B be the one-arrow string supported at vertices zero and four, with $a_0 = 1$ and every other arrow zero. Its presentation is

$$P_4 \oplus P_3 \xrightarrow{(b_0,\ a_0 a_4)} P_0 \longrightarrow B \longrightarrow 0.$$

The two images have disjoint normal-path support and span the kernel, because B retains exactly e_0, a_0. The next entries are a_4 on the first arm and b_3 on the second; continue by their unique forbidden successors. This gives an unbounded exact resolution by the same normal-path argument. Full arrow-commuting equations give

$$\dim \mathrm{Hom}(B, B) = 1, \quad \dim \mathrm{Hom}(S_0, B) = 0, \quad \dim \mathrm{Hom}(B, S_0) = 1, \quad \dim \mathrm{Hom}(S_4, B) = 1.$$

In the Hom complex for $\mathrm{Ext}^1(B, B)$ both adjacent differentials vanish, so $\mathrm{Ext}^1(B, B) \simeq \mathbb{Q}$. A concrete nonsplit self-extension has $\mathbb{Q}^2$ at zero and four, $a_0 = I_2$, $b_0 = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, all other arrows zero. The first-coordinate submodule and quotient are B. A section at vertex zero would lift the quotient basis to a vector with second coordinate one; b_0 sends it to a nonzero first-coordinate vector, contradicting commutation with the zero b_0 action on B. Thus B is not rigid in this actual ambient category. No almost-split universality is inferred from this nonsplitting.

For finite-dimensional N let $DN = \mathrm{Hom}_{\mathbb{Q}}(N, \mathbb{Q})$. On each finite idempotent projective the evaluation pairing identifies $P \otimes_A DN$ with $D\,\mathrm{Hom}_A(P, N)$. Dualization is exact on these finite-dimensional terms, so the resulting complexes give $\mathrm{Tor}_n^A(M, DN) \simeq D\,\mathrm{Ext}_A^n(M, N)$ for the stated finitely generated resolutions. In particular $\mathrm{Tor}_1^A(B, DB) \simeq \mathbb{Q}$. The exact program checks six source/target pairs through degree twelve, all five simple-target formulas and 250 normal-kernel samples. The unbounded resolution assertions rest on the normal-form proof, not on the finite cutoff. These new homological proofs are prose/exact calculations; full corresponding resolution objects are not yet Lean packaged.

A geometric path may label a string after specifying the ribbon structure, relations and category. Turning it into a geodesic needs a metric and appropriate boundary/stop data. Intersections compute morphisms only under the hypotheses of an actual graded/Fukaya comparison. The homologically smooth graded-gentle perfect-category theorems do not identify our entire ordinary or completed module category with the Fukaya category by inspection [93]. In particular the residue resolutions warn against silently imposing smoothness or finite projective dimension.

S20.3 Pants tensors, physical dilation and sewing boundaries

For $C = \mathbb{Q}[\epsilon]/\epsilon^2$, take $\varepsilon(1) = 0$, $\varepsilon(\epsilon) = 1$. In the ordered bases $(1, \epsilon)$ and $(1 \otimes 1, 1 \otimes \epsilon, \epsilon \otimes 1, \epsilon \otimes \epsilon)$, multiplication and comultiplication are

$$\mu = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix}, \qquad \Delta = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The pairing is $\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$, with determinant -1. Associativity follows from $\epsilon^2 = 0$; coassociativity sends one to the three placements of a single one among two ϵ's and sends ϵ to $\epsilon^{\otimes 3}$. On the four tensor basis vectors, all three maps in

$$\Delta\mu = (\mu \otimes I)(I \otimes \Delta) = (I \otimes \mu)(\Delta \otimes I)$$

give, respectively, $1 \otimes \epsilon + \epsilon \otimes 1$, $\epsilon \otimes \epsilon$, $\epsilon \otimes \epsilon$, and zero. This proves the Frobenius identity. The unit and counit identities follow directly on the same basis. These are exact oriented 2D TQFT gluing data [95], not a presumed positive Hilbert structure.

The handle map $H = \mu\Delta$ sends $1 \mapsto 2\epsilon$ and $\epsilon \mapsto 0$. Closed genus-g contractions $\varepsilon H^g \eta$ are $0, 2, 0, 0, \ldots$. The genus-two value really vanishes in this example. In characteristic two the torus value also vanishes, illustrating a meaningful coefficient change rather than an embedding-space relabelling. After extending to $\mathbb{C}$ with the Euclidean metric, $\Delta^*\Delta = \mathrm{diag}(2, 1)$ and $\Delta \neq \mu^*$; $\mu : \mathbb{C}^4 \to \mathbb{C}^2$ is not an isometry. The exact code constructs $K = \mu/\sqrt{2}$ and a 4×4 failure block L with $K^*K + L^*L = I_4$. Stacking them gives an isometry into $\mathbb{C}^2 \oplus \mathbb{C}^4$, extendable to a unitary with ancillas. Its success probabilities on the four basis states are $1/2, 1/2, 1/2, 0$. Postselection implements K with state-dependent success; normalizing the selected outcome is not the original linear map μ.

For a closed genus-$g \geq 2$ surface a pants decomposition has $2g - 2$ pieces and $3g - 3$ sewn curves. Conformal sewing adds moduli-dependent propagators and analytic consistency [96]; string amplitudes need worldsheet, ghost and integration data beyond this finite tensor. A pants network or a hyperbolic graph is likewise not an AdS/CFT duality or a Ryu–Takayanagi derivation [97]. Such interpretations remain specified research targets until their extra structure is constructed. Arithmetic/anabelian reconstruction additionally requires arithmetic profinite fundamental-group data and its Galois structure under the proper field/curve hypotheses. Neither a discrete F_2 graph nor a coefficient $\mathbb{F}_2$ module supplies that arithmetic extension [98].

S20.4 Quantum games and the Connes-embedding distinction

For uniform input bits x, y, CHSH accepts outputs u, v when $u \oplus v = xy$. With ± 1 observables its success probability is $1/2 + \langle T \rangle/8$, where

$$T = A_0 \otimes (B_0 + B_1) + A_1 \otimes (B_0 - B_1).$$

For a deterministic classical assignment one bracket vanishes and the other is ± 2, giving success at most $3/4$; all sixteen assignments are checked. On $|\Phi^+\rangle$, take $A_0 = Z, A_1 = X$, $B_0 = (Z + X)/\sqrt{2}, B_1 = (Z - X)/\sqrt{2}$. Then $T = \sqrt{2}(Z \otimes Z + X \otimes X)$ gives success $(2 + \sqrt{2})/4$. For arbitrary self-adjoint involutions, $T^2 = 4I - [A_0, A_1] \otimes [B_0, B_1]$ and each commutator has norm at most two. Hence $\|T\| \leq 2\sqrt{2}$, proving optimality. The exact matrix checker verifies the entire spectrum $\{-2\sqrt{2}, 0, 0, 2\sqrt{2}\}$, not just a numerical sample.

MIP*=RE, in its pinned v4, gives an external unrestricted correlation separation and a negative answer to Connes embedding [99]. We do not rerun its external formalization or claim a new proof of it. CHSH has its optimum on two qubits and the same commuting optimum by the preceding bound; it does not exhibit the general separation. A learned game strategy is useful for finite witness search and checked bounds, but unrestricted computability and dimension quantifiers cannot be settled by a finite game simulation.

S20.5 Tropical accelerator reference designs and reproducibility

The chip-design campaign pins twenty public repositories, twenty-four focused document ranges and thirteen primary papers. It covers AlphaChip/Circuit Training, independent macro-placement

benchmarks, OpenROAD-family flows, quantum layout and low-level control. The source ledger records what was actually read, not a full audit or reproduction of every model. An exact public identity for the phrase "bare-metal Qiskit" was not established; no repository was invented.

Three original executable references implement a masked signed max-plus tile, an ordered mixed MAC/max-plus scheduler and a fail-closed synthetic IQ steering controller. Wide addition precedes winner selection and final saturation; earliest-index tie handling is explicit. Empty masks remain invalid, not an extra candidate of value zero. The controller requires a current calibration epoch, enough samples and a valid margin/deadline before allowlisting a steering action. Checks include all 65,536 signed eight-bit additions, 262,144 masked two-lane cases, reordered completions and twelve negative controls. These are digital reference models, not synthesized RTL, routed chips, measured RF feedback, power/area estimates or universal hardware speedups. A tropical context block can avoid multiplications inside that primitive while its projections, scores and other layers still require their own arithmetic.

The Linux snapshot preserves the Mac project, source clones, histories, checkpoints and environment archives. Linux GPU execution uses the existing Torch CUDA environment and a separately approved isolated cuQuantum prefix; macOS binaries are not treated as a Linux installation. The conda lock, setup instructions, gate/head replay records and all failures are retained. The on-page book contains one canonical laboratory for this chapter and links all five new checkpoint architectures into the existing tensor inspector. New physics, width and homological results have their explicit prose/numerical/external status; selected earlier Lean components do not constitute a full formalization of this extension or a proof of GRH.

S21 Construction, curation and interpretation

S21.1 A record is an evidence-bearing physical task

Each default Parquet row has a persistent record identity, parent/similarity group, partition, evidence class, lossless scientific payload and typed numeric columns. Forward inputs have the prefix `condition_`; supervision uses `target_`; `evaluation_` fields are excluded from inputs. The full payload, source hash, record identifier and audit text are not automatically fed into a model. Otherwise an apparent condition/target mask could still leak the target through metadata or a duplicated representation. An inverse task needs an explicit desired-condition contract and role change.

The origin fields distinguish a new simulation from a reused planned-source identity or geometry with new labels. The source-anchor configuration preserves existing train assignments. New public PDB coordinates retain their own source identity rather than being called members of PDBbind or newly acquired experimental quantum measurements. All anchors are train only: they do not establish blind performance on a model that may have encountered the parent.

The actual native interfaces are tested without changing either model. Aster accepts condition-only Hamiltonian records through its existing tokenizer. Changing a hidden target leaves those input tokens unchanged. Mora-3's scalar graphs preserve computed/predicted evidence. Its existing scalar/array supervision interfaces reject those targets under their observed-only policy. That rejection is recorded, not bypassed by changing the evidence label. Source construction is therefore distinct from the owning model's future admission, scaling, objective and evaluation decisions.

S21.2 Effective material families and gap guards

The two-band QWZ model uses

$$H(k_x, k_y) = \sin k_x\, \sigma_x + \sin k_y\, \sigma_y + (m + \cos k_x + \cos k_y)\sigma_z.$$

Its continuum band gap is $2\min\{|m+2|, |m|, |m-2|\}$. To see this, put $u = \cos k_x$, $v = \cos k_y$. The squared positive energy is $2 + m^2 + 2m(u+v) + 2uv$, a bilinear function on $[-1,1]^2$; its minimum occurs at a corner. Thus an odd mesh that misses π cannot certify a gap near $m = 0$ or2. The corrected release uses this analytic guard, even32/64 meshes, random gauge checks and an unavailable label near closure. The occupied-band integer uses the oriented link-product convention of Fukui–Hatsugai–Suzuki [100]; the sign convention is retained in the row.

For the Rice–Mele pump, take $t_1 = 1 + \delta\cos\theta$, $t_2 = 1 - \delta\cos\theta$ and $\Delta = r\sin\theta$. With positive t_1, t_2, minimization over k first gives $(t_1 - t_2)^2 + \Delta^2$ at $k = \pi$; minimization over θ gives the exact gap $2\min(2\delta, r)$. Harper's direct gap has a conservative mesh/Lipschitz guard, explicitly distinguished from interval certification. The displayed bulk slice has $k_y = 0$; an open strip is evaluated separately at $k_y = .37$. A single strip is not a proof of its entire edge dispersion.

The SSH winding, Kitaev sign invariant and classical dessin RG retain their own hypotheses. Surface genus is not an electronic phase index. In the Kitaev row, a physical onsite defect changes particle and hole diagonals with opposite signs. Its gauge change has the form $\mathrm{diag}(G, \bar{G})$. These preserve the recorded particle–hole symmetry; arbitrary independent diagonal perturbations would not. A quarantined draft revealed this error before release. No failed draft replaces a verified artifact.

S21.3 Molecular geometry and forces

The molecular set contains explicit chemical graphs, atom order, hydrogens, stereochemistry, method-labelled optimized references and bounded coordinate perturbations. MMFF94 supplies empirical molecular-mechanics energies, analytic forces and partial charges [101]. It supplies neither ab-initio electron density nor experimental affinity, ADMET, redox or biological efficacy. An optimization convergence code is not a theorem of a unique global minimum. Absolute MMFF energies are not cross-molecule atomization energies.

The checker reconstructs the stored bond graph in its *actual atom order*. A canonical SMILES string determines chemical identity but need not give the coordinate-array order. Geometry-assigned stereochemistry is checked against that identity; mirror controls fail. Energies and forces are recalculated on every accepted record. Central energy differences check force signs/units, and proper rotations/translations check the expected scalar/vector behavior. Missing or unsupported labels are not imputed.

For the heavy-atom point cloud, the complete finite Vietoris–Rips2-skeleton gives H_0, H_1 over $\mathbb{F}_2$. All edges and triangles, including ties, are ordered explicitly. Independent boundary-rank calculations verify selected Betti numbers against the barcodes. These are geometric distance-filtration features; a molecular loop or persistence bar is not an electronic topological phase. MMFF and the idealized π-orbital Hamiltonian retain separate labels.

S21.4 What the split certificate proves

Let V be the finite parent set and connect two parents whenever a declared similarity test passes. Add edges for all shared source/variant lineages. Assign a partition to each connected component and inherit it for every row. If an accepted similarity edge joined two partitions, its endpoints would

lie in one component with two assignments, a contradiction. This proves the exact finite no-crossing property for the declared relation.

For materials, normalized parameter and relative finite-spectrum thresholds are family/size-specific. SSH hopping swaps, Kitaev chemical-potential sign and the classical RG global-sign symmetry are folded conservatively. For molecules, stereo-agnostic constitution, nonempty Murcko scaffold and radius-two Morgan Tanimoto threshold .65 form chemical components. A second all-conformer pass adds edges for equal heavy-element multisets with sorted pair-distance RMS at most .10Å. This descriptor is rotation/permutation invariant but is not a complete congruence invariant or a certified RMSD optimizer; conservative false-positive grouping is possible.

Whole-component allocation balances row counts and declared family/topology or molecular-size/persistence strata. It never breaks a component to manufacture an80/10/10 result. The large peptide component makes the molecular training partition much larger than80%; rare/held-out peptide coverage can consequently be absent. Exact distributions and cross-partition nearest witnesses accompany the release. Protein TM-score, homology separation and global overlap with the models' complete historical corpora are not certified by small-molecule fingerprints. Threshold separation cannot itself guarantee freedom from all leakage or overfitting.

S21.5 Motion, charge modulation and exchange

For normal free fermions, $P(t) = UP(0)U^*$ preserves Hermiticity, trace and the spectrum: $P^* = P$ follows by adjoints, trace follows by cyclicity, and $P(t) = UP(0)U^{-1}$ is a similarity. Thus $0 \le P \le I$ and projector occupancy remain valid. Computing the diagonal of $-i[H, P]$ gives

$$\dot{n}_i = 2\,\mathrm{Im}\sum_j H_{ij}P_{ji}, \qquad J_{ij} = -2\,\mathrm{Im}(H_{ij}P_{ji}) = -J_{ji}.$$

The currents therefore obey site continuity and conserve total particle number. Their arrows represent probability/particle flux, not measured trajectories of individual electrons. Charge modulation is $-[n_i(t) - n_i(0)]$ in the declared electron-charge convention.

For a paired model, put $\Psi = (c, c^\dagger)$ and $R = \langle \Psi\Psi^\dagger \rangle$. From an initial Slater projector use $R_0 = \mathrm{diag}(I - P_0, P_0^{\mathsf{T}})$. Evolve with the full BdG Hamiltonian and recover $P = I - R_{11}$, $F = R_{12}$. The independently computed pairing source is

$$S_i = -2\,\mathrm{Im}(\Delta F^*)_{ii}, \qquad \dot{n}_i = -\sum_j J_{ij} + S_i.$$

The commutator derivative and this separate source formula agree in the checks; the source is not defined as the residual of the desired equation. For a Gaussian state, Wick's rule gives, for $i \neq j$, $\mathrm{Cov}(n_i, n_j) = |F_{ij}|^2 - |P_{ij}|^2$; on the diagonal it is $n_i(1 - n_i)$. The negative term is an exchange hole, not a computed Coulomb exchange-energy functional. Explicit small Fock-space evolution independently checks the covariance, current and pairing conventions.

The interacting control is the two-site Hubbard model at two electrons:

$$H = -t\sum_\sigma (c_{0\sigma}^\dagger c_{1\sigma} + \mathrm{h.c.}) + U\sum_{a=0}^{1} n_{a\uparrow}n_{a\downarrow}.$$

In its singlet/doubly-occupied symmetric sector the matrix is $\left(\begin{smallmatrix} 0 & -2t \\ -2t & U \end{smallmatrix}\right)$; the triplet energy is zero. Diagonalizing gives the exact exchange gap $J = (\sqrt{U^2 + 16t^2} - U)/2$, approaching $4t^2/U$ for large positive U. The full six-state two-electron sector supplies density, spin-swap and current trajectories;

a separate16-state Jordan–Wigner calculation checks them. This finite dimer is not a thermodynamic phase classification.

For nuclei, velocity Verlet uses actual MMFF forces and masses, with explicit time/energy units, finite energy drift and momentum checks. A Grad-only RDKit loop initially reused stale distance information. Refreshing energy at the same coordinates before the gradient restored the expected second-order step-size behavior in independent refinement tests. Failed trajectories are retained and excluded. The finite isolated trajectories are not equilibrium sampling. MMFF charges remain fixed. The separate uncalibrated π graph has its own dimensionless electronic time; it is not self-consistent TDDFT coupled to the nuclear femtosecond trajectory.

S21.6 Source-backed peptides and genuine boundaries

Public wwPDB coordinates are supplied under CC0 [102]. The source packet includes Trp-cage1L2Y, villin1VII, historical Trp-zipper1HRW (superseded by 1LE0, with the actual file HEADER identity retained), pancreatic polypeptide1PPT, azurin1AZU, crambin1CRN, ubiquitin1UBQ and engrailed homeodomain1ENH. The full source file, checksum, DOI, first-model/chain selection and bounded residue-subsystem mapping remain explicit. An NMR ensemble's model order is not physical time. Azurin's electron-transport classification does not calibrate our coarse contact Hamiltonian or turn it into an electronic-structure result.

The CISS review and peptide-bridge charge-transfer preprint motivate future spin, environment and lead benchmarks [103, 104]; the source review is focused, not an independent reproduction of their experiments. A spinless contact graph cannot supply their spin polarization, electron-transfer rates or measured quantum-biological mechanism. Synthetic strain, vacancy, mirror, coupling and conformer variants have their own provenance and share the parent group. They are additional training candidates, not newly observed systems.

S21.7 Circuits and geometric visualization

For a normal one-particle unitary U, adjacent row elimination gives Givens matrices G_k and diagonal phases D with $U = G_1^* \cdots G_m^* D$. For a, b in one column, the eliminating block is

$$G = \frac{1}{\sqrt{|a|^2 + |b|^2}} \begin{pmatrix} \bar{a} & \bar{b} \\ -b & a \end{pmatrix}.$$

Direct multiplication gives $G(a,b)^{\mathsf{T}} = (r,0)^{\mathsf{T}}$ and $G^*G = I$. A two-mode second-quantized gate acts as 1 on the vacuum, as its one-particle block on the single-occupation sector and as the determinant on double occupation. The exported circuit verifies these matchgates and numerical recomposition. Browser replay applies the actual gates to the saved initial correlation matrix. BdG pairing needs a different Bogoliubov compiler; it is not passed by a number-conserving Givens circuit. The Hubbard example instead includes Jordan–Wigner parity and measured product-formula convergence.

Actual molecular structures use CPK colors and explicit atom/bond order, with SDF/XYZ views, ball-and-stick, sticks and space-filling representations. Effective material-site layouts are labelled schematic. Electron-colored Gaussian splats are declared kernels on discrete site values, not ab-initio volumetric density. For the symmetric graph Laplacian L, localized wavelets are $sLe^{-sL}\delta_i$; since $\mathbf{1}^{\mathsf{T}}L = 0$, they have zero sum. Their signed colors are not probabilities. Time as a fourth projection coordinate is a visualization operation, not a new spatial dimension.

The dataset page links the checked public releases and source code. The full new probabilistic/homological/physical interfaces are not yet packaged in Lean; the existing 360 printed reports

retain their earlier scope. Neither a simulated transport law, dataset split certificate nor a compiled finite circuit closes the arithmetic comparison required for GRH.

S22 Molecular structure, complex transport and learned chemistry

The complementary-data laboratory now separates deposited protein structure, actual finite quantum transport and learned conformation generation. Covalent bonds use explicit molecular graphs or qualified PDB templates, with depth ordering and visible atom/bond scales. The ribbon view reads deposited `HELIX`/`SHEET` records; short synthetic peptide backbones remain unassigned rather than acquiring invented secondary structure.

Five additional finite propagation examples compare matched SSH/QWZ topological and trivial controls and an SSH domain wall. Complex phase, probability, current, coherence and selected eigenstates are distinct views. Analytic hydrogen volumes illustrate nodes, quantum beats and a phase vortex; they are not molecular electronic-structure calculations. Eight further CPU runs train classical and two-qubit hybrid score/flow models on periodic MMFF butane/pentane torsion spaces. All checkpoints, losses, controls and positive noise endpoints are explicit. These examples neither establish quantum advantage nor close the independent arithmetic bridge for GRH.

S23 Geometry, orbitals and generative time

S23.1 Chemical connectivity and source secondary structure

An atomic coordinate cloud does not determine chemical bonds merely by looking connected. SDF graphs retain their explicit bond indices. For the PDB source view we select the first model, blank/A alternate locations and positive occupancy; solvent is omitted from the display. Standard amino-acid templates, explicit `CONECT`, qualified ligand/hydrogen proximity and geometrically valid peptide C–N links have separate provenance. The complete original file remains available. Display connectivity is not a newly measured bond label.

The parser follows the wwPDB3.3 record definitions [105]. A class-one `HELIX` is an alpha helix; class five is a 3_{10} helix, not another alpha helix. `SHEET` retains strand identity, sheet membership and parallel/antiparallel sense. Ribbon patches follow the actual CA trace and carbonyl orientation, with strand arrows and narrower coils. A peptide bond must connect neighboring residues before a patch or spline-control neighbor is allowed: spatially nearby disconnected fragments and TER-separated chains must not acquire a phantom ribbon. Regression tests include these controls. Backbone recognition in a short synthetic SDF peptide supplies a trace, not a DSSP assignment or evidence of a stable secondary structure.

The full-source geometry is distinct from the capped effective contact graph. In the full-source view, only mapped modeled CA sites carry its electronic field. Nuclear display interpolation adds no new integration steps. An NMR model index, a ribbon parameter and a generative time are not physical time.

S23.2 What the complex transport animation computes

For a finite Hermitian matrix $H = V\,\mathrm{diag}(E_k)V^*$ and a normalized initial vector ψ_0, the continuously evaluated browser state is

$$\psi_j(t) = \sum_k V_{jk} e^{-itE_k} (V^*\psi_0)_k.$$

Unitarity gives $\sum_j |\psi_j(t)|^2 = 1$. Differentiating gives $\dot{n}_i = 2\,\mathrm{Im}\sum_j H_{ij}\psi_j\bar{\psi}_i$ and hence $\dot{n}_i + \sum_j J_{ij} = 0$ for $J_{ij} = -2\,\mathrm{Im}(H_{ij}\psi_j\bar{\psi}_i)$. The actual complex one-particle coherence is $\rho_{ij} = \psi_i\bar{\psi}_j$. A common phase cancels from both current and coherence; node colors therefore use a stated phase reference. The probability height is a display encoding, not another physical coordinate. Current arrows are fluxes, not classical electron paths.

Two48-site SSH controls use the same initial endpoint orbital and exchanged strong/weak hoppings. Their periodic bulk winding is respectively zero and one for $q(k) = t_1 + t_2e^{ik}$, with gap $2|t_1 - t_2|$. A separate61-site domain-wall chain starts and ends with strong bonds and has two weak bonds adjacent to its central even site. Its sublattices have sizes31 and30. At zero energy, set the odd-site amplitudes to zero and solve the tridiagonal recurrence on even sites. The amplitudes decay away from the wall by the weak/strong ratio. The nonzero off-diagonal entries determine this vector up to scale, so the nullspace is one-dimensional. This proves a finite interface zero mode; it does not identify an arithmetic Hilbert–Pólya spectrum.

The QWZ examples have a 12×10 rectangle and two orbitals per site. They use the same unfiltered corner spinor for masses -1 and -3. Their bulk Chern numbers are -1 and0 in the existing link-product convention; 32/64 meshes and gauge controls agree. Finite boundary probability, spectral weights and coherence remain measurements of the particular finite state, not definitions or proofs of that bulk invariant. Independent matrix-exponential checks reproduce the five propagated states. The hydrogen phase winding below, chemical torsion torus, dessin genus and persistent-homology classes are also different invariants.

S23.3 An analytic orbital benchmark with actual nodes

In atomic units, take $h = -\frac{1}{2}\Delta - r^{-1}$ on the usual hydrogen domain. The displayed normalized eigenfunctions are

$$\phi_{1s} = \pi^{-1/2}e^{-r}, \qquad \phi_{2s} = \frac{(2-r)e^{-r/2}}{4\sqrt{2\pi}}, \qquad \phi_{2p_j} = \frac{x_j e^{-r/2}}{4\sqrt{2\pi}}.$$

Their energies are $E_1 = -1/2$, $E_2 = -1/8$. The angular integrals and $\int_0^\infty r^n e^{-ar}\,dr = n!a^{-n-1}$ give their unit norms; opposite angular parity gives orthogonality of $1s$ and $2p_z$. Consequently

$$\psi(t) = \sqrt{1-a}\,\phi_{1s} + \sqrt{a}\,e^{-3it/8}\phi_{2p_z}, \qquad 0 \le a \le 1,$$

has unit norm after removing its unobservable common ground-state phase. Its density has the interference term $2\sqrt{a(1-a)}\phi_{1s}\phi_{2p_z}\cos(3t/8)$: this is the actual quantum beat shown in the volume. The $2s$ radial node is $r = 2$; the $2p_z$ nodal plane is $z = 0$.

The state $(\phi_{2p_x} + i\phi_{2p_y})/\sqrt{2}$ is proportional to $(x + iy)e^{-r/2}$. On a circle avoiding the nodal axis its phase winds once, and $L_z\psi = \psi$. Its current is $\mathrm{Im}(\bar{\psi}\nabla\psi)$ and has an azimuthal component. This stationary density with nonzero current need not show a moving classical particle. A finite-radius ray-marched volume and its opacity/exposure curve are rendering choices, distinct from the analytic density and normalization. These are nonrelativistic isolated-hydrogen benchmarks, not many-electron molecular orbitals or ab-initio peptide density.

S23.4 A useful periodic chemical target

Butane has one selected torsion; pentane has two. All other internal coordinates are held at the recorded MMFF reference geometry. The finite grids have72 and 24^2 points, respectively. At300K define

$$p_k = Z^{-1}\exp[-(E_k - E_{\min})/(k_BT)], \qquad q_0(\theta) = \sum_k p_k\,K_{\sigma^2/2}(\theta - \theta_k), \quad \sigma = .05,$$

where the product wrapped heat kernel is used on $\mathbb{T}^d$ and $d = 1, 2$. Thus q_0 is an explicitly softened finite-grid Gibbs target. It is positive, smooth and normalized; it is not a full solvent or molecular-equilibrium distribution. No affinity, redox, efficacy or electronic-density label is inferred. The molecular animation rotates downstream atoms about each actual bond axis; its poses agree with independent RDKit dihedral assignments to numerical precision and preserve the bond graph and fixed bond lengths.

S23.5 Score diffusion on the torus

For the unit circle,

$$K_\tau(x) = \sum_{m\in\mathbb{Z}} \frac{e^{-(x+2\pi m)^2/(4\tau)}}{\sqrt{4\pi\tau}}, \qquad \partial_\tau K_\tau = \partial_x^2 K_\tau.$$

The product kernel gives $q_\tau = q_0 * K_\tau$ and $\partial_\tau q_\tau = \Delta q_\tau$. A forward sample is $X_\tau = \mathrm{wrap}(X_0 + \sqrt{2\tau}\,\xi)$. Conditioning the wrapped Gaussian mixture shows

$$\nabla \log q_\tau(x) = \mathbb{E}[-\xi/\sqrt{2\tau} \mid X_\tau = x].$$

Indeed, differentiate each image term and divide by the mixture density; the Gaussian score is the stated latent-noise target. Conditional least squares therefore identifies the marginal score. The implemented weighted DSM loss uses weight 2τ, not an unlabelled Euclidean proxy on cut angles. The periodic-input network is trained over the full normalized time interval.

With reverse clock u, the ideal SDE is $dY_u = 2\nabla \log q_{T-u}(Y_u)\,du + \sqrt{2}\,dW_u$. Its Fokker–Planck equation at density q_{T-u} is $-2\Delta q + \Delta q = -\Delta q$, as required. This connection to score SDEs is the standard method [106]; the periodic mixture here supplies an explicit control. Initialization is sampled from the actual q_T, with $T = 2$, rather than silently assuming it is uniform. Sampling stops at $\tau = .03$, so the ideal endpoint is $q_{.03}$, whose Gaussian component variance is $\sigma^2 + .06$. It is not q_0. The initial run exposed this distinction; its source and outputs are preserved. The corrected run uses the proper endpoint reference and no excluded small-time training interval.

Exact wrapped-mixture scores, finite-difference checks, nested64/128-step Brownian controls and stored learned trajectories accompany the experiment. A finite Euler diagnostic is neither a convergence theorem nor a guarantee that a learned score recovers equilibrium. Remaining noise and learned-field error are shown separately from the MMFF target.

S23.6 Wrapped-geodesic flow matching

Let X_0 be uniform on $\mathbb{T}^d$, independently let $X_1 \sim q_0$, put $D = \mathrm{wrap}(X_1 - X_0)$ and $X_t = \mathrm{wrap}(X_0 + tD)$. The cut locus has probability zero under these continuous laws. For every smooth periodic test function f,

$$\frac{d}{dt}\mathbb{E}f(X_t) = \mathbb{E}\langle \nabla f(X_t), D\rangle = \int \langle \nabla f(x), v_t(x)\rangle \rho_t(x)\,dx, \quad v_t(x) = \mathbb{E}[D \mid X_t = x].$$

This is the weak continuity equation with the correct uniform and q_0 endpoints. Writing $B = [-\pi, \pi]^d$, the conditional formulas are explicitly

$$\rho_t(x) = (2\pi)^{-d} \int_B q_0(x + (1-t)D)\,dD,$$
$$\rho_t(x)v_t(x) = (2\pi)^{-d} \int_B D\, q_0(x + (1-t)D)\,dD.$$

Because q_0 is smooth and strictly positive on the compact torus, these integrals are jointly smooth through both endpoints and $\rho_t \geq \min q_0 > 0$. Division gives a smooth bounded periodic v_t with a uniform spatial Lipschitz bound. In particular $v_1 = 0$ by symmetry. The usual unique flow of this globally Lipschitz time-dependent field transports the stated marginals. Conditional regression targets D and learns an approximation to v_t, following the flow-matching and manifold formulation [107, 108]. Neither the regression loss nor a visualization replaces endpoint-error measurement.

S23.7 Physical state preparation, RG and phase-conditioned design

The earlier seven-spin dessin experiment supplies an actual quantum Gibbs state and its partial trace; its learned circuits approximate specified state/field targets. The exact coarse action in the RG chapter is produced by integrating out physical spins. Here the chemical learner transports a classical probability law on internal coordinates. A quantum channel, an RG partial trace and a score/noise schedule are three different constructions, with different states, clocks and acceptance tests.

A concrete materials-design constraint already appears in the QWZ controls. Any continuous mass path from $m = -3$ to $m = -1$ passes through $m = -2$. The explicit bulk gap formula then gives zero gap. Thus such a generator cannot both change between these two Chern sectors and preserve the gap throughout that path. Sector-conditioned sampling, explicit transition states and gap/invariant checks are meaningful design tasks. Merely drawing a continuous latent interpolation does not meet them. The finite transport controls and the earlier learned thermal-state experiments provide distinct physics benchmarks for these requirements, rather than unverified predictions of a synthesized topological material.

S23.8 What the quantum hybrid actually contributes

Each hybrid has two re-uploading two-qubit layers: periodic R_z angle/time encoding, trainable R_y rotations and CZ, followed by eight actual Pauli expectations and a two-layer64-unit tanh classical readout. The angles and readout are trained together. Integer coordinate frequencies preserve 2π periodicity of the measured features, and composition preserves it in the vector field. These are classically simulated quantum-feature hybrids, not coherent quantum chemical evolution or quantum-assisted measurements. The matched classical controls use periodic first/second harmonics and the same readout width. Parameter counts and individual seeds remain explicit.

Eight2400-update CPU runs retain heldout objectives, checkpoint weights, periodicity errors, independent Kronecker/Pauli calculations, parameter finite differences, exact-score comparisons and actual endpoint histograms. The limited runs do not establish quantum advantage. Flow and diffusion have different positive-noise endpoint laws and must not be ranked as though their targets were identical. Generative interpolation time is not molecular time.

The chemical torus is a concrete internal-coordinate space. Drawing it next to a dessin, a noncommutative torus or a Fukaya construction does not construct a functor or prove homological mirror symmetry. Likewise none of these finite learners constructs the missing arithmetic self-adjoint determinant bridge. Adding heat/noise is also distinct from integrating out physical degrees of freedom in the earlier actual RG chapter; a noise schedule alone is not an RG construction. New source, operator-domain, arithmetic and Lean obligations remain explicit.

S24 Learning objectives with explicit analytic guarantees

A useful learned contribution must distinguish approximation of known L-functions from proof of their zero locations. We propose holomorphic coefficient/readout learning, certified contour losses, arithmetic trace and functional-equation residuals, Schatten-controlled operator refinement, certificate search for the original signed moment, and exact-algebra representation constraints. Each has a different acceptance test.

For an analytic target on $|s-c| \le r$, let $|F| \le M_r$ there and choose $R < \rho < r$. A degree-N holomorphic readout trained on $m > N$ equally spaced points of the radius-ρ circle has the explicit uniform error budget

$$\|p_k - F\|_{|s-c|\le R} \le E_{\mathrm{opt}}(k) + E_{\mathrm{oracle}} + E_{\mathrm{alias}} + E_{\mathrm{Taylor}}.$$

The following proofs give all four terms, a primitive-Dirichlet arithmetic bound for M_r, and an infinite-feature kernel limit with its actual rates. Thus known analytic families can be approximated to arbitrary requested accuracy with independently certified evaluation error. This does not prove GRH. Uniform approximation by an already-real-zero determinant class is a different, potentially GRH-equivalent requirement; it must not be assumed through an alleged universal-approximation theorem.

S25 A certifiable analytic learning lane

S25.1 Targets, fields and arithmetic scope

The output is a complex holomorphic function, not an unconstrained real MLP on $\mathbb{R}^2$. Fix a center c and a finite family of targets F_χ analytic on a neighborhood of the closed radius-r disk. Certified values of $M_{r,\chi} \ge \sup|F_\chi|$ and of an evaluation error η are inputs to the quantitative theorem. They must be obtained without GRH or the desired arithmetic positivity. A finite conductor family uses the maximum of its independently bounded constants; this is not a uniform theorem over all conductors. Arithmetic character data are exact discrete inputs, not rounded embedding labels.

For primitive nonprincipal Dirichlet characters, the completed function is entire and a direct bound is given below. Pole-cleared zeta/Dedekind targets and known holomorphic Hecke or elliptic targets require their own explicit normalization and growth bounds. General Artin holomorphy is not assumed. Approximating on a compact domain avoiding possible poles does not prove that those poles are absent. Real/complex analytic norms, finite-field algebra and tropical semirings must likewise retain their distinct meanings.

S25.2 A finite holomorphic readout theorem

Write $F(c+z) = \sum_{n\ge 0} a_n z^n$ and choose $0 < R < \rho < r$, integers $m > N \ge 0$ and sample angles $\theta_j = 2\pi j/m$. Let $|y_j - F(c+\rho e^{i\theta_j})| \le \eta$, including all quadrature and rounding errors. The model and empirical objective are

$$p_b(c+z) = \sum_{n=0}^{N} b_n (z/\rho)^n, \qquad \mathcal{L}(b) = \frac{1}{2m}\sum_{j=0}^{m-1}\left|\sum_{n=0}^{N} b_n e^{in\theta_j} - y_j\right|^2.$$

This is a fixed holomorphic feature layer and a learned complex linear readout. It is a genuine convex learning problem; no nonlinear feature-learning claim is attached to its guarantee.

Proposition S25.1 (Four-part uniform error bound)**.** *Put $u = R/\rho$, $v = R/r$ and $w = (\rho/r)^m$. Let $b_n^* = m^{-1}\sum_j y_j e^{-in\theta_j}$, and take gradient descent with $0 < \gamma < 2$. After k steps from b^0, on $|s - c| \le R$,*

$$|p_{b^k}(s) - F(s)| \le \frac{|1-\gamma|^k \|b^0 - b^*\|_2}{\sqrt{1-u^2}} + \eta\frac{1-u^{N+1}}{1-u} + M_r\frac{w}{1-w}\frac{1-v^{N+1}}{1-v} + M_r\frac{v^{N+1}}{1-v}.$$

The terms are optimization, certified sample error, Cauchy aliasing and analytic truncation, respectively.

Proof. Discrete Fourier orthogonality gives $A^*A/m = I$ for $A_{jn} = e^{in\theta_j}$, since $m > N$. In real and imaginary coordinates the gradient is $b - b^*$; hence $b^k - b^* = (1-\gamma)^k(b^0 - b^*)$. Cauchy–Schwarz bounds its evaluation by $\|b^k - b^*\|_2(\sum_{n=0}^N u^{2n})^{1/2}$, giving the first term. The displayed iteration is in exact arithmetic. A numerical FFT/GD implementation must also bound coefficient-rounding error; a coefficient error of norm d_k adds at most $d_k/\sqrt{1-u^2}$ to the evaluation budget. This is convergence to the projected least-squares optimum; the empirical residual need not vanish when the labels have components outside the model.

For exact samples, the Fourier coefficient divided by ρ^n is $a_n + \sum_{\ell\ge1} a_{n+\ell m}\rho^{\ell m}$. Uniform convergence on the sampling circle permits interchanging sums. Cauchy's estimate $|a_j| \le M_r r^{-j}$ bounds the alias by $M_r r^{-n} w/(1-w)$. Sample errors contribute at most $\eta\rho^{-n}$. Multiply by R^n, sum the two finite geometric series and add $\sum_{n>N} M_r(R/r)^n$. These are exactly the last three terms. □

Choose N, then $m > N$, then η and the optimization tolerance so that each displayed term is at most $\varepsilon/4$. All four can tend to zero; thus any fixed compact subdisk and finite independently bounded target family have arbitrary-accuracy approximation. This claim needs a certified oracle bound, not an empirical test-set MSE. The present high-precision demonstrator is explicitly numerical and does not relabel its floating evaluations as interval certificates.

S25.3 The infinite-width limit and its actual rate

Choose $R < \rho < r_0 < r$ and use features $\phi_n(z) = (z/r_0)^n$. Their kernel has the compact-uniform limit

$$K_N(z,w) = \sum_{n=0}^{N} \phi_n(z)\overline{\phi_n(w)} \longrightarrow K(z,w) = \frac{1}{1 - z\bar{w}/r_0^2}.$$

The geometric tail proves this limit directly; positivity follows because each quadratic form is a sum of squared feature combinations. The readout space is the sequence space ℓ^2 with bounded evaluation on $|z| < r_0$. On the radius-ρ circle its population coefficient Gram operator is diagonal with eigenvalues $\lambda_n = (\rho/r_0)^{2n}$.

Proposition S25.2 (Infinite-feature readout convergence)**.** *Suppose the initial coefficient error $e(0)$ is in ℓ^2, as for zero initialization and the analytic target above. Population gradient flow has $e_n(t) = e^{-\lambda_n t}e_n(0)$ and converges in ℓ^2, hence uniformly on every $|z| \le R < r_0$. If $\beta > 0$ and $\sum_n \lambda_n^{-2\beta}|e_n(0)|^2 \le B_\beta^2$, then*

$$\|p_t - F\|_{|z|\le R} \le \frac{B_\beta}{\sqrt{1-(R/r_0)^2}}\left(\frac{\beta}{et}\right)^\beta, \qquad t > 0.$$

For zero initialization this source condition holds whenever $(r_0/r)(r_0/\rho)^{2\beta} < 1$, with $B_\beta \le M_r[1 - ((r_0/r)(r_0/\rho)^{2\beta})^2]^{-1/2}$.

Proof. The diagonal loss gives the scalar ODE in each coordinate. Dominated convergence, with majorant $|e_n(0)|^2$, proves norm convergence. Evaluation is bounded by the geometric-series factor. Under the source condition, $\|e(t)\|_2 \le B_\beta \sup_{\lambda>0} \lambda^\beta e^{-t\lambda}$; the maximum occurs at $\lambda = \beta/t$. Cauchy's coefficient estimate gives $|a_n r_0^n| \le M_r (r_0/r)^n$, and summing the resulting geometric series proves the explicit source bound. □

There is no width-uniform exponential rate: $\lambda_n \to 0$. At finite degree the slowest mode has rate $(\rho/r_0)^{2N}$. Fourier whitening recovers the identity Gram matrix in Proposition S25.1, but its unbounded infinite-dimensional preconditioner must not be called a free uniform rate. Ridge adds a positive rate floor and a calculable bias; its coefficient bias factor is $\alpha/(\lambda_n + \alpha)$, and α must tend to zero for unbiased convergence. A standard Gaussian-process initialization need not lie in the kernel's RKHS; the stated ℓ^2 hypothesis cannot be silently applied to it. These exact fixed-feature statements are distinct from the nonlinear NTK limits of deep networks [110] and from the trained few-qubit feature-learning circuits.

S25.4 An unconditional primitive-Dirichlet size oracle

For a primitive nonprincipal character of conductor q, parity $a \in \{0, 1\}$ and root number $|\epsilon_\chi| = 1$, put $d = \pi/q$ and $\theta_\chi(t) = \sum_{n\ge1} \chi(n) n^a e^{-dn^2 t}$. The standard primitive theta transformation and Mellin identity give

$$\Lambda(s,\chi) = \int_1^\infty \theta_\chi(t) t^{(s+a)/2-1}\, dt + \epsilon_\chi \int_1^\infty \theta_{\bar\chi}(t) t^{(1-s+a)/2-1}\, dt.$$

The character conjugation and root number are retained. These standard functional/analytic inputs are unconditional in the stated Dirichlet scope [111]; this text does not claim a new Lean proof of Poisson summation. Exponential decay makes both integrals entire in s.

For $|s - c| \le r$, choose an integer $k \ge \max(0, (1 + |\operatorname{Re} c| + r + a)/2 - 1)$ and define

$$Q_k(x) = \sum_{j=0}^{k} \frac{k!}{(k-j)! x^{j+1}}.$$

Integration by parts gives $\int_1^\infty t^k e^{-xt}\, dt = e^{-x} Q_k(x)$. Consequently $M_r = 2\sum_{n\ge1} n^a e^{-dn^2} Q_k(dn^2)$ is a valid finite upper bound. For a computable finite version, choose $n_0 \ge 1$ and sum through $n_0 - 1$; the remaining one-integral sum is at most

$$C_a Q_k(dn_0^2) \frac{e^{-dn_0^2/2}}{1 - e^{-dn_0}}, \qquad C_0 = 1, \quad C_1 = (de)^{-1/2}.$$

Indeed $n^a e^{-dn^2/2} \le C_a$, Q_k decreases, and $(n_0 + j)^2 \ge n_0^2 + 2n_0 j$. Twice this tail covers both Mellin terms. The same bound controls theta truncation, with no assumption about zeros.

After truncating the n sum and the t integral, elementary interval exponential, logarithm, sine and cosine evaluation on a compact interval, with a bounded-derivative quadrature remainder, gives an arbitrary requested certified η. The explicit exponential tails control both truncations. This proves computability of the oracle; it does not assert that an ordinary high-precision special-function call is already a validated implementation. For a finite conductor cutoff, take the maximum of the computed bounds.

S25.5 Additional objectives and what they can certify

S25.5.1 Contour-aware approximation and active learning.

Augment the coefficient objective with independent interval enclosures of $|p - F|$ on contours, derivative-weighted losses and adaptive samples near small $|p|$. If $\sup_C |p-F| < \inf_C |p|$ on a Jordan contour, Rouché's theorem transfers the exact interior zero count of p to F. An interval root count and a strict margin are acceptance conditions. Sampling finitely many heights alone proves no exhaustion of the off-critical domain.

S25.5.2 Arithmetic and functional-equation residuals.

Use exact character masks, conductor dependence, primitive Euler factors, gamma normalization and the conjugate functional equation as structural features or independently verified residuals. On $\operatorname{Re} s \geq 1 + \delta$, with $\delta > 0$, for $N \geq 3$ the logarithmic-derivative Dirichlet tail has the elementary bound $N^{-\delta}(\log N/\delta + 1/\delta^2)$, from $\Lambda(n) \leq \log n$ and integral comparison. This is a useful certified training region; it does not extend absolute convergence into the critical strip. A functional equation alone also does not locate zeros.

S25.5.3 Operator and determinant refinement.

Learn compatible finite compressions, graph-norm/domain residuals, self-adjoint boundary conditions and uniform Schatten-tail bounds from the actual infinite dessin/order construction. Trace-norm convergence transfers ordinary Fredholm determinants, with the standard bound

$$|\det(I + A) - \det(I + B)| \leq \|A - B\|_1 e^{1+\|A\|_1+\|B\|_1}$$

[112]. Strong resolvent convergence alone does not supply this bound. A reciprocal critical-spectrum model typically needs a genus-one regularized determinant and Hilbert–Schmidt control; it must not silently be treated as trace class. The arithmetic determinant identification remains an independent obligation even when the learned operators are Hermitian.

S25.5.4 Certificate-guided moment and algebra learning.

For the actual original-family low-gcd moment, use adversarial legal masks, conductors, short rows and retained II–I covariance to search for a sharp envelope or counterexample. Rational/interval or exact symbolic checking must certify a proposed inequality uniformly. A sum-of-squares or dual certificate can prove its stated parameter box; extrapolating it to all conductors needs a separate analytic argument. Preserve exact locally gentle relations, surface-order completions, determinant semi-invariant weights and module category labels in the representation architecture. Small relation loss is not an exact functor or an arithmetic comparison.

S25.5.5 Quantum kernels, width and proof search.

Quantum feature Gram positivity is not Weil/arithmetic positivity. A proved finite Gram lower bound gives a readout rate, while growing width requires the actual limiting kernel and initialization hypotheses. Proof-guided co-training can learn proposed identities or compression maps, then export rational/algebraic candidates for independent Lean/interval validation. Validation, rather than a confidence score, decides which claims survive.

S25.6 A decisive obstruction to a false universal guarantee

Proposition S25.3. *No sequence of real-rooted polynomials converges locally uniformly on $\mathbb{C}$ to $1+z^2$. Every such polynomial has error at least $7/16$ somewhere on $|z-i|=1/4$.*

Proof. On that circle, $|1+z^2| = |z-i||z+i| \geq (1/4)(2-1/4) = 7/16$. A smaller uniform error would, by Rouché, force a zero of the polynomial inside the disk, whereas all of its zeros are real and the disk misses the real axis. The same argument excludes local-uniform convergence. □

Thus a claim that an everywhere-real-zero/self-adjoint determinant family universally approximates the completed arithmetic target may already contain the missing GRH bridge. Hurwitz transfers zero-freeness only after independent local-uniform convergence and a nonzero limit have been proved. The analytic learner above deliberately proves approximation without asserting that spectral restriction. Its learned functions are evidence-bearing candidates, not a completed GRH proof. All new analytic/probabilistic statements here are prose proofs and scoped computations; no complete new Lean layer is claimed.

S25.7 Exact rational constants without transcendental rounding

The next implementation step certifies growth and analytic-tail constants, while leaving sample evaluation and coefficient rounding separate. Put $\delta = 3/q < \pi/q$ and, for an integer $L \geq 1$, define

$$U_L(x) = (1+x/L)^{-L}, \qquad x > 0.$$

The binomial coefficients give $(1+x/L)^L \leq \sum_{j\geq 0} x^j/j! = e^x$, so $e^{-x} \leq U_L(x) < 1$. Every evaluation of U_L at a positive rational argument is exact rational. With the same Q_k as above, choose $n_0 \geq 1$ and let

$$\widehat{M} = 2\left[\sum_{n=1}^{n_0-1} n^a U_L(\delta n^2) Q_k(\delta n^2) + C_a Q_k(\delta n_0^2)\frac{U_L(\delta n_0^2/2)}{1-U_L(\delta n_0)}\right], \quad C_0 = 1, \quad C_1 = \frac{1+\delta^{-1}}{2}.$$

This rational number bounds $|\Lambda(s,\chi)|$ on the specified disk. Indeed decreasing d to δ increases the positive integral majorant. For $a=1$, the maximum of $xe^{-\delta x^2/2}$ is $(\delta e)^{-1/2} \leq \delta^{-1/2} \leq (1+\delta^{-1})/2$. Split the Gaussian exponent into two halves, use $(n_0+j)^2 \geq n_0^2 + 2n_0 j$, and sum the resulting geometric tail. Replacing its numerator and ratio by U_L increases the bound, since its denominator stays strictly positive. This proves the formula.

For the actual conductors $3,4,5$, center $1/2$, radius $12/5$, $n_0 = 8$ and $L = 16$, exact fraction arithmetic gives respectively $\widehat{M} < 4$, $\widehat{M} < 10$, $\widehat{M} < 6$. Using $\widehat{M}$ (or these integer upper bounds) and the rational radii $R = 7/10$, $\rho = 6/5$ makes the Cauchy-alias and Taylor terms exact rational certificates as well. Full numerator/denominator records and the executable `certify_learning_constants.py` are supplied. This certifies constants, not the complex sample oracle, floating optimizer or GRH.

S26 Learning nonlocal strategies and checking their certificates

A two-player nonlocal game samples questions x, y and scores outputs a, b. The players may share a state, but communicate neither questions nor outcomes during play. Our finite tensor strategies have $p(a,b|x,y) = \mathrm{Tr}[\rho(P_a^x \otimes Q_b^y)]$. Offline joint optimization is allowed and is distinct from communication during the game. We train local measurement frames and a Schmidt angle using the exact Born objective, and replay actual checkpoints, probabilities, correlations, word moments and simulated shots in the interactive laboratory.

For $0 \le \alpha \le 2$ use the tilted score

$$S_\alpha = \alpha\langle A_0\rangle + \langle A_0B_0 + A_0B_1 + A_1B_0 - A_1B_1\rangle.$$

Its classical maximum is $2+\alpha$; its quantum maximum is $\sqrt{8+2\alpha^2}$ [113]. At $\alpha = 0$ the uniform CHSH winning probability is $1/2 + S_0/8$. This conversion does not apply unchanged to the tilted objective. The three seeds for each of $\alpha = 0, 1, 3/2$ give nine trained runs. One $\alpha = 3/2$ run remains at the classical value 3.5; the other runs reach the stated quantum maxima numerically. This retained failure rules out a claim of universal convergence of the optimizer.

The magic-square task uses two Bell pairs, local four-dimensional joint measurements, row parity $+1$ and column parities $(+1, +1, -1)$. Its classical value is $8/9$ and its quantum value is 1, as proved below. Three further trained local-frame runs reach values within $3 \cdot 10^{-13}$ of one. All twelve runs are small CPU simulations with saved seeds and parameters, not hardware Bell experiments or evidence for quantum training advantage.

These examples belong to both the closure of finite tensor correlations C_{qa} and the commuting-operator set C_{qc}. The external theorem MIP* = RE separates these sets and implies the negative answer to Connes embedding [115]. We do not reproduce that theorem. Finite learned strategies, positive state Gram matrices and finite NPA constraints do not witness its separation.

S27 Exact game calculations and the embedding boundary

S27.1 CHSH bounds and a constructive tilted strategy

For deterministic signs, either $B_0 = B_1$ or $B_0 = -B_1$, so the CHSH term is at most 2; choosing $A_0 = 1$ gives the tilted classical maximum $2 + \alpha$. Convex mixtures cannot improve a linear objective. For self-adjoint involutions, with operators of different parties commuting, write $\mathcal{B} = A_0(B_0 + B_1) + A_1(B_0 - B_1)$. The exact identity

$$2\sqrt{2}I - \mathcal{B} = \frac{1}{\sqrt{2}}\left[\left(A_0 - \frac{B_0+B_1}{\sqrt{2}}\right)^2 + \left(A_1 - \frac{B_0-B_1}{\sqrt{2}}\right)^2\right]$$

proves the CHSH quantum bound by positivity. Expanding the squares uses only the involution and cross-party commutation relations.

For the tilted case take $|\psi_\theta\rangle = \cos\theta|00\rangle + \sin\theta|11\rangle$, $A_0 = Z$, $A_1 = X$ and $B_{0,1} = \cos\mu Z \pm \sin\mu X$. Then $S_\alpha = \alpha\cos 2\theta + 2\cos\mu + 2\sin\mu\sin 2\theta$. Choosing

$$\sin 2\theta = \sqrt{\frac{4-\alpha^2}{4+\alpha^2}}, \qquad \tan\mu = \sin 2\theta$$

with $0 \le \theta \le \pi/4$ and $0 \le \mu \le \pi/4$, gives $\sqrt{8+2\alpha^2}$ by direct substitution (including the endpoints). This verifies attainment; the universal tilted upper bound uses the sum-of-squares theorem in [113] for $\alpha < 2$; the endpoint follows by continuity since $\|\mathcal{B}_2 - \mathcal{B}_\alpha\| \le 2 - \alpha$. This is not an unproved inference from our restricted two-qubit parametrization.

S27.2 A full magic-square strategy

On two local qubits arrange the observables

$$\begin{pmatrix} Z\otimes I & I\otimes Z & Z\otimes Z \\ I\otimes X & X\otimes I & X\otimes X \\ Z\otimes X & X\otimes Z & Y\otimes Y \end{pmatrix} = (O_{ij}).$$

Entries commute within each row and column. Pauli multiplication gives row products $+I$ and column products $+I, +I, -I$. Alice measures a row, Bob the transpose of a column, and they share $|\Phi_4\rangle = \frac{1}{2}\sum_{k=0}^{3}|k,k\rangle$. The identity $\langle\Phi_4|A\otimes B|\Phi_4\rangle = \mathrm{Tr}(A^TB)/4$ gives correlation one at every intersection. The product identities force the required parities. Thus they win every question pair and the quantum value is one.

A classical perfect table would have total product $+1$ by multiplying the rows and -1 by multiplying the columns, a contradiction. At least one of the nine question pairs fails for any deterministic assignment. Conversely, Alice can output all $+1$; Bob outputs all $+1$ in the first two columns and $(-1,+1,+1)$ in the last, failing only one cell. The uniform classical value is exactly $8/9$; shared randomness cannot improve it. Our learner conjugates Alice's observables by one local unitary U_A and Bob's transposed observables by U_B, preserving each party's parity and commutation constraints at every optimization step. The exact score is $1/2 + \sum_{ij}\mathrm{Tr}(A_{ij}^TB_{ij})/72$.

S27.3 Which learning objectives carry proofs?

Finite strategies and noisy experimental laws. Unitary parametrization preserves valid projective measurements, so negative Born loss searches inside the chosen finite strategy class. Mixing the state with I/d with visibility $v \in [0,1]$ preserves positivity, normalization and no-signalling. For K joint outcomes, N ideal independent samples and $\delta \in (0,1)$, Hoeffding plus a union bound gives maximum frequency error at most $\sqrt{\log(2K/\delta)/(2N)}$ with probability at least $1-\delta$. Browser sampling is an illustration of this conditional IID statement. With dimension fixed, compactness and explicit matrix-product Lipschitz bounds permit exhaustive finite nets for arbitrary-accuracy global value approximation. Local gradient descent alone has no such guarantee.

Moment and dual-certificate learning. For words u, v in measurement generators, define $\Gamma_{u,v} = \langle\psi|u^*v|\psi\rangle$ (or its mixed-state trace). For every c, $c^*\Gamma c = \|\sum_v c_v v\psi\|^2 \geq 0$; the mixed case follows by convexity. We compute these actual finite word Gram matrices, not an optimized outer relaxation. NPA learning may propose PSD moment matrices obeying word relations, while an independently checked identity $\beta I - \mathcal{B} = \sum_j P_j^*P_j$ proves an upper bound. A small residual is not an exact certificate. Successive NPA levels converge to the commuting-operator value under the hierarchy's hypotheses [114]. Each level defines an outer relaxation; a validated optimized value or a dual-feasible certificate yields an upper bound. One feasible primal moment matrix alone does not bound the optimum and is not a realization theorem.

Tracial moments and Connes embedding. General vector-state moments need not be cyclic. A microstate learner for a tracial algebra must additionally preserve cyclic identities, positivity, norm bounds and compatibility across increasing word lengths, and compare with normalized matrix traces. Our displayed state moments do not fulfill that separate task merely because they are PSD. The external undecidability result [115] also precludes a single computable dimension/error schedule deciding all promised nonlocal games. It neither forbids certified solutions of specified finite games nor establishes any impossibility result for GRH. The valid connection to our program is certificate search with exact relation checks; neither Bell violation nor kernel positivity gives the missing arithmetic determinant/zero-location bridge. These new game proofs and computations are not claimed as complete Lean4 formalization.

S28 Hubs, spokes and the topology retained by a dessin

The hubs-and-spokes construction in the HoTT book, Sections 6.6–6.8 [109], gives a useful language for attaching cells. For a loop $f : S^1 \to X$, form the homotopy pushout $X \cup_f CS^1$: add a fresh hub and paths from every $f(u)$ to that hub, continuously indexed by $u : S^1$. This attaches a disk, rather than merely a finite graph of radial edges. The book's Remark 6.7.1 explains why identifying the hub with an existing boundary point requires extra coherence; Remark 6.7.2 distinguishes propositional from judgmental computation.

For our dessins the useful distinction is geometric. A vertex star has a disk neighborhood with a specified cyclic ordering of its incident bands. Removing the disk center gives a punctured disk, which retains its peripheral loop. Removing only the vertex of a finite star instead leaves disconnected open rays. These operations cannot be interchanged. In a cellular dessin, filling a face imposes its attaching-word relation; leaving its center punctured retains a boundary component of the ribbon neighborhood.



disk
coherent circle of spokes

punctured disk
peripheral loop

The shaded regions are two-dimensional; the displayed finite spokes sample them. The annulus shown on the right is a deformation retract of the punctured disk, not a missing vertex in a one-dimensional graph.

Proposition S28.1 (Face filling changes the covering problem)**.** *Let a connected cellular dessin be a spine of an oriented genus-g ribbon surface with $n > 0$ boundary components. Its graph fundamental group is free of rank $2g + n - 1$. Capping all boundary components gives the presentation*

$$\left\langle a_1, b_1, \ldots, a_g, b_g \;\middle|\; \prod_{i=1}^{g} [a_i, b_i] = 1 \right\rangle.$$

For the one-face clean dessins $K_{2,3}$ and $K_{2,5}$ used here, these free ranks are respectively 2 and 4, before capping.

Proof. A spanning-tree collapse of the connected spine leaves $E - V + 1$ circles. The ribbon surface retracts onto that spine and has Euler characteristic $2 - 2g - n = V - E$, proving the rank formula. Cutting a surface along a standard system of handle arcs gives the boundary word $\prod_i [a_i, b_i] \prod_j c_j$. Van Kampen therefore presents its group by these generators and this one relation. With a boundary present, eliminate c_n, leaving the stated free group. Capping attaches disks along the c_j, so van Kampen sets each $c_j = 1$, giving the closed-surface presentation. For our rotations on $K_{2,m}$, both black vertices have the same cyclic order on the m incident edges. Suppressing the bivalent white vertices gives dart rotations $(0 \cdots m-1)(m \cdots 2m-1)$ and involution $i \leftrightarrow i + m$. The face permutation is one $2m$-cycle for odd m. Thus $V = m + 2$, $E = 2m$, $F = 1$, and $g = (m-1)/2$, giving both examples. □

The once-punctured torus and closed torus both have $H_1 \cong \mathbb{Z}^2$, but their groups are F_2 and $\mathbb{Z}^2$; only the closed torus has $H_2 \cong \mathbb{Z}$. The analogous genus-two groups are F_4 and the four-generator

surface group, with $H_1 \cong \mathbb{Z}^4$ in both cases. Indeed the standard closed-surface cellular complex has one vertex, $2g$ edges and one face. The face word is a product of commutators, so its abelianized boundary is zero. Removing the face leaves the wedge of circles. This computes the stated homology and also explains why H_1 alone misses the face relation. The torus group is abelian while F_2 is not; in higher genus the closed surface has nonzero H_2 while the graph does not.

Consequently the graph's universal cover is a tree, whereas the universal cover of the filled genus-two surface includes lifted faces and is a disk. Collapsing a spanning tree changes the combinatorial cover graph; the Cayley tree of a free basis is not literally the original biregular dessin cover. This distinction matters when lifting strings, periodic paths and boundary conditions in the infinite locally gentle algebra.

The gluing language organizes local patches and their compatibility maps. It does not reconstruct the ribbon cyclic order, complex structure, Belyi map, quadratic zero relations, coefficient ring or polynomial/formal/operator completion from the homotopy type alone. These data are retained explicitly in the existing dessin and order constructions. Descent of modules, derived categories, Hom-control, AR meshes or Fukaya categories needs a separately defined functor and hypotheses; a homotopy pushout of underlying spaces is not such a theorem. Nor does it imply an arithmetic determinant identity.

The accompanying tool computes the same one-face genus-one/two/four dart families and draws local disk/annulus patches and their surface polygon presentations, peripheral loops and face words. These are exact finite combinatorial checks plus the topological proof above. HoTT is referenced as a conceptual foundation; no univalence or higher inductive-type axiom is added to the pinned Lean4 development. Its complete cell-attachment and categorical formalization remains an explicit task.

S29 Coherent electron-density clouds in an explicit spatial basis

The density viewer now evolves spatial functions, including interference terms, instead of drawing independent spheres around occupied sites. Let $g_i(\mathbf{r})$ be normalized real Gaussian basis functions, let $S_{ij} = \int g_i g_j$ be their positive-definite overlap, and set $L = S^{-1/2}$. For a Hermitian orthonormal-basis Hamiltonian h and a matrix C_0 of p orthonormal occupied columns, put

$$C(t) = e^{-ith}C_0, \qquad \rho(\mathbf{r},t) = \sum_{a=1}^{p} \left| \sum_i g_i(\mathbf{r})(LC(t))_{ia} \right|^2 .$$

This density is nonnegative and has integral p: unitarity preserves $C(t)^*C(t) = I$, and integration yields $\mathrm{Tr}(C(t)^*L^*SLC(t)) = p$. The cross terms carry coherence. Replacing them by a sum of atomic densities generally changes the model. The displayed gain/loss field is $\rho(t) - \rho(0)$, whose full-space integral is zero; positive and negative colors indicate redistribution, not particles with negative probability. Time interpolation of saved density frames is a display operation and is not an extra time-integration step.

For fixed-nuclei H_2^+ at separation $2a_0$, six normalized s Gaussians (exponents $0.3, 1.2, 4.8$ at each nucleus) give an explicitly integrated one-electron Coulomb Galerkin model. We start in one normalized left-center Gaussian, then evolve with its time-independent Hamiltonian. The electronic ground energy is approximately -1.08675615 Hartree; adding nuclear repulsion gives -0.58675615 Hartree. This small basis is not a chemical-accuracy or basis-convergence result. Independent Coulomb quadrature, kinetic-integral and matrix-exponential checks pass.

The second example uses the deposited Trp-cage CA geometry, a specified twenty-orbital contact Hamiltonian and ten occupied model orbitals. Its spatial Gaussians have exponent $0.25\,\text{Å}^{-2}$. The resulting field is the exact density of this finite effective basis model, not a calibrated protein electron

density, valence count or transfer-time prediction. The first example uses atomic units; the second retains its effective model time and length convention. These examples are separate from the fixed private dataset release and do not change its labels.

The viewer offers orbit/zoom, volume and isosurface rendering, time scrubbing, exposure and density/gain-loss palettes. The 64^3 grid and 41 saved frames are encoded as 8-bit square-root-density atlases. The maximum decoded integrated-charge errors are 0.00158 electron for H_2^+ and 0.00141 model particle for Trp-cage; analytical full-space charge conservation is exact. Rendering thresholds, finite boxes and quantization are explicit display limitations. The separate hydrogen viewer retains exact analytic hydrogen orbitals and their coherent time-dependent superpositions.

S30 A finite, validated approximation result

Let $\chi_3 = (0, 1, -1)$, $\chi_4 = (0, 1, 0, -1)$ and $\chi_5 = (0, 1, -1, -1, 1)$, extended periodically, with parities $a = 1, 1, 0$. We approximate the completed functions

$$\Lambda_q(s) = (q/\pi)^{(s+a)/2}\Gamma((s + a)/2)L(s, \chi_q).$$

The primitive theta/Mellin identity is the explicit classical analytic input; no hypothesis about zeros is used. These three examples do not assert a uniform result for all conductors, general Hecke functions or Artin holomorphy.

Theorem S30.1 (Scope of the supplied finite certificate)**.** *For each $q = 3, 4, 5$, the supplied exact-rational calculation constructs a complex polynomial P_q of degree 28 satisfying*

$$\sup_{|s-1/2|\le 7/10} |P_q(s) - \Lambda_q(s)| < 5 \cdot 10^{-15}.$$

The computation includes enclosures of all 64 true Fourier-circle nodes, completed-function sample error, analytic truncation, Fourier aliasing, the exact closed-form 180-step gradient-descent iterate and coefficient midpoint rounding. The Python implementation and informal analytic proof have independent checks; this is not a kernel-checked Lean theorem.

Proof. Write $F(c+z) = \sum_{n\ge 0} a_n z^n$, with $c = 1/2$, $R = 7/10$, $\rho = 6/5$, $r = 12/5$. The independently recomputed rational theta majorants are strictly below $M_q = 4, 10, 6$, respectively, on $|z| \le r$. Use true roots $\zeta_j = e^{2\pi i j/64}$ and rational labels y_j with proved error $|y_j - F(c + \rho\zeta_j)| \le \eta$. Use the real Euclidean gradient on complex coefficient coordinates for

$$\mathcal{L}(b) = \frac{1}{128}\sum_{j=0}^{63}\left|\sum_{n=0}^{28} b_n\zeta_j^n - y_j\right|^2.$$

Discrete Fourier orthogonality gives the least-squares optimum $b_n^* = 64^{-1}\sum_j y_j\bar{\zeta}_j^n$, $0 \le n \le 28$, and gradient $\nabla\mathcal{L}(b) = b - b^*$. For step $1/4$ and zero initialization, the exact 180-step iterate is $b_n^{180} = (1 - (3/4)^{180})b_n^*$. The code evaluates this closed form; it does not claim an explicit180-update optimizer run.

Set $u = R/\rho = 7/12$, $v = R/r = 7/24$ and $w = (\rho/r)^{64} = 2^{-64}$. Let $\widehat{b}_n$ be an outward rectangle enclosing b_n^* and ϵ_n the sum of the coordinate halfwidths of the rectangle for b_n^{180}. Choose each actual polynomial coefficient as that rectangle's exact rational midpoint c_n. The normalized basis is explicit:

$$P_q(s) = \sum_{n=0}^{28} c_n\left(\frac{s - 1/2}{6/5}\right)^n.$$

Thus the exported coefficient with index n multiplies the normalized variable, rather than the raw power $(s-1/2)^n$. Its error on the compact disk is bounded by

$$E_{\mathrm{opt}} = (3/4)^{180} \sum_{n=0}^{28} |\widehat{b}_n|_1 u^n, \qquad E_{\mathrm{arithmetic}} = \sum_{n=0}^{28} \epsilon_n u^n,$$

$$E_{\mathrm{oracle}} = \eta \frac{1-u^{29}}{1-u}, \qquad E_{\mathrm{alias}} = M_q \frac{w}{1-w} \frac{1-v^{29}}{1-v},$$

$$E_{\mathrm{Taylor}} = M_q \frac{v^{29}}{1-v}.$$

Here $|\widehat{b}_n|_1$ is the sum of the maximum absolute real and imaginary coordinates, a safe upper bound for every enclosed complex modulus. Cauchy's estimate $|a_n| \le M_q r^{-n}$ and the Fourier identity $b_n^* = a_n \rho^n + \sum_{\ell \ge 1} a_{n+64\ell} \rho^{n+64\ell}$ for exact labels prove the alias and Taylor bounds. Label perturbations contribute at most η per Fourier coefficient. The remaining two terms follow directly from the exact iterate and coefficient rectangles. Each term is rounded upward to a 180-bit dyadic rational; the reported total is the exact sum of those reported terms. The certified totals below are strictly below the asserted threshold. Detailed sample enclosures are proved in the extended edition and encoded with exact integer endpoints. □

Conductor	Maximum sample radius	Uniform polynomial bound
3	3.026×10^{-17}	1.786×10^{-15}
4	7.206×10^{-17}	4.455×10^{-15}
5	1.404×10^{-16}	2.906×10^{-15}

The displayed values are rounded upward; exact rational records establish the strict inequalities.

S31 What this does and does not certify

The final JSON records exact rational centers, interval endpoints, source hashes and all five error terms. Float values are display summaries only. A high-precision comparison is an independent diagnostic, not the reason an interval contains its mathematical target. Arithmetic uses outward integer operations, and invalid float/order inputs are rejected, including through typed caches. Earlier pilots and failed validation probes are retained.

This is a fixed finite tolerance/configuration. Increasing accuracy needs new truncation and precision choices; the fixed $N = 11, T = 128$ interface is not an arbitrary-accuracy oracle. The broader approximation existence theorem is separate. The certificate says nothing about the location of zeros: P_q is not required to be real-rooted or a self-adjoint spectral determinant. The independent arithmetic determinant identity, positivity and original signed short-row/covariance estimates remain open. No GRH proof or complete new Lean4 formalization is claimed.

S32 Validated primitive operations

Real intervals are closed dyadic intervals $[l2^{-224}, h2^{-224}]$ with integer endpoints; complex intervals are Cartesian products. Addition is exact at this scale. For multiplication take the minimum/maximum of all four integer endpoint products, divide by 2^{224} and round down/up. For a reciprocal whose interval omits zero, reverse the endpoint order and round $2^{448}/h$ down and $2^{448}/l$ up. Negative denominators and scalar division use the same endpoint rules. Each operation encloses all represented

real values by monotonicity; induction proves inclusion for every expression built from them. Rectangular complex multiplication uses these real rules. The sum of coordinate halfwidths bounds the complex distance from its exact rational center. Binary floating arguments are never silently converted.

For $0 < x < 1$, the alternating arctangent sum and the endpoint obtained by adding the next signed term enclose $\arctan x$. Sixty terms at $1/5$ and $1/239$, together with Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$, give the interval for π. The negative coefficient reverses the second interval. The identity follows from tangent addition and the range of the two angles; no decimal value of π is a trusted input.

For rational $x > 0$, range-reduce $x = 2^k v$, $1 \le v < 2$. With $z = (v-1)/(v+1) \le 1/3$,

$$\log v = 2\sum_{j=0}^{J-1} \frac{z^{2j+1}}{2j+1} + \mathcal{R}, \qquad 0 \le \mathcal{R} \le \frac{2z^{2J+1}}{(2J+1)(1-z^2)}.$$

The geometric tail bounds each remaining denominator below by $2J+1$. Use $J = 92$, the analogous expression for $\log 2$, and sign-aware interval multiplication by k. This covers negative k as well.

For complex z, repeatedly halve until its interval ℓ^1 magnitude bound is $A \le 1/2$. The degree 60 Taylor polynomial for e^z then has remainder at most

$$\frac{A^{61}}{61!}\frac{1}{1 - A/62}.$$

Successive absolute terms have ratio at most $A/62$, so geometric comparison proves this bound. Add its symmetric interval to each coordinate, then square outward the required number of times. The identity $e^{2z} = (e^z)^2$ and interval inclusion prove the final enclosure. Caches are type-sensitive: otherwise a cached exact argument can accidentally bypass rejection of an equal-valued float. No implementation change is allowed during the recorded run; the producer verifies source bytes again before emitting its certificate.

S33 Recomputing the compact growth constants

For the larger growth disk, $c = 1/2$, $r = 12/5$, take $k = 2$ when $a = 1$ and $k = 1$ when $a = 0$. Both real Mellin exponents on that disk are at most k. Define

$$Q_k(x) = \sum_{j=0}^{k} \frac{k!}{(k-j)!x^{j+1}}, \qquad U_{16}(x) = (1 + x/16)^{-16}, \qquad \delta = 3/q.$$

Integration by parts gives $\int_1^\infty t^k e^{-xt}\,dt = e^{-x}Q_k(x)$. The same Gaussian split and geometric-tail comparison used below, now with cutoff $n_0 = 8$, imply the exact rational majorant

$$\widehat{M}_q = 2\left[\sum_{n=1}^{7} n^a U_{16}(\delta n^2)Q_k(\delta n^2) + C_a Q_k(64\delta)\frac{U_{16}(32\delta)}{1 - U_{16}(8\delta)}\right], \quad C_0 = 1, \quad C_1 = (1+\delta^{-1})/2.$$

Replacing the actual $d = \pi/q$ by the smaller positive δ increases the integral majorant. Exact fraction comparison gives $\widehat{M}_3 < 4$, $\widehat{M}_4 < 10$, $\widehat{M}_5 < 6$. The producer recomputes these fractions, asserts the strict comparisons and exports them; the constants in the readout proof are thus checked rather than supplied by a floating growth estimate.

S34 The completed-function integral and its tails

Let $d = \pi/q$ and $\theta_q(t) = \sum_{n\ge1} \chi_q(n) n^a e^{-dn^2 t}$. The standard primitive theta transformation, followed by splitting the Mellin integral at1, gives

$$\Lambda_q(s) = \int_1^\infty \theta_q(t) t^{(s+a)/2-1}\,dt + \int_1^\infty \theta_q(t) t^{(1-s+a)/2-1}\,dt.$$

The second root-number coefficient is1 for these three characters: their finite Gauss sums are $i\sqrt{3}$, $2i$ and $\sqrt{5}$, respectively. For the last, $2\cos(2\pi/5) - 2\cos(4\pi/5) = \sqrt{5}$; the first two are direct conjugate-root differences. Division by $i^a\sqrt{q}$ gives the claimed normalization. Exponential decay makes the two integrals entire. This classical theta identity, not an assumption of Artin holomorphy or desired positivity, is the analytic input of the implementation.

The actual circle nodes are enclosed by computing $e^{2\pi ij/64}$ with the validated primitives. All input boxes fit $D = [-1,2] + i[-4/3, 4/3]$. Thus both exponents $p = (s+a)/2-1$ and $p' = (1-s+a)/2-1$ satisfy $|\Re p| \le 2$, $|\Im p| \le 2/3$. The domain guard precedes all term discards. Retain $1 \le n \le 11$ and integrate $1 \le t \le 128$.

Put $\delta = 3/q < d$, $U(x) = (1+x/128)^{-128}$ and $Q(x) = x^{-1} + 2x^{-2} + 2x^{-3}$. Since $(1+x/128)^{128} \le e^x$, $e^{-x} \le U(x) < 1$ for $x > 0$. Integration by parts gives $\int_1^\infty t^2 e^{-xt}\,dt = e^{-x}Q(x)$. With $C_0 = 1$, $C_1 = (1+\delta^{-1})/2$ the omitted-n tail of both integrals is

$$2C_a Q(\delta 12^2)\frac{U(\delta 12^2/2)}{1-U(12\delta)}.$$

Indeed $n^a e^{-\delta n^2/2} \le C_a$, the function Q decreases, and $(12+j)^2 \ge 12^2 + 24j$. These inequalities yield a geometric bound after splitting the exponent in halves. The retained-n time tail is at most

$$\sum_{n=1}^{11} 2n^a 128^3 U(128\delta n^2) Q(128\delta n^2).$$

It follows by substituting $t = 128u$ and majorizing $t^{\Re p}$ by t^2. The two tails are disjoint: omitted n use the full integral, whereas retained n contribute only $t > 128$.

S35 High-order integration with a certified remainder

Split each dyadic interval $[2^j, 2^{j+1}]$, $j = 0, \dots, 6$, into four equal panels. For its midpoint m and halfwidth h, $h/m \le 1/9$. For $f(z) = e^{-\lambda z} z^p$, $\lambda = \pi n^2/q$, the principal logarithm is analytic on $|z-m| \le \rho = m/2$. On that circle,

$$|f(z)| \le M_\rho = 3\max\{(3m/2)^2, (m/2)^{-2}, 1\} U(\delta n^2 m/2).$$

Here $\Re z \ge m/2$, $|\arg z| \le 1$ and $e^{|\Im p|} \le e^{2/3} \le 3$. The maximum handles both signs of $\Re p$. Cauchy's coefficient estimate and cancellation of odd powers on $[-h,h]$ therefore bound the error after degree $2K$ by

$$\frac{2hM_\rho (h/\rho)^{2K+2}}{(2K+3)(1-(h/\rho)^2)}.$$

Indeed integrate each remaining even term, replace its denominator by the first omitted denominator, then sum a geometric series. Here $K = 12$. Multiply by $2n^a$ for both Mellin terms and sum all 28 panels and retained n. Keeping this majorant even for later discarded terms only overcounts error.

To evaluate the polynomial stably, set $u = h/m$, $q_{-1} = 0$, $q_0 = 1$ and use

$$q_{j+1} = \frac{u(p - \lambda m - j)q_j - (\lambda m)u^2 q_{j-1}}{j+1}.$$

Multiplying $f' = (-\lambda + p/t)f$ by $t = m + x$ and matching coefficients proves this recurrence, with $q_j = f^{(j)}(m)h^j/(j!f(m))$. The integrated polynomial is $2he^{-\lambda m}m^p \sum_{j=0}^{K} q_{2j}/(2j+1)$; all its arithmetic is outward.

Fixed absolute exponential-rounding errors can be amplified by high-order coefficients in late panels. The code therefore first bounds a complete (n, panel) pair, including both powers, by

$$4hn^a(m+h)^2 U(\delta n^2(m-h)).$$

For real $t \geq 1$ its phase has modulus1 and $t^{\Re p} \leq (m+h)^2$, proving the bound. If it is below 2^{-90}, skip that pair entirely and add its bound to the final radius. This is controlled truncation, not numerical underflow. The final sample radius is the interval-sum halfwidth plus the omitted-n, time, Cauchy and discard bounds. All are recorded separately and checked as nonnegative. No high-precision special-function routine is trusted here.

S36 Reproduction and remaining interfaces

Run the preserved source on the existing Linux Python environment with no new packages: `python3` on `examples/analytic_oracle/theta_oracle.py`, with argument `-output NEW`. It refuses to overwrite an existing experiment directory. The final-v5 experiment contains 192 samples, three rational degree 28 polynomials, copied source and SHA-256 hashes. Independent validation checks rational endpoint arithmetic, transcendental enclosures, 1160 exact coefficient identities, high- precision completed-function comparisons and the exported component sums.

The fixed-parameter oracle solves a concrete evaluation/certification gap. It does not identify a positive arithmetic operator, force real-rooted polynomials, prove the signed original-family short-row bound or supply complete Lean4 verification. Surface/order and neural architectures may use these certificates as independent targets, but a small learned loss is not a substitute for the remaining arithmetic and categorical theorems.

S37 Requested-error readouts from preserved certified labels

The accepted final-v5 sample certificate is preserved unchanged. For each $q = 3, 4, 5$, it supplies $m = 64$ rational complex labels y_j, bounds $|y_j - \Lambda_q(c + \rho\zeta_j)| \leq \eta_j$, and outward rectangles enclosing the true roots $\zeta_j = e^{2\pi ij/m}$, where $c = 1/2$, $\rho = 6/5$. The classical primitive theta/Mellin identity and accepted growth bounds $M_q = 4, 10, 6$ on $|s - c| \leq r = 12/5$ remain the analytic inputs. This extension accepts an exact rational error request $\varepsilon > 0$ on the fixed compact $|s - c| \leq R = 7/10$. It changes the finite readout, while retaining the certified sample errors and the 64-node alias budget. It is not an arbitrary-accuracy theta-sampling interface.

Theorem S37.1 (Executed rational readout with explicit refusal)**.** *For each supported conductor and rational request, the finite-data selection below either returns a normalized rational polynomial P and an executed, exactly replayable gradient-descent trace with*

$$\sup_{|s-c| \leq R} |P(s) - \Lambda_q(s)| \leq \varepsilon,$$

or refuses the request with the exact minimum base budget over all 64 *allowed degrees. Refusal concerns attainability by this particular fixed-data majorant; it is not an impossibility theorem for approximation.*

Proof. Let B_n be an outward rectangle enclosing $b_n^* = m^{-1}\sum_j y_j\bar{\zeta}_j^n$, $0 \le n < 64$, computed from the preserved root rectangles. Put $t_n = \mathrm{mid}(B_n)$ and let e_n be the sum of its coordinate halfwidths. These are exact rationals, and $|t_n - b_n^*| \le e_n$. For a selected $d < 64$ execute, using exact rational operations,

$$b_n^0 = 0, \qquad b_n^{k+1} = \tfrac{3}{4}b_n^k + \tfrac{1}{4}t_n, \qquad 0 \le n \le d.$$

This is genuine gradient descent with step $1/4$ for the explicitly specified real Euclidean objective $J(b) = \frac{1}{2}\sum_{n=0}^{d}|b_n - t_n|^2$, whose gradient is $b - t$. All iterates and exact objective values are retained. The output is literally

$$P_k(s) = \sum_{n=0}^{d} b_n^k \left(\frac{s-c}{\rho}\right)^n.$$

Set $u = R/\rho = 7/12$, $v = R/r = 7/24$, $w = (\rho/r)^{64} = 2^{-64}$, $S_d(x) = \sum_{n=0}^{d} x^n$ and $\eta = \max_j \eta_j$. The five budgets are

$$E_{\mathrm{opt}}(k) = \sum_{n=0}^{d} |b_n^k - t_n|_1 u^n, \qquad E_{\mathrm{oracle}} = \eta S_d(u),$$

$$E_{\mathrm{alias}} = M_q \frac{w}{1-w} S_d(v), \qquad E_{\mathrm{Taylor}} = M_q \frac{v^{d+1}}{1-v},$$

$$E_{\mathrm{arithmetic}} = \sum_{n=0}^{d} e_n u^n.$$

Here $|x + iy|_1 = |x| + |y|$. Cauchy's estimate and true-root Fourier orthogonality give the alias and Taylor terms; certified label errors give the oracle term. Triangle inequality applied to $b_n^k - b_n^* = (b_n^k - t_n) + (t_n - b_n^*)$ gives the other two. Thus their exact sum bounds the compact-uniform error.

Define $C_d = E_{\mathrm{oracle}} + E_{\mathrm{alias}} + E_{\mathrm{Taylor}} + E_{\mathrm{arithmetic}}$. Select the smallest $d < 64$ with $C_d < \varepsilon$; if none exists, report $\min_{d<64} C_d$ and refuse. For a selected degree, $E_{\mathrm{opt}}(k) = (3/4)^k \sum_n |t_n|_1 u^n \longrightarrow 0$. Therefore executing updates until $E_{\mathrm{opt}}(k) \le \varepsilon - C_d$ terminates after finitely many steps. The serialized coefficients, budgets, targets and trace use exact rational endpoints; displayed decimal approximations are not proof inputs. □

S38 Objective semantics, precision floors and replay

The original true-circle sample-label least-squares objective is

$$\mathcal{L}_y(b) = \frac{1}{2m}\sum_{j=0}^{m-1}\left|\sum_{n=0}^{d} b_n\zeta_j^n - y_j\right|^2.$$

With $d < m$, Fourier orthogonality gives $A^*A = mI$ and the real Euclidean coordinate gradient $\nabla\mathcal{L}_y(b) = b - b^*$. The executed objective J instead has the exact rational midpoint target t. The difference $\nabla\mathcal{L}_y - \nabla J = t - b^*$ is enclosed coefficientwise by e_n. Consequently this is an actual rational GD run for J and a certified perturbed-gradient implementation for $\mathcal{L}_y$. It is not the exact

optimizer trajectory of the ideal-root objective, nor a replay of neural or quantum training. An interval enclosure of an ideal-root GD trajectory would be a different explicitly labelled object.

Each actual update contracts $b^k - t$ by $3/4$, so the exact stored losses satisfy $J(b^{k+1}) = (9/16)J(b^k)$. A replay recomputes every rational coordinate update, loss and weighted residual. The closed form $b^k = (1 - (3/4)^k)t$ is an independent check on the executed trace, not its producer. Exact rational loop coordinates introduce no further midpoint or floating export error. The arithmetic budget covers the difference between the certified Fourier rectangles and their midpoint targets, including the inherited root enclosure and outward interval operations. A subsequent float conversion requires its own exact conversion-error bound.

The normalized polynomial convention is essential: a stored coefficient with index n multiplies $((s - 1/2)/(6/5))^n$, not $(s - 1/2)^n$. The input certificate is pinned by SHA-256, including its source hashes, all 192 sample enclosures, root rectangles and analytic radius components. Producing source is captured at entry, checked unchanged before emission, and copied into a fresh output directory. An existing output is never overwritten. An independent replay should also check the fixed input hash, all dimensions, degree choice, the first stopping iterate, all five exact terms and their literal sum against the requested rational tolerance.

The attainable boundary of this majorant is

$$C_{\min,q} = \min_{0 \le d < 64} C_{q,d}.$$

Every request $\varepsilon > C_{\min,q}$ has strict budget slack for some degree and the proved finite stop rule. At or below that boundary this interface refuses. The degree minimizing C_d need not be 63: Taylor error decreases with degree while the oracle and arithmetic sums increase. In particular $C_d \ge \eta$ and $C_d \ge M_q 2^{-64}/(1 - 2^{-64})$. Increasing the number of optimization steps does not lower these fixed-data budgets. Approximate boundary values on the stated compact are $7.2929 \cdot 10^{-17}$, $1.7370 \cdot 10^{-16}$ and $3.3741 \cdot 10^{-16}$ for $q = 3, 4, 5$; only the exact exported rational minima are used for acceptance. These are certificate floors, not lower bounds on the actual errors.

A full adaptive theta oracle would separately need proof-preserving choices of omitted-n cutoff, time cutoff, quadrature order, whole-panel discard budget, primitive series lengths, interval precision and serialized-bound precision. The present $N = 11$, $T = 128$, 224-bit arithmetic, 60/92/60 primitive series lengths, 2^{-90} discard threshold and 180-bit rounded budgets cannot support every positive request. Increasing a nominal bit count alone leaves the fixed primitive remainders and analytic tails in place. The preserved labels are not recomputed here. The classical identity, finite arithmetic and informal proof are not new Lean declarations. No zero-location, all-conductor, positive arithmetic determinant, quantum-advantage or GRH conclusion follows.

The interactive replay is available at the certified readout laboratory. All decimal graphics are illustrative views of the linked exact records.

S39 Requested-error theta samples for three fixed conductors

For the primitive characters $\chi_3 = (0, 1, -1)$, $\chi_4 = (0, 1, 0, -1)$, $\chi_5 = (0, 1, -1, -1, 1)$ with parities $a = 1, 1, 0$, retain the classical primitive theta/Mellin identity and root number one already stated above. Put $D = [-1, 2] + i[-4/3, 4/3]$ and $\delta = 3/q$. The following adaptive extension is implemented with independent interval and sample checks; it is an informal analytic theorem, not a Lean result.

Theorem S39.1 (Finite adaptive sample specification)**.** *For every supported conductor, exact rational $s \in D$, and rational $\eta > 0$, the choices below terminate and give a rational complex midpoint y and exact rational radius $R \le \eta$ with $|y - \Lambda_q(s)| \le R$. The same conclusion holds for a specified true circle node when its root and node enclosures are refined on every precision retry.*

Proof. For $x \geq 0$ define the exact rational bound $E(x) = 2^{-\lfloor x \rfloor}$. Since $e > 2$, $e^{-x} \leq E(x)$; this bound is nonincreasing and tends to zero. Let $Q(x) = x^{-1} + 2x^{-2} + 2x^{-3}$ and $C_0 = 1$, $C_1 = (1 + \delta^{-1})/2$. For $N \geq 1$, $n_0 = N + 1$, $J \geq 1$, $T = 2^J$, use

$$B_n = 2C_a Q(\delta n_0^2) \frac{E(\delta n_0^2/2)}{1 - E(\delta n_0)},$$

$$B_t = \sum_{n=1}^{N} 2n^a T^3 E(\delta n^2 T) Q(\delta n^2 T).$$

Here $\delta n_0 \geq 6/5$ ensures a positive denominator; $E(x)$ equals one for $0 \leq x < 1$, so this safeguard is essential. On the four panels of each $[2^j, 2^{j+1}]$, write midpoint m, halfwidth h, $\rho = m/2$, $v = h/\rho \leq 2/9$, and $B(m) = \max\{(m + \rho)^2, (m - \rho)^{-2}, 1\}$. Use the Cauchy remainder

$$B_Q = \sum_{\text{panels}} \sum_{n=1}^{N} \frac{2n^a \, 2h \, 3B(m) E(\delta n^2 m/2) v^{2K+2}}{(2K + 3)(1 - v^2)}.$$

Choose in turn the first $N \geq 1$, first $J \geq 1$, and first $K \geq 0$ for which their respective budgets are at most $\eta/8$. Gaussian decay, then exponential time decay, then $v \leq 2/9$ prove termination. These are the accepted tail and Cauchy arguments with each true exponential separately replaced by its upper bound E.

There are $P = 4JN$ potential pairs. A nonzero-character pair may be omitted only if its entire two-power contribution has bound

$$b = 4hn^a (m + h)^2 E(\delta n^2 (m - h)) \leq \eta/(8P).$$

Add every omitted bound to B_{discard}, giving $B_{\text{discard}} \leq \eta/8$. Evaluate all other integrated Taylor polynomials by the stated exact coefficient recurrence using outward rectangles. Refine both interval precision and all primitive series remainders until the final rectangle's sum of coordinate halfwidths satisfies $R_{\text{arith}} \leq \eta/4$. For this fixed finite expression all input and primitive intervals shrink; continuity and outward arithmetic imply $R_{\text{arith}} \to 0$. Precision doubling therefore terminates, and the sum of the five raw components is at most $3\eta/4$. Choose an integer output scale B such that $5\,2^{-B} \leq \eta/8$ and round each component upward by $\text{up}_B(x) = 2^{-B}\lceil 2^B x \rceil$. The reported radius is their literal exact sum. With $\mathbf{b} = (R_{\text{arith}}, B_n, B_t, B_Q, B_{\text{discard}})$, it obeys

$$|y - \Lambda_q(s)| \leq \sum_{i=1}^{5} \text{up}_B(b_i) \leq \frac{3\eta}{4} + 5\,2^{-B} \leq \frac{7\eta}{8} < \eta.$$

The output retains exact rational midpoints, rounded components, B, raw interval radius, parameters, input metadata and source hashes. The raw analytic bounds are exactly reconstructible; they need not be serialized as potentially enormous fractions. The output precision varies with the request and introduces no fixed reporting floor. □

This fixed-conductor sample result uses no hypothesis on zeros. It does not remove a downstream fixed-node Fourier alias floor, supply an all-conductor theorem or a positive arithmetic determinant, or prove GRH.

S40 Adaptive sampling parameters and precision termination

The primitive theta/Mellin identity with root number one yields

$$\Lambda_q(s) = \sum_{n\ge 1} \chi_q(n) n^a \int_1^\infty e^{-\pi n^2 t/q} \big(t^{(s+a)/2-1} + t^{(1-s+a)/2-1}\big)\, dt.$$

On $D = [-1,2] + i[-4/3, 4/3]$, both exponents p satisfy $|\Re p| \le 2$ and $|\Im p| \le 2/3$. For exact $x \ge 0$, put $E(x) = 2^{-\lfloor x \rfloor}$. The elementary inequality $e > 2$ gives $e^{-x} \le e^{-\lfloor x \rfloor} \le E(x)$ and $E(x) \le 2e^{-x\log 2}$. Use this solely as a pointwise nonincreasing upper bound; it is not multiplicative and has no asserted exponential integral identity.

Integration by parts gives $\int_1^\infty t^2 e^{-xt}\, dt = e^{-x} Q(x)$ for $Q(x) = x^{-1} + 2x^{-2} + 2x^{-3}$. For $a = 0$ the bound $n^a e^{-\delta n^2/2} \le C_0 = 1$ is immediate. For $a = 1$, $n \le (1+n^2)/2$ and $n^2 e^{-\delta n^2/2} \le 2/(e\delta) \le 1/\delta$ give $C_1 = (1 + 1/\delta)/2$. For $n = n_0 + j$, the inequality $n^2 \ge n_0^2 + 2n_0 j$ therefore bounds the omitted-n sum by

$$2C_a Q(\delta n_0^2) \frac{e^{-\delta n_0^2/2}}{1 - e^{-\delta n_0}} \le 2C_a Q(\delta n_0^2) \frac{E(\delta n_0^2/2)}{1 - E(\delta n_0)} = B_n.$$

Take $n_0 = N + 1 \ge 2$; then $\delta n_0 \ge 6/5$ and the latter denominator is positive. Substituting $t = Tu$ in each retained-n time tail gives

$$B_t = \sum_{n=1}^{N} 2n^a T^3 E(\delta n^2 T) Q(\delta n^2 T).$$

The two tails are disjoint because the omitted-n tail uses the whole integral and the time tail uses only retained n.

For $j = 0, \dots, J-1$ and $\ell = 0, \dots, 3$ take

$$m = 2^j \left(1 + \frac{2\ell + 1}{8}\right), \qquad h = \frac{2^j}{8}, \qquad T = 2^J.$$

These panels cover $[1, T]$ and have $h/m \le 1/9$. On $|z - m| = \rho = m/2$, principal log is analytic, $\Re z \ge m/2$, and $|\arg z| \le 1$. For $f(z) = e^{-\pi n^2 z/q} z^p$, the angular multiplier is at most $e^{2/3} \le 3$, whence

$$|f(z)| \le 3B(m) E(\delta n^2 m/2), \qquad B(m) = \max\{(3m/2)^2, (m/2)^{-2}, 1\}.$$

Cauchy's coefficient estimate, cancellation of odd terms, and geometric summation yield the B_Q in Theorem S39.1. Including zero-character or later discarded terms in that sum is harmless. Gaussian, time, and geometric decay prove termination of the sequential first-success searches with $B_n, B_t, B_Q \le \eta/8$. For fixed q and $L = \max\{1, \log(1/\eta)\}$ their asymptotic scales can be chosen as $N = O(\sqrt{L})$, $T = O(L)$, $J = O(\log L)$, and $K = O(L)$. Polynomial prefactors in N, T, J contribute only $O(\log L)$ to the logarithm of the error estimates; this is a truncation-size statement, not a rational-arithmetic bit-complexity bound.

For real $t \in [m-h, m+h]$, $t \ge 1$ and $t^{\Re p} \le (m+h)^2$. The bound for a whole pair, including both powers, is therefore $4hn^a(m+h)^2 E(\delta n^2 (m-h))$. There are $P = 4JN$ potential pairs; summing only bounds at most $\eta/(8P)$ gives $B_{\text{discard}} \le \eta/8$. No partial recurrence or isolated exponential is discarded. For active pairs retain the coefficient recurrence

$$r_{-1} = 0, \quad r_0 = 1, \quad r_{j+1} = \frac{(h/m)(p - \lambda m - j) r_j - \lambda m (h/m)^2 r_{j-1}}{j+1}, \qquad \lambda = \pi n^2/q.$$

Coefficient comparison in $tf' = (-\lambda t + p)f$ proves that the integrated Taylor polynomial is

$$2he^{p\log m - \lambda m} \sum_{j=0}^{K} \frac{r_{2j}}{2j+1}.$$

Outward arithmetic encloses its exact value, including the signed character weight and both powers.

Use an isolated p-bit dyadic context, with all primitive remainders refined as p increases. Put $\delta_p = 2^{-p-16}$. Each arctangent series separately stops when its next-term bound is at most δ_p, giving total weighted analytic tail width at most $20\delta_p$ in Machin's formula, before outward dyadic rounding. Each reduced logarithm series separately has tail at most δ_p; for $x = 2^k v$, the combined bound is $(1 + |k|)\delta_p$, before rounding. For the complex exponential, after reduction to $A \le 1/2$, choose the Taylor order so the bound

$$\frac{A^{H+1}}{(H+1)!} \frac{1}{1 - A/(H+2)}.$$

is at most δ_p. Choose H by finite exact search; its denominator is positive. Add a symmetric exponential error enclosure to each coordinate before outward squaring. The fixed guard16 is not itself an acceptance bound: the final complete rectangle radius must satisfy $R_{\text{arith}} \le \eta/4$. All interval caches must include p and preserve exact input-type guards. The successful p and retry precisions/radii are recorded. Individual series lengths are determined by these, the exact inputs and the pinned source and are reconstructible by replay; they are not explicitly stored.

For fixed N,J,K and exact target point, the expression has finitely many operations and only nonzero fixed integer denominators. Each primitive input is bounded for sufficiently large p; exponential halving counts are consequently bounded. Exact primitive tails and dyadic rounding widths tend to zero. Induction through the finite continuous expression, including finitely many outward squarings, gives $R_{\text{arith}} \to 0$. Doubling p after a failed comparison terminates. The argument also covers true roots and nodes regenerated at every p, since their widths tend to zero. A fixed nondegenerate s rectangle has no corresponding shrinking-input claim.

A growing-order primitive estimate. To justify convergence despite a growing exponential Taylor order, write $\epsilon_p = 2^{-p}$, $\delta_p = 2^{-p-16}$, and let $d(X)$ be the sum of a rectangle's coordinate widths and $m(X)$ its ℓ^1 magnitude bound. The rounded reduced input W satisfies $a = m(W) \le 1/2 + 2\epsilon_p < 3/5$ and $d(W) \le 2^{-k} d(Z) + 4\epsilon_p$; the remainder is at most 2^{-H}, so the first accepted order satisfies $H \le p + 18$. For the computed term boxes, outward multiplication and integer division give $m_j \le a m_{j-1} + 6\epsilon_p$ and $d_j \le a d_{j-1} + m_{j-1} d(W) + 12\epsilon_p$. Thus $m_j \le a^j + 15\epsilon_p$ and $\sum_{j=0}^{H} d_j \le (25/4) d(W) + 75H\epsilon_p$; adding the remainder increases width by at most $4(\delta_p + \epsilon_p)$. The pre-squaring width is therefore $O(d(Z) + p2^{-p})$ and its magnitude is bounded (indeed below 3 for $p \ge 32$). Each outward square has width at most $2m(X)d(X) + 8\epsilon_p$; the bounded number of squarings consequently preserves convergence to zero. Finally, exact rational partial sums and endpoint rounding give Machin width at most $20\delta_p + 40\epsilon_p$, and logarithm width at most $(1 + |k|)(\delta_p + 2\epsilon_p)$ for fixed $x = 2^k v$. These estimates justify the primitive-width step without assuming a fixed Taylor order or asserting a global arithmetic complexity bound.

For serialization, initialize

$$B = \max\{32, \operatorname{bitlen}(\operatorname{den}\eta) - \operatorname{bitlen}(\operatorname{num}\eta) + 8\}$$

and increase B until $5\,2^{-B} \le \eta/8$. Write $\widehat{x} = 2^{-B}\lceil 2^B x \rceil$ for each of the five raw components. Their exact reported sum is at most $3\eta/4 + 5\,2^{-B} \le 7\eta/8$, and every component remains an upper bound. The output records B, the raw interval radius and complete rectangle, N/J/K, discard threshold/pair

counts, exact target metadata and source hashes. These reconstruct all raw analytic bounds and pair sets without serializing huge exact tail denominators. For the practical requests below B is87 and114, respectively. Exact endpoints remain rational; the mathematical specification assumes an integer representation sufficient for the requested size. Dynamic rounding removes the old fixed reporting floor. For rational s, validate its exact coordinates in closed D before conversion: an outward dyadic enclosure of $\Im s = 4/3$ extends beyond D, but the analytic bounds concern the true point. Circle nodes satisfy the strict exact bounds $-7/10 \le \Re s \le 17/10$, $|\Im s| \le 6/5$.

The exact parameter-design checker gives

η	q	N	J	T	K	active pairs
10^{-24}	3	12	7	128	19	71
10^{-24}	4	14	8	256	20	65
10^{-24}	5	16	8	256	20	111
10^{-32}	3	14	7	128	25	84
10^{-32}	4	16	8	256	26	72
10^{-32}	5	18	8	256	26	129

with exact rational budgets and no special-function oracle. The arithmetic precision is selected from the actual interval radius, not inferred from this table. This extension retains the existing classical theta identity; it adds no zero-location, positive determinant, all-conductor, Lean or GRH claim. The following end-to-end adaptive readout supplies its own node-count, alias, degree, Fourier and optimization choices.

S41 An end-to-end adaptive analytic readout

The new driver changes the sample truncations, interval precision, circle size and degree; it does not reuse the frozen labels to assert a smaller error. Its scope is still the three stated primitive quadratic characters and the fixed compact $|s - 1/2| \le 7/10$.

Theorem S41.1 (Requested-error construction in the stated finite family)**.** *Given a positive rational ε and $q \in \{3, 4, 5\}$, the specified exact-arithmetic algorithm terminates in the mathematical unbounded-resource model and returns a normalized rational polynomial P with*

$$\sup_{|s-1/2|\le 7/10} |P(s) - \Lambda_q(s)| \le \varepsilon.$$

Its sample enclosures, rational optimizer updates and all error budgets are explicit. This is an approximation theorem, with the stated classical theta identity as an input; it supplies no zero-location or positive determinant conclusion and is not a Lean theorem or a practical runtime bound.

Proof. Put $u = 7/12$, $v = 7/24$. Choose $d \ge 0$ so $10v^{d+1}/(1 - v) \le \varepsilon/8$, and choose $m > d$ so

$$10\frac{2^{-m}}{1 - 2^{-m}}\frac{1}{1 - v} \le \varepsilon/8.$$

The driver rounds m upward to a multiple of eight, which only decreases this bound. Set the requested sample error $\eta = \varepsilon(1 - u)/8$. The adaptive sampler gives all m true circle nodes and labels with error at most η. Form their outward Fourier rectangles through degree d, and refine the nodes and finite evaluation arithmetic if their weighted midpoint radii exceed $\varepsilon/8$. This refinement terminates: for these fixed finite d, m, node uncertainty and every finite-expression arithmetic width

tend to zero with precision. Merely lifting fixed boxes would not suffice, so the driver regenerates the nodes and samples on this retry.

Choose rational midpoint targets t_n and execute the previously specified quadratic-objective updates until $\sum_{n=0}^{d} |b_n^k - t_n|_1 u^n \le \varepsilon/4$. Their geometric contraction proves finite termination. The oracle, aliasing, Taylor and Fourier-midpoint terms are each at most $\varepsilon/8$, and the optimization term is at most $\varepsilon/4$. Hence the raw five-term sum is at most $3\varepsilon/4$. Choose a serialization precision B with $5\,2^{-B} \le \varepsilon/8$, round each term upward, then add the five reported rationals exactly. The printed bound is at most $7\varepsilon/8 \le \varepsilon$. The polynomial uses $((s-1/2)/(6/5))^n$, and the preceding Fourier/Cauchy proof gives its asserted compact-uniform bound. □

The trace is losslessly encoded by the fixed target vector and a rational scale extracted from each *executed* coefficient state, together with its objective, weighted residual and full-state hash. Every coefficient can be recovered exactly; a separate replay expands the states and checks every update. The producer does not generate its trace from a closed form. The saved objective remains the rational Fourier-midpoint quadratic, with its perturbation from ideal-root least squares explicitly budgeted.

S42 Termination, compression and the boundary of the guarantee

The degree and circle-size choices use only rational comparisons. Since $0 < v < 1$, the Taylor test succeeds at finite d; since $2^{-m} \to 0$, the alias test succeeds at finite $m > d$. The common growth bound10 is safe for all three families; the certificate also recomputes the previously proved exact conductor-specific majorants below $4, 10, 6$.

At fixed d, m, η, the analytic truncation parameters are fixed while precision is doubled. The computed sample centers stay bounded: their underlying finite expression is fixed and the true completed-function values are bounded on the sample circle. Its arithmetic widths tend to zero, as do the regenerated true-root enclosures. Fourier computation is a finite composition of additions, products and division by the positive integer m. Its midpoint-radius sum therefore tends to zero. This proves the driver's additional precision retry, without pretending that changing a box's nominal precision shrinks its uncertainty.

The accepted sampler error may remain above its actual arithmetic width because its analytic tails and output bounds are fixed for this η. That causes no problem: the oracle allocation uses the total certified sample radius, whereas the Fourier-midpoint allocation uses only finite coefficient-rectangle width. No analytic sample error is silently moved into the latter term or removed by numerical convergence.

For the stored training trace, the exact loop gives $b^{k+1} = 3b^k/4 + t/4$ from $b^0 = 0$. Inductively every state is a rational scalar multiple of the same target vector. The producer first executes the actual vector updates, then extracts this scalar from a nonzero target coordinate, checks all coordinates, and records a canonical full-vector SHA-256 hash. If the entire target vector is zero, its zero state needs only scale0. The compressed record is thus an exact representation of the executed states, not a floating-point sketch. Hashes are integrity aids; exact vector reconstruction and the independently checked recurrence establish the mathematical claim.

Coefficient export uses exact rational numerators and denominators. A chunked decimal integer encoder/parser handles endpoints beyond Python's usual digit-conversion guard without changing global interpreter settings. No fixed 180-bit serialization floor remains. The decimal graphics are separate approximate projections, not polynomial coefficients to which the exact certificate may be applied without a conversion-error bound.

All producing source bytes are captured, copied into fresh output paths and checked unchanged. Historical fixed producers are pinned and untouched. For the new modules the run also checks the import-time source snapshots, so a stale imported module cannot be attributed to later source bytes. After output creation, a failed evaluation, final provenance check or serialization emits uncertified failure metadata. Invalid requests and stale import-time sources are refused before creating output artifacts. Old outputs are never overwritten. Extreme requests can be computationally impractical: the termination proof gives no machine-time or memory guarantee, and finite test runs establish only their reported cases.

Nothing requires the output polynomials to be real-rooted, to equal an arithmetic Fredholm determinant, or to have a self-adjoint realization. Consequently arbitrary compact approximation of these known functions does not imply GRH. Uniformity in changing conductor, general Hecke/Dedekind/elliptic/Artin scopes, the independent determinant comparison, and actual analytic/category Lean objects remain separate obligations.

S43 An arithmetic-family moment with every original row retained

Let $K = \mathbb{Q}(\sqrt{-3})$, $\mathcal{O} = \mathbb{Z}[\omega]$, and write N for the ideal or element norm. Fix a finite-order Hecke character ν and a finite set S of prime ideals containing those above 6 and those dividing its conductor. For $0 \neq u \in \mathcal{O}$, let κ_u be the character of $K(u^{1/6})/K$ defined using arithmetic Frobenius, and put $\psi_u = \mathrm{prim}(\nu\kappa_u)$. Here prim means the underlying primitive Hecke character: it is formed before introducing any divisor variable or evaluating conductor-dependent zero extensions. The rows below are all actual Eisenstein elements, including units and prime powers supported in S.

The external inputs are abelian Kummer reciprocity and character uniqueness, the primitive Hecke functional equation and its usual vertical growth, Stirling's formula, scalar Dedekind-zeta convexity, and Blomer–Goldmakher–Louvel's large sieve for their specified sixth-order family. We use the pinned arXiv:1112.1650v1, Theorem 1.3, Lemma 2.1 and Appendix II; its Section 5 supplies the all-ideal decomposition whose fixed-twist proof is redone here. No statement about all primitive characters of comparable conductor, Artin holomorphy, Lindelöf bounds or zero locations is assumed.

Theorem S43.1 (The original-family numerator moment)**.** *For every $\varepsilon > 0$, uniformly for $H \geq 1$ and $t \in \mathbb{R}$,*

$$\sum_{0<Nu\leq H} |L(1/2+it,\psi_u)|^2 \ll_{\nu,S,\varepsilon} H^{1+\varepsilon}(1+|t|)^{1+\varepsilon}. \tag{8}$$

For every fixed $0 < \delta < 1/2$, the same estimate holds with $1/2+it$ replaced by $1/2+\delta+it$, with the constant also depending on δ. If the central bound has constant C_0 at an exponent $0 < \varepsilon_0 \leq 1/4$, an admissible off-central constant is

$$200C_0\left(\frac{3/2+\delta}{1/2-\delta}\right)^2. \tag{9}$$

Proof. For the canonical BGL character $\vartheta_{\mathfrak{a}}$, Appendix II gives a radicand $\alpha_{\mathfrak{a}}$ with $(\alpha_{\mathfrak{a}})_S = \mathfrak{a}\mathfrak{g}^{-6}$. Since $\mathcal{O}_S$ is principal, comparison with any literal u whose outside-S ideal is $\mathfrak{a}$ gives $\kappa_u = \eta\vartheta_{\mathfrak{a}}$ for one of finitely many fixed S-unit Kummer twists η. This is an equality of primitive characters, established outside a finite set and extended by uniqueness. For each outside-S ideal $\mathfrak{a}$ and S-supported integral ideal $\mathfrak{b}$, at most six elements generate $\mathfrak{a}\mathfrak{b}$. The resulting S-supported multiplicity sum converges at exponent $1+\varepsilon$.

For each fixed $\xi = \nu\eta$, split the canonical parameter as $\mathfrak{a} = \mathfrak{a}_0\mathfrak{a}_1\mathfrak{a}_2^6$, where $\mathfrak{a}_0$ is squarefree, $\mathfrak{a}_1$ is squarefull and sixth-power-free, and $(\mathfrak{a}_0, \mathfrak{a}_1) = 1$. The primitive conductor is at most a fixed multiple of $N(\mathfrak{a}_0\mathfrak{a}_1)$. The common-gamma approximate functional equation, with $X = H/(N\mathfrak{a}_2)^6$

and $T = 1 + |t|$, has common maximal length $B = X^{1/2}T(HT)^{\eta}$ for any sufficiently small fixed $\eta > 0$. Mellin inversion removes its row-dependent conductor weight. Split a column as $\mathfrak{b}_S\mathfrak{b}_1\mathfrak{b}_2$, with $\mathfrak{b}_S$ supported in S, $\mathfrak{b}_1$ squarefree and $\mathfrak{b}_2$ squarefull, the last two coprime and prime to S. The original primitive values at $\mathfrak{b}_S$ remain bounded row multipliers. The S-supported and squarefull column weights are summable at $1/2 + \eta$. Multiplying squarefree column coefficients by the fixed ξ preserves their squared norm, whatever its finite order. BGL's large sieve therefore bounds the energy for fixed $\mathfrak{a}_1, \mathfrak{a}_2$ by

$$(HT)^{O(\eta)}\{M + B + (MB)^{2/3}\}, \qquad M = X/N\mathfrak{a}_1.$$

Squarefull counting and summation give $X + B\sqrt{X} + (XB)^{2/3} \ll (HT)^{O(\eta)}XT$. The $\mathfrak{a}_2$ sum converges at exponent 6. Principal rows are separated before the pole-free functional equation and bounded by scalar zeta convexity. This proves (8).

For the off-central statement divide each $L(s, \psi)$ by $(s + 1)^{5/4}$ and, on principal rows, also multiply by $(s - 1)/(s + 1)$. These are bounded analytic functions on $\operatorname{Re} s > 1/2$, continuous on the boundary. Poisson representation and Jensen's inequality transfer the central squared moment. The convolution of $P_\delta(y) = \delta/[\pi(\delta^2 + y^2)]$ with $(1 + |t|)^{-3/2+\varepsilon_0}$ is at most eleven times that weight when $0 < \varepsilon_0 \le 1/4$. Restoring the denominator and the principal pole-removal factor gives (9). The extended edition gives the detailed derivation with all primitive local values. □

Separate fixed-unitary extension. The moment theorem also holds for any one fixed unitary Hecke character ν. For precision choose $\nu(\mathfrak{a}) = \nu_0(\mathfrak{a})(N\mathfrak{a})^{-i\tau}$, with angular parameter $m \in \mathbb{Z}$ and completed gamma factor

$$\gamma_\nu(s) = 2(2\pi)^{-(s+|m|/2+i\tau)}\Gamma(s + |m|/2 + i\tau).$$

All finite Kummer twists have this same infinity type. Use $T_\nu = 1 + |t + \tau| + |m|$, comparable to $1 + |t|$ with ν-dependent constants. Pure norm rows have $L(s, \psi) = \zeta_K(s + i\tau)$ and are separated before the pole-free functional equation; their pole at $1 - i\tau$ is removed by $(s - 1 + i\tau)/(s + 1 + i\tau)$. No uniformity in m, τ or the conductor of ν is asserted. This extension of the moment is separate from the following finite-order type-I residue formula.

S43.1 The exact masked remainder and the remaining loss

Retain the original zero-extended symbol $\chi_d(u) = (u/d)_6$ on good squarefree ideals. Let μ be the ideal Möbius function and set

$$\beta_U(d) = \sum_{\substack{ab=d \\ Na, Nb \le U}} \mu(a)\mu(b), \quad T(u, d) = S \cup \operatorname{supp}(u) \cup \operatorname{supp}(d).$$

Let $\psi_{u,2} = \operatorname{prim}(\psi_u^2)$, formed independently of squaring any zero sentinel. With $\mathfrak{f}_u$ and $\mathfrak{f}_{u,2}$ the primitive conductors, the full-mask series is

$$F_{u,d}(s) = \frac{L(s, \psi_u)}{L(2s, \psi_{u,2})} C_{T(u,d)}(s), \tag{10}$$

$$C_T(s) = \frac{\prod_{p \in T,\ p \nmid \mathfrak{f}_u}(1 - \psi_u(p)(Np)^{-s})}{\prod_{p \in T,\ p \nmid \mathfrak{f}_{u,2}}(1 - \psi_{u,2}(p)(Np)^{-2s})}.$$

Initially it equals $\sum_{c \text{ squarefree},\ (c,T)=1} \psi_u(c)(Nc)^{-s}$ for $\operatorname{Re} s > 1$. Every prime of T remains deleted, including those forgotten by either primitive conductor.

For finite-order ν, $D \geq 1$, $U \geq 1$, $U^2 \leq D$, and $W \in C_c^\infty((0,\infty))$ supported in $[1,2]$, put

$$\widehat{W}(s) = \int_0^\infty W(x) x^{s-1}\, dx, \qquad \mathcal{M}_4(W) = \sum_{j=0}^{4} \sup_x |(x\partial_x)^j W(x)|.$$

On $\sigma = 1/2 + \delta$, the exact residue-subtracted type-I remainder is

$$E_u = \frac{D^\delta}{2\pi} \int_{\mathbb{R}} \widehat{W}(\sigma + it) D^{it} \sum_{\substack{d \text{ squarefree} \\ (d,S)=1}} \frac{\beta_U(d)\nu(d)\chi_d(u)}{(Nd)^{\sigma+it}} F_{u,d}(\sigma + it)\, dt. \tag{11}$$

This definition retains the actual β_U, all phases, d-masks, conductors and the independently primitive square.

Theorem S43.2 (An improved masked residual estimate)**.** *Under the preceding finite-order hypotheses, for every $\varepsilon > 0$,*

$$\sum_{0<Nu\leq H} |E_u|^2 \ll_{\nu,S,\delta,\varepsilon} \mathcal{M}_4(W)^2 D^{2\delta} U^{2-4\delta} H (HD)^\varepsilon. \tag{12}$$

Proof. The same deleted set in both factors of (10) gives $|F_{u,d}(\sigma + it)| \ll (Nu\, Nd)^\eta |L(\sigma + it, \psi_u)|$: the reciprocal incomplete Euler product at $2\sigma > 1$ is bounded by $\zeta_K(2\sigma)$, and its remaining finite numerator is bounded by a small power of its radical. Apply Minkowski to (11) in the exact row space and use Theorem S43.1. The actual coefficient sum is at most

$$\sum_d \frac{|\beta_U(d)|}{(Nd)^{\sigma-\eta}} \leq \left(\sum_{Na\leq U} (Na)^{-\sigma+\eta} \right)^2 \ll U^{1-2\delta+2\eta}.$$

Four Mellin integrations by parts bound the remaining height integral by a constant times $\mathcal{M}_4(W)$. Square the result and absorb the small losses using $U^2 \leq D$. □

At $H = D^{3/4}$ and $U = D^a$ with fixed $0 < a \leq 1/2$, the upper bound's excess over H is $2\delta + (2 - 4\delta)a$, up to small losses; as $\delta \downarrow 0$ it still leaves $2a$. This is a limitation of the upper bound, not an energy lower bound. The extracted principal residue, the original short term at smaller scales and the signed type-II minus type-I covariance are not controlled by (12). The next arithmetic task is to estimate the complete d-weighted moment before its absolute coefficient sum, or establish sufficient signed covariance. No positive arithmetic determinant, zero-location theorem or GRH proof follows. The analytic argument uses the external classical inputs above; finite diagnostics and selected existing Lean local identities do not formalize the Hecke functions, approximate functional equation, Poisson transfer or these moment bounds.

S44 Family identification, the approximate functional equation and masks

We prove the analytic statements of Section S43 from the explicit classical inputs stated there. Write $I(S)$ for the nonzero integral ideals prime to S, and $\mathcal{O}_S$ for the ring of S-integers. Constants may depend on the fixed field, S, ν and any declared positive small exponent. No such constant is asserted uniform in the conductor or infinity type of a varying ν.

S44.1 A finite comparison with the literal Kummer rows

Use Appendix II of BGL, with $n = 6$. Its fixed local modulus and representatives may be chosen once, since S contains the primes above 6 and $\mathcal{O}_S$ has class number one. For each $\mathfrak{a} \in I(S)$ the construction writes

$$\mathfrak{a} = (x)E\mathfrak{g}^6, \qquad \alpha_{\mathfrak{a}} = xm_E, \qquad (m_E)_S = E,$$

so $(\alpha_{\mathfrak{a}})_S = \mathfrak{a}\mathfrak{g}^{-6}$. Choose $y \in K^\times$ generating $\mathfrak{g}$ as an $\mathcal{O}_S$-ideal. If $(u)_S = \mathfrak{a}$, then

$$e = \alpha_{\mathfrak{a}} y^6/u \in \mathcal{O}_S^\times, \qquad [\alpha_{\mathfrak{a}}] = [eu] \quad \text{in } K^\times/K^{\times 6}.$$

There is a precision here: Appendix II allows choosing $\mathfrak{g}$ relative to a tested denominator. Fix one admissible global decomposition first. Its radicand gives the canonical character $\vartheta_{\mathfrak{a}}$ on all primes outside the finite set $S \cup \operatorname{supp}(\mathfrak{a}\mathfrak{g})$ where that choice applies. Uniqueness of Hecke characters, equivalently Chebotarev in the finite abelian quotient, then identifies the underlying primitive characters everywhere. With matching arithmetic-Frobenius conventions,

$$\kappa_u = \kappa_{e^{-1}}\vartheta_{\mathfrak{a}}. \tag{13}$$

If the residue-symbol convention is conjugated, conjugate the canonical family and its coefficient sequence; the same large sieve follows.

The group $\mathcal{O}_S^\times/(\mathcal{O}_S^\times)^6$ is finite. Thus (13) uses one of a finite set of fixed Kummer twists η, all unramified outside S and of bounded S-conductor. This is a comparison of primitive characters, not a pointwise multiplication rule for independently zero-extended arrays at bad primes. In particular, a conductor drop must not be replaced by a product of artificial zeros.

Since $\mathcal{O}$ is principal with six units, for fixed $\mathfrak{a} \in I(S)$ and an S-supported integral ideal $\mathfrak{b}$ there are at most six u satisfying $(u) = \mathfrak{a}\mathfrak{b}$. For any nonnegative row quantity, the literal moment is consequently at most

$$6 \sum_{\substack{\mathfrak{b}\ S\text{-supported}\\ N\mathfrak{b}\le H}} \sum_{\eta} \sum_{\substack{\mathfrak{a}\in I(S)\\ N\mathfrak{a}\le H/N\mathfrak{b}}} \left|L(s, \operatorname{prim}(\nu\eta\vartheta_{\mathfrak{a}}))\right|^2. \tag{14}$$

An ideal bound $X^{1+\varepsilon}T^{1+\varepsilon}$ therefore transfers to all literal rows, since

$$\sum_{\mathfrak{b}\ S\text{-supported}} (N\mathfrak{b})^{-1-\varepsilon} = \prod_{p\in S}(1-(Np)^{-1-\varepsilon})^{-1} < \infty.$$

No quotient of the original unit rows or deletion of S-prime powers has occurred.

S44.2 Fixed twists, exact tame conductors and the squarefree sieve

Fix $\xi = \nu\eta$ and write $\rho_{\mathfrak{a}} = \operatorname{prim}(\xi\vartheta_{\mathfrak{a}})$. Its fixed conductor is supported in S before the Kummer twist. On columns prime to S, $|\xi(\mathfrak{b})| = 1$. BGL Theorem 1.3 thus gives, for any fixed $\mathfrak{a}_1$,

$$\begin{aligned}\sum_{\substack{\mathfrak{a}_0\in I(S)\ \text{squarefree}\\ N\mathfrak{a}_0\le M}} \left|\sum_{\substack{\mathfrak{b}_1\in I(S)\ \text{squarefree}\\ N\mathfrak{b}_1\le B}} \lambda_{\mathfrak{b}_1}\xi(\mathfrak{b}_1)\vartheta_{\mathfrak{a}_1}(\mathfrak{b}_1)\vartheta_{\mathfrak{a}_0}(\mathfrak{b}_1)\right|^2 \\ \ll (MB)^{\varepsilon}\{M+B+(MB)^{2/3}\}\sum_{\mathfrak{b}_1}|\lambda_{\mathfrak{b}_1}|^2.\end{aligned} \tag{15}$$

The first fixed multiplier preserves the coefficient norm; the second has modulus at most one. Restricting the positive outer row sum is harmless. A coprimality condition with another fixed ideal

belongs to λ. In particular, ν need not have order dividing six and is not asserted to belong to BGL's finite ray-class character set.

There is a unique valuation decomposition

$$\mathfrak{a} = \mathfrak{a}_0\mathfrak{a}_1\mathfrak{a}_2^6, \quad \mathfrak{a}_0 \text{ squarefree}, \quad \mathfrak{a}_1 \text{ squarefull and sixth-power-free}, \quad (\mathfrak{a}_0, \mathfrak{a}_1) = 1.$$

Indeed, for $v_p(\mathfrak{a}) = 6q + r$, $0 \le r \le 5$, put the residual prime in $\mathfrak{a}_0$ if $r = 1$, and its exponent r in $\mathfrak{a}_1$ if $2 \le r \le 5$. BGL Lemma 2.1 gives $\vartheta_{\mathfrak{a}\mathfrak{b}^6} = \vartheta_{\mathfrak{a}}$ and $\vartheta_{\mathfrak{a}_0\mathfrak{a}_1} = \vartheta_{\mathfrak{a}_0}\vartheta_{\mathfrak{a}_1}$ on ideals prime to S in this sixth-power-free case. Those are the identities needed here; no arbitrary multiplicativity assertion is substituted.

At a prime outside S, tame Kummer inertia at residual exponent r has order $6/\gcd(6, r)$. It is nontrivial for $r = 1, \dots, 5$ and has conductor exponent one. The fixed ξ is unramified there, so cannot cancel that inertia. At primes in S the local sixth-power classes are finite, hence only finitely many primitive conductor types occur after multiplication by the fixed ξ. Consequently

$$Q_{\rho_{\mathfrak{a}}} = |D_K| N\mathfrak{f}_{\rho_{\mathfrak{a}}} \le C_{\xi,S} N(\mathfrak{a}_0\mathfrak{a}_1), \tag{16}$$

independently of $\mathfrak{a}_2$. The equality of the conductor's outside-S part with $\mathrm{rad}(\mathfrak{a}_0\mathfrak{a}_1)$ does not identify that conductor with the original larger deleted set.

S44.3 Removing the row-dependent functional-equation weights

First take finite-order ν. All characters then have trivial infinity type at the complex place. The primitive completion may be normalized as

$$\Lambda(s, \rho) = Q_\rho^{s/2}\gamma_K(s)L(s, \rho), \qquad \gamma_K(s) = 2(2\pi)^{-s}\Gamma(s), \qquad \Lambda(s, \rho) = \epsilon_\rho\Lambda(1 - s, \bar{\rho}), \quad |\epsilon_\rho| = 1.$$

For nonprincipal rows the completed function is entire. Apply this functional equation to the Mellin integral with even auxiliary factor $G(z) = e^{z^2}$ and denominator z, shifting across $z = 0$. At $s = 1/2 + it$ this gives two weighted Dirichlet series, truncated below to polynomials; the dual prefactor has modulus one. Principal rows are excluded from this pole-free step and treated separately below.

For fixed $\mathfrak{a}_2$, let $X = H/(N\mathfrak{a}_2)^6 \ge 1$ and $T = 1 + |t|$. An explicit normalized kernel is

$$V_t(x) = \frac{1}{2\pi i}\int_{(\eta)} e^{z^2}\frac{\Gamma(s + z)}{\Gamma(s)}T^{-z}x^{-z}\,\frac{dz}{z}, \qquad x = \frac{2\pi N\mathfrak{b}}{\sqrt{Q_\rho}T}, \tag{17}$$

with $\eta > 0$ small. Its factors combine to $(\sqrt{Q_\rho}/(2\pi N\mathfrak{b}))^z$ in the actual completed-L kernel; there is no extra factor of T. The denominator z supplies the residue at zero. On $z = \eta + iy$, Stirling and $|t + y| \le T(1 + |y|)$ give

$$\left|\frac{\Gamma(1/2 + \eta + i(t + y))}{\Gamma(1/2 + it)T^{\eta+iy}}\right| \ll_\eta (1 + |y|)^{C_\eta}e^{\pi|y|/2}.$$

The factor e^{-y^2} makes this an integrable majorant uniformly in t. The analogous bound holds on every fixed positive shifted line and for the dual gamma kernel. Moving far enough to the right gives arbitrary power decay beyond a common total-ideal cutoff

$$B' = X^{1/2}T(HT)^\eta, \tag{18}$$

up to fixed ξ, S, K constants. For any requested $A > 0$, a sufficiently large positive kernel shift, depending also on η, makes the truncation error $O((HT)^{-A})$ uniformly in all rows in this block.

Ideal counting $O_K(Y)$ bounds the absolute tails. Such errors are negligible after the finite row sums by taking A larger if needed.

Insert (17) on $\operatorname{Re} z = \eta$. The factor $(\sqrt{Q_\rho}T)^z$ is independent of the column and is a row multiplier of modulus at most $(CXT^2)^{\eta/2}$. Its imaginary part is a phase. The remaining column coefficient is $(N\mathfrak{b})^{-1/2-\eta-it-iy}$, apart from fixed constants. Minkowski or Cauchy with the integrable gamma majorant therefore removes the row-dependent weight at cost $(HT)^{O(\eta)}$ in the squared norm. No independence of the root numbers is assumed or needed.

For each column use the unique decomposition

$$\mathfrak{b} = \mathfrak{b}_S\mathfrak{b}_1\mathfrak{b}_2, \quad \mathfrak{b}_S \ S\text{-supported}, \quad \mathfrak{b}_1 \text{ squarefree}, \quad \mathfrak{b}_2 \text{ squarefull}, \quad (\mathfrak{b}_1, \mathfrak{b}_2) = 1,$$

where $\mathfrak{b}_1, \mathfrak{b}_2 \in I(S)$; a prime of outside-$S$ valuation one goes in $\mathfrak{b}_1$, and a prime of valuation at least two goes with its full exponent in $\mathfrak{b}_2$. For fixed $\mathfrak{b}_S, \mathfrak{b}_2$ retain $\rho_{\mathfrak{a}}(\mathfrak{b}_S)$ as the original primitive row value. On the good factors the exact canonical identity applies. The remaining factors involving $\mathfrak{b}_2$, including $\vartheta_{\mathfrak{a}_0}(\mathfrak{b}_2)$, are bounded row multipliers. Their moduli are at most one; only then are they discarded in an inequality. In particular the value at $\mathfrak{b}_S$ is not replaced by the product of zero-extended primitive arrays.

For the $\mathfrak{b}_1$ polynomial the coefficient sequence is independent of $\mathfrak{a}_0$; it includes the indicator $(\mathfrak{b}_1, \mathfrak{b}_2) = 1$ and the common cutoff $N\mathfrak{b}_1 \le B'/N(\mathfrak{b}_S\mathfrak{b}_2)$. Padding it by zeros up to B' is legitimate. Its squared norm is at most $\zeta_K(1+2\eta)$. Minkowski over $\mathfrak{b}_S, \mathfrak{b}_2$ is legitimate because

$$\sum_{\mathfrak{b}_S, \mathfrak{b}_2} N(\mathfrak{b}_S\mathfrak{b}_2)^{-1/2-\eta} < \infty.$$

The finite Euler product handles $\mathfrak{b}_S$; the squarefull series converges above $1/2$. Apply (15) to each such polynomial and to the conjugate dual family. For fixed $\mathfrak{a}_1, \mathfrak{a}_2$ the resulting energy is

$$(HT)^{O(\eta)}\{M + B' + (MB')^{2/3}\}, \qquad M = X/N\mathfrak{a}_1. \tag{19}$$

The additional condition $(\mathfrak{a}_0, \mathfrak{a}_1) = 1$ simply restricts the positive row sum. The large-sieve small exponent can be chosen sufficiently small along with η and absorbed in the indicated small loss.

S44.4 All-ideal summation and principal rows

Every squarefull ideal $\mathfrak{c}$ has a unique expression $\mathfrak{d}^2\mathfrak{e}^3$ with $\mathfrak{e}$ squarefree. This follows primewise by using exponent zero or one in $\mathfrak{e}$ according to the parity of an exponent at least two. Therefore

$$\#\{\mathfrak{c} \text{ squarefull} : N\mathfrak{c} \le X\} \ll_K X^{1/2}, \qquad \sum_{\mathfrak{c} \text{ squarefull}} (N\mathfrak{c})^{-s} < \infty \quad (s > 1/2).$$

For the count, sum $O((X/(N\mathfrak{e})^3)^{1/2})$ and use $\zeta_K(3/2) < \infty$. Summing (19) over $\mathfrak{a}_1$ gives

$$(HT)^{O(\eta)}\{X + B'\sqrt{X} + (XB')^{2/3}\} \ll (HT)^{O(\eta)}XT,$$

since the squarefull series converges at 1 and $2/3$, $B' = \sqrt{X}\,T(HT)^\eta$, and $T^{2/3} \le T$ for $T \ge 1$. The $\mathfrak{a}_2$ sum is bounded by $H\sum_{\mathfrak{a}_2}(N\mathfrak{a}_2)^{-6} = H\zeta_K(6)$. Choosing all internal losses small for the requested final ε proves the fixed-twist all-ideal central moment.

There are $O_K(H)$ ideal rows in total. On any row where the primitive character is principal its L-function is ζ_K, and ordinary scalar convexity gives $|\zeta_K(1/2+it)|^2 \ll_\varepsilon T^{1+\varepsilon}$. This treats those rows without a pole-free approximate functional equation. Finally (14) proves (8) for every actual nonzero Eisenstein element.

S44.5 Poisson transfer with an explicit constant

For a nonprincipal row define

$$f_\psi(s) = \frac{L(s,\psi)}{(s+1)^{5/4}},$$

using the holomorphic logarithm on $\mathrm{Re}(s+1) > 0$. On a principal row additionally multiply by $(s-1)/(s+1)$. Analytic continuation and ordinary degree-two convexity, used separately at each fixed conductor, show that f_ψ is bounded analytic on the right half-plane $\mathrm{Re}\, s > 1/2$, with continuous boundary and decay at infinity. The principal pole is removed before applying Poisson representation to its real and imaginary parts. Jensen gives

$$|f_\psi(1/2+\delta+it)|^2 \le \int_{\mathbb{R}} P_\delta(t-v)|f_\psi(1/2+iv)|^2\,dv.$$

The exact row family is finite for fixed H, so its sum interchanges with the integral without a limiting-family issue. The boundary pole-removal factor has modulus at most one.

Take $0 < \varepsilon_0 \le 1/4$ and $\alpha = 3/2 - \varepsilon_0 \in [5/4, 3/2)$. Since $|3/2 + iv| \ge (1+|v|)/\sqrt{2}$, the central squared moment of the f_ψ is bounded by $3C_0H^{1+\varepsilon_0}w_\alpha(v)$, where $w_\alpha(v) = (1+|v|)^{-\alpha}$. For $|t| \ge 2$, split $|v-t| \le |t|/2$ from its complement. The near contribution to $P_\delta * w_\alpha$ is at most $2^\alpha w_\alpha(t) < 3w_\alpha(t)$. In the far region use $P_\delta(t-v) \le 4\delta/(\pi t^2)$ and $\int w_\alpha = 2/(\alpha-1)$, giving at most

$$\frac{2\delta 3^\alpha}{\pi(\alpha-1)} w_\alpha(t) \le 8w_\alpha(t).$$

Here $(1+|t|)^\alpha/t^2 \le 3^\alpha/4$ for $|t| \ge 2$. For $|t| \le 2$, the convolution is at most one and hence at most $3^\alpha w_\alpha(t) < 6w_\alpha(t)$. Thus $P_\delta * w_\alpha \le 11 w_\alpha$ uniformly for $0 < \delta < 1/2$. Restoring the denominator costs less than $6(1+|t|)^{5/2}$, and the inverse principal factor is bounded by $(3/2+\delta)/(1/2-\delta)$. The product $11 \cdot 3 \cdot 6 = 198 < 200$ proves (9). The constant is uniform as $\delta \downarrow 0$, but not as the contour approaches the principal pole line $\mathrm{Re}\, s = 1$. No zero-free hypothesis appears.

S44.6 The separately scoped unitary infinity type

Let $\nu(\mathfrak{a}) = \nu_0(\mathfrak{a})(N\mathfrak{a})^{-i\tau}$ be fixed and unitary, with angular parameter m and gamma factor γ_ν stated in Section S43. The external primitive functional equation is explicitly

$$Q_\rho^{s/2}\gamma_\nu(s)L(s,\rho) = \epsilon_\rho Q_\rho^{(1-s)/2}\gamma_{\bar\nu}(1-s)L(1-s,\bar\rho), \qquad |\epsilon_\rho| = 1.$$

A constant norm-twist phase in another completion convention is absorbed in ϵ_ρ. All finite Kummer twists share this infinity type. Replace T in the kernel by $T_\nu = 1 + |t+\tau| + |m|$. The numerator gamma ratio now has real base $1/2 + |m|/2 > 0$ and vertical parameter $t+\tau$; its conjugate dual has parameter $-(t+\tau)$. Stirling gives the same integrable Gaussian-majorized kernel on each fixed shifted line. Constants can depend on m, τ; T_ν is comparable to T for this one fixed ν. The finite-prime conductor analysis, modulus-one coefficient twist, all-ideal decomposition and all original primitive row values remain unchanged.

A pure norm row has $L(s,\rho) = \zeta_K(s+i\tau)$; exclude it from the pole-free approximate functional equation and use scalar convexity at $t+\tau$. In the Poisson step cancel its actual pole by

$$h_\tau(s) = \frac{s-1+i\tau}{s+1+i\tau}.$$

This factor is analytic in the relevant right half-plane. On its boundary $|h_\tau| \le 1$, while inside $|h_\tau^{-1}| \le (3/2+\delta)/(1/2-\delta)$. Hence the same transfer factor 200 applies to the now ν-dependent

central constant. If the opposite norm convention is chosen, replace τ by $-\tau$ in the gamma factor and pole alike. This proves the separate fixed-unitary moment assertion; it does not claim uniformity in a family of infinity types or replace the finite-order principal-residue expression below.

S44.7 Keeping the full type-I mask during transfer

For $\sigma = 1/2 + \delta$, define L^T by removing every prime of $T(u,d)$, not just conductor primes. The identity (10) is equivalently

$$F_{u,d}(s) = \frac{L(s,\psi_u)}{L^{T(u,d)}(2s,\psi_{u,2})} \prod_{p\in T(u,d),\ p\nmid \mathfrak{f}_u} (1-\psi_u(p)(Np)^{-s}).$$

Absolute convergence at $2\sigma > 1$ bounds the reciprocal incomplete product by $\zeta_K(2\sigma)$, without requiring any zero-free theorem in the critical strip. For every fixed $\eta > 0$,

$$\prod_{p\in T}(1+(Np)^{-\sigma}) \ll_{\sigma,\eta} (N\operatorname{rad} T)^{\eta}, \qquad N\operatorname{rad} T \le C_S Nu\, Nd.$$

Indeed $1 + q^{-\sigma} \le q^{\eta}$ beyond a fixed threshold, and only finitely many prime ideals lie below it. Therefore

$$|F_{u,d}(\sigma+it)| \ll_{\nu,S,\delta,\eta} (Nu\,Nd)^{\eta}|L(\sigma+it,\psi_u)|. \tag{20}$$

This is a bounded inequality after the exact mask identity, not an identity that removes original zeros. For example, with principal ν and $u = \pi^3$ or π^6 at $N\pi = 7$, the primitive local ratio, restoration and full masked factor at $s = 1$ are respectively

$v_\pi(u)$	primitive ratio	restoration	masked factor
3	48/49	49/48	1
6	8/7	7/8	1

The squared primitive character in the first row is unramified at π although the original primitive character is ramified there. Squaring a zero sentinel would give the wrong denominator.

For finite-order ν, the normalized type-I piece is the retained d sum of the smoothed squarefree series at $Y = D/Nd$, divided by $\sqrt{D}$. Its possible pole is only at $s = 1$, on principal rows. Contour shifting with the full mask gives $I_u = P_u + E_u$, with E_u exactly (11) and

$$P_u = \mathbf{1}_{\psi_u=1}\sqrt{D}\,\widehat{W}(1) \sum_{\substack{d \text{ squarefree}\\(d,S)=1}} \frac{\beta_U(d)\nu(d)\chi_d(u)}{Nd}\rho_{T(u,d)},$$

$$\rho_T = \frac{\operatorname{res}_{s=1}\zeta_K(s)}{\zeta_K(2)} \prod_{p\in T}(1+(Np)^{-1})^{-1}.$$

This does not impose positivity on P_u. Since β_U is supported on $Nd \le U^2 \le D$, every inner scale Y is at least one. Four integrations by parts give $|\widehat{W}(\sigma+it)| \ll_\sigma \mathcal{M}_4(W)(1+|t|)^{-4}$, which justifies the shifted integrals and their horizontal limits with the classical Hecke growth estimates.

Apply Minkowski to (11) in ℓ^2 of the actual u rows, then use (20) and the off-central moment. For sufficiently small internal losses this bounds $\|E\|_2$ by

$$CD^{\delta}H^{1/2+O(\eta)} \sum_d \frac{|\beta_U(d)|}{(Nd)^{\sigma-\eta}} \int_{\mathbb{R}} |\widehat{W}(\sigma+it)|(1+|t|)^{(1+O(\eta))/2}\,dt.$$

All original $\nu(d)\chi_d(u)$ have modulus at most one and remain inside the exact expression before this inequality. The actual truncated factorization gives, without replacing β_U by arbitrary coefficients,

$$\sum_d \frac{|\beta_U(d)|}{(Nd)^{\sigma-\eta}} \le \left(\sum_{Na\le U} (Na)^{-\sigma+\eta}\right)^2 \ll_{\delta,\eta} U^{1-2\delta+2\eta}.$$

Choose $\eta < \delta$ if necessary and use the fixed-field ideal count. The height integral is bounded by $C\mathcal{M}_4(W)$. Squaring and absorbing the small losses in $(HD)^{\varepsilon}$, using $U^2 \le D$, proves (12). Thus the formerly conditional numerator moment input in this residual transfer is discharged, while its positive-power cutoff loss is retained explicitly.

S44.8 What remains open and what has been verified

The new remainder envelope improves the old scalar-convexity H power $3/2 - \delta + \varepsilon$ to $1 + \varepsilon$. At fixed U one may choose δ for a requested small-loss exponent; for $U = D^a$ with fixed $a > 0$, the remaining excess tends to $2a$ as $\delta \downarrow 0$. Neither statement controls P_u, the other signed piece, nor their covariance. In the original Vaughan identity the short $2a_U$ term must be restored at smaller scales where its support can meet the annulus. Uniform $\mathcal{M}_4$ bounds do not justify interchanging a profile supremum with the row moment or assuming $U^2 \le D'$ after a scale D' is reduced.

The independent prose audit is recorded in `planning/analytic-oracle/arithmetic-review/adversarial-afm-review-20261104.md`. The source-hashed finite checker tests valuation/support decompositions, squarefull coordinates, height exponents and rational Poisson envelopes. Its finite success does not prove reciprocity, analytic continuation, the large sieve, the approximate functional equation or the moments. No new Lean declarations accompany these analytic theorems. Existing Lean local Euler identities are a separately limited verification layer. The independent positive arithmetic determinant comparison, original signed low-gcd short-row estimate, all required contour uniformities and GRH remain open; Dirichlet, Hecke, Dedekind and general Artin scopes are not identified with one another.

Pinned primary sources and external inputs. The principal analytic family input is Blomer–Goldmakher–Louvel, arXiv:1112.1650v1, Theorem 1.3, Lemma 2.1, Appendix II and Section 5. Our fixed-ν proof changes column coefficients and redoes the functional-equation reduction; it does not claim the published corollary directly states every fixed twist. For comparison, Gao–Zhao, arXiv:2201.01885v2, Lemma 2.9 and Section 3.2, uses the sextic large-sieve/moment interface. Its GRH-conditional central-density theorem is not invoked. Abelian Kummer identification uses the classical reciprocity theorem, with the project's pinned Milne *Class Field Theory*, version 4.03 (6 August 2020), Theorem 0.8 and Chapter I, Section 1, a reference: Milne's notes. The primitive degree-one Hecke continuation and completed functional equation are explicit classical external inputs; the project's pinned Kudla notes on Tate's thesis, Theorem 4.3 and Corollary 4.4, supply a reference. Their sketch and incomplete displayed epsilon formulas are not used as independently certified constants. Standard Stirling, Mellin inversion, Poisson representation and scalar convexity complete the external analytic dependencies. None is replaced by a finite diagnostic or a GRH-equivalent assumption.

S45 Recombining the actual weighted cutoff

Keep the finite-order Eisenstein family and original masks of Section S43. For $U \geq 1$ and a squarefree ideal m, put

$$A_U(m) = \sum_{d|m} \beta_U(d) = \sum_{\substack{ab|m \\ (a,b)=1,\ Na,Nb\leq U}} \mu(a)\mu(b).$$

Every summation variable here is an integral ideal. The last formula uses squarefreeness of m; an unconstrained square of a Möbius polynomial would also include forbidden overlapping supports.

Theorem S45.1 (Exact recombination and an absolute-half-plane rate)**.** *Write $T_0 = S \cup \operatorname{supp}(u)$ and $J_{u,U}(s) = \sum_{d \text{ squarefree},\ (d,S)=1} \beta_U(d)\nu(d)\chi_d(u)(Nd)^{-s}F_{u,d}(s)$. For $\operatorname{Re} s > 1$,*

$$J_{u,U}(s) = \sum_{m \text{ squarefree},\ (m,T_0)=1} A_U(m)\psi_u(m)(Nm)^{-s},$$
$$|A_U(m)| \leq 3^{\omega(m)}, \qquad A_U(m) = \mu(m) \quad (Nm \leq U).$$

For every $\kappa > 0$ and $0 < \epsilon < \kappa$, uniformly for $\operatorname{Re} s \geq 1 + \kappa$, all original rows and all $U \geq 1$,

$$\left| J_{u,U}(s) - \frac{1}{L^{T_0}(s,\psi_u)} \right| \leq U^{-\kappa+\epsilon}\{\zeta_K(1+\epsilon)^3 + \zeta_K(1+\epsilon)\}.$$

Here L^{T_0} removes every prime of T_0, including conductor-drop primes; the bound is in the absolute-convergence half-plane only.

Proof. On a surviving term $m = dc$, the two squarefree ideals c, d are coprime, and both avoid T_0. The original product $\nu(d)\chi_d(u)\psi_u(c)$ is then $\psi_u(m)$. Grouping by m gives A_U. Each prime of m has three possible positions in $(a, b, m/(ab))$, so the absolute sum is at most $3^{\omega(m)}$. When $Nm \leq U$ all divisors fit both cutoffs; the primewise sum is $1 - 1 - 1 = -1$, giving $\mu(m)$.

The same bound justifies absolute convergence and regrouping for $\operatorname{Re} s > 1$, because on squarefree ideals $3^{\omega(m)} = \tau_3(m)$ and $\sum_m \tau_3(m)(Nm)^{-v} = \zeta_K(v)^3$ for $v > 1$, where this latter sum runs over all nonzero integral ideals. Subtract the absolutely convergent Möbius series for $1/L^{T_0}$. Only $Nm > U$ remain. Rankin's inequality with exponent $\eta = \kappa - \epsilon$ bounds their absolute sum by

$$U^{-\eta} \sum_m \{\tau_3(m) + 1\}(Nm)^{-1-\epsilon},$$

which proves the stated constant independently of deleted primes, row multiplicities or character phases. □

For W supported in $[1, 2]$, the same exact regrouping gives the original normalized type-I polynomial

$$I_u = D^{-1/2} \sum_{m \text{ squarefree},\ (m,S)=1} A_U(m)\nu(m)\chi_m(u)W(Nm/D).$$

This formula retains all original zero sentinels. It permits a direct family large sieve before any absolute d sum; its quantitative moment is developed separately below. The displayed absolute-half-plane rate alone is not that critical-strip moment.

S46 Euler packaging, coprimality and the analytic boundary

For $p \notin T_0$, set $z_p = \psi_u(p)(Np)^{-s}$ and $t_p = z_p/(1+z_p)$. On $\operatorname{Re} s > 1$, deleting an additional surviving prime divides the squarefree Euler product by $1+z_p$. Consequently the exact retained expression is

$$J_{u,U}(s) = F_{u,1}(s)B_{u,U}(s), \qquad B_{u,U}(s) = \sum_{d \text{ squarefree},\ (d,T_0)=1} \beta_U(d) \prod_{p \mid d} t_p.$$

Original zeros enforce $(d,u)=1$ even when a sixth power drops from its primitive conductor. This factorization cannot restore a forbidden d row by replacing a zero sentinel with an unramified primitive value. The equality continues wherever both sides are holomorphic.

Without the cutoffs, $\beta_\infty(d) = \mu(d)2^{\omega(d)}$ on squarefree ideals. Absolute convergence justifies

$$B_{u,\infty} = \prod_{p \notin T_0} (1-2t_p) = \prod_{p \notin T_0} \frac{1-z_p}{1+z_p}, \qquad F_{u,1}B_{u,\infty} = \prod_{p \notin T_0} (1-z_p) = \frac{1}{L^{T_0}(s,\psi_u)}.$$

Theorem S45.1 gives a quantitative rate rather than an unsupported exchange of this limit with a critical-strip contour.

A local counterexample to the naive square. In a one-prime projection at a good Eisenstein prime of norm 7, take the principal local phase and $s=1$. Then $z=1/7$, $t=1/8$ and, for $U \geq 7$, the exact polynomial is $B_U = 1-2t = 3/4$. The unconstrained square $(1-t)^2 = 49/64$ differs by $1/64$: it admits the forbidden pair $a=b=p$. After multiplying by $F = 1+z = 8/7$ it gives $7/8$ instead of the correct $6/7$. This is an actual local Euler calculation, not an infinite-series or mean-square counterexample. It identifies the exact coprimality condition that a proposed mollifier-square simplification must preserve.

The exact intermediate coefficient band. For a squarefree m with $U < Nm \leq U^2$, one has $A_U(m) = -\mu(m)$. Indeed the unrestricted disjoint-assignment sum is $\mu(m)$. If $Na > U$, every $b \mid m/a$ automatically has $Nb \leq Nm/Na < U$, and $\sum_{b \mid m/a} \mu(b)$ vanishes unless $a = m$. Hence the excluded a assignments contribute exactly $\mu(m)$; the excluded b assignments contribute another $\mu(m)$. They cannot overlap because $NaNb > U^2 \geq Nm$ contradicts $ab \mid m$. Subtraction proves the claim. Together with the small-norm band, this gives $A_U = \mu$ for $Nm \leq U$ and $A_U = -\mu$ for $U < Nm \leq U^2$. The original smoothed columns have $Nm \geq D \geq U^2$, so the new band does not justify replacing their coefficients by $-\mu(m)$: generally they lie beyond it.

Theorem S46.1 (A stronger cutoff convergence would require zero-freeness)**.** *Fix a nonprincipal primitive ψ_u. If these actual $J_{u,U}$ converge locally uniformly on $\operatorname{Re} s > 1/2$ as $U \longrightarrow \infty$, then $L(s,\psi_u)$ has no zero there.*

Proof. Classical abelian Hecke continuation makes the numerator of $F_{u,1} = L^{T_0}(s,\psi_u)/L^{T_0}(2s,\psi_{u,2})$ holomorphic. Its denominator is nonzero on $\operatorname{Re} s > 1/2$ by the absolute Euler product at $2s$. Each $B_{u,U}$ is a finite sum with nonzero denominators $1+z_p$ on $\operatorname{Re} s > 0$. Thus every $J_{u,U}$ is holomorphic on the asserted half-plane. A locally uniform limit G is holomorphic there. On $\operatorname{Re} s > 1$, the already proved convergence gives $L^{T_0}(s,\psi_u)G(s) = 1$. The identity theorem extends this equality to the connected half-plane, precluding a zero of L^{T_0}. The finitely deleted factors $1-\psi_u(p)(Np)^{-s}$ do not vanish when $\operatorname{Re} s > 0$, so the same conclusion holds for the primitive L-function. □

This is an implication, not a proof that the stronger convergence holds, and it does not assert a converse. In particular, a family-averaged energy bound is not silently promoted to locally uniform convergence for each individual character. Principal rows have a pole at 1 and were excluded from this proposition. General Artin holomorphy is not an input.

Reproducible exact diagnostics. The new source `examples/arithmetic_cutoff/exact_recombination.py` uses actual split/inert prime ideals and raw sextic symbols in $\mathbb{Z}[\omega]$, with exact rational arithmetic. It checks the direct masked sum, coefficient regrouping, Euler packaging, both finite full-cutoff limits and original zeros, including third/sixth-power rows. Its overlapping- support negative controls are not tests of an infinite large sieve or of zero location. The analytic convergence argument above uses ordinary ideal Euler products and the explicitly stated Hecke continuation theorem; no new Lean declaration or complete GRH proof is claimed.

S47 A cutoff-uniform envelope for the original masked remainder

Fix $K = \mathbb{Q}(\omega)$, $\mathcal{O} = \mathbb{Z}[\omega]$, a finite-order Hecke character ν, and a finite set S containing the primes above 6 and the conductor of ν. All unspecified ideal sums below are over nonzero integral ideals. Let $\chi_m(u) = (u/m)_6$ be the original symbol, zero when $(m, u) > 1$, and put

$$\beta_U(d) = \sum_{\substack{ab=d \\ Na, Nb \le U}} \mu(a)\mu(b), \qquad A_U(m) = \sum_{d \mid m} \beta_U(d).$$

The columns d, c, m are squarefree and prime to S where indicated. Retain the primitive characters $\psi_u = \mathrm{prim}(\nu\kappa_u)$, $\psi_{u,2} = \mathrm{prim}(\psi_u^2)$, and the full deleted set $T(u, d) = S \cup \mathrm{supp}(u) \cup \mathrm{supp}(d)$. For $H, D, U \ge 1$, $U^2 \le D$, and $W \in C_c^\infty((0, \infty))$ supported in $[1, 2]$, the original normalized type-I piece is

$$I_u = D^{-1/2} \sum_{\substack{m \text{ squarefree} \\ (m,S)=1}} A_U(m)\nu(m)\chi_m(u) W(Nm/D). \tag{21}$$

Its exact principal residue is

$$P_u = \mathbf{1}_{\psi_u=1} \sqrt{D}\, \widehat{W}(1) \sum_{\substack{d \text{ squarefree} \\ (d,S)=1}} \frac{\beta_U(d)\nu(d)\chi_d(u)}{Nd} \rho_{T(u,d)},$$

$$\rho_T = \frac{\mathrm{res}_{s=1}\, \zeta_K(s)}{\zeta_K(2)} \prod_{p \in T} (1 + (Np)^{-1})^{-1}. \tag{22}$$

Define $E_u = I_u - P_u$; this is the same residue-subtracted remainder as the fully masked Mellin contour, with no cutoff limit or profile supremum.

Theorem S47.1 (Uniformity in the actual truncated cutoff)**.** *For every $\varepsilon > 0$,*

$$\sum_{0 < Nu \le H} |E_u|^2 \ll_{\nu,S,\varepsilon} \|W\|_\infty^2 (HD)^\varepsilon \{H + DH^{1/3} + (HD)^{2/3}\}. \tag{23}$$

The sum is over every actual nonzero Eisenstein element, including units, S-supported powers and sixth-power rows. The constant is independent of U under $U^2 \le D$. This is an application of the classical sextic large sieve with exact original coefficients, not a zero-location theorem.

Proof. On an absolutely convergent line, expand the restored squarefree series $F_{u,d}$. Its condition $(c, T(u,d)) = 1$ includes $(c,d) = 1$, and the original symbols multiply. Grouping $m = cd$ gives (21) by Mellin inversion. Each prime of m may belong to the first truncated Möbius divisor, to the second, or to neither. Consequently $|A_U(m)| \le 3^{\omega(m)} \ll_\eta (Nm)^\eta$, uniformly in U. The length-D coefficient sequence in (21) therefore has squared norm $\ll_\eta \|W\|_\infty^2 D^\eta$.

The all-ideal row norm in Section 3 of Blomer–Goldmakher–Louvel, arXiv:1112.1650v1 gives

$$B_2^1(X,D) \ll_\eta (XD)^\eta \{X + DX^{1/3} + (XD)^{2/3}\}.$$

For completeness, split $a = a_1 a_2^2 \cdots a_5^5 a_6^6$ dyadically; $a_1, \dots, a_5$ are squarefree and pairwise coprime. If their dyadic sizes are A_j, put $R = \prod_{j=1}^6 A_j$ and $M = \max(A_1, A_2)$. Pivot on the first or second variable, using the squarefree powered sieve proved simultaneously in that paper's Section 3.2. The other variables are bounded column multipliers. The block energy is at most

$$(XD)^\eta \{R + DR/M + D^{2/3}R/M^{1/3}\} \sum_m |\lambda_m|^2.$$

The inequalities $R \le X$, $R^3 \le XM^3$ and $R^3 \le X^2 M$ follow from $\prod_j A_j^j \le X$ and the pivot choice. Summing the logarithmically many blocks proves the displayed bound.

Finite S-unit and ray-class partitions convert the original literal symbol into the canonical column value $\vartheta_m(a)$ with fixed modulus-one coefficient multipliers. On noncoprime pairs both values are zero: squarefree m has outside-S conductor exactly m. This orientation preserves sixth-power zeros. At most six elements generate each outside-S ideal times an S-supported ideal; the S-supported sums converge at powers 1, 1/3 and 2/3. Thus (23) holds first with I_u in place of E_u.

A principal ψ_u requires every outside-S valuation of u to be divisible by 6. There are $O_{\nu,S}(H^{1/6})$ such literal rows. Equation (22), $|\rho_T| \ll_K 1$, and the actual coefficient sum

$$\sum_d \frac{|\beta_U(d)|}{Nd} \le \left(\sum_{Na \le U} \frac{1}{Na} \right)^2 \ll \log^2(2U)$$

give $\sum |P_u|^2 \ll DH^{1/6} \|W\|_\infty^2 \log^4(2U)$. This is absorbed by the right side of (23), using $U^2 \le D$ and a smaller internal exponent. Finally $|E_u|^2 \le 2|I_u|^2 + 2|P_u|^2$. The extended proof specifies the canonical-family transfer and the powered-sieve input precisely. □

Combining this with the previously proved fixed-twist AFM transfer yields

$$\sum_{0 < Nu \le H} |E_u|^2 \ll_{\nu,S,\delta,\varepsilon} \mathcal{M}_4(W)^2 (HD)^\varepsilon \min\{D^{2\delta} U^{2-4\delta} H,\ H + DH^{1/3} + (HD)^{2/3}\}. \tag{24}$$

At $H = D^{3/4}$ and $U = D^a$, $0 \le a \le 1/2$, the excess over H is at most $\min\{2\delta + (2-4\delta)a, 1/2\}$, up to arbitrarily small losses. The new envelope improves the old one for $a > 1/4$ as $\delta \downarrow 0$, but still leaves an exponent loss of 1/2. It does not settle the original signed short-row bound, type-II covariance, positive arithmetic determinant comparison or GRH. The principal-residue bound used here is sufficient for this envelope, not the stronger target needed to reassemble the full signed argument.

S48 All-ideal reciprocity and the surviving block obstruction

The proof uses no new zero-free input. Its external analytic ingredient is the pinned BGL large sieve and the powered estimates established in its proof; elementary ideal counting and the already restored Mellin identity supply the remaining interfaces. All constants may depend on fixed K, S, ν and declared small exponents, never on U.

S48.1 Exact recombination retains the primitive corrections

On $\operatorname{Re} s > 1$, start from the full defining series, with its original zero masks:

$$F_{u,d}(s) = \sum_{\substack{c \text{ squarefree} \\ (c,T(u,d))=1}} \nu(c)\chi_c(u)(Nc)^{-s}.$$

Its primitive ratio involves $L(s,\psi_u)$, the independently primitive $L(2s,\psi_{u,2})$ and every deleted Euler factor. Expanding this *restored* series gives the exact equality

$$\begin{aligned}\sum_{\substack{d \text{ squarefree} \\ (d,S)=1}} &\beta_U(d)\nu(d)\chi_d(u)(Nd)^{-s}F_{u,d}(s) \\ &= \sum_{\substack{m \text{ squarefree} \\ (m,S)=1}} A_U(m)\nu(m)\chi_m(u)(Nm)^{-s}. \end{aligned} \tag{25}$$

Indeed $(c,d)=1$ makes $m=cd$ squarefree, with $d \mid m$. If m meets u, both original symbols retain their zero. Only on the absolutely convergent line is this infinite series used. Mellin inversion there gives the finite smoothed polynomial (21). Nothing evaluates (25) as a Dirichlet series on $\operatorname{Re} s = 1/2+\delta$ or interchanges a cutoff limit with continuation across unknown zeros.

For squarefree m, a factorization $ab = d \mid m$ assigns each prime to a, to b, or to neither. The actual inequalities $Na, Nb \le U$ select a subset of these assignments. Thus

$$|A_U(m)| \le 3^{\omega(m)} \ll_\eta (Nm)^\eta.$$

The last divisor bound follows by choosing a fixed threshold so $3 \le (Np)^\eta$ beyond it, then absorbing the finitely many smaller prime ideals in a constant. When $Nm \le U$ all assignments occur, so $A_U(m) = \prod_{p\mid m}(1-1-1) = \mu(m)$; no such simplification is made for the length-D columns in general. With $\lambda_m = D^{-1/2}A_U(m)\nu(m)W(Nm/D)$, ideal counting and a smaller internal η give

$$\sum_m |\lambda_m|^2 \ll_\eta \|W\|_\infty^2 D^\eta, \qquad \operatorname{supp}\lambda \subseteq \{D \le Nm \le 2D\}. \tag{26}$$

S48.2 The powered squarefree input and all-ideal pivot

Let ϑ_a be BGL's canonical primitive sixth-order family. Their Section 3 defines B_1^j using squarefree row/column ideals and kernel $\vartheta_a(m)^j$, while B_2^j uses *all* ideal rows and kernel $\vartheta_m(a)^j$. Theorem 1.3 states the squarefree bound for $j=1$; its Section 3.2 proves the estimates simultaneously for $1 \le j \le 5$, because the recursion in Lemma 3.4 uses $j\ell$. In particular the two inputs needed here are

$$B_1^j(M,D) \ll_\eta (MD)^\eta\{M + D + (MD)^{2/3}\}, \qquad j = 1,2. \tag{27}$$

To see the final step in the published recursion explicitly, its $E_{4/3}$ estimate is $M^{4/3} + D + (MD)^{2/3}$. If $M \le D$, its first term is at most $(MD)^{2/3}$; if $M > D$, use reciprocity/duality (3.2) and the estimate with M, D reversed. The finite set of powers is treated together; no inference from a fixed conductor count to an arbitrary primitive-character sieve occurs.

Write each ideal row uniquely as

$$a = a_1 a_2^2 a_3^3 a_4^4 a_5^5 a_6^6,$$

where $a_1, \dots, a_5$ are pairwise-coprime squarefree ideals and a_6 is arbitrary. For valuation $6q+r$, $0 \le r \le 5$, place the prime in a_r if $r > 0$ and its power q in a_6. Dyadically let $A_j \le Na_j < 2A_j$,

with $A_j \ge 1$, and put $R = \prod_{j=1}^{6} A_j$. A nonempty block satisfies $\prod_j A_j^j \le X$ when $Na \le X$. Choose $\ell = 1$ if $A_1 \ge A_2$, and $\ell = 2$ otherwise; put $M = A_\ell$. Keep every other ideal fixed. Evaluation of the single *column* character gives

$$\vartheta_m(a) = \vartheta_m(a_\ell)^\ell \vartheta_m(a/a_\ell^\ell).$$

The second factor is a bounded column multiplier, including its original zero if m meets any fixed factor. It may depend on all the fixed ideals but not on the pivot row. Pairwise-coprimality and the exact row norm restriction merely select a subset of the positive pivot-row sum. Apply (27) to the first factor. There are $O_K(R/M)$ choices of the other ideals by ideal counting. The block energy is bounded by

$$(XD)^\eta \{R + DR/M + D^{2/3}R/M^{1/3}\} \sum_m |\lambda_m|^2. \tag{28}$$

All three monomial bounds used to sum these blocks are elementary. First $R \le \prod_j A_j^j \le X$. If $M = A_1$, then

$$(R/M)^3 = A_2^3 \cdots A_6^3 \le A_1 A_2^2 A_3^3 A_4^4 A_5^5 A_6^6 \le X,$$

since $A_2 \le A_1$ and $j \ge 3$ for the remaining factors. If $M = A_2$, use $A_1 \le A_2$ in the same comparison. Finally,

$$\frac{R^3}{M(\prod_j A_j^j)^2} = \frac{A_1}{M A_2 A_3^3 A_4^5 A_5^7 A_6^9} \le 1.$$

Thus $R/M \le X^{1/3}$ and $R/M^{1/3} \le X^{2/3}$. Insert these in (28) and sum the logarithmically many blocks. This proves the all-ideal estimate used in Theorem S47.1. It is precisely the mechanism of BGL Lemma 3.4 and (3.6)–(3.9), with the final powered bound inserted. The simpler fixed-squarefull-row argument would leave the weaker $D\sqrt{X}$ term.

S48.3 Literal rows, reciprocal orientation and principal count

Let $a = (u)_S$ denote the outside-S ideal. The finite S-unit comparison gives $\kappa_u = \eta_u \vartheta_a$ as primitive characters, with η_u in a finite fixed set. For $(a, m) = 1$, BGL reciprocity therefore gives

$$\chi_m(u) = \eta_u(m) C([a], [m]) \vartheta_m(a). \tag{29}$$

The notation is deliberate: the left side is the project's original $(u/m)_6$, not BGL's canonical character indexed by m. When $(a, m) > 1$, the left side is zero. The right side is also zero, because squarefree m has outside-S primitive conductor exactly m. Equation (29) hence extends to every pair. Using $\vartheta_a(m)$ on noncoprime pairs instead would be false: primitive ϑ_{b^6} has restored primes of b, whereas the original symbol must still vanish there.

Partition rows by η_u and $[a]$, both finite. On each part the factor $\eta_u(m)C([a], [m])$ is a fixed modulus-one column multiplier. A positive row sum over that part is bounded by the full B_2^1 row sum. For each S-supported ideal b and outside-S ideal a, at most six actual elements generate ab. Thus the required literal energy is bounded by the all-ideal norm summed over b, with $X = H/Nb$. The geometric products

$$\sum_{b \ S\text{-supported}} (Nb)^{-q} = \prod_{p \in S} (1 - (Np)^{-q})^{-1} < \infty, \qquad q = 1,\ 1/3,\ 2/3,$$

transfer the three terms without losing any unit or S-power row. Together with (26), this proves the claimed I energy.

If ψ_u is principal, ν is unramified outside S, and tame Kummer inertia at $v_p(u)$ has order $6/\gcd(6, v_p(u))$. It must be trivial, so each such valuation is divisible by 6. Writing $(u) = ba_6^6$ with b supported in S, ideal counting gives

$$\#\{0 < Nu \le H : \psi_u = 1\} \ll_K H^{1/6} \prod_{p \in S} (1 - (Np)^{-1/6})^{-1}.$$

This is only a necessary condition, sufficient for an upper count. The exact residue formula (22) and $|\widehat{W}(1)| \le \|W\|_\infty$ then give the principal energy bound used above. No sign or positivity of that residue is assumed.

S48.4 What the envelope still fails to prove

The $DH^{1/3}$ term is still larger than $H(HD)^\varepsilon$ at $H = D^{3/4}$. The monomial comparison itself is saturated by $A_1 = A_2 = X^{1/3}$ and $A_3 = \cdots = A_6 = 1$. This proves sharpness of that *particular comparison*, not an energy lower bound for actual arithmetic coefficients. In particular, no conclusion that the signed A_U cannot cancel is justified. The next load-bearing input must exploit those actual truncated Möbius coefficients inside the mixed moment, or establish the required signed type-II minus type-I covariance. The smaller-scale short term, principal contribution, full arithmetic determinant identification and GRH remain separate obligations.

Exact finite checks use the split Eisenstein primes generated by $3 + \omega$, $4 + \omega$, $5 + 2\omega$, $6 + \omega$, of norms $7, 13, 19, 31$. Their actual residue symbols are computed in $\mathbb{Q}(\zeta_6)$, retaining sixth-power zeros. Coefficient recombination and dyadic inequalities pass; they check finite algebra and bookkeeping, not the external analytic sieve or the contour theorem. No new Lean declarations accompany this argument.

S49 High cutoffs, actual bilinear factors and a signed covariance test

Retain the original Eisenstein family, fixed finite-order ν, fixed S, common $W \in C_c^\infty((0,\infty))$ supported in $[1, 2]$, and every raw shared-prime zero. Put

$$C_U(m) = \sum_{\substack{abc=m \\ Na>U,\, Nb>U}} \mu(a)\mu(b), \qquad r_U(m) = \#\{(a, b) : ab = m,\ Na, Nb > U\}.$$

All factors are ideals, squarefree and pairwise coprime; on the active annulus $A_U = C_U - \mu$. A coefficient identity is different from a row-energy bound.

Proposition S49.1 (The high-cutoff window)**.** *If $D > 1$ and $2D/7 \le U^2 \le D$, then for every surviving squarefree column $D \le Nm \le 2D$,*

$$C_U(m) = \mu(m) r_U(m), \qquad A_U(m) = \mu(m)(r_U(m) - 1), \qquad |A_U(m)| \ge 1.$$

For each fixed nonzero W, the normalized squared coefficient norm satisfies

$$D^{-1} \sum_{\substack{m \text{ sf} \\ (m,S)=1}} |A_U(m) W(Nm/D)|^2 \gg_{W,S} 1 \tag{30}$$

for large D, uniformly in this window. This is not a lower bound for row energy.

Proof. The least norm of a nonunit ideal outside S is at least 7. A long-long term with $c \neq 1$ would have $Nm > 7U^2 \geq 2D$. Thus $ab = m$, including both endpoints. The fixed-point-free swap $(a, b) \mapsto (b, a)$ makes r_U even, so A_U is an odd nonzero integer. The last claim follows by counting squarefree ideals on a fixed interval where $|W|$ is bounded below; the constant and error are proved in the extended proof. □

The exact remaining polynomial is genuinely bilinear, with $(a, b) = 1$, $U < Na, Nb < 2D/U$, $D \leq Na\,Nb \leq 2D$, and $x_{ab}(u) = x_a(u)x_b(u)$, including zeros. Both factors see the same row $u = \varepsilon\alpha\beta^2$. Cauchy followed by the accepted powered BGL bound gives only

$$Q_{B_X}(C_U, C_U) \ll_{\nu,S,\epsilon} (DX)^{\epsilon}\|W\|_{\infty}^2\{\sqrt{D}\,X^2 + DX + D^{5/6}X^{5/3}\}. \tag{31}$$

At $X \asymp D^{1/4}$ the exponents are $1, 5/4, 5/4$; the earlier regrouped bound has $1/2, 5/4, 13/12$ and is stronger. No BT_σ saving is obtained. Multiplying two marginal operator norms does not supply a coupled fourth moment.

A finite test in the original family. There is no automatic nonnegative signed covariance, even when $C_U(m)\mu(m) \geq 0$. Take $\nu = 1$, S the primes above $2, 3$, $D = 400$, $U = 17$, and all 72 literal rows of B_7. Choose the common nonnegative smooth profile constructed in the extended proof, with values 10 at 439/400, 1 at 475/400, and zero at every other integer norm divided by 400. There are exactly four active squarefree ideal columns, of norms $439, 439, 475, 475$. In that order,

$$A = (1, 1, 1, 1), \quad C = (0, 0, 2, 2), \quad \mu = (-1, -1, 1, 1), \qquad \Re G_{B_7} = \begin{pmatrix} 72 & 0 & 0 & 18 \\ 0 & 72 & 18 & 0 \\ 0 & 18 & 72 & 0 \\ 18 & 0 & 0 & 72 \end{pmatrix}.$$

Exact raw-symbol enumeration, independently reproducible from the ideal and residue-field definitions, gives

$$Q(C, \mu) = -\frac{27}{25}, \qquad Q(C, C) = \frac{36}{25}, \quad Q(\mu, \mu) = \frac{864}{25}, \quad Q(A, A) = \frac{954}{25}. \tag{32}$$

The real Gram matrix is positive definite, with eigenvalues $54, 90$; its positivity does not force a cross covariance to be positive. This fixed finite example refutes that shortcut, not the asymptotic BT_σ target or the original signed mean-square estimate.

The two-cutoff contour remains coupled. For $T_0 = S \cup \mathrm{supp}(u)$, $\psi = \psi_u$ and independently primitive $\psi_2 = \mathrm{prim}(\psi^2)$, put $q_p = \psi(p)(Np)^{-s}$, $\alpha_p = (Np)^{-z}$, $\beta_p = (Np)^{-w}$ outside T_0. The actual three-placement Euler factor is $1 + q_p - q_p\alpha_p - q_p\beta_p$. Its exact factorization is

$$\mathcal{F}(s, z, w) = \frac{L^{T_0}(s, \psi)}{L^{T_0}(s + z, \psi)L^{T_0}(s + w, \psi)} \prod_{p \notin T_0} \left(1 - \frac{q_p^2(1 - \alpha_p)(1 - \beta_p)}{(1 - q_p\alpha_p)(1 - q_p\beta_p)}\right). \tag{33}$$

Exact double Perron inversion is valid on positive auxiliary lines $c, d > 1 - \Re s$ for $1/2 < \Re s < 1$, retaining the original principal pole exclusion. These safe lines reproduce the previous $U^{2-4\delta}$ energy loss at $\Re s = 1/2 + \delta$, with the same common-profile derivative and fixed-margin dependencies. The correction product alone does not justify moving those lines through reciprocal-L poles. In fact $\mathcal{F}(s, z, 0) = 1/L^{T_0}(s + z, \psi)$ as a meromorphic identity. No zero-free region, reciprocal moment, analytic saving, new Lean formalization or GRH proof is supplied by this construction.

S50 Proof details for the high-window and two-cutoff tests

S50.1 The coefficient lower bound and its precise limitation

The element lattice $\mathbb{Z}[\omega] \subset \mathbb{C}$ has covolume $\sqrt{3}/2$. Counting disks with boundary error $O(\sqrt{x})$ and dividing by the six units, using that this ring is a PID, gives

$$N_K(x) = \kappa x + O(\sqrt{x}), \qquad \kappa = \frac{\pi}{3\sqrt{3}}.$$

Finite inclusion-exclusion at S gives $N_{K,S}(y) = \kappa y \prod_{p\in S}(1-(Np)^{-1}) + O_S(\sqrt{y})$. Insert $1_{\mathrm{sf}}(m) = \sum_{d^2\mid m}\mu(d)$. Absolute convergence at 2, the tail $\sum_{Nd>\sqrt{x}}(Nd)^{-2} = O(x^{-1/2})$, and $\sum_{Nd\le\sqrt{x}}(Nd)^{-1} = O(\log(2x))$ yield

$$N_{\mathrm{sf},S}(x) = c_S x + O_S(\sqrt{x}\log(2x)), \qquad c_S = \frac{\kappa}{\zeta_K(2)}\prod_{p\in S}(1+(Np)^{-1})^{-1} > 0.$$

These tail estimates follow by partial summation from the displayed lattice count. For a fixed nonzero W, choose $[a,b] \subset (1,2)$ with $|W| \ge \eta > 0$. Proposition S49.1 gives, for large D,

$$D^{-1}\sum |A_U(m)W(Nm/D)|^2 \ge \tfrac{1}{2}\eta^2 c_S(b-a).$$

There is no uniform lower constant over all profiles with one specified supremum norm. Nor can a coefficient lower bound be combined with an operator upper bound to produce a row-energy lower bound: no lower singular value has been established on the actual vector.

S50.2 What legal factor Cauchy actually proves

For dyadic $Na \asymp A, Nb \asymp B$, retain the exact $(a,b)=1$ mask, hyperbolic support and $W(Na\,Nb/D)$ in

$$T_a(u) = \sum_b \mu(b)\nu(b)W(Na\,Nb/D)x_b(u).$$

Cauchy gives

$$|\sum_a \mu(a)\nu(a)x_a(u)T_a(u)|^2 \ll A\sum_a |T_a(u)|^2.$$

For fixed a, fixed second row factor and fixed unit/ray classes, the b coefficients are independent of the pivot row and have squared norm $\ll B\|W\|_\infty^2$. The prior classical power-one/two BGL input has $\Delta(X,B) = X + B + (XB)^{2/3}$. Summing the $O(A)$ fixed columns and $O(X)$ other row factors gives

$$\mathcal{E}_{A,B} \ll_{\nu,S,\epsilon} (DX)^\epsilon \|W\|_\infty^2 \frac{A^2B}{D} X\Delta(X,B) \ll (DX)^\epsilon\|W\|_\infty^2\{AX^2 + DX + D^{2/3}A^{1/3}X^{5/3}\}.$$

In the high window $A, B \asymp \sqrt{D}$, and only boundedly many dyadic pairs occur. This proves (31); exchanging factors or row pivots leaves its DX term. A product of marginal second-moment bounds supplies no estimate for the two transforms on the same (r,s). Any useful saving must control that coupled fourth moment or the signed actual-coefficient covariance, with all masks retained.

S50.3 A smooth profile and the exact covariance computation

Let

$$\phi(t)=\begin{cases}\exp(1-1/(1-t^2)), & |t|<1,\\ 0, & |t|\ge 1,\end{cases}\qquad W_x(y)=x\phi(4(400y-439))+\phi(4(400y-475)).$$

For $x\ge 0$ this is nonnegative and smooth; its disjoint supports lie strictly inside $(1,2)$. For every integer n, $W_x(n/400)$ equals x at $n=439$, 1 at $n=475$ and zero otherwise. Thus the computation uses exact values of a genuine common C_c^∞ profile, rather than claiming a finite rational weight is smooth. At $x=10$, $\|W_x\|_\infty=10$.

The primary ideal generators, in the order used above, are

$$(-23,-18),\quad (-5,18),\quad (10,-15),\quad (25,15),$$

where (a,b) denotes $a+b\omega$. The first two have norm 439. The last two factor as $(-2,3)(-5,0)$ and $(-5,-3)(-5,0)$, with norms $19\cdot 25$. Since $19,25>17$, the long-factor count is 2 there and zero on the prime columns. All squarefree ideals at both selected norms are included. The complete B_7 uses the four good primes of norms $7,7,13,13$ for distinct ordered row factors and all six units. It has 72 distinct rows, maximum norm 2197; each is nonprincipal by its outside-S Kummer valuation 1 or 2 modulo 6, whose nontrivial tame character has conductor exponent 1.

For the raw Gram $G_{ij}=\sum_{u\in B_7} x_{m_i}(u)\overline{x_{m_j}(u)}$, exact finite-field exponentiation and lattice enumeration give the displayed real matrix. The weighted real vectors are $\ell=(0,0,2,2)$, $v=(-x,-x,1,1)$, and $a=(x,x,1,1)$. Therefore

$$Q(C_U,\mu)=D^{-1}\ell^t(\Re G)v=\frac{36-9x}{50}.$$

Its diagonal contribution is $36/50>0$ and its off-diagonal contribution is $-9x/50$. For $x=10$ this is $-27/25$ and gives (32). The identity $(\Re G-72I)^2=18^2I$ proves the real eigenvalues $54,90$; these are not being asserted as the eigenvalues of the full complex Gram. The exact source is `examples/arithmetic_cutoff/signed_covariance.py`. A second test retains raw zeros on the 246 rows $0<Nu\le 89$ whose outside-S part is nonunit. Its covariance at $x=10$ is $-69/50$. This subset is guaranteed nonprincipal; it is not called the entire nonprincipal family. Here $89=\lfloor 400^{3/4}\rfloor$. These are finite counterexamples to automatic covariance positivity, not asymptotic counterexamples to BT_σ or GRH.

S50.4 Double Perron without a divergent critical-strip series

For $T_0=S\cup\operatorname{supp}(u)$ retain $F_0(s)=L^{T_0}(s,\psi)/L^{T_0}(2s,\psi_2)$, with the independently primitive square character. Set $t_p=q_p/(1+q_p)$ outside T_0. The finite actual cutoff is

$$J_U(s,u)=F_0(s)B_U(s),\qquad B_U(s)=\sum_{\substack{a,b\ \text{sf},\ (a,b)=1\\ (ab,T_0)=1,\ Na,Nb\le U}}\mu(a)\mu(b)\prod_{p\mid ab}t_p.$$

For $\sigma=\Re s>1/2$, the series

$$\mathcal{G}(s,z,w)=\sum_{\substack{a,b\ \text{sf},\ (a,b)=1\\ (ab,T_0)=1}}\mu(a)\mu(b)\prod_{p\mid ab}t_p\,(Na)^{-z}(Nb)^{-w}=\prod_{p\notin T_0}(1-t_p(Np)^{-z}-t_p(Np)^{-w})$$

converges absolutely on $\sigma+\Re z>1$, $\sigma+\Re w>1$, uniformly in all vertical parts at fixed positive margins. This follows from $|t_p|\le (Np)^{-\sigma}/(1-(Np)^{-\sigma})$. There $\mathcal{F}=F_0\mathcal{G}$, without evaluating an infinite c series in the critical strip. Initial absolute convergence also proves (33) by the local identity

$$(1-q)(1+q-q\alpha-q\beta)=(1-q\alpha)(1-q\beta)-q^2(1-\alpha)(1-\beta).$$

The correction product is holomorphic on

$$\sigma+\Re z>0,\quad \sigma+\Re w>0,\quad 2\sigma+\min(\Re z,0)+\min(\Re w,0)>1.$$

On margins $\sigma+\Re z,\sigma+\Re w\ge\eta>0$ and last exponent $\ge 1+\epsilon$, its local deviation from 1 is at most $4(1-2^{-\eta})^{-2}(Np)^{-1-\epsilon}$. This proves normal convergence and a phase/deletion-uniform vertical upper bound. Nonvanishing of the correction is not asserted.

For $1/2<\sigma<1$, positive lines $c,d>1-\sigma$, and $V=\lfloor U\rfloor+1/2$, define

$$\mathcal{I}_{T_1,T_2}=\frac{1}{(2\pi i)^2}\int_{c-iT_1}^{c+iT_1}\int_{d-iT_2}^{d+iT_2}\mathcal{F}(s,z,w)\frac{V^{z+w}}{zw}\,dw\,dz.$$

The half-integer V includes the actual equality cutoff, since $Na\le U$ is exactly $Na<V$. Put $\Delta_V=\inf_{n\in\mathbb{Z}_{\ge1}}|\log(V/n)|\ge 1/(2V+1)$, and let $\mathcal{A}=\sum_{a,b}|\mu(a)\mu(b)\prod_{p\mid ab}t_p|(Na)^{-c}(Nb)^{-d}$ with the same disjoint support. For $c,T>0$ and $v>0$, $v\ne1$, define

$$K_{c,T}(v)=\frac{1}{2\pi i}\int_{c-iT}^{c+iT}\frac{v^z}{z}\,dz.$$

Finite-height termwise integration of the absolutely convergent $\mathcal{G}$ and the one-variable tail estimate $|K_{c,T}(v)-1_{v>1}|\le 2v^c/(\pi T|\log v|)$ give

$$|\mathcal{I}_{T_1,T_2}-J_U|\le|F_0|\mathcal{A}V^{c+d}\left\{\frac{2}{\pi T_1\Delta_V}+\frac{2}{\pi T_2\Delta_V}+\frac{4}{\pi^2T_1T_2\Delta_V^2}\right\}. \tag{34}$$

Thus $\mathcal{I}_{T_1,T_2}\to J_U$ as both heights tend independently to infinity, away from the original principal pole. This is an oscillatory limit, not an absolutely integrable infinite vertical integral.

Taking $T_1=T_2=V^2$ gives the safe upper bound $|J_U|\ll(Nu)^\eta|L(s,\psi)|V^{c+d}\log^2(2V)$, with constants retaining fixed margins, ν,S, and σ. Choosing c,d close to $1-\sigma$ reproduces $U^{2-2\sigma+\eta}$ in amplitude and $U^{2-4\delta+\eta}$ in energy on $\sigma=1/2+\delta$. Insertion into the original common-profile Mellin argument retains $M_4(W)^2$ and all previously required height/conductor uniformities. No derivative-free improvement follows from this Perron estimate.

The correction converges even on some negative auxiliary lines, but moving there can cross $z=\rho-s$ or $w=\rho-s$, from zeros of the primitive L-function. The exact meromorphic slice $\mathcal{F}(s,z,0)=1/L^{T_0}(s+z,\psi)$ forbids assuming universal cancellation of these poles. At a numerator zero the slice must be understood meromorphically, not by an unjustified pointwise substitution. Sequential shifts have single boundary integrals and crossed-zero residues as well as the kernel double residue; away from s zeros or poles that double residue is $1/L^{T_0}(s,\psi)$. A principal L pole supplies a reciprocal zero, not a reciprocal pole. Uniform control of actual reciprocal integrals/residues or the signed covariance remains a new input. Numerator AFM alone supplies neither.

S51 A quantitative covariance budget without a new exponent

Retain the original Eisenstein sextic symbols, fixed finite-order ν, fixed S, one possibly complex common $W\in C_c^\infty$ supported in $[1,2]$, all squarefree columns prime to S, and every raw shared-prime

zero. In the accepted window $2D/7 \le U^2 \le D$, put $r_m = r_U(m)$, $\lambda_m = \mu(m)\nu(m)W(Nm/D)$ and $v_m = \overline{\lambda_m}$. Thus $C_U = \mu r$ and $A_U = \mu(r-1)$ on the active annulus. For a finite literal row block B, let

$$G_{mn} = \sum_{u\in B} x_m(u)\overline{x_n(u)}, \qquad R = \operatorname{diag}(r_m), \qquad M_j = \sum_m r_m^j |W(Nm/D)|^2 \quad (0 \le j \le 2).$$

The primitive $\psi_u, \psi_{u,2}$ and full deleted sets retain their previous meanings; no conductor-drop value replaces a raw zero.

Proposition S51.1 (An actual-vector lower budget)**.** *If $0 \le G \le \Lambda_B I$, then*

$$\Re Q_B(C_U, \mu) \ge -\frac{\Lambda_B}{2D}\{\sqrt{M_0 M_2} - M_1\}. \tag{35}$$

In this high window, uniformly in U,

$$\sum_{D\le Nm\le 2D} r_m \ll_K D, \qquad \sum_{D\le Nm\le 2D} r_m^2 \ll_K D\log(2D). \tag{36}$$

Consequently

$$\Re Q_B(C_U, \mu) \ge -C_K \|W\|_\infty^2 \Lambda_B \sqrt{\log(2D)}.$$

The vector inequality is sharp among positive contractions with the specified v, Rv; no such sharpness is asserted for the arithmetic Gram. The moment bound counts actual balanced ideal factorizations, rather than assuming independence of the two row factors.

On B_X, inserting the previously accepted powered-BGL norm gives only

$$\Re Q_{B_X}(C_U, \mu) \ge -C_{\nu,S,\epsilon}\|W\|_\infty^2 (DX)^\epsilon \{X^2 + DX + D^{2/3}X^{5/3}\}.$$

At $X \asymp D^{1/4}$ the exponents remain $1/2, 5/4, 13/12$. At $H = D^{3/4}$ this budget still has the old $D^{1/2}$ excess over H; it supplies no BT_σ saving. A negative-part budget is weaker than $2\Re Q(C_U, \mu) \ge Q(C_U, C_U) + Q(\mu, \mu) - R_\sigma$.

For $S_R = \sqrt{R}$, the exact identity

$$\tfrac{1}{2}(RG + GR) = S_R G S_R + \tfrac{1}{2}[S_R, [S_R, G]]$$

localizes the commutator to unequal factor-count levels, with entry $(\sqrt{r_m} - \sqrt{r_n})^2 G_{mn}$. If B retains all six literal units, the actual unit sum further gives $G_{mn} = 0$ unless $Nm \equiv Nn \pmod{36}$. These are exact deletions, not a power saving. Even zero different-level pairings leave the nonnegative within-level energies $\sum_k (k-1)^2 Q_B(\mu \mathbf{1}_{r=k}, \mu \mathbf{1}_{r=k})$. The unequal-product coupled fourth moment and its mixed Möbius term remain unestimated. No new analytic exponent, energy lower bound, Lean formalization or GRH conclusion follows.

S52 Proof and arithmetic boundary of the covariance budget

S52.1 The positive-contraction calculation

Since G is the exact raw Gram, it is positive semidefinite. If $\Lambda_B > 0$, put $T = 2G/\Lambda_B - I$, a self-adjoint contraction. Then

$$\Re(v^* R G v) = \frac{\Lambda_B}{2}\{M_1 + \Re(v^* R T v)\} \ge \frac{\Lambda_B}{2}\{M_1 - \|Rv\|_2 \|v\|_2\},$$

which is (35); $\Lambda_B = 0$ is immediate. The Hermitian rank-two matrix $\{(Rv)v^* + v(Rv)^*\}/2$ has possible nonzero eigenvalues $(M_1 \pm \sqrt{M_0 M_2})/2$. Its negative spectral projection realizes the lower bound among all $0 \le G \le \Lambda_B I$, a linear-algebra sharpness statement only. If $a \le r_m \le b$ on the weighted support, $a + b > 0$, the pointwise inequality $r_m^2 \le (a+b)r_m - ab$ also gives

$$\Re Q_B(C_U, \mu) \ge -\frac{\Lambda_B M_0 (b-a)^2}{8D(a+b)}.$$

For $a = b = 0$ the covariance is zero. The constant follows by maximizing $\sqrt{(a+b)t - ab} - t$ on $[a, b]$.

S52.2 The balanced-divisor second moment

Set $T_0 = \sqrt{D}$. The high-window factor restrictions imply

$$\sqrt{2/7}\, T_0 < Na, Nb < \sqrt{14}\, T_0.$$

The elementary ideal count $\#\{a : Na \le Y\} \ll_K Y$ proves the first assertion of (36). For the second, two ordered factorizations $ab = cd = m$ have the unique form

$$a = gh, \quad c = gk, \quad b = k\ell, \quad d = h\ell, \qquad g = (a, c).$$

Squarefreeness makes g, h, k, ℓ pairwise coprime and squarefree. To upper-bound this positive count, enlarge to arbitrary ideals. For $M = \max(Nh, Nk)$ the individual bounds are

$$Ng \le \frac{\sqrt{14}\, T_0}{M}, \qquad N\ell \le \frac{\sqrt{14}\, T_0}{M}, \qquad 1 \le M \le \sqrt{14}\, T_0.$$

On $2^j \le M < 2^{j+1}$ there are $O_K(2^{2j})$ choices of (h, k) and $O_K((T_0/2^j)^2)$ choices of (g, ℓ), with bounded terminal ranges absorbed in the constant. Each dyadic range contributes $O_K(D)$; there are $O(\log(2D))$ ranges. This proves the second moment with no extra logarithm from counting only $NgN\ell$. Restricting to primes outside fixed S decreases the count. Hence

$$M_0, M_1 \ll_K \|W\|_\infty^2 D, \qquad M_2 \ll_K \|W\|_\infty^2 D \log(2D).$$

These constants are independent of the cutoff in the stated window. Only the common supremum norm is used, including for complex W.

S52.3 What the commutator actually controls

Write

$$t_{mn} = D^{-1} \Re\{\lambda_m \overline{\lambda_n} G_{mn}\}, \qquad \delta_{mn} = (\sqrt{r_m} - \sqrt{r_n})^2.$$

The exact commutator identity proves

$$\Re Q_B(C_U, \mu) \ge Q_B(\sqrt{r}\,\mu, \sqrt{r}\,\mu) - \tfrac{1}{2} \sum_{m,n} \delta_{mn} |t_{mn}|.$$

The displayed loss retains the actual character and profile phases before taking absolute values; it is at most the phase-blind loss obtained from $|t_{mn}| \le D^{-1} |\lambda_m \lambda_n G_{mn}|$. On the accepted four-column smooth B_7 fixture at amplitude $x \ge 0$, the positive term is $18/25$, this coefficient-adapted loss is $9x/50$, and the lower bound is attained. Purely imaginary entries of the full complex Gram make the phase-blind loss larger. This calibrates the new budget and does not give an asymptotic obstruction.

For the unit deletion let $\zeta_6 = 1+\omega$. Every good prime norm q is $1 \pmod 6$, and direct residue-field exponentiation gives $x_p(\zeta_6) = \zeta_6^{(q-1)/6}$. Multiplying over $p \mid m$ and reducing $Nm = \prod_{p|m}(1+6e_p)$ modulo 36 gives

$$x_m(\zeta_6) = \zeta_6^{(Nm-1)/6 \bmod 6}.$$

Multiplication of all literal rows by ζ_6 preserves their raw zeros and proves the norm-congruence deletion in the core. The vector budget may be applied separately on these six classes. No D-dependent number of classes or conductor saving is produced.

The moment proof above counts the coefficient diagonal $ab = cd$. It has not estimated the unequal-product character quadruples in the coupled row fourth moment. Replacing the inherited DX norm by $D^{1-\sigma}X$ merely because equal-level and different-class entries vanish is the unsupported step. Small negative commutator loss alone also leaves the within-level energies in the core. Principal residues are unchanged; this covariance is used on the nonprincipal blocks where the original residue is zero, not as a new centering prescription.

S52.4 The complete near-saturating face

For a six-factor row block let $R_0 = \prod_{j=1}^{6} A_j$, $M = \max(A_1, A_2)$ and $\prod_j A_j^j \le H$. The accepted exact-mask block norm is

$$\Lambda_A \ll_{\nu,S,\epsilon} (HD)^{\epsilon}\{R_0 + DR_0/M + D^{2/3}R_0/M^{1/3}\}.$$

The column moment and covariance proofs apply to every such block. Write $x_j = \log A_j$, $h = \log H$, $c_0 = \sum_j jx_j$ and $r_0 = \sum_j x_j$. Then, for $x_1 \ge x_2$,

$$h - 3(r_0 - \max(x_1, x_2)) = h - c_0 + (x_1 - x_2) + x_4 + 2x_5 + 3x_6;$$

for $x_2 \ge x_1$ replace $x_1 - x_2$ by $2(x_2 - x_1)$. All terms are nonnegative. Thus the middle monomial is saturated on

$$A_1 = A_2, \qquad A_1A_3 = H^{1/3}, \qquad A_4 = A_5 = A_6 = 1,$$

with full capacity. The balanced $A_3 = 1$ case is only an endpoint. The a_3 direction remains nonprincipal by its outside-S valuation $3 \pmod 6$. If $R_0/M \ge H^{1/3}D^{-\eta}$, the total displayed slack is at most $3\eta \log D$. Therefore a fixed power gain must be uniform on this full neighboring region. The proved budget still costs DR_0/M; it gives no global improvement over the accepted envelope.

The independent standard-library diagnostic checks nine ideal moment cases, eight raw Gram cases with 27 000 phase values and 5 184 raw zeros, exact radical commutator/vector budgets, and 227 exponent cases. Finite tables use common norm profiles realizable by smooth disjoint bumps; they are not tests of an infinite analytic estimate. Full derivation, source bindings and preserved checks are under `planning/covariance-budget-20261105/attempt/` and `outputs/checks/2026-11-05-covariance-budget/attempt/`.

S53 The fourth-moment conductor and its full raw zero set

Retain the original Eisenstein sextic symbols $x_m(u) = (u/m)_6$, fixed finite-order ν, fixed S, and one possibly complex common smooth W supported in $[1,2]$. Put $\lambda_m = \mu(m)\nu(m)W(Nm/D)$ and $r_m = r_U(m)$. Columns remain squarefree and prime to S. In $2D/7 \le U^2 \le D$, the actual polynomial is

$$\mathcal{C}_U(u) = \sum_{\substack{(a,b)=1,\ Na,Nb>U \\ D \le Na\,Nb \le 2D}} \mu(a)\mu(b)\nu(ab)W(Nab/D)x_a(u)x_b(u).$$

Its squared row energy has four factor indices a, b, c, d, with $(a, b) = (c, d) = 1$. These internal masks are essential. At each prime,

$$e_p = \mathbf{1}_{p|a} + \mathbf{1}_{p|b} - \mathbf{1}_{p|c} - \mathbf{1}_{p|d} \quad \text{belongs to } \{0, \pm 1\}.$$

Thus, for the two squarefree products $m = ab, n = cd$, the primitive oscillatory conductor outside fixed S is $q = mn/(m, n)^2$. The full raw zero-preserving period is $L = mn/(m, n)$:

$$x_m(u)\overline{x_n(u)} = \mathbf{1}_{(u,(m,n))=1}\theta_{m/(m,n),\, n/(m,n)}(u).$$

No common-prime zero is restored by taking the primitive character.

A generic fourth moment $|\sum_m \lambda_m x_m(u)|^4$ has four individually squarefree columns $m_1, \dots, m_4$ with no such internal coprimality. Its signed local exponent $e_p = \sum_{i=1}^{2} \mathbf{1}_{p|m_i} - \sum_{i=3}^{4} \mathbf{1}_{p|m_i}$ can also be ± 2, giving a nontrivial cubic component. For $L = \mathrm{rad}(m_1 m_2 m_3 m_4)$, the exact row-power interface is

$$Q_j = \prod_{\substack{p|L \\ je_p \not\equiv 0 \ (6)}} p, \qquad Z_j = L/Q_j, \qquad \prod_{i=1}^{2} x_{m_i}(v^j)\overline{\prod_{i=3}^{4} x_{m_i}(v^j)} = \theta_{j,e}(v)\mathbf{1}_{(v,Z_j)=1}.$$

Here $\theta_{j,e}$ is primitive on Q_j, with its own zero extension. At $j = 3$ an exponent ± 2 becomes primitive-principal but its raw zero remains. For the original two-squarefree-product expansion, $Q_j = q$ for all $1 \le j \le 5$, including the quadratic $j = 3$ direction. For the accepted row blocks $Nr_j \asymp A_j$, put $R_0 = \prod_j A_j$, $P = \max(A_1, \dots, A_5)$ and retain $\prod_j A_j^j \le H$. The full A_3 saturation face and all row-factor masks remain.

The small-conductor pair count recovers an already controlled high-gcd sector; it is not a new analytic advance. On the remaining original sector $Nq > P^2$, favorable smooth completion costs $\min(A_j, \sqrt{Nq}) = A_j$ for $1 \le j \le 5$. The literal sharp cutoff is not replaced by this smooth trial. The signed large-conductor fourth-moment and mixed Möbius sums remain open; no improved full exponent, BT_σ, new Lean theorem, zero-location result or GRH conclusion follows.

S54 Scope of the raw fourth-moment reduction

Write $c_{a,b} = \mu(a)\mu(b)\nu(ab)W(Nab/D)$ on the exact factor support in the core. Before absolute Cauchy, the normalized energy is

$$\frac{1}{D} \sum_{(a,b),(c,d) \text{ on that support}} c_{a,b}\overline{c_{c,d}} \sum_{u \in B} x_a(u)x_b(u)\overline{x_c(u)x_d(u)}.$$

Squarefreeness of ab and cd forces each positive and negative prime multiplicity to be at most one, proving $e_p \in \{0, \pm 1\}$. For the mixed term against the actual μ polynomial the exponent is $\mathbf{1}_{p|a} + \mathbf{1}_{p|b} - \mathbf{1}_{p|n}$, with the same conclusion and the same full union mask. The common possibly complex W, ν, hyperbolic support and all raw zeros stay in these identities.

In contrast, unrestricted four-column memberships $(1, 1, 0, 0)$ give $e_p = 2$, a cubic component; $(1, 1, 1, 0)$ give $e_p = 1$ although both product supports contain p. Neither the parity symmetric difference of four supports nor the symmetric difference of two nonsquarefree product supports gives the general conductor. For the interface in the core, its local character order is

j	$1, 5$	$2, 4$	3	6
$e_p = \pm 1$	6	3	2	1
$e_p = \pm 2$	3	3	1	1

Nontrivial entries have tame conductor exponent one; trivial entries remain deleted at $p \mid v$. Multiplying the interface over all six row factors, with the fixed unit/S phase, preserves every original zero, including those at sixth-power factors. All inter-factor, squarefree, ray and capacity restrictions remain external restrictions on the same row sum; this identity asserts no independence.

At a residue field of size Q, let χ have order six and ρ be nontrivial additive. The zero-extended complete sum $G_k(t) = \sum_{z \neq 0} \chi(z)^k \rho(tz)$ satisfies

	$t = 0$	$t \neq 0$
$k \not\equiv 0$ (6)	0	$\lvert G_k(t)\rvert = \sqrt{Q}$
$k \equiv 0$ (6)	$Q - 1$	-1.

For $t \neq 0$, substitution gives $G_k(t) = \overline{\chi(t)^k} G_k(1)$; for nontrivial k, expanding the norm square gives $(Q-1) - \sum_{a \neq 0,1} \chi(a)^k = Q$. The proof applies at split and inert primes. By CRT the full period-L transform vanishes if any local frequency on Q_j is zero; otherwise its magnitude is

$$(NQ_j)^{1/2} \prod_{p \mid Z_j} \left| -1 + Np\, \mathbf{1}_{\text{local frequency at } p \text{ is } 0} \right|.$$

Canceled common primes thus supply Ramanujan factors.

Exact deletion, with all remaining restrictions kept in F, gives

$$\sum_r \theta(r) F(r) \mathbf{1}_{(r,Z)=1} = \sum_{d \mid Z} \mu(d) \theta(d) \sum_z \theta(z) F(dz).$$

For a squarefree pivot let E be the radical of its fixed excluded primes, including Z and the relevant other row factors, with conductor primes omitted since θ already vanishes there. The additional squarefree identity is

$$\mu(r)^2 \mathbf{1}_{(r,E)=1} = \sum_{\substack{h^2 \mid r \\ (h,E)=1}} \mu(h) \sum_{\substack{d \mid E \\ d \mid r}} \mu(d).$$

Since $(d,h) = 1$, its lifted row sum has terms $\mu(d)\mu(h)\theta(d)\theta(h)^2 \sum_z \theta(z) F(dh^2 z)$. The scaled F retains the exact window, primary/ray restrictions and other masks. Divisor triangles cost at most $(HD)^\epsilon$; they do not restore a forbidden nonzero.

The previous fixed-radial-profile high-gcd proof (`research/short-row-high-gcd.md`, independently reviewed in `planning/reviews/book/short-row-formal.md`) already bounds its off-diagonal tail by $O_{\Phi,\epsilon}(D^{2+\epsilon}/G_0^2)$ for $Nh \geq G_0$. For annular $m = hr, n = ht$, $Nq \leq Y$ implies $Nh \geq D/\sqrt{Y}$, so the old smooth-scope bound is $O_{\Phi,\epsilon}(D^\epsilon Y)$. The present elementary count $O(D\sqrt{Y})$, factor multiplicities $r_U(m) r_U(n) \ll_\epsilon D^\epsilon$, and $\#B_A = O(R_0)$ give only $O(\|W\|_\infty^2 D^\epsilon R_0 \sqrt{Y})$ on a literal sharp block. This is a weaker conductor-form recovery in its own bookkeeping scope, not a new high-gcd theorem; the smooth estimate is not silently transferred to a selected sharp off-diagonal piece. The full A_3 face and neighborhood slack are also inherited.

Finally, primitive smooth Poisson completion has bound $O_V(\min(A_j, \sqrt{Nq}))$ on the original support. For $Nq > P^2$ and $1 \leq j \leq 5$, every scaled length $A_j/(Nd\,Nh^2)$ is below $\sqrt{Nq}$, so even this favorable trial takes its trivial length term. Its constant uses fixed smooth seminorms and need not stay bounded when a sharp boundary is approached. In the admissible subsector $(m,n) = 1$, $Nq = Nm\,Nn \asymp D^2$. A complete zero mean does not control the nonzero frequencies of the short original masked row window. The unresolved target is their joint signed sum with weights $r_U(m) r_U(n) \lambda_m \overline{\lambda_n}$ and $r_U(m) \lambda_m \overline{\lambda_n}$, before per-conductor absolute estimates. No new global analytic conclusion is claimed.

S55 A quadratic pivot on the cube factor

Retain the original zero-extended sextic symbols $\chi_m(u) = (u/m)_6$ over $K = \mathbb{Q}(\omega)$, the fixed S containing all primes above 6 and comparison moduli, and the squarefree good columns $D \le Nm \le 2D$. Write an outside-S row ideal uniquely as $r_1 r_2^2 r_3^3 r_4^4 r_5^5 r_6^6$: only $r_1, \dots, r_5$ are pairwise-coprime squarefree ideals; r_6 is arbitrary and may overlap them. Use primary generators. Fix one literal unit and one S-supported row factor, retaining their original values. Let $A_j \le Nr_j < 2A_j$, $A_j \ge 1$, $R_0 = \prod_j A_j$ and $\prod_j A_j^j \le H$; the actual capacity restriction on rows is retained. Coefficients $b_m \in \mathbb{C}$ are arbitrary and fixed independently of every variable row.

Proposition S55.1 (The cube-factor block norm)**.** *For every $\varepsilon > 0$, the raw Gram on this actual block obeys*

$$\sum_{u \text{ in the block}} \left| \sum_m b_m \chi_m(u) \right|^2 \ll_{\nu,S,\varepsilon} (HD)^{\varepsilon} \left\{ R_0 + \frac{DR_0}{A_3} \right\} \sum_m |b_m|^2.$$

The constant is uniform in the fixed unit/S-factor, frozen good factors, H, D and the coefficient sequence. There is no rational-norm-squarefree restriction. Consequently, with $M = \max(A_1, A_2)$, the old powered-BGL bound and this bound give

$$\Lambda_A \ll_{\nu,S,\varepsilon} (HD)^{\varepsilon} \left\{ R_0 + \min\left(\frac{DR_0}{M} + \frac{D^{2/3} R_0}{M^{1/3}}, \frac{DR_0}{A_3} \right) \right\}. \tag{37}$$

The external analytic input is Goldmakher–Louvel, arXiv:1112.1642v2, Corollary 1.2 with $n = 6$, already an explicit classical cube-power sieve. The refinement is this application, not a new large-sieve theorem. For the original normalized coefficients $b_m = D^{-1/2} A_U(m) \nu(m) W(Nm/D)$, the accepted divisor bound gives $\sum_m |b_m|^2 \ll_\eta \|W\|_\infty^2 D^\eta$, uniformly in $U \ge 1$ with $U^2 \le D$ (and the common W may be complex). Choosing smaller internal exponents yields the corresponding uniform block energy envelope. General cutoff or factor-count weights remain inside the arbitrary coefficient norm; that norm is not silently replaced by one. Summing S-supported row factors uses the reduced capacity $H/N\mathfrak{b}$ and the established convergent geometric sums.

On the complete old saturated face $A_1 = A_2 = A$, $A_3 = B$, $A_4 = A_5 = A_6 = 1$, $AB = H^{1/3}$, the new bound is $A^2 B + DA^2$ in place of the old middle monomial DAB. For $H = D^{3/4}$ put $A = D^a$, $B = D^{1/4-a}$, $0 \le a \le 1/4$. When $a < 1/8$, its leading exponent is $1 + 2a < 5/4$; the gain requires B/A to grow by a power for a power improvement. For $a \ge 1/8$, the combined bound still has leading exponent $5/4$. Thus the A_3-dominant part improves, while no new global exponent, BT_σ estimate, signed covariance theorem, Lean theorem or GRH conclusion follows.

Inspecting the surviving sector. The quadratic-pivot inspector plots the complete face, the five monomial powers and the remaining target gap. These are analytic bound exponents, not measured operator norms or trained-network predictions. The refined norm can be inserted in the earlier covariance budget; it does not remove its within-level energies.

A necessary near-face condition. Put $\mathcal{M}_A = DR_0 / \max(M, A_3)$. The combined envelope is at most $(HD)^{\varepsilon}\{R_0 + \mathcal{M}_A + D^{2/3} R_0 / M^{1/3}\}$. If $\mathcal{M}_A \ge DH^{1/3} D^{-\eta}$, then the earlier near-face slack is at most $3\eta \log D$ and, additionally, $A_3 \le MD^\eta$. Thus a uniform remaining leading-term estimate may focus on this smaller neighborhood; it must still cover the entire $B \le A$ part of the face. This is a necessary condition on explicit upper-bound monomials, not a lower bound for actual arithmetic energy.

S56 Proof and scope of the quadratic pivot

Freeze all factors except r_3, and write the actual literal row $u = Fr_3^3$. Multiplicativity of the original numerator gives the exact zero-preserving identity

$$\chi_m(u) = \chi_m(F)\chi_m(r_3)^3.$$

Put $c_m = b_m\chi_m(F)$, a coefficient sequence independent of r_3 with $\sum |c_m|^2 \le \sum |b_m|^2$. In particular the arbitrary sixth-power factor is the coefficient mask

$$\chi_m(r_6^6) = \mathbf{1}_{(m,r_6)=1},$$

not the constant one. No condition $(r_3, r_6) = 1$ is introduced. The actual capacity and coprimality of r_3 with r_1, r_2, r_4, r_5 only restrict this positive outer row sum; after freezing, it may be enlarged to all squarefree good pivot ideals in the dyad. The variable collision mask (m, r_3) remains in the original symbol.

Apply the previously proved literal comparison (29) to the primary generator of r_3 itself:

$$\chi_m(r_3)^3 = \eta_{r_3}(m)^3 C([r_3], [m])^3 \vartheta_m(r_3)^3.$$

The twist η_{r_3} ranges over a fixed finite set of S-unit Kummer characters, and $[r_3]$ over the fixed finite reciprocity group. Partition pivot rows by $(\eta_{r_3}, [r_3])$. On each part the first two factors are one fixed modulus-one column multiplier. Thus its coefficient sequence is independent of the pivot row and has squared norm at most $\sum |b_m|^2$. This finite partition, rather than an assertion that arbitrary row-dependent phases may be absorbed, resolves the fixed-S phase issue.

Goldmakher–Louvel Definition 1 and Remark 3 identify the cubic powers of a sixth-order Hecke family as a quadratic Hecke family; their Corollary 1.2 gives directly

$$\sum_{\substack{a \text{ sf,good} \\ Na \le 2A_3}} \left| \sum_{\substack{m \text{ sf,good} \\ Nm \le 2D}} z_m \vartheta_m(a)^3 \right|^2 \ll_{K,S,\eta} (A_3 D)^\eta (A_3 + D) \sum_m |z_m|^2.$$

This is the required column-indexed orientation and allows arbitrary complex coefficients and squarefree ideals of nonsquarefree rational norm. The canonical family has primitive trivial-infinity-type characters, a fixed finite reciprocity group, and the same-class primitive-product property required by Definition 1. At each $p \mid m$ outside S, the local tame sextic character has order six, so its cube has order two and conductor exponent one. Thus its outside-S primitive conductor is exactly m; the S-part has only finitely many fixed conductor types. Accordingly both sides of the comparison are zero at $(m, a) > 1$. No conductor-drop value replaces the raw zero, and no global formula for the full conductor is required. The fixed ν stays in coefficients.

Enlarge the positive row sum separately on each finite part and apply this sieve. Ideal counting gives $O_K(A_j)$ possibilities for each frozen r_j, including arbitrary r_6, hence $O_K(R_0/A_3)$ frozen tuples. Summing gives $(A_3D)^\eta(R_0 + DR_0/A_3)\sum |b_m|^2$. Since $A_3 \le H$, adjust η to obtain Proposition S55.1; literal unit multiplicity is at most six. Together with the prior block bound this proves (37). It does not change the actual coefficient normalization or principal residues.

On the face in the core, the old bound has exponents $a + 1/4$, $5/4$, $11/12 + 2a/3$, whereas the quadratic bound has $a + 1/4$ and $1 + 2a$. Its improvement for $a < 1/8$ leaves the balanced point and $a \ge 1/8$ with the old $5/4$ leading exponent. Even the improved endpoint $a = 0$ costs D, above the target $D^{3/4}$. The pinned GL PDF has SHA256 `5cc5f03388d3996b4333f07be51f989f`

60779fe172f8bd24c7aa6ddad64dd304. This proof invokes that classical result; finite exponent checks do not reprove it, and no new formalization is claimed.

For the last condition, use $R_0/M \le H^{1/3}$ from the exact capacity inequalities. The hypothesis implies $R_0/A_3 \ge H^{1/3}D^{-\eta}$, so $A_3/M = (R_0/M)/(R_0/A_3) \le D^{\eta}$. It also implies $R_0/M \ge H^{1/3}D^{-\eta}$, which is exactly the previous slack hypothesis. Finally $\min(a+b,c) \le \min(a,c)+b$ for nonnegative a,b,c proves the separated envelope, with no replacement of a signed moment by a positive lower bound.

S57 Joint completion before conductor absolute values

In the original C–C expansion write $m = hr, n = ht$ with h,r,t pairwise coprime squarefree, $q = rt \ne 1$, $L = hq$, $Q = Nq$, $\ell = NL$. Set $\theta = x_r\overline{x_t}$ and $\xi_q = x_q^3$. The original internal masks give $\theta^3 = \xi_q$ and primitive θ^j on q for $1 \le j \le 5$, while $\mu(hr)\mu(ht) = \mu(q)$. Fix generators α of h, β of q, choosing $\alpha = 1$ for $h = 1$. With $\delta = 1 + 2\omega$, put $e_\beta(z) = \exp(2\pi i \operatorname{Tr}(z/(\delta\beta)))$, $\tau_\beta(\eta) = \sum_{z \bmod q} \eta(z)e_\beta(z)$, and $c_h(k) = \prod_{p\mid h}(Np\mathbf{1}_{k\in p} - 1)$, $c_1 = 1$. For $f_j(v) = \theta(v)^j\mathbf{1}_{(v,h)=1}$, $1 \le j \le 5$, $f_6(v) = \mathbf{1}_{(v,L)=1}$ and $G_j(k) = \sum_{v \bmod L} f_j(v)e_{\alpha\beta}(kv)$, joint completion gives

$$\prod_{j=1}^{6} G_j(k_j) = Q^2\xi_q(-\alpha)\tau_\beta(\xi_q)\overline{\theta\Big(\prod_{j=1}^{5} k_j^j\Big)} \prod_{j=1}^{5} c_h(k_j)c_L(k_6). \tag{38}$$

This holds when the first five frequencies are q-units; otherwise the left side is zero. The mixed Gauss phase is orientation-independent, with every common-prime zero retained. The exact row Fourier weight still contains all squarefree, ray, inter-factor, capacity and profile restrictions, including the full A_3 face. Its weighted joint conductor/frequency estimate remains open. This identity and its finite Parseval budget give no improved global exponent or GRH conclusion.

S58 The surviving joint identity and its missing estimate

CRT gives $G_j(k) = \theta(\alpha)^j\tau_\beta(\theta^j)\overline{\theta(k)^j}c_h(k)$ for q-unit k, and $G_6 = c_L$. Pairing Gauss powers $1,5$ and $2,4$ yields $\prod_{j=1}^5 \tau_\beta(\theta^j) = Q^2\xi_q(-1)\tau_\beta(\xi_q)$; the CRT factors multiply to $\theta(\alpha)^{15} = \xi_q(\alpha)$. This proves (38), with all additive-generator phases explicit.

On a fixed literal class write $u = u_0\prod_{j=1}^6 v_j^j$ and let $F(v)$ be its exact finite row weight. Only $v_1,\dots,v_5$ are squarefree and pairwise coprime; v_6 is arbitrary with its permitted overlaps. Retain every original mask and the sharp capacity boundary in F. Define $\widehat{F}_L(k) = \sum_v F(v)e_{\alpha\beta}(-\sum_j k_jv_j)$ and, with $t = q/r$, put

$$\begin{aligned} a(r) &= r_U(hr)r_U(ht)\nu(r)\overline{\nu(t)} \\ &\qquad \times W(Nh\,Nr/D)\overline{W(Nh\,Nt/D)}\theta_{r,t}(u_0), \\ A(K) &= \sum_{r\mid q} a(r)\overline{\theta_{r,q/r}(K)}, \\ B(K) &= \sum_{\substack{k \bmod L,\ (k_1\cdots k_5,q)=1 \\ \prod_{j=1}^5 k_j^j = K \bmod q}} \widehat{F}_L(k)\prod_{j=1}^{5} c_h(k_j)c_L(k_6). \end{aligned}$$

Set $a(r) = 0$ at excluded orientations; the actual factor counts and common possibly complex W

remain. Fourier inversion gives exactly

$$E_{\mathcal{T}} = \frac{1}{D} \sum_{(h,q)\in\mathcal{T}} \mu(q) \frac{Q^2 \xi_q(-\alpha)\tau_\beta(\xi_q)}{\ell^6} \sum_{K\in(\mathcal{O}/q)^\times} B(K)A(K). \tag{39}$$

Sum the original u_0 classes and blocks. This retains low gcd, including $h = 1$, before conductor absolute values. For C–μ replace the second $r_U(ht)$ by 1. A column-orientation-dependent restriction cannot be hidden in a common F when applying the next identity.

Distinct orientations have nontrivial cubic quotient, so

$$\sum_K |A(K)|^2 = \varphi(q) \sum_{r|q} |a(r)|^2, \qquad |E_{h,q}| \le \frac{Q^{5/2}}{D\ell^6} \left(\varphi(q) \sum_{r|q} |a(r)|^2 \right)^{1/2} \|B\|_2.$$

The product-frequency fibers have size $\varphi(q)^4$ under uniform averaging. The actual B is not uniform: all orientations agree on the coherent set $x_p(K)^2 = 1$ for every $p \mid q$, of size $\varphi(q)/3^{\omega(q)}$. Centered representatives have $|k_j| \le \sqrt{\ell}$. Restricting to any frequency subset Ω has error bounded by the same displayed budget with $\|B - B_\Omega\|_2$; no small lower-height tail is proved.

The full face $A_1 = A_2 = A$, $AA_3 = H^{1/3}$, $A_4 = A_5 = A_6 = 1$ remains. At its pure A_3 endpoint, inversion in the fixed factors leaves $(\sum_r a(r)) \sum_v F_3(v)\xi_q(v)\mathbf{1}_{(v,h)=1}$: the actual row character there is orientation-independent. Literal length-one Fourier factors can undo complete phase balance. A quantitative signed estimate for the actual h, q, K sum is still required. Exact split/inert Gauss, full-mask CRT and orientation tests pass 991 assertions; receipt: `outputs/checks/2026-11-05-joint-dual/research/joint-gauss-check-v1.json`. These diagnostics do not prove an incomplete-moment saving or GRH.

S59 A radical-conductor refinement of the raw envelope

Retain the original literal symbols $\chi_m(u) = (u/m)_6$, extended by zero, the fixed good set S, and the squarefree ideal columns $D \le Nm \le 2D$. On an actual dyadic row block write

$$u = u_0 r_1 r_2^2 r_3^3 r_4^4 r_5^5 r_6^6, \qquad A_j \le N r_j < 2A_j, \qquad \prod_{j=1}^{6} A_j^j \le H.$$

Only $r_1, \dots, r_5$ are pairwise-coprime squarefree ideals; r_6 is arbitrary and may overlap them. Fix primary generators, a literal unit and an S-supported factor in u_0. Put $R = \prod_j A_j$, $M = \max(A_1, A_2)$ and $Q = R/A_6$. For arbitrary complex coefficients b_m, independent of every variable row, define

$$\mathcal{E}_A(b) = \sum_{u \text{ in the actual block}} \left| \sum_m b_m \chi_m(u) \right|^2, \qquad \Lambda_A = \sup_{b\ne 0} \frac{\mathcal{E}_A(b)}{\sum_m |b_m|^2}.$$

Proposition S59.1 (An ordinary conductor comparison)**.** *For every $\varepsilon > 0$,*

$$\Lambda_A \ll_{S,\varepsilon} (HD)^\varepsilon A_6 \{D + Q^2\}. \tag{40}$$

Consequently the original residue-subtracted remainder satisfies, uniformly for $H, D, U \ge 1$ and $U^2 \le D$,

$$\sum_{0<Nu\le H} |E_u|^2 \ll_{\nu,S,\varepsilon} \|W\|_\infty^2 (HD)^\varepsilon \{H + DH^{1/6} + (DH)^{2/3}\}. \tag{41}$$

Here ν, S are fixed and $W \in C_c^\infty((0,\infty))$ is one common, possibly complex profile supported in $[1,2]$; there is no profile supremum.

The input to (40) is the classical ordinary sieve for *all* primitive characters of conductor norm at most Q, with constant $D+Q^2$, not a higher-order-family theorem for arbitrary conductor labels. The following section gives its standard Gauss-sum and two-dimensional lattice proof, including prime-power conductors at S. For fixed r_6, the first five factors have primitive outside-S conductor $q = r_1 \cdots r_5$. The literal primitive character agrees with the raw symbol even at collisions with q; the sixth-power factor instead remains the coefficient mask $\mathbf{1}_{(m,r_6)=1}$. At most $5^{\omega(q)} \ll_\eta (Nq)^\eta$ tuples give the same primitive character. Applying the ordinary sieve and then counting r_6 proves (40). No rational-norm-squarefree restriction is imposed.

To obtain (41), combine this with the accepted powered-BGL block bound from Section S47. With $S_A = R/M$ and $T_A = D^{2/3}R/M^{1/3}$, capacity gives $R \le H$, $A_6 \le H^{1/6}$, $RS_A \le H$ and $T_A \le (DH)^{2/3}$. Hence

$$\begin{aligned}\min(R + DS_A + T_A, DA_6 + R^2/A_6) &\le R + T_A + DA_6 + \min(DS_A, H^2/S_A^2)\\ &\le H + DH^{1/6} + 2(DH)^{2/3}.\end{aligned}$$

Dyadic and fixed-S summation, the uniform actual coefficient norm, and the principal-residue estimate give the assertion for $E_u = I_u - P_u$.

At $H = D^{3/4}$ the new envelope has exponent $7/6$, improving the old $5/4$ by $1/12$; its excess above the target H is still $D^{5/12}$. On the *entire* old face $A_1 = A_2 = A$, $A_3 = B$, $A_4 = A_5 = A_6 = 1$, $AB = H^{1/3}$, one has $Q = A^2B = AH^{1/3} \le H^{2/3}$, so (40) costs $D^{1+\varepsilon}$ there. This includes $B \le A$. The new combined upper bounds reach $7/6$ at $(\log_D A_1, \log_D A_2) = (5/12, 1/6)$ with other coordinates zero; the pure point $\log_D A_1 = 3/4$ also retains the old BGL cross term. These are upper-envelope witnesses, not arithmetic energy lower bounds or counterexamples to signed cancellation.

The simpler proposed replacement of the radicand ab^2 by its radical ab inside BGL remains unjustified: after the existing fixed reciprocity phases, the mixed reciprocity quotient is $\vartheta_b(a')/\vartheta_a(b')$, which varies outside S. The present application does not require that replacement. No target signed moment, type-II covariance estimate, positive arithmetic determinant comparison, zero-location result, new Lean theorem or GRH conclusion follows from (41).

S60 Proof of the ordinary-conductor refinement

S60.1 Primitive Gauss sums at every conductor

Let $\mathcal{O} = \mathbb{Z}[\omega]$, $\delta = 1 + 2\omega$, and choose one generator f of each nonzero ideal $\mathfrak{f}$. Write $e_f(x) = \exp(2\pi i \operatorname{Tr}(x/(\delta f)))$, $U(\mathfrak{f}) = (\mathcal{O}/\mathfrak{f})^\times$, and $\varphi(\mathfrak{f}) = |U(\mathfrak{f})|$. This additive character is primitive, since the different is (δ). For a primitive multiplicative character ρ modulo $\mathfrak{f}$, extended by zero at nonunits, set $\tau_f(\rho) = \sum_{x \bmod \mathfrak{f}} \rho(x) e_f(x)$. We verify the full-modulus identity

$$|\tau_f(\rho)|^2 = N\mathfrak{f}, \qquad \sum_{x \bmod \mathfrak{f}} \rho(x) e_f(zx) = \overline{\rho(z)}\tau_f(\rho) \quad (z \in \mathcal{O}). \tag{42}$$

No squarefree-conductor hypothesis is used, so this includes wild prime powers in the fixed S-part.

The Ramanujan sum is

$$c_{\mathfrak{f}}(y) = \sum_{x \in U(\mathfrak{f})} e_f(xy) = \sum_{\mathfrak{d} \mid (\mathfrak{f}, (y))} N\mathfrak{d}\, \mu(\mathfrak{f}/\mathfrak{d}).$$

Indeed insert $\mathbf{1}_{(x,\mathfrak{f})=1} = \sum_{\mathfrak{b}\mid(x,\mathfrak{f})} \mu(\mathfrak{b})$; the complete additive sum with $\mathfrak{b} \mid x$ is $N(\mathfrak{f}/\mathfrak{b})$ when $\mathfrak{f}/\mathfrak{b} \mid (y)$ and zero otherwise. Changing variables $x = ay$ in the Gauss square now gives

$$\begin{aligned}|\tau_f(\rho)|^2 &= \sum_{a\in U(\mathfrak{f})} \rho(a) c_{\mathfrak{f}}(a-1) \\ &= \sum_{\mathfrak{d}\mid\mathfrak{f}} N\mathfrak{d}\,\mu(\mathfrak{f}/\mathfrak{d}) \sum_{\substack{a\in U(\mathfrak{f}) \\ a\equiv 1 \pmod{\mathfrak{d}}}} \rho(a) = N\mathfrak{f}.\end{aligned}$$

For every proper $\mathfrak{d} \mid \mathfrak{f}$, the reduction $U(\mathfrak{f}) \to U(\mathfrak{d})$ is surjective by local lifting and CRT. Primitivity says ρ is nontrivial on its kernel, so that inner sum vanishes. For $\mathfrak{d} = \mathfrak{f}$ it equals one. For unit z, substitution proves the second identity in (42). For all frequencies together, finite additive Parseval gives

$$\sum_{z \bmod \mathfrak{f}} \left|\sum_x \rho(x) e_f(zx)\right|^2 = N\mathfrak{f}\,\varphi(\mathfrak{f}).$$

The unit frequencies already exhaust this quantity by the first identity; every nonunit-frequency sum is therefore zero. This proves the asserted zero extension. For $\mathfrak{f} = \mathcal{O}$ there is one residue class and one character; take $\tau_1 = 1$ and the single frequency zero.

S60.2 The classical ordinary sieve in these lattice conventions

For $D, Q \geq 1$, coefficients z_α supported on any subset of $\{\alpha \in \mathcal{O} : |\alpha|^2 \leq C_0 D\}$ satisfy

$$\sum_{N\mathfrak{f}\leq Q} \sum_{\rho \text{ primitive mod } \mathfrak{f}} \left|\sum_\alpha z_\alpha \rho(\alpha)\right|^2 \ll_{\mathcal{O},C_0} (D+Q^2) \sum_\alpha |z_\alpha|^2. \tag{43}$$

Here the primitive characters may be restricted to the unit-trivial subfamily. We give the standard proof to make the conductor comparison explicit.

First the reduced fractions x/f, with $x \in U(\mathfrak{f})$ and $N\mathfrak{f} \leq Q$, are distinct modulo $\mathcal{O}$. Their denominator ideal is intrinsic: $\{a \in \mathcal{O} : a(x/f) \in \mathcal{O}\} = \mathfrak{f}$. Thus equality modulo $\mathcal{O}$ forces the same denominator ideal and residue numerator. For different fractions and every $k \in \mathcal{O}$,

$$\left|\frac{x}{f} - \frac{y}{g} - k\right| = \frac{|xg - yf - kfg|}{|fg|} \geq \frac{1}{Q},$$

because the numerator is a nonzero Eisenstein integer and $|fg| = (N\mathfrak{f}N\mathfrak{g})^{1/2} \leq Q$. The fixed codifferent factor $1/\delta$ changes only the lattice constants. The modulus-one frequency zero obeys the same separation from all others.

For completeness, a ρ_0-separated set t_r in a fixed two-dimensional torus obeys the additive bound

$$\sum_r \left|\sum_{|\alpha|\ll\sqrt{D}} z_\alpha \exp(2\pi i \operatorname{Tr}(\alpha t_r))\right|^2 \ll (D + \rho_0^{-2}) \sum_\alpha |z_\alpha|^2. \tag{44}$$

Use fixed linear coordinates to write the lattice as $\mathbb{Z}^2$ and its additive pairing as $e(n \cdot t)$. The support lies in $|n_i| \leq T$ with $T \asymp \max(1, \sqrt{D})$. To prove the dual inequality, majorize this box indicator by $e^{2\pi} \exp(-\pi\|n\|^2/T^2)$, expand the square, and apply Gaussian Poisson summation. Up to fixed constants the resulting kernel is

$$T^2 \sum_{\ell\in\mathbb{Z}^2} \exp(-cT^2\|t_r - t_s + \ell\|^2).$$

Take ρ_0 at most a fixed torus constant. The lifted points are $c'\rho_0$-separated, so disjoint disks bound their count in a disk of radius L by $O(1+(L/\rho_0)^2)$. Gaussian annular summation bounds every absolute kernel row sum by $O(T^2+\rho_0^{-2})$. Schur's inequality and finite-dimensional duality prove (44), with constants independent of the number of frequencies.

Put $\mathcal{S}(x/f)=\sum_\alpha z_\alpha e_f(\alpha x)$. By (42),

$$\rho(\alpha)=\tau_f(\overline{\rho})^{-1}\sum_{x\in U(\mathfrak{f})}\overline{\rho(x)}e_f(\alpha x).$$

After taking squared absolute values, enlarging the primitive-character sum to all characters of $U(\mathfrak{f})$ and using orthogonality yields

$$\sum_{\rho\ \text{primitive mod }\mathfrak{f}}\left|\sum_\alpha z_\alpha\rho(\alpha)\right|^2\le\frac{\varphi(\mathfrak{f})}{N\mathfrak{f}}\sum_{x\in U(\mathfrak{f})}|\mathcal{S}(x/f)|^2\le\sum_{x\in U(\mathfrak{f})}|\mathcal{S}(x/f)|^2.$$

Apply (44) with $\rho_0\gg Q^{-1}$ to prove (43).

Finally $\mathcal{O}$ has class number one. Choose one generator α_m of each ideal column, with $|\alpha_m|^2=Nm$; these are an injective subset of the lattice disk. A primitive trivial-infinity-type Hecke character is a unit-trivial primitive residue character on $\mathcal{O}/\mathfrak{f}$, and its ideal value equals its value on α_m. Descent of the residue character to a proper divisor would preserve unit-triviality and contradict primitive Hecke conductor. Thus (43) applies to arbitrary ideal coefficients of norm at most $2D$, including squarefree ideals of nonsquarefree rational norm.

S60.3 Raw zeros, tuple multiplicity and block transfer

Freeze r_6 and the literal unit/S-factor, putting

$$c_m=b_m\chi_m(u_0)\mathbf{1}_{(m,r_6)=1},\qquad\sum_m|c_m|^2\le\sum_m|b_m|^2.$$

This is independent of every first-five variable. In particular the arbitrary overlapping sixth-power factor is not replaced by one. Let $v=\prod_{j=1}^5 r_j^j$ denote the chosen literal product. Every outside-S valuation is between one and five. BGL, arXiv:1112.1650v1, Lemma 2.1, together with the established finite S-unit comparison in (29), gives its primitive literal Kummer character κ_v of trivial infinity type, with

$$q=r_1\cdots r_5\mid\operatorname{cond}(\kappa_v)\mid\mathfrak{c}_S q,\qquad\kappa_v(m)=\chi_m(v)\quad((m,S)=1).$$

The equality includes collisions with q: both sides are zero. The fixed $\mathfrak{c}_S$ includes all wild conductor types and literal comparison twists. No row-dependent phase enters the coefficients.

A fixed primitive character determines its outside-S conductor q. Each prime of q has at most five assignments to the pairwise-coprime $r_1,\dots,r_5$, so tuple multiplicity is at most $5^{\omega(q)}\ll_\eta(Nq)^\eta$. All conductors in the dyadic block have norm at most C_SQ. Enlarge the positive row sum, apply (43), and count $O(A_6)$ possibilities for the frozen arbitrary r_6. This proves (40); the actual capacity, squarefree, ray and inter-factor restrictions only shrink the positive sum. Equivalently $\mathcal{E}_A(b)\le\Lambda_A\sum_m|b_m|^2$; there is no coefficient norm factor in the definition of Λ_A.

S60.4 Uniform coefficient and principal-residue assembly

The other block estimate is exactly the accepted powered-BGL input (28), whose squarefree powers are justified in BGL Section 3.2. The capacity comparisons and elementary minimum in Section S59 give

$$\Lambda_A \ll (HD)^{\varepsilon}\{H + DH^{1/6} + (DH)^{2/3}\}.$$

Sum logarithmically many blocks with smaller internal exponents. An S-supported literal factor $\mathfrak{b}$ reduces capacity to $H/N\mathfrak{b}$; the required sums over $\mathfrak{b}$ converge at powers 1, 1/6, 2/3. Literal units have multiplicity at most six. This proves the raw bound with arbitrary fixed b_m.

For the actual coefficients use (26): $|A_U(m)| \le 3^{\omega(m)} \ll_\eta (Nm)^\eta$ gives

$$b_m = D^{-1/2}A_U(m)\nu(m)W(Nm/D), \qquad \sum_m |b_m|^2 \ll_\eta \|W\|_\infty^2 D^\eta,$$

uniformly for $U \ge 1$, $U^2 \le D$. The common W is not optimized separately for different rows. If the primitive ψ_u is principal, ν is unramified outside S, so tame Kummer inertia forces every outside-S valuation of u to be divisible by six. Ideal counting and the convergent S-supported sum at power 1/6 give $O_{\nu,S}(H^{1/6})$ such literal rows. In the exact formula (22), $|\rho_T| \ll_K 1$, $|\widehat{W}(1)| \le \|W\|_\infty$, and

$$\sum_d \frac{|\beta_U(d)|}{Nd} \le \left(\sum_{Na\le U}\frac{1}{Na}\right)^2 \ll \log^2(2U).$$

Thus $|P_u| \ll \sqrt{D}\|W\|_\infty \log^2(2U)$ on principal rows, and

$$\sum_{0<Nu\le H} |P_u|^2 \ll DH^{1/6}\|W\|_\infty^2 \log^4(2U).$$

This fits the new envelope. The exact identity $E_u = I_u - P_u$ and $|E_u|^2 \le 2|I_u|^2 + 2|P_u|^2$, with smaller internal exponents, prove (41). No cutoff limit or profile supremum occurs.

The ordinary comparison is an application refinement. A direct BGL radical relabelling would additionally require reciprocity of its mixed family, not just its conductor count: for pairwise-coprime a, b, a', b', the quotient after the original fixed ray phases is $\vartheta_b(a')/\vartheta_a(b')$, which varies outside S. GL, arXiv:1112.1642v2, Definition 1(2), is therefore not supplied by that shortcut. This does not rule out a different mixed-family theorem; it is simply unneeded for the proved weaker ordinary comparison. The remaining exponent gap and every independent signed, spectral, formal and GRH obligation stated in Section S59 remain open.

S61 Learning the actual signed arithmetic correlations

The unsigned refinement in Section S59 leaves a signed problem. Here is a separate, executed finite learning objective on the original correlations, with a certificate for every coordinate and observable. It is not a uniform estimate for an expanding family. For ordered distinct squarefree columns $m = hr, n = ht$, put $q = rt$ and

$$\Gamma_i = \sum_{u\in\mathcal{R}} \chi_r(u)\overline{\chi_t(u)}\mathbf{1}_{(u,h)=1}, \qquad i = (h, q, r).$$

The symbols vanish at every collision with q; the common-prime mask is retained separately. With principal ν, actual complementary divisor counts $r_U(m)$, and one common complex profile $W_m = W(Nm/D)$, define

$$d_i^{CC} = r_U(m)r_U(n)W_m\overline{W_n}, \qquad d_i^{C\mu} = r_U(m)W_m\overline{W_n}, \qquad \lambda_i = 1 + |d_i^{CC}|^2 + |d_i^{C\mu}|^2.$$

The exact diagonal Δ_X and these offdiagonal coordinates give

$$E_X = \Delta_X + D^{-1} \sum_i \mu(q_i) d_i^X \Gamma_i, \qquad X \in \{CC, C\mu\}. \tag{45}$$

Thus $E_{CC} = \sum_{u \in \mathcal{R}} |C_u|^2$ and $E_{C\mu} = \sum_{u \in \mathcal{R}} C_u \overline{M_u}$ for the actual normalized row sums. Neither the Möbius sign nor a complex conjugate is discarded.

Proposition S61.1 (A finite supervised certificate)**.** *Let g_i be complex readout coordinates and $L(g) = \sum_i \lambda_i |g_i - \Gamma_i|^2$. For $K_X = \sum_i |d_i^X|^2/\lambda_i$ and $\lambda_{\min} = \min_i \lambda_i$, the substituted readouts satisfy*

$$|\widehat{E}_X - E_X|^2 \le D^{-2} K_X L(g),$$
$$\max_i |g_i - \Gamma_i|^2 \le L(g)/\lambda_{\min}.$$

The executed metric-gradient update $g^{(t+1)} = (g^{(t)} + \Gamma)/2$ gives $L(g^{(t)}) = 4^{-t} L(g^{(0)})$. Therefore every positive rational requested tolerance on these three squared bounds is attained after finitely many updates on this fixed exact instance, with unrestricted resources.

The implementation's resource guard refuses requests beyond its update budget; termination in this proposition is not a practical runtime guarantee. The labels Γ_i are computed exactly before training, so no speedup over exact label evaluation or generalization to unseen conductors is claimed.

The supplied instance uses $D = 6000, U = 60, H = 625$, S above $2, 3$, principal ν, and all 15 good squarefree ideal columns at activated norms $7507, 7519, 8281, 10507$ (respectively $2, 4, 1, 8$ ideals). One explicitly specified common smooth four-bump profile is zero at every other integer norm. Both the 2262 literal rows and the separately labelled 2148 guaranteed nonprincipal subset are evaluated; the latter is not asserted to be the whole nonprincipal family. There are 95 offdiagonal (h, q) blocks, 65 with $h = 1$, and 210 ordered learned coordinates. Every row case retains 4530 raw column zeros. Real and complex profiles give four requested-10^{-12} certificates, using $53, 55, 53, 55$ metric updates respectively: 216 executed updates, with every full rational state saved. A separate 200 executed ordinary coordinate updates are budget-limited and carry no requested-error claim. The standalone replay is linked from the book and project page. No analytic saving, positive determinant, zero-location assertion, upstream quantum training, new Lean declaration or GRH conclusion follows.

S62 Finite phase features, gradients and exact verification

For each block let $G_q = (\mathbb{Z}/6\mathbb{Z})^{\omega(q)}$. A literal admissible row determines local phases e_p at every prime of q; rows with a q-collision or a common-prime collision with h are deleted before forming the exact sparse histogram $n(e)$. Orientation r has

$$\theta_r(e) = \zeta_6^{\sum_{p|q} s_p e_p}, \qquad s_p = 1 \ (p \mid r), \quad s_p = -1 \ (p \mid t), \qquad \Gamma_r = \sum_e n(e) \theta_r(e).$$

Different active orientations have distinct exponent vectors modulo 6, so their characters are orthonormal for the uniform measure on G_q. The model and target projection are

$$F_g(e) = \sum_{r \in J_q} g_r \overline{\theta_r(e)}, \qquad F_\Gamma(e) = \sum_{r \in J_q} \Gamma_r \overline{\theta_r(e)}.$$

Indeed F_Γ is the orthogonal projection of $|G_q| n$ onto this character subspace, and $\mathbb{E}_{G_q} F_g \theta_r = g_r$. The positive spectral metric assigns weight λ_r to this orthonormal basis. Its squared error is exactly

$\sum_r \lambda_r |g_r - \Gamma_r|^2$. Thus the run evaluates the full finite feature-projection loss by Parseval; it does not claim to have executed a sampled or stochastic training loss. This projection need not be a nonnegative density. A Hermitian readout matrix need not be positive semidefinite.

Gradient and contraction. Use ζ_6 coordinates $g = a + b\zeta_6$, so $|g|^2 = a^2 + ab + b^2$ and $M = \left(\begin{smallmatrix} 1 & 1/2 \\ 1/2 & 1 \end{smallmatrix}\right)$. For the objective $L/2$, the ordinary real-coordinate gradient in block i is $\lambda_i M(x_i - x_i^*)$. In the block metric $\lambda_i M$, the gradient is $x_i - x_i^*$. A metric step of size $1/2$ therefore halves every complex error and quarters L. Conjugate initial coordinates and this step preserve the Hermitian reverse orientations. The ordinary trajectory treats all ordered coordinates independently; asymmetric weights need not preserve that symmetry. For ordinary steps $\eta = 2/(3\lambda_{\max})$, the Hessian eigenvalues lie between $\lambda_{\min}/2$ and $3\lambda_{\max}/2$, whence

$$L_{t+1} \le \left(1 - \frac{\lambda_{\min}}{3\lambda_{\max}}\right)^2 L_t.$$

This last bound is conservative; the saved 50 ordinary steps per case are not asserted to attain the requested tolerance.

Proof of Proposition S61.1. Subtract (45) with the same exact diagonal. Weighted Cauchy–Schwarz, applied only to the error, gives

$$\left|\sum_i \mu(q_i) d_i^X (g_i - \Gamma_i)\right|^2 \le \left(\sum_i \frac{|d_i^X|^2}{\lambda_i}\right)\left(\sum_i \lambda_i |g_i - \Gamma_i|^2\right).$$

Here $|\mu(q_i)| = 1$ because q_i is squarefree; the predicted sum itself keeps its signs and phases. Each summand in the second factor is nonnegative, which proves the coordinate bound. The exact affine update gives $g^{(t)} - \Gamma = 2^{-t}(g^{(0)} - \Gamma)$, proving contraction. For finite rational positive λ_i, K_X, L_0 and $\tau > 0$, some integer t makes all three bounds at most τ^2. The zero-error case stops immediately. No varying-family constants enter this finite assertion. □

The activated original family. Let

$$\phi(t) = \begin{cases} \exp(1 - 1/(1-t^2)), & |t| < 1, \\ 0, & |t| \ge 1, \end{cases} \qquad W(y) = \sum_{N \in \{7507,7519,8281,10507\}} w_N \phi(4(6000y - N)).$$

These disjoint bumps are supported in $(1, 2)$ and $\phi(0) = 1$. For the real case $w_N = 1$; for the complex case $w_N = 1 + \zeta_6(4N - 30000)/6000$. At integer norms $W(n/6000) = w_n$ at the four activated norms and is zero elsewhere. Hence all active original squarefree columns are present, rather than an undeclared column submatrix. No exponential evaluation is needed at these exact nodes. The profile is a fixed finite demonstration and does not establish a profile-uniform analytic estimate. The counts r_U are $0, 2, 4, 2$ at the four norms. The guaranteed nonprincipal subset removes precisely those literal rows whose outside-S norm is 1; no classification of the omitted 114 rows as a whole is asserted. Every nonunit outside-S factor has a prime exponent from 1 to 3, since its prime norm is at least 7 and $7^4 > 625$; its nonzero residue modulo 6 gives a nontrivial local character. All six units and all original S-power rows belong to the separately saved full literal case.

The seed 20261105 selects deterministic rational conjugate initial coordinates by SHA256 of the ordered column identifiers. There is no random sampling. Exact $\mathbb{Q}(\zeta_6)$ arithmetic stores every metric and ordinary state. Training accepts only when both observable certificates and the

uniform-coordinate certificate are at most τ^2, preventing a scalar cancellation from masquerading as accurate correlation learning. The real full-row targets are $E_{CC} = 2393/125$ and $E_{C\mu} = 431/50$; the guaranteed-subset targets are 4293/250 and 771/100. Complex-profile targets and the complete phase histograms are saved alongside. Each case has 64 nonzero negative-$\mu(q)$ C–C target orientations. Replacing all signs by plus is an explicit failing control, not an estimator.

Reproduction and scope. The complete run is reproducible.

The source is `examples/arithmetic_learning/learn.py`; its README states the exact smooth profile, CLI, resource guard and loss convention. The accepted immutable output is `outputs/experiments/2026-11-05-arithmetic-learning-v2/`. All source/helper hashes, exact input bytes and artifact hashes are bound before and after execution and checked on replay. Existing outputs, nonpositive or binary-float tolerances, and stale source/input/case bytes are refused. The preserved v1 logging failure is not the accepted run. The focused verifier independently reconstructs 31668 local symbols and 450 Gram targets, then checks the full saved metric trajectory. Independent mathematical review and exact code checks are recorded separately; finite arithmetic checks are not a proof of an asymptotic signed estimate. Full exact training labels are supervision, so this is a certified finite reference architecture, without an efficiency, generalization, PSD, zero-location or GRH claim. Analytic and categorical Lean remain unfinished.

S63 Why the common profile must stay fixed

The unsigned envelope of Section S59 and the unresolved signed estimate have different profile quantifiers. We give an obstruction to an H-sized bound uniform through the sup norm alone. It uses the *actual* coefficients and nonprincipal rows, rather than replacing the columns by arbitrary weights. It does not contradict an estimate for one fixed smooth profile or one with adequate derivative dependence.

Fix $K = \mathbb{Q}(\omega)$, $\omega^2 + \omega + 1 = 0$, principal $\nu = 1$, and a finite good set S containing the primes above $2, 3$. Write $\zeta = 1 + \omega$, $\delta = 1 + 2\omega$, so ζ has order six and $\delta^2 = -3$. Retain the literal raw symbols and normalization

$$I_u(W) = D^{-1/2} \sum_{m \text{ squarefree } (m,S)=1} A_U(m)\chi_m(u)W(Nm/D), \qquad E_u = I_u - P_u.$$

Here and below squarefree means *ideal* squarefree. For $U^2 = D$ the accepted high-window identity gives $A_U(m) = \mu(m)(r_U(m) - 1)$ on $D < Nm < 2D$, with $r_U(m)$ even and $|A_U(m)| \geq 1$. The multiset of prime-ideal norms of a squarefree ideal is determined by its rational norm. Thus $A_U(m) = A_N$ depends only on $N = Nm$, including divisor-norm multiplicities. Moreover

$$\eta(m) := \chi_m(\zeta) = \zeta^{(Nm-1)/6}, \qquad (m, S) = 1. \tag{46}$$

The primitive character η has conductor $(2)^2(\delta)^3$, of norm 432; its powers $1, 5$ have conductor norm 432, powers $2, 4$ norm 27, and power 3 norm 16. All five powers are nonprincipal. These row conductors do not remove the raw zero at any prime dividing a sixth factor.

Put $\phi(t) = \exp(1 - 1/(1 - t^2))$ for $|t| < 1$ and zero otherwise. For integer D set $w_N = \operatorname{sgn}(A_N)\overline{\zeta^{(N-1)/6}}$ when a good squarefree ideal of norm N exists and $D < N < 2D$, and zero otherwise. Define one common, complex profile

$$W_D(y) = \sum_{D<N<2D} w_N \phi(4(Dy - N)). \tag{47}$$

Its disjoint supports lie strictly inside $(1,2)$, its sup norm is one for large D, and $\|W_D^{(j)}\|_\infty \ll_j D^j$. In particular the endpoints have value zero. This is a profile varying with scale, independent of the row, not a separate choice for each ideal or each row.

Proposition S63.1 (A moving-profile lower bound)**.** *For every fixed $0 < \theta < 1$, take $H = D^\theta$, $U = \sqrt{D}$, and $X = H^{1/6}$. As integer $D \to \infty$,*

$$\sum_{0<Nu\le H} |E_u(W_D)|^2 \ge (c_S + o(1))DH^{1/6}, \qquad c_S > 0. \tag{48}$$

This lower bound already comes from literal rows $u_v = \zeta\beta_v^6$, where v is an arbitrary good ideal of norm at most X and β_v is a generator. These rows have nonprincipal primitive character η, so $P_{u_v} = 0$. Their exact values are

$$E_{u_v}(W_D) = D^{-1/2} \sum_{\substack{m \text{ squarefree, } (m,Sv)=1 \\ D<Nm<2D}} |A_U(m)|. \tag{49}$$

Every collision with v remains deleted, even when the primitive conductor drops to 432. No squarefree condition is imposed on v.

At $H = D^{3/4}$ the lower exponent is $9/8$, whereas the current upper exponent is $7/6$. Thus this does not establish sharpness of $7/6$ or a new upper saving. For $0 < \theta < 2/3$, the term $DH^{1/6}$ dominates the current unsigned envelope; it is necessary up to subpower losses in this *varying-profile* class, even on nonprincipal rows.

For $k \ge 0$ put $M_k(W) = \sum_{j=0}^k \|(y\partial_y)^j W\|_\infty$. If, for every $\varepsilon > 0$, a bound uniform over all common profiles with constants independent of W, D were to assert

$$\sum_{0<Nu\le H} |E_u(W)|^2 \ll_\varepsilon M_k(W)^\alpha (HD)^\varepsilon H, \qquad \alpha \ge 0,$$

then necessarily $k\alpha \ge 1 - 5\theta/6$. At $\theta = 3/4$ this is $k\alpha \ge 3/8$. For a single fixed ε the necessary condition is only $k\alpha + (1+\theta)\varepsilon \ge 1 - 5\theta/6$. These are necessary cost comparisons, not sufficient estimates. A fixed-W result, a class with bounded seminorms, or the existing $M_4(W)^2$ dependence is not refuted.

The same rows expose a concrete centering error in the joint completion: removing the orientation mean of the actual correlation can delete a nonzero signed contribution on *nonprincipal* rows. The extended proof retains the literal unit phase, gives an exact four-orientation example, and explains why orientation projection alone supplies no saving. The standalone profile and coherent-mean inspector shows the complete finite norm groups, arbitrary sixth-factor masks and exact residual quantities. Its finite tables are diagnostics of this construction, not asymptotic estimates, training or GRH evidence.

S64 Proof and joint-mean boundary of the moving-profile construction

S64.1 Norm groups and primitive conductors

For a split rational prime p, a squarefree ideal contains zero, one or both conjugate primes. Its rational norm exponent is respectively $0, 1, 2$; at exponent one there are two choices, and at exponent two its divisor norms include two copies of p. For an inert prime, the exponent is zero or two, and

the nontrivial choice is the single prime ideal of norm p^2. Thus the rational norm determines μ, the divisor-norm multiset and r_U, even though it need not determine the ideal. For example norm 49 has squarefree ideal divisor norms $1, 7, 7, 49$, whereas the inert prime of norm 25 has divisor norms $1, 25$. Additional fixed S deletions remove some ideals without changing this constancy among those remaining. The high-window formula and its parity are the already proved facts of Section S49, not a new coefficient estimate.

At a good prime ideal $\mathfrak{p}$ of norm q, Euler's definition gives $\chi_{\mathfrak{p}}(\zeta) = \zeta^{(q-1)/6}$. All such norms are 1 modulo 6; for $a = 1 + 6r, b = 1 + 6s$ one has $(ab-1)/6 = r + s + 6rs$. Multiplicativity proves (46). Write $\chi_4(N) = (-1)^{(N-1)/2}$ for odd N. On good ideals,

$$\eta = \chi_4(N)\omega^{(N-1)/3},$$
$$\eta^2 = \omega^{2(N-1)/3} = \omega^{-(N-1)/3}, \qquad \eta^3 = \chi_4(N).$$

The inverse orientation in the middle expression is essential.

For completeness the local conductors are exact, not merely upper bounds. The quadratic norm character factors modulo $(2)^2 = (4)$. Since $\mathrm{Tr}(\delta\gamma) \in 3\mathbb{Z}$ for every $\gamma \in \mathcal{O}$, expanding the norm gives $N(\alpha + \delta^3\beta) \equiv N\alpha \pmod 9$. The cubic norm character therefore factors modulo δ^3. The coprime local witnesses

$$\alpha_2 = 13 + 6\omega, \quad N\alpha_2 = 127, \quad \alpha_2 - 1 = 2\omega\delta^3,$$
$$\alpha_3 = -11, \quad N\alpha_3 = 121, \quad \alpha_3 - 1 = 4\delta^2.$$

show minimality. The first is 1 modulo 2, is $1 + 2\omega$ modulo 4, and is 1 modulo δ^3; its η value is -1. The second is 1 modulo 4 and $-2 = 1 + \delta^2$ modulo δ^3, but not 1 modulo δ^3; its η value is ω and its square is ω^2. Thus the quadratic local exponent is two and the cubic one is three. The two moduli are coprime, giving the stated norms $432, 27, 16$ for the nontrivial powers. Global units have norm one, so these residue characters are unit trivial and descend to ideal Hecke characters in this class-number-one field. A norm-seven prime has value ζ, proving order six. A larger finite S does not principalize the character: CRT gives representatives of the same nontrivial ray classes avoiding the extra excluded primes. Primitive conductor reduction never restores raw primes excluded by S or the sixth factor.

S64.2 Exact alignment, with arbitrary overlapping sixth factors

The bumps in (47) have radius $1/(4D)$ and center N/D. Distinct supports are disjoint. At an active integer node their value is exactly w_N, and all endpoints are zero. Differentiation gives $\|W_D^{(j)}\|_\infty \le (4D)^j \|\phi^{(j)}\|_\infty$. Expanding $(y\partial_y)^j$ into ordinary derivatives and using $1 < y < 2$ gives $M_k(W_D) \ll_k D^k$.

Every nonzero ideal v has a generator β_v. The literal element $u_v = \zeta\beta_v^6$ is independent of that choice, since all units have sixth power one; distinct ideals yield distinct rows. Its norm is $(Nv)^6$. For squarefree good m, exact raw multiplicativity says

$$\chi_m(u_v) = \eta(m)\mathbf{1}_{(m,v)=1}.$$

Repeated primes of v are allowed and its collisions with m remain zero. The primitive row character is always η, so the actual principal indicator in (22) is zero when $\nu = 1$. Substituting the nodal profile proves (49) with no factor six, unit averaging or residue replacement.

S64.3 The coprime-triple density and the selected $h = 1$ block

Put $\kappa = \pi/(3\sqrt{3})$. Lattice counting, with six unit generators per ideal, gives

$$A_K(Y) := \#\{a : Na \le Y\} = \kappa Y + O(\sqrt{Y} + 1), \qquad A_K(Y) \le CY \quad (Y \ge 1).$$

This is the existing elementary ideal count. Let $T(D, X)$ count ordered triples (m, n, v) with m, n squarefree good, $D < Nm, Nn < 2D$, $Nv \le X$, v good and arbitrary, and the three ideals pairwise coprime. For $D, X \to \infty$, $X \le D$, we claim

$$\begin{aligned} T(D, X) &= (\kappa^3 \mathcal{C}_S + o(1))D^2 X, \\ \mathcal{C}_S &= \prod_{\mathfrak{p} \in S} (1 - t_{\mathfrak{p}})^3 \prod_{\mathfrak{p} \notin S} (1 - 5t_{\mathfrak{p}}^2 + 6t_{\mathfrak{p}}^3 - 2t_{\mathfrak{p}}^4) > 0, \\ t_{\mathfrak{p}} &= (N\mathfrak{p})^{-1}. \end{aligned} \tag{50}$$

Here a product over S means its prime ideals. To prove this, first impose the local restrictions at a fixed finite set of primes containing S. They are finite Boolean combinations of divisibility by $\mathfrak{p}$ and $\mathfrak{p}^2$ in m, n, and by $\mathfrak{p}$ in v. Finite inclusion-exclusion and the count for fixed ideal multiples give the product of the local densities, with relative error tending to zero. At a good prime the allowed cases are no positive valuation, a single valuation one in m or n, or an arbitrary positive valuation in v. The factor is therefore

$$(1 - t)^3 \left(1 + 2t + \frac{t}{1 - t}\right) = (1 - t)^2(1 + 2t - 2t^2) = 1 - 5t^2 + 6t^3 - 2t^4.$$

At S all three valuations must vanish, giving $(1 - t)^3$. The geometric series for v is essential; v is not squarefree.

An omitted failure belongs to one of five events: $\mathfrak{p}^2 \mid m$, $\mathfrak{p}^2 \mid n$, $\mathfrak{p} \mid (m, n)$, $\mathfrak{p} \mid (m, v)$ or $\mathfrak{p} \mid (n, v)$. Divisibility counts and $A_K(Y) \le CY$ bound each by $O(D^2X/(N\mathfrak{p})^2)$, uniformly; if a quotient scale is below one its count is zero. Thus the tail after norm P is

$$O\left(D^2 X \sum_{N\mathfrak{p} > P} (N\mathfrak{p})^{-2}\right) = o_P(D^2 X).$$

The series converges by the ideal count (or $\zeta_K(2) < \infty$). Take the large-scale limit first at fixed P, then $P \to \infty$. Each local factor is positive and differs from one by $O(t^2)$, proving (50). Strict annulus endpoints have no effect on the main term by the same lattice count. The secondary range $X/2 \le Nv < X$ has half this limiting volume, though it may be empty at small X.

On the selected literal unit class every expanded pair term in (49) is nonnegative. Keeping only $(m, n) = 1$ gives its exact $h = 1$ A–A contribution

$$\mathcal{E}^{AA}_{\text{selected},h=1} = D^{-1} \sum_{v \text{ good } Nv \le X} \sum_{\substack{m,n \text{ squarefree,good } D < Nm, Nn < 2D \\ (m,n)=(m,v)=(n,v)=1}} |A_U(m) A_U(n)| \ge D^{-1} T(D, X).$$

Consequently, with $X = H^{1/6}$,

$$\sum_{0 < Nu \le H} |E_u|^2 \ge \sum_{Nv \le X,\ v \text{ good}} |E_{u_v}|^2 \ge \mathcal{E}^{AA}_{\text{selected},h=1} \ge (\kappa^3 \mathcal{C}_S + o(1))DX.$$

The first inequality is positivity of the *outer squared row sum*. The selected $h = 1$ block does not lower-bound the signed $h = 1$ sum over other literal rows or a unit-complete family. No positive-semidefinite claim about a gcd-truncated Gram matrix is needed. This is A–A, not a positive-density assertion for C–C long-long weights, which may vanish. This proves Proposition S63.1 with $c_S = \kappa^3 \mathcal{C}_S$. The derivative tradeoff follows by comparing powers in $DH^{1/6} \ll D^{k\alpha + (1+\theta)\varepsilon} H$ and then letting $\varepsilon \downarrow 0$ only when the proposed bound holds for every positive ε.

S64.4 Projection preserves the coupled problem, not a saving

Use the exact joint notation of Section S58. For fixed h, q with $q \neq 1$ (including $h = 1$), let $n = \varphi(q)$, $\ell = N(hq)$, $Q = Nq$, and

$$c = \frac{Q^2 \xi_q(-\alpha)\tau_\beta(\xi_q)}{\ell^6},$$
$$\Gamma_r^F = c \sum_K B_F(K)\overline{\theta_r(K)}, \qquad A(K) = \sum_{r \in J} a_r \overline{\theta_r(K)}.$$

The active orientations J include all original restrictions, and the literal unit phase belongs to a_r. The distinct θ_r are orthonormal for $n^{-1}\sum_K$. If P projects onto their span, then

$$PB_F = (cn)^{-1} \sum_{r \in J} \Gamma_r^F \theta_r, \qquad \sum_K |PB_F(K)|^2 = (|c|^2 n)^{-1} \sum_r |\Gamma_r^F|^2.$$

Replacing B_F by PB_F in the full bilinear pairing is exact. Cauchy–Schwarz is precisely $|c\sum_K B_F A| \leq \|a\|_2 \|\Gamma^F\|_2$. The conductor and group-size factors cancel. Projection rank alone gives no arithmetic saving for this particular coupled vector.

For $s = \omega(q)$ let $C = \{K : x_{\mathfrak{p}}(K)^2 = 1 \text{ for all } \mathfrak{p} \mid q\}$. All orientations agree on C. If M_C denotes multiplication by its indicator, the matrix in the orthonormal orientation basis is

$$PM_C P = 3^{-s}\mathbf{1}\mathbf{1}^*, \qquad \lambda = |J|/3^s, \qquad \|(1-P)M_C P\|^2 = \lambda(1-\lambda).$$

Thus coherent restriction does not commute with projection. For example

$$\left| \sum_C (B_F - PB_F)A \right| \leq \sqrt{\lambda(1-\lambda)} \|(1-P)B_F\|_2 \|A\|_2,$$

where all norms in this inequality are counting ℓ^2 norms. Normalized norms would require an additional factor n on the right. Counting a coherent subset alone is not a lower bound for its actual signed pairing.

S64.5 A nonzero orientation mean on actual nonprincipal rows

For the selected rows let V be any finite set of good ideals, put the first five row factors equal to one, and retain the sixth-factor mask. For each h, q define $Z_{h,q} = \#\{v \in V : (v, hq) = 1\}$. After removing the separately displayed literal unit phase, the exact correlation is

$$\Gamma_r^F = Z_{h,q} = c \sum_K B_V(K)\overline{\theta_r(K)} \quad \text{for every active orientation } r.$$

To pass from the published C–C identity to A–A replace each r_U by $r_U - 1$. At $h = 1$, $m = r, n = q/r$, the coefficient is

$$a_r^{AA} = (r_U(m) - 1)(r_U(n) - 1) W_m \overline{W_n} \theta_{m,n}(\zeta), \qquad \mu(q) a_r^{AA} = |A_U(m) A_U(n)|.$$

This retains $\mu(q)$, the common complex profile and its conjugation, all deleted common primes, and the literal ζ phase. If $Z_{1,q} > 0$ and at least one active interior A–A coefficient is nonzero, the actual pairing is positive, whereas $\sum_r |\Gamma_r^F - \operatorname{mean} \Gamma^F|^2 = 0$. An orientation-variance-only replacement with no mean term is therefore false. The mean is not a principal residue: these rows have character η, with nonprincipal powers. This does not refute a properly formulated fixed-profile signed estimate.

A fresh exact example uses $D = 4000$, $U = 63$ and chosen prime ideals of norms $7, 13, 79, 67, 97$. Their product has four active orientations: the pairs 6097/7663, 7189/6499 and both reversals. Here $r_U = 2$, and A_U is -1 on the three-prime columns and $+1$ on the two-prime columns. The four good ideals with $10 \le Nv \le 20$ have norms $13, 13, 19, 19$. Exactly one collides with the chosen 13-prime, so $Z = 3$ for all four orientations. The exact selected $h = 1$ contribution is $12/4000 = 3/1000$ despite zero orientation variance. Dropping the literal ζ phase incorrectly changes the coefficient sum from 4 to -1, giving $-3/4000$. The coherent fraction is $4/243$ and the squared leakage is $956/59049$. The residue group and infinite Fourier tail are not estimated by this finite check. The fixed row conductor 432 must be distinguished from the pair-column conductor $q = mn$, whose norm here and in the asymptotic annulus is comparable to D^2.

The exact implementation at `examples/profile_uniformity/` keeps complete norm groups and arbitrary sixth factors, including distinct ideals of norm 49 with different masks. The saved six-scale examples, exact joint fixture and their independent checks certify finite identities only. Full-union masks, coherent means, fixed-profile uniformity and the remaining mixed/pure upper-envelope neighborhoods still require a new signed estimate. No determinant comparison, zero location, analytic or categorical Lean theorem, quantum result or GRH proof is obtained.

S65 A bound for the signed coprime slice and an exact character twist

The coprime-column contribution can be negative, so it needs its own bound. Fix the original ν, S, one common possibly complex smooth W supported in $[1, 2]$, $D > 1$, $U \ge 1$ and $U^2 \le D$. Consider an actual block

$$u = u_0 v_1 v_2^2, \quad A_j \le N v_j < 2A_j \ (j = 1, 2), \quad Nu \le H, \qquad R = A_1 A_2, \quad M = \max(A_1, A_2).$$

The good ideals v_1, v_2 are squarefree and coprime, with their literal primary/ray representatives; all other good factors are the unit ideal. Let F be the actual indicator, including capacity and every row mask. In the remaining neighborhoods the first factor is nonunit, so the primitive row character is nonprincipal and $E_u = I_u$ there. For $X, Y \in \{A_U, C_U, \mu\}$ define the actual signed slice

$$S_1^{XY} = \frac{1}{D} \sum_u F(u) \sum_{\substack{m,n \text{ squarefree,good} \\ (m,n)=1}} X(m) Y(n) \nu(m) \overline{\nu(n)} \chi_m(u) \overline{\chi_n(u)} W(Nm/D) \overline{W(Nn/D)}. \quad (51)$$

All columns in this formula are the original annular columns, including raw zero sentinels. No positive-semidefinite claim about the coprime truncation is made.

Proposition S65.1 (A bound for the actual signed $h = 1$ slice)**.** *Uniformly in the original cutoff and this actual block,*

$$|S_1^{XY}| \ll_{\nu,S,\epsilon} \|W\|_\infty^2 (HD)^\epsilon \min\left\{ R + \frac{DR}{M} + \frac{D^{2/3} R}{M^{1/3}},\ D + R^2 \right\}. \quad (52)$$

The exact identity underlying the estimate is

$$\begin{aligned} D S_1^{XY} &= \sum_{\substack{d \text{ squarefree} \\ (d,S)=1}} \mu(d) \sum_{u \text{ in the actual block} \atop (d,u)=1} T_d^X(u) \overline{T_d^Y(u)}, \\ T_d^X(u) &= \sum_{\substack{a \text{ squarefree} \\ (a,dS)=1}} X(da) \nu(a) \chi_a(u) W(Nd\,Na/D). \end{aligned} \quad (53)$$

In particular $X(da)$ is not replaced by $X(d)X(a)$, and the mask $(d,u)=1$ remains.

The proof applies the accepted block sieves inside each positive row norm in (53), then explicitly loses the d signs in an absolute triangle. At $H=D^{3/4}$ the two remaining power-scale neighborhoods $A_1 \asymp D^{5/12}, A_2 \asymp D^{1/6}$ and $A_1 \asymp D^{3/4}, A_2 \asymp 1$, with feasible capacity constants, both retain power $7/6$. The desired $3/4$ still needs a signed saving of $5/12$. This proves a separate signed-slice bound; it does not improve the published global envelope or supply the missing cancellation.

There is also an exact arithmetic description of the two-pivot phase. Put $\delta = 1+2\omega$ and, at each good prime ideal p, set $j(p) = J_p(\chi_p, \chi_p^2)$. Extend j multiplicatively on squarefree columns. Let $\Psi(a)$ be the unique generator of an ideal prime to 2δ satisfying $\Psi(a) \equiv 1 \pmod{2\delta}$.

Proposition S65.2 (The actual Jacobi column twist)**.** *The map Ψ is an algebraic Hecke character with exact finite conductor (2δ), of norm 12, and principal convention $\Psi((\alpha)) = \alpha$ for $\alpha \equiv 1 \pmod{2\delta}$ (infinity type $(1,0)$). For good squarefree columns,*

$$(j(p)) = p, \quad -j(p) = \Psi(p), \qquad \mu(m)j(m) = \Psi(m). \tag{54}$$

Define $\vartheta(m) = \Psi(m)/\sqrt{Nm}$ and $\rho(m) = j(m)/\sqrt{Nm} = \mu(m)\vartheta(m)$ on those columns. In the high window the actual gauged coefficients are $b_A = (r_U - 1)\vartheta\nu W$, $b_C = r_U\vartheta\nu W$ and $b_\mu = \vartheta\nu W$.

This does not replace the original row kernel. Its compensating inverse gauge remains, so all ordinary coefficient/operator norms are unchanged. The new character has nonzero infinity type and is not silently covered by a theorem stated only for finite-order twists. The identities in (54) involving μ must not be extended to arbitrary nonsquarefree sixth factors.

A fresh fixed-profile example takes $\nu = 1$, S the primes above $2, 3$, literal unit one and primary row generators 1 modulo 3. It lies in an actual $A_6 = 1$ block at $D = 60000$, $U = 244$, $H = 3833$, $A_1 = 37$, $A_2 = 7$. All 20 rows of the stated literal class are nonprincipal. A selected q has exactly two legal column orientations, of norms $97051/96091$ and its reversal. Their correlations in the ω basis are $(-4,-2)$ and $(-2,2)$, with mean -3. Their selected contribution is $S_{1,q}^{AA} = -1/10000$; deleting the mean gives zero. The centered squared norm is 6, not zero, and the proposed variance-only Cauchy bound would incorrectly require $36 \le 12$. This refutes that centering rule in a fixed-profile row block, not the asymptotic target. Other q contributions remain. The coprime-kernel inspector shows the actual rows, raw zeros, two-column Gram matrix and compensated gauge; no training or arithmetic spectrum is inferred from that finite view.

S66 Proof of the signed-slice budget and its surviving analytic burden

S66.1 Gcd inversion before an absolute bound

Insert $\mathbf{1}_{(m,n)=1} = \sum_{d \mid m,\ d \mid n} \mu(d)$ in (51), and write $m = da, n = db$. Squarefreeness forces $(a,d) = (b,d) = 1$. The common factors give $\nu(d)\overline{\nu(d)} = 1$ and $\chi_d(u)\overline{\chi_d(u)} = \mathbf{1}_{(d,u)=1}$. This proves (53), retaining the original coefficient $X(da)$ and the full row-zero mask. For $X = Y$ each inner squared row sum is nonnegative; its $\mu(d)$-weighted sum need not be.

Put $L = D/Nd$. Whenever a column survives, $L \ge 1/2$. The actual divisor-assignment bound and ideal counting give

$$\sum_{\substack{a \text{ squarefree} \\ (a,dS)=1}} |X(da)\nu(a)W(Nd\,Na/D)|^2 \ll_\eta D^\eta \|W\|_\infty^2 L.$$

Indeed $N(da) \le 2D$, $|X(da)| \ll_\eta D^\eta$ uniformly in d, U, and there are $O(L)$ ideals of norm at most $2L$ for $L \ge 1/2$. This includes C_U by $C_U = A_U + \mu$ on the active annulus. There is no multiplicativity assertion for $X(da)$. For $L \ge 1$, the powered block bound and the ordinary-conductor comparison of Section S59, followed by Cauchy for the cross term, imply

$$|G_d^{XY}| \ll (HD)^\eta \|W\|_\infty^2 L \min\left\{R + \frac{LR}{M} + \frac{L^{2/3}R}{M^{1/3}},\ L + R^2\right\}, \tag{55}$$

where G_d^{XY} denotes the inner row sum in (53). The raw (d, u) restriction is only enlarged inside these positive norms; u_0 is a fixed column multiplier of modulus at most one. For $1/2 \le L < 1$, only the unit column can survive. The direct row count is $O(R)$, and the right side of (55) is at least a constant times R since $R \ge 1$ and $L \ge 1/2$. Thus this endpoint does not require a sieve with an unstated subunit length.

Now take the absolute triangle in d explicitly. The two estimates may be summed separately and their minimum taken. The elementary ideal sums

$$\sum_{Nd \le 2D} \frac{1}{Nd} \ll \log(2D), \qquad \sum_d (Nd)^{-2} \ll 1, \qquad \sum_d (Nd)^{-5/3} \ll 1$$

turn the first estimate into $R\log(2D) + DR/M + D^{2/3}R/M^{1/3}$ after division by D; the second gives $D + R^2\log(2D)$. Absorb the logarithms with smaller internal exponents to obtain (52). All raw zeros, literal/ray/squarefree/capacity restrictions were retained before the stated positive enlargements. No bound for an indefinite submatrix was inferred from positivity of a larger Gram matrix.

At the mixed power-scale point the powered exponents are $7/12, 7/6, 10/9$, and the ordinary exponents are $1, 7/6$. At the pure point they are $3/4, 1, 7/6$ versus $1, 3/2$. These are upper-bound exponents with feasible dyadic constants, not actual energy lower bounds. A sufficient $h = 1$ target for the A–A combination is

$$\left|\sum_d \mu(d) G_d^{AA}\right| \ll_{\nu,S,W,\epsilon} DH(HD)^\epsilon. \tag{56}$$

At $H = D^{3/4}$ this asks for power $7/4$ before division by D, while (52) gives $13/6$, leaving $5/12$. This is sufficient for this slice, not a necessary bound for every separate C–C/C–μ term if their signed assembly cancels further. The quantities G_d depend on prime supports, $X(da)$ and row deletions; they are not an independent smooth function of Nd to which a scalar Möbius partial-sum estimate can simply be applied.

S66.2 The correctly oriented Jacobi generator

At a good prime ideal p, let $n = Np$ and $\chi = \chi_p$ be the literal positive sextic residue character. In $\mathcal{O}/p$, its value reduces to $x^{(n-1)/6}$. For $j(p) = J(\chi, \chi^2)$, the elementary Gauss–Jacobi identity and the three nontrivial Gauss norms give $|j(p)|^2 = n$. Reduction in that *same* residue field gives

$$j(p) \equiv \sum_x x^{(n-1)/6}(1-x)^{(n-1)/3} = 0 \pmod p.$$

Every exponent after expansion lies strictly between 0 and $n-1$, so the finite-field power sums vanish. Hence $(j(p)) = p$. This includes an inert prime of norm p_0^2, and inverting the literal character conjugates the ideal at a split prime.

This raw sum equals the old $J(\chi^2,\chi^3)$: the quotient of the two Gauss–Jacobi expressions is $G(\chi)G(\chi^5)/G(\chi^3)^2=1$. It differs from the theta-curve Frobenius normalization by its explicitly stated factor $-\chi^3(-1)$; that sign is not imported silently. No new Gauss or Jacobi norm theorem is claimed.

Let $f=(2\delta)$. The six units map bijectively onto $(\mathcal{O}/f)^\times$: the components at 2 and δ have orders three and two, with complementary kernels. Since $\mathcal{O}$ is a PID, every ideal prime to f has a unique generator $\Psi(a)\equiv 1 \pmod f$. Uniqueness gives multiplicativity. If $u_f(\alpha)$ is the unit having the residue of α modulo f, then

$$\Psi((\alpha))=\alpha/u_f(\alpha), \qquad |\Psi(a)|^2=Na.$$

This explicitly defines the finite part and the principal infinity-type convention. The finite conductor divides f. A proper divisor omits a component: the unit -1 is 1 modulo 2, and ω is 1 modulo δ. A smaller defining modulus would require the unit ideal to have value -1 or ω. Thus the finite conductor is exactly f.

To identify the actual $j(p)$, now reduce in the *character-value* ring, a different operation from reduction modulo p above. Modulo δ, a sixth root v satisfies $v=v^3$ and $v^2=1$. The zero branches at $x=0,1$ are kept, so $j(p)\equiv\sum_{x\neq 0,1}\chi^3(x)=-1 \pmod\delta$. Modulo 2, $v=v^4$, and $J(\chi,\chi^2)\equiv J(\chi^4,\chi^2)=-\chi^4(-1)=-1$. The inverse-character Jacobi identity here follows by the bijection $x/(1-x)$ onto the nonzero field elements other than -1. Consequently $-j(p)\equiv 1 \pmod f$, proving $-j(p)=\Psi(p)$. Multiplication proves (54) on squarefree columns. For nonsquarefree ideals μ vanishes, so that last identity must not be extended to arbitrary sixth factors. This construction establishes the fixed character directly; it assumes no elliptic modularity, spectral comparison, or distribution of prime arguments.

S66.3 The actual two-factor completion, not a new row kernel

Freeze the literal u_0,v_3,v_4,v_5,v_6 and put $Z=u_0v_3^3v_4^4v_5^5v_6^6$. The last factor is arbitrary, with its allowed overlaps. For $h=1$, write $q=mn\neq 1$, $Q=Nq$, $\theta=\chi_m\overline{\chi_n}$ and $\xi_q=\theta^3$. For $q=(\beta)$ use the original inverse-different pairing $e_\beta(z)=\exp(2\pi i\,\mathrm{Tr}(z/(\delta\beta)))$. Let $F(v_1,v_2)$ be the exact common finite weight, retaining the first-five coprimality, primary/ray and capacity restrictions. It must be independent of the column orientation. Define

$$\widehat{F}_q(k_1,k_2)=\sum_{v_1,v_2}F(v_1,v_2)e_\beta(-k_1v_1-k_2v_2),$$
$$T_{q,F}(\theta)=\sum_{k_1,k_2\in(\mathcal{O}/q)^\times}\widehat{F}_q(k_1,k_2)\overline{\theta(k_1k_2^2)}.$$

Periodization sums every actual representative; no injectivity assumption is needed. The primitive powers θ,θ^2 have zero Gauss sums at nonunit frequencies, including at inert primes. Exact inversion gives

$$\sum F\theta(v_1)\theta(v_2)^2=Q^{-2}\tau_\beta(\theta)\tau_\beta(\theta^2)T_{q,F}(\theta)=Q^{-2}J_q(\theta,\theta^2)\tau_\beta(\xi_q)T_{q,F}(\theta).$$

Jacobi sums CRT without additive-generator multipliers, so $J_q=j(m)\overline{j(n)}=\sqrt{Q}\,\rho(m)\overline{\rho(n)}$. The quadratic central phase is the already recorded period-four law

$$\gamma(\beta)=\tau_\beta(\xi_q)/\sqrt{Q}=\tfrac12\{1+i^{-b}+i^a+i^{b-a}\}, \qquad \beta=a+b\omega.$$

For composite squarefree q, identifying $\tau_\beta(\xi_q)$ with $\sum_x e_\beta(x^2)$ requires CRT expansion of the local root counts; $1+\xi_q$ is not the root-count function on a composite ring. The known quadratic Gaussian evaluation is reused, not retested as a new phase theorem. Its pairing is exactly the displayed e_β.

With $b_X = \rho X \nu W$, the actual two-pivot identity is

$$DS_1^{XY} = \sum_q \frac{\gamma(\beta_q)}{Q} \sum_{mn=q} b_X(m)\overline{b_Y(n)}\theta_{m,n}(Z)T_{q,F}(\theta_{m,n}). \tag{57}$$

The two Gauss magnitudes, equivalently the central Gauss and Jacobi magnitudes, give exactly $1/Q$. There is no extra $\xi_q(-1)$ in this two-factor identity; the full six-factor formula of Section S57 retains that sign. If β changes to $u\beta$, both γ and T acquire the same quadratic sign $\xi_q(u)$, which cancels in their product. If Z shares any prime with q, $\theta(Z) = 0$, including at a sixth-power collision. No dropped primitive-conductor prime is restored.

Multiplying columns by ρ changes the kernel by its compensating inverse unitary gauge. It preserves operator norms and eigenvalues even for an indefinite coprime-truncated matrix. A fixed-character coefficient estimate cannot replace a bound for the actual q-dependent $T_{q,F}$. At the pure endpoint $v_2 = c$ with $(c, q) = 1$, exact summation gives

$$T_{q,F}(\theta) = \theta(c)^2\tau_\beta(\overline{\theta^2}) \sum_{k_1 \in (\mathcal{O}/q)^\times} \widehat{F}_1(k_1)\overline{\theta(k_1)}.$$

Since $\theta^2(-1) = 1$, the extra Gauss pair multiplies to Q, returning the original one-pivot scalar $\tau_\beta(\theta)/Q$. A nonunit c gives the original zero. Thus a putative central-phase gain cannot be retained after discarding a literal length-one Fourier factor.

S66.4 A central reciprocal remains after the gauge

On squarefree columns put $B_U = \mu A_U$ and $\vartheta = \Psi/\sqrt{N}$. The original ternary assignment gives exactly

$$B_U(m) = \sum_{\substack{abc=m \\ Na,Nb \le U}} \mu(c), \qquad \rho(m)A_U(m) = \vartheta(m)B_U(m).$$

All factors are squarefree and pairwise coprime. Fix a nonzero numerator z and keep the full deleted set $T = S \cup \operatorname{supp}(z)$. Write Ξ for the unitary primitive character of $\vartheta\nu\kappa_z$, with these raw deletions retained. Its nonzero infinity type prevents a silent reuse of a finite-order-only moment theorem. For $\Re s > 1$,

$$J_U^{\sharp,T}(s) = \sum_{m \text{ squarefree},\ (m,T)=1} B_U(m)\Xi(m)(Nm)^{-s} = N_U^T(s)/L^T(s,\Xi),$$

$$N_U^T(s) = \sum_{\substack{a,b \text{ squarefree} \\ (a,b)=1,\ (ab,T)=1 \\ Na,Nb \le U}} \frac{\Xi(ab)}{(Nab)^s} \prod_{p \mid ab} (1 - \Xi(p)(Np)^{-s})^{-1}.$$

Fixing a, b leaves the Möbius Euler product in c; this proves the identity by absolute convergence. The numerator is finite and analytic for $\Re s > 0$. At $U = 1$ it is one, so the series is exactly $1/L^T(s,\Xi)$. That boundary identity is not a high-window counterexample. It shows why cancellation of central reciprocal poles cannot simply be inferred from the column-character identification.

For $x_p = \Xi(p)(Np)^{-s}$, $\alpha_p = (Np)^{-v}$ and $\beta_p = (Np)^{-w}$, the gauged three-placement Euler factor is $1 + x_p(\alpha_p + \beta_p - 1)$, with the opposite signs from the ungauged factor in Section S49. In the initial absolute domain it gives

$$\prod_{p \notin T} \{1 + x_p(\alpha_p + \beta_p - 1)\} = \frac{L^T(s+v,\Xi)L^T(s+w,\Xi)}{L^T(s,\Xi)} \prod_{p \notin T} H_p,$$

where

$$H_p = \frac{\{1 + x_p(\alpha_p + \beta_p - 1)\}(1 - x_p\alpha_p)(1 - x_p\beta_p)}{1 - x_p}.$$

Expansion gives

$$H_p - 1 = \frac{x_p^2(\alpha_p + \beta_p - \alpha_p^2 - \beta_p^2 - \alpha_p\beta_p)}{1 - x_p} + \frac{x_p^3\alpha_p\beta_p(\alpha_p + \beta_p - 1)}{1 - x_p}.$$

Thus the correction converges normally in the open region $\Re s > 0$ where the real parts of all of

$$2s + v,\ 2s + w,\ 2s + 2v,\ 2s + 2w,\ 2s + v + w,$$
$$3s + 2v + w,\ 3s + v + 2w,\ 3s + v + w$$

exceed one. In particular, for $\Re s \geq 1/2 + \eta$ and $\Re v, \Re w \geq 0$, its terms differ from one by $O_\eta((Np)^{-1-2\eta})$. Deleted primes only remove factors. Finite correction factors may vanish; no zero-free product is asserted. The displayed factorization is first an absolute-domain identity and is used meromorphically only where the functions have been continued. At $v = w = 0$ it simplifies to $L^T(s, \Xi)/L^T(2s, \Xi^2)$, a special slice cancellation which does not remove the general central reciprocal. No finite-height inversion is declared exact without truncation errors; no new contour, reciprocal family moment or analytic saving is proved. Most importantly, this fixed-z coefficient series is not the actual q-dependent joint weight in (57).

S66.5 The fixed-profile remaining-sector diagnostic

Use the fixed smoothstep $s(t) = e^{-1/t}/(e^{-1/t} + e^{-1/(1-t)})$ for $0 < t < 1$, extended by zero and one at the two sides, and set $W(y) = s(8(y-1))s(8(2-y))$. It is smooth, supported in $[1, 2]$, and identically one on $[9/8, 15/8]$, independently of D. Take $\nu = 1$ and S the primes above $2, 3$. At $D = 60000, U = 244, H = 3833$, select the actual block $A_1 = 37, A_2 = 7$, literal unit one and primary row generators 1 modulo 3. Complete enumeration gives 44 candidate pairs, with two coprimality and 22 capacity rejections, leaving 20 distinct nonprincipal rows. The first outside-S valuation is one in each row.

Choose $\mathfrak{p}_p = (p, \omega - \lambda_p)$ for

$$(p, \lambda_p) \in \{(37, 10), (43, 6), (61, 13), (307, 17), (313, 98)\}.$$

Their product q has exactly two legal annular column orientations: 97051/96091 and its reversal. Direct small- and long-divisor assignments give $r_U = (0, 2)$, $\mu = (-1, 1)$, $C_U = (0, 2)$ and $A_U = (1, 1)$. Both profile values are exactly one. Twelve orientation/row evaluations are raw zeros. Exact residue-field exponentiation yields

$$\Gamma_1 = -4 - 2\omega, \qquad \Gamma_2 = -2 + 2\omega, \qquad \Gamma_{\mathrm{av}} = -3.$$

Here Γ_{av} is the orientation average. With $a^{AA} = (-1, -1)$ and $\mu(q) = -1$,

$$S_{1,q}^{AA} = -6/60000 = -1/10000, \qquad \sum_r a_r(\Gamma_r - \Gamma_{\mathrm{av}}) = 0.$$

The centered norm is $\sum_r |\Gamma_r - \Gamma_{\mathrm{av}}|^2 = 6$ and $\sum_r |a_r|^2 = 2$, whereas $|\sum_r a_r\Gamma_r|^2 = 36 > 12$. A variance-only Cauchy bound therefore fails even with nonzero variance. It is not a negative full row energy or an asymptotic counterexample.

The corresponding *selected two-column* Gram matrix has diagonal $14, 20$ and the displayed conjugate off-diagonal entries. Its eigenvalues are $17 \pm \sqrt{21}$ and its A quadratic form is 28 before division by D. Removing the diagonal for coprimality gives eigenvalues $\pm 2\sqrt{3}$ and the A form -6. Other original columns and q values have not been suppressed in a full-family claim: this display is a selected-q contribution with its complete stated row block. All finite data and scalar identities have independent exact checks; no finite spectrum or learned label implies a determinant or GRH theorem. The next estimate must retain the signed mean, reciprocal burden and actual coupled q weights, closing the still-open 5/12 gap.

S67 The signed divisor mean and a complete-column laboratory

The preceding signed-slice estimate loses the sign of the common ideal divisor. We now retain that variable in an exact analytic reduction and in a complete finite computation. Neither supplies the missing 5/12 saving at $H = D^{3/4}$. The computations below use a new fixed profile; they do not fill in the uncomputed columns of the earlier broad-plateau profile.

Let $\mathcal{C}$ be original squarefree good ideal columns in $[D, 2D]$, and let $\mathcal{R}$ be an actual finite literal row block, with nonnegative multiplicities w_u. All shared-prime zeros of the original $\chi_m(u)$ remain. For arbitrary complex coefficient vectors define the linear-first form

$$DQ_1(x, y) = \sum_{u \in \mathcal{R}} w_u \sum_{\substack{m,n \in \mathcal{C} \\ (m,n)=1}} x_m \overline{y_n} \chi_m(u) \overline{\chi_n(u)}. \tag{58}$$

The original component is obtained with $x_m = X_U(m)\nu(m)W(Nm/D)$ and the corresponding y_m. On the nonprincipal row blocks used below the original residue term vanishes. Equation (58) is a raw coprime form on a generic block; principal rows would still require their separate residue assembly.

For every ideal divisor of an active column put

$$F_d(x; u) = \sum_{m \in \mathcal{C},\ d|m} x_m \chi_m(u), \qquad Z_d(x, y) = \sum_u w_u F_d(x; u) \overline{F_d(y; u)}.$$

The exact identity is

$$DQ_1(x, y) = \sum_d \mu(d) Z_d(x, y). \tag{59}$$

There is no extra $\mu(q)$: the vectors already contain the original coefficients. Full divisors $d = m$ are essential to cancel self-pairs. Divisors and squarefreeness refer to ideals, including distinct ideals with the same rational norm.

Proposition S67.1 (A weighted divisor-incidence lift)**.** *Suppose $\lambda_d \geq 0$ is an arbitrary-coefficient bound*

$$\sum_u w_u |F_d(x; u)|^2 \leq \lambda_d \sum_{m \in \mathcal{C},\ d|m} |x_m|^2.$$

With $B(m) = \sum_{d|m} \lambda_d$ and $\|x\|_B^2 = \sum_m B(m)|x_m|^2$, one has

$$|DQ_1(x, y)| \leq \|x\|_B \|y\|_B. \tag{60}$$

For a finite literal phase table one may take the exactly computed

$$\lambda_d = E_d := \sum_u w_u \#\{m \in \mathcal{C} : d \mid m,\ \chi_m(u) \neq 0\}.$$

This last choice is a Frobenius certificate, not a new analytic estimate. For $x = t + e$ it gives the finite perturbation bound

$$|Q_1(x,x) - Q_1(t,t)| \le D^{-1}(2\|t\|_B\|e\|_B + \|e\|_B^2). \tag{61}$$

Proof. Insert ideal-divisor inversion in (58). Apply row Cauchy to each Z_d and weighted Cauchy to the resulting sum over d. Then $\sum_d \lambda_d\|x_d\|_2^2 = \|x\|_B^2$. The squared Frobenius norm of the actual F_d matrix is exactly E_d. Expand $Q_1(t+e,t+e) - Q_1(t,t)$ to obtain the last assertion. Positivity of the coprime-truncated matrix was never assumed. □

Using an accepted positive-row sieve bound for λ_d only recovers the old exponent up to ideal-divisor factors. It does not estimate the signed sum before the triangle in (59).

There is also a genuinely signed scalar expansion. Fix squarefree residual ideals a, b, which may overlap, and an actual row u. Keep the full deleted set $\mathcal{T} = S \cup \operatorname{supp}(abu)$ and set

$$\mathcal{Z}_{U;a,b,\mathcal{T}}(s) = \sum_{\substack{d \text{ squarefree} \\ (d,\mathcal{T})=1}} \frac{\mu(d)A_U(da)A_U(db)}{(Nd)^s}, \qquad \Re s > 1.$$

For four positive cutoff variables put $P_p = 1-(Np)^{-v_1}-(Np)^{-v_2}$ and $Q_p = 1-(Np)^{-w_1}-(Np)^{-w_2}$. The common-divisor Euler factor is $1-(Np)^{-s}P_pQ_p$. Its first-order factorization has four numerator and five denominator zeta factors:

$$\prod_{p\notin\mathcal{T}}(1-(Np)^{-s}P_pQ_p) = \frac{\prod_{i=1}^2 \zeta_K^{\mathcal{T}}(s+v_i)\zeta_K^{\mathcal{T}}(s+w_i)}{\zeta_K^{\mathcal{T}}(s)\prod_{i,j=1}^2 \zeta_K^{\mathcal{T}}(s+v_i+w_j)}\,\mathcal{H}_{\mathcal{T}}(s;v,w). \tag{62}$$

The correction is normally convergent at $\Re s \ge \frac12+\eta$, $\Re v_i, \Re w_j \ge 0$, for each fixed $\eta > 0$. This says nothing about zeros or residues of the zeta ratio. Finite-height Perron kernels, with their explicit endpoint/error costs, recover the cutoffs by successive limits.

A direct finite cutoff expansion gives more explicit information:

$$\mathcal{Z}_{U;a,b,\mathcal{T}}(s) = \frac{\mathcal{N}_{U;a,b,\mathcal{T}}(s)}{\zeta_K^{\mathcal{T}}(s)}. \tag{63}$$

The numerator is a finite sum over eight nonzero joint short-divisor states, holomorphic on $\Re s > 0$. Uniformly for $\sigma \ge \sigma_0 > 0$,

$$|\mathcal{N}_{U;a,b,\mathcal{T}}(\sigma+i\tau)| \ll_{K,\sigma_0,\epsilon} (Na\,Nb)^{\epsilon}U^{4\max(1-\sigma,0)+\epsilon}.$$

The costs near $\sigma = 1/2$ persist. The remaining long state has the Möbius sign, but the eight finite short states cannot be omitted. A fixed-residual reciprocal zeta factor also does not prove an uncancelled pole in the complete coherent sum.

Exact complete-column reference. Choose once $\delta = 1/240000$ and $W_\delta(y) = s((y-1)/\delta)s((2-y)/\delta)$, where s is the smooth step from zero to one. For the finite grid $D \le 60000$ every strict integer norm node has $W_\delta = 1$; endpoints vanish. Its derivatives cost $O_k(\delta^{-k})$. The three cases have different declared row blocks, not a common asymptotic scaling experiment. They use principal ν, S above $2, 3$, literal class one and nonprincipal actual rows.

D	columns	divisors	rows	S_1^{AA}
144	41	53	5	$-19/36$
576	164	212	11	$-221/288$
60000	16953	21750	20	$-5119/5000$

In the last case the exact even/odd Möbius masses are 543416 and 604844, giving numerator -61428 before division by D. All 339060 raw column–row values, including 47908 zeros, and all nine cross components are retained. In particular 1361 columns have nonsquarefree rational norm while remaining squarefree as ideals. The signed-divisor inspector exposes the actual states and dimensions. These finite negative slices are neither whole-family asymptotic counterexamples nor a positive arithmetic determinant or GRH result.

S68 Proofs and limits of the signed-divisor development

S68.1 Actual divisor incidence and a finite error certificate

The identity $\mathbf{1}_{(m,n)=1} = \sum_{d|m,\ d|n} \mu(d)$ is an identity of ideal divisors. Expanding the finite sums proves (59). When $m = da$ is squarefree, $(d, a) = 1$. If $(d, u) = 1$, multiplication by $\chi_d(u)$ is a rowwise unit phase; if $(d, u) > 1$ it is the original zero. For original $x_m = X_U(m)\nu(m)W(Nm/D)$, precisely $F_d(x; u) = \nu(d)\chi_d(u)T_d^X(u)$ on $(d, u) = 1$, and it is zero otherwise. Thus their row norms agree on the actual restricted support. For general vectors the coefficient x_{da} remains unfactored. No identity $X_U(da) = X_U(d)X_U(a)$ is used. Residual a, b may overlap.

Write $\|x_d\|_2^2 = \sum_{d|m} |x_m|^2$. The hypotheses of Proposition S67.1 give

$$\begin{aligned} |DQ_1(x, y)| &\le \sum_d |Z_d(x, y)| \le \sum_d \lambda_d \|x_d\|_2 \|y_d\|_2 \\ &\le \left(\sum_d \lambda_d \|x_d\|_2^2\right)^{1/2} \left(\sum_d \lambda_d \|y_d\|_2^2\right)^{1/2} = \|x\|_B \|y\|_B. \end{aligned}$$

This is a divisor-incidence lift of positive-row bounds to an indefinite coprime form. It is not an inference from positivity of a larger Gram matrix. In particular $B(m) \le \lambda_{\max} \tau_{\text{ideal}}(m)$, with $\tau_{\text{ideal}}(m) = 2^{\omega(m)}$ on squarefree ideals. The subpolynomial divisor bound recovers the existing exponent; this triangle does not supply the desired signed cancellation.

For the original unweighted two-pivot block ($w_u = 1$), the accepted arbitrary-coefficient inputs at $L = D/Nd \ge 1$ allow

$$\lambda_d \ll_\epsilon (HD)^\epsilon \min\left\{R + \frac{LR}{M} + \frac{L^{2/3}R}{M^{1/3}},\ L + R^2\right\}, \qquad R = A_1A_2, \quad M = \max(A_1, A_2).$$

Their constants are classical-input constants, not computed numbers. Any removal of a d-row mask occurs only inside the positive norm. For $1/2 \le L < 1$, $Na \in [L, 2L]$ permits at most the unit ideal; its operator norm is bounded directly by the actual row measure. For an empty subsequence take $\lambda_d = 0$.

For the fully computable choice $\lambda_d = E_d$, the matrix entries are $\sqrt{w_u}\chi_m(u)$ on d-divisible columns. The squared Frobenius norm is precisely the stated raw nonzero-entry count. It is valid for arbitrary coefficient vectors, not just the saved trajectory. No unknown sieve constant enters that finite certificate. For approximations t_x, t_y with errors $e_x = x - t_x, e_y = y - t_y$,

$$|Q_1(x, y) - Q_1(t_x, t_y)| \le D^{-1}(\|e_x\|_B \|y\|_B + \|t_x\|_B \|e_y\|_B).$$

The AA specialization follows by expanding three terms. To verify it without floating square roots, put $T = \|t\|_B^2, E = \|e\|_B^2$ and $L = D|Q_1(x, x) - Q_1(t, t)| - E$. If $L > 0$, the certificate is the equivalent real inequality $L^2 \le 4TE$; if $L \le 0$ it is immediate. In the saved exact $\mathbf{Z}[\omega]$ computations with rational weights, these are integer/rational checks rather than floating-root tests. These are finite coefficient-error statements. They do not prove a varying-family remainder estimate, spectral positivity or a determinant comparison.

S68.2 A signed common-divisor Euler product

From here restore ALL original annular squarefree good columns, not an arbitrary subset from the vector lemma, and use the actual unweighted raw row block of Section S65. On the stated nonprincipal blocks it is the corresponding component of the residue-subtracted moment. On a block containing principal rows the raw identity remains valid, but the original principal correction is additional and is not discarded. Expand the exact d identity while keeping its sign. For fixed a, b, u the inner sum is

$$\sum_{\substack{d \text{ squarefree} \\ (d,Sabu)=1}} \mu(d) A_U(da) A_U(db) W(Nd\,Na/D)\overline{W(Nd\,Nb/D)}. \tag{64}$$

The exact outer coefficient is $\nu(a)\overline{\nu(b)}\chi_a(u)\overline{\chi_b(u)}$. The residual a, b are individually squarefree but need not be coprime. If their quotient character loses a common prime, that prime remains in $\mathcal{T} = S \cup \operatorname{supp}(abu)$ and in every original zero. The common d character cancels only to $\mathbf{1}_{(d,u)=1}$.

For $\Re s > 1$ the series $\mathcal{Z}$ is absolutely convergent, with coefficient majorant $3^{\omega(a)+\omega(b)}9^{\omega(d)}$, uniformly in U. Use four positive Perron variables. Write $\mathcal{E}_{\mathcal{T}}(s; v, w) = \prod_{p \notin \mathcal{T}}(1-(Np)^{-s}P_pQ_p)$. The fixed residual prefactor is $P_a(v)Q_b(w) = \prod_{p|a} P_p(v) \prod_{p|b} Q_p(w)$; a common prime of a, b contributes both factors. The common d prime has nine joint cutoff positions and the sign $\mu(p) = -1$, giving local factor $1 - zPQ$ with $z = (Np)^{-s}$, $\alpha_i = (Np)^{-v_i}$ and $\beta_j = (Np)^{-w_j}$. Thus the exact correction in (62) is

$$H_p = \frac{(1 - zPQ)\prod_{i=1}^{2}(1 - z\alpha_i)\prod_{j=1}^{2}(1 - z\beta_j)}{(1 - z)\prod_{i,j=1}^{2}(1 - z\alpha_i\beta_j)}. \tag{65}$$

The linear coefficient of numerator and denominator is $-1 - \sum_{i,j} \alpha_i\beta_j$, fixing all nine zeta signs. At all four auxiliary variables zero the product is $1/\zeta_K^{\mathcal{T}}(s)$ and $H_p = 1$. At $v_2 = w_2 = 0$ it is $1/\zeta_K^{\mathcal{T}}(s + v_1 + w_1)$, again with $H_p = 1$. These are generating-function checks, not finite-cutoff identifications.

For $\Re s \geq 1/2 + \eta$, $\Re v_i, \Re w_j \geq 0$, the degree-five numerator and denominator of (65) have equal constant and linear coefficients. The sums of absolute coefficient majorants in degrees two through five are at most 146 and 26. Since $|PQ| \leq 9$ and $|\alpha_i|, |\beta_j| \leq 1$,

$$|H_p - 1| \leq C_\eta (Np)^{-1-2\eta}, \qquad C_\eta = \frac{172}{(1 - 7^{-1/2-\eta})^5}. \tag{66}$$

Indeed the five denominator factors have modulus at least $1 - |z|$; good prime ideal norms are at least seven. Ideal counting proves normal convergence, uniform in deleted primes and vertical phases, and

$$|\mathcal{H}_{\mathcal{T}}| \leq \exp\left(C_\eta \sum_p (Np)^{-1-2\eta}\right).$$

Local correction factors can vanish; no inverse-correction bound is asserted. Normal convergence controls neither the four numerator nor five reciprocal zeta factors in the critical region. Their poles, zeros and crossed residues cannot be suppressed.

S68.3 Finite Perron heights and the exact eight-state numerator

Set $V = \lfloor U \rfloor + 1/2$ and $\Delta_V = \min_{n \geq 1} |\log(V/n)| \geq 1/(2V+1)$. For four finite vertical heights T_j on positive real lines c_j, let $I_T(s)$ be the fourfold integral of $P_aQ_b\mathcal{E}_{\mathcal{T}}(s; v, w)$ against $\prod_j V^{z_j}\, dz_j/(2\pi i z_j)$.

In the initial half-plane $\sigma = \Re s > 1$, finite-height Fubini is justified by the absolute placement majorant. Put

$$A_{\mathrm{abs}}(\sigma, c) = \prod_{p|a}(1 + q^{-c_1} + q^{-c_2}) \prod_{p|b}(1 + q^{-c_3} + q^{-c_4}) \\ \cdot \prod_{p \notin \mathcal{T}} \{1 + q^{-\sigma}(1 + q^{-c_1} + q^{-c_2})(1 + q^{-c_3} + q^{-c_4})\}, \quad q = Np.$$

Then

$$|I_T(s) - \mathcal{Z}_{U;a,b,\mathcal{T}}(s)| \le V^{\sum c_j} A_{\mathrm{abs}}(\sigma, c) \left\{ \prod_{j=1}^{4} \left(1 + \frac{2}{\pi T_j \Delta_V}\right) - 1 \right\}, \tag{67}$$

and $A_{\mathrm{abs}} \le 3^{\omega(a)+\omega(b)} \zeta_K(\sigma)^9$. For one short integer norm n, the indicator is at most $(V/n)^c$, and the truncated Perron error is at most $2(V/n)^c/(\pi T \Delta_V)$. Expand the difference of four products and sum every nonempty error subset to prove (67). Consequently successive Perron limits recover the exact cutoffs. The half-integer gap also gives a height-uniform $C_U(V/n)^c$ bound, allowing dominated convergence in the initial absolutely convergent series. Four infinite Perron integrals are not declared absolutely convergent or interchanged silently with Mellin inversion.

There is an entirely finite alternative. Partition $a = a_0 a_1 a_2$ and $b = b_0 b_1 b_2$, internally coprime in each column. Cross-column overlaps remain allowed. A common d prime occupies one state $(i, j) \in \{0, 1, 2\}^2$. Separate its $(0, 0)$ part e from eight pairwise coprime squarefree atomic ideals h_{ij} avoiding $\mathcal{T}$, and let h be their product. Define

$$h_i^L = \prod_{j=0}^{2} h_{ij} \ (i = 1, 2), \qquad h_j^R = \prod_{i=0}^{2} h_{ij} \ (j = 1, 2).$$

An atomic h_{ij} with both indices nonzero belongs to one short factor in each column. These four short factors must not be forced globally disjoint. Let $\mathcal{A}_U$ comprise precisely these tuples satisfying $N(a_i h_i^L) \le U$ and $N(b_j h_j^R) \le U$ for $i, j = 1, 2$. The numerator in (63) is exactly

$$\mathcal{N}_{U;a,b,\mathcal{T}}(s) = \sum_{\mathcal{A}_U} \frac{\mu(a_1 a_2)\mu(b_1 b_2)\mu(h_{11}h_{12}h_{21}h_{22})}{(Nh)^s} \prod_{p|h}(1 - (Np)^{-s})^{-1}. \tag{68}$$

At a common prime the sign is $(-1)^{1+\mathbf{1}_{i>0}+\mathbf{1}_{j>0}}$. It is negative for $(0, 0)$, positive for the four single-short states, and negative for the four double-short states. Summing the remaining squarefree e, coprime to $\mathcal{T}h$, gives $1/\zeta_K^{\mathcal{T}}(s)$ times the local product in (68). This proves the exact identity, including its sign.

Multiplying the four cutoff inequalities yields $Nh \le U^4$; double states are counted twice in that product. Hence the numerator is finite and holomorphic on $\Re s > 0$. At $U = 1$ it equals one: this boundary is not a high-window counterexample or a global pole obstruction. For $\sigma \ge \sigma_0 > 0$ and every $\epsilon > 0$,

$$|\mathcal{N}_{U;a,b,\mathcal{T}}(\sigma + i\tau)| \ll_{K,\sigma_0,\epsilon} (Na\, Nb)^{\epsilon} U^{4\max(1-\sigma,0)+\epsilon}. \tag{69}$$

There are at most $3^{\omega(a)+\omega(b)}$ residual partitions and $8^{\omega(h)}$ atomic placements at fixed h. The finite local product is $\ll_{\sigma_0,\epsilon} (Nh)^{\epsilon}$. Apply the ideal-divisor bound and ideal counting up to U^4, using smaller internal exponents; a logarithm at the transition is absorbed in U^{ϵ}. Uniformity as $\sigma_0 \to 0$ is not claimed. For $\sigma = 1/2 + \eta$ with $0 < \eta < 1/2$, the old-type cutoff cost is $U^{2-4\eta+\epsilon}$, not a new saving.

S68.4 A uniform common-profile Mellin embedding of the actual mean

For residual a, b put $M = \max(Na, Nb)$ and $\varrho = \min(Na, Nb)/M$. At $\varrho \le 1/2$ the product of the two profile values vanishes, including equality because of endpoint vanishing. Otherwise put

$$\Phi_\varrho(t) = W(t)\overline{W(\varrho t)}$$

when $Na \ge Nb$, and use its conjugate for the reversed orientation. It is supported in $[1/\varrho, 2] \subset [1, 2]$. With $M_k(W) = \max_{j \le k} \|(t\, d/dt)^j W\|_\infty$,

$$|\widehat{\Phi}_\varrho(\sigma + i\tau)| \le C_{I,k} M_k(W)^2 (1 + |\tau|)^{-k} \tag{70}$$

uniformly for σ in any fixed compact strip I and all $\varrho \in [1/2, 1]$. Leibniz and log-coordinate integration by parts prove this bound; no ratio-dependent change of W is made. The transform is entire. For $c > 1$, ordinary Mellin inversion after the finite numerator identity gives the exact scalar (64) as

$$\frac{1}{2\pi i} \int_{\Re s = c} \widehat{\Phi}_\varrho(s) (D/M)^s \frac{\mathcal{N}_{U;a,b,\mathcal{T}}(s)}{\zeta_K^{\mathcal{T}}(s)}\, ds. \tag{71}$$

This integral is vertically absolutely convergent in its initial half-plane. The original profile forces $M \le 2D$, so the residual a, b and actual row sums are finite.

Keep the full local deletion factor

$$\frac{1}{\zeta_K^{\mathcal{T}}(s)} = \frac{1}{\zeta_K(s)} \prod_{p \in \mathcal{T}} (1 - (Np)^{-s})^{-1}.$$

At fixed $\sigma_0 > 0$ it costs $\ll_{S,\sigma_0,\epsilon} (N \operatorname{rad}(abu))^\epsilon$, hence at most $(HD^2)^\epsilon$ on actual support. This estimates the finite mask dependence, not the central reciprocal. Multiplying (71) by every original outer coefficient and summing the actual rows and residuals gives

$$DS_1^{AA} = \frac{1}{2\pi i} \int_{\Re s = c} \frac{D^s \mathcal{A}_{D,H,U,W}(s)}{\zeta_K(s)}\, ds. \tag{72}$$

Explicitly, with each profile orientation retained,

$$\mathcal{A}_{D,H,U,W}(s) = \sum_{u \in \mathcal{R}} \sum_{\substack{a,b \text{ squarefree,good} \\ Na, Nb \le 2D,\ \varrho > 1/2}} \nu(a)\overline{\nu(b)}\chi_a(u)\overline{\chi_b(u)}\, M^{-s}\, \widehat{\Phi}_\varrho(s) \cdot \mathcal{N}_{U;a,b,\mathcal{T}}(s) \prod_{p \in \mathcal{T}} (1 - (Np)^{-s})^{-1}.$$

This is a *finite coherent sum*, not an independent marginal moment or a positive sum. All raw zeros remain in its outer coefficients. For fixed finite D, H, U it is holomorphic on $\Re s > 0$ and rapidly decreasing on fixed vertical strips there, with constants depending on those finite parameters. No uniform target estimate for it is supplied. This is an exact signed embedding, not a zero-location theorem.

S68.5 A nonzero short-state correction and the unproved step

An actual annular control uses good prime ideals of norms $7, 13, 19, 31, 43$ with primary generators $(1, 3), (4, 3), (-2, 3), (1, 6), (7, 6)$ in the $1, \omega$ basis. Take $D = 1500, U = 25$, $d = p_7$, $a = p_{13}p_{19}$, $b = p_{13}p_{31}$ and the nonprincipal row $u = p_{43}$ in the pure block $A_1 = 37, A_2 = 1, H = 241$, literal

class one. The full raw supports are disjoint from u, and $N(da) = 1729, N(db) = 2821$ lie in the annulus. A single fixed smooth profile with plateau $[1.1, 1.9]$ equals one at both nodes. Direct assignments give $A_U(a) = A_U(b) = -1$, $A_U(da) = 1$, $A_U(db) = -1$. The actual signed d coefficient is $+1$. The $(0,0)$ state alone would give -1; states $(1,0)$ and $(2,0)$ each contribute $+1$. Dropping all short states therefore misses a nonzero $e = 1$ correction. This illustrates the new eight-state identity, not a new discovery of coefficient nonmultiplicativity.

The original pair has gcd p_7p_{13}. Its contributions across $d = 1, p_7, p_{13}, p_7p_{13}$ cancel exactly. Thus a fixed-residual reciprocal does not prove an uncancelled pole or an obstruction in the complete coherent mean. The missing step is to estimate that full mean by $DH(HD)^{\epsilon}$, retaining fixed-profile/seminorm dependence, raw masks and conductor/ray/capacity restrictions. Scalar Mertens cancellation in the long state, a normal correction product, or an absolute cutoff numerator bound does not prove it. Principal and short assembly, independent positive determinant/zero-location bridges and separate Dirichlet, Hecke, Dedekind, elliptic and Artin obligations remain open. No holomorphy or desired zero location is assumed to close them.

S68.6 Exact arithmetic and source-bound computational scope

The evaluator streams each actual row and accumulates original $x_m\chi_m(u)$ into every ideal-divisor bucket $d \mid m$. It then forms all nine cross products for A_U, C_U, μ and sums with $\mu(d)$. Write $n_{\mathrm{col}} = |\mathcal{C}|$, $I = \sum_m \tau_{\mathrm{ideal}}(m)$, K for the distinct divisors and r for the listed rows. This bucket stage costs $O(r(I + K))$ arithmetic operations with a fixed three channels. Its streaming storage is $O(K)$ plus incidences and either a streamed row or separately retained phase table. Character construction, factorization and integer bit lengths are additional costs, not declared free. For coefficients bounded by C, bucket coordinates are $O(n_{\mathrm{col}}C)$, products $O(n_{\mathrm{col}}^2C^2)$ and accumulated values $O(rn_{\mathrm{col}}^2C^2)$ for unit row weights. For compressed integer multiplicities replace r in this bit-size bound by $\sum_u w_u$ and account for weight multiplication.

For $D = 60000$, the actual phase table has shape 20×16953; the incidence has 73420 nonzeros over 21750 divisor buckets, and F_d has $20 \times 21750 \times 3$ complex states. There are 271688516 ordered coprime column pairs. The naive all-pair row loop would visit 5748084180 pairs and was *not executed* at this size. This is an operation count, not an unexecuted timing benchmark. Independent direct pair evaluation covers the two complete small families; the large family is checked linearly from literal field arithmetic and divisor-poset inversion.

The exact large results also include

$$S_1^{CC} = \frac{661}{7500}, \qquad S_1^{C\mu} = \frac{13931}{30000}, \qquad S_1^{\mu\mu} = -\frac{229}{1250}.$$

For the ordered vectors (A_U, C_U, μ) the full pairing table is

$$\frac{1}{30000}\begin{pmatrix} -30714 & -11287 & 19427 \\ -11287 & 2644 & 13931 \\ 19427 & 13931 & -5496 \end{pmatrix}.$$

This is the table of three original coefficient vectors, not a spectral carrier for GRH. The conjugate μC entry and the AA identity retain their actual linear-first convention. All three saved computations use source v1. A separate v2 helper fixes its unused $N = U = 1$ high-window assertion; every one of the 17158 executed columns has $N > U$, so no accepted row/table or source snapshot was silently replaced. The fixed-profile flat-node implementation refuses requests above its stated $D \le 60000$ certificate. Neither these three different finite blocks nor their measured runtimes imply an asymptotic saving.

S69 A target-sized budget for rough-prime masks

The actual signed divisor mean admits a useful quantitative simplification: the rough-prime coprimality mask can be restored with an explicit correction bound. The remaining main term still retains the Möbius sign and the coherent row mean; it is not estimated by this result.

Work over $\mathbb{Z}[\omega]$. Fix a finite S containing the primes above $2, 3$, a unitary multiplicative ν on good ideals, and one common complex profile W supported on $[1, 2]$, smooth and zero at both endpoints. Put $B = \|W\|_\infty$ and assume

$$1 \le U, \qquad U^2 \le D, \qquad 2D \le 7U^2, \qquad H \ge 1. \tag{73}$$

Let $\mathcal{R}$ be the actual finite family of nonzero literal rows, with original weights $w_u \ge 0$, $1 \le Nu \le H$, and $R_w = \sum_u w_u$. All original internal squarefree/coprime conditions, unit and S-ray classes, capacity and sixth-factor masks remain in this family. Write S_1^{AA} for the raw form (58) with $x_m = A_U(m)\nu(m)W(Nm/D)$, where

$$A_U(g) = \sum_{\substack{xy \mid g,\ (x,y)=1 \\ Nx \le U,\ Ny \le U}} \mu(x)\mu(y). \tag{74}$$

Here the columns in (58) are the COMPLETE original good squarefree annulus $[D, 2D]$, with the common W as its norm cutoff and the original full-good-ideal/ν, S formulation. No selected-column indicator or additional column-ray filter is allowed in this reduction; all actual row unit/ray/capacity filters remain. Principal-row residues, if present, require their separate assembly. For the intended $A_1 > 1$ blocks the defining conductor of ν is supported in S, and the rows are nonprincipal.

Let f range over good squarefree U-smooth ideals, put $F = Nf$, and set

$$T_f(t; u) = \sum_{\substack{a \text{ good squarefree} \\ (a,f)=1}} A_U(fa)\nu(a)\chi_a(u)W(FtNa/D). \tag{75}$$

Here $\chi_a(u) = 0$ on $(a, u) > 1$. For a prime ideal p, let $T_f^{(p)}(t; u)$ impose the additional condition $(a, p) = 1$. The smooth factor f is distinct from the original column-pair gcd sector $h = 1$. Define the following *unnormalized* expressions:

$$K_0 = \sum_u w_u \sum_{\substack{f \text{ good SF, } U\text{-smooth} \\ (f,u)=1}} \mu(f)|T_f(1; u)|^2, \tag{76}$$

$$K_T = \sum_u w_u \sum_{\substack{f \text{ good SF, } U\text{-smooth} \\ (f,u)=1}} \mu(f) \sum_{\substack{p \notin S,\ Np > U \\ p \nmid u}} |T_f^{(p)}(Np; u)|^2, \tag{77}$$

$$K_S = \sum_u w_u \sum_{\substack{f \text{ good SF, } U\text{-smooth} \\ (f,u)=1}} \mu(f) \sum_{p \notin S,\ Np > U} |T_f(Np; u)|^2. \tag{78}$$

Throughout, p denotes a prime ideal. None of these expressions is declared positive, since the $\mu(f)$ signs remain.

The exact decomposition of the common ideal divisor $d = fe$ has at most one rough prime when the residual pair is nonunit. The unit residual pair also permits two distinct rough primes, necessarily with $f = 1$. Its exact correction is

$$C_2 = \sum_{\substack{\{p,q\} \text{ unordered, } p \ne q \\ p,q \notin S,\ Np, Nq > U}} |W(NpNq/D)|^2 \sum_{\substack{u \in \mathcal{R} \\ (u,pq)=1}} w_u. \tag{79}$$

Conjugate primes of equal rational norm are included. One has

$$DS_1^{AA} = K_0 - K_T + C_2. \tag{80}$$

In K_S, terms with $p \mid a$ or $p \mid u$ are explicitly auxiliary: fpa can fail to be squarefree. They use $A_U(fa)$ as written, not a new original column or a substitution for $A_U(fpa)$.

Theorem S69.1 (Explicit rough-prime mask budget)**.** *Put* $\kappa_U(u) = \#\{p \notin S : p \mid u,\ Np > U\}$ *and* $\Delta = (K_S - K_T)/D$. *Under* (73),

$$|\Delta| \le B^2 \left(229376 \sum_u w_u \kappa_U(u) + 2048 R_w \right), \tag{81}$$

$$0 \le C_2/D \le 1792 B^2 R_w. \tag{82}$$

Consequently

$$S_1^{AA} = \frac{K_0 - K_S}{D} + E, \tag{83}$$

$$|E| \le B^2 \left(229376 \sum_u w_u \kappa_U(u) + 3840 R_w \right). \tag{84}$$

If $H \le D^{3/4}$, *then every row has* $\kappa_U(u) \le 1$, *and in particular*

$$|E| \le 233216 B^2 R_w \le 2^{18} B^2 R_w. \tag{85}$$

Only the rough-prime deletion has been restored; the smooth-divisor mask, literal row family and original coefficients remain. The explicit bound uses the exact restricted identity

$$A_U(g) = \mu(g)(2\mathbf{1}_{Ng \le U} - 1) \quad (Ng \le 7U,\ g \text{ good squarefree}).$$

Every active term of $T_f(Np; u)$ has $N(fa) < 7U$, hence $|A_U(fa)| = 1$ there. This is not a shrinkage or proxy for the whole original coefficient family. Only absolute correction bounds use it; the signed coefficients in the clean main remain unchanged.

For general H, one has $\kappa_U(u) \le 2\log(2H)/\log(2U)$, also at $U = 1$. For an actual two-pivot block with $R_w \ll A_1 A_2 \le H$ and $H \le D^{3/4}$, (85) gives an $O(B^2 H)$ error with no divisor-power or profile-derivative loss. The remaining target problem for this raw A–A slice is the clean signed main $(K_0 - K_S)/D$. It has not been bounded here; the 7/6 envelope and missing 5/12 improvement for the complete original remainder are unchanged. The argument neither deletes a coherent mean nor treats the coprime-truncated form as positive.

Three exact finite controls. These use principal ν, S above $2, 3$, literal unit class one and complete original good squarefree annuli. The first two row variables are squarefree, good and coprime and may include the unit ideal; all other factors equal one.

D	U	H	A_1	A_2	columns	rows
200	10	53	13	1	52	5
512	16	107	19	1	139	7
900	30	384	7	7	250	2

Put $\eta_0(t) = \exp(-1/t)$ for $t > 0$ and $\eta_0(t) = 0$ for $t \le 0$, and define

$$s(t) = \frac{\eta_0(t)}{\eta_0(t) + \eta_0(1-t)}.$$

Choose once $\delta = 1/4096$ and $W_\delta(t) = s((t-1)/\delta)s((2-t)/\delta)$. For integer $D \le 1024$, strict integer norm nodes have weight one and endpoints have weight zero; its derivatives cost $O_k(4096^k)$. The following numbers are RAW, before division by D:

D	K_0	K_S	K_T	E_{row}	E_{res}	C_2	DS_1^{AA}
200	339	235	209	20	6	21	151
512	366	781	711	48	22	89	−256
900	436	410	366	0	44	28	98

In every case $K_S - K_T = E_{\mathrm{row}} + E_{\mathrm{res}}$ and $DS_1^{AA} = K_0 - K_T + C_2$. The displayed nonnegative values of $E_{\mathrm{row}}, E_{\mathrm{res}}$ are finite observations; no general nonnegativity is assumed for either signed correction. Only C_2 is nonnegative by definition. The first two capacities are $\lfloor D^{3/4} \rfloor$. The third $H = 384$ is a declared noncritical fixture capacity, not $\lfloor 900^{3/4} \rfloor$; both of its rows have $\kappa_U = 0$. All ideal and raw-zero masks, unit residual states and explicitly auxiliary non-squarefree p^2 terms are retained. The exact control summary records the scopes and exact fractions after one division by D. These are finite identities, not a proof or numerical fit of the analytic bounds, and do not estimate the signed clean main.

S70 Proof of the rough-prime mask budget

S70.1 The common ideal divisor and the exact rough part

Here (58) uses the complete original good squarefree annulus and common W, without an additional selected-column or column-ray indicator. The actual row filters are unchanged. The identity $\mathbf{1}_{(m,n)=1} = \sum_{d \mid m,\ d \mid n} \mu(d)$ in (58) is an identity of ideal divisors. Write $m = da, n = db$. The residual ideals a, b are squarefree but may overlap; squarefreeness of the original columns requires $(d, ab) = 1$. On $(d, u) = 1$ the original common phases satisfy

$$|\nu(d)|^2 \chi_d(u)\overline{\chi_d(u)} = 1.$$

On $(d, u) > 1$ the product is the original raw zero. Hence the full deleted set is $\mathcal{T} = S \cup \mathrm{supp}(abu)$, including residual primes where $\chi_a\overline{\chi_b}$ might become imprimitive. With

$$F_{a,b}(x) = W(xNa/D)\overline{W(xNb/D)}, \qquad M = \max(Na, Nb), \quad \alpha = \min(Na, Nb),$$

the inner signed sum is

$$\sum_{\substack{d \ \mathrm{SF} \\ (d,\mathcal{T})=1}} \mu(d) A_U(da) A_U(db) F_{a,b}(Nd). \tag{86}$$

If its profile is nonzero, then

$$D/\alpha < Nd < 2D/M. \tag{87}$$

All endpoints have zero profile value. In particular the profile is identically zero if $\alpha \le M/2$. No independent marginal row family or changed profile has been substituted.

Factor $d = fe$ uniquely, with f squarefree U-smooth and e squarefree with every prime norm greater than U. Whenever ez is squarefree,

$$A_U(ez) = A_U(z). \tag{88}$$

Indeed neither short divisor in (74) can contain a prime of e; deleting e preserves the two cutoffs and both Möbius signs in a bijection of the original assignments. There is no factor $\mu(e)$ in (88). This invariance is specific to A_U; it does not hold unchanged for the other coefficient channels. The values $A_U(fa), A_U(fb)$ are not factorized.

If $M > 1$, goodness gives $M \geq 7$. From (73) and (87), $Nd \leq 2D/M \leq U^2$. Thus e is either 1 or one prime ideal, and (86) is exactly

$$\sum_{\substack{f \text{ good SF, } U\text{-smooth} \\ (f,\mathcal{T})=1}} \mu(f) A_U(fa) A_U(fb) \left(F_{a,b}(Nf) - \sum_{\substack{p \notin \mathcal{T} \\ Np > U}} F_{a,b}(NfNp) \right). \tag{89}$$

The rough part here belongs to the *common divisor* d; the full columns can have additional rough primes in a or b.

If $M = 1$, then $a = b = 1$. Three rough prime ideals would have norm product strictly greater than $7U^2 \geq 2D$: two contribute more than U^2, and the third good prime has norm at least 7. Two rough primes are possible, but then a nonunit f would similarly give $Nf\,Ne > 7U^2 \geq 2D$. Hence that case has $f = 1$ only. Its sign is $\mu(pq) = 1$ and its coefficient is $A_U(pq) = A_U(1) = 1$. Summing this case over the unchanged literal rows gives (79); summing all other cases of (89) over a, b gives the completed squares in (76)–(77). This proves (80). The unit ideal among residuals and conjugate split primes must both be retained. The condition $f = 1$ is not a termwise assertion that the original pair has gcd 1: the Möbius expansion cancels noncoprime original pairs across its divisor sectors.

S70.2 An exact disjoint error identity

We restore only $(p, abuS) = 1$ to $p \notin S$. In particular the conditions $(f, abuS) = 1$ and every raw $\chi_a(u)$ remain in the two residual sums. The full deleted union has the disjoint partition

$$\mathbf{1}_{p \mid abu} = \mathbf{1}_{p \mid u} + \mathbf{1}_{p \nmid u} \mathbf{1}_{p \mid a \text{ or } p \mid b}. \tag{90}$$

Therefore $K_S - K_T = E_{\text{row}} + E_{\text{res}}$, where the errors are *unnormalized*:

$$E_{\text{row}} = \sum_u w_u \sum_{\substack{f \text{ good SF, } U\text{-smooth} \\ (f,u)=1}} \mu(f) \sum_{\substack{p \notin S,\ Np > U \\ p \mid u}} |T_f(Np; u)|^2, \tag{91}$$

$$E_{\text{res}} = \sum_u w_u \sum_{\substack{f \text{ good SF, } U\text{-smooth} \\ (f,u)=1}} \mu(f) \sum_{\substack{p \notin S,\ Np > U \\ p \nmid u}} \left(|T_f(Np; u)|^2 - |T_f^{(p)}(Np; u)|^2 \right). \tag{92}$$

The two errors are signed; their bounds below use absolute values, and neither is assumed nonnegative. At $p \mid u$, all residuals divisible by p already have raw character zero. This is consistent with the full square in (91); it does not make that square zero. These formulas keep the complete row/residual union without double counting.

Suppose a nonzero omitted residual has $p \mid a$, and write $a = pc$. Its profile requires $Nf\,(Np)^2 Nc < 2D$. If fc is nonunit, then

$$Nf\,(Np)^2 Nc > 7U^2 \geq 2D,$$

a contradiction. Thus necessarily $f = c = 1$ and $a = p$. This is a support statement about the explicitly auxiliary extension; the formal column $fpa = p^2$ is not an original squarefree column. With $P = Np$, set

$$z_p(u) = \nu(p) \chi_p(u) W(P^2/D), \qquad T_p(u) = T_1(P; u). \tag{93}$$

Here $A_U(p) = 1$, since $P > U$. The difference $T_f - T_f^{(p)}$ vanishes unless $f = 1$, when it equals z_p. Consequently

$$E_{\mathrm{res}} = \sum_u w_u \sum_{\substack{p \nmid u \\ p \notin S,\ Np > U}} \left(2\,\mathrm{Re}(T_p(u)\overline{z_p(u)}) - |z_p(u)|^2 \right). \tag{94}$$

The subtraction $-|z_p|^2$ is the exact intersection $p \mid a$ and $p \mid b$; omitting it would count that intersection twice. Complex profiles and all remaining literal phases are allowed. If $z_p \neq 0$, then

$$\sqrt{D} < P < \sqrt{2D}. \tag{95}$$

The condition $P > U$ is automatic here from $U^2 \leq D$.

S70.3 Elementary ideal and coefficient estimates

We give all counting inputs; no prime distribution theorem or Mertens estimate is needed. The Eisenstein norm is Euclidean. After rounding both real coordinates of a complex quotient in the basis $1, \omega$, the error $s + t\omega$ has $|s|, |t| \leq 1/2$ and norm at most $3/4 < 1$. A least-norm nonzero element of any ideal therefore generates it by Euclidean division. Multiplication by $a + b\omega$ has determinant $a^2 - ab + b^2$, giving the principal ideal norm. Since $a^2 - ab + b^2 \geq (a^2 + b^2)/2$, counting all possible generators, with harmless overcounting by units, proves for $X \geq 1$ that

$$\#\{a : Na \leq X\} \leq (2\sqrt{2X} + 1)^2 < 16X. \tag{96}$$

An annulus $L < Na < 2L$ therefore contains at most $32L$ ideals when $2L > 1$. If $2L \leq 1$ its profile contribution is zero, including the unit endpoint when $2L = 1$. This proves the needed uniform bound even when $1/2 < L < 1$; the unit ideal must not be dropped in that range.

There is an exact coefficient identity on the restricted support:

$$A_U(g) = \mu(g)\big(2\mathbf{1}_{Ng \leq U} - 1\big) \quad (g \text{ good squarefree},\ Ng \leq 7U). \tag{97}$$

Every proper divisor of a nonunit good g has norm at most $Ng/7 \leq U$. If $Ng \leq U$, all ternary assignments are allowed and their sum is $\prod_{p \mid g}(1 - 1 - 1) = \mu(g)$. If $Ng > U$, only the assignments $(x, y) = (g, 1), (1, g)$ are omitted; each has sign $\mu(g)$, leaving $-\mu(g)$. The unit ideal gives $A_U(1) = 1$ and is included. No approximation or multiplicativity of the cutoff coefficient is used.

In an active term of (75), $P = Np > U$ gives

$$N(fa) < 2D/P < 2D/U \leq 7U.$$

Therefore $|A_U(fa)| = 1$ exactly there. This statement concerns only the rough-prime support, not the entire original coefficient family. With $L = D/(FP)$, the ideal count and $|\chi_a(u)| \leq 1$ now give

$$|T_f(P; u)| \leq 32B\,\frac{D}{FP}. \tag{98}$$

The same statement holds if the sum is empty, and covers $L < 1$ by the preceding endpoint discussion. A dyadic use of (96) gives

$$\sum_f \frac{1}{(Nf)^2} \leq \sum_{j \geq 0} 16\,2^{j+1} 2^{-2j} = 64, \tag{99}$$

where the left side may even include all ideals. The smooth squarefree restriction only decreases this bound.

S70.4 The normalized row-mask budget

Apply triangle only to (91), use (98), and then (99). The raw square contributes D^2/P^2; dividing by D once gives

$$\frac{|E_{\mathrm{row}}|}{D} \le 65536B^2 \sum_u w_u \sum_{\substack{p\mid u,\ p\notin S\\ Np>U}} \frac{D}{(Np)^2}. \tag{100}$$

By (73), $D/(Np)^2 \le D/U^2 \le 7/2$. Let $\kappa_U(u)$ count the distinct good rough prime ideals dividing u. Each $P = Np$ is at least 7 and greater than U, so $P^2 > 7U \ge 2U$. The complete ideal norm of the literal row therefore gives

$$\kappa_U(u) \le \frac{2\log(Nu)}{\log(2U)} \le \frac{2\log(2H)}{\log(2U)}. \tag{101}$$

The denominator is positive also at $U = 1$; unit rows have $\kappa_U = 0$. All sixth-factor divisors remain in this bound, and no rational-norm gcd is substituted for an ideal mask. Using $D/(Np)^2 \le 7/2$ in (100) gives the sharper form

$$\frac{|E_{\mathrm{row}}|}{D} \le 229376B^2 \sum_u w_u \kappa_U(u). \tag{102}$$

The logarithmic bound (101) is available for arbitrary H.

S70.5 The residual intersection budget

Use (94), $|z_p| \le B$, and (98) at $f = 1$. After the single division by D,

$$\frac{|E_{\mathrm{res}}|}{D} \le B^2 R_w \sum_{\sqrt{D}<Np<\sqrt{2D}} \left(\frac{64}{Np} + \frac{1}{D}\right). \tag{103}$$

Here absolute values bound only the correction; the exact identity (94) retains its original phases. Since $D \ge 1$ and $Np \le \sqrt{2D} \le 2D$, one has $1/D \le 2/(Np)$. Also

$$\sum_{\sqrt{D}<Np<\sqrt{2D}} \frac{1}{Np} \le 16\sqrt{2} \tag{104}$$

by (96). Thus

$$|E_{\mathrm{res}}|/D \le 1056\sqrt{2}\, B^2 R_w \le 2048B^2 R_w.$$

Together with (102), this proves (81).

S70.6 The exact unit-residual correction and the clean main

In (79), each contributing pair has $P, Q > U$ and $PQ \le 2D$. Hence both prime norms are less than $2D/U$. There are at most $32D/U$ such prime ideals by (96), so the number of unordered distinct pairs is at most $512D^2/U^2$. Only inside this *nonnegative* expression enlarge the masked row count to R_w and the profile to B^2. Thus

$$0 \le \frac{C_2}{D} \le 512\frac{D}{U^2}B^2 R_w \le 1792B^2 R_w,$$

proving (82). Conjugate primes of equal rational norm remain included; no rational-squarefree restriction or new diagonal exclusion is imposed. This is a bookkeeping bound for the exact

correction, not an oscillatory saving or a lower bound for the full signed slice. Other residual sectors can cancel it.

Now (80) gives (83) with $E = \Delta + C_2/D$, and the explicit bounds give (84). Suppose $H \le D^{3/4}$. If two distinct good rough primes divided a row, then $H \ge Nu \ge (Np)(Nq) \ge 49$, but also

$$H \ge Nu \ge (Np)(Nq) > U^2 \ge 2D/7.$$

Combining with $H \le D^{3/4}$ gives $D^{1/4} < 7/2$ and hence $H < (7/2)^3 = 343/8 < 49$, a contradiction. Thus $\kappa_U(u) \le 1$ at every scale in this capacity range. Substitution in (84) gives $|E| \le 233216B^2R_w \le 2^{18}B^2R_w$, proving (85). This is an exact endpoint argument, not an asymptotic approximation.

For a two-pivot block, counting each of its two dyadic ideal variables by (96) gives $R_w \ll A_1A_2$ when the original fixed unit classes and multiplicities are bounded. Internal coprimality and capacity reduce that count; grouping literal fibers preserves total multiplicity. Under $A_1A_2 \le H \le D^{3/4}$ the correction is therefore $O(B^2H)$. The exact local coefficient identity and these correction bounds do not estimate the remaining signed main $(K_0 - K_S)/D$. Its original $A_U(fa)$, coherent rows, common fixed profile and Möbius mean remain the analytic obligation; no new global exponent follows.

S70.7 Finite control scope

The tables in the shared core retain complete original squarefree ideal columns and raw character zeros. Their residual-prime extension is a separate auxiliary sum, including the non-squarefree p^2 terms needed for (94); those are not original columns. The exact intersection subtraction and unit residual states with $1/2 < L < 1$ are checked separately. The common profile has fixed $\delta = 1/4096$ and large fixed derivative constants. The capacities and row blocks differ between cases, and the $H = 384$ case is outside the critical capacity range. These finite equalities neither determine the analytic constants nor imply an asymptotic saving.

S70.8 A coherent prime-substitution budget, and its remaining cost

This subsection requires $U \ge 7$. It does not extend the prime-density argument to $U = 1$, although the coefficient and mask lemmas above hold for all $U \ge 1$. For real $t \ge U$, put

$$G_f(t) = \sum_{\substack{u \in \mathcal{R} \\ (f,u)=1}} w_u |T_f(t;u)|^2, \qquad M_1(W) = \max\{B, \|xW'(x)\|_\infty\}.$$

The active residuals satisfy $Na \le 2D/(FU)$, so their union is finite; G_f is smooth and vanishes for $t \ge 2D/F$. Since the rough prime in K_S is unmasked, it enters only this radial common profile. Let

$$E_S(t) = \pi_K^S(t) - \mathrm{Li}(t), \qquad \delta_S(U) = \sup_{t \ge U} \frac{|E_S(t)|}{t}, \qquad \mathcal{J} = \sum_f \mu(f) \int_U^\infty \frac{G_f(t)}{\log t}\, dt,$$

where π_K^S counts prime *ideals* outside the fixed finite S with multiplicity, including inert prime norms and distinct conjugate ideals. Stieltjes partial summation gives the exact coherent identity

$$\begin{aligned} K_S &= \mathcal{J} + \mathcal{E}_{\mathrm{PIT}}, \\ \mathcal{E}_{\mathrm{PIT}} &= \sum_f \mu(f) \left(-G_f(U)E_S(U) - \int_U^\infty E_S(t) G_f'(t)\, dt \right). \end{aligned} \tag{105}$$

The lower term uses $\pi_K^S(U)$, so the condition $Np > U$ is exact even when a prime norm equals U. A conjugate pair contributes a jump of two. This equality precedes any absolute sum over f.

Let $\lambda(L)$ be a TRUE arbitrary-coefficient squared norm bound for the specified positive row operator on good squarefree ideals of norm in $[L, 2L]$, including the actual weights. For $1/2 \le L < 1$, set $\lambda_*(L) = R_w$ by the direct unit-column bound; for $L \ge 1$, use $\lambda_*(L) = \lambda(L)$. No columns are active for $L < 1/2$. With $L = D/(Ft)$, differentiate only the common W: multiplication by $t\,d/dt$ replaces $W(x)$ by $xW'(x)$. The exact restricted identity (97), ideal counting and weighted row Cauchy give

$$G_f(t) + |tG_f'(t)| \ll M_1(W)^2 \lambda_*(L) L.$$

Any enlargement of a common f-row mask here occurs only inside these positive norm estimates. Substitution in (105), with $L_f = D/(FU)$, proves

$$\frac{|\mathcal{E}_{\mathrm{PIT}}|}{D} \ll M_1(W)^2 \delta_S(U) \sum_{\substack{f \text{ good SF, } U\text{-smooth} \\ F < 2D/U}} \frac{1}{F} \left(\lambda_*(L_f) + \int_{1/2}^{L_f} \lambda_*(L)\, \frac{dL}{L} \right). \tag{106}$$

The original Möbius sign remains in K_S and $\mathcal{J}$ themselves; the absolute f sum appears only in this error majorant.

Specialize now ONLY to the established positive input for the actual $A_6 = 1$ unweighted two-pivot block, or bounded natural literal weights for which the same input has been established. Put

$$R = A_1 A_2, \qquad M_{\mathrm{row}} = \max(A_1, A_2), \qquad A_1 A_2^2 \le H.$$

At $L \ge 1$ that input is

$$\lambda(L) \ll_\epsilon (HD)^\epsilon \min \left\{ R + \frac{LR}{M_{\mathrm{row}}} + \frac{L^{2/3} R}{M_{\mathrm{row}}^{1/3}},\ L + R^2 \right\}. \tag{107}$$

It is not asserted for arbitrary nonnegative weights, enlarged primitive conductor families or arbitrary sixth-factor enlargements. Using its first branch in (106) gives

$$\frac{|\mathcal{E}_{\mathrm{PIT}}|}{D} \ll_\epsilon (HD)^\epsilon M_1(W)^2 \delta_S(U) \mathcal{B}_{\mathrm{op}}, \tag{108}$$

$$\mathcal{B}_{\mathrm{op}} = R \log^2(2D) + \frac{D}{U} \frac{R}{M_{\mathrm{row}}} + \left(\frac{D}{U} \right)^{2/3} \frac{R}{M_{\mathrm{row}}^{1/3}}. \tag{109}$$

The logarithmic term uses $\sum_{Nf \le X} (Nf)^{-1} \ll \log(2X)$; the other terms use convergent ideal sums of powers 2 and 5/3. The same positive bounds give $|\mathcal{J}|/D \ll_\epsilon (HD)^\epsilon B^2 \mathcal{B}_{\mathrm{op}} / \log U$. These are inherited majorants, not a new signed gain or an estimate of $K_0 - \mathcal{J}$.

Capacity implies

$$\mathcal{B}_{\mathrm{op}} \ll H \log^2(2D) + (D/U) H^{1/3} + (D/U)^{2/3} H^{2/3}.$$

At $H = D^{3/4}$ and $U \asymp \sqrt{D}$, these terms have powers $3/4, 3/4, 5/6$. The last power occurs in this majorant already on pure scales $A_1 \asymp H, A_2 = 1$ with fixed capacity slack. It is not an actual-error lower bound.

For the fixed field, Thorner–Zaman, *Algebra & Number Theory* 13 (2019), 1039–1068, Theorem 1.1, applied to the trivial extension K/K with absolute degree $[K : \mathbb{Q}] = 2$ and absolute discriminant 3, supplies for sufficiently large U a bound

$$\delta_S(U) \ll_{K,S} \exp(-c_K \sqrt{\log U}).$$

The possible exceptional real zero is fixed: retain its secondary $\mathrm{Li}(x^\beta)$ term, or absorb it only into fixed-field constants and the threshold. Finite S deletions are included. No prime theorem uniform in the growing deleted set or row conductor is used. The DOI, publisher URL and pinned reading-cache hashes are in the public citation record.

A fixed-field subpower prime error does not make the available $D^{5/6+\epsilon}\delta_S(U)$ majorant certify the target $D^{3/4+\epsilon}$ for every $\epsilon > 0$. Actual signed errors might cancel. Even an exact prime density would leave the coherent signed $K_0 - \mathcal{J}$ unestimated. The elementary correction theorem uses only B, whereas the prime-substitution error requires $M_1(W)^2$. The complete original remainder's $7/6$ envelope remains unchanged.

S71 The coherent signed main and its remaining inputs

The physical sixth-power-free comparison in Section S73 provides a new positive upper bound. We now collect the exact coherent reductions that it can enter. They preserve the original mean and all raw masks; none proves the target by a scalar cancellation assertion.

Use the complete original good squarefree annulus and common complex profile of Section S69. Write $\mathcal{U}$ for the actual nonzero literal rows, $w_u \ge 0$, $R_w = \sum_u w_u$ and $B = \|W\|_\infty$. Retain fixed S, ν, all internal row masks and (73). Every identity below is a raw A–A identity; principal residues still require their separate assembly. Classical operator specializations are only for the original unweighted $A_6 = 1$ rows, or their already justified bounded natural multiplicities.

First there is a useful distinction between two auxiliary kernels. Let K_{all} be K_S of (78) with the U-smooth restriction on its squarefree f removed, retaining $(f,u) = 1$ and exactly $A_U(fa)$ in every T_f. Set

$$\begin{aligned} P_2 &= \sum_u w_u \sum_{\substack{p\notin S\\ Np>U}} \sum_{\substack{q\notin S,\ Nq>U\\ q\nmid u}} |W(NpNq/D)|^2, \\ K_S &= K_{\mathrm{all}} + P_2, \qquad 0 \le P_2/D \le 3584 B^2 R_w. \end{aligned} \tag{110}$$

Here (p,q) is ordered, $p = q$ is allowed, and only q has the stated row mask. This is distinct from the original unordered, distinct-prime C_2 in (79). In particular

$$S_1^{AA} = \frac{K_0 - K_{\mathrm{all}}}{D} + \Delta + \frac{C_2 - P_2}{D}. \tag{111}$$

At $H \le D^{3/4}$ the error in this *new* auxiliary main is at most $236800 B^2 R_w$. The old $233216 B^2 R_w$ bound for $(K_0 - K_S)/D$ is unchanged. The physical comparison in (133) estimates the new combined main, in its stated $A_6 = 1$ scope, without estimating its two summands separately.

The prime-discrepancy clauses below additionally require $U \ge 7$. Define

$$E_S(t) = \pi_K^S(t) - \mathrm{Li}_2(t), \qquad \mathrm{Li}_2(t) = \int_2^t \frac{dv}{\log v}, \qquad M_1(W) = \max\{B, \|xW'(x)\|_\infty\}.$$

The count is of prime ideals outside fixed S, including distinct conjugate ideals and inert prime norms. Put

$$Q(t) = \sum_{\substack{f \text{ good SF}\\ f\ U\text{-smooth}}} \mu(f) \sum_{\substack{u\in\mathcal{U}\\ (f,u)=1}} w_u |T_f(t;u)|^2, \qquad \mathcal{J} = \int_U^\infty \frac{Q(t)}{\log t}\, dt.$$

Proposition S71.1 (A joint form retaining its mean)**.** *On the finite space of good squarefree g with $Ng \le 2D/U$, let*

$$b_g(t) = A_U(g)\nu(g)W(Ng\,t/D), \quad (\mathsf{R}c)(u) = \sqrt{w_u}\sum_g c_g\chi_g(u), \quad (\mathsf{P}_f c)_g = \mathbf{1}_{f|g}c_g.$$

With ordinary counting ℓ^2 norms,

$$Q(t) = b(t)^*\mathcal{M}_U b(t), \qquad \mathcal{M}_U = \sum_{f\ U\text{-smooth}} \mu(f)\mathsf{P}_f\mathsf{R}^*\mathsf{R}\mathsf{P}_f, \quad t \ge U. \tag{112}$$

The matrix is self-adjoint, with no positivity assertion. The exact prime substitution is

$$K_S = \mathcal{J} + \mathcal{E}_{\mathrm{PIT}}, \qquad \mathcal{E}_{\mathrm{PIT}} = -E_S(U)Q(U) - \int_U^\infty E_S(t)Q'(t)\,dt. \tag{113}$$

The endpoint excludes $Np = U$ exactly. It must remain.

For any true arbitrary-coefficient row norm $\lambda_*(L)$, use the direct unit-column value R_w on $1/2 \le L < 1$. On the active range $1 \le L \le D/U$, the current original two-pivot bound is the minimum in Proposition S73.4; it is not a theorem for arbitrary row weights. Let $\lambda^{\#}(D/T)$ dominate this norm on the two adjacent active length dyads. Then

$$\frac{|\mathcal{E}_{\mathrm{PIT}}|}{D} \ll_\epsilon (HD)^\epsilon M_1(W)^2\left\{\frac{|E_S(U)|}{U}\lambda_*(D/U) + \sum_T \mathfrak{d}_2(T)\lambda^{\#}(D/T)\right\}, \tag{114}$$

where $T = 2^jU < 2D$ and $\mathfrak{d}_2(T)^2 = \int_T^{2T}|E_S(t)|^2t^{-3}\,dt$. This is a sufficient-input budget, not an asserted power estimate for $\mathfrak{d}_2$ or for the endpoint. The pointwise version retains $\lambda_*(D/U) + \int_{1/2}^{D/U}\lambda_*(L)\,dL/L$, with the current minimum inside that integral.

On good rough-prime columns the exact quadratic-form compression is $\Pi_{\mathcal{P}}\mathcal{M}_U\Pi_{\mathcal{P}} = \Pi_{\mathcal{P}}\mathsf{R}^*\mathsf{R}\Pi_{\mathcal{P}}$. This is not an invariant-subspace assertion: off-block couplings remain and can cancel its contribution. Radial Mellin Parseval likewise retains equal-norm ideal correlations, including their derivatives; it neither orthogonalizes orientations nor permits deleting their mean. Even exact prime density leaves $K_0 - \mathcal{J}$ to estimate.

Proposition S71.2 (A sufficient smooth-divisor tail cutoff)**.** *For the original unweighted $A_6 = 1$ two-pivot rows put $R = A_1A_2$, $M_{\mathrm{row}} = \max(A_1, A_2)$. Restrict the smooth f sums in K_0, K_S to $Nf > Y \ge 1$. The inherited positive input gives*

$$\begin{aligned}\frac{|K_0^{>Y}|}{D} &\ll_\epsilon (HD)^\epsilon B^2\left\{R + \frac{DR}{M_{\mathrm{row}}Y} + \frac{D^{2/3}R}{M_{\mathrm{row}}^{1/3}Y^{2/3}}\right\},\\ K_S^{>Y} &= 0 \qquad (Y \ge 2D/U).\end{aligned} \tag{115}$$

If $R \ll H$, a sufficient target-size tail cutoff is

$$Y \ge \max\left\{1, \frac{2D}{U}, \frac{DR}{M_{\mathrm{row}}H}, \frac{DR^{3/2}}{M_{\mathrm{row}}^{1/2}H^{3/2}}\right\}.$$

This is a coarse inherited sufficient bound, not signed cancellation or an optimal localization. The smooth incidence divisor f is not the original column-pair gcd h.

Theorem S71.3 (A conditional input diagnostic)**.** *Fix S and $0 < \alpha < 1/2$. If for every $\epsilon > 0$,*

$$\mathfrak{d}_2(T) \ll_{\epsilon,K,S} T^{-\alpha+\epsilon}$$

holds for all sufficiently large real T, then

$$\zeta_K(s) \neq 0 \qquad (\operatorname{Re} s > \max\{1-\alpha, 1/2\},\ s \neq 1).$$

This is a strict-region implication from an explicitly hypothetical input, not a new zero-free theorem or a necessary condition for the actual signed target.

With the current physical input the pure witness has the sufficient PIT majorant 19/24, and the mixed witness retains the smaller 7/9 from the minimum. Independent scalar powers $U^{-1/12}$ and $U^{-1/18}$ would suffice for those majorants, with compatible endpoint control; their all-scale mean-square versions would entail the strict strips 11/12 and 17/18 above. None is asserted available. The old pure 5/6 budget and $U^{-1/6}$ input, with conditional strip 5/6, are historical values for the former operator bound. Proposition S73.2 gives the current global 9/8 comparison, and Proposition S73.3 the scoped $A_6 = 1$ comparison 13/12. The target, principal/short assembly beyond the stated scopes, independent determinant/zero-location bridges and full analytic/category Lean formalization remain open.

S72 Proofs for the coherent main reductions

Throughout, squarefree and prime mean ideal squarefree and prime. The original symbol is extended by zero, the complete good annulus is unchanged, and the same complex W is used for every row. The common smooth divisor f is an incidence variable after ideal Möbius inversion; it is not the original column-pair gcd sector $h = 1$.

S72.1 Removing only the auxiliary smooth restriction

The high-window support of a rough-prime term satisfies

$$N(fa), N(fb) < 2D/U \leq 7U.$$

The accepted exact coefficient identity (97) therefore applies to these squarefree products, including the unit:

$$A_U(g) = \mu(g)(2\mathbf{1}_{Ng\leq U} - 1), \qquad Ng \leq 7U. \tag{116}$$

For completeness, every proper divisor of a nonunit good g has norm at most $Ng/7 \leq U$. The unrestricted ordered disjoint-divisor sum is $\prod_{p|g}(1-1-1) = \mu(g)$. If $Ng > U$, precisely the two assignments $(g,1),(1,g)$ are omitted, each with weight $\mu(g)$. If $Ng \leq U$ none is omitted; $g = 1$ is direct. This does not simplify the general annular K_0 coefficients, whose norms can reach $2D$.

For a nonunit residual pair put $M = \max(Na, Nb) \geq 7$. On active rough-prime support, with $F = Nf$ and $P = Np > U$, $FMP < 2D \leq 7U^2$, hence $F < U$. Such f is automatically U-smooth. Thus dropping its smoothness changes only $a = b = 1$. There $F < 2D/P < 7U$. If a squarefree good f contains a prime q with $Nq > U$, any additional nonunit factor would give $F > 7U$. The omitted nonsmooth f is consequently exactly q, with $\mu(q) = -1$ and $|A_U(q)| = 1$. This proves the sign and all masks in (110). The outer p was already deliberately unmasked in K_S; only $q \nmid u$ comes from the original f-row mask. In particular $p = q$ gives a permitted auxiliary p^2 term, not an original squarefree column.

A nonzero pair has both prime norms below $2D/U$. Using (96), the number of ordered pairs is at most $(32D/U)^2$. Therefore

$$P_2/D \le 1024B^2R_wD/U^2 \le 3584B^2R_w.$$

Substitute $K_S = K_{\mathrm{all}} + P_2$ into (83). The old critical error $233216B^2R_w$ and the new charged $3584B^2R_w$ give $236800B^2R_w$ in (111). The original C_2 remains unordered and distinct, with both row masks; it is not this ordered correction. No lower bound for the full indefinite raw slice is inferred from either nonnegative correction.

S72.2 An exact two-jump masked Mertens functional

Expand the nonunit residual part of K_S before any absolute sum in f, a conductor or a row. Retain

$$\mathcal{T} = S \cup \mathrm{supp}(abu), \qquad \eta_{a,b}(u) = \nu(a)\overline{\nu(b)}\chi_a(u)\overline{\chi_b(u)}.$$

The ideals a, b are individually squarefree but may overlap. Every common support prime remains in $\mathcal{T}$, even if their quotient character loses that prime. The rough p in K_S may belong to $\mathcal{T}$; this is its defined auxiliary enlargement. Only the common f phases cancel, leaving the full $(f, \mathcal{T}) = 1$ mask. By (116), the remaining coefficient is

$$\mu(f)A_U(fa)A_U(fb) = \mu(f)\mu(a)\mu(b)(2\mathbf{1}_{FNa\le U} - 1)(2\mathbf{1}_{FNb\le U} - 1).$$

Put $\alpha = \min(Na, Nb)$, $M = \max(Na, Nb)$ and

$$X = \frac{D}{PM}, \quad r = \frac{\alpha}{M}, \quad c_0 = \frac{U}{M}, \quad d_0 = \frac{U}{\alpha}.$$

At $r \le 1/2$ the profile product vanishes, including equality. Otherwise let $\Phi_r(v) = W(v)\overline{W(rv)}$ when $Na = M$, and use its conjugate in the reversed orientation. Set $h(t) = \Phi_r(t/X)$, supported on $[X/r, 2X]$ with zero endpoints. The exact cutoff product is

$$(2\mathbf{1}_{FNa\le U} - 1)(2\mathbf{1}_{FNb\le U} - 1) = 1 - 2\mathbf{1}_{c_0<F\le d_0}.$$

No half-weight occurs at a norm cutoff. With the closed-prefix sum

$$M_{\mathcal{T}}(y) = \sum_{\substack{Nf\le y\\(f,\mathcal{T})=1}} \mu(f), \qquad M_{\mathcal{T}}(y) = 0 \quad (y < 1),$$

the scalar factor, multiplied outside by $\mu(a)\mu(b)\eta_{a,b}(u)$, is

$$\mathcal{I}_{a,b,u}(P) = \sum_{(f,\mathcal{T})=1} \mu(f)h(Nf)(1 - 2\mathbf{1}_{c_0<Nf\le d_0}). \tag{117}$$

Ordinary finite Abel summation gives, for every real y,

$$\sum_{\substack{Nf\le y\\(f,\mathcal{T})=1}} \mu(f)h(Nf) = h(y)M_{\mathcal{T}}(y) - \int_0^y M_{\mathcal{T}}(t)h'(t)\,dt.$$

Subtracting the two closed prefixes proves the endpoint-sensitive identity

$$\begin{aligned}\mathcal{I}_{a,b,u}(P) = &-\int_0^\infty M_{\mathcal{T}}(t)h'(t)\,dt + 2\int_{c_0}^{d_0} M_{\mathcal{T}}(t)h'(t)\,dt\\ &+ 2h(c_0)M_{\mathcal{T}}(c_0) - 2h(d_0)M_{\mathcal{T}}(d_0).\end{aligned} \tag{118}$$

When $c_0 = d_0$ all extra terms cancel. A boundary outside the profile has $h = 0$. Combining the two integrals gives a multiplier of absolute value one almost everywhere, rather than three separate derivative costs. Log-coordinate differentiation yields

$$\int |h'(t)|\,dt \le 2\log 2\, M_1(W)^2, \qquad 2|h(c_0)| + 2|h(d_0)| \le 4B^2.$$

Consequently

$$|\mathcal{I}_{a,b,u}(P)| \le 6M_1(W)^2 \sup_{X\le t\le 2X} |M_{\mathcal{T}}(t)|. \tag{119}$$

The full outer residual and row sum is still coherent.

A useful explicit uniformity quantity is

$$m_Q(Z) = \sup_{\substack{E \text{ finite good prime ideals} \\ N\prod_{p\in E} p\le Q}} \sup_{t\ge Z} \frac{|M_{S\cup E}(t)|}{t}.$$

It always satisfies $m_Q(Z) \le 16$ by (96). For the original residuals,

$$N\,\mathrm{rad}(abu)_{S^c} \le Na\,Nb\,Nu \le (2D/P)^2 H \le 14DH.$$

Restrict the expanded nonunit K_S to $X \ge Z \ge 1/2$. This cuts the residual f-length X, not $F = Nf$ in the tail below. Equation (119) bounds each scalar by $12M_1(W)^2 X m_{14DH}(Z)$. For fixed P put $V = D/P$. Ideal counting gives

$$\#\{(a,b) : \max(Na, Nb) \le t\} \le 256t^2, \qquad \sum_{\max(Na,Nb)\le T} \frac{1}{\max(Na,Nb)} \le 512T.$$

The second inequality follows by partial summation and is vacuous for $T < 1$. Since $M \le V/Z$, the absolute residual cost is at most $6144M_1(W)^2 m_{14DH}(Z)V^2/Z$. Also $\sum_{Np>U}(Np)^{-2} \le 32/U$, again by ideal counting. Only now taking the residual, prime and row triangle proves

$$\frac{|K_S[\text{nonunit}; X \ge Z]|}{D} \le 196608M_1(W)^2 R_w \frac{D}{UZ}\, m_{14DH}(Z). \tag{120}$$

No arbitrary-weight sieve is hidden here: the only row input is $|\eta_{a,b}(u)| \le 1$ and R_w. A *hypothetical* uniform estimate $|M_{S\cup E}(t)| \ll_{\theta,\epsilon,S} (N\prod_{p\in E} p)^{\epsilon} t^{\theta}$, $0 < \theta \le 1$, would give, for $Z \ge 1$, the restricted bound

$$\frac{|K_S[\text{nonunit}; X \ge Z]|}{D} \ll M_1(W)^2 (DH)^{\epsilon} R_w (D/U) Z^{\theta-2}.$$

No $\theta < 1$ hypothesis is assumed. On the region $2X \le 7$, every nonunit good f has zero profile (the possible norm-7 endpoint is zero). Hence only $f = 1$ survives, and (117) is exactly $h(1)(1-2\mathbf{1}_{c_0<1\le d_0})$. For instance $a = b$ gives $|W(1/X)|^2$. Thus even a strong long-sum Mertens estimate cannot supply cancellation in this one-term region after the outer residual triangle. This is a limitation of that scalar route, not a lower bound for the complete signed sum; other residual and row sectors may cancel it.

S72.3 Deleted-prime transfer and its fixed-field subpower input

The deletion has an exact fixed-zeta interpretation. Put $M_K(x) = \sum_{Na\le x} \mu(a)$, and let $\mathcal{D}_{\mathcal{T}}$ be *all* ideals supported on $\mathcal{T}$, including prime powers. For $\sigma > 0$ set

$$E_{\mathcal{T}}(\sigma) = \prod_{p\in\mathcal{T}} (1 - (Np)^{-\sigma})^{-1}.$$

Absolute Euler multiplication initially on $\operatorname{Re} s > 1$ gives the coefficient identity and hence the finite convolution

$$M_{\mathcal{T}}(x) = \sum_{\substack{d \in \mathcal{D}_{\mathcal{T}} \\ Nd \le x}} M_K(x/Nd). \tag{121}$$

Restricting d to squarefree ideals would be false. This is an artificial finite Euler deletion, not a new primitive Hecke conductor. Its finite factors have possible poles only on $\operatorname{Re} s = 0$.

Choose C_K with $A_K(x) \le C_K x$, and define $\rho_K(y) = \sup_{v \ge y} |M_K(v)|/(C_K v) \le 1$. For $x \ge 1$ and $0 < \delta < 1$, split (121) at $Nd = \sqrt{x}$. On the first part use $\rho_K(\sqrt{x})$ and $\sum (Nd)^{-1} \le E_{\mathcal{T}}(1-\delta)$. Rankin's inequality bounds the second positive sum by $x^{-\delta/2} E_{\mathcal{T}}(1-\delta)$. Thus

$$|M_{\mathcal{T}}(x)| \le C_K x E_{\mathcal{T}}(1-\delta)\{\rho_K(\sqrt{x}) + x^{-\delta/2}\}. \tag{122}$$

Its minimum with the trivial $C_K x$ bound is always valid.

For clarity, the fixed-field analytic input used to estimate ρ_K is separate from this elementary transfer. The already pinned Thorner–Zaman, Theorem 3.1 gives the classical Dedekind zero-free region, with a possible single fixed exceptional real zero. Specialize only to the trivial extension K/K, of absolute degree 2 and discriminant 3. Shrink the fixed constant so any fixed exceptional zero is outside the chosen region, retaining its dependence in the fixed threshold. For $|t| \ge 2$, the Euler bound just right of 1, fixed-degree functional equation/convexity in a strip of width $O(1/\log(|t|+3))$, and Borel–Carathéodory for an analytic logarithm on smaller zero-free disks give

$$|1/\zeta_K(\sigma + it)| \ll_K \log^A(|t|+3), \qquad \sigma \ge 1 - c_K/\log(|t|+3).$$

Indeed the upper bound for $\operatorname{Re} \log \zeta_K$ and its value at a center to the right of 1 are $O(\log\log(|t|+3))$; a smaller disk has the same bound for its absolute logarithm. Bounded t is handled in a fixed zero-free neighborhood of 1, where the reciprocal is holomorphic also at the pole of ζ_K. No logarithm is continued through that pole.

Here is the Perron error needed for the deduction. The planar fundamental-cell count in this class-number-one field gives $A_K(x+h) - A_K(x) \ll_K h + \sqrt{x} + 1$ for $h \ge 0$. Apply truncated Perron to $1/\zeta_K$ at $x' = \lfloor x \rfloor + 1/2$, whose summatory value equals $M_K(x)$. The coefficients are bounded in absolute value by the ideal counts. Dyadic annuli around x' control the near-integer terms by $x \log^A(2x)/T + \sqrt{x}\log^A(2x)$; the remote terms have the first cost. Shift to $1 - c_K/\log(T+3)$, which lies inside the shrunken zero-free region for every $|t| \le T$. There is no residue at 1, since the reciprocal has a zero there. The vertical and horizontal bounds give

$$|M_K(x)| \ll_K x \log^A(2x) \left\{ \exp\left(-\frac{c_K \log x}{\log(T+3)} \right) + T^{-1} \right\} + \sqrt{x}\log^A(2x).$$

Choosing $T = \exp(c'_K \sqrt{\log(2x)})$ and absorbing logarithms proves

$$|M_K(x)| \ll_K x \exp(-c''_K \sqrt{\log(2x)}). \tag{123}$$

Small x is absorbed in the fixed constant. This is fixed-field subpower cancellation; it asserts no fixed power or uniformity over varying primitive conductors or fields.

The remaining deletion factor can be quantified elementarily. Put $R_{\mathcal{T}} = N \prod_{p \in \mathcal{T} \setminus S} p$ and $Z_0 = \log(e^3 R_{\mathcal{T}}) \ge 3$. For $0 < \delta \le 1/2$, split the prime sum for $\log E_{\mathcal{T}}(1-\delta)$ at Z_0^2. Ideal counting bounds the small part by $O_K(Z_0^{2\delta} \log Z_0)$. For the large part use $\sum_{p \in \mathcal{T} \setminus S} \log Np \le \log R_{\mathcal{T}}$ and the monotonicity of $v^{1-\delta}\log v$; its bound is $O(\log R_{\mathcal{T}}/(Z_0^{2-2\delta}\log Z_0)) = O(1)$. Fixed S factors cost a fixed constant. Hence

$$\log E_{\mathcal{T}}(1-\delta) \le C_{K,S}\{1 + Z_0^{2\delta}\log Z_0\}.$$

If $\delta \log Z_0 \le 1$, this is $E_{\mathcal{T}}(1-\delta) \ll Z_0^A$. Taking $\delta = \min\{c/\sqrt{\log(2x)}, 1/\log Z_0, 1/2\}$ in (122) proves, for $x \ge 3$,

$$|M_{\mathcal{T}}(x)| \ll_{K,S} xZ_0^A \exp\left(-c\min\left\{\sqrt{\log(2x)}, \frac{\log(2x)}{\log Z_0}\right\}\right). \tag{124}$$

For $R_{\mathcal{T}} \le D^C$ and $x \ge D^v$, fixed $C, v > 0$, this gives a uniform $x\log^A(2D)\exp(-c\sqrt{\log(2x)})$ bound. It is useful on long scalar ranges, but gives no power saving and no cancellation in the unit-f range above. The exact product bound (122) retains every raw deletion before any convenient larger estimate.

S72.4 The joint matrix, raw zeros and its compression

All spaces here carry ordinary counting ℓ^2 norms. On the fixed finite short-column space of Proposition S71.1, $g = fa$ is squarefree precisely when $(f, a) = 1$. For $(f, u) = 1$, multiplicativity gives

$$(\mathsf{R}\mathsf{P}_f b)(u) = \sqrt{w_u}\nu(f)\chi_f(u)T_f(t;u).$$

If $(f, u) > 1$, every g in that projected column support has the original zero $\chi_g(u) = 0$. Thus its row is exactly zero, rather than being restored through a primitive character. Squaring proves $G_f(t) = \|\mathsf{R}\mathsf{P}_f b(t)\|^2$ and (112). The Gram entries remain

$$(\mathsf{R}^*\mathsf{R})_{g,h} = \sum_u w_u\overline{\chi_g(u)}\chi_h(u).$$

No coherent orientation mean is removed.

The divisor sum in each matrix entry is the exact Schur factor

$$C_U(g,h) = \sum_{\substack{f \mid (g,h)\\ f\ U\text{-smooth}}} \mu(f) = \mathbf{1}_{(g,h)\text{ has no prime norm } \le U}.$$

On active $t \ge U$ support, $Ng, Nh < 7U$. Any such g containing a rough prime must be that prime alone. It follows that

$$C_U(g,h) = \mathbf{1}_{(g,h)=1} + \mathbf{1}_{g=h=p,\ p\notin S,\ Np>U}. \tag{125}$$

For the projection $\Pi_{\mathcal{P}}$ onto rough-prime columns, $\mathsf{P}_f\Pi_{\mathcal{P}} = 0$ for every nonunit smooth f and $\mathsf{P}_1\Pi_{\mathcal{P}} = \Pi_{\mathcal{P}}$. Therefore

$$\Pi_{\mathcal{P}}\mathcal{M}_U\Pi_{\mathcal{P}} = \Pi_{\mathcal{P}}\mathsf{R}^*\mathsf{R}\Pi_{\mathcal{P}}. \tag{126}$$

This is a quadratic-form *compression*, not an invariant restriction: in general $\mathcal{M}_U\Pi_{\mathcal{P}}$ and $\Pi_{\mathcal{P}}\mathcal{M}_U$ differ. Off-block couplings and other columns can cancel its contribution. It is no lower bound for full Q, the prime error or the original raw slice.

On any common active length dyad $[L, 2L]$, a true squared row norm $\lambda_*(L)$ gives, for arbitrary coefficients c there,

$$|c^*\mathcal{M}_U c| \le \sum_f \|\mathsf{R}\mathsf{P}_f c\|^2 \le \lambda_*(L)\sum_g d_U(g)|c_g|^2, \qquad d_U(g) = \#\{f \mid g : f\ U\text{-smooth}\}.$$

This is a positive incidence-frame majorant, not Möbius cancellation. Since g is squarefree, $d_U(g) \le 2^{\omega(g)} \ll_\eta (Ng)^\eta$. Self-adjointness bounds the compressed operator norm by the same $\lambda_*(L)\max d_U$ and permits polarization for cross terms. For $L = D/t$, the exact short coefficient and ideal counting give

$$\|b(t)\|^2 \ll B^2L, \qquad \|tb'(t)\|^2 \ll M_1(W)^2L.$$

The direct unit endpoint covers $1/2 \le L < 1$; no column survives below $1/2$. Differentiating the fixed matrix, not a moving support matrix, gives $Q'(t) = 2\,\mathrm{Re}(b'(t)^* \mathcal{M}_U b(t))$ and

$$|Q(t)| + |tQ'(t)| \ll_\eta (HD)^\eta M_1(W)^2 \lambda_*(D/t) D/t. \tag{127}$$

Only here may the classical specialization be inserted, in exactly its original natural-row scope. For arbitrary weights a true weighted norm is needed; the elementary Frobenius alternative is $\lambda_*(L) \ll R_w L$.

S72.5 Exact Stieltjes boundary and squarefunction criterion

The finite Q vanishes for $t \ge 2D$, including its endpoint. Stieltjes integration over (U, ∞) counts precisely $Np > U$; comparing its prime measure with $dt/\log t$ proves (113). Changing the normalization of Li by a constant changes the two boundary/integral terms by opposite amounts and changes no identity. We fix Li_2 here. The original raw slice is consequently

$$S_1^{AA} = \frac{K_0 - \mathcal{J}}{D} + \Delta + \frac{C_2}{D} - \frac{\mathcal{E}_{\mathrm{PIT}}}{D}.$$

It keeps the original masks and the full mean. The prime measure contains only fixed S, never a growing deleted set $S \cup \mathrm{supp}(abu)$.

Let $\delta_S(U) = \sup_{t \ge U} |E_S(t)|/t$. Use (127) and substitute $L = D/t$ to obtain

$$\frac{|\mathcal{E}_{\mathrm{PIT}}|}{D} \ll_\epsilon (HD)^\epsilon M_1(W)^2 \delta_S(U) \left\{ \lambda_*(D/U) + \int_{1/2}^{D/U} \lambda_*(L)\, \frac{dL}{L} \right\}. \tag{128}$$

One may keep $\max d_U$ explicitly instead of absorbing it into a subpower; no uniform logarithmic improvement is thereby claimed. With the current physical input of Proposition S73.4, the minimum of its three branches remains inside this integral. All constants and internal small exponents are chosen uniformly on the neighboring capacity-feasible blocks.

For $I_T = [T, 2T] \cap [U, 2D]$, let $\lambda^\#(D/T)$ dominate $\lambda_*(L)$ on $L \in [D/(2T), D/T]$, using the direct endpoint when necessary. Equation (127) implies

$$\|Q'\|_{L^2(I_T)} \ll (HD)^\eta M_1(W)^2 \lambda^\#(D/T) D/T^{3/2}.$$

Our weighted squarefunction convention gives $\|E_S\|_{L^2(I_T)} \le (2T)^{3/2} \mathfrak{d}_2(T)$. Cauchy on each dyad and the retained endpoint prove (114). The weighted definition is comparable by absolute constants to $T^{-3/2}\|E_S\|_{L^2([T,2T])}$. Global Cauchy across unrelated dyads need not improve this bound. The sufficient target condition is that its braces be $\ll H(HD)^\epsilon$, including the pointwise term. It is an explicit input to investigate, not an established estimate or an independence assumption between E_S and the signed derivative.

S72.6 Mellin Parseval retains equal-norm coherence

Extend only the same finite short-column vector to $t > 0$ for this identity; below U it need not equal the unrestricted original T_f with additional longer columns. No signed equality for K_S is enlarged below U. For fixed f, u let $c_g = A_U(g)\nu(g)\chi_g(u)\mathbf{1}_{f|g}$, and use

$$\widehat{W}(i\tau) = \int_0^\infty W(x) x^{i\tau} \frac{dx}{x}.$$

Ordinary Mellin Plancherel for the finite amplitude gives exactly

$$\begin{aligned}&\int_0^\infty \left|\sum_g c_g W(Ng\,t/D)\right|^2 \frac{dt}{t}\\ &= \frac{1}{2\pi}\int_{\mathbb{R}} |\widehat{W}(i\tau)|^2 \left|\sum_g c_g (D/Ng)^{i\tau}\right|^2 d\tau.\end{aligned} \tag{129}$$

For the derivative of this *amplitude*, replace $|\widehat{W}|^2$ by $\tau^2|\widehat{W}|^2$. This does not assert a diagonal Mellin formula for the square of the quadratic derivative Q'. Finite signed summation in f and the actual rows preserves the matrix form at the norm-twisted coefficient vector. No absolute-value or centering operation is inserted into that equality.

If distinct ideals g, h have $Ng = Nh$, their radial profiles and all their t derivatives are identical. Their radial cross correlation is the same nonzero profile norm as the diagonal when $W \neq 0$. The Mellin phases $(Ng)^{-i\tau}$ also coincide; distinct conjugate split ideals provide actual such orientations. Thus neither radial Parseval nor the derivative removes their coherent mean. The safe decomposition is

$$\sum_r a_r \Gamma_r = \Gamma_{\mathrm{av}} \sum_r a_r + \sum_r a_r(\Gamma_r - \Gamma_{\mathrm{av}}).$$

Both terms remain. The already recorded fixed-profile example in Section S65 has centered variance sum 6, coefficient norm squared 2, but full scalar squared $36 > 12$; it is not rerun or identified with this prime error. It illustrates why a variance-only replacement is invalid. Applying an existing arbitrary-coefficient norm at every τ in (129) leaves its operator power unchanged; the twist has modulus one.

S72.7 Proof of the inherited smooth-divisor localization

Now specialize only to the original unweighted $A_6 = 1$ two-pivot rows. The exact K_0 positive summand is

$$G_f(1) = \sum_{\substack{u\in\mathcal{U}\\(f,u)=1}} |T_f(1;u)|^2.$$

Here $A_U(fa)$ is in the general annular domain and stays unfactored. The divisor-assignment bound and ideal counting give its coefficient squared norm $\ll_\eta B^2 D^\eta L$, $L = D/F$. The accepted powered subsequence norm (55) gives

$$G_f(1) \ll (HD)^\eta B^2 \left\{ LR + \frac{L^2 R}{M_{\mathrm{row}}} + \frac{L^{5/3} R}{M_{\mathrm{row}}^{1/3}} \right\}.$$

For $1/2 < L < 1$ count the unit residual directly. Its $O(RD^\eta B^2)$ cost is covered by the same expression up to a constant; for $L = 1/2$ the endpoint is zero. No subunit sieve is used. Now take the triangle only in the tail $F > Y$, and enlarge the nonnegative squarefree smooth-ideal upper sums to all ideals. Partial summation from ideal counting yields

$$\sum_{Y<F<2D} F^{-1} \ll \log(2D), \quad \sum_{F>Y} F^{-2} \ll Y^{-1}, \quad \sum_{F>Y} F^{-5/3} \ll Y^{-2/3}.$$

Division by D and absorption of logarithms and internal assignment powers prove (115). Every rough-prime K_S term has $FPNa > FU$, so $F > Y \geq 2D/U$ forces zero profile even in its auxiliary nonsquarefree cases. The C_2 and P_2 unit corrections are not silently removed by this tail split.

When $R \ll H$, solving the last two monomial inequalities for a sufficient Y gives the displayed cutoff in Proposition S71.2. At the mixed witness it is enough to take $Y = D^{13/24}$ with fixed capacity slack, giving tail exponents $7/12, 5/8, 3/4$. At the pure witness $Y = D^{5/8}$ gives $3/4, 3/8, 3/4$. Both exceed the rough support exponent $1/2$ for large D. These are inherited sufficient upper cutoffs, not energy lower bounds or assertions of current optimality after the new physical input. All small smooth divisors, including $f = 1$, remain; no signed main estimate follows from the positive tail. The current whole-form comparison in Proposition S73.3 is a different consequence of the new external norm input.

S72.8 Proof of the conditional zero-free input diagnostic

Assume exactly the all-scale, all-ϵ hypothesis of Theorem S71.3. For $\operatorname{Re} s > 1$ put

$$F_S(s) = \int_2^\infty E_S(t) t^{-s-1}\, dt.$$

On a dyad, Cauchy gives

$$\int_T^{2T} |E_S(t)| t^{-\sigma-1}\, dt \le \mathfrak{d}_2(T) \left(\int_T^{2T} t^{1-2\sigma}\, dt \right)^{1/2} \le C_\sigma \mathfrak{d}_2(T) T^{1-\sigma}.$$

At $\sigma = 1$ use the logarithmic limit of the finite integral. On a compact subset of $\sigma > 1 - \alpha$, choose ϵ smaller than its distance to that boundary. The dyadic majorant is a summable geometric series, also after any fixed power of $\log t$. The same integral therefore converges normally, with all holomorphic derivatives, on $\sigma > 1 - \alpha$. Its bounded initial interval is entire. No contour has crossed an unknown zero.

For $\sigma > 1$, the prime-ideal series satisfies

$$P_S(s) = \sum_{p \notin S} (Np)^{-s} = L_0(s) + sF_S(s), \qquad L_0(s) = \int_2^\infty \frac{t^{-s}}{\log t}\, dt.$$

In this field $\pi_K^S(2) = 0$ and $\operatorname{Li}_2(2) = 0$, so no lower boundary atom is missing. Direct differentiation gives $L_0'(s) = -2^{1-s}/(s-1)$. Only this meromorphic derivative is continued; no single-valued logarithmic primitive is asserted through $s = 1$. Consequently

$$-P_S'(s) = \frac{2^{1-s}}{s-1} - (sF_S(s))'.$$

For $\zeta_K^S(s) = \zeta_K(s) \prod_{p \in S} (1 - (Np)^{-s})$, its initial logarithmic derivative is

$$-\frac{(\zeta_K^S)'(s)}{\zeta_K^S(s)} = -P_S'(s) + H_S(s), \quad H_S(s) = \sum_{p \notin S} \sum_{k \ge 2} \log(Np)(Np)^{-ks}.$$

The higher-prime-power series is normally holomorphic for $\sigma > 1/2$: its geometric tail is bounded on compact subsets by $C \sum_p \log(Np)(Np)^{-2\sigma}$, an absolutely convergent ideal series. Inert prime ideals are included, not mistaken for absent rational primes. The right side has no possible pole in $\sigma > \max\{1 - \alpha, 1/2\}$ except $s = 1$. Standard continuation of the fixed Dedekind zeta, or $\zeta_K(s) = \zeta(s) L(s, \chi_{-3})$, identifies it with the actual logarithmic derivative. A zero elsewhere would give a pole with nonzero integer residue, impossible there. Finite deleted Euler factors vanish only on $\sigma = 0$, so they change no zero in this region. This proves Theorem S71.3. A fixed exceptional

real zero, if inside the strict region, would be excluded by the hypothetical input itself; it is not assumed absent.

For orientation, the old and current sufficient scalar power accounting is recorded below. Every input and ensuing strip in the table is conditional, and endpoint control is additional to a mean-square norm. Here $\sigma = \operatorname{Re} s$ and the input column means $U^{-\alpha}$.

certificate	budget	input α	strict strip
old pure (historical)	5/6	1/6	$\sigma > 5/6$
current pure	19/24	1/12	$\sigma > 11/12$
current mixed	7/9	1/18	$\sigma > 17/18$

The current values use Proposition S73.4, not the historical pure 5/6 operator budget. Since $U \asymp D^{1/2}$, subtracting the target 3/4 from the respective PIT powers and doubling gives 1/12 and 1/18. A pointwise all-scale power for δ_S would imply the corresponding squarefunction power; the converse and its pointwise endpoint are not supplied. The accepted fixed-field subpower PIT is not one of these independent powers.

The new physical input does prove its stated global 9/8 and $A_6 = 1$ signed/combined 13/12 comparisons. The coherent reductions above do not improve them to the target, nor establish desired positivity, general Artin holomorphy, determinant/zero-location bridges, a varying-profile fixed-seminorm theorem or new Lean results. Finite formal controls and source-bound reviews check bookkeeping; they do not replace any infinite analytic hypothesis. Full primary reading caches remain local: only public citation/version/hash metadata accompanies these proofs.

S73 A sieve comparison indexed by the physical radicand

This section applies a new external preprint input to the original rows. It is an application of that theorem, not a new proof of a large sieve, a signed target estimate, or an analytic Lean result. Throughout, $K = \mathbb{Q}(\omega)$, S is fixed and contains the primes above $2, 3$ and the original conductor exclusions, and ν is the original fixed finite-order Hecke column character. Let $D, H \geq 1$, $U \geq 1$, $U^2 \leq D$, and let W be one common possibly complex smooth profile supported on $[1, 2]$, zero at its endpoints. Write $B_W = \|W\|_\infty$. The estimates below concern the original unweighted literal rows, or already justified bounded natural multiplicities. They do not insert arbitrary positive row weights into an unweighted theorem.

Pinned external input. De Faveri's arXiv:2610.04045v1, Theorem 1.1, bounds the squared row norm $\Theta_n(A, B)$ for the Section 2.3 family of order-dividing-n Hecke characters, with both indices nth-power-free ideals prime to fixed S:

$$\Theta_n(A, B) \ll_\epsilon (AB)^\epsilon (A + B + A^{1-1/n} B^{2/n} + A^{2/n} B^{1-1/n}). \tag{130}$$

Here $n \geq 3$, K contains μ_n, and the chosen fixed S-integer ring has class number one. The constants may depend on the fixed family construction. We use $n = 6$, not a conjectural $A + B$ bound; the additional μ_{2n} condition of that paper's Theorem 1.4 is not a hypothesis of Theorem 1.1. Set

$$\Phi_6(A, B) = A + B + A^{5/6} B^{1/3} + A^{1/3} B^{5/6},$$
$$\Phi_{\mathrm{all}}(H, D) = H + DH^{1/6} + H^{5/6} D^{1/3} + H^{1/3} D^{5/6}.$$

Proposition S73.1 (Exact raw-family identification)**.** *There is an allowed fixed choice of the data in* (130) *for which, for every good ideal* $\mathfrak{a}$*, with unique generator* $\alpha_{\mathfrak{a}} \equiv 1 \pmod 3$*,*

$$\chi_{\mathfrak{a}}(\mathfrak{m}) = \left(\frac{\alpha_{\mathfrak{a}}}{\mathfrak{m}}\right)_6$$

including the original zero at each common prime. In particular, the row index is the physical sixth-power-free ideal, including valuations $2,3,4,5$*, rather than its radical conductor.*

The extended proof specifies the data and removes the possible unit ambiguity. No extra column-ray selection is made. For the unique outside-S decomposition of an actual literal row,

$$\begin{aligned} u &= u_0 \alpha_{\mathfrak{a}} \alpha_{\mathfrak{r}_6}^6, & \mathfrak{a} &= \mathfrak{r}_1 \mathfrak{r}_2^2 \mathfrak{r}_3^3 \mathfrak{r}_4^4 \mathfrak{r}_5^5, \\ \chi_{\mathfrak{m}}(u) &= \chi_{\mathfrak{m}}(u_0)\chi_{\mathfrak{a}}(\mathfrak{m})\mathbf{1}_{(\mathfrak{m},\mathfrak{r}_6)=1}, & Nu &= Nu_0\, N\mathfrak{a}\, (N\mathfrak{r}_6)^6. \end{aligned} \tag{131}$$

The first five factors are squarefree and pairwise coprime; $\mathfrak{r}_6$ is arbitrary and may overlap them. All original union, common-prime, literal, ray and capacity restrictions remain.

Proposition S73.2 (Physical-row comparison)**.** *For the original raw sums* I_u *of* (21)*, and for the original residue-subtracted* $E_u = I_u - P_u$ *with fixed* ν, S*,*

$$\sum_{0<Nu\le H} |I_u|^2 + \sum_{0<Nu\le H} |E_u|^2 \ll_{\nu,S,\epsilon} B_W^2 (HD)^{\epsilon} \Phi_{\mathrm{all}}(H,D).$$

For $1 \le H \le D$ *one has*

$$\Phi_{\mathrm{all}}(H,D) \le 4DH^{1/6}. \tag{132}$$

Thus the critical positive envelope at $H = D^{3/4}$ *is* $D^{9/8+\epsilon}$*, improving the previous* $7/6$ *comparison by* $1/24$ *while still exceeding the* $3/4$ *target by* $3/8$*.*

The agreement with the varying-profile obstruction of Section S63 concerns only a supremum-norm-controlled class allowing W_D to vary. It is not a lower bound or a sharpness assertion for one fixed W, nor a GRH counterexample.

Proposition S73.3 (The actual $A_6 = 1$ signed slice)**.** *For an original unweighted two-pivot block, retaining all its actual filters and the complete annular columns, the raw form* S_1^{AA} *of* (51) *satisfies*

$$|S_1^{AA}| \ll_{\nu,S,\epsilon} B_W^2 (HD)^{\epsilon} \Phi_6(H,D).$$

The same bound holds after summing the fixed-S parts of such rows: the D *term costs a fixed polylogarithmic count, absorbed in* $(HD)^{\epsilon}$*, not a convergent exponent-zero Euler sum. At* $H = D^{3/4}$ *the available power is* $13/12$*, an improvement of* $1/12$ *from the old* $7/6$ *majorant, but still* $1/3$ *above the target.*

The proof inserts (130) into the exact original common-divisor identity (53); $A_U(dn)$ is not factored and is not put into the rough short-coefficient domain. Under the additional high-window and capacity hypotheses of Section S71, its exact ordered rough correction gives

$$\left|\frac{K_0 - K_{\mathrm{all}}}{D}\right| \le |S_1^{AA}| + 236800 B_W^2 R_w. \tag{133}$$

This estimates the actual combined form through its exact identity; it does not take separate main-term triangles in K_0, K_{all}, and does not prove signed cancellation at the target scale.

Proposition S73.4 (Updated short positive certificate)**.** *On an original unweighted block, put* $R = A_1A_2$, $M = \max(A_1, A_2)$ *and* $H_b = \min\{H/Nu_0, 8A_1A_2^2\}$ *for fixed literal/S factor* u_0*. Empty blocks contribute zero; a nonempty block has* $H_b \geq 1$*. For* $1 \leq L \leq D$ *the actual squared row-operator norm has the bound*

$$\lambda_{\mathrm{now}}(L) \ll (HD)^{\epsilon} \min \begin{Bmatrix} R + \dfrac{LR}{M} + \dfrac{L^{2/3}R}{M^{1/3}}, & L + R^2, \\ \Phi_6(H_b, L) \end{Bmatrix}.$$

For $1/2 \leq L < 1$ *use the direct unit-column norm* R_w*. This new norm may be substituted into the mean-retaining joint Abel criterion of Proposition S71.1, with its lower endpoint intact. At* $H = D^{3/4}$, $U \asymp D^{1/2}$*, its pure witness costs* $19/24$*; at the displayed mixed witness the minimum retains the old* $7/9$ *bound, which beats the generic* $19/24$*.*

In this positive certificate sufficient independent scalar discrepancy powers are respectively

$$U^{-1/12+\epsilon} \quad \text{and} \quad U^{-1/18+\epsilon}. \tag{134}$$

If imposed as all-scale mean-square inputs, these would entail the conditional zero-free strips $\Re s > 11/12$ and $\Re s > 17/18$ of Theorem S71.3. Neither input nor strip is proved here; they are not necessary conditions for the actual signed target. The endpoint and coherent $K_0 - J$ remain. Smooth-tail localization (Proposition S71.2) does not remove $f = 1$ or supply this estimate.

S74 Proof of the physical-row sieve application

S74.1 The precise literature input and the primary generator

The external input is de Faveri, *Optimal large sieve for fixed order characters*, arXiv:2610.04045v1, 2 October 2026, Theorem 1.1, printed page 2. Input (130) is invoked with the family in its Sections 2.2–2.5; its deep proof is not reproduced or claimed as a new sieve or Lean theorem here. The primary PDF SHA-256 is the concatenation of these two lines:

`97720b3cacccbedb9c2014520ad4d128`
`ae23afd0a781bdeff27dab4e86b7c726`

The public citation receipt records the version and text hash. Full PDF/text reading snapshots are local cache and excluded from public closure.

The ring $\mathcal{O} = \mathbb{Z}[\omega]$ is a PID. Its six units map bijectively onto $(\mathcal{O}/(3))^{\times}$, which has six elements. Consequently every fractional ideal prime to 3 has a unique generator $\alpha_{\mathfrak{a}} \equiv 1 \pmod 3$, interpreted locally at 3. This choice is multiplicative, since a product of two such generators is again 1 modulo 3 and generates the product ideal.

Choose the source's fixed modulus $\mathfrak{c}_0$ sufficiently divisible at S that its stipulated local sixth-power condition holds, and choose it divisible by (3). In the source's Section 2.3 choose ordinary ideal representatives E_0 prime to S, and set $m_{E_0} = \alpha_{E_0}$. Global principality makes these allowed generators after S-localization too. The prescribed products then give $m_E = \alpha_E$ for each representative E. For coprime good ideals $\mathfrak{a}, \mathfrak{m}$, the construction gives

$$\mathfrak{a} = (x)E\mathfrak{g}^6, \qquad x \equiv 1 \pmod{\mathfrak{c}_0}, \qquad (\mathfrak{g}, \mathfrak{m}) = 1, \qquad \chi_{\mathfrak{a}}(\mathfrak{m}) = \left(\frac{xm_E}{\mathfrak{m}}\right)_6.$$

Choose an ordinary fractional generator $\gamma_{\mathfrak{g}}$ of $\mathfrak{g}$. Ordinary ideal equality implies

$$\alpha_{\mathfrak{a}} = \eta x m_E \gamma_{\mathfrak{g}}^6, \qquad \eta \in \mu_6. \tag{135}$$

All factors are units at 3, and $\gamma_{\mathfrak{g}}^6 \equiv 1 \pmod 3$ because the residue unit group has order six. The other three displayed generators are 1 modulo 3. Hence $\eta \equiv 1 \pmod 3$ and injectivity on μ_6 forces $\eta = 1$. Taking the sixth-power residue symbol proves the exact identification in Proposition S73.1. For a common prime the source's induced character and the original raw symbol are both zero. Thus no primitive-conductor transfer can silently restore a nonunit value.

The field contains μ_6; it need not contain μ_{12} for the Theorem 1.1 input. Its characters have order dividing six and trivial infinite type, so physical valuations $2, 3, 4, 5$ are included. They are not replaced by a squarefree radical or indexed by its norm. The fixed choice of $S, \mathfrak{c}_0, E, m_E$ is independent of D, H, U and of all coefficient masks below. We fix the complex embedding to agree with the original sextic symbol. This is a deliberate allowed construction, not an assertion that every noncanonical choice in the source is identical.

S74.2 The complete raw row family and the sixth factor

At each prime outside S, divide its valuation in (u) by six. The remainder defines the sixth-power-free ideal $\mathfrak{a}$, and the quotient defines $\mathfrak{r}_6$. The quotient may exceed one and may be positive at a prime of $\mathfrak{a}$. Thus $\mathfrak{r}_6$ is neither assumed squarefree nor made coprime to the first five factors. With the multiplicative primary generators, the quotient $u_0 = u/(\alpha_{\mathfrak{a}}\alpha_{\mathfrak{r}_6}^6)$ is an integral S-supported element, including its literal unit. This decomposition is unique and has the norm relation in (131).

For a squarefree good column $\mathfrak{m}$, the sixth-power factor has symbol one when $(\mathfrak{m}, \mathfrak{r}_6) = 1$ and zero otherwise. The exact family identification gives (131), even when $\mathfrak{a}, \mathfrak{r}_6$ overlap. In particular a prime common to both factors is still an original raw zero, not two independent masks. For fixed $u_0, \mathfrak{r}_6$ set

$$c_{\mathfrak{m}} = A_U(\mathfrak{m})\nu(\mathfrak{m})\chi_{\mathfrak{m}}(u_0)W(N\mathfrak{m}/D)\mathbf{1}_{(\mathfrak{m},\mathfrak{r}_6)=1}.$$

These coefficients are independent of the remaining row $\mathfrak{a}$. They vanish outside the COMPLETE original good squarefree annulus; putting zero on the extra sixth-power-free indices is permitted. The actual row restrictions can only decrease this positive norm, and

$$\sum_{\mathfrak{m}} |c_{\mathfrak{m}}|^2 \ll_\epsilon B_W^2 D^{1+\epsilon}. \tag{136}$$

Indeed $|A_U(\mathfrak{m})| \le 3^{\omega(\mathfrak{m})}$ uniformly in U, and there are $O_K(D)$ ideal columns. This is the original assignment bound; it does not assert $A_U(dn) = A_U(d)A_U(n)$ or use the rough short-coefficient formula on K_0.

The fixed ν and literal/S phase have modulus one on good columns. A row-independent unitary Jacobi factor of nonzero infinity type could also be included in these arbitrary coefficients. This does not turn it into an order-six row character, supply a theorem for varying infinite types, or improve the operator norm by a gauge. Arbitrary positive row weights are not part of this application; they require a true weighted operator bound in the exact identities of Proposition S71.1.

S74.3 The raw global sum and the principal assembly

For fixed $u_0, \mathfrak{r}_6$, use (130) with

$$A = \frac{H}{Nu_0(N\mathfrak{r}_6)^6}, \qquad B = 2D.$$

If $A < 1$ there is no remaining ideal row. If $A = 1$ the unit row remains; it is not dropped as a subunit endpoint. Equations (131) and (136), and the factor D^{-1} in $|I_u|^2$, give a bound by

$B_W^2(HD)^\epsilon\Phi_6(A,D)$. Now sum all $\mathfrak{r}_6$ of norm at most $(H/Nu_0)^{1/6}$. The $A, A^{5/6}D^{1/3}, A^{1/3}D^{5/6}$ terms use respectively

$$\sum_{\mathfrak{r}_6}(N\mathfrak{r}_6)^{-6}, \qquad \sum_{\mathfrak{r}_6}(N\mathfrak{r}_6)^{-5}, \qquad \sum_{\mathfrak{r}_6}(N\mathfrak{r}_6)^{-2},$$

which converge. The D term instead uses the ideal count $O_K((H/Nu_0)^{1/6})$. Thus the four resulting terms are

$$\begin{aligned} &H(Nu_0)^{-1}, \qquad DH^{1/6}(Nu_0)^{-1/6}, \\ &H^{5/6}D^{1/3}(Nu_0)^{-5/6}, \qquad H^{1/3}D^{5/6}(Nu_0)^{-1/3}. \end{aligned} \tag{137}$$

Every exponent on the fixed finite S-supported semigroup is positive, so these sums converge geometrically, with the six literal units included. This proves the raw part of Proposition S73.2, with every arbitrary sixth-factor overlap still represented by its coefficient mask.

For the residue-subtracted family, use the original formula (22). Its deleted Euler residue is bounded by a fixed constant and

$$\sum_d \frac{|\beta_U(d)|}{Nd} \le \left(\sum_{Na\le U}\frac{1}{Na}\right)^2 \ll_K \log^2(2U).$$

Therefore $|P_u| \ll_{\nu,S} B_W\sqrt{D}\log^2(2U)$ on a principal row. Outside S, such a row has all valuations divisible by six, since the original ν has its defining conductor in S. The already established count is $O_{\nu,S,\epsilon}(H^{1/6}(HD)^\epsilon)$. Consequently

$$\sum_u |P_u|^2 \ll_{\nu,S,\epsilon} B_W^2(HD)^\epsilon DH^{1/6}. \tag{138}$$

Using $|I-P|^2 \le 2|I|^2+2|P|^2$ proves the stated E_u bound. No Artin holomorphy, spectral positivity or zero-location assertion is used. For $1 \le H \le D$, each of the four terms in Φ_{all} is at most $DH^{1/6}$, proving (132). The varying-W_D construction of Section S63 gives a compatible supremum-norm-class obstruction; it does not make a fixed-W lower bound.

S74.4 The signed $A_6 = 1$ common-divisor form

For a fixed u_0 the first-five physical ideal is sixth-power-free. For the two-pivot block it is exactly $v_1v_2^2$, of norm at most H/Nu_0, not Nv_1Nv_2. Apply the positive norm (130) inside the exact common-divisor identity (53). For each squarefree good d, put $L = D/Nd$ and keep the original coefficient $A_U(dn)$, the column mask $(n, dS) = 1$, and the raw row mask $(d, u) = 1$. The row mask may be enlarged only inside the positive bound. The assignment bound and ideal count give coefficient squared norm $\ll_\epsilon B_W^2D^\epsilon L$ for $L \ge 1$. For $1/2 \le L < 1$ the only possible good residual is the unit; its direct row count is bounded by the same polynomial up to an absolute factor. There are no active columns for $L < 1/2$. Thus, only now taking the triangle over d,

$$|S_1^{AA}| \ll B_W^2(HD)^\epsilon \sum_{Nd\le 2D}\frac{1}{Nd}\Phi_6(H/Nu_0, D/Nd).$$

The four ideal sums have exponents $1, 2, 4/3, 11/6$. The first is logarithmic, and the others converge, which proves the fixed-u_0 assertion of Proposition S73.3. If all S-supported u_0 are included, the terms with positive Nu_0 exponent sum geometrically. The D term has exponent zero and costs

only the finite count $O_S((\log(2H))^{|S|})$, absorbed into $(HD)^\epsilon$. It is not summed as a convergent exponent-zero Euler product.

The accepted identity of Section S71 gives

$$S_1^{AA} = \frac{K_0 - K_{\mathrm{all}}}{D} + \mathcal{E}_{\mathrm{corr}}, \qquad |\mathcal{E}_{\mathrm{corr}}| \le 236800 B_W^2 R_w$$

in its stated high window and at $H \le D^{3/4}$. This retains its ordered (p,q) correction, including $p = q$ and the asymmetric restored prime mask; it is distinct from the unordered unit correction C_2. Taking this already bounded correction gives (133). Neither K_0 nor K_{all} is separately bounded by a substituted marginal mean. The resulting 13/12 bound is an improved positive comparison, not the target signed estimate.

S74.5 The updated short certificate and its unresolved input

For a nonempty block $H_b \ge 1$. At $1 \le L \le D$, the two already established block bounds and the new physical-row bound are all valid positive estimates of the SAME actual unweighted operator. Their minimum is Proposition S73.4. The factor 8 in H_b is the literal dyadic bound $Nv_1(Nv_2)^2 < 8A_1A_2^2$, with the original capacity retained. No radical conductor is substituted for that physical norm. At $1/2 \le L < 1$ retain the actual unit-column norm. For the exact mean-retaining Abel identity of Proposition S71.1, write $X_0 = D/U$ and $M_1(W) = \max(\|W\|_\infty, \|xW'\|_\infty)$. Its sufficient positive discrepancy certificate is

$$\frac{|\mathcal{E}_{\mathrm{PIT}}|}{D} \ll (HD)^\epsilon M_1(W)^2 \delta_S(U) \left\{ \lambda_{\mathrm{now},*}(X_0) + \int_{1/2}^{X_0} \lambda_{\mathrm{now},*}(L) \frac{dL}{L} \right\}.$$

Here $U \ge 7$ is retained for the Li/Stieltjes definition, and the lower endpoint $E_S(U)Q(U)$ is still part of the exact identity. The minimum must stay inside the integral for general neighboring capacity-feasible blocks. It is not a variance-only substitution. At the critical line, the four new short monomial powers are $3/4, 1/2, 19/24, 2/3$. The pure minimum is 19/24; in the mixed $A_1 \asymp D^{5/12}, A_2 \asymp D^{1/6}$ neighborhood the old 7/9 entry remains smaller. Equating those majorant powers with the 3/4 target gives the sufficient scalar exponents in (134). These are not actual-error lower bounds. The conditional strip implication in Theorem S71.3 therefore has boundaries 11/12 and 17/18 for these proposed inputs; no such input or strip has been established here. In particular, pointwise endpoint control and the coherent signed $K_0 - J$ are not supplied by an independent mean-square assumption alone.

The new primary input improves a positive norm certificate. Equal-norm ideal orientations, coherent means, original common profiles, full-union raw zeros and all fixed-conductor qualifications remain. The actual signed target, independent determinant bridge and analytic/category Lean work remain open.

S75 Coherent dual bands, actual cutoffs and arithmetic localization

The physical-row comparison in Section S73 leaves a signed problem. We now retain its exact primitive roots and sharp row capacity, prove quantitative finite-prefix and divisor-band comparisons, and localize a coherent sector that can already be controlled. The remaining unequal-norm signed moment is stated explicitly. These results do not improve the whole signed power 13/12 or the global unsigned power 9/8. The deep sieve input remains the pinned external theorem (130); its cubic specialization is used below.

S75.1 The original form and the scope of the dual representation

Let $K=\mathbb{Q}(\omega)$, $\mathcal{O}=\mathbb{Z}[\omega]$ and $\delta=1+2\omega=i\sqrt{3}$. The fixed set S contains the primes above $2,3$ and the original exclusions. All columns are the complete good squarefree *ideal* annulus $D\le N\mathfrak{m}\le 2D$, $D>1$. Distinct ideals of the same norm remain distinct. Put

$$\begin{aligned} A_U(\mathfrak{m}) &= \sum_{\substack{\mathfrak{r}\mathfrak{s}\mid\mathfrak{m}\\(\mathfrak{r},\mathfrak{s})=1\\ N\mathfrak{r},N\mathfrak{s}\le U}} \mu(\mathfrak{r})\mu(\mathfrak{s}),\\ c_{\mathfrak{m}} &= A_U(\mathfrak{m})\nu(\mathfrak{m})R_{\mathfrak{m}}(u_0)W(N\mathfrak{m}/D). \end{aligned} \tag{139}$$

Here $1\le U\le\sqrt{D}$, ν is the original fixed unitary finite-order column character, and W is one common possibly complex smooth profile supported on $[1,2]$, zero at its endpoints. Write $B_W=\|W\|_\infty$. For a good ideal $\mathfrak{v}$, let $\alpha_{\mathfrak{v}}\equiv 1 \pmod 3$ be its unique primary generator and set $R_{\mathfrak{m}}(\mathfrak{v})=(\alpha_{\mathfrak{v}}/\mathfrak{m})_6$. The literal symbol is zero at a common prime. A primitive extension at an S-argument below is a different object, not restoration of a deleted raw row value.

Let $A_1,A_2\ge 1$. Only the original unweighted two-pivot rows are used:

$$u=u_0\alpha_{\mathfrak{v}_1}\alpha_{\mathfrak{v}_2}^2,\qquad A_i\le N\mathfrak{v}_i<2A_i,\qquad Nu_0N\mathfrak{v}_1(N\mathfrak{v}_2)^2\le H.$$

The ideals $\mathfrak{v}_1,\mathfrak{v}_2$ are good, squarefree and coprime; all actual literal and ray restrictions remain. The other four factors are the unit ideal. The fixed u_0 is literal unit/S-supported, without good-prime support. Extra good frozen factors, a sum over varying u_0, and arbitrary row weights are not included. Write $R=A_1A_2$ and let R_w be the actual number of rows. The normalization is

$$\begin{aligned} \mathcal{P}_D &= DS_1^{AA}\\ &= \sum_{(\mathfrak{v}_1,\mathfrak{v}_2)}\ \sum_{\text{actual}\ (\mathfrak{m},\mathfrak{n})=1} c_{\mathfrak{m}}\overline{c_{\mathfrak{n}}}R_{\mathfrak{m}}(\mathfrak{v}_1\mathfrak{v}_2^2)\overline{R_{\mathfrak{n}}(\mathfrak{v}_1\mathfrak{v}_2^2)}. \end{aligned} \tag{140}$$

The original Möbius signs are already in the two A_U coefficients; no additional $\mu(\mathfrak{m}\mathfrak{n})$ is inserted. This raw coprime form is not a positive energy or principal-term assembly.

S75.2 Primitive normalization and an exact finite-cutoff kernel

For each source class A, use the fixed decomposition $\widetilde{\chi}_{\mathfrak{m}}=\xi_{\mathfrak{m}}\psi_A$, $\mathrm{cond}(\xi_{\mathfrak{m}})=\mathfrak{m}$, and $\beta_{\mathfrak{m}}=\epsilon(\xi_{\mathfrak{m}})\psi_A(\mathfrak{m})$. Individual ξ and ψ may have nonzero fixed angular and radial unitary types; they are not declared sixth-order. The finite source reciprocity factors and the *whole* original row-ray projector are expanded in their complete finite character basis, with coefficients $h_{AB\rho}(\mathfrak{v}_2)$. On coprime class-$A,B$ columns the resulting primitive character is

$$\begin{aligned} &\sigma=\mathrm{prim}(\rho\psi_A\overline{\psi_B}),\quad \mathfrak{f}_S=\mathrm{cond}(\sigma),\\ &\eta=\xi_{\mathfrak{m}}\overline{\xi_{\mathfrak{n}}}\sigma,\qquad\qquad F=N\mathfrak{f}_S\,N\mathfrak{m}\,N\mathfrak{n}. \end{aligned} \tag{141}$$

Its total infinity type is zero. The actual S conductor, including any drop, is retained.

Proposition S75.1 (Primitive roots with their types retained)**.** *For a primitive unitary Hecke character $\chi((\alpha))=\theta(\alpha)\,\mathrm{phase}(\alpha)^kN(\alpha)^{it}$, with conductor (γ), let*

$$\tau_\gamma(\theta)=\sum_{x \bmod \gamma}\theta(x)\exp\bigl(2\pi i\,\mathrm{Tr}(x/(\delta\gamma))\bigr).$$

Then

$$\epsilon(\chi) = (3N\mathfrak{f})^{it} i^{-|k|} \operatorname{phase}(\delta\gamma)^k \frac{\tau_\gamma(\theta)}{\sqrt{N\mathfrak{f}}}. \tag{142}$$

The unit-conductor/principal formula has its usual meromorphic terms. For pairwise coprime primitive conductors,

$$\epsilon\Big(\prod_j \chi_j\Big) = i^{\sum_j |k_j| - |\sum_j k_j|} \prod_j \epsilon(\chi_j) \prod_{i \neq j} \chi_i(\mathfrak{f}_j). \tag{143}$$

Consequently the actual character in (141) *satisfies*

$$\begin{aligned} \epsilon(\eta) &= \kappa_{AB\rho} u_{\mathfrak{m}} \overline{v_{\mathfrak{n}}}, \\ u_{\mathfrak{m}} &= \beta_{\mathfrak{m}} \xi_{\mathfrak{m}}(\mathfrak{f}_S) \rho(\mathfrak{m}), \\ v_{\mathfrak{n}} &= \beta_{\mathfrak{n}} \xi_{\mathfrak{n}}(\mathfrak{f}_S) \overline{\rho(\mathfrak{n})}. \end{aligned} \tag{144}$$

The scalar $\kappa_{AB\rho}$ is fixed. These are two distinct unitary column gauges. The primitive values $\sigma(\mathfrak{e})$ and $\sigma(\mathfrak{h})$ at S, including zero, remain separately in the kernel; they cannot be replaced by ρ there.

This closes the normalization hypothesis for these stated conventions. A bounded, invertible change of phases gives no power saving by itself.

For a fixed actual $\mathfrak{v}_2$, define the exact integer endpoints

$$a = \lceil A_1 \rceil, \qquad b = \min\{\lceil 2A_1 \rceil - 1, \lfloor H/(Nu_0(N\mathfrak{v}_2)^2) \rfloor\}.$$

An empty interval contributes zero. Choose one fixed nondecreasing smooth step s_0, equal to zero on $(-\infty, 0]$ and one on $[1, \infty)$, and put

$$G_{\mathfrak{v}_2}(x) = s_0(4(A_1 x - a + 1/2)) s_0(4(b + 1/2 - A_1 x)). \tag{145}$$

It is the exact row indicator at every ideal norm node, has amplitude at most one, support in $[1/2, 5/2]$ and total variation at most two. Its derivatives cost powers of A_1; it is not the fixed column W. Set $C_0 = 3/(4\pi^2)$ and

$$\dot{G}(X) = C_0^{-1/2} \int_0^\infty G(t) J_0(2\sqrt{Xt/C_0})\, dt. \tag{146}$$

Proposition S75.2 (The actual signed kernel before absolute norms)**.** *For every nonzero class/ray component, the good conductor of η is $\mathfrak{mn}$, so η is nonprincipal. Put $P_S = \prod_{p \in S} p$ and*

$$L_{\mathfrak{e}} = \frac{A_1}{(N\mathfrak{k})^2 N\mathfrak{l} N\mathfrak{e}}, \qquad K_{\mathfrak{l}} = \sqrt{b/N\mathfrak{l}}.$$

Squarefree-row inversion, finite S deletion and primitive Poisson give (140) *by summing the following kernel:*

$$\begin{aligned} &c_{\mathfrak{m}} \overline{c_{\mathfrak{n}}} R_{\mathfrak{m}}(\mathfrak{v}_2^2) \overline{R_{\mathfrak{n}}(\mathfrak{v}_2^2)} h_{AB\rho}(\mathfrak{v}_2) \\ &\quad \cdot \mu(\mathfrak{k}) \mu(\mathfrak{l}) \mu(\mathfrak{e}) \eta(\mathfrak{k}^2 \mathfrak{l} \mathfrak{e}) \epsilon(\eta) \frac{L_{\mathfrak{e}}}{\sqrt{F}} \sum_{\mathfrak{y}} \dot{G}_{\mathfrak{v}_2}(L_{\mathfrak{e}} N\mathfrak{y}/F) \overline{\eta(\mathfrak{y})}. \end{aligned} \tag{147}$$

The outer sums are over every actual $\mathfrak{v}_2$, every class/ray component and original coprime column pair, all $\mathfrak{l} \mid \mathfrak{v}_2$ with $N\mathfrak{l} \le b$, all good squarefree $\mathfrak{k}$ with $(\mathfrak{k}, \mathfrak{v}_2) = 1$ and $N\mathfrak{k} \le K_{\mathfrak{l}}$, and every $\mathfrak{e} \mid P_S$. The dual sum is over all *nonzero integral ideals, including S-supported ones. It converges at each fixed parameter set with the actual derivative constants of G. The original pre-S cutoff is never extended in this identity.*

All signs, means, product norms and raw zeros occur before a triangle. A higher induced modulus does not have a unit Gauss root automatically: its nonprimitive additive frequencies may vanish.

S75.3 What finite prefixes and actual divisor bands now satisfy

Write

$$\begin{aligned}\Phi(Z,L) &= Z + LZ^{1/6} + Z^{5/6}L^{1/3} + Z^{1/3}L^{5/6},\\ I(t) &= \sum_{j\ge 0}\frac{t^j}{(j!)^2}.\end{aligned} \tag{148}$$

Let $\mathcal{P}_{\le X}$ retain all terms of (147) with $N\mathfrak{y} \le X$, including every source/ray/S component.

Proposition S75.3 (All-component exact finite prefixes)**.** *For every finite $X \ge 1$,*

$$\frac{|\mathcal{P}_{\le X}|}{D} \ll B_W^2(DXA_2)^{\epsilon}\frac{A_1A_2}{D}\Phi(X,D). \tag{149}$$

A separate exact all-orders estimate is

$$|\mathcal{P}_{\le X}| \ll B_W^2(DXA_2)^{\epsilon}A_1A_2\Phi(X,D)I\left(\frac{(5/2)A_1X}{C_0D^2}\right). \tag{150}$$

No completed Euler model, low-frequency approximation or $X \le \sqrt{H}$ hypothesis is used in (149)*. On a fixed same-source-class/$\rho = 1$ component, the additional critical-half band estimate is*

$$\frac{|\mathcal{P}_{C,1}(Y,2Y]|}{D} \ll B_W^2(DYA_2)^{\epsilon}A_2\sqrt{A_1/Y}\,\Phi(Y,D)(1+\log(2A_1))^2. \tag{151}$$

Constants depend on the fixed S, ray/source data, ν, step and ϵ, not on any hidden uniform smoothness of the sharp G.

At $H \asymp D^{3/4}$, the mixed witness $(A_1,A_2) \asymp (D^{5/12},D^{1/6})$ has a target-sized full finite prefix $X \asymp D$. The pure prefix at $X = D$ has upper power $11/12$. At $X = D^{3/8}$ the pure/mixed powers are $13/16$ and $31/48$. These growing bounds cannot be sent to $X = \infty$.

Proposition S75.4 (Finite transfer of the actual cutoff and profile)**.** *Let $X \ge 2$, $1 \le U \le X$, and let $K = (K_{\mathfrak{d}\mathfrak{a}})$ be a finite matrix indexed by good squarefree ideals of norms at most X, zero unless $(\mathfrak{d},\mathfrak{a}) = 1$ and $N\mathfrak{d}N\mathfrak{a} \le X$. All additional masks may remain in K. Set*

$$\begin{aligned}G_{\mathcal{I}\mathcal{J}} &= \max_{\mathfrak{d}\in\mathcal{I}} 3^{\omega(\mathfrak{d})}\max_{\mathfrak{a}\in\mathcal{J}} 3^{\omega(\mathfrak{a})},\\ c_P &= 1/\log(4X), \qquad L_T = \operatorname{asinh}(T/c_P)/\pi.\end{aligned}$$

Empty matrices are zero. For every finite $T \ge 8X$,

$$\|[A_U(\mathfrak{d}\mathfrak{a})K_{\mathfrak{d}\mathfrak{a}}]\|_{\mathrm{op}} \le G_{\mathcal{I}\mathcal{J}}\left(e^2L_T^2 + \frac{32X^{3/2}}{T}\right)\|K\|_{\mathrm{op}}. \tag{152}$$

For $T = (4X)^4$ this is $O_\epsilon(X^\epsilon\log^2(4X))\|K\|$. The common complex multiplier $W(N\mathfrak{d}N\mathfrak{a}/D)$ costs

$$\begin{aligned}M(W) &= \frac{1}{2\pi}\int_{\mathbb{R}}|\widehat{W}(it)|\,dt, \qquad \widehat{W}(s) = \int_0^\infty W(x)x^s\,\frac{dx}{x},\\ M(W) &\le \frac{\log 2}{\pi}(B_W + \|xW'(x) + x^2W''(x)\|_\infty).\end{aligned} \tag{153}$$

Two occurrences in a bilinear/Gram estimate cost the square of the transfer norm. This is not $A_U(\mathfrak{d}\mathfrak{a}) = A_U(\mathfrak{d})A_U(\mathfrak{a})$ or coefficient shrinkage.

Proposition S75.5 (A cubic comparison for an actual auxiliary band)**.** *Fix one same-source-class/$\rho = 1$ component of the auxiliary individual-column extension of* (147)*. Insert the exact coprime identity $\sum_{\mathfrak{d}|(\mathfrak{m},\mathfrak{n})} \mu(\mathfrak{d})$, and retain the* identity-source-class *divisors $E < N\mathfrak{d} \le 2E$, $1 \le E \le D/2$. Denote the resulting exact signed dual band by $\mathcal{P}^{\mathrm{id}}_{E,C}(Y, 2Y]$. Put $L = D/E$ and $\Theta_3(E, L) = E + L + (EL)^{2/3}$. For fixed $0 < \lambda < 1/2$,*

$$\begin{aligned}\frac{|\mathcal{P}^{\mathrm{id}}_{E,C}(Y, 2Y]|}{D} &\ll M(W)^2 (DYA_2)^{\epsilon} A_2 A_1^{1-\lambda} D^{2\lambda-1} Y^{-\lambda} \\ &\quad \cdot \min\{\Phi(Y, L),\ (Y/E)\Theta_3(E, L)\}.\end{aligned} \tag{154}$$

The arithmetic identity behind the new branch is

$$\beta_{\mathfrak{d}\mathfrak{a}} = \epsilon(\widetilde{\chi}_{\mathfrak{d}})\beta_{\mathfrak{a}}\alpha(1, C)\widetilde{\chi}_{\mathfrak{a}}(\mathfrak{d})^2, \qquad [\mathfrak{d}] = 1, \quad [\mathfrak{a}] = C. \tag{155}$$

It exposes the physical cubic residue matrix in $\mathfrak{d}, \mathfrak{a}$. The external input is de Faveri, Theorem 1.1 and Example 2.2, now at $n = 3$, not a conductor-parametric sieve.

This improves a *scoped norm comparator*: at $E \asymp D^{2/3}$ the second branch is $O(Y)$. It still uses $|\mu(\mathfrak{d})| \le 1$ at its final Cauchy step. Such $\mathfrak{d}$ are auxiliary divisors, not the original column-pair gcd h. All divisor classes, small $\mathfrak{d}$ and the full high-dual sum are not bounded here.

S75.4 Coherent means that can be localized, and those that remain

Proposition S75.6 (The full equal-norm sector)**.** *Original coprime squarefree columns with $N\mathfrak{m} = N\mathfrak{n} = N > 1$ satisfy $\mathfrak{n} = \overline{\mathfrak{m}}$, with rational-squarefree N supported on split rational primes. The original good masks require* both *primes above each such rational prime to avoid S; S need not be enlarged or conjugation-stable. There are $O(D)$ ordered annular pairs. Define S_{eq} as the normalized contribution to S_1^{AA}, namely D^{-1} times their full primal contribution to $\mathcal{P}_D$ in* (140)*. Then*

$$|S_{\mathrm{eq}}| \ll B_W^2 R_w D^{\epsilon} \ll B_W^2 R (DH)^{\epsilon}. \tag{156}$$

Before the factor D^{-1}, their ordered orientation sum at norm N is

$$\begin{aligned}a_U(N)^2 |W(N/D)|^2 \prod_{p|N} \{a_p \chi_p(t) + \overline{a_p \chi_p(t)}\}, \\ t = N u_0 N \mathfrak{v}_1 (N\mathfrak{v}_2)^2,\end{aligned} \tag{157}$$

where $a_p = \nu(\pi_p)\overline{\nu(\overline{\pi}_p)}$, $\chi_p(t) = (t/\pi_p)_6$, and $a_U(N) = \sum_{rs|N,(r,s)=1,r,s\le U} \mu_{\mathbb{Z}}(r)\mu_{\mathbb{Z}}(s)$. This includes all ordered orientations and every raw collision zero. For the all-component primitive-ideal prefix/band,

$$\frac{|\mathcal{P}_{\mathrm{eq}}(\le Y)| + |\mathcal{P}_{\mathrm{eq}}(Y, 2Y]|}{D} \ll B_W^2 (DYA_2)^{\epsilon} RY/D. \tag{158}$$

The signed complement is bounded by the whole primal term minus that prefix, not by an absolute high-mode summation.

Since $R \ll H$ on nonempty blocks, the desired full fixed-W target is equivalent, up to (156), to that for unequal norms. Its desired constants may depend on fixed W or appropriate specified seminorms. Formula (157) is neither identically zero nor positive in general.

Proposition S75.7 (Literal-ray recombination in the pure sector)**.** *In the pure case $\mathfrak{v}_2 = \mathcal{O}$, choose an S-supported ray modulus $\mathfrak{c} = (\gamma_c)$ divisible by (6). For an original coprime pair put $\mathfrak{q} = \mathfrak{mn}$, $Q = N\mathfrak{q}$, $\gamma_q = \alpha_{\mathfrak{m}}\alpha_{\mathfrak{n}}$, and let $\vartheta_{\mathfrak{q}}(x) = (x/\mathfrak{m})_6\overline{(x/\mathfrak{n})_6}$ be the primitive finite character modulo $\mathfrak{q}$. Let $P_{\mathfrak{k}^2}$ be the full literal primary/good-S/ray indicator on $\mathcal{O}/\mathfrak{c}$ after $v_1 = k^2 z$, with finite additive transform $\widehat{P}$. The exact common-modulus Fourier coefficient is*

$$\widehat{F}_{\mathfrak{k}^2}(\beta) = \tau_{\mathfrak{q}}\vartheta_{\mathfrak{q}}(\gamma_c)\overline{\vartheta_{\mathfrak{q}}(\beta)}\widehat{P}_{\mathfrak{k}^2}(\beta\gamma_q^{-1} \bmod \mathfrak{c}). \tag{159}$$

Its literal-lattice radial Poisson prefactor is $L_{\mathfrak{k}}/(N\mathfrak{c}Q)$, $L_{\mathfrak{k}} = A_1/(N\mathfrak{k})^2$. The S-local Fourier-mean sublattice $\beta = \gamma_c\zeta$ has coefficient $r_{\mathfrak{k}^2}\tau_{\mathfrak{q}}\overline{\vartheta_{\mathfrak{q}}(\zeta)}$, where $r = \sum P$. The six-unit projection annihilates every nontrivial unit restriction and retains the unit-trivial branch, whose condition is

$$N\mathfrak{m} \equiv N\mathfrak{n} \pmod{36}. \tag{160}$$

For the basic S above $2, 3$ with no extra ray condition, $\mathfrak{c} = (6)$ has $r = 3$ and ideal coefficient $6r/N\mathfrak{c} = 1/2$. This is not the global $\beta = 0$ mode, which vanishes. All S-supported dual ideals remain, with values defined by any generator once the character is unit-trivial.

Literal additive bands and the earlier primitive-ideal bands have different fixed-S frequency support/rescalings; no unmatched windows are identified. Unequal norms congruent modulo 36 and all other literal residue channels remain. Finite ray recombination supplies no automatic zero mean or signed target estimate.

Remaining arithmetic obligation. A finite-prefix comparison, the sparse equal-norm sector and the cubic auxiliary-band comparison do not bound the complete signed unequal-norm kernel. The required additional estimate must retain all small divisors/source classes and the literal S mean together with complementary residues before the final conductor/divisor/dual triangle. No desired positivity, energy lower bound, new zero-free strip, determinant comparison, analytic Lean or GRH claim follows.

S76 Proofs for the coherent-band comparisons

These proofs use de Faveri's pinned arXiv:2610.04045v1, Theorem 1.1 and its stated character family as external input. The primitive Poisson convention agrees with its Lemma 3.1; the formulas below also derive the normalization. This is not a proof of that deep large-sieve theorem. The source version and independent proof reviews are bound by the accompanying source map; full third-party reading caches are not public source dependencies.

S76.1 Finite Fourier transforms and the primitive Gaussian

For $\mathfrak{f} = (\gamma)$ use the primitive additive pairing

$$e_\gamma(x) = \exp\bigl(2\pi i \operatorname{Tr}(x/(\delta\gamma))\bigr), \qquad \delta = 1 + 2\omega.$$

Let θ be primitive modulo $\mathfrak{f}$, extended by zero at nonunits of this *actual* conductor. For every $\beta \in \mathcal{O}$,

$$\sum_{x \bmod \gamma} \theta(x)e_\gamma(\beta x) = \overline{\theta(\beta)}\tau_\gamma(\theta). \tag{161}$$

At a unit this is change of variable. At a nonunit frequency the additive character factors through a proper conductor quotient. Primitivity supplies a unit in the kernel of unit reduction on which

θ is nontrivial. Multiplication by that unit leaves the additive factor unchanged and forces the sum to vanish. Unit reduction is surjective prime by prime and hence by CRT. Parseval for this nondegenerate finite additive pairing gives

$$\sum_{\beta \bmod \mathfrak{f}} \left| \sum_x \theta(x) e_\gamma(\beta x) \right|^2 = N\mathfrak{f}\,\varphi(\mathfrak{f}), \qquad |\tau_\gamma(\theta)|^2 = N\mathfrak{f}.$$

For the unit conductor use its one-element convention $\tau = 1$. For coprime γ_1, γ_2, the two CRT frequencies are γ_2^{-1} and γ_1^{-1}, so

$$\tau_{\gamma_1\gamma_2}(\theta_1\theta_2) = \theta_1(\gamma_2)\theta_2(\gamma_1)\tau_{\gamma_1}(\theta_1)\tau_{\gamma_2}(\theta_2). \tag{162}$$

The product is primitive because each ramified local conductor occurs in only one factor.

Here are the archimedean factors that a raw Gauss sum alone would miss. Put $d = |k|$ and

$$P_k(z) = \begin{cases} z^k, & k \geq 0, \\ \bar{z}^{-k}, & k < 0. \end{cases}$$

With Euclidean Fourier transform $\widehat{g}(w) = \int_{\mathbb{C}} g(z) e^{-2\pi i \operatorname{Re}(z\bar{w})}\, dz$, differentiating the two-dimensional Gaussian gives

$$\widehat{P_k(z)e^{-\pi b|z|^2}}(w) = (-i)^d b^{-d-1} P_k(w) e^{-\pi|w|^2/b}, \qquad b > 0. \tag{163}$$

Holomorphic or antiholomorphic harmonicity removes the lower contraction terms. The sign is $(-i)^{|k|}$ for either sign of k. Let $\chi_0 = \chi N^{-it}$ and $a_\gamma = 2/(\sqrt{3}|\gamma|)$. Its theta series is

$$\Theta_{\chi_0}(t) = \sum_{\alpha \in \mathcal{O}} \theta(\alpha) P_k(\alpha) e^{-\pi a_\gamma t |\alpha|^2}.$$

The lattice $\gamma\mathcal{O}$ has covolume $(\sqrt{3}/2)|\gamma|^2$ and dual lattice $2/(\bar{\gamma}\bar{\delta})\mathcal{O}$. Write a dual point as $w = 2\bar{\beta}/(\bar{\gamma}\bar{\delta}) = c\bar{\beta}$. Then $|c| = a_\gamma$ and $\operatorname{phase}(c) = \operatorname{phase}(\delta\gamma)$. Equations (161) and (163), applied in each residue class, yield

$$\Theta_{\chi_0}(t) = i^{-d} \operatorname{phase}(\delta\gamma)^k \frac{\tau_\gamma(\theta)}{\sqrt{N\mathfrak{f}}} t^{-d-1} \Theta_{\bar{\chi}_0}(1/t). \tag{164}$$

The inverse different and the conductor generator have not been replaced by a raw character value at δ.

Use $\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s}\Gamma(s)$ and

$$\Lambda(s, \chi) = (3N\mathfrak{f})^{s/2} \Gamma_{\mathbb{C}}(s - it + d/2) L(s, \chi).$$

Every nonzero ideal has six generators; their contributions agree because $\theta(u)u^k = 1$. The Mellin transform of the theta series, with exponent $s + d/2$, is therefore exactly $3(3N\mathfrak{f})^{d/4}\Lambda(s, \chi_0)$. The same constant occurs on the conjugate side. Mellin integration of (164) proves the root formula for $t = 0$. At nonunit conductor the zero frequency vanishes by primitivity. At unit conductor and $k \neq 0$, $P_k(0) = 0$. At unit conductor and $k = 0$, retain the Gaussian zero term when splitting its Mellin integral at one; this gives the usual meromorphic polar terms and root one. They are not dropped from the functional equation.

Changing γ to $u\gamma$ multiplies its Gauss sum by $\theta(u)$ and its angular factor by u^k, whose product is one. Thus the formula is independent of the conductor generator. Complex conjugation gives

$\tau_\gamma(\bar\theta) = \theta(-1)\overline{\tau_\gamma(\theta)}$, where $\theta(-1) = (-1)^k$; together with the angular factor this proves $\epsilon(\bar\chi) = \overline{\epsilon(\chi)}$. Finally

$$\Lambda(s, \chi_0 N^{it}) = (3N\mathfrak{f})^{it/2}\Lambda(s - it, \chi_0), \qquad \epsilon(\chi_0 N^{it}) = (3N\mathfrak{f})^{it}\epsilon(\chi_0).$$

This proves (142) also for radial unitary twists.

In (162), the angular cross factors $\mathrm{phase}(\gamma_2)^{k_1}\,\mathrm{phase}(\gamma_1)^{k_2}$ convert the finite residue values into ideal values $\chi_1(\mathfrak{f}_2)\chi_2(\mathfrak{f}_1)$. The remaining i factor is $i^{|k_1|+|k_2|-|k_1+k_2|}$. Radial cross factors satisfy

$$\prod_j (3N\mathfrak{f}_j)^{it_j} \prod_{i\neq j} (N\mathfrak{f}_j)^{it_i} = \left(3N\prod_j \mathfrak{f}_j\right)^{i\sum_j t_j}.$$

Iteration proves (143). Coprimality is a hypothesis on the *actual primitive ideal conductors*, not on rational norms or higher induced moduli.

S76.2 Transport to every original class and ray component

Let the types of $\xi_\mathfrak{m}, \xi_\mathfrak{n}$ be $(k_A, t_A), (k_B, t_B)$. The character σ in (141) has types $(k_B - k_A, t_B - t_A)$. The three primitive conductors $\mathfrak{m}, \mathfrak{n}, \mathfrak{f}_S$ are pairwise coprime; no good local ramification can cancel against the other two. Applying (143) gives

$$\begin{aligned} \epsilon(\eta) &= \gamma_{AB\rho}\epsilon(\xi_\mathfrak{m})\overline{\epsilon(\xi_\mathfrak{n})}\xi_\mathfrak{m}(\mathfrak{n}\mathfrak{f}_S)\overline{\xi_\mathfrak{n}(\mathfrak{m}\mathfrak{f}_S)}\sigma(\mathfrak{m}\mathfrak{n}), \\ \gamma_{AB\rho} &= i^{|k_A|+|k_B|+|k_B-k_A|}\epsilon(\sigma). \end{aligned} \tag{165}$$

Source reciprocity is $\widetilde{\chi}_\mathfrak{m}(\mathfrak{n})\overline{\widetilde{\chi}_\mathfrak{n}(\mathfrak{m})} = \alpha(A, B)$. Substituting $\widetilde{\chi} = \xi\psi$ in (165) cancels all crossed ψ values and proves (144), with $\kappa_{AB\rho} = \alpha(A, B)\gamma_{AB\rho}$. In particular the remaining factor is $\rho(\mathfrak{m})\rho(\mathfrak{n})$, not an assumed class constant; ρ need not factor through the source class group.

For good dual $\mathfrak{z}$, write $\widetilde{\chi}_\mathfrak{m}(\mathfrak{z}) = \gamma_A(\mathfrak{z})R_\mathfrak{m}(\mathfrak{z})$, with the fixed finite reciprocity phase in that class. Then

$$\begin{aligned} \overline{\sigma(\mathfrak{z})\xi_\mathfrak{m}(\mathfrak{z})}\xi_\mathfrak{n}(\mathfrak{z}) &= \Omega_{AB\rho}(\mathfrak{z})\overline{R_\mathfrak{m}(\mathfrak{z})}R_\mathfrak{n}(\mathfrak{z}), \\ \Omega_{AB\rho} &= \bar\sigma\,\psi_A\bar\psi_B\,\bar\gamma_A\gamma_B, \qquad |\Omega_{AB\rho}(\mathfrak{z})| = 1. \end{aligned} \tag{166}$$

The equality includes raw collision zeros. This is a fixed unitary *row* phase for each component, not equality of the two column classes. At S the separate scalar $\sigma(\mathfrak{k}^2\mathfrak{l}\mathfrak{e})\overline{\sigma(\mathfrak{h})}$ is kept, including zero. There is no division by such a scalar, and no substitution of $\rho(\mathfrak{e}/\mathfrak{h})$ at a conductor drop. Off the original coprime column support, products of individual column gauges below are only an auxiliary extension. They are not roots or primitive conductors of overlapping pairs.

S76.3 The sharp capacity, square-divisor signs and exact Poisson

The endpoints in (145) are integers. The two smooth transitions lie between ideal norm nodes, so its value at $x = N\mathfrak{v}_1/A_1$ is exactly one or zero according to the original interval and capacity. For nonempty fibres these transitions are disjoint. Their monotonicity gives total variation two; their jth derivatives cost powers of A_1. The elementary identity

$$\sum_{\mathfrak{v}\ \text{good}} \mu(\mathfrak{v})^2\mathbf{1}_{(\mathfrak{v},\mathfrak{v}_2)=1}B(\mathfrak{v}) = \sum_{\substack{\mathfrak{k}\ \text{good,SF}\\(\mathfrak{k},\mathfrak{v}_2)=1}} \mu(\mathfrak{k}) \sum_{\mathfrak{l}\mid\mathfrak{v}_2} \mu(\mathfrak{l}) \sum_{\mathfrak{z}\ \text{good}} B(\mathfrak{k}^2\mathfrak{l}\mathfrak{z}) \tag{167}$$

follows by expanding the squarefree indicator and then the remaining coprimality indicator. Auxiliary $\mathfrak{z}$ need not be squarefree, and may overlap $\mathfrak{k}$ or $\mathfrak{l}$. The original norm support fixes $N\mathfrak{l} \le b$ and

$N\mathfrak{k} \le \sqrt{b/N\mathfrak{l}}$ *before* Poisson. Including more $\mathfrak{k}$ would add zero primal terms whose individual dual windows need not be zero.

On good $\mathfrak{v}_1$, expand exactly

$$\mathbf{1}_{\mathrm{ray}} R_{\mathfrak{m}}(\mathfrak{v}_1)\overline{R_{\mathfrak{n}}(\mathfrak{v}_1)} = \sum_{\rho} h_{AB\rho}(\mathfrak{v}_2)\eta_{\mathfrak{m},\mathfrak{n},\rho}(\mathfrak{v}_1).$$

This includes every reciprocity/ray component, not a selected same class term. With $L = A_1/((N\mathfrak{k})^2 N\mathfrak{l})$, good-$S$ deletion is the *finite* identity

$$\sum_{\mathfrak{z}\ \mathrm{good}} G(N\mathfrak{z}/L)\eta(\mathfrak{z}) = \sum_{\mathfrak{e}\mid P_S} \mu(\mathfrak{e})\eta(\mathfrak{e}) \sum_{\mathfrak{y}} G(N\mathfrak{y}/(L/N\mathfrak{e}))\eta(\mathfrak{y}). \tag{168}$$

Primitive zeros at S suppress the corresponding terms automatically. They do not assign forbidden raw values to an original row.

For a nonprincipal primitive type-zero η of conductor norm F, its normalized functional equation and ordinary Mellin inversion give

$$\sum_{\mathfrak{y}} G(N\mathfrak{y}/L)\eta(\mathfrak{y}) = \epsilon(\eta)\frac{L}{\sqrt{F}} \sum_{\mathfrak{y}} \dot{G}(LN\mathfrak{y}/F)\overline{\eta(\mathfrak{y})}. \tag{169}$$

To see the field constant explicitly, put

$$\gamma_K(s) = C_0^{s-1/2}\frac{\Gamma(s)}{\Gamma(1-s)}, \qquad \widehat{G}(s) = \int_0^\infty G(t)t^s\,\frac{dt}{t}.$$

The Mellin transform of (146) is $\gamma_K(s)\widehat{G}(1-s)$. The Bessel identity may first be justified absolutely in $0 < \operatorname{Re} s < 1/4$; smooth compact support and repeated Hankel integration by parts continue it to $\operatorname{Re} s > 0$ at each fixed G. Mellin inversion of the primal ideal sum, its primitive functional equation, and the change $s \mapsto 1-s$ now give (169) with exactly $L/\sqrt{F}$. There is no principal residue since η has nonempty good ramification. The intermediate contour moves involve the entire nonprincipal completed function and the fixed smooth transform, not an inverse L function or an assumed zero-free strip. This also recovers the pinned source Lemma 3.1 convention.

At a good prime ideal of norm P, a nontrivial primitive character modulo p lifted to p^j, $j \ge 2$, has raw Gauss sum zero for a primitive additive frequency. More generally it vanishes unless the frequency has valuation $j-1$, when its value is $P^{j-1}\bar{\chi}(\beta/\pi^{j-1})\tau_p(\chi)$ with compatible additive normalizations. This follows by summing the lifts first. Thus neither lower character order nor a higher physical modulus licenses a unit root. Applying (169) only at the actual F in (141), and inserting (168) into (167), proves the exact signed kernel. Splitting every dual ideal uniquely as $\mathfrak{y} = \mathfrak{h}\mathfrak{z}$, with $\mathfrak{h}$ supported on S and $\mathfrak{z}$ good, retains all modes.

S76.4 The joint bilinear estimate behind finite prefixes

The external sixth-order norm gives the following arbitrary-column- coefficient comparison, for every $Z, L \ge 1$:

$$\sum_{\substack{\mathfrak{z}\ \mathrm{good}\\ N\mathfrak{z}\le Z}} \left| \sum_{\substack{\mathfrak{a}\ \mathrm{SF,good}\\ N\mathfrak{a}\asymp L}} b_{\mathfrak{a}} R_{\mathfrak{a}}(\mathfrak{z}) \right|^2 \ll (ZL)^{\epsilon}\Phi(Z,L)\sum_{\mathfrak{a}} |b_{\mathfrak{a}}|^2. \tag{170}$$

For completeness, extract $\mathfrak{z} = \mathfrak{r}\mathfrak{s}^6$ with sixth-power-free $\mathfrak{r}$. Freeze $\mathfrak{s}$ as the original column collision mask $(\mathfrak{a},\mathfrak{s}) = 1$ and apply (130) to $\mathfrak{r}$. The four sums have factors $\sum (N\mathfrak{s})^{-6}$, the count $N\mathfrak{s} \le Z^{1/6}$,

$\sum(N\mathfrak{s})^{-5}$ and $\sum(N\mathfrak{s})^{-2}$, respectively. Ideal counting proves (170). This is not an arbitrary *row*-weight theorem or an extension of a residue-subtracted energy estimate. For $1/2 \le L < 1$ use the direct unit residual column; below $1/2$ there is no annular residual column. Primitive ξ is not treated as sixth-order: convert to raw R inside each fixed residual-class positive norm, where its common ψ row phase cancels.

At $s = \lambda + i\tau$, use (144) to form the two *distinct* vectors

$$\begin{aligned} b_{\mathfrak{m}}(s) &= c_{\mathfrak{m}} u_{\mathfrak{m}} R_{\mathfrak{m}}(\mathfrak{v}_2^2)\xi_{\mathfrak{m}}(\mathfrak{k}^2\mathfrak{l}_{\mathfrak{e}})\overline{\xi_{\mathfrak{m}}(\mathfrak{h})}(N\mathfrak{m})^{s-1/2}, \\ d_{\mathfrak{n}}(s) &= c_{\mathfrak{n}} v_{\mathfrak{n}} R_{\mathfrak{n}}(\mathfrak{v}_2^2)\xi_{\mathfrak{n}}(\mathfrak{k}^2\mathfrak{l}_{\mathfrak{e}})\overline{\xi_{\mathfrak{n}}(\mathfrak{h})}(N\mathfrak{n})^{s-1/2}. \end{aligned} \tag{171}$$

The pairing is $b_{\mathfrak{m}}(s)\overline{d_{\mathfrak{n}}(\bar{s})}$, not a PSD form at s. The exact remaining primitive-conductor factor is

$$(N\mathfrak{f}_S)^{s-1/2}\left(\frac{A_1}{N\mathfrak{l}N_{\mathfrak{e}}}\right)^{1-s}, \tag{172}$$

with scalar $\sigma(\mathfrak{k}^2\mathfrak{l}_{\mathfrak{e}})\bar{\sigma}(\mathfrak{h})$ and finite $\mathfrak{k}$ weight $(N\mathfrak{k})^{-2+2s}$. Its S exponent is $s - 1/2$, not its inverse. On $0 < \lambda < 1/2$ its modulus is at most one.

Insert exact gcd inversion before comparison and put $\mathfrak{m} = \mathfrak{d}\mathfrak{a}$, $\mathfrak{n} = \mathfrak{d}\mathfrak{b}$. Residual $\mathfrak{a}, \mathfrak{b}$ may overlap each other but are individually coprime to $\mathfrak{d}$. Raw multiplicativity gives

$$\overline{R_{\mathfrak{d}\mathfrak{a}}(\mathfrak{z})}R_{\mathfrak{d}\mathfrak{b}}(\mathfrak{z}) = \mathbf{1}_{(\mathfrak{d},\mathfrak{z})=1}\overline{R_{\mathfrak{a}}(\mathfrak{z})}R_{\mathfrak{b}}(\mathfrak{z}).$$

Their source classes are $A[\mathfrak{d}]^{-1}$ and $B[\mathfrak{d}]^{-1}$, generally different. Bilinear Cauchy in the row variable, including (166), gives two separate positive norms. Only *inside* them may the raw $(\mathfrak{d}, \mathfrak{z})$ mask be dropped. Uniformly in the fixed column phases and τ,

$$\sum_{\mathfrak{a}} |b_{\mathfrak{d}\mathfrak{a}}(s)|^2, \quad \sum_{\mathfrak{b}} |d_{\mathfrak{d}\mathfrak{b}}(\bar{s})|^2 \ll B_W^2 D^{2\lambda+\epsilon}/N\mathfrak{d}. \tag{173}$$

Indeed $|A_U(\mathfrak{m})| \le 3^{\omega(\mathfrak{m})}$ and an annular subsequence has $O(D/N\mathfrak{d})$ columns. Neither $A_U(\mathfrak{d}\mathfrak{a})$ nor $\beta_{\mathfrak{d}\mathfrak{a}}$ is factored. For the small residual interval the direct unit count has the same bound.

Apply (170) to the two separate vectors. The divisor sums have exponents $1, 2, 4/3, 11/6$:

$$\sum_{N\mathfrak{d}\le 2D} \frac{1}{N\mathfrak{d}}\Phi(Z, D/N\mathfrak{d}) \ll \log(2D)\Phi(Z, D), \tag{174}$$

with the direct endpoint understood. For $Y < N\mathfrak{h}N\mathfrak{z} \le 2Y$, the weight modulus is at most $Y^{-\lambda}$ and the legal positive row prefix is $N\mathfrak{z} \le 2Y/N\mathfrak{h} \ge 1$. The four positive row exponents in Φ give

$$\sum_{\mathfrak{h}\ S\text{-supported},\, N\mathfrak{h}\le 2Y} \Phi(2Y/N\mathfrak{h}, D) \ll_S \Phi(Y, D).$$

This is a summation over *all* S-supported modes, not an exponent-zero convergent sum or a missing raw S value.

The sharp G requires an honest transform bound. Integration by parts against its finite variation gives

$$\widehat{G}(1-s) = -\frac{1}{1-s}\int t^{1-s}\, dG(t).$$

For fixed $0 < \lambda < 1/2$, Stirling's ratio estimate is $|\gamma_K(\lambda + i\tau)| \ll_\lambda (1 + |\tau|)^{2\lambda-1}$. Thus

$$\int_{\mathbb{R}} |\widehat{G}(1-\lambda-i\tau)\gamma_K(\lambda+i\tau)|\, d\tau \ll_\lambda 1, \tag{175}$$

using the integrable tail $(1+|\tau|)^{2\lambda-2}$ and $\mathrm{TV}(G)\le 2$. No derivative seminorm growing with A_1 is hidden here. Only after the joint estimate may the original finite $\mathfrak{k}$ sum be bounded by $\zeta_K^S(2-2\lambda)$; the identity has not been Euler-completed. Finite $\mathfrak{e}$, the original $\mathfrak{l}\mid\mathfrak{v}_2$ divisor sum, and unweighted actual $\mathfrak{v}_2$ counting cost a fixed constant and $A_2(DA_2)^{\epsilon}$. Equations (173)– (175) give

$$\frac{|\mathcal{P}_D(Y,2Y]|}{D}\ll_\lambda B_W^2(DYA_2)^{\epsilon}A_2A_1^{1-\lambda}D^{2\lambda-1}Y^{-\lambda}\Phi(Y,D). \tag{176}$$

Every finite source/ray component is reconstructed using the distinct gauges and bilinear argument, rather than a same-class inference.

For $0<\lambda<1/6$, each of the four powers of Y after multiplication by $Y^{-\lambda}$ is positive. Finite dyadic summation, a last partial band and the dual unit separately therefore give the same bound with $Y=X$ for every finite prefix $X\ge 1$. For a requested final loss $\epsilon_0>0$, fix $\lambda=\min\{1/12,\epsilon_0/4\}$, independently of all scales, and use the positive-norm loss $\epsilon_0/2$. Since $A_1,X\ge 1$,

$$D^{2\lambda}(A_1X)^{-\lambda}\le D^{\epsilon_0/2},\qquad (DXA_2)^{\epsilon_0/2}D^{\epsilon_0/2}\le (DXA_2)^{\epsilon_0}.$$

This proves (149); neither endpoint $\lambda=0$ nor $\lambda=1/2$ is invoked.

For the independent all-orders bound, expand J_0 exactly at every finite dual prefix. The jth moment satisfies $|M_j|=|\int t^jG(t)dt|\le 2(5/2)^j$. The resulting original-column factors $(D/N\mathfrak{m})^j$ and $(D/N\mathfrak{n})^j$ are contractions, including on $\mathfrak{m}=\mathfrak{d}\mathfrak{a}$; they are not replaced by $(D/N\mathfrak{a})^j$. The exact S factor is $(N\mathfrak{f}_S)^{-j-1/2}$, and the finite $\mathfrak{k}$ norm exponent is $2+2j$ with the *same* character phase $\eta(\mathfrak{k})^2$ as before. After gcd inversion, weighted bilinear Cauchy may bound $(N\mathfrak{h}N\mathfrak{z})^j\le X^j$ inside each positive norm; the product of the square roots costs X^j, not X^{2j}. All remaining divisor/S/subsequence constants are uniform in j. The jth joint bound is the prefactor of (150) times $((5/2)A_1X/(C_0D^2))^j/(j!)^2$. Summing this absolutely convergent series proves (150) without an infinite-dual interchange.

For completeness, the scoped critical-half estimate uses $H(v)=e^{v/2}G(e^v)$. Its Fourier transform is $\widehat{G}(1/2-i\tau)$. The fixed smooth step and its two transitions give $\|H\|_1+\|H'\|_1\ll 1$ and $\|H''\|_1\ll 1+A_1$. The minimum of the three Fourier bounds $O(1)$, $O(1/|\tau|)$ and $O(A_1/\tau^2)$ has integral $O(1+\log(2A_1))$. On the half line $|\gamma_K(1/2+i\tau)|=1$ exactly. The original finite square-divisor sum is harmonic, $\sum_{N\mathfrak{k}\le K_{\mathfrak{l}}}(N\mathfrak{k})^{-1}\ll 1+\log(2A_1)$; it is not an infinite $\zeta_K(1)$. The same joint argument in the stated same-class/ $\rho=1$ component proves (151). This estimate is not summed over all dual bands to infer a whole target.

S76.5 Finite-height Schur transfer, including the operator error

For $x\ne 1$ define

$$I_T(x)=\frac{1}{2\pi}\int_{-T}^{T}\frac{x^{c_P+it}}{c_P+it}\,dt.$$

Fourier inversion of $e^{-c_Pv}\mathbf{1}_{v\ge 0}$ gives its full vertical limit $\mathbf{1}_{x>1}$. Integration by parts in each oscillatory tail gives

$$|I_T(x)-\mathbf{1}_{x>1}|\le\frac{2x^{c_P}}{\pi T|\log x|}. \tag{177}$$

There is no pole-free inverse-L contour assertion. Set $U^{\#}=\lfloor U\rfloor+1/2$. Every divisor norm $n\le X$ is a positive integer, so

$$|\log(U^{\#}/n)|\ge\frac{1}{2X+1}\ge\frac{1}{3X},\qquad (U^{\#}/n)^{c_P}\le e.$$

The indicator is exactly $n \le U$, even when U is an integer. Since $6e/\pi < 6$, (177) is at most $e_T = 6X/T \le 3/4$. A product of the two indicators therefore has error at most $2e_T + e_T^2 \le 18X/T$.

For squarefree $\mathfrak{m}$ put

$$H_{\mathfrak{m}}(v,w) = \prod_{p\mid\mathfrak{m}} (1 - (Np)^{-v} - (Np)^{-w}).$$

There are exactly $3^{\omega(\mathfrak{m})}$ ordered disjoint divisor assignments. On the supported coprime products only, $H_{\mathfrak{d}\mathfrak{a}} = H_{\mathfrak{d}} H_{\mathfrak{a}}$, so

$$\begin{aligned} A_U(\mathfrak{d}\mathfrak{a}) &= \frac{1}{(2\pi i)^2} \int_{c_P - iT}^{c_P + iT} \int_{c_P - iT}^{c_P + iT} \frac{(U^{\#})^{v+w}}{vw} H_{\mathfrak{d}}(v,w) H_{\mathfrak{a}}(v,w)\, dv\, dw + R_T(\mathfrak{d},\mathfrak{a}), \\ |R_T(\mathfrak{d},\mathfrak{a})| &\le (18X/T) 3^{\omega(\mathfrak{d})} 3^{\omega(\mathfrak{a})}. \end{aligned} \tag{178}$$

Forbidden entries of K remain zero. The integral is the actual operator $\operatorname{diag}(H_{\mathfrak{d}}) K \operatorname{diag}(H_{\mathfrak{a}})$, with an ordinary right diagonal, not a Hermitian assumption. The two diagonal norms cost $G_{\mathcal{I}\mathcal{J}}$, $(U^{\#})^{2c_P} \le e^2$, and each normalized absolute vertical integral is L_T.

A uniform entry error alone does not multiply operator norms. Instead, with counting ℓ^2 norms,

$$\|R_T \circ K\|_{\mathrm{op}} \le \|R_T \circ K\|_F \le \max|R_T|\, \|K\|_F \le \max|R_T| \sqrt{\min(\#\mathcal{I}, \#\mathcal{J})}\, \|K\|_{\mathrm{op}}.$$

The accepted fixed-field ideal bound $\#\{N\mathfrak{a} \le X\} \le 2X$ gives an error at most $18\sqrt{2} X^{3/2} G_{\mathcal{I}\mathcal{J}}/T$, less than the displayed 32 constant. This proves (152). Splitting the finite set of prime ideals $Np < 3^{1/\eta}$ from the rest proves $3^{\omega(\mathfrak{m})} \ll_\eta (N\mathfrak{m})^\eta$. Use $\eta = \epsilon/2$ and $T = (4X)^4$ to obtain the subpower/log bound. This is a mathematical finite-matrix transfer, not a claim of fast numerical Perron evaluation at that height.

Let $F_0(v) = W(e^v)$. Two integrations by parts give $|\widehat{F_0}(t)| \le \min\{\|F_0\|_1, \|F_0''\|_1/t^2\}$. Split at $|t| = 1$ to get

$$M(W) \le (\|F_0\|_1 + \|F_0''\|_1)/\pi.$$

Here $F_0''(v) = xW'(x) + x^2 W''(x)$ and its support has length $\log 2$, proving (153). Absolute Fourier inversion expresses $W(N\mathfrak{d}N\mathfrak{a}/D)$ as an integral of two unitary norm diagonals and a scalar. Its Schur multiplier norm is at most $M(W)$. Compose with (178); two coefficient occurrences cost the square. All original zeros remain throughout, and no factorization of a primitive $\beta_{\mathfrak{d}\mathfrak{a}}$ is supplied by this lemma.

S76.6 The cubic transposition in an identity divisor class

Fix the scope of Proposition S75.5. On good arguments the primary-generator source construction is index-multiplicative. For $[\mathfrak{d}] = 1$, comparison with the unit source class removes the S conductor of $\widetilde{\chi}_{\mathfrak{d}}$; its type is zero. For $[\mathfrak{a}] = C$ and $(\mathfrak{d},\mathfrak{a}) = 1$,

$$\xi_{\mathfrak{d}\mathfrak{a}} = \widetilde{\chi}_{\mathfrak{d}} \xi_{\mathfrak{a}}.$$

The primitive conductors are coprime and their angular correction in (143) is one. Multiplication by the actual $\psi_C(\mathfrak{d}\mathfrak{a})$ yields

$$\beta_{\mathfrak{d}\mathfrak{a}} = \epsilon(\widetilde{\chi}_{\mathfrak{d}}) \beta_{\mathfrak{a}} \widetilde{\chi}_{\mathfrak{d}}(\mathfrak{a}) \widetilde{\chi}_{\mathfrak{a}}(\mathfrak{d}).$$

Source reciprocity is $\widetilde{\chi}_{\mathfrak{d}}(\mathfrak{a}) \overline{\widetilde{\chi}_{\mathfrak{a}}(\mathfrak{d})} = \alpha(1, C)$, proving (155). In the product for residuals $\mathfrak{a}, \mathfrak{b}$, their common ϵ and α cancel:

$$\beta_{\mathfrak{d}\mathfrak{a}} \overline{\beta_{\mathfrak{d}\mathfrak{b}}} = \beta_{\mathfrak{a}} \overline{\beta_{\mathfrak{b}}} (\widetilde{\chi}_{\mathfrak{a}} \overline{\widetilde{\chi}_{\mathfrak{b}}})^2(\mathfrak{d}). \tag{179}$$

If $\mathfrak{g} = (\mathfrak{a}, \mathfrak{b})$, the primitive cubic conductor is $(\mathfrak{a}/\mathfrak{g})(\mathfrak{b}/\mathfrak{g})$, nonprincipal when $\mathfrak{a} \neq \mathfrak{b}$. The same source class cancels its S component. Common good primes have dropped only from that primitive conductor; the full $\mathfrak{ab}$ union still remains excluded from $\mathfrak{d}$.

The other common $\mathfrak{d}$ phases from ν, u_0 and the $\mathfrak{k}, \mathfrak{l}, \mathfrak{v}_2, \mathfrak{e}, \mathfrak{h}$ modulations cancel in the pair. Their raw zeros, together with the $\mathfrak{z}$ factor, retain

$$(\mathfrak{d}, S\mathfrak{a}\mathfrak{b}\mathfrak{k}\mathfrak{v}_2\mathfrak{z}) = 1.$$

This notation means coprimality with the union of the displayed prime supports. Primitive values at $\mathfrak{e}, \mathfrak{h}$ are legitimate unitary phases of a good-conductor character. The radial factor $(N\mathfrak{d})^{2s-1}$ is kept, not called a cancelled phase.

At fixed $\mathfrak{z}, s$ the core matrix is

$$K_{\mathfrak{d}\mathfrak{a}}^{(\mathfrak{z})} = \mathbf{1}_{[\mathfrak{d}]=1,[\mathfrak{a}]=C,(\mathfrak{d},\mathfrak{k}\mathfrak{v}_2\mathfrak{z})=1}\widetilde{\chi}_{\mathfrak{a}}(\mathfrak{d})^2\overline{R_{\mathfrak{a}}(\mathfrak{z})}. \tag{180}$$

Its cubic character already has the $(\mathfrak{d}, \mathfrak{a})$ zero. All additional fixed residual phases are column contractions. On good arguments the squared sextic symbol is the actual primary cubic symbol $(\alpha_{\mathfrak{a}}/\mathfrak{d})_3$. The pinned external Example 2.2 constructs precisely this family with primary generators congruent to one modulo 3. Both variables are squarefree and hence cube-free; source-sixth-class selectors and deleting the prime above 2 only restrict the positive matrix. Theorem 1.1 at $n = 3$, or its transpose, therefore gives

$$\|K^{(\mathfrak{z})}\|_{\mathrm{op}}^2 \ll_\eta (EL)^\eta \Theta_3(E, L). \tag{181}$$

The lengths are physical ideal norms E, L, not varying conductor labels.

The original annular condition implies $L/2 \leq N\mathfrak{a} \leq 2L$, a bounded number of dyads with lower endpoint at least one. On the encompassing rectangle $N\mathfrak{d}N\mathfrak{a} \leq 4D$, apply Proposition S75.4 with $X = 4D$. This is the necessary repair: $A_U(\mathfrak{d}\mathfrak{a})$ depends on *both* matrix indices and cannot be put into a fixed coefficient vector by fiat. Composing the common W costs $M(W)$. With $0 < \lambda < 1/2$, the squared row radial factor is $O(E^{2\lambda-1})$, and the column squared norm is

$$\sum_{L/2\leq N\mathfrak{a}\leq 2L} (N\mathfrak{a})^{2\lambda-1} \ll_\lambda L^{2\lambda}.$$

The product is $D^{2\lambda}/E$. Thus the two Mellin vectors have squared norms at most

$$M(W)^2 D^{2\lambda+\eta}\Theta_3(E, L)/E.$$

Cauchy in $\mathfrak{d}$ now uses $|\mu(\mathfrak{d})| \leq 1$, explicitly losing rather than exploiting its sign. A final triangle over $N\mathfrak{z} \leq Z$ gives the new inner branch

$$|T_E(Z; s)| \ll M(W)^2 D^{2\lambda+\eta}(Z/E)\Theta_3(E, L). \tag{182}$$

The accepted separate positive-divisor argument gives $|T_E(Z; s)| \ll M(W)^2 D^{2\lambda+\eta}\Phi(Z, L)$: there are $O(E)$ divisors and each actual subsequence norm is $O(M(W)^2 D^{2\lambda+\eta}/E)$. Both estimates apply to the same quantity, so their minimum is legal.

For $Y < N\mathfrak{h}N\mathfrak{z} \leq 2Y$, put $Z_{\mathfrak{h}} = 2Y/N\mathfrak{h} \geq 1$. The new branch uses the convergent positive exponent-one sum $\sum_{\mathfrak{h}} Z_{\mathfrak{h}} \ll_S Y$; the old branch uses its four positive Φ exponents. Sum each branch to obtain the minimum after summation. The same finite $\mathfrak{k}$ majorant, $\mathfrak{e}/\mathfrak{l}$ bounds, actual A_2 count and fixed-line BV integral as above now prove (154), including the once-only $1/D$ normalization. No all-divisor or whole-dual conclusion follows.

On rough prime divisors $\mathfrak{d} = p$, $Np > U$, no short divisor can contain p, so $A_U(p\mathfrak{a}) = A_U(\mathfrak{a})$ exactly. The remaining fixed-residual arithmetic sum has the shape

$$-A_U(\mathfrak{a})A_U(\mathfrak{b}) \sum_{\substack{E<Np\le 2E,\ Np>U,\ [p]=1\\ p\notin S\cup\mathrm{supp}(\mathfrak{a}\mathfrak{b}\mathfrak{k}\mathfrak{v}_2\mathfrak{z})}} \lambda_{\mathfrak{a}\mathfrak{b}}(p)(Np)^{2s-1} \cdot W(NpN\mathfrak{a}/D)\overline{W(NpN\mathfrak{b}/D)}, \tag{183}$$

up to the already fixed residual phases, with $\lambda_{\mathfrak{a}\mathfrak{b}} = (\widetilde{\chi}_{\mathfrak{a}}\overline{\widetilde{\chi}_{\mathfrak{b}}})^2$. Here $\mu(p) = -1$ is constant. A fixed-field scalar Mertens estimate is not a uniform bound for these varying cubic prime twists. The auxiliary principal case $\mathfrak{a} = \mathfrak{b}$ may cancel against other divisor sectors; it is no physical lower obstruction.

S76.7 Classification and proof of the full equal-norm sector

Factor the common rational norm prime by prime. At an inert prime there is only one ideal of norm p^2; equal norms force the same squarefree valuation in both columns and coprimality forces it to zero. The same argument excludes a ramified prime. At a split prime write the two pairs of valuations as $(x, y), (z, w) \in \{0, 1\}^2$. Then $x + y = z + w$ and $xz = yw = 0$. Sum two is forbidden; sum one forces opposite choices. Hence $\mathfrak{n} = \bar{\mathfrak{m}}$ and the common norm is rational-squarefree supported on split primes. Both primes above each such prime must already be good. This does not assume that the original S is conjugation-stable or enlarge it. Each eligible $\mathfrak{m}$ has exactly one partner, so the ordered pair count is at most $\#\{N\mathfrak{m} \le 2D\} \le 4D$. All other rational-nonsquarefree norms remain in the *full* annulus; only this sector has been classified.

The ternary bound is uniform in the original cutoff: $|A_U(\mathfrak{m})| \le 3^{\omega(\mathfrak{m})} \ll_\eta D^\eta$. For each actual row the absolute triangle in this sparse sector costs $O(B_W^2 D^{1+\epsilon})$ before normalization. Divide once by D and sum the actual rows to obtain (156). Original row masks give $R_w \ll R$, and a nonempty capacity block has $Nu_0A_1A_2^2 \le H$, hence $R \le H/(Nu_0A_2) \le H$. For each fixed W, $S_1^{AA} = S_{\mathrm{eq}} + S_{\mathrm{neq}}$ proves the target equivalence stated in the core, with constants allowed to depend on that fixed profile.

Conjugating the residue-field definition gives, including zeros,

$$R_{\bar{\mathfrak{m}}}(x) = \overline{R_{\mathfrak{m}}(\bar{x})}, \qquad R_{\mathfrak{m}}(x)\overline{R_{\bar{\mathfrak{m}}}(x)} = R_{\mathfrak{m}}(Nx). \tag{184}$$

For a split rational p, $p \mid Nx$ exactly when x meets at least one of its two conjugate primes, so the right side retains the original *union* zero. For an orientation of rational- squarefree norm N, restriction to rational integers gives a primitive Dirichlet character $\psi_{\mathfrak{m}} \pmod N$: $\mathbb{Z}/N\mathbb{Z} \longrightarrow \mathcal{O}/\mathfrak{m}$ is an isomorphism, and each local character has exact order six. Its ideal norm pullback has conductor $(N) = \mathfrak{m}\bar{\mathfrak{m}}$ and infinity type zero. There is no smaller radical or common truncated-frequency assertion.

Choose one prime π_p above each admissible split rational p. Every divisor of an orientation $\mathfrak{m}$ has a unique rational divisor norm and the same Möbius parity. Thus $A_U(\mathfrak{m}) = a_U(N)$ with the *two original* cutoffs, and the partner has the same coefficient. At $t = Nu_0N\mathfrak{v}_1(N\mathfrak{v}_2)^2$, (184) makes the local orientation factors $a_p\chi_p(t)$ and its conjugate. Expanding their product gives all $2^{\omega(N)}$ ordered orientations exactly, proving (157). Both reversals are included, not divided by two. The common complex profile gives $|W(N/D)|^2$. For $\nu = 1$ and no raw collision the character factors are $2\,\mathrm{Re}\,\mu_6 \in \{2, 1, -1, -2\}$, so they need not vanish or be positive; a_U or W can still vanish. This is not a row-energy lower bound.

For (158), use the exact primitive representation *restricted first* to these $O(D)$ pairs. Its conductor norm is $N\mathfrak{f}_S N^2 \ge D^2$, and $|\dot{G}| \le 2C_0^{-1/2}$ by $|J_0(t)| \le 1$ on real t. The Poisson amplitude

is at most $A_1/((N\mathfrak{k})^2 N\mathfrak{l} N\mathfrak{c}D)$. There are $O(Y)$ integral dual ideals in a prefix or band, counting every S-supported ideal once. The finite square-divisor sum is bounded by its convergent positive Nk^{-2} majorant; finite S/ray and $\mathfrak{l}$ divisor costs give $A_2(DA_2)^\epsilon$. With the sparse pair count and coefficient product $B_W^2 D^\epsilon$, the numerator is $O(B_W^2(DYA_2)^\epsilon RY)$. Division by D proves the bound. This uses no general row-weight norm and no root cancellation.

At fixed parameters, exact Poisson gives $DS_{\mathrm{eq}} = \mathcal{P}_{\mathrm{eq}}(\le Y) + \mathcal{P}_{\mathrm{eq}}(> Y)$, with its actual smooth derivative constants. Therefore

$$\frac{|\mathcal{P}_{\mathrm{eq}}(> Y)|}{D} \ll B_W^2\{D^\epsilon R_w + (DYA_2)^\epsilon RY/D\}. \tag{185}$$

This is a *signed* whole-minus-prefix identity and triangle, not absolute summation of high modes. The condition $RY/D \le H$ gives pure $Y \asymp D$ and mixed $Y \asymp D^{7/6}$ target-sized prefix/complement sectors. Unequal norms and arbitrary larger individual windows remain unestimated.

S76.8 A common literal modulus and the surviving ray mean

This last construction concerns the pure block only. The primary section on good ideals is multiplicative because $\mu_6 \to (\mathcal{O}/3)^\times$ is bijective. Let $\mathfrak{c} = (\gamma_c)$ be fixed, S-supported and divisible by (6), large enough for all actual row-ray congruences. For $\mathfrak{q} = \mathfrak{m}\mathfrak{n}$ coprime to it, put

$$\vartheta_{\mathfrak{q}}(x) = (x/\mathfrak{m})_6\overline{(x/\mathfrak{n})_6}, \qquad \tau_{\mathfrak{q}} = \sum_{x \bmod \mathfrak{q}} \vartheta_{\mathfrak{q}}(x) e_{\gamma_q}(x).$$

Every good local factor is nontrivial sextic, so its finite conductor is exactly $\mathfrak{q}$, including nonunit frequency zeros. It need not yet be unit-trivial or an ideal character. The finite indicator $P_{\mathfrak{k}^2}$ imposes residual generator $x \equiv 1 \pmod 3$, original S-goodness and the actual ray condition on $\alpha_{\mathfrak{k}}^2 x$. It is independent of $\mathfrak{m}, \mathfrak{n}$. Literal square-divisor inversion is

$$\mathcal{P}_D = \sum_{(\mathfrak{m},\mathfrak{n})=1} c_{\mathfrak{m}}\bar{c}_{\mathfrak{n}} \sum_{\substack{\mathfrak{k}\ \text{good,SF} \\ N\mathfrak{k} \le \sqrt{b}}} \mu(\mathfrak{k})\vartheta_{\mathfrak{q}}(\alpha_{\mathfrak{k}}^2) \sum_{x \in \mathcal{O}} P_{\mathfrak{k}^2}(x \bmod \mathfrak{c})\vartheta_{\mathfrak{q}}(x) G(Nx/L_{\mathfrak{k}}). \tag{186}$$

The original $(\mathfrak{k}, \mathfrak{q})$ zero is kept. Auxiliary lattice x need not generate a squarefree ideal.

Use the same positive additive-pairing sign as above. CRT separates the common-modulus Fourier transform into the $\mathfrak{q}$ sum at frequency $\beta\gamma_c^{-1}$ and the $\mathfrak{c}$ sum at $\beta\gamma_q^{-1}$. Primitive finite Gauss Fourier proves (159) for *every* β, including its zero at $(\beta, \mathfrak{q}) > 1$. This is the finite recombination of all ray components; it is not a unit root at the larger induced modulus $\mathfrak{c}\mathfrak{q}$. For $g_L(x) = G(|x|^2/L)$ the Euclidean radial transform is

$$\widehat{g}_L(w) = \pi L \int_0^\infty G(t) J_0(2\pi\sqrt{Lt}|w|)\, dt.$$

At the common dual vector $w = 2\bar{\beta}/(\bar{\delta}\bar{\gamma}_c\bar{\gamma}_q)$ this equals $(\sqrt{3}/2)L\dot{G}(LN\beta/(N\mathfrak{c}Q))$. The lattice covolume is $(\sqrt{3}/2)N\mathfrak{c}Q$. Consequently, writing $F_{\mathfrak{k}^2} = P_{\mathfrak{k}^2}\vartheta_{\mathfrak{q}}$,

$$\sum_{x \in \mathcal{O}} F_{\mathfrak{k}^2}(x) G(Nx/L) = \frac{L}{N\mathfrak{c}Q} \sum_{\beta \in \mathcal{O}} \widehat{F}_{\mathfrak{k}^2}(\beta)\dot{G}(LN\beta/(N\mathfrak{c}Q)). \tag{187}$$

No class/root phase is discarded: the original finite function itself has been transformed. Primitive-conductor descriptions induced to the common modulus must reproduce precisely this frequency support, rather than a falsely flat lifted Gauss sum.

Combining the formulas gives the explicit signed literal additive band

$$\mathcal{B}_Y = \sum_{(\mathfrak{m},\mathfrak{n})=1} c_{\mathfrak{m}}\bar{c}_{\mathfrak{n}} \sum_{\substack{\mathfrak{k}\ \text{good,SF}\\ N\mathfrak{k}\le\sqrt{b}}} \frac{\mu(\mathfrak{k})L_{\mathfrak{k}}\tau_{\mathfrak{q}}\vartheta_{\mathfrak{q}}(\gamma_c)}{N\mathfrak{c}Q}\vartheta_{\mathfrak{q}}(\alpha_{\mathfrak{k}}^2) \cdot \sum_{Y<N\beta\le 2Y} \overline{\vartheta_{\mathfrak{q}}(\beta)}\widehat{P}_{\mathfrak{k}^2}(\beta\gamma_q^{-1})\dot{G}(L_{\mathfrak{k}}N\beta/(N\mathfrak{c}Q)). \tag{188}$$

All finite $\mu(\mathfrak{k})$, the original unfactored coefficients and the full pair/ray/raw masks precede any absolute norm. There is no gcd triangle or positive completion in this identity. The frequency band is literal and unit-invariant; it is not simply the old primitive-ideal band under a new name.

On $\beta = \gamma_c\zeta$, the fixed-residue Fourier frequency is zero. The factor $\vartheta_{\mathfrak{q}}(\gamma_c)$ cancels and

$$\widehat{F}_{\mathfrak{k}^2}(\gamma_c\zeta) = r_{\mathfrak{k}^2}\tau_{\mathfrak{q}}\overline{\vartheta_{\mathfrak{q}}(\zeta)}.$$

This S-local mean is not the global $\beta = 0$, which is zero by primitive nontriviality. A six-unit orbit sum in this radial band vanishes if $\vartheta_{\mathfrak{q}}|_{\mu_6}$ is nontrivial. Otherwise it is six times the sum over *all* integral ideals; values there are defined with any generator, not a nonexistent primary generator for an S-supported ideal. At $\varepsilon = 1+\omega$, local residue powers give

$$\vartheta_{\mathfrak{q}}(\varepsilon) = \varepsilon^{(N\mathfrak{m}-N\mathfrak{n})/6}.$$

Every good prime-ideal norm is one modulo six, and $(\prod Np-1)/6 \equiv \sum(Np-1)/6 \pmod 6$, proving (160). In the surviving case $\vartheta_{\mathfrak{q}}$ is a primitive type-zero ideal character and $\epsilon_{\mathfrak{q}} = \tau_{\mathfrak{q}}/\sqrt{Q}$. The exact retained mean channel is

$$\mathcal{B}_Y^{S\text{-mean}} = \sum_{\substack{(\mathfrak{m},\mathfrak{n})=1\\ N\mathfrak{m}\equiv N\mathfrak{n}\ (36)}} c_{\mathfrak{m}}\bar{c}_{\mathfrak{n}} \sum_{\substack{\mathfrak{k}\ \text{good,SF}\\ N\mathfrak{k}\le\sqrt{b}}} \mu(\mathfrak{k})\vartheta_{\mathfrak{q}}(\alpha_{\mathfrak{k}}^2)\frac{6r_{\mathfrak{k}^2}}{N\mathfrak{c}}\frac{L_{\mathfrak{k}}}{\sqrt{Q}}\epsilon_{\mathfrak{q}} \cdot \sum_{Y/N\mathfrak{c}<N\mathfrak{z}\le 2Y/N\mathfrak{c}} \overline{\vartheta_{\mathfrak{q}}(\mathfrak{z})}\dot{G}(L_{\mathfrak{k}}N\mathfrak{z}/Q). \tag{189}$$

This includes nonunit and S-supported dual modes. The $\mathfrak{k}$ phase is cubic, $\vartheta_{\mathfrak{q}}(\mathfrak{k})^2$, not one, and its original cutoff has not been Euler-completed.

For the basic S above $2,3$ and no additional ray condition, modulo (6) the allowed coordinates $a+b\omega$ are $(1,0),(1,3),(4,3)$. The fourth primary residue $(4,0)$ is excluded by the original 2-adic coprimality condition. Thus $r=3$, $N\mathfrak{c}=36$, and $6r/N\mathfrak{c} = 1/2$ exactly. Dropping that condition would give the incorrect $2/3$. The new exact local Fourier check has $\sum|\widehat{P}|^2 = 36\cdot 3 = 108$; it is a finite S-ring consistency check, not an arithmetic-family or asymptotic band test.

Proposition S75.6 controls the whole primal equal-norm sector. That alone does not control this sector in an individual literal band or S-mean channel. To formulate a literal unequal-norm problem, impose $N\mathfrak{m} \ne N\mathfrak{n}$ on the original pair sum before applying (186) and (187). The projected formula retains unequal norms congruent modulo 36 as well as all complementary literal residues. Neither the full $\mathcal{B}_Y$ nor its separately projected unequal-norm version is proved small here. At a pure critical block a sufficient band estimate would be

$$|\mathcal{B}_Y| \ll_{S,\nu,W,\epsilon} B_W^2 DA_1(DA_1Y)^{\epsilon},$$

with the fixed-profile dependence retained. This is not necessary for the whole kernel, where bands may cancel. Finite ray/root recombination alone does not provide it; nor does a nonzero local mean give a lower bound for the whole band. This identifies the remaining coherent unequal-norm arithmetic problem without deleting its means or changing any original conductor, capacity or profile.

S77 An arithmetic Gauss tail with the original finite cutoff

The coherent kernel of Section S75 contains a difficult root-weighted mean. A large part of its *short-divisor expansion* can now be bounded by retaining a cubic interaction before taking a triangle. This gives a signed cutoff comparison, not a smaller norm for the original coefficients. A direct exact-discrete-frequency version is included as well. The remaining small-short-divisor moment and the separate discrete-frequency error are stated below. The whole signed power 13/12 and global unsigned power 9/8 do not change.

S77.1 Original objects and a fixed common profile

Keep $K = \mathbb{Q}(\omega)$, $\mathcal{O} = \mathbb{Z}[\omega]$, $\delta = 1 + 2\omega$, and the fixed exclusions S of Section S75.1. A good ideal is prime to S. All columns are the complete good squarefree ideal annulus $D \le N\mathfrak{m} \le 2D$, $D > 1$. Equal norms do not identify ideals. For a real cutoff $1 \le T \le \sqrt{D}$ put

$$\begin{aligned} A_T(\mathfrak{m}) &= \sum_{\substack{\mathfrak{a}\mathfrak{b}\mid\mathfrak{m}\\(\mathfrak{a},\mathfrak{b})=1\\ N\mathfrak{a},N\mathfrak{b}\le T}} \mu(\mathfrak{a})\mu(\mathfrak{b}),\\ c_f(\mathfrak{m}) &= f(\mathfrak{m})\nu(\mathfrak{m})R_{\mathfrak{m}}(u_0)W(N\mathfrak{m}/D). \end{aligned} \tag{190}$$

Here ν is the original fixed unitary finite-order character; u_0 is the fixed literal unit/S-supported factor with no good support. The actual coefficient is c_{A_U}, $1 \le U \le \sqrt{D}$. The function W is one fixed, possibly complex, smooth common profile supported on $[1, 2]$, zero at the endpoints. Define

$$\begin{aligned} B_W &= \|W\|_\infty, \qquad \widehat{W}(t) = \int_0^\infty W(x)x^{-it}\,\frac{dx}{x},\\ \mathcal{M}(W) &= \frac{1}{2\pi}\int_{\mathbb{R}} |\widehat{W}(t)|\,dt. \end{aligned} \tag{191}$$

This explicit seminorm is finite and controls B_W. It is a seminorm of the column profile, not of the sharp nodal row function.

Only the original unweighted *pure* two-pivot block is used: $A_2 = 1$, $\mathfrak{v}_2 = \mathcal{O}$, and the other four factors are units. The first row is good and squarefree, with the original literal/ray conditions and capacity. Its integer endpoints are

$$a = \lceil A_1 \rceil, \qquad b = \min\{\lceil 2A_1 \rceil - 1, \lfloor H/Nu_0 \rfloor\}, \qquad K = \sqrt{b}. \tag{192}$$

An empty interval contributes zero throughout. Otherwise $b \ge a \ge 1$. Use the exact G of (145), its transform $\dot{G}$ in (146), and $C_0 = 3/(4\pi^2)$. In particular $0 \le G \le 1$, $\operatorname{supp} G \subset [1/2, 5/2]$, and $\int G \le 2$. The original *pre-S* cutoff K is never recomputed after another deletion.

Let $\mathcal{I}$ be the original ordered coprime column pairs. One may also impose the *original* unequal-norm projection; the same choice is then used in every expression below. For each source pair A, B and original ray character ρ use the primitive η, $\mathfrak{f}_S$ and F of (141). Thus $F = N\mathfrak{f}_S N\mathfrak{m}N\mathfrak{n}$, and η has total infinity type zero. Put

$$\begin{aligned} T_q &= S \cup \operatorname{supp}(\mathfrak{m}\mathfrak{n}),\\ Q_s^{T_q}(K;\eta^2) &= \sum_{\substack{\mathfrak{k}\ \text{good,SF}\\ N\mathfrak{k}\le K,\ (\mathfrak{k},T_q)=1}} \frac{\mu(\mathfrak{k})\eta(\mathfrak{k})^2}{(N\mathfrak{k})^s},\\ P_S(\eta) &= \sum_{\mathfrak{e}\mid P_S} \frac{\mu(\mathfrak{e})\eta(\mathfrak{e})}{N\mathfrak{e}}, \qquad P_S = \prod_{p\in S} p. \end{aligned} \tag{193}$$

The symbol $(\mathfrak{k}, T_q) = 1$ means avoidance of every prime in that set. Squaring η never restores a deleted prime. Values at $\mathfrak{e}$ and at any later S-supported frequency are separate primitive values, including zero.

S77.2 The actual retained mean and its finite leading model

In the exact primitive kernel (147), write each dual ideal uniquely as $\mathfrak{y} = \mathfrak{h}\mathfrak{r}\mathfrak{s}^6$: $\mathfrak{h}$ is S-supported, $\mathfrak{r}$ is good and sixth-free, and $\mathfrak{s}$ is good. Overlap of $\mathfrak{r}, \mathfrak{s}$ is allowed. For $X \geq 64$ set

$$V = X^{1/6}/2, \qquad b_{\mathfrak{w}} = (X/N\mathfrak{w})^{1/6}, \qquad t_{\mathfrak{w}} = \big((N\mathfrak{w})^{-1/6} - 1/2\big)_+. \tag{194}$$

The top band is $V < N\mathfrak{s} \leq 2V$, $N\mathfrak{y} \leq X$. Replacing its $\mathfrak{s}$ count by its principal fixed-ray density defines the retained mean M_{top}. This operation keeps, rather than deletes, its coherent main term. The selected branches are $\rho^6 = 1$. At this top window $N\mathfrak{h}N\mathfrak{r} < 64 < 7^6$, so $\mathfrak{w} = \mathfrak{h}\mathfrak{r}$ runs over *all* integral ideals of norm below 64, including S-supported ideals.

Set

$$\kappa_S = \frac{\pi}{3\sqrt{3}} \prod_{p \in S} (1 - (Np)^{-1}), \qquad g(\mathfrak{m}) = \prod_{p \mid \mathfrak{m}} (1 - (Np)^{-1}). \tag{195}$$

On original coprime support the density is $\kappa_S g(\mathfrak{m}) g(\mathfrak{n})$. For $\theta = (A, B, \rho, \mathfrak{m}, \mathfrak{n}, \mathfrak{e}, \mathfrak{w})$ in the preceding sets, with $\rho^6 = 1$, define

$$\begin{aligned} d_\theta[f_1, f_2] = \frac{\kappa_S A_1}{D} h_{AB\rho}(\mathcal{O}) c_{f_1}(\mathfrak{m}) \overline{c_{f_2}(\mathfrak{n})} \frac{g(\mathfrak{m}) g(\mathfrak{n}) \epsilon(\eta)}{\sqrt{F}} \\ \cdot \frac{\mu(\mathfrak{e}) \eta(\mathfrak{e})}{N\mathfrak{e}} \overline{\eta(\mathfrak{w})}, \qquad \lambda_\theta = \frac{A_1 N\mathfrak{w}}{N\mathfrak{e} F}. \end{aligned} \tag{196}$$

The exact definition of the normalized retained mean is

$$\begin{aligned} \mathcal{M}_{\mathrm{top}} := \frac{M_{\mathrm{top}}}{D} = \sum_\theta d_\theta[A_U, A_U] \sum_{\substack{\mathfrak{k} \text{ good,SF} \\ N\mathfrak{k} \leq K,\ (\mathfrak{k}, T_q) = 1}} \frac{\mu(\mathfrak{k}) \eta(\mathfrak{k})^2}{(N\mathfrak{k})^2} \\ \cdot \int_V^{b_{\mathfrak{w}}} \dot{G}\left(\frac{\lambda_\theta t^6}{(N\mathfrak{k})^2} \right) dt. \end{aligned} \tag{197}$$

All sums here are finite. In particular there is no $\mathfrak{k} \mid \mathfrak{w}$ or $(\mathfrak{k}, \mathfrak{w}) = 1$ condition. The finite original source/ray coefficients $h_{AB\rho}(\mathcal{O})$ have not been centered. An argument at $\mathfrak{e}$ is not divided by a value at $\mathfrak{w}$.

Put $g_0 = C_0^{-1/2} \int G$ and define

$$\mathcal{L}_0[f_1, f_2] = g_0 X^{1/6} \sum_\theta d_\theta[f_1, f_2]\, t_{\mathfrak{w}} Q_2^{T_q}(K; \eta^2). \tag{198}$$

Replacing both column functions here is an explicitly defined auxiliary model, not a change to the original primal data.

Proposition S77.1 (Matched leading model with its original cutoff)**.** *For the actual coefficients and every finite $X \geq 64$,*

$$|\mathcal{M}_{\mathrm{top}} - \mathcal{L}_0[A_U, A_U]| \ll B_W^2 (DX)^\epsilon \frac{A_1^2 X^{7/6}}{D^2}. \tag{199}$$

The constant uses no derivative bound for the sharp G. Both interval endpoints and the original finite K occur in $\mathcal{L}_0$. This estimates the model error, not the leading signed model.

S77.3 The root is an incomplete cubic Gauss weight

For each good ideal use its primary generator $\alpha_{\mathfrak{a}} \equiv 1 \pmod 3$ and write $\chi_{j,\mathfrak{a}}$ for its positive residue character of order $j = 3, 6$. For good squarefree $\mathfrak{m}$ put

$$\begin{aligned} G_{\mathfrak{m}} &= \frac{1}{\sqrt{N\mathfrak{m}}} \sum_{x \bmod \alpha_{\mathfrak{m}}} \chi_{3,\mathfrak{m}}(x) \exp\Big(2\pi i \operatorname{Tr} \frac{x}{\delta \alpha_{\mathfrak{m}}}\Big), \\ g_{\mathfrak{m}}(z) &= \overline{\chi_{3,\mathfrak{m}}(z)} G_{\mathfrak{m}}. \end{aligned} \tag{200}$$

The second expression is the normalized finite Gauss sum at frequency z, including zero when $(\mathfrak{m}, z) > 1$. For $\mathfrak{q} = \mathfrak{m}\mathfrak{n}$ put $\theta_q = \chi_{6,\mathfrak{m}} \overline{\chi_{6,\mathfrak{n}}}$. Let θ_S be the *actual finite S part* of the total type-zero η, at its own primitive conductor $\mathfrak{f}_S = (\gamma_S)$. Then

$$\mathcal{S}_{\mathfrak{m},\mathfrak{n},\rho} = \theta_S(\alpha_{\mathfrak{m}}\alpha_{\mathfrak{n}}) \theta_q(\gamma_S) \frac{\tau_{\gamma_S}(\theta_S)}{\sqrt{N\mathfrak{f}_S}}. \tag{201}$$

The unit-conductor convention is one. This scalar has modulus one; it is not obtained by discarding the infinity type of a constituent. The accepted Jacobi character in (54) gives $\vartheta = \Psi/\sqrt{N}$, with conductor (2δ) of norm 12 and nonzero infinity type. The exact root formula is

$$\begin{aligned} \epsilon(\eta) &= \mathcal{S}_{\mathfrak{m},\mathfrak{n},\rho} \Gamma(\alpha_{\mathfrak{m}}\alpha_{\mathfrak{n}}) \mu(\mathfrak{m}) \mu(\mathfrak{n}) \vartheta(\mathfrak{m}) \overline{\vartheta(\mathfrak{n})}\, \overline{G_{\mathfrak{m}}} G_{\mathfrak{n}}, \\ \Gamma(a + b\omega) &= \tfrac{1}{2}\{1 + i^{-b} + i^{a} + i^{b-a}\}. \end{aligned} \tag{202}$$

No extra quadratic sign is inserted. In the original kernel, $\eta(\mathfrak{k}^2)$ and the second-row phase, when present, produce the raw cubic frequency $\overline{g_{\mathfrak{m}}(\alpha_{\mathfrak{k}}\alpha_{\mathfrak{v}_2})} g_{\mathfrak{n}}(\alpha_{\mathfrak{k}}\alpha_{\mathfrak{v}_2})$; here $\mathfrak{v}_2 = \mathcal{O}$. This incomplete weighted frequency set is not a complete residue average.

S77.4 A signed short-divisor tail from a physical cubic matrix

For squarefree $\mathfrak{m}$ there is an exact sign-preserving assignment identity:

$$\mu(\mathfrak{m}) A_T(\mathfrak{m}) = \sum_{\substack{\mathfrak{a}\mathfrak{b}\mathfrak{c} = \mathfrak{m} \\ N\mathfrak{a}, N\mathfrak{b} \le T}} \mu(\mathfrak{c}). \tag{203}$$

The factors are pairwise coprime and squarefree. For $Y \le U$ the difference of cutoff boxes has two disjoint pieces. After interchanging $\mathfrak{a}, \mathfrak{b}$ in the second and setting $\mathfrak{z} = \mathfrak{b}\mathfrak{c}$, the exact coefficient is

$$T_U(\mathfrak{z}) + T_Y(\mathfrak{z}), \qquad T_L(\mathfrak{z}) = \sum_{\substack{\mathfrak{b} \mid \mathfrak{z} \\ N\mathfrak{b} \le L}} \mu(\mathfrak{z}/\mathfrak{b}). \tag{204}$$

It is independent of $\mathfrak{a}$. The conjugate Gauss CRT identity retains the cross interaction

$$\overline{G_{\mathfrak{a}\mathfrak{z}}} = \overline{G_{\mathfrak{a}} G_{\mathfrak{z}}} \chi_{3,\mathfrak{a}}(\alpha_{\mathfrak{z}}), \qquad (\mathfrak{a}, \mathfrak{z}) = 1. \tag{205}$$

The raw cubic entry is zero at a common prime.

The deep input is explicitly external: de Faveri, arXiv:2610.04045v1, Theorem 1.1 and Example 2.2, at $n = 3$. For the physical primary cubic matrix on good squarefree ideals of norms comparable to A, Z it gives

$$\big\| [\chi_{3,\mathfrak{a}}(\alpha_{\mathfrak{z}})] \big\|_{\mathrm{op}}^2 \ll_{\epsilon} (AZ)^{\epsilon} \{A + Z + (AZ)^{2/3}\}. \tag{206}$$

Both indices are actual ideals and hence cube-free, not arbitrary primitive-conductor labels. This is the same pinned input as the cubic comparison in Proposition S75.5.

Theorem S77.2 (Actual Gauss-weighted cutoff tail)**.** *For any bounded multiplicative ideal function λ on good ideals, with $|\lambda| \le 1$, and $1 \le Y \le U \le \sqrt{D}$,*

$$\begin{aligned}\left|\sum_{\mathfrak{m}\ \mathrm{good,SF}} \frac{\mu(\mathfrak{m})(A_U - A_Y)(\mathfrak{m})\overline{G_{\mathfrak{m}}}\lambda(\mathfrak{m})}{\sqrt{N\mathfrak{m}}} W(N\mathfrak{m}/D)\right| \\ \ll_\epsilon \mathcal{M}(W) D^\epsilon \{D^{1/3} + \sqrt{D/Y}\}.\end{aligned} \tag{207}$$

The constant is uniform in the conductor or raw deletion set in λ. For the actual pure leading model, with every original finite K, class/ray/root/S component and coprime pair retained,

$$|\mathcal{L}_0[A_U, A_U] - \mathcal{L}_0[A_Y, A_Y]| \ll B_W \mathcal{M}(W) D^\epsilon A_1 X^{1/6}\{Y^{-1/2} + D^{-1/6}\}. \tag{208}$$

The original unequal-norm projection is also allowed.

This uses the cubic interaction before the short-divisor triangle. It does not assert pointwise or coefficient-norm shrinkage of A_U, nor a separate cancellation theorem for $\mu(\mathfrak{c})$.

S77.5 What has been reduced, and what remains

Combining Proposition S77.1 and Theorem S77.2 gives

$$\begin{aligned}|\mathcal{M}_{\mathrm{top}} - \mathcal{L}_0[A_Y, A_Y]| \ll (DX)^\epsilon \Big\{ B_W^2 \frac{A_1^2 X^{7/6}}{D^2} \\ + B_W \mathcal{M}(W) A_1 X^{1/6}(Y^{-1/2} + D^{-1/6})\Big\}.\end{aligned} \tag{209}$$

At $A_1 \asymp H = D^{3/4}$, $X = D^{3/8}$ and $Y = \min\{U, D^{1/8}\}$, the errors are target-sized. If $Y = U$ the coefficient tail is exactly zero; otherwise the three displayed powers are $-1/16$, $3/4$ and $31/48$. The remaining exact coefficients satisfy

$$\mu(\mathfrak{m})A_Y(\mathfrak{m}) = \sum_{\substack{\mathfrak{abc}=\mathfrak{m}\\ N\mathfrak{a}, N\mathfrak{b} \le Y}} \mu(\mathfrak{c}), \qquad N\mathfrak{c} \ge D/Y^2 \ge D^{3/4}. \tag{210}$$

The small-short-divisor, long residual-Möbius cubic Gauss bilinear moment $\mathcal{L}_0[A_Y, A_Y]$ is still unestimated at the target.

S77.6 The exact discrete top prefix, with every ray retained

There is also a direct reduction that does not introduce the principal-density split. Its exact support is

$$\mathcal{T}_X = \{(\mathfrak{w}, \mathfrak{s}) : N\mathfrak{w} < 64,\ \mathfrak{s} \text{ good},\ V < N\mathfrak{s} \le 2V,\ N\mathfrak{w}(N\mathfrak{s})^6 \le X\}. \tag{211}$$

There is no squarefree restriction on $\mathfrak{s}$. Overlap of $\mathfrak{w}, \mathfrak{s}$ and of $\mathfrak{k}, \mathfrak{s}$ remains allowed. Let $\widetilde{\Theta}$ be the same outer set as θ above, but with *every* original ρ, and put

$$\begin{aligned}\widetilde{d}_\theta[f_1, f_2] = \frac{A_1}{D} h_{AB\rho}(\mathcal{O}) c_{f_1}(\mathfrak{m})\overline{c_{f_2}(\mathfrak{n})}\frac{\epsilon(\eta)}{\sqrt{F}} \\ \cdot \frac{\mu(\mathfrak{e})\eta(\mathfrak{e})}{N\mathfrak{e}}\overline{\eta(\mathfrak{w})}.\end{aligned} \tag{212}$$

This is an explicit definition, not division by a density or a primitive value. With the original $\mathfrak{k}$ support as in (193), define the normalized exact piece

$$\begin{aligned}\mathcal{P}_X[f_1,f_2]=\sum_{\theta\in\widetilde{\Theta}}\widetilde{d}_\theta[f_1,f_2]\sum_{\mathfrak{k}}\frac{\mu(\mathfrak{k})\eta(\mathfrak{k})^2}{(N\mathfrak{k})^2}\\ \cdot\sum_{\mathfrak{s}:\ (\mathfrak{w},\mathfrak{s})\in\mathcal{T}_X}\bar{\rho}(\mathfrak{s})^6\mathbf{1}_{(\mathfrak{s},\mathfrak{mn})=1}\dot{G}\left(\frac{\lambda_\theta(N\mathfrak{s})^6}{(N\mathfrak{k})^2}\right).\end{aligned}\tag{213}$$

Then $\mathcal{P}_X[A_U,A_U]$ is the actual original top primitive-prefix contribution divided by D. It is a signed representation-fixed piece, not a positive energy or the whole infinite dual sum.

Let $\mathcal{L}_X^K[f_1,f_2]$ be exactly (213) with $\dot{G}$ replaced by g_0. Only in that leading model can the finite $\mathfrak{k}$ sum be factored as $Q_2^{T_q}(K;\eta^2)$. The discrete $\mathfrak{s}$ sum and its endpoints remain. Unlike $\mathcal{L}_0$, this model has no $\kappa_S g(\mathfrak{m})g(\mathfrak{n})$ density, no $\rho^6=1$ projection and no integral replacing the $\mathfrak{s}$ family.

Proposition S77.3 (Direct original-prefix cutoff reduction)**.** *Put $\Lambda=A_1X/D^2$. With precisely the original support,*

$$|\mathcal{P}_X[A_U,A_U]-\mathcal{L}_X^K[A_U,A_U]|\ll B_W^2(DX)^\epsilon A_1V\Lambda,\tag{214}$$

and

$$\begin{aligned}|\mathcal{L}_X^K[A_U,A_U]-\mathcal{L}_X^K[A_Y,A_Y]|\ll B_W\mathcal{M}(W)(DX)^\epsilon A_1V\\ \cdot\{Y^{-1/2}+D^{-1/6}\}.\end{aligned}\tag{215}$$

In particular the actual original piece satisfies

$$\begin{aligned}|\mathcal{P}_X[A_U,A_U]-\mathcal{L}_X^K[A_Y,A_Y]|\ll(DX)^\epsilon A_1V\{B_W^2\Lambda\\ +B_W\mathcal{M}(W)(Y^{-1/2}+D^{-1/6})\}.\end{aligned}\tag{216}$$

The original unequal-norm projection is allowed.

At the critical pure parameters used above the three powers are again $-1/16$, $3/4$ and $31/48$, with exact zero coefficient tail when $Y=U$. This is a reduction of the *actual discrete* piece to a different, still unestimated, small-short-divisor model. That model retains every ρ, every nonunit S mode, the exact $\mathfrak{w},\mathfrak{s}$ support, root/conductor data, original finite K and both long residual-Möbius column coefficients.

Separate open errors and scope. The accepted fixed-ray disk count gives the distinct top-band certificate

$$\frac{|\mathcal{P}_{\mathrm{top}}-M_{\mathrm{top}}|}{D}\ll B_W^2(DX)^\epsilon A_1X^{1/18}\{1+\sqrt{A_1X/D^2}\}.\tag{217}$$

Its pure critical power is $37/48>3/4$; this is an upper certificate, not a lower bound for the actual error. The direct reduction in Proposition S77.3 does not introduce that split: its discrete model retains those effects. No improved bound for the separate discrepancy, for the remaining small-cutoff models, or for the other original sectors is proved here. The result is an application of a named external cubic sieve to an actual arithmetic tail, not a new sieve, a complete analytic Lean formalization, a determinant/zero-location bridge or a proof of GRH. Fixed-profile constants and the explicit $\mathcal{M}(W)$ cannot be replaced by a uniform supnorm-only claim for varying profiles.

S78 Proofs for the finite-cutoff Gauss tail

The arithmetic norm input is the explicitly pinned de Faveri, arXiv:2610.04045v1, Theorem 1.1 and Example 2.2, at order three. It is used on its actual primary cube-free ideal family. We do not reprove that deep theorem. The primitive root conventions are those already established in Proposition S75.1, and the Jacobi orientation is that of (54). The accompanying source map binds the accepted research proofs and their independent reviews. Full third-party reading caches are not public proof inputs.

S78.1 The coherent main retains all sixth-part and ray data

For a good ideal $\mathfrak{s}$, the original fixed-source transport in (141) gives

$$\overline{\eta(\mathfrak{s}^6)} = \overline{\rho(\mathfrak{s})}^6 \mathbf{1}_{(\mathfrak{s},\mathfrak{mn})=1}. \tag{218}$$

This is an assertion about the total character on good arguments, not about the order of the individual possibly angular/radial $\xi_{\mathfrak{m}}$. Write $\psi = \bar{\rho}^6$, and let δ_ψ be one if it is principal on good ideals and zero otherwise. The conductor of ψ belongs to fixed S data.

Here is the fixed-ray counting input, with its full deletion. For $t \geq 0$, the classical disk estimate gives

$$\begin{aligned} N_{\psi,q}(t) &:= \sum_{\substack{\mathfrak{s} \text{ good} \\ N\mathfrak{s}\leq t,\ (\mathfrak{s},\mathfrak{q})=1}} \psi(\mathfrak{s}) \\ &= \delta_\psi \kappa_S g(\mathfrak{q}) t + E_{\psi,q}(t), \\ |E_{\psi,q}(t)| &\ll_{S,\psi} t^{1/3} \sum_{\mathfrak{d}|\mathfrak{q}} (N\mathfrak{d})^{-1/3} \ll_\epsilon D^\epsilon t^{1/3}, \qquad N\mathfrak{q} \leq 4D^2. \end{aligned} \tag{219}$$

We recall a proof to specify the uniformity being used. For a fixed ideal modulus, count elements in each translated lattice coset inside a disk of radius R. Convolve the disk indicator with a fixed compact smooth nonnegative mollifier of radius Δ. The smoothed disks of radii $R-\Delta$ and $R+\Delta$ sandwich the original disk. Their zero modes differ from its area by $O(R\Delta + \Delta^2)$.

For a nonzero dual vector ξ, polar integration gives

$$\widehat{\mathbf{1}_{|z|\leq R}}(\xi) = \frac{RJ_1(2\pi R|\xi|)}{|\xi|}, \qquad |J_1(u)| \ll \min\{1, u^{-1/2}\}. \tag{220}$$

The decay follows directly from the angular integral for J_1: for $u \geq 1$, remove arcs of length $u^{-1/2}$ around the two stationary points. Their total contribution is $O(u^{-1/2})$. On the complement integrate once by parts in the phase; the boundary terms and the integral of the derivative of the reciprocal phase derivative are also $O(u^{-1/2})$. For $u \leq 1$ the angular formula gives the bounded estimate. Thus no unproved cancellation of lattice points is inserted. The mollifier supplies arbitrarily rapid decay beyond $|\xi| \asymp \Delta^{-1}$. Counting points of the fixed dual lattice in unit-width radial shells yields

$$\sum_{\xi \neq 0} |\widehat{\mathbf{1}_{|z|\leq R}}(\xi)\widehat{\varphi}(\Delta\xi)| \ll R^{1/2}\Delta^{-1/2}. \tag{221}$$

The translation only changes unit phases. Choose Δ comparable to $R^{-1/3}$ for large R. The boundary and dual errors are both $O(R^{2/3})$. Bounded R is absorbed into the fixed constant.

A type-zero finite ray character is a fixed periodic generator function, trivial on the six units. Apply the preceding count to its finitely many residue cosets, and only then form the possibly

complex weighted sum. Division by six and the covolume $\sqrt{3}/2$ of $\mathcal{O}$ give the coefficient $\pi/(3\sqrt{3})$. The principal coset average is the good-S density; a nonprincipal character has mean zero. For $t \ge 1$ this proves the undeleted error $O(t^{1/3})$. For $0 < t < 1$ the ideal count is zero and $t \le t^{1/3}$, so the same ideal estimate remains valid; $t = 0$ is exact.

Now use the *entire* finite inclusion–exclusion:

$$N_{\psi,q}(t) = \sum_{\mathfrak{d}\mid\mathfrak{q}} \mu(\mathfrak{d})\psi(\mathfrak{d}) N_{\psi,1}(t/N\mathfrak{d}). \tag{222}$$

Divisors with $N\mathfrak{d} > t$ are still present, with their subunit arguments. For principal ψ the main sum is $\kappa_S t \prod_{p\mid q}(1-(Np)^{-1})$; otherwise it is zero. The error in (219) follows. Its final divisor product is $O_\epsilon(D^\epsilon)$ by separating finitely many small primes and using $1 + P^{-1/3} \le P^\eta$ at all sufficiently large P. This is ideal-squarefreeness, not rational-squarefreeness.

For fixed $\mathfrak{h}, \mathfrak{r}$, the original sum over $V < N\mathfrak{s} \le b_{\mathfrak{h}\mathfrak{r}}$ has weight $\dot{G}(c\,t^6)$. In the pure block the scale is

$$c = \frac{A_1 N\mathfrak{h} N\mathfrak{r}}{(N\mathfrak{k})^2 N\mathfrak{e} F}. \tag{223}$$

Closed prefixes at both endpoints give the exact error formula

$$w(b)E_{\psi,q}(b) - w(V)E_{\psi,q}(V) - \int_V^b E_{\psi,q}(t)w'(t)\,dt, \qquad w(t) = \dot{G}(ct^6). \tag{224}$$

There are no half-weights at ideal norm ties. The angular representations of J_0, J_1 imply, with $\Lambda = A_1 X/D^2$,

$$\|\dot{G}\|_\infty \ll 1, \qquad |x\dot{G}'(x)| \ll \sqrt{x}, \qquad \int_V^b |w'(t)|\,dt \ll \sqrt{\Lambda}. \tag{225}$$

Indeed $ct^6 \le \Lambda$ and $b/V \le 2$; differentiating the Bessel factor, not G, proves the last assertion. Consequently the error at each fixed outer set is $O(D^\epsilon V^{1/3}(1+\sqrt{\Lambda}))$.

At the top window $N\mathfrak{h}N\mathfrak{r} < 64$. Every good ideal of norm below 64 is sixth-free, because a nontrivial good prime has norm at least seven. Therefore the unique decomposition into its S part and good part runs exactly over all $\mathfrak{w}$ in (197). The principal condition is precisely $\rho^6 = 1$. On the original coprime support $g(\mathfrak{m}\mathfrak{n}) = g(\mathfrak{m})g(\mathfrak{n})$. This proves the stated finite formula for M_{top}/D.

For the separate error, $|A_U| \le 3^{\omega(\mathfrak{m})} \ll_\epsilon D^\epsilon$, and there are $O(D)$ original annular ideals. After normalizing once by D, $A_1/(D\sqrt{F}) \le A_1/D^2$ cancels the two column counts. The S/ray/$\mathfrak{e}$ sets are fixed, the top $\mathfrak{w}$ set is finite, and the *original* finite $\mathfrak{k}$ sum is majorized by $\sum (N\mathfrak{k})^{-2}$. Since $V^{1/3} \asymp X^{1/18}$, this proves (217). Only its error uses these triangles; the mean in (197) stays signed.

S78.2 Exact good-modulus Gauss factors and separate primitive S values

The finite cubic sum at frequency z is

$$\frac{1}{\sqrt{N\mathfrak{m}}} \sum_{x \bmod \alpha_{\mathfrak{m}}} \chi_{3,\mathfrak{m}}(x) \exp\!\left(2\pi i \operatorname{Tr} \frac{zx}{\delta\alpha_{\mathfrak{m}}}\right) = \overline{\chi_{3,\mathfrak{m}}(z)} G_{\mathfrak{m}}. \tag{226}$$

For a unit frequency this is change of variable. At a collision, one local additive frequency is zero and its nonprincipal local cubic character has zero sum. CRT proves the assertion for every good squarefree ideal, including inert primes and nonsquarefree rational norms. The normalized sum has modulus one at unit frequencies.

The total η has type zero, so the previously proved root formula is its finite Gauss sum divided by $\sqrt{F}$. CRT at the coprime primitive conductors $\mathfrak{f}_S, \mathfrak{q}$ gives exactly (201) times $\tau_q(\theta_q)/\sqrt{Nq}$.

Although their product is unit-trivial, the finite S and good-q characters need not be unit-trivial separately. No extra unit average is taken. The actual finite S character, including its conductor drop, is used at its own primitive modulus. At unit conductor its Gauss sum is one.

On good squarefree q, all local powers $\theta_q, \theta_q^2, \theta_q^3$ are nonprincipal and primitive. The Gauss–Jacobi identity therefore gives

$$\tau_q(\theta_q)\tau_q(\theta_q^2) = J_q(\theta_q, \theta_q^2)\tau_q(\theta_q^3). \tag{227}$$

Over a residue field this follows by setting $x + y = t$ in the product of Gauss sums; the zero-t term vanishes because the product character is nonprincipal. Multiplication by CRT proves the composite identity: its additive cross multipliers on the left combine to those of θ_q^3 on the right. The Jacobi sum itself has no additive-generator multiplier.

The correctly oriented result already proved in Section S65 is

$$\begin{aligned} J_q(\theta_q, \theta_q^2) &= j(\mathfrak{m})\overline{j(\mathfrak{n})}, \\ j(\mathfrak{m})/\sqrt{N\mathfrak{m}} &= \mu(\mathfrak{m})\vartheta(\mathfrak{m}), \qquad \vartheta = \Psi/\sqrt{N}, \quad \operatorname{cond}\Psi = (2\delta). \end{aligned} \tag{228}$$

Its sign is fixed by $-j(p) \equiv 1 \pmod{2\delta}$ and $(j(p)) = p$ for the positive sextic character. The reduction argument includes inert primes. Unique generators congruent to one modulo 2δ then identify $-j(p)$ with $\Psi(p)$. Multiplication yields the displayed identity only on squarefree columns. In particular Ψ has its actual infinity type $(1, 0)$; it is not a finite-order gauge.

Cubic reciprocity for the same primary section and the fact that $\chi_3(-1) = 1$ yield

$$\begin{aligned} \chi_{3,\mathfrak{m}}(\alpha_{\mathfrak{n}}) &= \chi_{3,\mathfrak{n}}(\alpha_{\mathfrak{m}}), \\ \tau_q(\theta_q^2)/\sqrt{Nq} &= G_{\mathfrak{m}}\overline{G_{\mathfrak{n}}}. \end{aligned} \tag{229}$$

The cubic cross factors cancel, while $\tau(\bar{\chi}_3) = \overline{\tau(\chi_3)}$. The quadratic Gaussian evaluation in the same two-factor completion is $\tau_q(\theta_q^3)/\sqrt{Nq} = \Gamma(\alpha_{\mathfrak{m}}\alpha_{\mathfrak{n}})$, with the period-four formula in (202). For composite q, this uses that Gauss evaluation with its exact pairing; $1 + \theta_q^3$ is not the root-count function of a composite residue ring. Combining these facts with (227) proves (202). There is no additional $\theta_q^3(-1)$ factor.

For the original good $\mathfrak{k}$,

$$\begin{aligned} \eta(\mathfrak{k}^2) &= \theta_S(\alpha_{\mathfrak{k}})^2\chi_{3,\mathfrak{m}}(\alpha_{\mathfrak{k}})\overline{\chi_{3,\mathfrak{n}}(\alpha_{\mathfrak{k}})}, \\ \overline{G_{\mathfrak{m}}}G_{\mathfrak{n}}\chi_{3,\mathfrak{m}}(z)\overline{\chi_{3,\mathfrak{n}}(z)} &= \overline{g_{\mathfrak{m}}(z)}g_{\mathfrak{n}}(z). \end{aligned} \tag{230}$$

Before the pure specialization the row factor $R_{\mathfrak{m}}(\mathfrak{v}_2^2)\overline{R_{\mathfrak{n}}(\mathfrak{v}_2^2)}$ makes $z = \alpha_{\mathfrak{k}}\alpha_{\mathfrak{v}_2}$. All raw collisions are zeros. The separate original $(\mathfrak{k}, \mathfrak{v}_2) = 1$ condition is not replaced by a frequency assertion. The primitive factors at $\mathfrak{e}$ and $\mathfrak{w}$ are $\eta(\mathfrak{e})$ and $\overline{\eta(\mathfrak{w})}$. When expressed with generators, their S parts are the separate factors $\theta_S(\gamma_{\mathfrak{e}})$ and $\overline{\theta_S(\gamma_{\mathfrak{w}})}$. If either is zero the component vanishes. No quotient of these arguments, division by a primitive zero, or primary section for an S-supported ideal is used. Changes of these generators cancel in the total unit-trivial character.

S78.3 Matched finite endpoint models and a real-axis remainder

For $z \geq 0$ and an integer $J \geq 0$, the angular cosine formula and its real Taylor remainder give

$$\left| J_0(2\sqrt{z}) - \sum_{j=0}^{J} \frac{(-z)^j}{(j!)^2} \right| \leq \frac{z^{J+1}}{((J+1)!)^2}. \tag{231}$$

Indeed use the cosine Taylor polynomial through degree $2J+1$ inside its angular integral. The remainder is at most $|2\sqrt{z}\cos\theta|^{2J+2}/(2J+2)!$, and the normalized angular moment of $\cos^{2j}\theta$ is $(2j)!/(4^j(j!)^2)$. This proves both the coefficients and the bound for every nonnegative z, without an exponential majorant or a sign assertion about the error.

Set

$$m_j=\int_0^\infty v^jG(v)\,dv,\qquad I_j(\mathfrak w)=\frac{b_{\mathfrak w}^{6j+1}-V^{6j+1}}{6j+1}=\int_V^{b_{\mathfrak w}}t^{6j}\,dt. \tag{232}$$

The exact order-J signed model of (197) is

$$\mathcal L_J[f_1,f_2]=C_0^{-1/2}\sum_\theta d_\theta[f_1,f_2]\sum_{j=0}^{J}\frac{(-1)^jm_j\lambda_\theta^jI_j(\mathfrak w)}{C_0^j(j!)^2}\,Q_{2+2j}^{T_q}(K;\eta^2). \tag{233}$$

The character in every Q_s is still η^2. Both finite endpoints, every primitive root and every original finite $\mathfrak k$ cutoff remain. In particular $I_0=X^{1/6}t_{\mathfrak w}$ and (233) at $J=0$ is exactly (198).

Applying (231) before the original $\mathfrak k$ sum and the t integral gives a per-θ bound

$$\frac{C_0^{-1/2}m_{J+1}(\lambda_\theta/C_0)^{J+1}I_{J+1}(\mathfrak w)}{((J+1)!)^2}\sum_{\mathfrak k\ \text{original}}(N\mathfrak k)^{-4-2J}. \tag{234}$$

Only this certified remainder takes absolute values. On the full support $\lambda_\theta b_{\mathfrak w}^6=A_1X/(N\mathfrak eF)\le\Lambda$. Since $m_{J+1}\le 2(5/2)^{J+1}$ and $\sum(N\mathfrak k)^{-4-2J}$ is bounded by one constant independent of J and all deletions, put $\Omega=(5/2)\Lambda/C_0$. Counting the two actual annular column sets only in the error yields

$$|\mathcal M_{\mathrm{top}}-\mathcal L_J[A_U,A_U]|\ll B_W^2(DX)^\epsilon A_1X^{1/6}\frac{\Omega^{J+1}}{((J+1)!)^2}. \tag{235}$$

Here the two $O(D)$ column counts cancel $1/(D\sqrt{F})\le D^{-2}$, the S/source/ray sets are fixed, the $\mathfrak w$ set has bounded size, and each interval has length at most V. The constant in this inequality is independent of J and of derivatives of G. At $J=0$ it proves (199). The infinite Bessel hierarchy was already available; the statement used here is the finite real-axis certificate. It bounds the remainder, not the signed polynomial.

S78.4 All source and ray phases give bounded multiplicative columns

Freeze $\mathfrak n,\mathfrak e,\mathfrak w,\mathfrak k$ and one source/ray component in the pure model. After (202), every $\mathfrak m$ dependence apart from $\mu(\mathfrak m)f(\mathfrak m)\overline{G_{\mathfrak m}}W(N\mathfrak m/D)/\sqrt{N\mathfrak m}$ is a product of bounded multiplicative functions, except for two fixed finite selectors. The multiplicative factors include ϑ,ν,g, fixed-numerator residue symbols at $u_0,\gamma_S,\mathfrak k,\mathfrak e,\mathfrak w$, the value $\theta_S(\alpha_{\mathfrak m})$, and the original $(\mathfrak m,\mathfrak n)=1$ deletion. They include zeros: a numerator valuation divisible by six does not restore a collision. Primary generators multiply on the required coprime products.

Let $\mathcal C$ be the fixed source group and $\mathcal H=(\mathcal O/(4))^\times$, of order 12. Finite character expansions give

$$\begin{aligned}\mathbf 1_{[\mathfrak m]=A}&=\frac{1}{|\mathcal C|}\sum_{\chi\in\widehat{\mathcal C}}\overline{\chi(A)}\chi([\mathfrak m]),\\ \Gamma(x\alpha_{\mathfrak n})&=\sum_{\chi\in\widehat{\mathcal H}}a_{\chi,\mathfrak n}\chi(x),\qquad\sum_\chi|a_{\chi,\mathfrak n}|\le\sqrt{12}.\end{aligned} \tag{236}$$

The first coefficient mass is one. The second follows by normalized Parseval and Cauchy since the function has modulus one on $\mathcal H$. These bounds are independent of $\mathfrak n,\mathfrak k,D$. Extending a fixed formula

off its source class only extends a coefficient function; the finite selector sum restores the original class. No primitive root of a wrong-class or overlapping-conductor pair is asserted.

Consequently every actual fixed-other-column expression is a finite linear combination, of fixed total coefficient mass, of the linear forms in (207). The associated λ may have growing conductor and other-column deletion, but $|\lambda| \le 1$ and multiplicativity on coprime products suffice. This does not flatten the original roots or make the coprime matrix positive. It is not a silent extension of a same-class, $\rho = 1$ formula.

S78.5 Two exact cutoff boxes expose a physical cubic matrix

To prove (203), in each term of (190) set $\mathfrak{c} = \mathfrak{m}/(\mathfrak{a}\mathfrak{b})$. Squarefreeness makes the factors pairwise coprime and $\mu(\mathfrak{m})\mu(\mathfrak{a})\mu(\mathfrak{b}) = \mu(\mathfrak{c})$. The cutoff difference is the exact disjoint assignment

$$\begin{aligned}&\mathbf{1}_{N\mathfrak{a},N\mathfrak{b}\le U} - \mathbf{1}_{N\mathfrak{a},N\mathfrak{b}\le Y}\\ &\quad = \mathbf{1}_{Y<N\mathfrak{a}\le U,\, N\mathfrak{b}\le U} + \mathbf{1}_{N\mathfrak{a}\le Y,\, Y<N\mathfrak{b}\le U}.\end{aligned} \tag{237}$$

Swap $\mathfrak{a}, \mathfrak{b}$ in the second piece and put $\mathfrak{z} = \mathfrak{b}\mathfrak{c}$ in both. On coprime primary $\mathfrak{a}, \mathfrak{z}$, Gauss CRT has the cross $\chi_{3,\mathfrak{a}}(\alpha_{\mathfrak{z}})^2$. Conjugation and cubic reciprocity give exactly (205). Thus the left side of (207), before taking its absolute value, is

$$\begin{aligned}\sum_{\substack{\mathfrak{a},\mathfrak{z}\ \text{good,SF}\\ Y<N\mathfrak{a}\le U}} &\frac{\overline{G_{\mathfrak{a}}}\lambda(\mathfrak{a})}{\sqrt{N\mathfrak{a}}}\,\frac{\overline{G_{\mathfrak{z}}}\lambda(\mathfrak{z})[T_U(\mathfrak{z}) + T_Y(\mathfrak{z})]}{\sqrt{N\mathfrak{z}}}\\ &\cdot \chi_{3,\mathfrak{a}}(\alpha_{\mathfrak{z}})W(N\mathfrak{a}N\mathfrak{z}/D).\end{aligned} \tag{238}$$

At a common prime the cubic entry is zero. Hence this encompassing SF rectangle encodes the original squarefreeness of $\mathfrak{m} = \mathfrak{a}\mathfrak{z}$, without inventing a primitive overlapping root. Also $\mathbf{1}_{(\mathfrak{a}\mathfrak{z},\mathfrak{n})=1} = \mathbf{1}_{(\mathfrak{a},\mathfrak{n})=1}\mathbf{1}_{(\mathfrak{z},\mathfrak{n})=1}$, and the other fixed-numerator collision masks split in the same way. In particular the full other-column mask survives. There is no assumption that A_U or T_L is multiplicative. The elementary estimate $|T_U(\mathfrak{z}) + T_Y(\mathfrak{z})| \le 2\tau(\mathfrak{z})$ is uniform in both cutoffs.

S78.6 Cubic Cauchy before the distinguished-divisor triangle

Use dyads $N\mathfrak{a} \asymp A$ in $(Y, U]$. Only $N\mathfrak{z} \asymp Z$, $Z \asymp D/A$, can meet the original profile. Both indices in (238) are good squarefree ideals, hence cube-free. Primary cubic reciprocity identifies the entry with the physical family of (206), or its transpose. Additional row/column masks are coefficient contractions inside that arbitrary-coefficient norm. No character indexed only by a growing conductor is substituted.

Ideal counting and the divisor estimate show, with arbitrarily small exponent losses,

$$\begin{aligned}\sum_{N\mathfrak{a}\asymp A}\left|\frac{\overline{G_{\mathfrak{a}}}\lambda(\mathfrak{a})}{\sqrt{N\mathfrak{a}}}\right|^2 &\ll 1,\\ \sum_{N\mathfrak{z}\asymp Z}\left|\frac{\overline{G_{\mathfrak{z}}}\lambda(\mathfrak{z})[T_U(\mathfrak{z}) + T_Y(\mathfrak{z})]}{\sqrt{N\mathfrak{z}}}\right|^2 &\ll_\epsilon D^\epsilon.\end{aligned} \tag{239}$$

These are coefficient bounds after the exact signed grouping, not a new bound for the norm of A_U.

The exact Mellin separation of the one common occurrence of W is

$$W(N\mathfrak{a}N\mathfrak{z}/D) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{W}(t)D^{-it}(N\mathfrak{a})^{it}(N\mathfrak{z})^{it}\,dt. \tag{240}$$

Its powers are unitary on this line, so it costs $\mathcal{M}(W)$ once. The arbitrary-coefficient cubic operator norm and (239) therefore give

$$|\text{one dyadic form}| \ll_\epsilon \mathcal{M}(W) D^\epsilon \sqrt{A + D/A + D^{2/3}}. \tag{241}$$

This is a bilinear matrix inequality, not positivity of the original coprime form. Summing the finite dyads with $A \geq Y/2$ and $A \leq U \leq \sqrt{D}$ gives

$$\mathcal{M}(W) D^\epsilon \{\sqrt{U} + \sqrt{D/Y} + D^{1/3}\} \ll \mathcal{M}(W) D^\epsilon \{\sqrt{D/Y} + D^{1/3}\}. \tag{242}$$

Dyadic logarithms and divisor losses are absorbed by distributing a prescribed ϵ among the preceding bounds. Partial dyads and real cutoff ties are literal coefficient indicators. Empty tails, including $Y = U$, are exactly zero. This proves (207).

S78.7 Full pair support and the original finite k reassembly

Fix the other column $\mathfrak{n}$. Its raw coprimality and numerator deletions are already in λ by Section S78.4. An optional condition $N\mathfrak{m} \neq N\mathfrak{n}$ is not multiplicative. The number of ideals of a fixed rational norm N is at most the integer divisor function: at a split prime an exponent e gives at most $e+1$ choices, at an inert prime at most one, and at a ramified prime one. Consequently there are $O_\epsilon(D^\epsilon)$ ideals in the forbidden norm fibre, with all orientations separate. Since $|A_U| + |A_Y| \ll_\epsilon D^\epsilon$, subtracting it costs

$$O_\epsilon(B_W D^{-1/2+\epsilon}) \quad \text{in the fixed-other-column sum.} \tag{243}$$

This is below (207); $\mathcal{M}(W) \geq B_W$ by Fourier inversion. No whole equal-norm energy is subtracted from a different frequency sector.

The untouched original column has the elementary bound

$$\sum_{\mathfrak{n} \text{ original}} \frac{|A_T(\mathfrak{n}) W(N\mathfrak{n}/D)|}{\sqrt{N\mathfrak{n}}} \ll_\epsilon B_W D^{1/2+\epsilon}, \qquad T \in \{U, Y\}. \tag{244}$$

Take this triangle only after (207). In the leading model all S, source/ray and $N\mathfrak{w} < 64$ coefficient masses are fixed; g_0 is bounded independently of capacity derivatives, and $F^{-1/2}$ supplies the two square-root norm weights already used. The normalization is once by $1/D$. For each actual fixed $\mathfrak{k}$, the column estimate is uniform in its bounded multiplicative numerator phase. Thus its *original finite* sum is bounded by $\sum (N\mathfrak{k})^{-2} < \infty$, with no $(N\mathfrak{k})^\epsilon$ or altered conductor cost. One does not need to complete that sum.

Equivalently, define $\mathcal{G}_D^K$ by $\mathcal{L}_0[f_1, f_2] = A_1 X^{1/6} \mathcal{G}_D^K[f_1, f_2]$. The preceding estimates give

$$|\mathcal{G}_D^K[A_U - A_Y, A_T]| \ll B_W \mathcal{M}(W) D^\epsilon \{D^{-1/6} + Y^{-1/2}\}. \tag{245}$$

For a tail on the second column, conjugate the scalar expression and reverse the roles. This sends $G_\mathfrak{n}$ to $\overline{G_\mathfrak{n}}$, W to $\bar{W}$ and the finite coefficients to their conjugates. The conjugate cubic matrix has the same norm, and $\mathcal{M}(\bar{W}) = \mathcal{M}(W)$. No Hermitian assertion about an individual source block is required. Finally use the exact bilinear identity

$$\begin{aligned} \mathcal{G}_D^K[A_U, A_U] - \mathcal{G}_D^K[A_Y, A_Y] = {} & \mathcal{G}_D^K[A_U - A_Y, A_U] \\ & + \mathcal{G}_D^K[A_Y, A_U - A_Y]. \end{aligned} \tag{246}$$

This proves (208) with all original finite cutoffs, primitive S zeros and pair masks.

S78.8 The critical reduction is not a bound for its residual mean

Equations (199) and (208) prove (209). When $U \geq D^{1/8}$, substituting $A_1 \asymp D^{3/4}$, $X = D^{3/8}$, $Y = D^{1/8}$ gives the exact exponent calculations

$$\begin{aligned} 2(3/4) + (7/6)(3/8) - 2 &= -1/16, \\ 3/4 + (1/6)(3/8) - (1/2)(1/8) &= 3/4, \\ 3/4 + (1/6)(3/8) - 1/6 &= 31/48. \end{aligned} \tag{247}$$

If $U < D^{1/8}$ choose $Y = U$; the coefficient difference then vanishes identically, rather than replacing $Y^{-1/2}$ by a smaller number. The small-cutoff moment remains exactly the auxiliary expression with that Y. Since both short divisors have norms at most Y, their residual in (210) has norm at least D/Y^2. All other-column and finite-frequency couplings remain in its two-column sum.

The current separate discrepancy has critical power

$$3/4 + (1/18)(3/8) = 37/48 > 3/4. \tag{248}$$

This upper certificate is not a noncancellation theorem. A bound for the small-cutoff leading model alone does not cancel or improve it. Both it and the remaining original sectors require additional signed arithmetic.

S78.9 Direct discrete frequencies require no principal-density approximation

For each original dual ideal, its good valuations determine uniquely $\mathfrak{y} = \mathfrak{w}\mathfrak{s}^6$ with good $\mathfrak{s}$ and good part of $\mathfrak{w}$ sixth-free. At $V = X^{1/6}/2$, the lower-open condition $N\mathfrak{s} > V$ implies $N\mathfrak{w} < 64$. Conversely every ideal of norm below 64 has sixth-free good part, since the least good prime norm is seven. This proves the exact support (211) with all its nonunit S modes and possible overlaps. No squarefreeness of $\mathfrak{s}$ or new coprimality with $\mathfrak{k}$ is introduced.

Equation (218) holds for *every* original ρ. Retaining that factor and the actual discrete $\mathfrak{s}$ family in (147) gives (213). The prefactor is exactly $A_1/(D\sqrt{F})$, including the once-only normalization. All primitive factors at $\mathfrak{e}, \mathfrak{w}$ are still separate, and a good collision remains zero even when its exponent is a sixth power.

The pointwise real-axis remainder (231) at $J = 0$ implies

$$|\dot{G}(x) - g_0| \leq C_0^{-3/2} x \int_0^\infty vG(v)\, dv \leq 3C_0^{-3/2} x, \qquad x \geq 0. \tag{249}$$

Every argument in (213) is at most $\Lambda = A_1 X/D^2$. Count only in this error: there are $O(D^2)$ annular column pairs, $A_1/(D\sqrt{F}) \leq A_1/D^2$, the original coefficient products are $O(B_W^2 D^\epsilon)$, and the original finite $\mathfrak{k}$ absolute budget is bounded by $\sum(N\mathfrak{k})^{-2}$. The $\mathfrak{e}$/source/ray and $\mathfrak{w}$ sets have fixed finite mass. Since $V \geq 1$, the exact good $\mathfrak{s}$ annulus contains $O(V)$ ideals. This proves (214), without a ray count, interpolated endpoint, density or arbitrary row-weight norm.

It remains to justify applying the column-tail theorem to every original ray. The good-q powers in the proof of (202) are local powers $1, 2, 3$ of θ_q; that proof never uses $\rho^6 = 1$. For every original component the total η is type zero and its finite θ_S is fixed at its actual primitive S conductor. Hence the finite selector expansion in (236) and its coefficient mass are valid on all these components. No new assertion of sixth order for an individual ξ or replacement of a primitive S Gauss sum is made.

Now freeze $\mathfrak{s}$ as well as $\mathfrak{n}, \mathfrak{k}, \mathfrak{e}, \mathfrak{w}$. Relative to Section S78.4, the only new $\mathfrak{m}$ factor is $\mathbf{1}_{(\mathfrak{m},\mathfrak{s})=1}$. It is a bounded multiplicative ideal function and splits exactly in the grouped column as

$$\mathbf{1}_{(\mathfrak{a}\mathfrak{z},\mathfrak{s})=1} = \mathbf{1}_{(\mathfrak{a},\mathfrak{s})=1} \mathbf{1}_{(\mathfrak{z},\mathfrak{s})=1}. \tag{250}$$

The $\mathfrak{n}/\mathfrak{s}$ mask is a scalar which may be zero. The fixed $\bar{\rho}(\mathfrak{s})^6$ phase is another bounded scalar. All earlier full-union raw deletions and separate primitive S values remain. Thus (207) is uniform in $\mathfrak{s}$ as well as in $\mathfrak{n}, \mathfrak{k}$, with no $(N\mathfrak{s})^{\epsilon}$ or $(N\mathfrak{k})^{\epsilon}$ cost. Their growing deleted primes are coefficient diagonals in the same physical cubic matrix.

There is no density in the new exact discrete model. Use the underlying arbitrary-λ theorem with the factor $g(\mathfrak{m})$ omitted, rather than dividing the earlier density-weighted moment by $g(q)$. If a $\mathfrak{k}$ domain has been written with q deletion, first express it on the common original good-SF set $N\mathfrak{k} \le K$ with its exact raw $\eta(\mathfrak{k})^2$ zero. This permits a fixed $\mathfrak{k}$ column argument, restores no prime and adds no $(\mathfrak{k}, \mathfrak{s})$ condition.

Apply the signed column estimate before the other column triangle. The latter is (244); after once-only $1/D$ it yields $B_W \mathcal{M}(W) D^{\epsilon}(Y^{-1/2} + D^{-1/6})$ for each fixed $\mathfrak{s}, \mathfrak{w}, \mathfrak{k}, \mathfrak{e}$ and finite component, with the original $A_1 g_0$ factor. Sum the original finite $\mathfrak{k}$ norm weights and the fixed $\mathfrak{e}, \mathfrak{w}$ components. Only now use the $O(V)$ count of the exact $\mathfrak{s}$ family. Reversed tails use conjugation, as in Section S78.7, and the original unequal-norm selector uses the same separate fixed-norm fibre bound (243). The exact identity

$$\begin{aligned} \mathcal{L}_X^K[A_U, A_U] - \mathcal{L}_X^K[A_Y, A_Y] = \mathcal{L}_X^K[A_U - A_Y, A_U] \\ + \mathcal{L}_X^K[A_Y, A_U - A_Y] \end{aligned} \tag{251}$$

then proves (215). Combining it with the one Bessel error proves (216).

This direct proof does not assert a smaller value for the separate $37/48$ certificate in the continuous mean route; it never introduces that split. All of its nonprincipal-ray and exact-discrete effects remain in $\mathcal{L}_X^K[A_Y, A_Y]$. At $Y \le D^{1/8}$ the residual column norms are at least $D/Y^2 \ge D^{3/4}$, just as in (210). The new model is still unbounded at target size. It is different from the principal-density $\mathcal{L}_0[A_Y, A_Y]$.

A capacity-empty integer interval, or empty exact $\mathcal{T}_X$, gives zero by definition. Absence of actual good squarefree rows in a *nonempty* integer interval alone does not imply that each dual prefix piece is zero; cancellation with other frequencies can be required. No full dual complement, other block, whole signed assembly or determinant/zero-location obligation is resolved by this direct small-cutoff reduction.

Why the completed shortcuts do not supply that arithmetic. The inverse-different cubic Gauss frequency is incomplete: expanding it leaves the actual weighted primary $\mathfrak{k}$ set, not a complete residue system modulo $\mathfrak{m}\mathfrak{n}$. The bounded number of $\mathfrak{w}$ modes supplies no such completion. Likewise, expanding both short divisors and taking their absolute triangles before the cubic cross does not give the proved estimate. Even an explicitly hypothetical uniform inner exponent $\alpha < 1$ would produce the diagnostic

$$D^{\alpha - 1/2} \left(\sum_{N\mathfrak{a} \le U} (N\mathfrak{a})^{-\alpha} \right)^2 \ll_{\alpha} D^{\alpha - 1/2} U^{2(1-\alpha)}. \tag{252}$$

At $U \asymp \sqrt{D}$ this is $D^{1/2}$. It is a conditional failure of that method, not an available smooth-coefficient Gauss theorem or an impossibility result. The actual proof above retains the physical cubic cross until its joint matrix norm. It bounds one specified signed tail and leaves the long residual-Möbius, finite-$\mathfrak{k}$ small-short moment as the next arithmetic problem. No whole signed, global unsigned, determinant, zero-location, analytic Lean or GRH conclusion is implied.

S79 Ramified theta transforms and the original pure dual prefix

The exact discrete prefix introduced in Section S77 can be estimated with its original coefficient A_U. The argument constructs the previously missing finite-character transform at the ramified prime and groups all original short and Heath–Brown factors before taking a short-factor triangle. The resulting power is 101/144 for this specified pure finite dual prefix. The whole signed comparison 13/12, the global unsigned comparison 9/8, and the remaining infinite-frequency assembly retain their earlier status.

The retained principal-density mean in the earlier discussion is a different expression with its own comparison problem. The result here uses every original ray component and the actual discrete sixth-part set. It supplies a new proof for the actual finite discrete prefix rather than modifying the historical cutoff stages.

S79.1 The normalized object and its original support

Put $\omega = e^{2\pi i/3}$, $K = \mathbb{Q}(\omega)$, $\mathcal{O} = \mathbb{Z}[\omega]$ and $\lambda = 1 + 2\omega = i\sqrt{3}$. A good ideal avoids the original fixed set S, containing the primes above $2, 3$. All ideal columns are good and squarefree; different ideals of equal norm remain different columns. Use the unique multiplicative primary generator $\alpha_{\mathfrak{a}} \equiv 1 \pmod 3$. The original character ν has fixed finite order, with its conductor exclusions in S; no order-dividing-six hypothesis is imposed on ν.

Assume

$$D \geq 2^{64}, \quad 1 \leq U \leq \sqrt{D}, \quad A_1 \asymp H = D^{3/4}, \quad X = D^{3/8}, \quad V = X^{1/6}/2. \tag{253}$$

Only the original unweighted pure two-pivot block is asserted: $A_2 = 1$, its second good factor and the other four good factors are units, and u_0 is the fixed literal unit/S-supported factor with no good prime support. Keep its original capacity endpoints, nodal function G, and pre-S cutoff $K_{\mathrm{cut}} = \sqrt{b}$. An empty capacity interval gives zero. None of these data is recomputed during a coefficient decomposition.

Let W be one fixed possibly complex smooth function supported on $[1, 2]$, with $W(1) = W(2) = 0$, and write

$$\begin{aligned} B_W &= \|W\|_\infty, \qquad \widehat{W}(t) = \int_0^\infty W(x) x^{-it} \frac{dx}{x}, \\ \mathcal{M}(W) &= \frac{1}{2\pi} \int_{\mathbb{R}} |\widehat{W}(t)|\, dt, \qquad\qquad V_1(W) = B_W + \int_1^2 |W'(x)|\, dx. \end{aligned} \tag{254}$$

These are seminorms of the common column profile. No uniform derivative hypothesis on the sharp capacity interpolant is added. The exact frequency set is

$$\mathcal{T}_X = \left\{ (\mathfrak{w}, \mathfrak{s}) : \begin{array}{l} \mathfrak{w} \text{ integral, } N\mathfrak{w} < 64, \quad \mathfrak{s} \text{ good integral,} \\ V < N\mathfrak{s} \leq 2V, \quad N\mathfrak{w}(N\mathfrak{s})^6 \leq X \end{array} \right\}. \tag{255}$$

There is no squarefreeness requirement on $\mathfrak{s}$. The permitted overlaps among $\mathfrak{k}, \mathfrak{s}, \mathfrak{w}$ remain permitted, including nonunit S-supported $\mathfrak{w}$.

Here is a full definition of the piece being bounded. Let $\Omega = (A, B, \rho, \mathfrak{m}, \mathfrak{n})$ range over the original finite source/ray components and their ordered coprime good squarefree columns in $[D, 2D]$, with the original source classes. An original unequal-norm projection may be imposed consistently. Use the actual primitive character η_Ω of (141), its actual S-conductor $\mathfrak{f}_S$, and

$$F_\Omega = N\mathfrak{f}_S\, N\mathfrak{m}\, N\mathfrak{n}, \qquad \mathfrak{q} = \mathfrak{m}\mathfrak{n}, \qquad c_U(\mathfrak{m}) = A_U(\mathfrak{m}) \nu(\mathfrak{m}) R_{\mathfrak{m}}(u_0) W(N\mathfrak{m}/D).$$

The numerator value $R_{\mathfrak{m}}(u_0)$ includes its original raw zeros. Put $P_S = \prod_{\mathfrak{p}\in S}\mathfrak{p}$. With $C_0 = 3/(4\pi^2)$, the normalized prefix is

$$\begin{aligned}\mathcal{P}_X[A_U, A_U] = \frac{A_1}{D}\sum_{\Omega} h_{AB\rho}(\mathcal{O})c_U(\mathfrak{m})\overline{c_U(\mathfrak{n})}\frac{\varepsilon(\eta_\Omega)}{\sqrt{F_\Omega}} \\ \cdot \sum_{\mathfrak{e}\mid P_S}\frac{\mu(\mathfrak{e})\eta_\Omega(\mathfrak{e})}{N\mathfrak{e}}\sum_{\substack{\mathfrak{k}\ \text{good,SF}\\ N\mathfrak{k}\le K_{\mathrm{cut}}}}\frac{\mu(\mathfrak{k})\eta_\Omega(\mathfrak{k})^2}{(N\mathfrak{k})^2} \\ \cdot \sum_{(\mathfrak{w},\mathfrak{s})\in\mathcal{T}_X}\overline{\eta_\Omega(\mathfrak{w})}\,\overline{\rho(\mathfrak{s})}^{\,6}\mathbf{1}_{(\mathfrak{s},\mathfrak{q})=1}\dot{G}\left(\frac{A_1 N\mathfrak{w}(N\mathfrak{s})^6}{(N\mathfrak{k})^2 N\mathfrak{e}F_\Omega}\right).\end{aligned} \tag{256}$$

All sums are finite. The ray label ρ here is distinguished from the Jacobi column weight below. Values at $\mathfrak{e}$ and $\mathfrak{w}$ are separate primitive values, including zero; a conductor drop does not restore a deleted prime. The original $1/D$ occurs once. The raw numerator is $D\mathcal{P}_X$.

Define $\mathcal{L}_X^{K_{\mathrm{cut}}}$ by replacing $\dot{G}$ in (256) by $g_0 = C_0^{-1/2}\int_0^\infty G(v)\,dv$, with every other datum fixed. For the original coefficients the established real-axis estimate is

$$|\mathcal{P}_X - \mathcal{L}_X^{K_{\mathrm{cut}}}| \ll B_W^2(DX)^{\epsilon}A_1V\Lambda, \qquad \Lambda = A_1X/D^2. \tag{257}$$

It uses $0 \le G \le 1$, its fixed compact support and a pointwise Bessel remainder. The value g_0 may vanish and is never divided out.

S79.2 Actual finite characters and a sharp ramified row

The actual root calculation leaves one angular Jacobi gauge and a cubic Gauss root in each varying column. On good squarefree ideals the Jacobi function is $\rho_J = \mu\vartheta$, where $\vartheta = \Psi/\sqrt{N}$ has infinity type $(1,0)$; here $\Psi(\mathfrak{a})$ is the chosen generator 1 mod 2λ. The arithmetic μ belongs in the coefficient opening. There is exactly one varying ϑ, or its conjugate; an additional finite-order assertion about ρ_J would be false.

Write $\chi_{3,\mathfrak{a}}$ for the positive cubic residue character, with its zero at every shared prime. The source Gauss root is

$$G_{\mathfrak{a}}^{\mathrm{src}} = \frac{1}{\sqrt{N\mathfrak{a}}}\sum_{x \bmod \alpha_{\mathfrak{a}}}\chi_{3,\mathfrak{a}}(x)e^{2\pi i\,\mathrm{Tr}(x/\alpha_{\mathfrak{a}})}. \tag{258}$$

The actual root uses $\lambda\alpha_{\mathfrak{a}}$ in this additive denominator. The normalization of the angular factor is explicit:

$$\begin{aligned}\Theta_\ell(\mathfrak{a}) &= (\alpha_{\mathfrak{a}}/|\alpha_{\mathfrak{a}}|)^\ell, \\ \vartheta(\mathfrak{a}) &= \Theta_1(\mathfrak{a})\overline{\chi_{3,\mathfrak{a}}(2)}, \\ G_{\mathfrak{a}}^{\mathrm{act}} &= \chi_{3,\mathfrak{a}}(\lambda)G_{\mathfrak{a}}^{\mathrm{src}}.\end{aligned} \tag{259}$$

Thus the two fixed corrections may be written as source frequency $12t$. The factor 12 belongs to the fixed finite character data; $12t$ is not declared a good squarefree moving level.

After exact product-character expansion of the joint finite S/source/Γ selectors, every remaining own-variable phase has the form $\Theta_{\pm1}\kappa$, where κ belongs to a fixed finite residue family, with its original modulus-zero extension. It includes higher 3-primary conductors of ν. The finite source quotient is a fixed ray quotient; fixed numerator symbols are finite Kummer characters. The full Fourier sums retain their actual coefficient masses and reconstruct the primitive $\mathfrak{e}/\mathfrak{w}$ zeros. Growing other-column deletions remain literal masks.

For a physical good squarefree frequency $\mathfrak{t}$, an exact interval $I \subset [T, 2T]$, and a finite good deletion ideal Q, write the actual row in either required conjugate orientation as

$$\mathcal{F}_\psi(T; \mathfrak{t}; Q, I) = \sum_{\substack{\mathfrak{a} \text{ good,SF} \\ N\mathfrak{a} \in I,\ (\mathfrak{a}, Q)=1}} \psi(\mathfrak{a}) \overline{G_{\mathfrak{a}}^{\mathrm{act}}} \chi_{3,\mathfrak{a}}(\alpha_{\mathfrak{t}}) \frac{W(N\mathfrak{a}/T)}{\sqrt{N\mathfrak{a}}}. \tag{260}$$

Here ψ is from this actual fixed family, including its angular factor. It is not an arbitrary bounded multiplicative function.

Theorem S79.1 (The actual fixed-family sharp row)**.** *For every such member, both conjugate orientations and every physical good squarefree* $\mathfrak{t}$*, including the unit,*

$$|\mathcal{F}_\psi(T; \mathfrak{t}; 1, I)| \ll_{\mathrm{fixed},\epsilon} V_1(W)(2 + TN\mathfrak{t})^\epsilon \min\{\sqrt{T},\ T^{1/4}(N\mathfrak{t})^{3/8}\}, \qquad T \geq 1. \tag{261}$$

The constant may depend on the fixed conductor/character family. A literal unit calculation extends the estimate to $1/2 \leq T < 1$*; for* $T \leq 1/2$ *the row is zero with the stated endpoint convention.*

Proof. Use the unconditional theta modularity and known 27 integral cusps in Dunn–Radziwiłł, arXiv:2109.07463v3, Section 5 and Appendix A. Its weight-zero cubic theta θ is automorphic with Kubota multiplier χ on $\Gamma_2 = \mathrm{SL}_2(\mathbb{Z})\Gamma_1(3)$, where $\Gamma_1(3)$ is the principal congruence subgroup of $\mathrm{SL}_2(\mathcal{O})$. Fix its known representatives γ_j and $F_j = \theta \circ \gamma_j$. The correct ramified coefficient projector acts on $\lambda^3\mu$. Its primary selector is $\mathbf{1}_{\lambda^3\mu \equiv 1 \bmod 3}$. A finite Fourier expansion writes it, and the away-3 periodic twist, as actual rational translates of theta.

For $a = A/C$ reduced, choose $\gamma = \left(\begin{smallmatrix} A & B \\ C & D_0 \end{smallmatrix}\right)$ of determinant one, and choose j so that $\delta = \gamma\gamma_j^{-1} \in \Gamma_2$. Use $U(x) = \left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right)$ and $W_C = \left(\begin{smallmatrix} 0 & -1/C \\ C & 0 \end{smallmatrix}\right)$. Then

$$\gamma^{-1}U(a) = U(-D_0/C)W_{-C}, \qquad \theta(U(a)w) = \chi(\delta)F_j(U(-D_0/C)W_{-C}w).$$

The multiplier is its literal Kubota value. Pure nonzero angular derivatives kill the pulled-back radial constants. Their Mellin identity has factor $(NC)^{1-2s}$, including the Jacobian. The explicit cusp coefficients have absolute Dirichlet norm at most

$$3^8 \zeta_K^{(\lambda)}(\sigma)\zeta_K^{(\lambda)}(3\sigma - \tfrac{1}{2}), \qquad 1 < \sigma < 13/12.$$

Normalized finite Fourier mass is $O_{\mathrm{fixed}}(\sqrt{N\mathfrak{t}})$ and each reduced denominator has norm $O_{\mathrm{fixed}}(N\mathfrak{t})$. This gives the conservative dual cost $(N\mathfrak{t})^{3/2+\epsilon}$.

Coefficientwise cube completion retains the raw $L(3s - \frac{1}{2}, \kappa^3\Theta_{3\ell})$, with every stipulated Euler omission. Apply convexity to this entire completed numerator first. Its reciprocal is used only where $\Re(3s - \frac{1}{2}) > 1$, by absolutely convergent Euler products. Exact half-integer Perron thresholds and partial summation give (261). The optimized height is $(T/(N\mathfrak{t})^{3/2})^{1/4}$ when this is at least one; the elementary count gives the other branch. The extended edition retains the constants and the zeroth-angular residues explicitly. □

S79.3 The full original coefficient and its finite prefix

On the actual contributing annulus $N\mathfrak{m} > U$, put

$$B_U(\mathfrak{m}) := \mu(\mathfrak{m})A_U(\mathfrak{m}) = \sum_{\substack{\mathfrak{abc}=\mathfrak{m} \\ N\mathfrak{a}, N\mathfrak{b} \leq U}} \mu(\mathfrak{c}). \tag{262}$$

This B_U is not the project's original C_U. Equation (262) is not asserted for arbitrary small columns or the unit. Unit factors inside an annular factorization remain present.

Theorem S79.2 (Original pure finite dual-prefix bound)**.** *Under* (253)*, with the complete original support and fixed data of* (256)*,*

$$|\mathcal{P}_X[A_U, A_U]| \ll_{\mathrm{fixed},\epsilon} (DXA_1)^{\epsilon} A_1 V\{B_W[\mathcal{M}(W) + V_1(W)]D^{-1/9} + B_W^2[\Lambda + D^{-1}]\}. \tag{263}$$

In particular its normalized leading power is

$$13/16 - 1/9 = 101/144 < 3/4, \qquad 3/4 - 101/144 = 7/144. \tag{264}$$

The raw numerator has one additional factor D.

Proof. Open (262) and the exact $J = 5$ finite Heath–Brown identity at $z_0 = (2D)^{1/5}$. There are at most eleven factors, including the original $\mathfrak{a}, \mathfrak{b}$. Choose their grouping from fixed independent dyadic scales. A balanced whole-factor partition has a physical cubic matrix bound $\mathcal{M}(W)D^{1/3+\epsilon}$, using the pinned de Faveri, arXiv:2610.04045v1, Theorem 1.1 and Example 2.2, at order three.

An unbalanced box has one long unrestricted coefficient-one factor $\mathfrak{v}$. At $D \geq 2^{64}$ neither an original short factor nor a short Möbius factor can occupy that scale. The product $\mathfrak{p}$ of all other factors has norm $O(D^{1/3})$ and its intrinsic signed assignment coefficient is divisor-bounded. This retains the $j = 1$ boxes, which are all balanced.

For $N\mathfrak{p} > Z$ the cubic matrix gives $\mathcal{M}(W)D^{\epsilon}[D^{1/3} + \sqrt{D/Z}]$. For $N\mathfrak{p} \leq Z$, freeze the original opposite column $\mathfrak{n}$ first. Keep $\mathbf{1}_{(\mathfrak{p},\mathfrak{n})=1}$ and all prefix $\mathfrak{k}/\mathfrak{s}/\mathfrak{w}$ zeros in its actual coefficient. The remaining long-variable deletion is $Q = \mathrm{rad}_{\mathrm{good}}(\mathfrak{p}\mathfrak{n}\mathfrak{k}\mathfrak{s}\mathfrak{w})$. Its exact divisor $\mathfrak{h}$ changes frequency to $\mathfrak{p}\mathfrak{k}\mathfrak{h}$ and length to $D/(N\mathfrak{p}N\mathfrak{h})$. Including the original prefix weight, Theorem S79.1 gives

$$\frac{1}{\sqrt{N\mathfrak{p}N\mathfrak{h}}}\left(\frac{D}{N\mathfrak{p}N\mathfrak{h}}\right)^{1/4}(N(\mathfrak{p}\mathfrak{k}\mathfrak{h}))^{3/8} = D^{1/4}(N\mathfrak{k})^{3/8}(N\mathfrak{p})^{-3/8}(N\mathfrak{h})^{-3/8}. \tag{265}$$

Thus the small-prefix column costs $V_1(W)D^{\epsilon}D^{1/4}(N\mathfrak{k})^{3/8+\epsilon}Z^{5/8}$. Choose $Z = D^{2/9}$; both column costs have power $7/18$.

Only after that bound take the opposite-column elementary $B_W D^{1/2+\epsilon}$ count. Use the original A_1/D once, sum the original finite $\mathfrak{k}$ with the convergent positive majorant $\sum(N\mathfrak{k})^{-13/8+\epsilon}$, and count the exact $\mathfrak{s}$ family by $O(V)$. Optional unequal norms cost the same fixed-norm fibre bound $O(B_W^2 A_1 V D^{-1+\epsilon})$. Finally (257) applies directly to A_U, A_U. No A_U-to-A_Y replacement or common-long-divisor opening is used. □

The estimate has fixed-conductor and explicit profile-seminorm dependence. It does not supply a supremum-norm estimate for varying W_D, an arbitrary weighted-row operator, the infinite dual complement, mixed or neighboring blocks, short/principal assembly, an independent positive determinant, or a zero-location theorem. The Dirichlet, Hecke, Dedekind, elliptic and general Artin obligations remain separate. No analytic/category Lean formalization or GRH proof is claimed.

S80 Proofs of the ramified transform and full-coefficient prefix bound

The external theta input is the unconditional Section 5 and Appendix A of Dunn–Radziwiłł, *Bias in cubic Gauss sums: Patterson's conjecture*, arXiv:2109.07463v3, 14 May 2024. Its GRH-conditional prime theorems are not used. The matrix input is the already admitted physical ideal family in de Faveri, *Optimal large sieve for fixed order characters*, arXiv:2610.04045v1, Theorem 1.1 and Example 2.2, at order three. The theta modularity and deep cubic norm remain external theorems. The constructions and estimates below specify how they apply to the actual finite-character family.

S80.1 Fixed residue data, angular order and raw evaluations

The full varying-column phase list, after the arithmetic μA_U opening, consists of one ϑ or its conjugate, the cubic Gauss root, the original finite-order ν, finite source/ray factors, the mod-4 Fourier factors of Γ, the actual primitive S-type factors, and residue symbols at the fixed numerators $u_0, \gamma_S, \mathfrak{e}, \mathfrak{w}$. The set $N\mathfrak{w} < 64$, the S-types and the $\mathfrak{e}$ set are fixed finite sets. The growing $\mathfrak{k}$ numerator belongs to the cubic frequency instead of this list.

There is a common fixed residue modulus for this list. A finite-order Hecke character has infinity type zero and factors through its actual conductor. The source group is explicitly a ray quotient: Section 2.3.2 of de Faveri defines $\mathcal{C} = H_{\mathfrak{c}} \otimes \mathbb{Z}/6\mathbb{Z}$ for one sufficiently deep S-supported ray modulus $\mathfrak{c}$. The chosen primary generators realize these ray characters as finite residue characters. The Γ factors are characters of $(\mathcal{O}/(4))^{\times}$.

The local sixth-power tests also have a fixed finite modulus. For a prime $\mathfrak{p}$, Hensel's criterion at 1 gives $1 + \mathfrak{p}^e \subset (K_{\mathfrak{p}}^{\times})^6$ when $e > 2v_{\mathfrak{p}}(6)$. One may use $e_{(2)} = 3$ and $e_{(\lambda)} = 5$. These are moduli for testing local classes, not replacements for primitive character evaluation moduli. For a fixed nonzero integral numerator ζ, its n-th residue function, $n \mid 6$, is a finite Kummer character for $K(\zeta^{1/n})/K$. A safe sextic conductor bound is supplied by the polynomial discriminant ideal $(6)^6(\zeta)^5$. A field/order discriminant divides this separable polynomial discriminant, including when a proper factor supplies the field. A fractional numerator can be cleared by a fixed sixth power, retaining its original numerator and denominator exclusions. This uses finite one-dimensional Kummer characters, without a general Artin-holomorphy hypothesis.

A joint finite S-type or primitive-value selector is not assumed multiplicative. If $\mathcal{H}$ is its fixed finite residue group, write its actual function as

$$\begin{aligned} f(x,y) &= \sum_{\chi,\psi \in \widehat{\mathcal{H}}} b_{\chi,\psi}\chi(x)\psi(y), \\ b_{\chi,\psi} &= |\mathcal{H}|^{-2} \sum_{x,y \in \mathcal{H}} f(x,y)\overline{\chi(x)\psi(y)}, \\ \sum_{\chi,\psi} |b_{\chi,\psi}| &\le |\mathcal{H}|\,\|f\|_{\infty}. \end{aligned} \tag{266}$$

Normalized Parseval and Cauchy prove the last line. This expansion precedes freezing a primitive type or separating column factors. Forbidden primitive $\mathfrak{e}/\mathfrak{w}$ values are zero in f; individual Fourier terms can be nonzero there, and only their full sum reconstructs the zero. No term is individually renamed an original nonzero row.

Thus one may use a common fixed raw modulus F_{raw} and functions

$$\kappa_{\mathrm{raw}}(\mathfrak{a}) = \kappa_{\mathrm{prim}}(\alpha_{\mathfrak{a}})\mathbf{1}_{(\mathfrak{a},F_{\mathrm{raw}})=1}. \tag{267}$$

They are imprimitive residue characters on that modulus. All stipulated prime zeros survive a primitive conductor drop.

The Jacobi function $\rho_J = \mu\vartheta$ is not one of these finite characters. Its μ has already entered the arithmetic coefficient B_U, and its unique remaining angular factor is ϑ, with order $+1$ or -1 in the two conjugate orientations. Common or prefix Jacobi factors stay in their actual outer coefficients when those variables are frozen. An arbitrary bounded multiplicative function, a growing opposite column, a Möbius coefficient or the nonperiodic density $\prod_{\mathfrak{p}\mid\mathfrak{a}}(1-(N\mathfrak{p})^{-1})$ is not made into a member of this fixed family. That density does not occur in $\mathcal{P}_X$.

S80.2 Known integral cusps and an explicit absolute coefficient norm

Let $\Gamma = \mathrm{SL}_2(\mathcal{O})$, $\Gamma_1(3) = \ker(\Gamma \to \mathrm{SL}_2(\mathcal{O}/(3)))$ and $\Gamma_2 = \mathrm{SL}_2(\mathbb{Z})\Gamma_1(3)$. The source theta function is a weight-zero function on hyperbolic three-space with $\theta(\delta w) = \chi(\delta)\theta(w)$ for $\delta \in \Gamma_2$. The multiplier χ is the source Kubota character. The quotient $\Gamma_2\backslash\Gamma$ has 27 cosets. One can see the finite count from $\mathcal{O}/(3) \simeq \mathbb{F}_3[\varepsilon]/(\varepsilon^2)$: the reduction kernel in SL_2 has 3^3 elements, and $\mathrm{SL}_2(\mathbb{F}_3)$ has 24.

Fix the published integral representatives γ_j and put $F_j(w) = \theta(\gamma_j w)$. For the exact chosen function define its actual Fourier coefficient $a_j(\mu)$ by

$$F_j(z,v) = \sigma_j v^{2/3} + \sum_{\mu\neq 0} a_j(\mu) v K_{1/3}(4\pi|\mu|v) e(\mathrm{Tr}(\mu z)), \qquad e(x) = e^{2\pi i x}. \tag{268}$$

For those representatives $\sigma_j = 3^{5/2}/2$ for $j \le 9$ and zero for $j \ge 10$. The support lies in $\lambda^{-4}\mathcal{O}$. The symbol a_j always means this actual coefficient; a source table using $d_j(\mu)$, $d_j(-\mu)$ or a conjugated function is converted before using an exact phase formula.

The source formulas (5.7), (5.13), (5.14) and Appendix A express these coefficients through τ, τ_1, τ_2, unit rotations, fixed unit factors and additive phases. The first nine families have the τ absolute norm and the other eighteen have the τ_1, τ_2 norms. Rotations, conjugation and index reversal preserve absolute norm.

Proposition S80.1 (Known-cusp absolute norm)**.** *For $1 < \sigma < 13/12$, every one of these actual cusp families satisfies*

$$\mathcal{A}_j(\sigma) := \sum_{\mu\neq 0} \frac{|a_j(\mu)|}{(N\mu)^\sigma} \le 3^8 \zeta_K^{(\lambda)}(\sigma)\zeta_K^{(\lambda)}(3\sigma - \tfrac{1}{2}). \tag{269}$$

The right side is independent of a moving good level or a translation. Here $\zeta_K^{(\lambda)}$ omits the prime above 3.

Proof. In the τ formula write $c, d \equiv 1 \pmod 3$, with c squarefree and d arbitrary, possibly sharing primes with c. For the first family $\mu = \pm\epsilon\lambda^{3n-4}cd^3$, $n \ge 1$, there are at most six sign/unit branches and

$$|\tau(\mu)| \le 3^{n/2+2}\sqrt{Nd}.$$

For the second family $\mu = \pm\lambda^{3n-3}cd^3$, $n \ge 0$, there are at most two branches and $|\tau(\mu)| \le 3^{n/2+5/2}\sqrt{Nd}$. These follow from the literal unnormalized primitive Gauss magnitude $|g(a,c)| = \sqrt{Nc}$ and the source factor $|d/c|$; the fixed Gauss numerators are units modulo good c. Retain the separate c and cube variables. A generic $(N\mu)^{1/6}$ coefficient bound would lose convergence near 1.

Put $z = 3^{1/2-3\sigma}$ and $Z_{\mathrm{SF}}(\sigma) = \sum_{c\ \text{primary,SF}} (Nc)^{-\sigma}$, with the prime λ excluded. Then

$$\mathcal{A}_\tau(\sigma) \le \left\{ \frac{6\,3^{2+4\sigma}z + 2\,3^{5/2+3\sigma}}{1-z} \right\} Z_{\mathrm{SF}}(\sigma)\zeta_K^{(\lambda)}(3\sigma - \tfrac{1}{2}). \tag{270}$$

The τ_1, τ_2 families each have three unit branches at $\lambda^{-4}cd^3$, with magnitude at most $9\sqrt{Nd}$; their possible opposite signs do not change the norm. Consequently

$$\mathcal{A}_{\tau_i}(\sigma) \le 27\,3^{4\sigma} Z_{\mathrm{SF}}(\sigma)\zeta_K^{(\lambda)}(3\sigma - \tfrac{1}{2}), \qquad i = 1, 2. \tag{271}$$

In the stated range $z \le 1/9$, $3^{2+4\sigma} \le 3^7$, $3^{5/2+3\sigma} \le 3^6$, and $27\,3^{4\sigma} \le 3^8$. The brace in (270) is at most $3^8/2$. Finally $Z_{\mathrm{SF}}(\sigma) \le \zeta_K^{(\lambda)}(\sigma)$. The actual coefficient convention and all unit phases therefore give (269). □

For later comparison, an affine state $F_j(Az+B,|A|v)$, $A\neq 0$, has index and coefficient

$$\nu = A\mu, \qquad a_{j,A,B}(\nu) = |A|\, a_j(\mu) e(\mathrm{Tr}(\mu B)). \tag{272}$$

Its absolute Dirichlet norm is exactly $|A|^{1-2\sigma}\mathcal{A}_j(\sigma)$. For the ramified diagonal $A=\lambda^{-2B_0}$, this factor is $3^{B_0(2\sigma-1)}$. Commuting a moving translation past that diagonal changes $U(-\lambda^4 c/r)$ to $U(-\lambda^{4+2B_0}c/r)$. Together with $\nu=\lambda^{-2B_0}\mu$ its phase is still $e(-\mathrm{Tr}(\lambda^4\mu c/r))$. This explains why an omitted ramified scale changes actual phases, even though the eventual absolute bound is fixed-data controlled.

S80.3 Exact rational reduction and the actual Kubota multiplier

For $a=A/C\in K$ reduced, choose $B_0,D_0\in\mathcal{O}$ with $AD_0-B_0C=1$ and put $\gamma_a=\left(\begin{smallmatrix}A & B_0\\ C & D_0\end{smallmatrix}\right)$. The Euclidean property of $\mathcal{O}$ supplies this choice. Let

$$U(x)=\begin{pmatrix}1 & x\\ 0 & 1\end{pmatrix}, \qquad W_C=\begin{pmatrix}0 & -1/C\\ C & 0\end{pmatrix}.$$

Integral a is included by $C=1$. Direct multiplication proves

$$\gamma_a^{-1}U(a)=\begin{pmatrix}D_0 & 1/C\\ -C & 0\end{pmatrix}=U(-D_0/C)W_{-C}. \tag{273}$$

Choose j with $\delta_a=\gamma_a\gamma_j^{-1}\in\Gamma_2$. Then the exact function identity is

$$\theta(U(a)w)=\chi(\delta_a)F_j(U(-D_0/C)W_{-C}w). \tag{274}$$

It uses known integral cusp functions rather than an unconstructed new cusp coefficient family.

The phase is computable, not an unspecified unit set to one. Find a rational integral $\sigma\in\mathrm{SL}_2(\mathbb{Z})$ with the same reduction modulo 3 as δ_a. Then $\delta'=\sigma^{-1}\delta_a\in\Gamma_1(3)$. If $\delta'=\left(\begin{smallmatrix}a' & b'\\ c' & d'\end{smallmatrix}\right)$, the source rule is

$$\chi(\delta_a)=\chi(\delta')=\begin{cases}(c'/a')_3, & c'\neq 0,\\ 1, & c'=0.\end{cases} \tag{275}$$

Here $a'\equiv 1 \pmod 3$ is nonzero and coprime to c'. The index j can be found by reducing modulo 3 and testing the 24-element rational subgroup against the 27 fixed cosets. Equivalently, a finite traversal of elementary generators constructs representatives. A representative differing from the published one changes its function by the fixed multiplier just computed. Its Fourier coefficients and constants carry that multiplier too.

One can also display the fixed ramified states explicitly. If a moving good level r survives the fixed denominator reduction, write $C=c_0r$, with c_0 in a fixed finite set and $(c_0,r)=1$. The matrix identity

$$U(A/C)W_r=\gamma_a U(-D_0/C)\,\mathrm{diag}(r/C,C/r)$$

and $D_0=br+9c_0d$ give the fixed state $F_j|U(-b/c_0)\,\mathrm{diag}(c_0^{-1},c_0)$ followed by the moving translation $U(-9c_0^2d/r)$. The general formula (274) also handles cancelled or nonunit fixed Fourier residues directly; this special display does not impose a primitive Fourier-support condition on them.

S80.4 Radial cusp constants, angular derivatives and the Mellin Jacobian

Write $w = (z, v)$. The action in (274) is

$$z' = -D_0/C - \frac{\bar{z}}{C^2(|z|^2 + v^2)}, \qquad v' = \frac{v}{NC(|z|^2 + v^2)}. \tag{276}$$

The pulled-back constant of F_j is therefore

$$\chi(\delta_a)\sigma_j(NC)^{-2/3}v^{2/3}(|z|^2 + v^2)^{-2/3}. \tag{277}$$

It is radial in z. Every nonzero pure Wirtinger derivative ∂_z^m or $\partial_{\bar{z}}^m$ at $z = 0$ kills it. At large v the original translated theta constant is also independent of z. The known nonconstant lattice support gives exponential decay after angular differentiation at both ends, with the reciprocal-end rate depending on NC.

For completeness, the zeroth-angular constants have not disappeared. For a finite combination $\sum_a c(a)\theta(z + a, v)$ put

$$C_\infty = \frac{3^{5/2}}{2}\sum_a c(a), \qquad C_0^* = \sum_a c(a)\chi(\delta_a)\sigma_{j(a)}(NC_a)^{-2/3}. \tag{278}$$

Its regularized Mellin transform with weight v^{2s-2} has residues

$$\operatorname*{Res}_{s=1/6} \Lambda(s, 0) = -C_\infty/2, \qquad \operatorname*{Res}_{s=5/6} \Lambda(s, 0) = C_0^*/2. \tag{279}$$

These signs follow by integrating the powers at infinity and zero. A projector with zero sum kills C_∞, but need not kill C_0^*. No finite noncubic label implies such a cancellation.

For $\ell > 0$ set $\mathcal{D}_\ell = (2\pi i)^{-\ell}\partial_z^\ell$; for $\ell < 0$ use $\mathcal{D}_\ell = (2\pi i)^{-|\ell|}\partial_{\bar{z}}^{|\ell|}$. The angular Mellin transform is

$$\Lambda_a(s, \ell) = \int_0^\infty \mathcal{D}_\ell[\theta(z + a, v)]_{z=0} v^{2s+|\ell|-2}\, dv. \tag{280}$$

For $\ell \neq 0$ it is entire by the endpoint bounds just proved.

Proposition S80.2 (Exact angular rational-cusp identity)**.** *For every reduced rational translate and $\ell \neq 0$,*

$$\Lambda_a(s, \ell) = \chi(\delta_a)(-1)^{|\ell|}(C/\bar{C})^\ell(NC)^{1-2s}\Lambda_{j,-D_0/C}(1 - s, -\ell). \tag{281}$$

The right Mellin transform uses the actual known function $F_j(z - D_0/C, v)$.

Proof. For $\ell > 0$, set $\bar{z} = 0$ in the full-coordinate chain rule. Then z' and v' in (276) are constant as functions of z, while $\bar{z}'$ is affine with derivative $-1/(\bar{C}^2v^2)$. Thus the pure derivative has exactly the factor $\chi(\delta_a)(-1)^\ell\bar{C}^{-2\ell}v^{-2\ell}$ and the opposite angular derivative of F_j at $(-D_0/C, 1/(NCv))$. There are no lower angular or radial terms. The negative sign case is the conjugate-coordinate calculation.

In the integral substitute $y = 1/(NCv)$. Its Jacobian and power weight give $(NC)^{-2s+\ell+1}y^{\ell-2s}\, dy$. Combining the factor $\bar{C}^{-2\ell}$ produces precisely $(C/\bar{C})^\ell(NC)^{1-2s}$. The exponent $1 - 2s$, rather than $-2s$, is essential. This proves (281) for both signs. □

If the real part of the dual Dirichlet variable exceeds one, its series is

$$\sum_{\mu\neq 0} a_j(\mu)e(-\operatorname{Tr}(\mu D_0/C))(\mu/|\mu|)^{-\ell}(N\mu)^{-(1-s)}. \tag{282}$$

The additive and angular phases remain literal; their absolute values are one only after this exact expression has been established. Proposition S80.1 controls the series.

S80.5 The integral-index primary projector and raw cube completion

Let κ be any one fixed residue member on the primary section. Use a fixed modulus containing its actual conductor and all original fixed exclusions. Separate its λ-part κ_λ and away-λ part κ_0. Choose $f \geq 2$ containing the λ-conductor, and put

$$\Phi(x) = \kappa_\lambda(x)\mathbf{1}_{x\equiv 1 \bmod \lambda^2} \quad (x \bmod \lambda^f), \qquad \Phi(x) = 0 \text{ on nonunits.} \tag{283}$$

The entire theta support has $\lambda^3\mu \in \mathcal{O}$. The correct argument is this integral index, not $\lambda^4\mu$ for a ramified character.

From the two τ families above, $\Phi(\lambda^3\mu)$ selects exactly

$$\mathcal{S} = \{\lambda^{-3}cd^3 : c, d \equiv 1 \pmod 3,\ c \text{ squarefree}\}. \tag{284}$$

Indeed the first family has positive λ-valuation after multiplication by λ^3, as do the second-family terms with $n \geq 1$. They are killed as nonunits. At $n = 0$ the negative branch is -1 modulo 3, and only the positive branch is 1. This proves the unit-state selection as well as its valuation condition.

For a moving good squarefree level $\mathfrak{t}$, let $\psi_\mathfrak{t}$ be either cubic orientation on its literal residue modulus. Let F_0 be the fixed away-λ raw modulus. The product $\psi_M(x) = \kappa_0(x)\psi_\mathfrak{t}(x)$ is a bounded periodic function on the valid period $M = F_0\mathfrak{t}$. This period is not asserted to be its primitive conductor. It is allowed to be imprimitive and to have overlapping $F_0, \mathfrak{t}$. No value $\kappa(\mathfrak{t})$ is divided out.

Define the coefficient-projected theta function using $\Phi(\lambda^3\mu)\psi_M(\lambda^3\mu)$. Its infinite constant is zero. On $\mathcal{S}$, $\tau(\mu) = 3^{5/2}\widetilde{g}(c)\sqrt{Nd}$. Multiplicativity on the primary section gives the exact cube identity

$$\Phi(\lambda^3\mu)\psi_M(\lambda^3\mu) = \kappa(c)\kappa(d)^3\psi_\mathfrak{t}(c)\mathbf{1}_{(d,\mathfrak{t})=1}. \tag{285}$$

Write

$$\mathcal{R}_{\kappa,\mathfrak{t},\ell}(s) = \sum_{c \text{ primary,SF}} \frac{\widetilde{g}(c)\kappa(c)\psi_\mathfrak{t}(c)\Theta_\ell(c)}{(Nc)^s}, \tag{286}$$

with every fixed raw zero in κ. Then initially $\Re s > 1$,

$$D_{\Phi,\psi}(s,\ell) = 3^{5/2+3s}i^\ell L^{\mathrm{raw}}(3s - \tfrac{1}{2}, \kappa^3\Theta_{3\ell})\mathcal{R}_{\kappa,\mathfrak{t},\ell}(s). \tag{287}$$

The factor i^ℓ is the actual angle of λ^{-3}. The cube series has denominator $(Nd)^{3s-1/2}$, and L^{raw} retains all original exclusions above $\lambda F_0\mathfrak{t}$, even after primitive conductor drop. There is no $\kappa(\lambda)$ factor or untwisted-zeta substitution. Conjugating the whole construction gives the actual conjugate Gauss root and opposite angular/cubic orientations.

The full ramified projector has an exact finite translation formula. The trace dual of $\mathcal{O}$ is $\lambda^{-1}\mathcal{O}$, so

$$\begin{aligned} b_h &= 3^{-f}\sum_{x \bmod \lambda^f} \Phi(x)e(-\operatorname{Tr}(xh/\lambda^{f+1})), \\ \Phi(x) &= \sum_{h \bmod \lambda^f} b_h e(\operatorname{Tr}(xh/\lambda^{f+1})), \qquad \sum_h |b_h| \leq 3^{f/2}. \end{aligned} \tag{288}$$

For the away period use normalized Fourier coefficients q_u with positive inversion sign and pairing $e(\operatorname{Tr}(xu/\alpha_M))$. Since M avoids λ, that pairing is perfect, and

$$\psi_M(x) = \sum_{u \bmod M} q_u e(\operatorname{Tr}(xu/\alpha_M)), \qquad \sum_u |q_u| \leq \sqrt{NM}. \tag{289}$$

Both mass bounds are Parseval followed by Cauchy. They hold for arbitrary bounded periodic functions; no primitive Fourier-support condition is inserted.

Consequently the exact projected function is a finite sum of $\theta(z + a_{h,u}, v)$ with

$$a_{h,u} = h\lambda^{2-f} + \lambda^3 u/\alpha_M, \qquad \sum_{h,u} |b_h q_u| \le 3^{f/2}\sqrt{NM}. \tag{290}$$

The inverse different explains the exponent $2 - f$. Every raw zero is reconstructed by this full sum. Each reduced denominator $C_{h,u}$ divides a fixed λ-power times M; hence $1 \le NC_{h,u} \ll_{\text{fixed}} N\mathfrak{t}$. Nonunit Fourier residues may lower that denominator. They remain terms of the exact transform, rather than being discarded.

S80.6 The coarse dual bound, safe convexity and literal Perron endpoints

Let $R = N\mathfrak{t}$, and set

$$E(s) = L^{\text{raw}}(3s - \tfrac{1}{2}, \kappa^3\Theta_{3\ell})\mathcal{R}_{\kappa,\mathfrak{t},\ell}(s).$$

For $\ell \neq 0$ the finite sum of entire angular Mellins is entire. The source Fourier–Mellin calculation has factor

$$\gamma_\ell(s) = \frac{1}{4}(2\pi)^{-2s-|\ell|}\Gamma(s + \tfrac{|\ell|}{2} - \tfrac{1}{6})\Gamma(s + \tfrac{|\ell|}{2} + \tfrac{1}{6}). \tag{291}$$

Thus division by γ_ℓ, whose reciprocal is entire, and the nonzero elementary factor in (287) makes E entire as well.

On $s = -\eta + i\tau$, $0 < \eta < 1/12$, (281), (269) and (290) give

$$\begin{aligned} \sum_{h,u} |b_h q_u|(NC_{h,u})^{1+2\eta}\mathcal{A}_{j(h,u)}(1+\eta) &\ll_{\text{fixed},\eta} R^{3/2+2\eta}, \\ |E(-\eta + i\tau)| &\ll_{\text{fixed},\ell,\eta} R^{3/2+O(\eta)}(1+|\tau|)^{2+O(\eta)}. \end{aligned} \tag{292}$$

The height exponent follows from the quotient of the two Gamma products. The exact multiplier $\chi(\delta_{h,u})(-1)^{|\ell|}(C_{h,u}/\bar{C}_{h,u})^\ell$ has been written before bounding its modulus. This is a conservative $R^{3/2+\epsilon}$ cost. No refined $R^{1/2+\epsilon}$ Ramanujan estimate is used.

The line $\Re s = 1 + \eta$ is absolutely bounded uniformly in R, by the original Gauss and cube coefficients. For a requested final $\epsilon > 0$, choose $\eta > 0$ sufficiently small in terms of ϵ; the constants may depend on that choice. The finite-order continuation from the theta Mellin representation permits the damped strip maximum principle. Applying it to the entire numerator, then allocating small exponents, gives

$$|E(\sigma + i\tau)| \ll_{\text{fixed},\ell,\epsilon} (R(2+|\tau|))^\epsilon R^{\frac{3}{2}(1-\sigma)}(1+|\tau|)^{2(1-\sigma)} \tag{293}$$

on the safe half-strip $1/2 + \eta \le \sigma \le 1 + \eta$. Only there divide by L^{raw}: the real part of $3s - \frac{1}{2}$ exceeds one, so its inverse is bounded by absolutely convergent Euler products, uniformly in all raw omissions. No inverse-L continuation through unknown zeros or zero-free strip is asserted. Equation (293) therefore bounds $\mathcal{R}$ on that half-strip.

Here are the endpoint details needed for Theorem S79.1. Replace each real endpoint by the half-integer threshold selecting exactly the same integer norms. All ideals in a repeated norm fibre are counted. Such a fibre has divisor-bounded multiplicity in this quadratic field. For an unnormalized sharp prefix at length T, truncated Perron at height $\mathcal{H} \ge 1$ has error

$$O_\epsilon\big(T^\epsilon[1 + (T/\mathcal{H})\log(2T)]\big). \tag{294}$$

This follows by summing $\min(1, T/(\mathcal{H}|n-y|))$ over integer distances from the half-integer endpoint y. No norm tie receives a half-weight.

At $\sigma = 1/2 + \eta$ the vertical contribution is bounded by

$$(RT)^{\epsilon} T^{1/2} R^{3/4} \mathcal{H}.$$

If $T \geq R^{3/2}$, choose $\mathcal{H} = (T/R^{3/2})^{1/4}$. On horizontal sides the bound is at most $T/\mathcal{H}$, up to allocated subpowers, since $R^{3/2}\mathcal{H}^2 \leq T$. The vertical integral, truncation and horizontal contributions are then $O((RT)^{\epsilon} T^{3/4} R^{3/8})$. An interval is a difference of two prefixes. If $T < R^{3/2}$, the elementary $O(T)$ count is the better branch.

Finally apply partial summation to the same $T^{-1/2} W(x) x^{-1/2}$, $x = N\mathfrak{a}/T$. Its supremum and total variation are bounded by $O(T^{-1/2} V_1(W))$. This proves (261) with the exact profile. For $1/2 \leq T < 1$ only the unit can occur; its coefficient and profile are literal and satisfy the same bound with a fixed endpoint constant. For $T \leq 1/2$ there is no contribution. This extension uses no analytic theorem at a fictitious length.

S80.7 The actual annular coefficient and exact eleven-factor expansion

The opened coefficient $B_U = \mu A_U$ in (262) is asserted on the actual annulus $N\mathfrak{m} > U$. It is distinct from original C_U. No identity for $B_U(\mathcal{O})$ is needed or asserted. Unit factors within the factorization of an annular column are essential and stay in every convolution below.

Let $\mathbf{1}$ be the constant-one function on good ideals, $\delta_{\mathcal{O}}$ the ideal-convolution identity, and $M_z(\mathfrak{a}) = \mu(\mathfrak{a}) \mathbf{1}_{N\mathfrak{a} \leq z_0}$, $z_0 = (2D)^{1/5}$. The exact good-ideal convolution identity is

$$\mu = \sum_{j=1}^{5} (-1)^{j-1} \binom{5}{j} \mathbf{1}^{*(j-1)} * M_z^{*j} + \mathbf{1}^{*4} * (\mu - M_z)^{*5}. \tag{295}$$

Indeed $\mathbf{1} * \mu = \delta_{\mathcal{O}}$, so expand $\mu * [\delta_{\mathcal{O}} - (\delta_{\mathcal{O}} - \mathbf{1} * M_z)^{*5}]$; also $\delta_{\mathcal{O}} - \mathbf{1} * M_z = \mathbf{1} * (\mu - M_z)$. Each factor of $\mu - M_z$ has norm strictly above z_0. The last term in (295) is supported strictly above $z_0^5 = 2D$ and vanishes at every residual $\mathfrak{c}$ under consideration.

The j-th term has j truncated Möbius factors and $j-1$ unrestricted coefficient-one factors. Together with the two original short factors it has $2j+1 \leq 11$ labels. The $j=1$ term and unit labels remain. Restricting the total column to squarefree support forces all factors to be squarefree and pairwise coprime.

Use independent dyads $[M_i, 2M_i)$, $M_i \geq 1$, with the original own-factor cutoffs at U, z_0. If the box meets the annulus,

$$D/2^{11} \leq P_{\mathrm{box}} := \prod_i M_i \leq 2D. \tag{296}$$

Choose the partition from these scales only. If some $M_i \in [P_{\mathrm{box}}^{1/3}, P_{\mathrm{box}}^{2/3}]$, it is a balanced subset. If all $M_i < P_{\mathrm{box}}^{1/3}$, multiply successive scales until first reaching $P_{\mathrm{box}}^{1/3}$; the result is below $P_{\mathrm{box}}^{2/3}$. If no balanced subset exists, a unique scale exceeds $P_{\mathrm{box}}^{2/3}$; two such scales would have product above P_{box}. The product of all remaining scales is below $P_{\mathrm{box}}^{1/3}$.

At $D \geq 2^{64}$,

$$P_{\mathrm{box}}^{2/3} \geq D^{2/3}/2^{22/3} > \sqrt{D} \geq U, \qquad P_{\mathrm{box}}^{2/3} > (2D)^{1/5}.$$

Therefore an unbalanced unique long factor is genuinely one of the unrestricted coefficient-one factors. It cannot be an original short factor or a truncated Möbius factor. The actual whole prefix

$\mathfrak{p}$, the product of all remaining factors, satisfies $N\mathfrak{p} \le 2^{10} P_{\mathrm{box}}^{1/3} \ll D^{1/3}$. The $j = 1$ term has no unrestricted factor and is automatically balanced; its unit assignments have not been discarded.

For a fixed box define each macro coefficient by summing all its signed assignments within its assigned factor group. On a squarefree ideal the number of at most eleven labelled factorizations is at most $11^{\omega(\mathfrak{a})}$. Fixed binomial factors and dyadic logarithms thus give a uniform D^{ϵ} pointwise bound, with every own cutoff retained. There is no separate count over $N\mathfrak{a}, N\mathfrak{b} \le U$.

S80.8 Whole-factor cubic matrices and the large-prefix part

A balanced partition gives two physical squarefree macro ideals $\mathfrak{u}, \mathfrak{v}$, each with scale in fixed multiples of $[D^{1/3}, D^{2/3}]$ and product comparable to D. Their own coefficients contain the original cutoffs and signs. Cubic Gauss CRT gives on coprime products

$$\overline{G_{\mathfrak{u}\mathfrak{v}}^{\mathrm{act}}} = \overline{G_{\mathfrak{u}}^{\mathrm{act}}}\,\overline{G_{\mathfrak{v}}^{\mathrm{act}}}\,\chi_{3,\mathfrak{u}}(\alpha_{\mathfrak{v}}). \tag{297}$$

The raw matrix entry is zero at a collision, so summing all squarefree macro pairs implements the original squarefree product. No primitive root at an overlapping product is invented.

The external physical cubic norm is

$$\big\|[\chi_{3,\mathfrak{u}}(\alpha_{\mathfrak{v}})]_{N\mathfrak{u}\asymp P,\, N\mathfrak{v}\asymp L}\big\|_{\mathrm{op}} \ll_{\epsilon} (PL)^{\epsilon}\sqrt{P + L + (PL)^{2/3}}. \tag{298}$$

The rows and columns are actual good squarefree ideals. All raw fixed/growing multiplicative masks are coefficient diagonals in this norm, after finite-selector expansion. Ideal counting gives squared coefficient norms D^{ϵ} when each coefficient is weighted by the inverse square root norm. Separate the one occurrence of the common W by

$$W(N\mathfrak{u}N\mathfrak{v}/D) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{W}(t) D^{-it} (N\mathfrak{u})^{it} (N\mathfrak{v})^{it}\, dt.$$

The powers are unitary diagonals, costing $\mathcal{M}(W)$ once. Balanced boxes therefore cost $\mathcal{M}(W) D^{1/3+\epsilon}$.

For a Type-I box write $\mathfrak{m} = \mathfrak{p}\mathfrak{v}$, where $\mathfrak{v}$ has coefficient one and the intrinsic $H_{\mathfrak{p}}$ contains all remaining assignments. Then $N\mathfrak{p} \ll D^{1/3}$. The portion $N\mathfrak{p} > Z$, with its literal own-variable indicator, has dyadic bound

$$\mathcal{M}(W) D^{\epsilon}\sqrt{P + D/P + D^{2/3}},$$

and hence total bound

$$\mathcal{M}(W) D^{\epsilon}[D^{1/3} + \sqrt{D/Z}]. \tag{299}$$

Macro own-factor constraints remain coefficients. A chosen equivalent row-dependent norm interval costs the reviewed logarithmic interval-Schur factor, preserving complete norm fibres. It is not assigned operator norm one.

S80.9 The full original opposite-column mask in the small-prefix row

Freeze the original $\mathfrak{n}$, the exact $\mathfrak{k}, \mathfrak{s}, \mathfrak{w}, \mathfrak{e}$, and a finite source component before estimating a Type-I row. The original pair mask is exactly

$$\mathbf{1}_{(\mathfrak{p}\mathfrak{v},\mathfrak{n})=1} = \mathbf{1}_{(\mathfrak{p},\mathfrak{n})=1}\mathbf{1}_{(\mathfrak{v},\mathfrak{n})=1}. \tag{300}$$

The first factor stays in the actual prefix coefficient, along with all original prefix/$\mathfrak{k}, \mathfrak{s}, \mathfrak{w}$ collisions, fixed numerator zeros and prefix phases. The intrinsic arithmetic $H_{\mathfrak{p}}$ still depends only on its own

factor assignments. Multiplication by these outer factors preserves its divisor bound. A forbidden prefix is zero before a row bound is applied.

The remaining long-variable deletion is

$$Q = \mathrm{rad}_{\mathrm{good}}(\mathfrak{p}\mathfrak{n}\mathfrak{k}\mathfrak{s}\mathfrak{w}).$$

It is not a substitute for the first factor in (300), and the growing $\mathfrak{n}$ is not part of a fixed character conductor. On the nonzero support $\mathfrak{p}, \mathfrak{k}$ are squarefree and coprime, by the original $\eta(\mathfrak{k})^2$ zero. The physical cubic frequency is $\mathfrak{t} = \mathfrak{p}\mathfrak{k}$.

The exact deletion formula is

$$\begin{aligned}\mathcal{F}_\psi(T; \mathfrak{t}; Q, I) = \sum_{\substack{\mathfrak{h} \mid Q \\ (\mathfrak{h},\mathfrak{t})=1}} & \frac{\mu(\mathfrak{h})\psi(\mathfrak{h})\overline{G_{\mathfrak{h}}^{\mathrm{act}}}\chi_{3,\mathfrak{h}}(\alpha_{\mathfrak{t}})}{\sqrt{N\mathfrak{h}}} \\ & \cdot \mathcal{F}_\psi(T/N\mathfrak{h}; \mathfrak{h}\mathfrak{t}; 1, I/N\mathfrak{h}).\end{aligned} \tag{301}$$

Insert $\sum_{\mathfrak{h} \mid (Q,\mathfrak{v})} \mu(\mathfrak{h})$ and set $\mathfrak{v} = \mathfrak{h}\mathfrak{a}$. Gauss CRT and reciprocity combine the actual crosses into $\chi_{3,\mathfrak{a}}(\alpha_{\mathfrak{h}\mathfrak{t}})$. This raw entry is zero at $(\mathfrak{a}, \mathfrak{h}) > 1$ and encodes the squarefree-product condition. The outer raw entry kills $(\mathfrak{h}, \mathfrak{t}) > 1$. No zero is inverted, including if a numerator exponent becomes divisible by three. Original interval endpoints divide literally by $N\mathfrak{h}$.

Here $T = D/N\mathfrak{p}$, and the exact product is

$$(T/N\mathfrak{h})N(\mathfrak{h}\mathfrak{t}) = DN\mathfrak{k}. \tag{302}$$

There is no common-long-divisor decomposition between the two columns in this proof. The other original column remains fixed.

Using the sharp bound and the original prefix factor gives (265). For $T/N\mathfrak{h} \in [1/2, 1)$ use the actual unit contribution; smaller arguments contribute zero, with the profile endpoint rule. Large frequencies are allowed by the proven all-level sharp bound, with its trivial fallback. Thus

$$\sum_{\mathfrak{h} \mid Q} (N\mathfrak{h})^{-3/8} \le \tau(Q) \ll_\epsilon (NQ)^\epsilon.$$

On the exact support, $N\mathfrak{n} \le 2D$, $N\mathfrak{s} \le 2V$, $N\mathfrak{w} < 64$, so $NQ \ll DZVN\mathfrak{k}$. Summing $(N\mathfrak{p})^{-3/8}$ over $N\mathfrak{p} \le Z$ by ideal counting proves the small-prefix column bound

$$V_1(W)(DXA_1)^\epsilon D^{1/4}(N\mathfrak{k})^{3/8+\epsilon} Z^{5/8}. \tag{303}$$

Every prefix zero was retained until that positive estimate. No coherence or orientation mean has been removed.

S80.10 The exact discrete prefix and the final normalization

Combine the balanced, large-prefix and small-prefix bounds, and put

$$Z = D^{2/9}, \qquad \sqrt{D/Z} = D^{1/4}Z^{5/8} = D^{7/18}. \tag{304}$$

The untouched original A_U column has the elementary bound $B_W D^{1/2+\epsilon}$. Apply it after the left estimate and use the original A_1/D once. All finite source/ray/primitive $\mathfrak{e}/\mathfrak{w}$ coefficient masses stay fixed, and the actual $\mathfrak{s}$ family is counted by $O(V)$ only at the end. The scalar g_0 is bounded and is never divided out.

The original finite $\mathfrak{k}$ sum is unchanged in the identity. Its small-prefix positive majorant is

$$\sum_{\substack{\mathfrak{k}\ \text{good,SF}\\ N\mathfrak{k}\le K_{\mathrm{cut}}}} (N\mathfrak{k})^{-13/8+\epsilon} \le \zeta_K(13/8-\epsilon), \qquad \epsilon<5/8. \tag{305}$$

Product and divisor losses can be allocated smaller exponents first; their $N\mathfrak{k}$ factors are absorbed by this convergent series. Balanced and large-prefix pieces use the easier exponent 2. This is a bound on a finite sum, not an Euler replacement of the model.

For the optional original unequal-norm projection, a fixed rational norm has $O_\epsilon(D^\epsilon)$ ideal coordinates. The actual A_U and every fixed HB component are divisor-bounded. Removing that same fixed-$\mathfrak{n}$ fibre costs $O(B_W D^{-1/2+\epsilon})$ in the left column, and therefore $O(B_W^2 A_1 V D^{-1+\epsilon})$ in the normalized form. Conjugate ideals remain different, and a norm inequality has not been made multiplicative.

This proves the leading-model part of (263). For the original exact prefix, all its Bessel arguments are at most $\Lambda = A_1 X/D^2$, and

$$|\dot G(x) - g_0| \le 3C_0^{-3/2} x.$$

There are $O(D^2)$ annular column pairs and $A_1/(D\sqrt{F_\Omega}) \le A_1/D^2$. The coefficient product is $O(B_W^2 D^\epsilon)$, the original finite $\mathfrak{k}$ absolute budget is bounded by $\sum (N\mathfrak{k})^{-2}$, and the actual sixth-part band has $O(V)$ ideals. Only in this remainder are these absolute counts taken. They give (257) directly for A_U, A_U. No A_U-to-A_Y error, principal density or scalar ray-count split is introduced.

Finally the exact powers are

$$\begin{aligned} &3/4+1/16-1/9 = 101/144, && 3/4-101/144 = 7/144,\\ &3/4+1/16-1/6 = 31/48, && 13/16+(3/4+3/8-2) = -1/16. \end{aligned} \tag{306}$$

The same norm-fibre correction has power $-3/16$. This completes the pure finite-prefix theorem with its once-only normalization and all original finite cutoffs.

S80.11 Verification scope and the remaining arithmetic comparison

The accepted evidence includes exact rational cusp matrices and finite quotient/coset calculations, angular-coordinate and Mellin checks, literal primary-support cases, affine coefficient transport, and synthetic scale-composition/exponent checks. The typed exponent grid is a formal subfamily; it is not exhaustive physical dyad coverage after the constants in (296). The proof uses the separate exact $D \ge 2^{64}$ inequalities for those constants. None of these checks is actual-family energy data, training, asymptotics or a kernel-checked Lean proof.

All original sources, primary versions, raw outputs and failed preliminary harnesses remain preserved. The new theorem is a prose proof with the identified external theta and cubic-norm inputs. Its domain is the representation-fixed pure $\mathcal{T}_X$ prefix in (256). The infinite dual complement, other pure sectors, mixed and neighboring blocks, short/principal assembly, independent positive arithmetic determinant and zero-location comparisons remain open. The Dirichlet, Hecke, Dedekind, elliptic and general Artin scopes retain their separate hypotheses; no Artin holomorphy, desired positivity, zero location or GRH premise has been inserted. The whole signed and global unsigned comparisons have not changed.

S81 Paired cutoffs, exact divisor signs and proved regimes

The comparison in this section concerns a specified part of the signed coprime-column form for the original unweighted, nonprincipal, two-pivot Eisenstein rows. We use the physical ideal norms

and the original ordered cutoff. Let D, U, H be the column, cutoff and capacity parameters, with $U^2 \le D \le 7U^2/2$ and $A_1 A_2^2 \ll H \le D$. The first two row ideals have dyadic norms A_1, A_2, $R = A_1 A_2$, and the actual primitive conductor of row u has norm $Q_u \asymp R$. The finite bad-prime set S, row type, and twist are fixed. The same smooth complex profile W supported on $[1, 2]$ is used on both sides. Ideal labels, including conjugate primes of the same norm, are retained individually.

S81.1 The selected pairing

For each good squarefree intermediate Möbius divisor d, put $E = Nd$. Let V_d be its original row amplitude and $\mathcal{E}_d$ the equal-norm part of the original middle Poisson window. We study selections of

$$\frac{2}{D} \operatorname{Re} \sum_d \mu(d) \langle \mathcal{E}_d, V_d - \mathcal{E}_d \rangle. \tag{307}$$

Here the counting inner product is linear in the first argument. This is one term in the expansion of the original signed norm. In particular the intermediate d is different from the gcd of the final column pair.

On an original allocation let $z = e_0 z_0$ with $J \le Nz < 2J$, let $e_1 \mid \operatorname{rad}(Sd)$ be the other deletion label, and put $h = re_0$, with r the original squarefree short divisor. Set

$$X = \frac{D}{E(Nr)^2}, \qquad X' = \frac{X}{(Ne_0)^2}, \qquad M = \frac{R\, Ne_0\, Ne_1\, J}{X}.$$

The original middle window has $X' > B_*$ and $K_0 < Nk \le MD^\eta$, for its specified B_*, K_0, η. The primal test discards the guaranteed zero cells $X/((Ne_0)(Ne_1)J) \le 1/2$, and the original weights retain any further zeros. An allocation with $X' \le B_*$ belongs to the direct component. Its re-expressed Poisson sum stays in the complement in (307).

On the equality side write $z_{0,0} = g\ell$ and $k_0 = g\bar{\ell}$, so $Nz_{0,0} = Nk_0$. On the partner side the selected changes in the good exponents consist of erasing the squarefree cutoff part x, erasing a part of the good conjugate overlap, and inserting allowed positive increments at other good primes outside d. A conjugate erasure removes the full original exponent, which can be two. At an insertion prime the old residue plus the increment is at most five. Changes at the good primes of d keep their exact new residues. For fixed other labels write $z_{0,1} = z_* x$ and $k = Ik_*$. The physical insertion ideal I keeps each exponent $1, \dots, 5$ and is disjoint from the selected erasure supports and d. Thus $Nx \asymp B$ and $NI \asymp K$ are physical lengths. The two varying raw phases are $\psi_u(x)\overline{\psi_u(I)}$, or their conjugates if the pairing is conjugated. The fixed row factors retain the union of deleted primes, the literal section, the twist and the primitive S-values.

The subtraction on the partner imposes $Nz_* Nx = Nk_* NI$ only in its original middle window. At fixed x this fixes at most one integer norm for I, and the reverse count fixes at most one norm for x. An integer norm supports at most divisor-many ideals on the polynomial clip. With the physical inverse $1/Nx$, the paired count therefore costs $\min(1, K/B)$ up to subpowers. The two half radial powers leave

$$(B/K)^{1/2} \min(1, K/B) = \sqrt{\min(B/K, K/B)}.$$

The remaining harmonic weights give $O(R/E)$ for this subtraction at each d, with the inherited profile and subpower costs. Both original endpoints remain in the count. Equal norms in the direct and other components remain in the pairing.

S81.2 Target bounds on the selected cross

The bounds have the common cost $O_{\epsilon,S,\nu,u_0}((DH)^{\epsilon}\mathcal{M}_{24}(W)^2)$, where $\mathcal{M}_{24}$ controls the indicated smooth profile derivatives. The live partner primal test supplies $E \ll D/B$ and partner radial scale $y_1' \gg Nk/R \gg K/R$. The proof first sums each actual column array. It then uses the row comparisons and the original label sums.

At the mixed scales $A_1 \asymp D^{5/12}$, $A_2 \asymp D^{1/6}$, the selected normalized cross has upper power $57/80$ over all physical ratios and all d. If $H \ll D^{3/4}$, a perturbed primitive sharp moment gives the target power $3/4$ for the same selected cross on

$$\frac{7}{24} \le \log_D A_1 \le \frac{439}{912}, \qquad A_1 A_2^2 \ll H.$$

The second pivot need only satisfy the stated original admissibility and capacity. The conductor comparison uses the actual R.

At the pure critical scales $A_1 \asymp R \asymp D^{3/4}$, $A_2 \asymp 1$, the target power $3/4$ holds over all d in each of the following physical sectors:

- $K \gtrsim D^{1089/1420}$, with any erasure length B.
- $B \gtrsim D^{1355/1482}$, with any insertion length K.
- $B \gtrsim R$ and $K \lesssim D^{921/1652}$.

The flexible central windows give further short and long threshold curves. There are also separate comparisons for a small fixed residual label product and for a small d-prefix. The comparable pure all-divisor comparison has power $55/68$, and the target holds on $Nd \le D^{9/68}$ in that sector. Each assertion concerns the selected changes described above. The remaining pure rectangles and the coherently averaged signed divisor sum are open.

S81.3 Ordered assignment signs

Fix an ordered factorization $\zeta = \zeta_1\zeta_2$ of the actual squarefree cutoff label $\zeta = e_0 z_0$ on either side, with $(d,\zeta) = 1$ and $N\zeta_i \le U$. Write $T_i = U/N\zeta_i$. The assignment sum multiplying $\mu(\zeta)$ is

$$F_d = \sum_{\substack{a,b \mid d,\ (a,b)=1 \\ Na \le T_1,\ Nb \le T_2}} \mu(a)\mu(b).$$

Complete the b-sum and then subtract its excluded terms. With $m = d/b$ in the subtraction, squarefreeness gives the exact identity

$$\frac{F_d}{\mu(d)} = \mathbf{1}_{E \le T_1} + \mathbf{1}_{E \le T_2} - 1 - \sum_{\substack{m \mid d \\ T_1 < Nm < E/T_2}} \sum_{\substack{a \mid m \\ Na \le T_1}} \mu(m/a). \tag{308}$$

The complete b-sum forces $a = d$. The unit $m = 1$ gives the indicator at T_2, while the nonunit terms with $Nm \le T_1$ vanish by the full inner Möbius sum. Both inequalities of the remaining band are strict. Below the product cap the band is empty. The three possible assignment values are $\mu(d), 0, -\mu(d)$ according as both, just one, or neither individual capacity reaches E.

Fix the row, both ordered factorizations and the primal and dual labels. Select both e_1 labels to be fixed S-supported ideals, take the good changed-residue ideal on d to be the unit, and require d to avoid the full fixed prime union of the row and the primal and dual labels. When $EN\zeta \le U^2$ on both

sides, their capacity step functions multiply the original outer Möbius sign. The fixed-field deleted Möbius estimate supplies subpower cancellation for the resulting fixed-label scalar on $Nd \asymp Y$ with $Y \geq D^{\kappa}$ for a fixed $\kappa > 0$. In the pure comparable block, a physical Nx-interval of width $O(BD^{-1/8})$ gives the target after the full divisor sum by a separate sparse erased-column estimate. This includes the narrow interval immediately above the product cap on the actual partner allocation. The wide correction and the remaining coherent signed estimate require additional arguments. The whole signed and unsigned powers 13/12 and 9/8, the independent determinant comparison, and the scope-specific zero-location obligations remain as in the preceding sections.

S82 Sharp insertion and ordered cutoff estimates

Throughout this section, x denotes the erased squarefree cutoff factor, I the positive insertion ideal, and B, K their respective physical dyadic lengths. The exact partner norms are $Nz = Nz_*Nx$ and $N\mathfrak{k} = Nk_*NI$. Here $\mathfrak{k}$ is the original dual ideal, and k_* is its fixed residual factor in a given cell. Write

$$E = Nd, \qquad A = A_1, \qquad R = A_1A_2, \qquad a = \log_D A, \quad r = \log_D R, \quad b = \log_D B, \quad k = \log_D K.$$

At the critical pure scales $a = r = 3/4$ at the level of powers, and at the critical mixed scales $a = 5/12,\ r = 7/12$. Fixed comparison constants in the scales are allowed throughout. A bound in this section includes the common factor $(DH)^{\epsilon}\mathcal{M}_{24}(W)^2$ for the paired cross, with the original fixed-data constants. In row moments the appropriate one-profile factor is understood separately.

The estimates concern the selected pairing with both original erasures, all positive off-d increases whose sum with the old residue is at most five, and the exact new residues on d. The old cutoff and conjugate erasures, including old exponent two in the conjugate erasure, remain in the phase. The cutoff is $Z = U^2$, with $U^2 \leq D \leq 7U^2/2$. The rows are the original unweighted literal nonprincipal rows. We keep the common W, fixed S, ν, u_0, literal ray, capacity and squarefree selectors, and the actual primitive $Q_u \asymp R$, Gauss root and raw zeros. The physical sieve is the order-six application in Section S74. Its exponent at column length D^t, relative to A, is

$$\Delta_a(t) = \max\bigl(0, t - a, t/3 - a/6, 5t/6 - 2a/3\bigr). \tag{309}$$

Its growth slope is at most one. A live subunit dyad is treated at constant scale.

S82.1 The middle subtraction and scalar comparisons

First remove the negligible inherited tail $N\mathfrak{k} > D^5$. Its absolute estimate applies to each selection below. Fix the original allocation and dyad on the partner side. The compact logarithmic separation of the partner profile uses $\log(NI/K) - \log(Nx/B) - \log(Q_u/R)$. Its weighted Fourier absolute integral, with the extra factor $1 + |\xi|$ in log frequency ξ, is bounded by a constant times $\mathcal{M}_{24}(W)(1 + y_1')^{-4}$ at the actual dyadic profile scale y_1'. This follows by extending the profile on a fixed-width log interval and integrating by parts. The phase factors then separate on the erased array, insertion array and row. There is no condition on B/K in this separation. Sharp subinterval partial summation removes the insertion norm twist with the erased array and every other row multiplier fixed.

For the original middle equality, the equation $Nz_*Nx = Nk_*NI$ fixes at most one integer norm on either side given the other ideal. There are divisor-many ideals of each integer norm on the clipped support. This counts the entire insertion ideal, including its powerful part. The physical $1/Nx$ yields weighted mass $O((DH)^{\epsilon}\min(1, K/B))$. The two half radial powers supply $(B/K)^{1/2}$,

giving the factor $\sqrt{\min(B/K, K/B)} \le 1$. The remaining label sums give $O(R/E)$. Apply this subtraction only in the original middle window with its exact two endpoints. Equal norms in the re-expressed direct, low and far pieces remain there.

The original live primal test gives

$$E \ll D/B, \qquad D/(EE_1(Nh_1)^2) > Nz/4, \qquad y_1' \gg N\mathfrak{k}/R \gg K/R. \tag{310}$$

For each side $i = 0, 1$ write $h_i = r_i e_{0,i}$ for the original grouped square-divisor label and $E_i = Ne_{1,i}$ for its deletion norm. The good part of $e_{1,i}$ is a divisor of d, and its other part is supported on S. Write $h_0 = fa_{\mathrm{id}},\ h_1 = fb_{\mathrm{id}}$ with $f = (h_0, h_1)$. The three ideals $f, a_{\mathrm{id}}, b_{\mathrm{id}}$ are pairwise coprime and squarefree. The ideal b_{id} is distinguished from the length exponent b. On the equality side $z_{0,0} = g\ell,\ \mathfrak{k}_0 = g\bar{\ell}$. For the good part $\bar{\ell}_{\mathrm{g}}$ of the conjugate define

$$j = (\bar{\ell}_{\mathrm{g}}, d), \quad c = (z_*, a_{\mathrm{id}}, \bar{\ell}_{\mathrm{g}}), \quad a_0 = a_{\mathrm{id}}/c, \quad \ell_0 = \ell/(\ell, z_* b_{\mathrm{id}}).$$

The good conjugate erasure support $\mathfrak{q}_\ell$ is a squarefree divisor of $\bar{\ell}_{\mathrm{g}}$ avoiding $a_{\mathrm{id}}, b_{\mathrm{id}}, d$. Its full erased factor is $\mathfrak{e}_\ell = \prod_{p|\mathfrak{q}_\ell} p^{1+\mathbf{1}_{p|z_*}}$. Let $\widetilde{q}_d$ be the actual sixth-free new representative at the good primes of d. With k_S the original S-part and v_6 the full sixth-power repetition, the exact partner residual is

$$\begin{aligned} k_* = k_S v_6^6 \frac{z_* a_{\mathrm{id}}^4 b_{\mathrm{id}}^2 \ell_0^6}{\ell\, c^6} \\ \times \frac{(\bar{\ell}_{\mathrm{g}}/j)\widetilde{q}_d}{\mathfrak{e}_\ell}. \end{aligned} \tag{311}$$

The quotient follows from the good exponent difference of the two primitive phases. Away from d, before changing any residue, that difference has representative exponent

$$v_p(z_{0,1}) + 4v_p(a_{\mathrm{id}}) + 2v_p(b_{\mathrm{id}}) - v_p(\ell) + v_p(\bar{\ell}_{\mathrm{g}}) \pmod 6.$$

At ℓ_0 one adds six to the negative exponent, and at c one subtracts the overflow six. The selected cutoff erasure removes x from this product. The selected conjugate erasure divides by $\mathfrak{e}_\ell$, and the d-part uses $\widetilde{q}_d$ in place of the matched residue. The positive increases multiply this exact residual by I. The norm and raw value of v_6^6 are retained, including any overlap with the other factors. On the original admissible pairs these operations give the literal product $\psi_u(x)\overline{\psi_u(I)}$ times the fixed row phases, with the original raw zeros.

Let D^ρ be the product of the two row norms relative to $R\sqrt{K/B}$, after the projectors and powerful sums. The full cutoff inverse norm is included in these norms. For the exact paired scalar, start with

$$\frac{D}{E^2 E_0 E_1\, Ng\, N\ell\, Nz_*(Nf)^4(Nc)^2(Na_0)^2(Nb_{\mathrm{id}})^2}.$$

Up to the fixed dyadic constants its profile scales are

$$\begin{aligned} y_0 &= \frac{D}{EE_0R(Nf\, Nc\, Na_0)^2}, \\ y_1' &\asymp \frac{D}{EE_1R}\frac{(Na_0)^4}{(Nf\, Nc)^2}(N\ell_0 Nv_6)^6 \frac{N\widetilde{q}_d}{Nj}\frac{Nk_S K}{B\, N\mathfrak{e}_\ell}. \end{aligned}$$

The actual profile uses Q_u, the physical ideal norms and the original fixed S-factors. Multiplying by $y_0^{-\alpha}(y_1')^{-\theta}$ gives the following weights before the relative row moment:

$$\begin{aligned} &D^{1-\alpha-\theta}R^{\alpha+\theta}E^{\alpha+\theta-2}E_0^{\alpha-1}E_1^{\theta-1}(B/K)^\theta \\ &\quad\times \frac{(Nf)^{2\alpha+2\theta-4}(Nc)^{2\alpha+2\theta-2}(Na_0)^{2\alpha-4\theta-2}}{Ng\, N\ell\, Nz_*(Nb_{\mathrm{id}})^2(N\ell_0 Nv_6)^{6\theta}(Nk_S)^\theta}(N\mathfrak{e}_\ell)^\theta(Nj/N\widetilde{q}_d)^\theta. \end{aligned} \tag{312}$$

The scalar is accompanied by the retained decay $(1+K/R)^{-(4-\theta)}$ from (310). Every absolute comparison follows the exact pairing with those labels.

In the harmonic ℓ-count the fixed $\bar{j} \mid \ell$ pays Nj to power one. A conjugate erasure prime p uses a distinct $\bar{p} \mid \ell$. Its conjugate lies in exactly one of $\ell_0, z_*, b_{\mathrm{id}}$, paying a second inverse norm. In the old exponent-two case, the original $p \mid z_*$ pays a third inverse norm. Thus the resulting local exponents are at most $-2+\theta$ and $-3+2\theta$, respectively. When Nc or its half power is needed, charge $c \mid z_*$. This charge is disjoint from both conjugate charges since $\mathfrak{q}_\ell \perp a_{\mathrm{id}}$, its conjugates lie in ℓ, and $c \mid a_{\mathrm{id}}, \bar{\ell}_{\mathrm{g}}$. The other sums below are harmonic on polynomial support, convergent inverse-power sums, or fixed-divisor choices.

For $1/6 \le \theta \le 1$, take $\alpha = 1-\theta$. The harmonic comparison at each d is

$$\frac{R}{E} D^{\rho} (B/K)^{\theta-1/2} (1+K/R)^{-(4-\theta)}. \tag{313}$$

The two three-half choices give the comparisons

$$\begin{aligned} &\frac{R^{3/2} D^{\rho}}{\sqrt{D}\sqrt{E}}, && (\alpha,\theta) = (1,1/2), \\ &\frac{R^{3/2} D^{\rho} (K/B)^{1/6}}{\sqrt{D} E^{1/3}}, && (\alpha,\theta) = (7/6,1/3), \end{aligned}$$

with the corresponding retained decays. In the second choice $E_0^{1/6} \ll_S E^{1/6}$. With $(\alpha,\theta) = (1,1/4)$, the E_0 cancels, the weights in $f, a_0, \ell_0 v_6$ have exponents $-3/2, -1, -3/2$, and the disjoint charge on c pays its half power. The resulting comparison is

$$\frac{R^{5/4} D^{\rho}}{D^{1/4} E^{3/4}} (K/B)^{1/4} (1+K/R)^{-15/4}. \tag{314}$$

By the first inequality in (310), summing the $E^{-3/4}$ costs $O((D/B)^{1/4})$ on its live range. The two three-half choices on $E \le D^e$ cost $D^{e/2}$ and $D^{2e/3}$, respectively.

The higher choice $(\alpha,\theta) = (5/2,1)$ leaves $(Nf)^3 (Nc)^5$ in (312). The exact primal restriction gives

$$(Nf)^2 (N b_{\mathrm{id}})^2 N z_* N x \ll D/(E E_1), \qquad (Nf\, Nc)^4 \ll (D/(E E_1 B))^4.$$

Write the remaining numerator as $(Nf)^{-1} Nc\, (Nf\, Nc)^4$ and pay the first two factors by the preceding harmonic counts. The additional E^{-4} and $E_0 \ll_S E$ give the harmonic comparison

$$\frac{R^{7/2} D^{3/2+\rho}}{B^4 E} (B/K)^{1/2} (1+K/R)^{-3}. \tag{315}$$

There is also a fixed-label selection for $(\alpha,\theta) = (1,0)$. For $p \mid a_0$, the original eligibility excludes b_{id}, $\ell, d, x, \mathfrak{q}_\ell$, and at most one of $z_*, \bar{\ell}_{\mathrm{g}}$ contains p. The exact residual has valuation four or five at this prime before adding the full sixth repetition. Thus $a_0^4 v_6^6 \mid k_*$. On $(N a_0)^4 (N v_6)^6 \ll D^{\xi}$, the joint count of a_0, v_6 is $O(D^{\xi/4})$, by summing $(N v_6)^{-3/2}$. Formula (312) then gives $R D^{\rho+\xi/4} (K/B)^{1/2}/E$. This selection is sufficient when $N k_* \ll D^{\xi}$. The comparison uses the full residual ideal, with its S-part and repetitions.

S82.2 The sharp moment and the grouped sparse tail

Write $I = I_1 P$ uniquely, where I_1 is squarefree and the nonzero valuations of P are $2,3,4,5$. We have $I_1 \perp P$. Fix P at first. The common-prime projector for x, I_1 uses $s_1 \mid (x, I_1)$, with $\mu(s_1)$

and the two raw values on s_1. Require s_1 itself to satisfy the original fixed prime eligibility of I_1, including $s_1 \perp P$, before stripping it. The residual I_1' avoids s_1 by its squarefreeness, along with the original forbidden union. Keep the original fixed P-deletion on the erased array. After stripping s_1, the squared erased mass is $O((DH)^\epsilon/(BNs_1))$, including its actual $1/Nx$. The insertion length is $K/(NPNs_1)$. A dyad $NP \asymp M$ has $O(\sqrt{M})$ powerful ideals, by the unique representation $P = a_{\mathrm{p}}^2 b_{\mathrm{p}}^3$ with b_{p} squarefree and the convergent $\sum_{b_{\mathrm{p}}} (Nb_{\mathrm{p}})^{-3/2}$. Overlap is allowed at exponent five. This pays the $M^{-1/2}$ shortening in the product of the two moments.

For the sharp insertion bound at length L, expand the fixed deletion and squarefree selectors in the order $I_1' = e(r')^2 m$, where $e = e_f e_P$ divides the full forbidden union and r' avoids that union. All Möbius signs and fixed raw factors are retained. At fixed P, e, r' the remaining sharp interval is an unweighted m-interval after the exact partial summation. Its endpoints can be treated as sharp disk endpoints. Smooth a disk of norm length L at radial width proportional to $\delta\sqrt{L}$. The normalized shell row norm is bounded by a subpower times

$$D^{\Delta_a(l)/2}\big(\sqrt{\delta L} + L^{1/6}\big), \qquad l = \log_D L.$$

To see the additive term, split each ideal uniquely as $\mathfrak{s}\, b_0 v^6$ with $\mathfrak{s}$ supported on S and b_0 good sixth-free. At fixed $\mathfrak{s}, v$, the shell contains divisor-many ideals per integer norm and has norm width $O(\delta L/(N\mathfrak{s}(Nv)^6))$. Apply the physical sieve to b_0. The square roots of the width weights sum absolutely. The constant atoms sum over $N\mathfrak{s}(Nv)^6 \ll L$, giving $L^{1/6}$ up to subpowers. This also accounts for open or closed endpoints.

The smoothed primitive Poisson transform has the original $Q_u^{-1/2}$ and primitive Gauss root. On a dual dyad T put $Y = \sqrt{LT/R}$. The disk transform is L times the Fourier transform of the unit disk's chord length $2\sqrt{1-v^2}$ at frequency comparable to Y. Integrating once by parts, separating the two endpoint intervals of width $1/Y$, and integrating by parts again on the interior gives $O(L(1+Y)^{-3/2})$. The endpoint derivative integral is $O(Y^{-1/2})$; the interior boundary and second derivative terms are $O(Y^{1/2})$ before the second division by Y. The same argument with the chord length multiplied by v bounds the frequency derivative, so its logarithmic derivative costs one power of $1+Y$. The convolution bump and its log derivative provide $(1+\delta Y)^{-M_0}$ for arbitrarily large fixed M_0. Consequently the cutoff log function m and m' have L^2 bounds $O(L(1+Y)^{-3/2}(1+\delta Y)^{-M_0})$ and $O(L(1+Y)^{-1/2}(1+\delta Y)^{-M_0})$, respectively. Splitting the Fourier integral at $1 + \|m'\|_2/\|m\|_2$ and applying Cauchy and Plancherel gives absolute log Fourier mass $O(L(1+Y)^{-1}(1+\delta Y)^{-M_0})$. Multiplying by $\sqrt{T}/\sqrt{Q_u}$ yields

$$O_{M_0}\left(\sqrt{L}(1+\delta\sqrt{LT/R})^{-M_0}\right).$$

The zero frequency is absent by nonprincipality. With $\delta \asymp D^{\lambda-\Delta_a(l)}$ when $\Delta_a(l) > \lambda$, the effective dual exponent is

$$t = \max(0, r - l + 2(\Delta_a(l) - \lambda)). \tag{316}$$

The main shell term has row RMS $O(\sqrt{L}D^{\lambda/2})$. The additive term has the same bound whenever $\Delta_a(l) \le 2l/3$. The smoothed term has that bound if $\Delta_a(t) \le \lambda$. Above the cutoff the arbitrary smoothing decay pays the degree-one growth of the sieve and the divisor subpowers. The ordinary estimate applies directly if $\Delta_a(l) \le \lambda$.

In pure use $\lambda = 0$ for $l \le 4/5$. For $l \ge 3/8$ the convex function in (316) before truncation is at most $3/8$ by the endpoints. In mixed use $\lambda = 1/20$ for $l \le 5/8$. The smoothing branch requires $l > 43/120$, and the same convex comparison gives $t \le 43/120$. The additive term meets the stated criterion in both cases. The exact deletion e-sum has divisor cost at fixed P, the r'-sum is harmonic,

and the P-count pays the shortening above. The resulting sharp rectangle loss is $(\Delta_a(b)+\lambda)/2$. It uses the physical ratio $\sqrt{K/B}$ even when B and K are unequal. The ordinary two-moment alternative has loss $(\Delta_a(b)+\Delta_a(k))/2$.

The coupled comparison keeps a sharper tail for $Nr' \ge D^q$ at $NP \asymp M$. Freeze e_f, expand the remaining $(x,P)=1$ by $s_2 \mid (x,P)$, with $s_2 \perp s_1$, and fix the powered portion P_s on s_2. Set $P = P_sP'$ and group the insertion coefficients by the actual $P'e_P(r')^2m$, of length $K' = K/(Ns_1\,Ne_f\,NP_s)$. The coefficients are divisor bounded and signed. They include the original full P-dyad, threshold, old residue test and clipped physical interval. For the sixth-free reduction of this grouped ideal, write it as $\mathfrak{s}\,b_0v^6$, and split $P' = P_0P_1,\ r' = r_0r_1$ by the support of v. The original $r' \perp P$ gives $P_1r_1^2 \mid b_0$. The two inverse-norm sums in the support count are bounded by

$$\min(1,(NP_sNP_0/M)^{1/2}), \qquad \min(1,Nr_0D^{-q}).$$

The first is the powerful reciprocal dyad count and the second the inverse-square ideal tail. The resulting squared mass at fixed $\mathfrak{s},v,P_0,r_0$ is at most their product times $K'/(N\mathfrak{s}(Nv)^6)$, up to subpowers. After square roots the v-sum converges, since $(NP_0)^{1/4}(Nr_0)^{1/2}(Nv)^{-3} \le (Nv)^{-5/4}$. The S-part sum also converges. Finally $NP_s \ge (Ns_2)^2$ makes the product of the insertion shortening and the erased s_2-shortening harmonic. Thus the relative norm saving in this branch is $M^{-1/4}D^{-q/2}$. It holds at the actual physical lengths for both ratio orientations.

S82.3 Coupling the dual to the shorter ordered factor

Open x into its actual two ordered cutoff factors. Fix the assignments of the cutoff base and of s_1, and retain the individual cutoff bounds. Before expanding their remaining coprimality, both factors are individually squarefree and keep all their fixed deletions. Use the common squarefree w projector, with pre-stripping eligibility in both arrays and with both residual factors avoiding w after shortening. The fixed phase is $\mu(w)\psi_u(w)^2$. Sharp norm product tests separate with polylogarithmic Fourier mass on the clipped polynomial ranges, keeping their endpoints. The auxiliary primitive insertion dual has the same raw orientation as x. Multiply it by the shorter cutoff factor as an actual ideal before forming row norms. The grouped coefficients have divisor bound and no added coprimality with the auxiliary frequency. With shortened lengths X_s, X_l the physical product is $X_sX_l \asymp B/(Ns_1(Nw)^2)$. The actual cutoff inverse norm and the smoothed transform mass $\sqrt{L}$ therefore give the factor

$$\frac{\sqrt{K/B}}{Ns_1Nw\,Nr'\sqrt{Ne}\sqrt{M}}.$$

The P-count pays the $M^{-1/2}$, and the s_1, w, r' sums remain harmonic.

For reference, set $n = \log_D Ns_1,\ j = \log_D Ne,\ p = \log_D Nr',\ v = \log_D M$, and write $\ell = k - v - n - j - 2p$ for the ordinary insertion log length. In each coupled application choose the relative target loss ρ, then use the sharp branch for large n and the extra deletion shortening $D^{-j/2}$ for large j. Put

$$\sigma = \Delta_a(b) + \Delta_a(k) - 2\rho, \qquad q = \max(0, \sigma - v/2).$$

The grouped branch $p \ge q$ pays the ordinary excess down to ρ. In the remaining branch take $\delta \asymp D^{-\beta}$. With $\beta = \max(0, \Delta_a(b-n) + \Delta_a(\ell) - j - 2\rho)$ the main shell meets the chosen loss directly. The additive term has exponent $(\Delta_a(b-n)+\Delta_a(\ell))/2 - \ell/3 - j/2$. The smoothed dual exponent is

$$t = \max(0, r - k + v + n + j + 2p + 2\beta).$$

Its two sieve losses are evaluated at $u_s + t, u_l$, where $u_i = \log_D X_i$, $u_s \le u_l \le 1/2$, and $u_s + u_l \le b - n$. The degree bound pays all larger dual dyads using the same smoothing decay. The following computations specify the small-p shell and cutoff choices in the regions of interest.

S82.4 Pure central and comparable regions

Set $a = r = 3/4$. On $2/3 \le b \le a,\ 1/2 \le k \le 31/40$ the coupled loss is the smaller of $\Delta_a(b)/2$ and

$$\rho_c = \begin{cases} (25b - 2k)/204 - 7/272, & k \le a, \\ (25b + 22k)/204 - 31/272, & k \ge a. \end{cases}$$

To verify the second choice when it is smaller, put $j_0 = \Delta_a(b) - 2\rho_c,\ n_0 = 3j_0$. Use the sharp branches at $n \ge n_0$ or $j \ge j_0$, as described above. In the remaining small-p range choose

$$\beta = \sigma - v/3 - 2n/3 - 4j/3 - 2p/3, \qquad 6\rho_c = (a+b)/2 - k + 8\sigma/3.$$

Since $v + 2p \le 2\sigma$, the lower bounds for ℓ, β on the two k-pieces are, respectively,

	ℓ	β
$k \le a$	$47/68 - 9b/17 + 11k/51$	$73/408 - 9b/34 + 8k/153$
$k \ge a$	$13/34 - 9b/17 + 32k/51$	$-97/204 - 9b/34 + 283k/306.$

Their minima are $41/102, 1/153$ and $31/68, 1/51$. Thus $\ell > 3/8$ and $\beta > 0$. The lower slope $1/3$ gives $\Delta_a(\ell) \le \Delta_a(k) - (k - \ell)/3$; the main shell costs at most ρ_c. For $k \le a$ the additive excess is at most $-\ell/6 - n/6 - j/2$. For $k \ge a$ it is at most $13/96 - 31/204 < 0$.

The dual satisfies $t \le a - k + 8\sigma/3$. In the two k-pieces, $b/2 - k + 8\sigma/3$ is $(25b - 2k - 18)/34$ and $(25b + 22k - 36)/34$, respectively, both negative on the stated rectangle. Thus $u_s + t < a$. The geometric mean excess is bounded by

$$\tfrac{1}{6}\max(1/8, (a+b)/2 - k + 8\sigma/3) = \rho_c.$$

Indeed both sieve arguments use the lower slope, with $u_s \le b/2 - n/2,\ u_s + u_l \le b - n$ and $u_l \le 1/2$.

Write ρ for the smaller loss. Substituting it into the harmonic comparison at $\theta = 1/6$ gives the short target for $57/80 \le b \le a,\ k \le (172b + 21)/264$. The ordinary sharp loss already covers $k \le b/2 + 3/16$. At $\theta = 1$ and $k \ge a$, the retained decay gives the long target for $273/404 \le b \le a,\ k \ge (508b + 1743)/2768$. The sharp comparison covers $k \ge 5/8 + 4b/21$ through $4/5$, and the ordinary long insertion comparison decreases above $4/5$. The fixed product selection of Section S82.1 instead gives $k \le (308b + 21 - 204\xi)/400$ on the lower k-piece. The two divisor-prefix choices with partner exponents $1/2, 1/4$, retaining the actual partner decay, give target through

$$e \le \max\left\{\tfrac{1}{4} - 2\rho + 7(k-a)_+,\ \tfrac{1}{4} - 4\rho + b - k + 15(k-a)_+\right\}, \qquad Nd \le D^e.$$

For $B \asymp K$ and $b \le a$, the central formula gives $\rho \le 1/17$, with the sharp branch sufficient below its central rectangle. For comparable $b \ge a$, the shifted comparison below has loss at most $1/17 + (1/51 + 5/12)(b - a)$ until the long target applies. The partner decay in the half split pays this additional loss. The two half powers and the one-plus-half powers then give $D^{55/68+\epsilon}$ over all d and the target on $Nd \le D^{9/68}$, respectively. The latter follows from $5/8 + 1/17 + 9/136 = 3/4$.

S82.5 Pure long insertion for every erasure length

Take $k = a + y,\ 0 \le y \le 1/56$. Work first at the reference erased exponent a, and put

$$\rho_0 = 1/17 + y/51, \quad n_0 = 6(1/16 - \rho_0), \quad j_0 = n_0/3, \quad \sigma = 9/68 + 27y/34.$$

The sharp and sparse branches use these reference thresholds, with the actual insertion length. Their erased ordinary excess increases by at most $5(b - a)_+/6$, before its square root. In the

remaining branch use $\beta = \Delta_a(a-n) + \Delta_a(\ell) - j - 2\rho_0$. Here $\ell \geq 31/68 - 22y/51 > 3/8$ and $\beta \geq 1/51 - 16y/153 > 0$. The main shell has reference loss ρ_0; the additive term has excess at most $1/8 + 5y/12 - (31/68 - 22y/51)/3 < 0$.

If $\ell \leq a$, then $\beta = \sigma - y/2 - v/3 - 2n/3 - 4j/3 - 2p/3$. If $\ell \geq a$, then $\beta = \sigma - 5v/6 - 7n/6 - 11j/6 - 5p/3$. The first gives $t \leq 8\sigma/3 - 2y$, and the second $t \leq 2\sigma - y \leq 8\sigma/3 - 2y = 6\rho_0$. At $b > a$, reduce u_s to at most $(a-n)/2$ and then u_l only enough to make the sum at most $a - n$. The reference pair remains ordered and nonnegative. The total reduction is at most $b - a$, either by the shorter-factor cap alone or by the original bound on the sum. The reference grouped length is below a, since $3/8 + 6\rho_0 < a$. Restoring the removed lengths adds at most $5(b-a)/12$ in the geometric mean, by the $5/6$ Lipschitz bound through actual exponent one. Thus the relative loss is $\rho_0 + 5(b-a)_+/12$.

For $b \leq a$ use (313) at $\theta = 1$; for $b \geq a$ use (314) with the live divisor cap. The respective exponents after summation are at most

$$a + \rho_0 + (b-a)/2 - 7y/2, \qquad a + \rho_0 - (b-a)/12 - 7y/2.$$

Both meet target at $y \geq 6/355$. For $k \geq a + 1/56$ the sharp loss $\Delta_a(b)/2$ already suffices through $4/5$, by the same two radial comparisons. Above $4/5$ the ordinary insertion adds $\Delta_a(k)/2$; at the endpoint $1/16 + 1/12 - 7/40 < 0$, and the bound decreases thereafter. This gives $K \gtrsim D^{1089/1420}$ over all B and d.

S82.6 Pure long erasure and the intermediate gap

For $a \leq b \leq 1,\ k \leq 4/5$, the sharp loss is $\rho_{\mathrm{sh}} = 5b/12 - 1/4$. The three-half choice with partner exponent $1/3$, summed over $E \ll D/B$, has exponent $a + 5(1-b)/12 - (a-k)_+/6 - 7(k-a)_+/2$. The high radial comparison (315) has exponent $31/8 - 37b/12 - k/2 - 3(k-a)_+$. For $k \leq a$ the two reach target when

$$k \leq L_b = -7/4 + 5b/2 \quad \text{or} \quad k \geq H_b = 25/4 - 37b/6.$$

They overlap for $b \geq 12/13$. For $k \geq 4/5$ the ordinary insertion in the high comparison adds $\Delta_a(k)/2$. Its endpoint exponent $409/120 - 37b/12$ is below target for $b \geq 9/10$, and decreases with k. For the small residual product selection, the low endpoint is instead $1/2 + b/6 - \xi/2$; it overlaps the high endpoint at $b \geq 69/76 + 3\xi/38$.

For $1355/1482 \leq b \leq 12/13$, work in the gap $L_b \leq k \leq H_b$. Define

$$g = \min((k - L_b)/6, (H_b - k)/2), \qquad \rho = \rho_{\mathrm{sh}} - g, \qquad C_b = 17/4 - 4b.$$

The two losses agree at C_b, where $\rho_C = 3b/2 - 5/4$ is the minimum of ρ. It suffices to achieve the chosen ρ, by using the smaller radial comparison. Use $j_0 = \Delta_a(b) - 2\rho,\ n_0 = 6j_0/5,\ \sigma = j_0 + \Delta_a(k)$, and the exact nonnegative width exponent from Section S82.3. The two sharp branches and the grouped tail apply with these choices. The minimum of $k - 2\sigma - n_0 - j_0$ occurs at C_b and equals $(233b - 202)/30 > 1/3$; this bounds ℓ from below. The shell additive excess is at most $1/5 - 1/9 < \rho_C$.

When $\ell \geq 3/8$ and $\beta > 0$, the slopes on $b - n$ and on $[\ell, k]$ bound t by $t_1 = a - k + 8\sigma/3$. When $\beta = 0$, the bound is $t_2 = a - k + 2\sigma + n_0 + j_0$. For $\ell < 3/8,\ \beta > 0$ use $\beta = j_0 - 5n/6 - j$, which again gives $t \leq t_2$. Both upper expressions peak at C_b, as is seen from the slopes $1/3, -1$ of j_0. At C_b their difference is $(438b - 397)/180 \geq 0$, so $t \leq t_* = (95 - 96b)/18$.

For $u_s + u_l \leq b$, convexity and monotonicity of the sieve losses reduce the geometric mean maximum to $(u_s, u_l) = (b - 1/2, 1/2)$ and $(b/2, b/2)$. Indeed the maximizing u_l at fixed u_s is $\min(1/2, b - u_s)$. Before the breakpoint use monotonicity and after it use convexity. At the two

endpoints the shifted length is between a and one. The second endpoint is larger by $(1-b)/8$, with value

$$7b/24 + 5t_*/12 - 5/16 = 815/432 - 139b/72 \le \rho_C.$$

The last inequality is exactly $b \ge 1355/1482$. This closes the gap for every K after the two radial comparisons and the independent middle subtraction.

S82.7 Pure short insertion over large erasures

For $b \ge a$ and $k < a$, the harmonic choice $\theta = 1/6$ and the quarter choice with the primal divisor cap meet target if the rectangle loss is at most

$$\rho = \max\{(b-k)/3,\ b/2 - k/4 - 3/16\}.$$

With the sharp loss this covers $b \le 3 - 4k$ or $b \ge 3k - 3/4$, which is exhaustive for $k \le 15/28$. For $15/28 \le k \le 921/1652$, work inside the remaining triangle and split it at $b = 9/8 - k/2$. Use the same j_0, n_0, σ and exact width as in the intermediate gap. Here

$$j_0 \le 3/236, \quad b - n_0 \ge 3117/4130 > a, \quad \ell \ge 3159/8260 > 3/8, \quad \rho \ge 1/16.$$

These bounds follow by the affine formulas on the two triangles. The main shell meets ρ, and its additive term is at most $(7/26 + 1/16)/2 - 1/8 < 1/16$. The same t_1, t_2 estimates apply, with $t_1 - t_2 \ge 67/4130$. Thus it is enough to evaluate $(\Delta_a(u_s + t_1) + \Delta_a(u_l))/2 - \rho$ at the two cutoff endpoints of the previous proof. On each triangle these expressions are convex. Their vertex values in the two endpoint orders are

k	b	$(b - 1/2, 1/2)$	$(b/2, b/2)$
15/28	6/7	−5/168	−1/48
921/1652	318/413	−83/9912	−83/9912
921/1652	699/826	−51/3304	0
921/1652	381/413	−787/19824	−85/2832.

The sharp branches, sparse tail and smoothed dual therefore meet ρ throughout. This proves the target for $K \lesssim D^{921/1652}$ and $B \gtrsim R$ over all d.

S82.8 Mixed ratios and the critical-capacity strip

At $a_* = 5/12,\ r_* = 7/12$ use the mixed sharp insertion loss $\lambda/2 = 1/40$ through $k_0 = 5/8$. Denote Δ_{a_*} by Δ. For $k \ge b,\ k \le k_0$, the harmonic $\theta = 1$ comparison is at most $r_* + \Delta(k)/2 + 1/40 - 3(k - r_*)_+$, maximal at $k = r_*$. For $k \le b \le r_*$ use $\theta = 1/6$ and the sharp moment, with the same bound 57/80. For $k \le b,\ b \ge r_*$ and $k \le k_0$ the quarter comparison and live divisor cap give exponent

$$23/48 + \Delta(b)/2 + 1/40 + (k-b)/4 + (1-b)/4 - 15(k - r_*)_+/4.$$

At fixed b this is maximal at $k = r_*$. It equals 57/80 at $b = r_*$, decreases with slope $-1/12$ through 5/6, and is constant thereafter. For $k > k_0,\ k \ge b$ the ordinary $\theta = 1$ exponent is at most $r_* + \Delta(k) - 3(k - r_*)$, maximal at k_0 with value 101/144. For $k > k_0,\ k \le b$ use $\theta = 1/6$. The exponent decreases in k; at $k = k_0$ it is convex in b on $[k_0, 1]$, with endpoint values $2/3, 205/288$. Both are below 57/80. Adding the separate harmonic equality count gives 57/80 over all physical ratios and divisors.

Under $A_1A_2^2 \ll H \ll D^{3/4}$, the capacity bounds $r \le (3/4+a)/2$ at the level of exponents. The actual Q_u is still used in each moment. On the left, put $a = a_* - z,\ 0 \le z \le 1/8$. Use $\lambda = 1/20 + 3z/5$ through $k_0 = 5/8$. At $l = 43/120$ the three nonconstant monomials of $\Delta_a(l)$ are $-7/120 + z,\ 1/20 + z/6,\ 1/48 + 2z/3$, each at most λ. The smoothing branch is therefore longer than this l. The reference convex dual expression has endpoint maximum $31/90$, and the capacity and $\Delta_a(l) \le \Delta_{a_*}(l) + z$ give $t \le 31/90 + 3z/10$. The three nonconstant monomials at that upper bound are

$$-13/180 + 13z/10, \quad 49/1080 + 4z/15, \quad 1/108 + 11z/12,$$

each at most λ by the endpoints in z. The additive term is controlled since $3a \ge 7/8$. The erased and sharp insertion half-losses together increase by at most $4z/5$. In the harmonic branches the capacity prefactor saves $z/2$, while in the quarter branch it saves $5z/8$. In both ordinary branches $k > 5/8 > r_*$, so the active partner decay and capacity prefactor pay the increase of the two sieve factors. This gives $57/80 + 3z/10 \le 3/4$.

On the right take $a = a_* + x,\ 0 \le x \le 59/912$, and use $\lambda = 1/20 + x/4$ through $k_0 = 31/48$. On the smoothing branch $l > 43/120$. In (316) the possible r-increase of $x/2$ is paid by $2(\lambda - 1/20)$. The reference convex function on $[43/120, 31/48]$ has endpoint values $9/40, 43/120$. Thus $\Delta_a(t) \le 1/20$; the additive term is controlled by $k_0 < 3a$. In the ratio comparisons take the capacity upper bound $r_b = r_* + x/2$. The scalar estimates increase with r, while the sharp proof has already retained the actual conductor. In the sharp ranges the maximum is $r_b + (\Delta_a(r_b) + \lambda)/2 = 57/80 + x/2$. For the quarter range, after setting $k = r_b$, the b-dependent term $(\Delta_a(b) - b)/2$ is nonincreasing, giving the same maximum. In the ordinary ranges the $\theta = 1$ endpoint is $21/32 + 4x/3$. The $\theta = 1/6$ bound at $k = k_0$ has b-endpoint values $29/48 + 7x/4$ and $373/576 + 19x/12$; convexity bounds its interior. All are at most $3/4$, with the last endpoint attaining that value at $x = 59/912$. The union is the first-pivot strip $7/24 \le a \le 439/912$, subject to the original capacity and with no independent closeness requirement on A_2.

All comparisons here preserve the original $\mu(d)$ in the identity and use an absolute divisor bound at the end. The remaining pure rectangles, the coherent divisor sum, other signed components and the determinant and zero-location comparisons remain open.

S83 Wide cutoff corrections and full-frequency compensation

Keep the pairing (307), the exact ordered cutoff identity (308), and all the original allocation, primitive-conductor and raw-zero conditions. At one live allocation put $h = re_0$, $E_1 = Ne_1$, and write the actual cutoff ideal as $\zeta = e_0 z_0 = \zeta_1\zeta_2$. Thus $T_i = U/N\zeta_i \ge 1$, $E = Nd$, and

$$\Omega = \frac{E}{T_1T_2} = \frac{EN\zeta}{U^2} < \frac{14}{E_1Nh}.$$

Let $\mathcal{P}$ consist of the good prime ideals of norms seven or thirteen, with the original S-deletion. These are at most four separate ideal labels. Put $s(E) = \mathbf{1}_{E\le T_1} + \mathbf{1}_{E\le T_2} - 1$. The exact wide correction is

$$\begin{aligned} F_d(T_1,T_2) &= \mu(d)\big(s(E) + C_d(T_1,T_2)\big), \\ C_d(T_1,T_2) &= \sum_{\substack{mn=d \\ Nm>T_1,\ Nn>T_2}} \left(1 - \sum_{\substack{p\in\mathcal{P} \\ p\mid m}} \mathbf{1}_{Nm>T_1Np}\right). \end{aligned} \tag{317}$$

The two factors are coprime because d is squarefree. The strict tests retain the original cutoff endpoints. For $E_1Nh \geq 2$ only the unweighted factorization count remains. For $E_1Nh \geq 14$ the correction vanishes.

At fixed row and fixed primal and dual labels, prescribe the exact intersection δ of d with the full forbidden prime union, including $\mathcal{P}$. In a divisor dyad $Y \leq Nd < 2Y$ put $Y' = Y/N\delta$. For fixed $\kappa > 0$ and $Y' \geq D^{\kappa}$, the fixed-label scalar after taking out its fixed coefficient and row phase has bound

$$\ll_{\kappa} \mathcal{M}_{24}(W)^2 \frac{4^{\omega(\delta)} \log^C(2D)}{Y\, N\delta} \exp(-c_{\kappa}\sqrt{\log(2Y')}). \tag{318}$$

The estimate includes both wide corrections, with the original full prime deletions. It gives subpower cancellation before a coherent average over the rows and other labels.

For each $p \in \mathcal{P}$, replacing the partner coefficient by its signed p-subtraction in (317) gives a zero primal amplitude $V_d^{[p]}$. The exact full-frequency complement is $V_d^{[p]} - \mathcal{E}_d^{[p]} = -\mathcal{E}_d^{[p]}$, where $\mathcal{E}_d^{[p]}$ is its middle equality projection. The original elementary equality norm gives a contribution of order at most $R(DH)^{\epsilon}\mathcal{M}_{24}(W)^2$ after the full divisor sum. This comparison requires all partner modes, including the re-expressed direct component.

In the pure comparable sector $A_1 \asymp R \asymp D^{3/4}$, $A_2 \asymp 1$, $B, K \asymp R$, the equality-side wide correction reaches $D^{3/4+\epsilon}$ on $Nd \leq D^{13/56}$ with the inherited profile cost. Its partner can carry the original coefficient, the step part, or the correction part. Thus this prefix includes their double correction. The remaining coherent wide sum and the other signed components remain open.

S84 Derivations for the wide cutoff correction

S84.1 The strict band and its prime terms

Use the original r, e_0, e_1, J, z_0 on each side. The primal-zero removal at the dyad $J \leq N\zeta < 2J$ implies $D/(EE_1(Nh)^2) > Nz_0/4$. Hence

$$\frac{EN\zeta}{U^2} < \frac{4D}{U^2}\frac{Ne_0}{E_1(Nh)^2} \leq \frac{14}{E_1Nh},$$

using $e_0 \mid h$ and $2D \leq 7U^2$. For a band label $T_1 < Nm < E/T_2$ in (308), $m \neq 1$, so the full inner Möbius sum vanishes. In its omitted part use $c = m/a$. The inner sum in (308) becomes

$$-\sum_{\substack{c \mid m \\ Nc < Nm/T_1}} \mu(c).$$

Here $1 < Nm/T_1 < \Omega < 14$. In the Eisenstein field the primes above two and three lie in S. The remaining inert primes have norm at least 25, and the only possible good prime norms below 14 are $7, 13$. Two good prime factors already have norm at least 49. Consequently this sum includes the unit and precisely the primes specified in (317). Substituting $n = d/m$ turns $Nm < E/T_2$ into $Nn > T_2$. The upper bound on Ω also proves the two stated conditions for vanishing of the prime term and the band.

If a p-subtraction is present, $Nm > T_1Np$ and $Nn > T_2$ give $EN\zeta > NpU^2 \geq 2D$. The original primal profile for that individual cutoff is

$$W\left(\frac{EE_1(Nh)^2Nz_0\,Nb}{D}\right), \qquad Nb \geq 1.$$

Its argument exceeds two, since $E_1(Nh)^2 \geq Ne_0$. The individual cutoff has zero primal contribution even if its dyad survived the original primal-zero removal. Its Poisson components are still present in the exact paired selection.

S84.2 The fixed-intersection scalar estimate

Fix the actual row, primal and dual labels, ordered factorizations, original dyads, and both deletion allocations. Let $\mathcal{L}$ be the union of the primes of S, of the row, of all these labels, and of $\mathcal{P}$. Its radical has polynomial norm after the inherited $N\mathfrak{k} \leq D^5$ clip and the original support bounds. Choose the exact intersection of d with $\mathcal{L}$ as a squarefree ideal δ, subject to all required memberships and deletions. The good e_1 factors, conjugate d-overlaps, and actual new d-residue supports are fixed by these choices. Thus the original row phase is fixed on the free part of d, and an original zero stays zero. The free part avoids the full union $\mathcal{L}$.

In the product of the two copies of (317), their real factors $\mu(d)$ leave the original outer $\mu(d)$. The step-step term needs one ideal variable and a step-correction term uses $d = mn$. For two corrections, use the four cells of primes in d specified by their membership in the two factors m_0, m_1. Both m_i's and their complements are products of the appropriate cells. This is an exact squarefree factorization with pairwise coprime cells, including units. Expand the finite small-prime choices and assign the fixed primes of δ among the cells, with at most $4^{\omega(\delta)}$ assignments. The tests $p \mid m_i$ are thereby fixed. The cell ideals have form $\delta_j q_j$, and the sign is $\mu(\delta)\prod_j \mu(q_j)$.

Select the largest free cell by its norm, with deterministic index order on ties, and fix the other free cells. If its norm is t, then $t \geq (Y')^{1/4}$. Its interval lies on the scale $L = Y'/\prod_{\text{other}} Nq_j$, with $L \gg (Y')^{1/4}$ for every nonempty selection. Its deleted set $\mathcal{T}$ consists of $\mathcal{L}$ and the primes in the other free cells, and still has polynomial radical norm. Every factorization inequality is an interval test in t, since each m_i, n_i either includes the selected cell or is fixed. The tie order changes only an endpoint. The original primal and direct tests and the middle cutoffs also give finitely many interval tests. At fixed labels X' is proportional to $1/E$, the middle parameter M is proportional to E, and the integer norm equality is fixed. The original strict and nonstrict endpoints remain in the scalar sum.

For the fixed field $\mathbb{Q}(\omega)$, write $M_{\mathcal{T}}(x) = \sum_{Nc\leq x,\,(c,\mathcal{T})=1} \mu(c)$. The fixed-field deleted Möbius input used here is

$$|M_{\mathcal{T}}(x)| \ll_\kappa x \log^C(2D) \exp(-c_\kappa \sqrt{\log(2x)})$$

for $x \geq D^\kappa$ and polynomial radical norm. The underlying fixed-field zero-free input is Thorner–Zaman, *Algebra & Number Theory* 13 (2019), 1039–1068, Theorem 3.1, specialized to the trivial extension of this one fixed field. The classical truncated Perron shift for the reciprocal zeta function gives $M_\varnothing(x) \ll x\exp(-c\sqrt{\log(2x)})$. A possible fixed exceptional zero affects the fixed starting threshold. Artificial deletions are transferred exactly by

$$M_{\mathcal{T}}(x) = \sum_{\substack{Na\leq x\\ \operatorname{supp} a\subseteq\mathcal{T}}} M_\varnothing(x/Na).$$

All powers of the deleted primes occur in this convolution. Split at $Na = \sqrt{x}$. In the first range use the fixed-field bound. In the second use the elementary ideal count and $1/Na \leq x^{-\eta/2}(Na)^{-1+\eta}$, with $\eta \asymp 1/\sqrt{\log(2x)}$. The finite reciprocal Euler product at $1-\eta$ costs a power of $\log(2D)$. Indeed, if the deleted radical has norm $\ll D^{C_0}$, split its primes at $\log^2(2D)$. Below that threshold the ideal count bounds the reciprocal-power sum by $O((\log(2D))^{2\eta}\log\log(3D))$. Above it use the bound $O(\log(2D))$ for the sum of their log norms. For $x \geq D^\kappa$, these estimates give the claimed uniform logarithmic Euler factor and subpower decay. The underlying analytic zeta function stays fixed.

After factoring out the fixed scalar and phase, the weight is $E^{-2}G_W(c_0/E)\overline{G_W(c_1/E)}$ for fixed positive c_i. Its supremum plus integrated absolute derivative is $O(\mathcal{M}_{24}(W)^2Y^{-2})$ on each remaining interval. This follows from boundedness of the radial transform and its first log derivative, and the fixed dyadic range of E. Partial summation gives

$$\ll_\kappa \mathcal{M}_{24}(W)^2Y^{-2}L\log^C(2D)\exp(-c_\kappa\sqrt{\log(2Y')}).$$

Here $L \gg (Y')^{1/4}$ permits the deleted bound with a fixed smaller exponent in its polynomial lower cutoff. The remaining free cells sum harmonically by the factors $1/Nq_j$ in L. Together with the fixed-prime assignments this proves (318). It applies to a scalar with the row and all other ideal labels fixed.

S84.3 The complete partner sum and the equality prefix

For each $p \in \mathcal{P}$, use the coefficient

$$-\mu(\zeta)\mu(d) \sum_{\substack{mn=d,\ p|m\\ Nm>T_1Np,\ Nn>T_2}} 1$$

on the partner and sum its original ordered factorizations. This coefficient is independent of the partner frequency and divisor bounded. The individual primal zero above proves $V_d^{[p]} = 0$. The exact original Poisson identity therefore gives the full complement $-\mathcal{E}_d^{[p]}$, with the same middle projection. Recombining all modes includes the re-expressed direct part and preserves every original raw value.

For the elementary equality bound, at fixed r, e_0, e_1, J put $L_0 = D/(EE_1(Nh)^2)$ and $J' = J/Ne_0$. The equal-norm ideals have $z_0 = g\ell$, $\mathfrak{k} = g\bar{\ell}$. There are $O(J'\log(2+2J'))$ choices on the dyad, up to the original constant-scale convention. Their norm Nz_0 appears inversely in the Poisson scalar, and the equal-norm profile scale is comparable to L_0/R. Thus the pointwise row bound is at most a subpower times $\mathcal{M}_{24}(W)L_0R^{-1/2}(1+L_0/R)^{-4}$. The $O(R)$ original rows give squared counting norm $O((DH)^\epsilon\mathcal{M}_{24}(W)^2RL_0)$. Summing the square roots over the original r, e_0, e_1, J uses the harmonic factors $1/(Nr\,Ne_0)$ and fixed divisor costs. After division by D, the squared norm is $O((DH)^\epsilon\mathcal{M}_{24}(W)^2R/E)$. The same proof handles the modified coefficient because it is divisor bounded and row independent at each allocation. Apply Cauchy to the original and modified middle projections and sum harmonically in d. This proves the target-size full-frequency compensation of Section S83.

For the pure comparable prefix, keep the correction on the equality side. Its coefficient is divisor bounded on its fixed labels. A partner correction, when included, is a coefficient on x alone after fixing the other labels. The insertion interval remains complete inside the original eligibility. The sharp insertion proof with powerful freezing and exact common-factor eligibility applies. Its relative loss is $D^{1/16}$ from the erased sieve factor $\Phi(A_1, B/Ns_1)/A_1 \ll D^{1/8}$. A nonzero equality correction implies $E_0Nh_0 < 14$, hence $E_0(Nh_0)^2 < 196$. Therefore the equality radial scale is $y_0 = D/(EE_0(Nh_0)^2R) \gg D/(ER)$. Use both half radial powers and keep the remaining equality-side $7/2$ decay. The fixed-d rectangle is bounded by a constant times

$$(DH)^\epsilon\mathcal{M}_{24}(W)^2\frac{R}{E}D^{1/16}\left(1+\frac{D}{ER}\right)^{-7/2}.$$

Count the exact original middle subtraction independently at the same inherited R/E cost with the common factors. For $E \le D^{13/56}$, $D/(ER) \gg D^{1/56}$ at the pure scales. The remaining decay pays

$D^{1/16}$ because $(7/2)(1/56) = 1/16$. Harmonic summation gives the stated prefix. The partner-only argument retains its own ratio $N\mathfrak{k}/Nz$, so the same automatic lower bound on its radial scale is unavailable. The coherent masked wide divisor sum remains open.

References

[1] A. Schreiber, *Surface Algebras I: Dessins D'enfants, Surface Algebras, and Dessin Orders*, arXiv:1810.06750v5 (2023).

[2] A. Schreiber, *Surface Algebras and Surface Orders II: Affine Bundles on Curves*, arXiv:1812.00621v1; record 2018, downloaded PDF internally dated November 27, 2024.

[3] A. Schreiber, *Multiparameter Persistent Homology—Generic Structures and Quantum Computing*, arXiv:2210.11433v1 (2022).

[4] C. Bessenrodt and T. Holm, *Weighted locally gentle quivers and Cartan matrices*, arXiv:math/0511610v1 (2005).

[5] S. Ford, A. Oswald and J.J. Zhang, *Homological conditions on locally gentle algebras*, arXiv:2409.08333v3 (5 September 2026).

[6] S. Opper, P.-G. Plamondon and S. Schroll, *A geometric model for the derived category of gentle algebras*, arXiv:1801.09659v7 (10 July 2025).

[7] W. Chang, *Geometric models for the algebraic hearts in the derived category of a gentle algebra*, arXiv:2305.03998v2 (4 September 2024).

[8] Y.-Z. Liu, C. Zhang and H. Zhang, *Constructing projective resolution and taking cohomology for gentle algebras in the geometric model*, arXiv:2308.07220v1 (2023).

[9] Z. Li, Y. Qiu and Y. Zhou, *A geometric realization of Koszul duality for graded gentle algebras*, arXiv:2403.15190v1 (2024).

[10] W. Chang, H. Jin and S. Schroll, *Recollements of partially wrapped Fukaya categories and surface cuts*, arXiv:2206.11196v2 (2023).

[11] B. Dequêne and M. Schoonheere, *Resolving subcategories for gentle algebras I: Monogeneous resolving subcategories for gentle trees*, arXiv:2502.20994v3 (2026).

[12] D.A. Buchsbaum and D. Eisenbud, *What makes a complex exact?*, J. Algebra 25 (1973), 259–268, primary paper.

[13] W. Bruns, *The existence of generic free resolutions and related objects*, Math. Scand. 55 (1984), 33–46.

[14] D. Eisenbud, *Homological algebra on a complete intersection, with an application to group representations*, Trans. AMS 260 (1980), 35–64.

[15] A. Beauville and Y. Laszlo, *Un lemme de descente*, C. R. Acad. Sci. Paris 320 (1995), 335–340, author text.

[16] P. Magyar, *Affine Schubert varieties and circular complexes*, arXiv:math/0210151v1 (2002).

[17] F.F. Knudsen and D. Mumford, *The projectivity of the moduli space of stable curves I: Preliminaries on 'det' and 'Div'*, Math. Scand. 39 (1976), 19–55, doi:10.7146/math.scand.a-11642.

[18] M. Breuning, *Determinant functors on triangulated categories*, arXiv:math/0610435v1 (2006).

[19] D. Eriksson and G. Freixas i Montplet, *Deligne–Riemann–Roch and intersection bundles*, J. Éc. polytech. Math. 11 (2024), 247–361, doi:10.5802/jep.254.

[20] P. Deligne, *La conjecture de Weil I*, Publ. Math. IHÉS 43 (1974), 273–307, primary text.

[21] P. Deligne, *La conjecture de Weil II*, Publ. Math. IHÉS 52 (1980), 137–252, primary text.

[22] P. Deligne, *Cohomologie étale (SGA* $4\frac{1}{2}$*)*, Lecture Notes in Mathematics 569, Springer-Verlag, 1977, doi:10.1007/BFb0091516.

[23] G. Tabuada, *Noncommutative Weil conjecture*, arXiv:1808.00950v7 (6 December 2019).

[24] C. Deninger, *Arithmetic geometry and analysis on foliated spaces*, arXiv:math/0505354v1 (2005).

[25] D.B. Ray and I.M. Singer, *R-Torsion and the Laplacian on Riemannian manifolds*, Advances in Math. 7 (1971), 145–210, doi:10.1016/0001-8708(71)90045-4.

[26] F. Bornemann, *On the numerical evaluation of Fredholm determinants*, arXiv:0804.2543v1; Math. Comp. 79 (2010), 871–915.

[27] B. Dequêne and M. Schoonheere, *Resolving subcategories for gentle algebras II: Resolving subcategories for gentle trees*, arXiv:2509.10120v2 (3 October 2025).

[28] C. Procesi, *The invariant theory of* $n \times n$ *matrices*, Advances in Math. 19 (1976), 306–381, doi:10.1016/0001-8708(76)90027-X.

[29] L. Le Bruyn and C. Procesi, *Semisimple representations of quivers*, Trans. Amer. Math. Soc. 317 (1990), 585–598, doi:10.1090/S0002-9947-1990-0958897-0.

[30] A. Craw and R. Yamagishi, *The Le Bruyn–Procesi theorem following Lusztig*, arXiv:2312.08527v4 (14 November 2025).

[31] P. Etingof, O. Golberg, S. Hensel, T. Liu, A. Schwendner, D. Vaintrob and E. Yudovina, *Introduction to Representation Theory*, with historical interludes by S. Gerovitch, Student Mathematical Library 59, American Mathematical Society, 2011, doi:10.1090/stml/059; author-hosted text.

[32] K. Akin, D.A. Buchsbaum and J. Weyman, *Schur functors and Schur complexes*, Advances in Math. 44 (1982), 207–278, doi:10.1016/0001-8708(82)90039-1.

[33] Q. Jiang, *Derived Grassmannians and derived Schur functors*, arXiv:2212.10488v3 (6 July 2023).

[34] A. Price, A. Stelzer and A. Yong, *Representations from matrix varieties, and filtered RSK*, arXiv:2403.09938v2 (6 October 2025).

[35] H. Derksen and J. Weyman, *Semi-invariants of quivers and saturation for Littlewood–Richardson coefficients*, J. Amer. Math. Soc. 13 (2000), 467–479, primary author text.

[36] A. Schofield and M. Van den Bergh, *Semi-invariants of quivers for arbitrary dimension vectors*, arXiv:math/9907174v1 (27 July 1999); Indag. Math. 12 (2001), 125–138.

[37] J.S. Milne, *Reductive Groups*, author's course notes, version 2.00 (10 March 2018), §5.18.

[38] W. Crawley-Boevey, *Classification of modules for infinite-dimensional string algebras*, Trans. Amer. Math. Soc. 370 (2018), 3289–3313, doi:10.1090/tran/7032; arXiv:1308.6410v2 (4 March 2016).

[39] M. Auslander and I. Reiten, *Almost split sequences for Cohen–Macaulay-modules*, Math. Ann. 277 (1987), 345–349, doi:10.1007/BF01457367.

[40] R. Bautista, S. Liu and C. Paquette, *Representation Theory of an Infinite Quiver*, arXiv:1109.3176v1 (14 September 2011); published as *Representation theory of strongly locally finite quivers*, Proc. London Math. Soc. 106 (2013), 97–162.

[41] C. Paquette, *Irreducible morphisms and locally finite dimensional representations*, arXiv:1508.04353v3 (4 January 2016).

[42] S. Liu, P. Ng and C. Paquette, *Almost split sequences and approximations*, arXiv:1203.5009v1 (19 March 2012).

[43] C.M. Ringel, *The Minimal Representation-Infinite Algebras which are Special Biserial*, author-hosted ICRA Tokyo 2010 proceedings text, 57 pages, retrieved 3 November 2026.

[44] G. Puninski and M. Prest, *Ringel's conjecture for domestic string algebras*, arXiv:1407.7470v2 (13 September 2015).

[45] R. Laking, M. Prest and G. Puninski, *Krull–Gabriel dimension of domestic string algebras*, arXiv:1506.07703v2 (29 June 2015).

[46] H.-L. Huang, Z. Lin and X. Su, *The Auslander–Reiten quivers of string algebras of affine type $\widetilde{C}$ and a conjecture by Geiss–Leclerc–Schröer*, arXiv:2110.02777v2 (16 November 2021).

[47] G. Abrams and G. Aranda Pino, *The Leavitt path algebra of a graph*, arXiv:math/0509494v1 (21 September 2005).

[48] T. Bates, D. Pask, I. Raeburn and W. Szymański, *The C^*-algebras of row-finite graphs*, New York J. Math. 6 (2000), 307–324, primary journal text.

[49] N.J. Fowler and I. Raeburn, *The Toeplitz algebra of a Hilbert bimodule*, arXiv:math/9806093v3 (23 June 1998).

[50] N. Brownlowe, M. Laca, D. Robertson and A. Sims, *Reconstructing directed graphs from generalised gauge actions on their Toeplitz algebras*, arXiv:1812.08903v1 (21 December 2018).

[51] E.T.A. Kakariadis and O.M. Shalit, *On operator algebras associated with monomial ideals in noncommuting variables*, arXiv:1501.06495v3 (21 November 2018).

[52] O.M. Shalit and B. Solel, *Subproduct systems*, arXiv:0901.1422v3 (26 May 2009).

[53] A. Connes and C. Consani, *Knots, primes and class field theory*, arXiv:2501.06560v1 (11 January 2025).

[54] A. Connes, C. Consani and H. Moscovici, *Zeta zeros and prolate wave operators*, arXiv:2310.18423v2 (4 May 2024).

[55] A. Connes, C. Consani and H. Moscovici, *On q-series and the moment problem associated to local factors*, arXiv:2403.01247v1 (2 March 2024).

[56] A. Connes, C. Consani and H. Moscovici, *Zeta Spectral Triples*, arXiv:2511.22755v1 (27 November 2025).

[57] A. Connes and W.D. van Suijlekom, *Quadratic Forms, Real Zeros and Echoes of the Spectral Action*, arXiv:2511.23257v1 (28 November 2025).

[58] A. Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, arXiv:2602.04022v1 (3 February 2026).

[59] A. Connes and C. Consani, *On the Jacobian of* $\overline{\mathrm{Spec}\,\mathbb{Z}}$, arXiv:2602.15941v1 (17 February 2026).

[60] C. Deninger, *Dynamical systems for arithmetic schemes*, arXiv:1807.06400v4 (7 February 2024); Indag. Math. 37 (2026), 25–136, doi:10.1016/j.indag.2024.05.007.

[61] C. Deninger, *Primes, knots and periodic orbits*, arXiv:2301.11643v1 (27 January 2023).

[62] C. Deninger, *There is no "Weil-"cohomology theory with real coefficients for arithmetic curves*, arXiv:2204.02714v2 (27 January 2023).

[63] R. Ponge and Y. Tian, *Nonclassical Weyl laws and Connes' Integration for weak Lorentz ideals, I*, arXiv:2605.25591v2 (28 September 2026).

[64] F.W.J. Olver et al. (eds.), *NIST Digital Library of Mathematical Functions*, finite summation by parts, Eq. 2.10.9 and Dirichlet L-functions, §25.15, consulted 3 November 2026.

[65] A. Chirvasitu, P.M. Hajac and M. Tobolski, *Non-surjective pullbacks of graph C*-algebras from non-injective pushouts of graphs*, arXiv:1907.10260v3 (31 December 2019), Theorems3.4 and4.3; doi:10.1112/blms.12389.

[66] Y. Lekili and A. Polishchuk, *Derived equivalences of gentle algebras via Fukaya categories*, arXiv:1801.06370v5 (27 August 2019), Definition3.1.1, Lemma3.1.3 and Theorem3.2.2.

[67] M. Tomforde, *Uniqueness theorems and ideal structure for Leavitt path algebras*, arXiv:math/0612628v6 (27 February 2007), Theorem7.3, pp.29–30.

[68] A. Weil, *Numbers of solutions of equations in finite fields*, Bull. Amer. Math. Soc.55 (1949),497–508, especially pp.499–502 and Eq.(4), doi:10.1090/S0002-9904-1949-09219-4; primary-paper scan.

[69] The Stacks Project, Tag0FJ9 and Tag0GSX (retrieved 3 November 2026).

[70] J. Alper, *Good moduli spaces for Artin stacks*, Ann. Inst. Fourier63 (2013),2349–2402, Theorem10.3, pp.2386–2387, doi:10.5802/aif.2833.

[71] The Stacks Project, Tag04V8: Taking roots of line bundles (retrieved 3 November 2026).

[72] G. Zwara, *Smooth morphisms of module schemes*, Proc. London Math. Soc. (3) 84 (2002), 539–558, primary publication.

[73] C. Riedtmann and G. Zwara, *Orbit closures and rank schemes*, Comment. Math. Helv. 88 (2013), 55–84, primary publication, Section6.

[74] Y. Liu, N. Li, B. Xing and P. Chen, *A lecture note on covering theory in representation theory of algebras*, arXiv:2508.01563v5, 28 May 2026.

[75] Q. Dong, Z. Lin, M. Lu and S. Ruan, *Representation-tame Geiß–Leclerc–Schröer algebras and a revised GLS conjecture on root systems*, arXiv:2609.26503v1, 22 September 2026.

[76] S. Schroll, *On geometric models in representation theory*, Proceedings of ICM 2026, vol.3, pp.250–271, doi:10.1137/25M1810647; author preprint arXiv:2601.14396v1.

[77] B. Keller, *Introduction to A_∞ algebras and modules*, arXiv:math/9910179v2 (2001), especially §§3.1,3.3,6.2.

[78] The Stacks Project Authors, *The Koszul complex*, Tag 0621, especially Definitions15.29.1–2 and Lemma15.29.5; snapshot November3,2026, hash recorded in the repository.

[79] The Stacks Project Authors, *Regular sequences are Koszul regular*, Tag 062F, Lemma15.31.2; snapshot November3,2026, hash recorded in the repository.

[80] S. Datta, S. Howard and D. Cochran, *Geometry of the Welch Bounds*, arXiv:0909.0206v2 (2012), Section2.1, Theorem2.2.

[81] K.G. Wilson and J. Kogut, *The renormalization group and the epsilon expansion*, Physics Reports12(1974),75–199, DOI10.1016/0370-1573(74)90023-4; pinned institutional scan and review scope in `planning/ACTUAL_RG_SOURCES.json`.

[82] M. Levin and C.P. Nave, *Tensor renormalization group approach to2Dclassical lattice models*, arXiv:cond-mat/0611687v2 (2008); Phys. Rev. Lett.99(2007),120601.

[83] J. Kim et al., *Pure Transformers are Powerful Graph Learners*, arXiv:2207.02505v2 (2022). Our finite measured-feature prototype does not reproduce its full benchmarks.

[84] S. Liu et al., *Unsupervised Discovery of Steerable Factors When Graph Deep Generative Models Are Entangled*, arXiv:2401.17123v1 (2024); scoped primary reading in the computational source manifest.

[85] S. Lahlou et al., *A theory of continuous generative flow networks*, arXiv:2301.12594v2 (2023); state measures and trajectory balance.

[86] M. Gross and B. Siebert, *Mirror Symmetry via Logarithmic Degeneration Data II*, arXiv:0709.2290v2; used as a geometric research context, not an implemented mirror comparison.

[87] B. Green and T. Tao, *Quadratic uniformity of the Möbius function*, arXiv:math/0606087v2, Section4, Lemma4.1, printed pages12–13. Focused identity reading only; no nilsequence estimate is asserted for the moving sextic family here.

[88] S.S. Kudla, *Tate's thesis*, expanded Jerusalem lecture notes (2001), Theorem4.3 and Corollary4.4, pinned primary notes.

[89] J.S. Milne, *Class Field Theory*, version4.03 (6August2020), Theorem0.8 and ChapterI,§1, author notes.

[90] A. Jacot, F. Gabriel and C. Hongler, *Neural Tangent Kernel: Convergence and Generalization in Neural Networks*, arXiv:1806.07572v4; specified classical scaling hypotheses.

[91] J. Liu et al., *Representation Learning via Quantum Neural Tangent Kernels*, arXiv:2111.04225v3; focused version/abstract review.

[92] M. Kontsevich and Y. Soibelman, *Affine structures and non-archimedean analytic spaces*, arXiv:math/0406564v1; valued and integral-affine data.

[93] Y. Lekili and A. Polishchuk, *Derived equivalences of gentle algebras via Fukaya categories*, arXiv:1801.06370v5; category and smoothness hypotheses retained.

[94] Z. Cui, P. Zhang and Y. Tang, *Quantum Flow Matching*, arXiv:2508.12413v1; abstract-level comparison, not an upstream reproduction.

[95] L. Abrams, *Two-dimensional topological quantum field theories and Frobenius algebras*, J. Knot Theory Ramifications 5 (1996), 569–587, author-hosted paper.

[96] G. Segal, *The definition of conformal field theory*, in *Topology, Geometry and Quantum Field Theory* (2004); primary publisher metadata and sewing-scope comparison.

[97] S. Ryu and T. Takayanagi, *Holographic Derivation of Entanglement Entropy from AdS/CFT*, arXiv:hep-th/0603001v2; external setting only.

[98] S. Mochizuki, *The Absolute Anabelian Geometry of Hyperbolic Curves*, author-hosted paper, introduction and Theorem 1.3.4; arithmetic extension hypotheses retained.

[99] Z. Ji, A. Natarajan, T. Vidick, J. Wright and H. Yuen, *MIP*=RE*, arXiv:2001.04383v4 (16 October 2026); external theorem and version record, not rerun locally.

[100] T. Fukui, Y. Hatsugai and H. Suzuki, *Chern Numbers in Discretized Brillouin Zone*, arXiv:cond-mat/0503172v2, 2005.

[101] RDKit contributors, *MMFF94 force fields and ETKDG conformer APIs*, `https://www.rdkit.org/docs/`; executed version2026.03.1.

[102] Worldwide Protein Data Bank, *Usage Policies*, CC0-1.0 coordinate archive, `https://www.wwpdb.org/about/usage-policies`.

[103] F. Evers et al., *Theory of Chirality Induced Spin Selectivity: Progress and Challenges*, arXiv:2108.09998v1, 2021.

[104] A. Privitera et al., *Direct detection of spin polarization in photoinduced charge transfer through a chiral bridge*, arXiv:2205.05353v1, 2022.

[105] wwPDB, Atomic Coordinate Entry Format3.3, secondary structure and coordinate sections, accessed2026-11-04. `https://www.wwpdb.org/documentation/file-format-content/format33/sect5.html`.

[106] Y. Song et al., Score-Based Generative Modeling through Stochastic Differential Equations, arXiv:2011.13456v1 (2020).

[107] Y. Lipman et al., Flow Matching for Generative Modeling, arXiv:2210.02747v1 (2022).

[108] R. T. Q. Chen and Y. Lipman, Flow Matching on General Geometries, arXiv:2302.03660v1 (2023).

[109] The Univalent Foundations Program, *Homotopy Type Theory: Univalent Foundations of Mathematics*, first-edition-257-g5561b73 (2013), Sections 6.6–6.8, especially Remarks 6.7.1–6.7.2, pinned official edition.

[110] A. Jacot, F. Gabriel and C. Hongler, *Neural Tangent Kernel: Convergence and Generalization in Neural Networks*, arXiv:1806.07572v4 (2020). This is background; our diagonal holomorphic readout limit is derived separately.

[111] NIST Digital Library of Mathematical Functions, *Dirichlet L-functions*, Section 25.15, equations 25.15.1–25.15.5, primary reference, accessed 4 November 2026.

[112] F. Bornemann, *On the Numerical Evaluation of Fredholm Determinants*, arXiv:0804.2543v2 (2008), Section 4, equation (4.1); Math. Comp. 79 (2010), 871–915.

[113] C. Bamps and S. Pironio, *Sum-of-squares decompositions for a family of CHSH-like inequalities and their application to self-testing*, arXiv:1504.06960v1 (2015).

[114] M. Navascués, S. Pironio and A. Acín, *A convergent hierarchy of semidefinite programs characterizing the set of quantum correlations*, arXiv:0803.4290v1 (2008).

[115] Z. Ji, A. Natarajan, T. Vidick, J. Wright and H. Yuen, *MIP*=RE*, arXiv:2001.04383v4 (16 October 2026). External theorem, not re-proved or newly formalized here.